

微网与外部电网电力调控策略

——基于共形预测与滚动随机优化的分层调度与安全闭环控制模型

摘 要

微网通过在电价低谷或光伏富余时段充电、在电价高峰时段放电降低购电费用，但计划与执行受光伏功率、小区负荷与外网电价三类不确定性共同影响，紧急购电与调整购电之间的结算倍数差异使预测偏差表现为不同经济代价。本文以 2025 年全年 10 分钟粒度数据为对象，建立“数据清洗—共形预测—随机优化—滚动调整—安全闭环”的多时间尺度框架。

问题一以电价与负载相同、光伏发电功率已知为背景，建立含充放电损耗的时域扩展网络流线性规划模型，将储能调度转化为**广义凸网络流问题**。在一定条件下，通过构造保持储能状态不变的消环操作证明**充放电约束可精确松弛**，同时建立效率损耗和退化折旧对放电死区宽度的拓宽机制，作为全局理想参考基线。代表性日优化结果显示，储能使全天购电费用降至 35 126.95 元，相较无储能情形下降 26.90%，说明该确定性模型能为后续不确定性调度提供稳定成本基准。

问题二中进一步考虑小区负载与光伏发电功率随时间动态随机波动的贴近实际情况，构建出由最大熵残差建模、自适应共形校准、报童临界分位、最坏情形日前防守调度以及跨模块信息传递组成的**五层闭环决策架构**。在调度模型中嵌入动态防守包络下界，并配合不确定性场景与补购补救机制，以降低规划不可行风险。

问题三在第二问日前调度基础上，结合分时光伏预报与历史残差场景更新光伏不确定性，建立考虑非对称调整费用的**滚动调度模型**，冻结已执行动作，并以当前储电量为起点优化当日剩余时段，同时推导内部无调整带，解释原计划保持与购电调整的条件，并以 NAV 刻画同一信息条件下的理论重优化价值，通过实验分析预测更新带来的费用改善。

问题四利用 Student-T-Copula 构造负荷偏高、光伏不足和高电价共同出现的联合风险场景。构建融合偏差跟踪、CVaR 风险和软约束的**实时 MPC 理论模型**，并引入**CBF 安全过滤**与**动态事件触发机制**，分别承担经济优化、安全修正与指令更新功能。在 70 日计划层风险试验中，CVaR 由 89 773.78 元降至 79 809.97 元；另组 70 日独立执行试验未出现储电越界，指令下发次数减少 52.18%。两组试验分别对风险权重与执行安全进行检验。

基于 2025 年微网全年回测，储能在确定性场景下可通过跨时段能量转移显著降低购电成本，模型则通过**风险防守**、**信息价值判断**和**安全控制**逐层扩展决策能力。由此，四个问题在统一储能物理基座上，对信息由确定性到不确定性、由日前到实时逐层处理，最终形成连续决策体系。

关键词：共形预测；最大熵建模；两阶段随机规划；模型预测控制；条件风险价值；控制屏障函数

1 引言

1.1 研究背景与意义

随着光伏等分布式能源不断接入配电系统，微网成为实现新能源就地消纳和提高用电经济性的重要形式。由于光伏功率具有明显的日内波动特征，而居民负荷通常呈现早晚用电高峰，二者在时间上的不匹配使得部分光伏电能难以直接消纳。储能设备能够通过跨时段充放电调节源荷差异，使微网在满足负荷需求的同时合理安排购电，从而降低外部电网购电成本。

然而，实际调度受到光伏出力和负荷波动的影响，日前计划与实际运行之间不可避免地存在偏差。因此，调度策略不能只追求预测条件下的最低购电费用，还需要在经济性与供电风险之间进行权衡。

于是本题的核心问题可以概括为：在未来信息存在不确定性的情况下，如何制定购电与储能计划，并判断新的预测信息是否值得引起计划调整。问题 1 从确定性日前预测出发，问题 2 进一步考虑光伏与负荷的实际变化，问题 3 引入一天内多次光伏预报更新，问题 4 则进一步考虑外部电网电价的实时波动，形成由确定性调度向动态不确定性调度逐步扩展的问题体系。

因此，本文的研究意义主要体现在两个方面：一方面，通过协调外部电网购电与储能充放电，在满足负荷及储能运行约束的前提下降低购电支出；另一方面，通过分析预测误差和信息更新对既定计划的影响，研究何时调整以及调整多少能够获得实际经济收益。同时，题目对储能容量、充放电功率、荷电状态范围和充放电效率作出了明确规定，为调度策略提供了物理边界。

1.2 文献综述

围绕微网经济调度与新能源不确定性，已有研究主要从能量管理、不确定性建模和滚动优化等方面展开。微网能量管理中，模型预测控制和滚动优化能够利用未来预测信息动态调整储能与电网功率。文献 [2] 通过软约束处理光伏预测误差对储能荷电状态的影响，文献 [9] 则利用 MPC 协调储能与电网以维持微网能量平衡，说明储能既能够降低购电成本，也能够吸收新能源出力波动。

针对预测不确定性，文献 [1] 利用 Wasserstein 距离构造分布模糊集，并通过分布鲁棒机会约束降低传统鲁棒方法的保守性；文献 [12] 进一步将类似方法用于联网微网的可靠性调度。这类研究能够利用历史数据刻画新能源的不确定性，但风险水平通常需要预先设定，尚未直接体现不同预测偏差所对应的经济代价。与此同时，文献 [10] 利用自适应共形预测获得具有覆盖率保证的光伏预测区间，说明预测可靠性需要与区间锐度共同考虑。

在预测信息持续更新方面，文献 [7] 提出动态滚动时域方法，根据不同时间段的信息增益选择更有价值的更新时刻，为本文问题 3 分析多时刻光伏预报提供了参考。文献 [8] 进一步将预测过程与调度成本联系起来，强调预测结果的价值最终应通过下游控制性能和经济收益体现，而不能仅以预测误差衡量。这说明，对于微网调度而言，预测精度本身并不是最终目标，关键在于预测信息能否有效改善购电决策。

总体而言，现有研究分别从预测、风险和调度等角度提供了较为成熟的处理思路，但针对本题中特定的计划购电、紧急购电及调整购电结算机制，仍需要进一步建立预测不确定性与实际调度成本之间的联系。因此，本文在借鉴上述研究思路的基础上，结合题目给定的费用规则、储能约束和信息结构，建立适用于本题的微网购电调控模型。

1.3 本文工作与主要贡献

本文以 2025 年全年 10 分钟粒度数据为对象，建立并验证了一套完整的微网电力调控模型，主要工作包括：

- 建立含充放电损耗的时域扩展网络流线性规划模型，证明充放电互斥约束可以精确松弛，使储能调度问题在无需整数变量的条件下保持全局可解性；进一步利用离散动态规划独立验证最优解及步长收敛性质，为后续随机调度提供确定性基准。
- 针对光伏与负荷预测误差，建立基于最大熵残差建模与自适应共形校准的动态不确定性刻画方法，并结合报童模型推导保守度的经济最优设定。由调节与紧急供能的边际代价比得到一般临界分位 $p^* = 1 - \lambda_{\text{调节}} / \lambda_{\text{紧急}}$ ，其中 0.80 与 0.70 是单时段净需求的经济参照，0.25 为计划层标定值；储能耦合和反馈后的实际分位不由该公式直接保证最优。
- 针对预测信息分时到达所产生的计划调整问题，建立当日剩余时域的滚动 MPC 与计划调整模型，利用 Lead-time-aware ACI-CP 和场景生成方法更新未来不确定性，并结合非对称调整费用推导无调整带；进一步以同场景 NAV 定义理论优化价值，以事后费用差衡量实际增量价值，通过 S0-S5 子集实验比较不同预报时刻在夜间与日间场景下的作用，从而给出是否有必要增加预报时刻的定量判断。
- 针对实时电价及负荷、光伏联合波动，建立日前购电计划与十分钟级储能执行之间的时间尺度映射，并利用 Student- t Copula 描述高负荷、低光伏与高电价同时发生时的尾部相关风险。在此基础上构建含 CVaR 的软跟踪实时 MPC，分别从时间接口守恒与可实现性、尾部相关结构、风险优化模型的可解性与唯一性、风险加权无调整带，以及 CBF 安全过滤和 DETC 事件触发机制等方面给出理论条件，给出理论扩展的保证条件；实际执行实验以既有轨迹为名义动作，尚未实现六步 MPC。

需要说明的是，本文并未引入 GPU 算子加速与强化学习：决策规模为单日 144 时段、单次两阶段线性规划在数万非零元量级，向量化 CPU 实现已能秒级求解，不存在可支撑 GPU 收益的热点算子；储能虽具有序贯状态，但已知物理模型可直接优化，本题尚无学习控制独立收益证据。相关候选方案的适配性判定与未启用理由在第模型评价章统一登记，以保证文中结论均有实测证据支撑。

2 问题重述

本文研究微网在不同信息条件下的购电与储能调度问题。题目由确定性日前调度逐步扩展至预测不确定性、预测更新和波动电价场景，要求在满足负荷供电及储能运行约束的同时，降低正常购电、计划调整和紧急购电等综合成本。具体而言，需要解决以下四个问题：

1. **问题一：确定条件下的日前购电与储能调度。**在电价、负荷和光伏出力已知的条件下，从每日 0:00 制定全天购电与储能充放电计划，使全天购电成本最小，并满足储能容量、功率、效率及日初日末储电量约束。
2. **问题二：负荷与光伏波动条件下的日前计划。**考虑负荷和光伏预测误差，在每日 0:00 利用已有信息制定日前购电计划；若实际运行中供电不足，则进行高价紧急购电，并依据全年实际数据评价计划的经济性与风险。

3. **问题三：预测信息更新下的购电策略调整。**考虑 0:00、6:00、12:00 和 18:00 获得的新光伏预测，在不改变已执行计划的前提下判断新信息是否具有调整价值，并通过滚动优化确定后续购电计划，同时比较不同预测时刻的信息价值。
4. **问题四：波动电价下的购电与储能调度。**进一步考虑电价、负荷和光伏的联合波动，重新研究日前计划、日内调整及实时储能控制，并结合尾部风险控制和安全执行机制评价动态电价下的调度效果。

2.1 数据来源与预处理

本文使用的数据均来自题目附件及附录。附件 1 给出某一代表性日的电价、小区负荷与光伏预测功率，时间粒度为 10 分钟；附件 2 给出 2025 年 1 月 1 日至 12 月 31 日的小区负荷与光伏实际功率；附件 3 给出该年每天 0:00、6:00、12:00 和 18:00 发布的未来 24 小时整点光伏发电预报；附件 4 给出全年各时段电价。储能设备参数由附录 1 给定：最大容量 12 000 kWh，最大充放电功率 5 000 kW，2025 年 1 月 1 日 0:00 储电量为 6 000 kWh，储电量范围为 1 200–10 800 kWh，充放电效率均为 90%。

数据质量检查发现，光伏实测序列存在结构性的夜间零漂噪声，即实际夜间出力应为零，而原始数据在部分夜间时段出现集中小幅非零值；同时存在少量明显超出正常范围的异常点。固定时钟窗口难以适应季节性日照变化：本数据中核夜间时长的月中位数由 6 月的 10.33h 变化至 12 月的 14.33h，日出时刻的月中位数由 6 月的 5:10 推迟至 12 月的 7:10，季节间变化约 4h。

因此，本文依据每日实测光伏序列自适应确定夜间窗口。对第 d 日取

$$\theta_d = \max\{0.01 \max_k P_{d,k}^{PV}, 1.0\},$$

其中 θ_d 的单位为 kW。根据实测序列首次和末次越过该阈值的时段确定日间区间，并将全天划分为核夜间 (CORE)、肩部 (SHOULDER) 和日间 (DAY) 三个区域。该窗口仅由当日实测光伏确定，不使用预测信息，也不跨日借用数据。

在此基础上，按照附录表中的 12 条值级清洗规则 (NR1–NR12) 进行处理，包括核夜间非零小值归零、异常大值标记为缺失并隔离、肩部亚千瓦级噪声归零，以及日间短缺失段线性插值；连续缺失超过 6 个时段不进行插值。电价与负荷仅进行异常标记和完整性检查，不直接修改原始取值。所有修改均保留审计记录，并在导出阶段检查 I1–I8 八项不变量。

清洗共生成审计记录 6 229 条，隔离异常数据点 129 个，核夜间零值格数由 23 316 增至 25 766，净增加 2 450 格；08:00–16:00 区间的零值数量仍为 0，低于 0.1 元/kWh 的近零电价数据标记 28 格。单次完整清洗耗时 0.159s，峰值内存占用 48.7 MB。

清洗后的主要时空特征见附录：光伏整体呈夏季日照时段较宽、冬季较窄的单峰形态；电价具有明显的早晚双峰特征，夏季午间低谷更深；净负荷在春夏部分正午时段出现负值，说明存在可由储能吸收的光伏富余电量。

对于问题 2–4，统一采用 2025 年 2 月 1 日至 12 月 31 日作为回测区间，共 334 天。2025 年 1 月作为唯一可用的初始校准期，仅用于生成初始预测结果及共形校准尺度，不参与后续回测结果的统计与评价。

2.2 模型假设

为统一各问题的物理口径并避免信息泄漏，作如下假设；其中可扰动参数通过灵敏度分析进一步检验。

H1. 时段口径一致。附件时间标签视为对应十分钟时段的右端点，第 k 个时段为 $(10(k-1), 10k]$ min，功率按时段平均值计。若实际采用左端点口径，则仅产生一个时间步的整体平移，不改变全天总量。

H2. 储能采用等效效率模型。仅考虑充放电转换损耗， $\eta_c = \eta_d = 0.9$ ，储能状态满足

$$E_{t+1} = E_t + \eta_c c_t - \frac{d_t}{\eta_d}.$$

忽略自放电以及温度、倍率等因素造成的效率和可用容量变化，容量、功率及安全区间均采用题设参数。

H3. 充放电互斥采用连续松弛。不显式引入 0-1 变量，而利用 $0 < \eta_c \eta_d < 1$ 、允许弃电等条件下的最优解性质保证至少存在互斥最优解；在正购电时段，任意最优解均不会同时充放电，因此加入显式互斥约束不改变最优值。若出现负电价或强制充放电，则重新检验该性质。

H4. 信息集与回测纪律。在线模型要求只使用已发布或已兑现信息。Q4-2/4-3 按题目给定价格路径重算；价格未知的六步控制另作理论扩展。归档预报清洗 NR7 使用目标日实测推断的夜间窗口，含后验信息，故现有 Q3 及 Q4-3 费用属于清洗数据上的条件回测，不能称为严格在线成绩；统计历史窗早于当日并不能消除这一预处理问题。

H5. 日循环与跨日状态连续。问题 1-3 采用

$$E_T = E_0 = 6000 \text{ kWh},$$

避免通过消耗初始储电量人为降低购电成本。进入全年回测后，计划层每日求解以 6000 kWh 初始化，并强制同值终点，因而其理想模型内与逐日传递等价；Q4 触发后实际轨迹不保证这一等价性。该终端条件作为统一基准口径，并在后续分析中考察其影响。

2.3 总体建模思路

四个问题形成由确定性到不确定性、由日前计划到日内控制的递进建模链条：问题 1 建立含储能损耗的确定性调度基准；问题 2 引入预测误差并形成具有风险防守能力的日前计划；问题 3 利用分时预测更新，按启用发布时次进行滚动调整；问题 4 进一步考虑波动电价及联合不确定性，在日前计划基础上引入风险控制与实时安全执行。各问题的模型、算法与验证方式见表1，总体结构见图1。

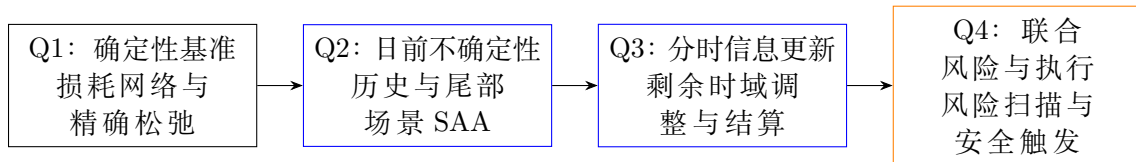


图 1 四问的建模关系。箭头表示问题设置的扩展；各问信息条件与统计周期不同，不表示总费用必然单调下降。六步软 MPC 为理论扩展。各层电量均按十分钟网格记录，Q4 通过除以 δ 转为功率。

3 符号说明与变量定义

全文符号统一遵循“能量用时段电量、状态用边界时刻储电量、执行层使用功率、其余调度层使用时段电量”的约定；功率换算为电量时一律乘以 Δt 。表 2 给出全文符号的唯一口径，第 4 章及其后各章的记号均与此一致。

表 1 各问题的建模方法与验证方式

问题	建模目标	主要模型与求解方法	验证方式
1	确定性日循环调度	含损耗的时域扩展网络流 LP、对偶分析	动态规划互证、基线对比、约束核验
2	随机条件下的日前计划	历史残差情景 + 两阶段随机规划、报童分位标定	覆盖率检验、基准对照、完美信息界
3	预测更新下的计划调整	四时次滚动 MPC、非对称调整成本与信息价值分析	S0-S5 实验、NAV 分析、逐日结算复算
4	波动电价下的风险控制	计划 CVaR 扫描、CBF 与 DETC；联合场景 MPC 为扩展	帕累托前沿、参数扫描、消融及安全审计

表 2 全文符号说明

符号	含义	单位	类型
集合与索引			
k, d, j, s	时段 ($k = 1, \dots, T$)、日期 (334 天)、预报时次 (0/6/12/18 时)、情景索引	—	索引
$R(d), \mathcal{C}_d, \mathcal{D}_d$	决策日 d 的历史信息窗口 $\{d' : \max(0, d - 30) \leq d' < d\}$ 、核夜间与日间时段集合	—	集合
参数与常量 (括号内为本题取值)			
$T, \Delta t$	每日时段数与时段长度 (144, 1/6)	—, h	常量
$\pi_k, L_k, P_k^{\text{PV}}$	第 k 时段电价、负荷功率、光伏功率	元/kWh, kW	参数
N_k	净负荷功率 $N_k = L_k - P_k^{\text{PV}}$	kW	派生参数
$E^{\text{rated}}, E_{\min}, E_{\max}, S_0$	额定容量、储电量上下界、初始储电量 (12000, 1200, 10800, 6000)	kWh	常量
$\bar{P}, \bar{C}, \eta_c, \eta_d, r$	最大充放电功率与时段电量上限、充放电效率与往返效率 (5000, 2500/3, 0.9, 0.9, 0.81)	kW, kWh	常量
p^*	计划层或调整层的风险分位数 (Q2 为 0.80; Q3 为 0.25/0.70)	—	标定量
$\lambda_{\text{emg}}, \lambda_{\text{adj}}, \theta_{\text{over}}$	紧急购电价、调整价、超计划违约相对电价的倍数 (5, 1.5, 0.5)	—	常量
σ_j, α, γ	共形尺度、名义显著性水平、ACI 反馈步长 ($\gamma = 0.02$)	kW, —	参数
$\alpha_{\text{cv}}, \beta, \rho_1, \rho_2$	CVaR 置信水平 (0.90)、风险权重、松弛惩罚与控制平滑权重	—, 元/kWh	参数
决策变量 (均为时段电量或边界时刻储电量, 单位 kWh)			
q_k	计划购电量	kWh	决策变量
$q_k^{\text{plan}}, q_k^{\text{adj}}$	计划购电量与调整购电量	kWh	决策变量
c_k, d_k	储能充电量与放电量, $0 \leq c_k, d_k \leq \bar{C}$	kWh	决策变量
e_k	第 k 时段末储电量, $e_k \in [E_{\min}, E_{\max}]$	kWh	状态变量
w_k, g_k, ξ_k	未利用供给电量、紧急购电量 (情景相关记为 $g_k^{(s)}$)、软约束松弛量	kWh	决策/追索变量
统计量、对偶量与评价指标			
$\varepsilon_{d,k}^{(j)}, \hat{Q}_\alpha(\cdot)$	第 j 时次第 d 天第 k 时段的光伏预报残差; 有限样本修正的分位数算子	kW	统计量
μ_k, λ_k	购电边际价值、储电量机会价值	元/kWh	对偶量
$CURI, CURI_\sigma, NAV$	区间宽度比、残差尺度比与同场景优化价值	—, 元	指标
ζ, z_ω	CVaR 线性化中的分位辅助变量与超额损失变量	元	辅助变量
问题 4 十分钟执行层 ($k = 0, \dots, 5$ 为局部提前步, τ 为全局时段)			
$S_\tau, S_\tau^{\text{ref}}$	全局十分钟时段末的实际储电量与上层下发的参考储电量	kWh	状态/参考
P_τ^{contract}	生效十分钟合同电量除以时段长度	kW	派生参数
$u_\tau = (P_\tau^{\text{ch}}, P_\tau^{\text{dis}})$	十分钟充/放电力指令 (集合 $\mathcal{U}_\chi(S)$ 为满足 CBF 与功率界的安全集合)	kW	决策变量
$b_\tau \in \{0, 1\}$	充放电模式变量; $u_\tau^{\text{last}}, u_\tau^{\text{safe}}, u_\tau^{\text{real}}$	—, kW	决策/中间量
$Z_{k,a} (a = 1, 2, 3)$	方向统一的负荷、光伏、电价残差	kW, 元/kWh	随机变量
R_C, ν, φ	Student- t Copula 尺度相关矩阵、自由度、时间自相关参数	—	估计参数
$\lambda_U, \lambda_L, \rho$	上下尾依赖系数与尺度相关	—	统计量
$r_k, \Delta E_k$	供需偏差与储电量偏差; $x_k^{\text{dev}} = (r_k/P_0, \Delta E_{k+1}/E_0)$	kW, kWh	中间量
$s_{k,\omega}^P, s_k^E, b_P, b_E$	软约束松弛量与跟踪容许偏差	kW, kWh	松弛变量/参数
$\chi, \sigma_\eta, \gamma_0, \lambda_\eta$	CBF 余量消耗率、DETC 权重、阈值系数与衰减率; η_τ^{tr} 为内部触发状态	—	参数/状态
$A_\omega(P), m_w, \tilde{\pi}, \bar{w}_\omega$	场景下的调整费、有效权重下的边际收益与风险价格及加权重	元, 元/kWh	中间/对偶量

4 模型建立

本章以共同物理约束为起点，依次给出确定性调度、日前两阶段计划、分时滚动调整与实时安全控制。

4.1 共同基础

4.1.1 时段能量、决策时序与费用

一天含 $T = 144$ 个十分钟时段， $\mathcal{T} = \{1, \dots, T\}$ ， $\Delta t = 1/6$ h。输入功率换算为电量： $\ell_t = L_t \Delta t$ 、 $v_t = P_t^{\text{PV}} \Delta t$ 。各问沿用计划购电 q_t 、充放电 c_t, d_t 、期末储电量 e_t ；紧急购电记为 g_t ，均为 kWh。基线只计题设购电与调整费用，吞吐退化单价取零；有依据的退化定价作为附录扩展。储能可行轨迹集合 $\mathcal{X}(a, b; s)$ 由下式在 $t = a, \dots, b$ 上定义：

$$e_t = e_{t-1} + \eta_c c_t - d_t / \eta_d, \quad 0 \leq c_t \leq \bar{C}, \quad 0 \leq d_t \leq \bar{D}, \quad (4.1)$$

$$E_{\min} \leq e_t \leq E_{\max}, \quad e_{a-1} = s, \quad e_b = S_0. \quad (4.2)$$

其中 $\eta_c = \eta_d = 0.9$ 、 $\bar{C} = \bar{D} = 5000 \Delta t$ ， $(E_{\min}, E_{\max}, S_0) = (1200, 10800, 6000)$ kWh。问题一的日循环是题设要求；问题二三延续它作为比较基准，计划每日从 6000 kWh 起算并回到同值，理想模型内等价于跨日传递。问题四的执行层保持硬安全界，直接报告未修补终端偏差。滚动调用前还需检查终端可达性： $-(b-a+1)\bar{D}/\eta_d \leq S_0 - s \leq (b-a+1)\eta_c \bar{C}$ 。在合法初值、无额外爬坡约束下，该条件也是可达的充分条件（将总增量均匀分配即可）；不满足时不能强行宣称日循环可行。给定输入时，供需关系为

$$q_t + v_t + d_t - \ell_t - c_t = w_t \geq 0. \quad (4.3)$$

w_t 是未利用供给，允许在富余时退出系统。正电价下，确定性最优解不会同时购电和弃用供给；但随机模型的计划已预付，实际出现多购电并不违反这一确定性结论。

4.1.2 统计接口与保证范围

残差统一定义为实现值减发布时预测值：第二问对象是净负荷，第三问对象是光伏，第四问再将残差方向变换为越大越不利。带日期的预测与得分只在实际值兑现后进入校准窗。固定评分规则下的无权重秩分位可给出交换性覆盖保证；工程使用的时间加权分位及投影 ACI 则分别作为经验估计和在线反馈，不能直接套用该保证。MaxEnt 用于描述边缘分布形状，经济分位用于解释代价权衡；两阶段 SAA 直接通过场景及其概率计入风险，不再把双侧区间下界当成另一个同时生效的硬防守约束。

4.1.3 求解策略总览

第一问含 575 个变量，30 个场景。问题二包含 4895 个变量，不能把前者规模用于所有模型。六步软 MPC 保留互斥模式，是尚未接入生产求解的混合整数凸二次规划；其理论可解性与实测耗时分别报告。

具体而言，首先对各时段的供需平衡、储能状态及购电约束进行稀疏矩阵装配，再调用 HiGHS 求解器进行线性规划求解。对于问题一，进一步读取供需平衡约束与储能动态约束的对偶变量 μ_k, λ_k ，用于分析储能充放电的边际价值；同时采用独立的储电量离散动态规划进行交叉验证。对于问题二到四，则按照决策时点实际可获得的信息逐日

构造预测情景，并采用确定性的情景分层与削减方法完成全年回测。整个计算流程固定数据处理顺序和参数，不依赖随机采样，以保证输入与算法口径可复现；退化线性规划可能有多条等价最优轨迹，不要求跨求解环境逐元素一致。

在实现层面，数据清洗与异常审计采用 C++17 完成，预测、优化、滚动控制、结算及结果统计采用 Python 实现。各问题的单日求解过程均封装为独立计算模块，全年回测通过统一接口循环调用，并在求解后自动执行约束、边界和结果一致性检查。由此形成稀疏建模–独立验证–批量回测–自动审计的计算流程。

4.1.4 复杂度与计算架构

问题 1 的规划规模随时间段数线性增长，单次求解不足 1s；问题 2 引入情景后，模型规模近似随情景数线性增加，在 $S = 30$ 时单日求解约 19.6 ms。动态规划的基本复杂度为

$$O(TN_s|A|),$$

其中 $\Delta e = 15 \text{ kWh}$ 时储能状态数为 641。

全年计算包括 334 次问题 2 两阶段线性规划、问题 3 的滚动重优化，以及问题 4 的 70 天十分钟闭环执行。单机 32 核 CPU 环境下，各阶段提供代码包的历史记录总耗时约 96 s，原记录完整流程墙钟时间约 2 min。其中单日线性规划已达到毫秒级，因此虽然独立日任务具备并行条件，但进程调度和数据复制的额外开销会削弱并行收益，主流程仍采用串行执行。对于本题规模，矩阵维度和非零元数量均较小，GPU 启动与数据传输成本高于计算收益，故不引入 GPU 加速。

在实现层面，数据清洗采用 C++17 完成，预测、优化、滚动结算和结果统计采用 Python 实现；情景生成、分层削减和日级求解均采用确定性流程，不依赖随机抽样。由此既保留了计算机实现上的效率优势，也避免将与模型结论无直接关系的程序细节作为算法贡献。

4.2 问题 1：确定性调度、求解算法与结果分析

4.2.1 问题 1 模型：确定性调度与储能边际价值

给定代表日负荷、光伏和电价，全天购电模型为

$$\min_{q,c,d,e} C = \sum_{t=1}^T \pi_t q_t, \quad (4.4a)$$

$$q_t + v_t + d_t - \ell_t - c_t \geq 0, \quad q_t \geq 0, \quad (4.4b)$$

$$e_t = e_{t-1} + \eta_c c_t - d_t / \eta_d, \quad (4.4c)$$

$$(c, d, e) \in \mathcal{X}(1, T; S_0). \quad (4.4d)$$

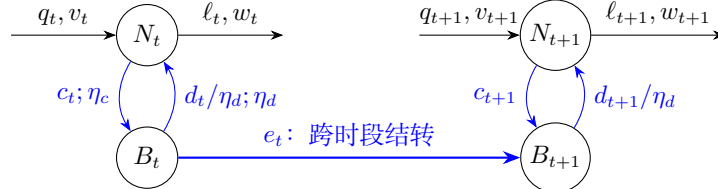
图 2 将同一模型写成带效率的时域网络：母线 N_t 与储能节点 B_t 通过充放电弧连接，储能节点通过跨时段弧保留电量。网络中的局部损耗环解释了为何可以求解连续线性规划。

命题 4.1 (互斥最优解与精确松弛). 若 $\pi_t > 0$ 、 $0 < r = \eta_c \eta_d < 1$ ，允许富余供给退出且无强制充放电下限，则至少存在满足 $c_t d_t = 0$ 的最优解；任意最优解在正购电时段均互斥。加入互斥约束不改变最优值。

证明的关键是同时减少充电 δ 与母线侧放电 $r\delta$: 储能状态不变, 净供给增加 $(1-r)\delta$ 。正购电时取足够小的步长, 有

$$\Delta C = -(1-r)\pi_t\delta = -0.19\pi_t\delta < 0. \quad (4.5)$$

零购电时通过增加富余量等成本消环。因此求解后出现退化循环应作可行消环处理, 不能立即判定定理被违反。若加入正的吞吐退化价格, 消环在零购电时也严格降本; 相应加强结论及完整网络定义见附录。



消环: $(c_t, d_t) \mapsto (c_t - \delta, d_t - r\delta)$; 状态不变, 净供给增加 $(1-r)\delta$

图 2 储能时域网络与消环机制。弧旁的“流量; 增益”按发送端定义, 放电弧到达母线的电量为 d_t 。跨时段弧连接储能节点; 允许弃用供给时, 损耗环可从最优解中消除。图为理论示意, 不是仿真结果。

命题 4.2 (日循环能量核算). 所有可行日循环轨迹满足

$$\sum_t d_t = r \sum_t c_t, \quad \sum_t q_t = \sum_t (\ell_t - v_t) + \sum_t w_t + (1-r) \sum_t c_t. \quad (4.6)$$

该恒等式可将购电变化分解为净需求、富余供给与储能损耗。

取供需、储能等式的乘子为 μ_t, λ_t 。KKT 条件给出

$$0 \leq \mu_t \leq \pi_t, \quad q_t > 0 \Rightarrow \mu_t = \pi_t, \quad w_t > 0 \Rightarrow \mu_t = 0, \quad (4.7)$$

$$a_t = \mu_t - \eta_c \lambda_t, \quad b_t = -\mu_t + \lambda_t / \eta_d. \quad (4.8)$$

充电约化成本 a_t 为正、负、零时, 分别取 $c_t = 0, \bar{C}$ 或允许内点; 放电按 b_t 同理。特别地, 零动作要求

$$\eta_c \lambda_t \leq \mu_t \leq \lambda_t / \eta_d. \quad (4.9)$$

这不是只按外网电价划分的静态峰谷规则: 弃光时母线电能价值可为零, 储能价值还受跨时段容量约束影响。当储电量上下界乘子为 $\underline{\nu}_t, \bar{\nu}_t$ 时, 内部时刻有

$$\lambda_{t+1} - \lambda_t = \bar{\nu}_t - \underline{\nu}_t. \quad (4.10)$$

因此机会价值在容量不受限时保持不变, 在触及边界时才可能跳变。若在时段末注入外生储电量 ξ_t , 最优值可微时 $\partial C^* / \partial \xi_t = -\lambda_t$; 不可微时用值函数次梯度解释。一笔从 i 到 j 的可行储能转移只有在 $\pi_j > \pi_i / r$ 时有利, 该局部条件不能代替全天优化。

4.2.2 问题 1 求解算法: 稀疏线性规划与动态规划互证

问题 1 采用“稀疏线性规划求解 + 充放电性质检查 + 离散动态规划复算”的三步策略。首先根据式 (4.1)–(4.3) 装配稀疏约束矩阵并求解全天最优调度; 随后检查命题 4.1 所给出的充放电互斥性质; 最后利用离散动态规划从另一条计算路径独立复算最优费用, 并比较不同离散步长下的收敛结果。

4.2.3 问题 1 结果分析：代表性日的最优计划

在附件 1 的代表性日上求解模型 (4.4a)–(4.4d)，得全天购电量 59 482.70 kWh、购电费 35 126.95 元。题目规定格式的指定时段购电量如表 3 所示，充放电量与 0:00、24:00 储电量如表 4 所示（两端均为 6 000.00 kWh）。

表 3 微网在指定时间段的购电量及全天的购电量和购电费（问题 1，题目表 1 格式）

时间段	购电量 (kWh)	时间段	购电量 (kWh)	时间段	购电量 (kWh)
10:00-10:10	0.00	12:00-12:10	486.40	—	—
14:00-14:10	0.00	16:00-16:10	394.93	—	—
18:00-18:10	636.98	20:00-20:10	0.00	—	—
全天购电量 (kWh)		59 482.70		全天购电费 (元): 35 126.95	

表 4 储能设备在指定时间段的充放电量及 0:00、24:00 储电量（问题 1，题目表 2 格式）

时间段	充电量	放电量	时间段	充电量	放电量
0:00-4:00	4 500.00	0.00	4:00-8:00	833.33	6 365.84
8:00-12:00	4 787.96	1 703.00	12:00-16:00	5 286.04	91.10
16:00-20:00	0.00	5 780.13	20:00-24:00	5 333.33	2 859.87
0:00 储电量: 6 000.00		24:00 储电量: 6 000.00		单位: kWh	

该策略的运行特征如图 3 所示。由于核夜间电价处于全天较低水平，储能主要在低价时段充电，并在后续高价时段释放电能。统计结果表明，全天充电量的 51.4% 集中于核夜间，放电量的 76.0% 同样发生在该时段附近，说明在本题参数下，储能的主要经济作用并非单纯消纳光伏，而是利用电价差异实现电能的跨时段转移。

这一结论也可以由模型的对偶变量得到验证。图 3(c) 给出了各时段的边际电价 μ_k ：在购电时段有 $\mu_k = \pi_k$ ，而在放电时段满足 $\pi_k \geq \mu_k$ 。表明储能是否充放电取决于当前购电价格与储能电量边际价值之间的比较关系，与式 (4.9) 所给出的充放电阈值条件一致。因此，LP 的最优调度结果不仅给出了具体的充放电方案，也揭示了储能进行跨时段套利的价格驱动机制。

4.3 问题 2：净负荷不确定性下的日前两阶段调度、求解算法与结果分析

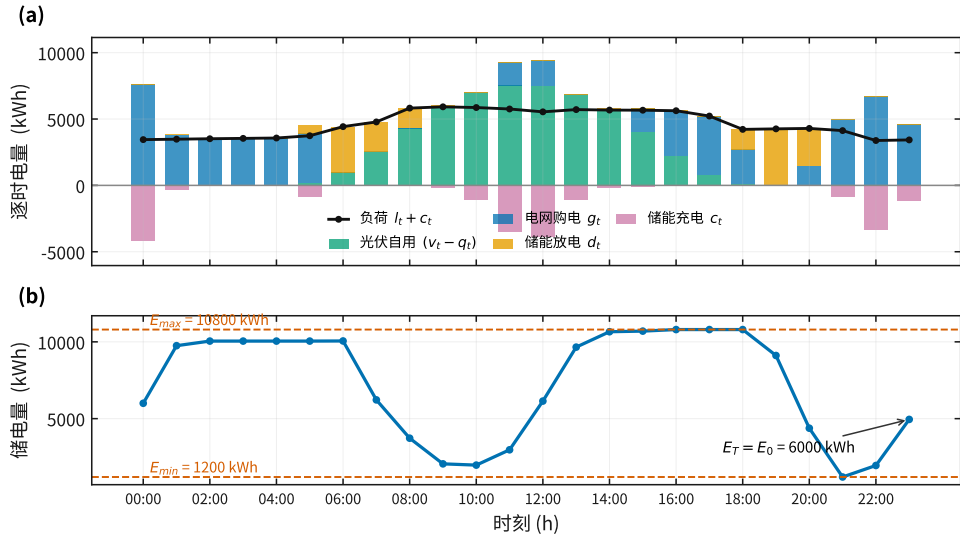
4.3.1 问题 2 模型：净负荷不确定性下的日前两阶段调度

日前计划需要权衡预付电费与事后 5 倍紧急购电费。主模型在 0:00 确定购电和储能轨迹，实际净负荷兑现后只补足电量缺口；采用场景平均近似 (SAA) 把这一两阶段决策写成 LP。本问的决策不调用附件 3；代码中的光伏下界输出是另行诊断，不进入本问 LP。历史净负荷 $N_{d,t} = L_{d,t} - P_{d,t}^{\text{PV}}$ 来自附件 2，预测与场景仅使用决策日前已完成的日期。

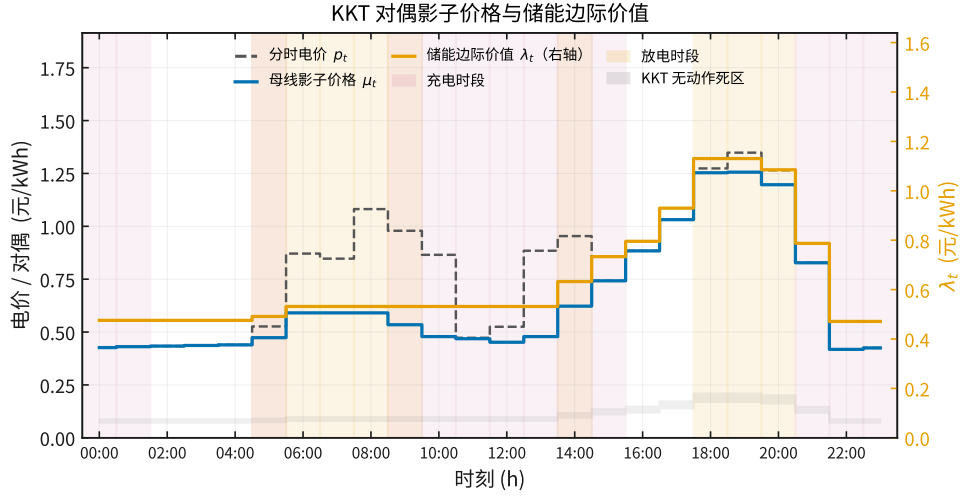
残差分布、校准与经济分位。 以当前因果窗口 $R(d)$ 计算 $\hat{N}_{d,t}$ ；重采样偏差为 $\epsilon_{d',t|d} = N_{d',t} - \hat{N}_{d,t}$ ， $d' \in R(d)$ 。它与历史发布预测的误差 $\epsilon_{d',t} = N_{d',t} - \hat{N}_{d',t|d',0}$ 不同：前者用于重建历史场景，后者用于检验真正的预测覆盖。不可用重新拟合的历史误差替代发布时误差。对任一明示的残差样本，选定历史支撑 $[-B, B]$ 并归一化为 $X = \epsilon/B$ ，用前 M 阶矩 $\hat{m}_j$ 建立

$$\max_{f \geq 0} - \int_{-1}^1 f \log f dx, \quad \int f dx = 1, \quad \int x^j f(x) dx = \hat{m}_j \quad (j = 1, \dots, M). \quad (4.11)$$

日前能量平衡与 SOC 轨迹



(a) 日前能量平衡与储电量轨迹



(b) KKT 对偶影子价格与储能边际价值

图 3 问题 1 的物理机制与最优性证据。(a) 逐时能量平衡（上）与储电量轨迹（下），全程位于 $[1200, 10800]$ kWh 且满足 $E_T = E_0 = 6000$ kWh；(b) 母线影子价格 μ_t 、储能边际价值 λ_t （右轴）与充放电时段，KKT 死区以解析边界给出。

命题 4.3 (最大熵分布及得分映射). 若矩向量位于 $[-1, 1]$ 的矩体内部, 则最优密度唯一, 且

$$f^*(x) = Z(\theta^*)^{-1} \exp[-\theta^{*\top} v(x)], \quad Z(\theta) = \int_{-1}^1 e^{-\theta^\top v(x)} dx, \quad (4.12)$$

$$p^*(\varepsilon) = B^{-1} f^*(\varepsilon/B) = \exp[-\lambda_0 - \sum_{j=1}^M \lambda_j \varepsilon^j], \quad |\varepsilon| \leq B. \quad (4.13)$$

这里 $v(x) = (x, \dots, x^M)^\top$, $\lambda_0 = \log[BZ(\theta^*)]$, $\lambda_j = \theta_j^*/B^j$ 。对偶函数 $\Phi = \log Z + \theta^\top \hat{m}$ 满足

$$\nabla \Phi = \hat{m} - \mathbb{E}_\theta v(X), \quad \nabla^2 \Phi = \text{Cov}_\theta[v(X)]. \quad (4.14)$$

绝对得分 $R = |\varepsilon|$ 的分布为

$$G(s) = F_\varepsilon(s) - F_\varepsilon((-s)^-), \quad s \geq 0. \quad (4.15)$$

变分

指数族是密度的解析形式, 乘子仍需数值求解。支撑由历史冻结, 不是未来误差的物理硬界;

模型阈值 $G^{-1}(1 - \alpha)$ 与经验共形阈值不是同一个保证。后者在无权重、固定评分规则下取

$$j = \lceil (n+1)(1 - \alpha) \rceil, \quad b^{\text{CP}} = \begin{cases} R_{(j)}, & j \leq n, \\ +\infty, & j = n+1. \end{cases} \quad (4.16)$$

对固定时段, 以 i 编号已经兑现的检验事件, $I_i = \mathbb{I}\{Y_i \notin \mathcal{C}_i\}$ 。对于逐次检验或按日平均误覆盖 $I_i \in [0, 1]$, 反馈形式为

$$\alpha_{i+1} = \Pi_{[\alpha_{\min}, \alpha_{\max}]}[\alpha_i + \gamma(\alpha^* - I_i)]. \quad (4.17)$$

命题 4.4 (ACI 路径恒等式及其边界). 记投影改变量为 r_i^Π , 则

$$\frac{1}{n} \sum_{i=1}^n I_i - \alpha^* = -\frac{\alpha_{n+1} - \alpha_1}{\gamma n} + \frac{1}{\gamma n} \sum_{i=1}^n r_i^\Pi. \quad (4.18)$$

无投影且边界外分别采用全域/空集时, 状态有界, 平均误覆盖趋于目标值; 投影实现必须保留修正项, 不能无条件宣称精确跟踪。

MaxEnt 刻画形状, ACI 根据兑现的覆盖误差调节阈值; 两者均不自动赋予整日联合覆盖保证。时间衰减权重与有限输出回退按经验规则记录, 不能继承式 (4.16) 的秩保证。

单时段净需求为 D 时, 期望费用 $J(x) = \pi x + 5\pi \mathbb{E}(D - x)_+$ 。连续分布内点满足 $J'(x) = \pi - 5\pi[1 - F_D(x)] = 0$, 故

$$F_D(x^*) = 1 - \frac{1}{5} = 0.80. \quad (4.19)$$

这是单侧需求的经济参照。双侧 80% 绝对残差区间不能直接解释为 20% 的光伏下尾分位; 储能耦合下也不能把 0.80 当作各时段最优覆盖定理。SAA 通过费用函数直接权衡风险, 分位诊断不重复叠加为硬约束。

图 4 将历史残差分布与 MaxEnt 拟合密度对比, 说明 Q2 的场景生成没有采用正态假设, 而是用矩约束刻画残差形状; 其作用是支持“残差形状可检验”, 而非宣称真实尾部已被精确识别。

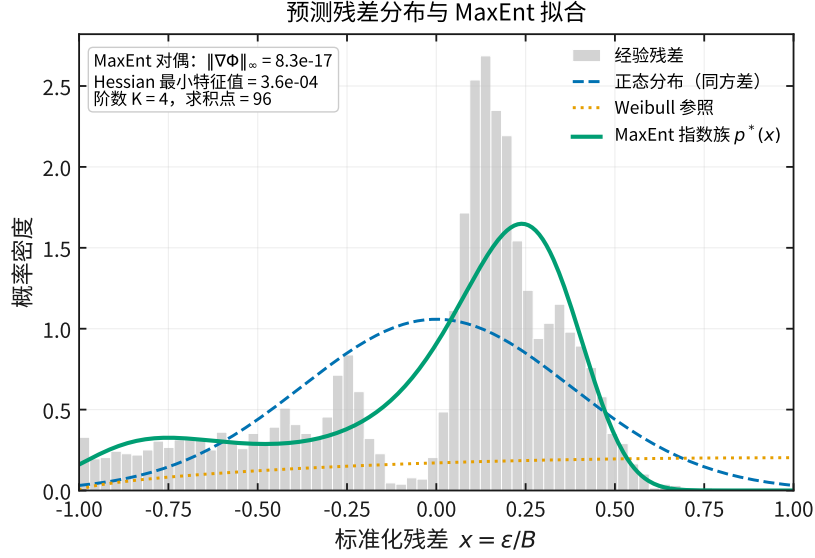


图 4 预测残差分布与 MaxEnt 拟合。净负荷残差直方图上叠加极大熵指数族密度 $p^*(x)$ 与同方差正态参照。MaxEnt 由前四阶矩以凸对偶求解，334 个因果窗口的梯度中位数 $1.9e-16$ 、最大 $1.2e-5$ 。

两阶段 SAA 与实际结算 保留历史日内整条净负荷路径，按日总残差排序确定性选取 29 条，再加入一条 MaxEnt 有符号尾部路径，共 30 条等权场景 ($\omega_s = 1/30$)。具体将各时段残差按历史 90% 绝对分位尺度标准化并池化，以 99.5% 绝对分位确定支撑，截尾后拟合四阶矩；同一形状分位乘回各时段尺度，构成尾部路径。这条共同尾部路径是压力场景，不是从真实联合分布独立抽取的样本，等权赋值也不等于已识别尾事件概率。令 $n_t^{(s)} = N_t^{(s)} \Delta t$ 。主模型为

$$\min_{q,c,d,e,g} \sum_t \pi_t q_t + 5 \sum_s \omega_s \sum_t \pi_t g_t^{(s)}, \quad (4.20a)$$

$$q_t + g_t^{(s)} + d_t - c_t \geq n_t^{(s)}, \quad q_t, g_t^{(s)} \geq 0, \quad (4.20b)$$

$$(c, d, e) \in \mathcal{X}(1, T; S_0). \quad (4.20c)$$

q, c, d, e 对所有场景共用，因此发布的储能动作不依赖未来实现；第二阶段只有 $g^{(s)}$ ，不允许场景各自选择一条事先不可知的充放电轨迹。MaxEnt 尾部路径实际进入优化；未经单独消融不能把全部降本归因于它。代码以均值 LP 储能轨迹 (c^D, d^D) 构造阈值 $b_t = Q_{\bar{p}}((N_t^{(s)} \Delta t + c_t^D - d_t^D)_s)$ ，却用 $N_t^{\text{act}} \Delta t > b_t$ 的日均指示量更新 α 。两侧储能项不一致，因此本文将这一路反馈记作经验裕度调节，不称为执行缺额覆盖率；严格覆盖校准应以同一目标量评分并重新回测。

命题 4.5 (SAA 可解性与可执行追索). 有限场景、正电价、合法初值和无额外购电上限下，式 (4.20) 存在有限最优解，且至少存在共同储能轨迹互斥的最优解。实际执行固定日前轨迹，紧急购电和结算为

$$g_t^{\text{act}} = [N_t^{\text{act}} \Delta t + c_t - d_t - q_t]_+, \quad C_d^{\text{act}} = \sum_t \pi_t q_t + 5 \sum_t \pi_t g_t^{\text{act}}. \quad (4.21)$$

经验场景外也可补购，因此物理供电可行不等于费用风险受到某一概率上界控制。问题三重新利用附件 3 发布的光伏预报制定 0:00 计划；它继承统计方法和储能状态，不把问题二的数值计划当作不可更改输入。

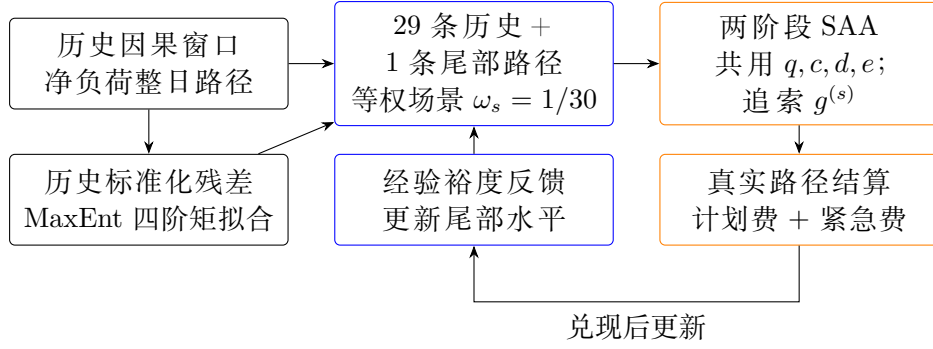


图 5 Q2 实际决策链：历史代表路径与 MaxEnt 尾部路径共同进入 LP。兑现后的经验反馈调节尾部水平；独立覆盖诊断不构成储能执行缺额的概率保证。

4.3.2 问题 2 求解算法：334 天信息约束回溯

问题 2 需要针对每天的预测状态独立构造不确定性情景并求解两阶段随机规划。为避免回溯过程中引入未来信息，情景构造严格限定在**决策时点已经获得的历史数据**内。对于第 d 天，仅利用此前已经完成的样本计算预测残差，不读取当日及未来日期的实际运行结果，从而保证回溯过程与实际决策的信息条件一致。

为提高计算效率并保证结果可重复，历史残差按照每日总残差水平排序，再采用等距分位点进行确定性分层，保留 29 条历史路径并增加 1 条 MaxEnt 尾部路径，等权进入 LP，不进行随机抽样。0.80 作为初始经济参照，尾部水平由经验反馈动态调整；实现层则在情景集上求解两阶段线性规划，其储能轨迹跨场景共用，只有紧急补购随场景变化。

全年回溯按照统一流程逐日执行：

信息截取 → 残差构造 → 情景分层 → 两阶段 LP 求解 → 实际结算 → 覆盖率更新。

其中第 d 天可使用的历史样本限定为

$$[\max(0, d - 30), d),$$

即最多使用决策日前连续 30 天的已完成数据。每日优化完成后，以当天实际净负荷进行事后结算，并将覆盖结果反馈至 ACI 更新模块。

为检验算法结果的稳定性，设置无裕度计划、解析报童方案、终端价值对照和完美信息等基准进行比较；同时将连续发生的紧急购电时段合并统计，以区分零散触发与持续性供电缺额。

算法的关键在于将**信息截取、情景生成、优化决策、实际结算与反馈更新**组织为统一的日级回溯流程，而非简单重复调用优化器，从而保证模型评价严格遵循实际可获得的信息顺序。

4.3.3 问题 2 结果分析：全年日前计划与紧急购电

在 2025-02-01 至 2025-12-31 的 334 天上逐日回溯，全年购电费用为 17 469 237.66 元（其中计划购电费 16 055 010.43 元、紧急购电费 1 414 227.23 元），日均 52 303.11 元；全年紧急购电 363 622.65 kWh，时段触发率 18.62%，共 247 天发生紧急购电（平均 7.53 块/日）。0.80 是单时段经济参照，18.62% 的实际补购触发率受到储能耦合、情景构造

及反馈共同影响，不能据其接近 20% 证明报童模型或共形覆盖。策略较无裕度对照节省 744.45 万元；与完美信息参照的 522.39 万元差额同时包含信息限制与策略近似。

题目表 3 指定的四个日期，其指定时段计划购电量如表 5 所示，4 小时块的充放电量与 0:00、24:00 储电量如表 6 所示，紧急购电的聚合结果如表 7 所示（完整的逐块明细见结果文件的“紧急购电量”工作表）。

表 5 问题 2：四个指定日期的指定时段计划购电量与全天合计

日期	10:00-10:10	12:00-12:10	14:00-14:10	16:00-16:10	18:00-18:10	20:00-20:10	全天购电量	全天购电费
2025-03-20	0.00	688.77	0.00	550.21	723.27	0.00	76 674.41	47 017.78
2025-06-21	0.00	740.08	0.00	540.75	711.80	62.57	78 910.55	48 900.38
2025-09-23	0.00	531.80	0.00	463.27	806.88	86.80	73 263.26	46 030.02
2025-12-21	0.00	950.44	0.00	798.25	731.36	0.00	94 711.59	60 429.31

表 6 问题 2：四个指定日期的 4 小时块充放电量与储电量

日期	0:00-4:00	4:00-8:00	8:00-12:00	12:00-16:00	16:00-20:00	20:00-24:00	0:00 / 24:00 储电量
2025-03-20	4 500/0	1 667/4 721	5 858/4 564	5 941/947	0/5 705	5 333/2 935	6 000/6 000
2025-06-21	4 500/0	833/4 656	5 833/3 444	5 833/1 350	0/5 833	5 333/2 807	6 000/6 000
2025-09-23	4 500/0	833/6 459	6 051/2 181	4 976/292	0/5 833	5 333/2 807	6 000/6 000
2025-12-21	4 500/0	1 667/1 508	5 833/7 807	7 740/2 354	0/5 664	5 333/2 976	6 000/6 000

表 7 四个指定日期与全年的紧急购电结果

问题	日期	块数	合计电量 (kWh)	最大单块 (kWh)	最长时段	首块起始时刻
2	2025-03-20	7	183.1	63.9	18:40-19:10	0:20
	2025-06-21	0	0.0	—	—	—
	2025-09-23	6	1 337.0	498.4	6:00-8:00	6:00
	2025-12-21	23	1 386.1	848.4	10:20-11:50	1:10
	全年 (247 天)	2 514	363 622.6	10 154.4	1:00-17:30 (990 min)	—
3	2025-03-20	9	1 194.9	489.7	12:30-13:50	6:00
	2025-06-21	2	26.7	20.9	4:00-5:00	4:00
	2025-09-23	2	44.5	40.9	17:00-17:50	5:50
	2025-12-21	1	240.3	240.3	11:00-11:50	11:00
	全年 (321 天)	1 117	79 350.8	715.8	6:00-9:10	—

由表 7 可见，问题 2 的紧急购电在四个指定日期上差异极大：2025-06-21 无紧急购电，而 2025-12-21 出现 23 个块、合计 1 386.1 kWh（最长块跨越 10:20–11:50），与冬季光伏出力窄且弱、日间缺口长时段存在的季节特征一致。对照实验说明保守度的价值：无风险裕度计划 24 913 778.12 元、解析报童策略 18 534 378.22 元、完美信息下界 12 245 382.22 元；自由终端加终端价值函数对照的期望费用 15 999 584.11 元，日循环期望费用为 16 909 040.51 元。二者相差 90.95 万元，但目标含不同终端信用，不能直接解释为跨日实际节省。

图 6 补充两个机制证据。图 6(a) 给出计划分位扫描下“期望费用—实测覆盖率”的权衡曲线：曲线在 $F^* = 0.85$ 附近最平，理论锚点 0.80 与样本内最优 0.85 的费用差 0.24%，说明费用对锚点不敏感但覆盖率对锚点敏感，故主结果仍取理论值 0.80 以避免样本内过拟合。图 6(b) 给出 334 个实测日的主策略费用分布与两类对照均值：主策略均值较低，但原数据未提供两类对照的完整逐日分布，不能比较其尾部宽度，单日最大紧急购电量为 13 825 kWh。

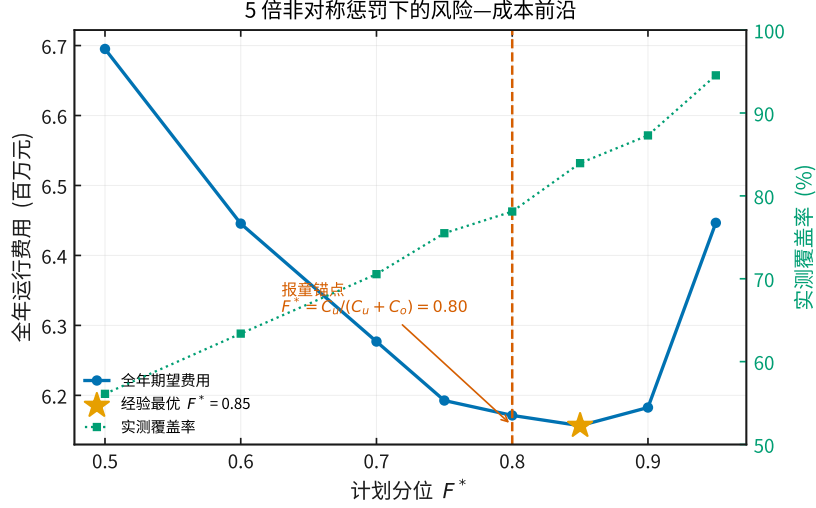


图 6 5 倍非对称惩罚下的风险—成本前沿。分位 F^* 扫描下的期望成本与缺失率权衡；星标为报童最优 $F^* = 0.80$ ，并给出 5 倍应急惩罚 ($C_u = 4p$ 、 $C_o = p$) 的解析位置。

图 6 给出不同分位下的期望成本与缺失率关系，用于解释 Q2 为何选取 $F^* = 0.80$ 作为计划层经济参照，而非盲目取最保守值；曲线在 $F^* = 0.85$ 附近最平，说明费用对锚点不敏感但覆盖率对锚点敏感。

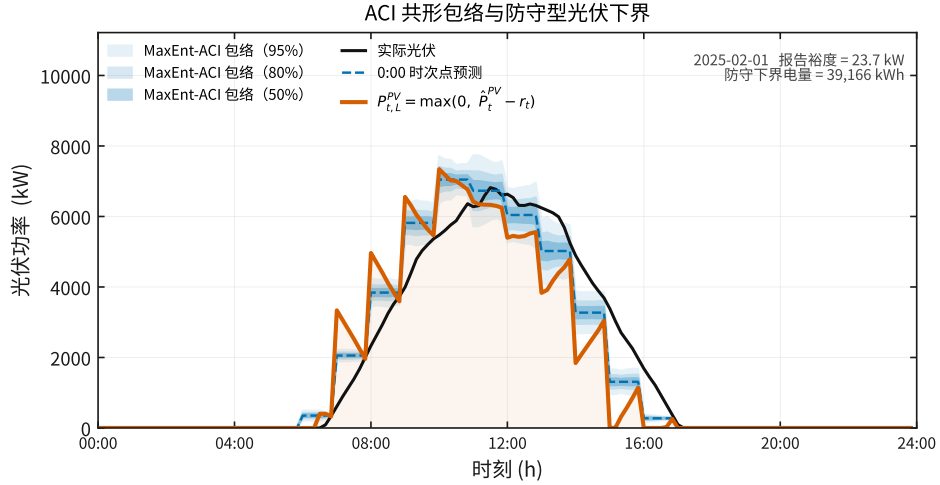


图 7 ACI 共形包络与防守型光伏下界。代表性日 ACI 共形预测包络；粗红实线为报童 $F^* = 0.80$ 对应的物理下界 $P_L = \max(0, \hat{P} - r)$ ，夜间严格截断为 0。

图 7 给出共形预测包络与防守型光伏下界，展示预测不确定性如何转化为可用于调度的物理边界，并体现保守程度与购电风险之间的权衡；该下界用于构造净需求输入，不单独作为 SAA 的硬约束。

4.4 问题 3：分时预测下的滚动调整、求解算法与结果分析

4.4.1 问题 3 模型：分时预测下的滚动调整与信息价值

本问按主比较口径将负荷视为已知，只对光伏预报误差建模。以 $j = 0, 1, 2, 3$ 表示 0、6、12、18 时发布， $\kappa_j = 36j + 1$ 为更新后首个十分钟时段，另记 $\kappa_4 = T + 1$ 。发布的预测覆盖未来 24 h，但当前基线仅优化当日剩余集合 $\mathcal{H}_j = \{\kappa_j, \dots, T\}$ ，并执行块 $\mathcal{B}_j = \{\kappa_j, \dots, \kappa_{j+1} - 1\}$ 。过去动作冻结，以实际储电量 $e_{\kappa_j-1}^{\text{act}}$ 起算，日末仍取 $e_T = S_0$ 。固定 24 h 优化与终端价值是附录 C.4 中的扩展，不用于解释现有费用。

提前量校准与剩余时域决策 预报的小时提前量先展开到十分钟网格，功率乘 Δt 才进入电量模型。对发布 j 、提前步 h 分组，历史得分为

$$R_{i,j,h} = |P_{i,j,h}^{\text{PV}} - \hat{P}_{i,j,h}^{\text{PV}}|, \quad I_{i,j,h} = \mathbb{I}\{P_{i,j,h}^{\text{PV}} \notin \mathcal{C}_{i,j,h}\}. \quad (4.22)$$

i 按已经兑现的检验事件编号；18 h 提前量的误差不能在下一次 6 h 更新时提前使用。理论上的分组校准可独立更新投影 ACI，形式同式 (4.17)。分位阈值 $b_{j,h}$ 给出

$$\mathcal{C}_{j,h} = [\hat{P}_{j,h}^{\text{PV}} - b_{j,h}, \hat{P}_{j,h}^{\text{PV}} + b_{j,h}] \cap [0, P_{\text{cap}}^{\text{PV}}]. \quad (4.23)$$

提前量分组避免混用近、远期误差尺度；不要求每次远期区间一定更宽，也不保证更新后区间必定收缩。有限样本与延迟兑现规则见附录 C.1。

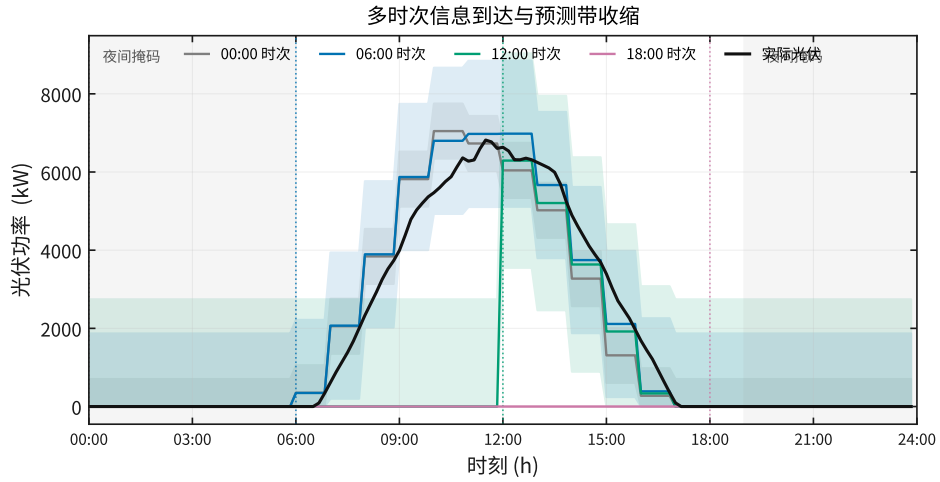


图 8 多时段信息到达与预测带收缩。0:00/6:00/12:00/18:00 四个信息节点的预测带收缩：新预报到达后提前量归零，不确定性带显著收窄；夜间段因光伏恒零而掩码。

图 8 展示四个预报时次到达后预测带的变化，说明新信息只对剩余时域有效，预测带随提前量减小而收缩；该图用于解释 Q3 的信息结构，而不表示更新后区间必然变窄。

所报滚动 LP 采用有符号光伏残差分位 $Q_{t,j}$ ；令 $u_{d,j} = \min\{1, \lceil 31(1 - \alpha_{d,j}) \rceil / 30\}$ ，以互补水平修正净负荷：

$$\hat{n}_{t,j} = [L_t - \hat{P}_{t,j}^{\text{PV}} - Q_{t,j}(1 - u_{d,j})]\Delta t. \quad (4.24)$$

0:00 和调整层的初始净需求目标分别为 0.25、0.70；经有限样本修正，首日实际光伏残差水平分别为 22/30、8/30，方向与净需求相反。代码每天用 $\hat{I}_d = T^{-1} \sum_t \mathbb{I}\{N_t^{\text{act}} \Delta t > q_t^0\}$ 同时更新两层状态： $\alpha_{d+1}^{\text{plan}} = \Pi[\alpha_d^{\text{plan}} + 0.02(0.75 - \hat{I}_d)]$ ， $\alpha_{d+1}^{\text{adj}} = \Pi[\alpha_d^{\text{adj}} + 0.02(0.30 - \hat{I}_d)]$ ，投影区间为 $[0.001, 0.999]$ 。它不是逐提前量、逐层独立校准；末期两状态分别达到 0.999 与 0.001，不能据单独诊断的覆盖率宣称调度层同时校准成功。代码未对加分位后的光伏代理值再作物理截断，该量仅作为净需求构造输入。

每个启用时次求解

$$\begin{aligned} \min_{q,c,d,e} \quad & \sum_{t \in \mathcal{H}_j} \pi_t q_{t,j}, \\ \text{s.t.} \quad & q_{t,j} + d_{t,j} - c_{t,j} \geq \hat{n}_{t,j}, \quad q_{t,j} \geq 0, \\ & (c_j, d_j, e_j) \in \mathcal{X}(\kappa_j, T; e_{\kappa_j-1}^{\text{act}}). \end{aligned} \quad (4.25)$$

该 LP 是分位驱动的代理决策，不是对全部调整和紧急费用的精确期望最优化。0:00 输出 q^0 ，后续仅覆盖尚未执行部分，形成最终生效轨迹 q^{act} 。按相对 0:00 计划的终值偏差作主结算：

$$C_d = \sum_t \pi_t q_t^0 + 5 \sum_t \pi_t g_t^{\text{act}} + \sum_t \pi_t \{1.5[q_t^{\text{act}} - q_t^0]_+ + 0.5[q_t^0 - q_t^{\text{act}}]_+\}. \quad (4.26)$$

其中 $g_t^{\text{act}} = [(L_t - P_t^{\text{PV,act}})\Delta t + c_t - d_t - q_t^{\text{act}}]_+$ 。若采用每次变更逐笔计费，应另建账本；来回改约费用一般不等于终值偏差费用，不能在结果中混算。

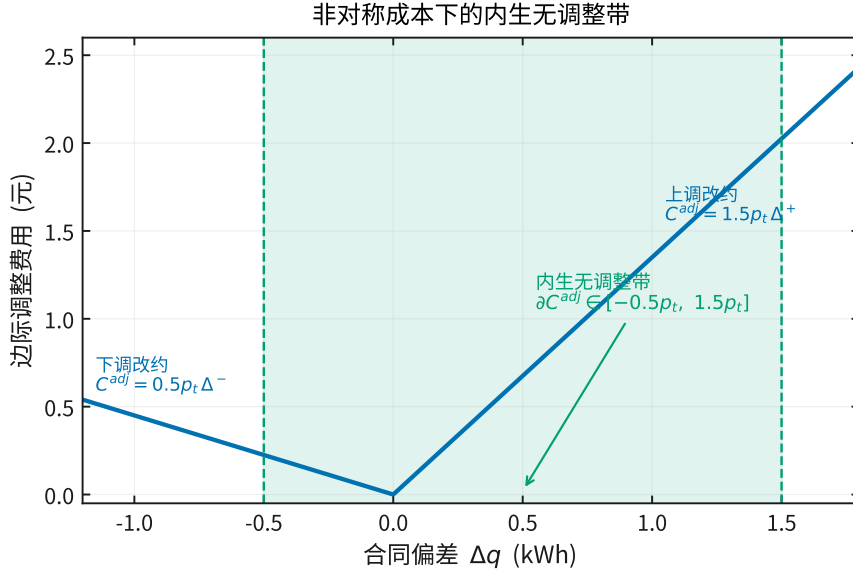


图 9 非对称成本下的内生无调整带。分段线性的契约调整边际成本：下调段 $-0.5p_t$ 、上调段 $+1.5p_t$ ；次梯度区间 $\partial C^{\text{adj}} \in [-0.5p_t, 1.5p_t]$ 内保持契约不变，形成无人工阈值的阻尼带。

图 9 给出非对称调整费用下的边际成本分段结构，说明“是否调整”可由次梯度区间判断，无需人为设定阈值；该带位于边际价值轴，不位于调整量轴，当前合同不等于初始合同时不能无条件平移。

经济参照、无调整带与反事实价值 为解释调整代价，单时段净需求近似 $J(x) = 1.5\pi x + 5\pi\mathbb{E}(D - x)_+$ 给出

$$J'(x) = 1.5\pi - 5\pi[1 - F_D(x)], \quad F_D(x^*) = 0.70. \quad (4.27)$$

因此 0.70 数值本身可以成立，但它不由 $1.5p = 5(1 - p)$ 得到，也不直接证明上述光伏分位代理策略最优。

在固定剩余时域、当前状态和场景下，定义完整经济评价模型：

$$\mathcal{J}_j = \sum_{t \in \mathcal{H}_j} A_t(q_t; b_t) + 5 \sum_s \omega_s \sum_{t \in \mathcal{H}_j} \pi_t g_t^{(s)}, \quad A_t(q; b) = \pi_t \{1.5[q - b]_+ + 0.5[b - q]_+\}. \quad (4.28)$$

约束沿用 SAA 供需式 (4.20b) 及 $\mathcal{X}(\kappa_j, T; e_{\kappa_j-1}^{\text{act}})$ ，购电和储能轨迹跨场景共用。主结算的基准 $b = q^0$ 固定；逐次计费扩展才取变更前合同为基准。该模型用于以下理论判据；未独立求解它时，不把旧代理 LP 输出当作它的最优解。

命题 4.6 (无调整带及适用位置). 固定其余合同分量，经优化消除储能与追索变量的费用为凸函数 $F(q)$ 。若在合同基准 $q = b$ 可双向扰动且 F 可微，令 $m = -F'(b)$ ，则

$$q = b \text{ 最优} \iff -0.5\pi \leq m \leq 1.5\pi. \quad (4.29)$$

该区间位于边际价值轴，不位于调整量轴；当当前合同 $\bar{q} \neq b$ 时，应检验 $0 \in \partial F(\bar{q}) + \partial A(\bar{q}; b)$ ，不能把同一带无条件平移至 $\bar{q}$ 。多时段联合保持还需完整向量 KKT 条件，单坐标判断不替代全局重优化。

命题 4.7 (同场景反事实价值). 令 J_j^{keep} 为式 (4.28) 加上 $q = \bar{q}$ 的最优值， J_j^{reopt} 为不加此限制的最优值。若保持方案可行，两者同场景、同目标且均精确求解，则

$$NAV_j = J_j^{\text{keep}} - J_j^{\text{reopt}} \geq 0. \quad (4.30)$$

样本外预测子集价值另记 $\Delta V_j = C_{S_{j-1}}^{\text{act}} - C_{S_j}^{\text{act}}$ ，可以为负；它包含预测误差和执行结果，不能用 NAV 非负性背书。理论区间缩减指标记为 $CURI_j = \log(U_j^{\text{old}}/U_j^{\text{new}})$ ， U_j 是在相同目标时段、相同权重下的平均区间宽度；仅在新旧宽度均为正时计算，对零宽退化窗口单列报告。现有代码记录的是相同剩余时域的历史残差标准差之比 $CURI_\sigma = \log(\sigma_{\text{old}}/\sigma_{\text{new}})$ ，不等于上述宽度指标。程序记录的 NAV 实为事后剩余费用差，且接受标记未参与调度；下文结果统一按事后价值解释。由此，区间是否收缩、模型是否建议调整和调整是否实际节省，是三个分别检验的问题。

4.4.2 问题 3 求解算法：滚动 MPC 与子集实验

问题 3 采用“计划—调整—结算”的滚动求解结构。计划层根据初始预测分位构造保守净负荷并求解全时域 LP；调整层在 6:00、12:00 和 18:00 三个时次，根据新到达的预测信息重新优化剩余时段；执行层则依据实际光伏出力完成逐日结算。

为评价不同预测时次的实际价值，设置 S0-S5 六组信息子集实验。各组实验仅改变允许使用的预测时次，其余数据、模型参数和求解流程保持一致，费用差异衡量给定策略利用新增信息后的效果，包含重优化与预测误差的共同作用。实现上，以时次掩码控制预测信息是否进入调整层，避免为不同子集重复编写模型。

问每次获得新预测后，将整点预测展开至 144 个十分钟时段，并对已发生时段和核夜间时段进行相应处理；调整层以上一阶段实际到达的储能状态作为新的初始状态，在剩余时域上进行链式重优化。最终按照题目规定的超计划与欠计划结算系数计算费用，并分别统计夜间和日间的信息价值。

4.4.3 问题 3 结果分析：滚动调整与预报时刻的信息价值

四时次滚动 MPC 的全年结算费用为 14 705 624.88 元，其中紧急购电 79 350.77 kWh、超计划违约费 15 276.48 元、欠计划违约费 3 384 327.71 元。欠计划违约费远高于超计划违约费，是式 (4.26) 不对称计价的直接结果：计划层初始净需求分位为 0.25，倾向于较低备货，缺额由 1.5 倍的调整层补齐，因而上调成为费用主要来源。四个指定日期的计划与调整购电量如表 8 所示，充放电量与储电量如表 9 所示。

表 8 问题 3：四个指定日期的指定时段计划购电量 / 调整购电量

日期	10:00-10:10	12:00-12:10	14:00-14:10	16:00-16:10	18:00-18:10	20:00-20:10
2025-03-20	0.00 / 0.00	0.00 / 539.91	0.00 / 0.00	556.07 / 510.34	682.35 / 682.35	0.00 / 0.00
2025-06-21	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 111.87	350.32 / 424.02	0.00 / 0.00
2025-09-23	0.00 / 0.00	0.00 / 480.12	0.00 / 0.00	395.72 / 535.25	732.60 / 759.48	0.00 / 0.00
2025-12-21	0.00 / 0.00	751.66 / 926.24	0.00 / 0.00	701.05 / 829.77	712.63 / 712.63	0.00 / 0.00

考虑到是否需要引入其他时刻的预报，比较六组信息结构的结算费用与边际价值 (表 10)。以仅用 0:00 预报的 S0 (21 289 146.32 元) 为基准，引入 6:00 预报使费用降至

表 9 问题 3：四个指定日期的 4 小时块充放电量与储电量

日期	0:00-4:00	4:00-8:00	8:00-12:00	12:00-16:00	16:00-20:00	20:00-24:00	0:00 / 24:00 储电量
2025-03-20	3785/0	833/5356	5181/2706	6021/433	0/5710	5333/2930	6000/6000
2025-06-21	0/498	124/3822	6062/100	4605/0	0/6154	5333/2486	6000/6000
2025-09-23	3060/0	1667/4586	5853/3445	5883/984	0/5603	5333/3037	6000/6000
2025-12-21	4500/0	1667/1508	5833/7807	8333/2835	0/5797	5333/2843	6000/6000

15 159 618.68 元、边际价值 612.95 万元；再引入 12:00 预报降至 14 696 300.89 元、边际价值 46.33 万元；继续引入 18:00 预报反而升至 14 705 624.88 元、边际价值 -0.93 万元。仅用 12:00 与 18:00 的 S4 为 17 059 426.95 元，说明信息价值高度依赖时次的先后顺序；完美预报下界 12 245 382.22 元与主策略之间仍有 246.02 万元的可改进空间。

表 10 问题 3：预报时次子集实验的全年费用与边际价值

子集	可用预报时次	全年费用 (元)	边际价值 (元)	价值口径
S0	仅 0:00	21 289 146.32	—	无调整基线
S1	0:00 + 6:00	15 159 618.68	+6 129 527.64	6:00 时次价值
S2	0:00 + 6:00 + 12:00	14 696 300.89	+463 317.79	12:00 时次价值
S3	0:00 + 6:00 + 12:00 + 18:00	14 705 624.88	-9 323.99	18:00 时次价值
S4	0:00 + 12:00 + 18:00	17 059 426.95	+4 229 719.37	时次组合对照
S5	完美预报 (界)	12 245 382.22	+9 043 764.10	信息价值上界

按核夜间与日间拆分后，6:00 时次的价值几乎全部来自日间 (+629.57 万元，核夜间部分 -16.61 万元)，12:00 时次的日间部分为 +44.57 万元，18:00 时次的日间部分仅为 +0.004 万元。18 时仅剩六小时，终端约束与预测误差都可能影响增量价值，单一子集差额不能识别唯一原因。本样本优先支持检验上午与午后新增发布的价值，尚不能断言更多发布必然有益。图10中的费用差为事后量，不是在线 NAV 触发证据；程序对启用时次均作调整。无调整带的理论判据见正文命题，现有代码未提取对应约化边际价值。

S0-S5 子集下的契约与 SOC 轨迹

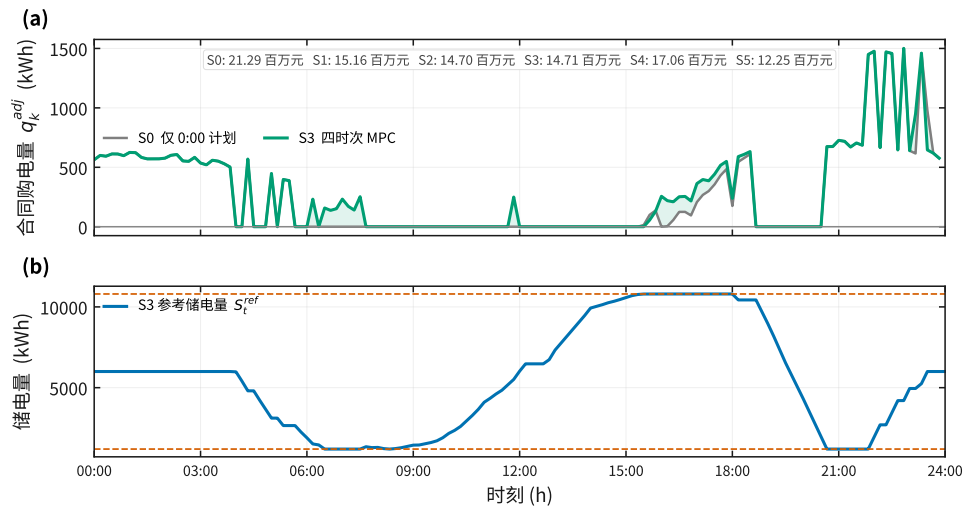


图 10 既有子集实验的合同与储电量轨迹。曲线显示 S0/S3 的代表性轨迹，图内列出各子集全年费用。子集费用差按实际结算计算，可以为负，不用于验证同场景 NAV 非负命题。

图 10 给出滚动闭环的实际执行形态：0:00 计划与四时次 MPC 的调整轨迹在更新时刻发生跳变，跳变幅度反映新信息对剩余时域的影响；对应的储电量轨迹始终保持在安全域内，说明链式重优化未破坏物理可行性。

图 11 对比预测精度增益与净调整价值，说明预测更准并不自动等于决策更优，只

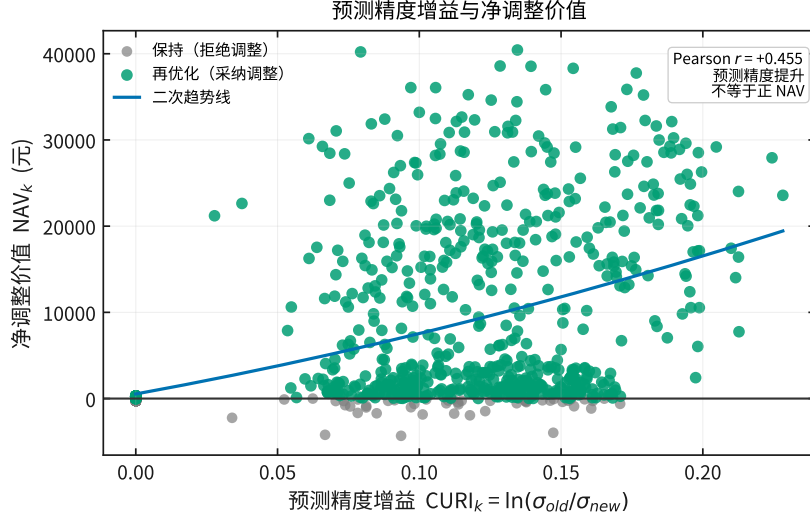


图 11 预测精度增益与净调整价值。散点展示预测精度增益 $CURI_k = \ln(\sigma_{old}/\sigma_{new})$ 与净调整价值 NAV_k 的关系：精度提升并不自动带来正 NAV，需决策阈值过滤。含正 / 负 NAV 分色与零轴参考。有能改变后续决策的信息才具有实际价值；该图用于支撑问题三的核心判断，而不替代式 (4.30) 的同场景 NAV 定义。

4.5 问题 4：波动电价下的多时间尺度风险闭环模型、求解算法与结果分析

4.5.1 问题 4 模型：波动电价下的多时间尺度风险闭环模型

六步实时控制作为理论扩展，采用因果信息集描述尚未兑现的负荷、光伏与价格。两个实验分别报告费用，不把给定价格重算称为在线价格预测效果。已运行的执行实验以既有轨迹为名义动作，经过安全投影和事件触发执行，未逐十分钟求解六步 MPC。

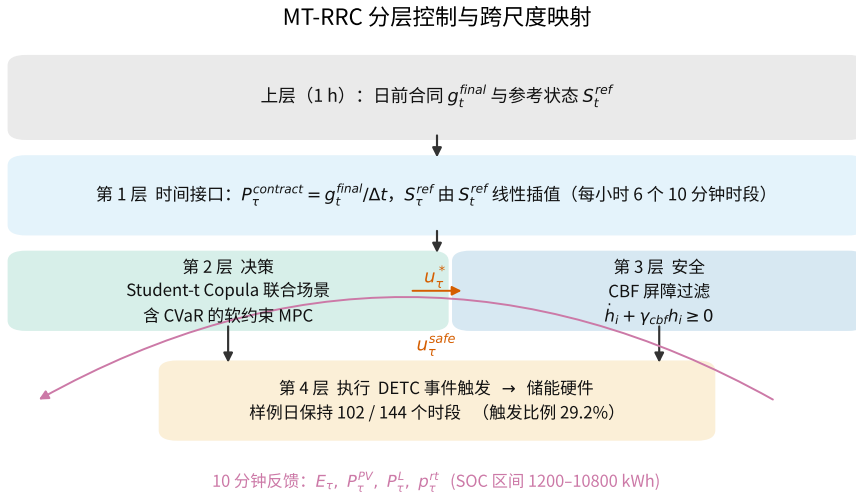


图 12 MT-RRC 分层控制与跨尺度映射。1 小时 $\rightarrow$ 10 分钟映射：日前契约 g_t^{final} 与参考 S_t^{ref} 经接口层展开为 $P_{\tau}^{contract}$ 与 S_{τ}^{ref} ，依次经 Student-t Copula 场景层、软 MPC 决策层、CBF 安全层与 DETC 执行层。

图 12 给出 MT-RRC 的分层控制与跨尺度映射，说明上层小时合同如何展开为十分钟参考，再依次经过场景层、MPC 层、安全层与执行层；图中 DETC 执行层显示样例日保持 102/144 个时段，触发比例 29.2%。

从购电计划到十分钟控制参考。 设十分钟时段 $\tau = 1, \dots, 144$ 、长度 $\delta = \Delta t$ ， S_τ 为该时段结束时的储电量， S_0 为日初状态。若上层下发的正是同一十分钟网格上生效的合同 q_τ^{act} 与参考储电量 S_τ^{plan} ，执行层直接取 $P_\tau^{\text{contract}} = q_\tau^{\text{act}}/\delta$ 、 $S_\tau^{\text{ref}} = S_\tau^{\text{plan}}$ ，完整保留原计划的逐时段形状。若实现端只提供小时总量 q_t^{hr} ($t = 1, \dots, 24$) 与 25 个小时端点储电量 S_t^{hr} ，则用分段线性接口映射到十分钟网格（命题 4.8，式 (D.1)–(D.3) 见附录 D.1）。

命题 4.8 (参考映射的守恒与可实现条件). 小时映射保持每小时计划电量，即 $\sum_{m=1}^6 \delta P_{6(t-1)+m}^{\text{contract}} = q_t^{\text{hr}}$ ；插值结果落在 $[E_{\min}, E_{\max}]$ 内

该命题不保证初始偏差可被精确跟踪，也不保证按小时重新平均后费用不变（重分配仍可能改变 $\sum_\tau \pi_\tau g_\tau$ ），故原生十分钟合同可得时优先使用直接映射。

供需与电价的联合场景：Student-t Copula。 波动电价下，光伏、负荷与电价之间的联合变化会放大尾部费用。在控制节点 τ 展望未来 $H_{\text{rt}} = 6$ 个时段，以 $k = 0, \dots, 5$ 表示局部提前步、省略条件记号 $(\cdot)_{\tau+k|\tau}$ ，把三类残差统一为越大越不利的方向：

$$\mathbf{Z}_k = \left(P_k^L - \hat{P}_k^L, \hat{P}_k^{\text{PV}} - P_k^{\text{PV}}, \pi_k - \hat{\pi}_k \right)^\top, \quad (4.31)$$

即负荷偏高、光伏偏低与电价偏高都对应正残差。记条件边缘分布为 $F_{k,a}$ ，对连续情形做概率积分变换 $U_{k,a} = F_{k,a}(Z_{k,a})$ ，再用 t Copula 描述其依赖结构：

$$\begin{aligned} F_{\mathbf{Z}_k}(\mathbf{z}) &= C_{\nu, R_C}(F_{k,1}(z_1), F_{k,2}(z_2), F_{k,3}(z_3)), \\ C_{\nu, R_C}(\mathbf{u}) &= T_{\nu, R_C}(t_\nu^{-1}(u_1), t_\nu^{-1}(u_2), t_\nu^{-1}(u_3)), \end{aligned} \quad (4.32)$$

其中 $R_C \succ 0$ 且 $\text{diag}(R_C) = \mathbf{1}$ 为潜在 t 向量的尺度相关矩阵。需要强调： R_C 不是原始功率与电价的 Pearson 相关矩阵；选择 t Copula 也不意味着三个物理边缘服从 t 分布（边缘可与极大熵或经验分布自由组合）。

命题 4.9 (尾部关联及其解释范围). 对连续边缘、有限的 $\nu > 0$ 与 $-1 < \rho < 1$ 的二元 t Copula，其上、下尾依赖系数相等，

$$\lambda_U = \lambda_L = 2 t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right) > 0. \quad (4.33)$$

该性质提供“共同极端偏差”的结构依据（非退化相关系数下高斯 Copula 的尾依赖为 0），但不能证明真实数据一定具有这种关联。原诊断错误地重排了变量与时间轴，且使用多元 t 密度代替 Copula 密度，故撤回原 $\nu = 7$ 、 $\rho = 0.815$ 、尾依赖 0.393 的拟合结论。本次以同期三元残差、反向光伏误差、历史日间筛选和秩伪似然重新诊断：5115 个样本上，网格最优 $\nu = 25$ 、负荷—价格潜在相关 0.8025、模型尾依赖 0.1034。这是固定相关阵下的描述性拟合，含并列秩且未做样本外模型选择，不能直接证明真实尾依赖或经济收益。原 79.35 元/70 日差额来自历史配对与固定错位配对，不是 t 场景与统计独立抽样的比较。

条件风险价值目标。 风险重算对总购电费用加入 CVaR，作为 Q2 场景模型的扩展；下面的实时控制目标则只对补购费用加风险项，二者需分别标明。本文以条件风险价值 (CVaR) 度量尾部风险，并采用 Rockafellar–Uryasev 线性化：

$$\min (1 - \beta) \sum_s \omega_s C_s + \beta \left(\zeta + \frac{1}{1 - \alpha_{\text{cv}}} \sum_s \omega_s z_s \right) \quad (4.34)$$

$$\text{s.t. } C_s = \sum_k \pi_k q_k + 5 \sum_k \pi_k g_k^{(s)}, \quad z_s \geq C_s - \zeta, \quad z_s \geq 0, \quad (4.35)$$

其中 $\alpha_{cv} = 0.90$ 为置信水平, $\beta \in [0, 1)$ 为风险权重。 $\beta = 0$ 时模型退化为 Q2 的风险中性 SAA 形式, $\beta \rightarrow 1$ 时退化为纯 CVaR 最小化。

命题 4.10 (CVaR 线性化的精确性). 对任意有限情景集且风险权重 $\beta > 0$, 式 (4.34)–(4.35) 中尾部项的最小值恰为 $\text{CVaR}_{\alpha_{cv}}(C)$; 可选取 $\zeta^* = \text{VaR}_{\alpha_{cv}}(C)$, 但最优阈值未必唯一。 $\beta = 0$ 时删除无效的尾部辅助变量。

函数 $F(\zeta) = \zeta + \frac{1}{1-\alpha} \mathbb{E}[(C - \zeta)^+]$ 关于 ζ 凸、一般不可微, 其极小点集合为符合分位条件的阈值集合, 即广义 α 分位区间、极小值为 $\text{CVaR}_\alpha(C)$, 引入 z_s 写成线性约束不改变等价性。

命题 4.11 (名义优化的可解性与唯一性边界). 设情景有限、 $\pi_{k,\omega} \geq 0$ 、初始电量在物理范围内, 且允许题设的紧急购电与剩余电量处置, 则对任一固定模式向量 $\mathbf{b}$, 式 (??) 是可行且能取到最优解的凸二次规划, 其控制序列唯一; 完整模型含 6 个二元变量, 是小规模混合整数凸二次规划, 也可枚举 2^6 个固定模式 QP 取得全局最优值, 但跨模式的最优控制不保证唯一。

名义首步 $u_\tau^* = u_0^*$ 进入下面的安全与执行模块。

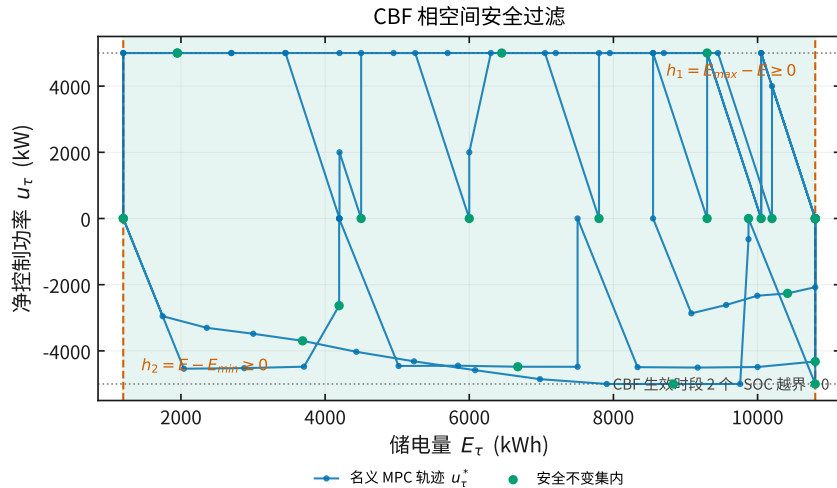


图 13 CBF 相空间安全过滤。(储能电量 E_τ , 控制量 u_τ) 相平面: 阴影为满足 $\dot{h} + \gamma h \geq 0$ 的安全不变集; 箭头示意名义 MPC 指令被二次规划投影回安全域的拉回效应。

图 13 在相平面上展示安全不变集与名义指令的投影过程, 说明 CBF 过滤如何在重新优化经济目标的前提下保证储能物理安全; 该过滤保留互斥与功率界, 但不保证经济最优或日末恢复。

合同边际价值与风险加权无调整带。 设授权节点内某一交付时段的合同功率由 P^{old} 改为 P , 则该时段的场景调整费为

$$A_\omega(P) = \delta \pi_\omega \{1.5 [P - P^{\text{old}}]^+ + 0.5 [P^{\text{old}} - P]^+\}, \quad (4.36)$$

多时段调整应逐段求和。若沿用式 (4.26) 的终值结算, 以下 P^{old} 应取初始合同 q^0/δ ; 它仅在逐次计费扩展下才等于刚执行的旧合同。本节另给出一个授权改约节点的风险

扩展：把调整费与补购费一起放入尾部损失 $L_\omega = A_\omega + C_\omega^{\text{em}}$ ，风险项改为 $\beta \text{CVaR}_\vartheta(L)$ ；它是明确标注的口径扩展，不是全部现金费用的 CVaR。

CVaR 的对偶集合为 $\mathcal{W} = \{w \geq 0 : \sum_\omega w_\omega = 1, w_\omega \leq \omega_\omega / (1 - \vartheta)\}$ ，其中 $0 < \vartheta < 1$ ， $\omega_\omega > 0$ 且原概率之和为 1。对偶权重给出风险加权重与风险价格

$$\bar{w}_\omega = \frac{\omega_\omega + \beta w_\omega}{1 + \beta}, \quad \tilde{\pi} = \sum_\omega \bar{w}_\omega \pi_\omega, \quad \sum_\omega \bar{w}_\omega = 1. \quad (4.37)$$

只用 w 平均价格会遗漏期望项。对固定模式与固定有效权重，令 $\Phi_w(P)$ 为剔除式 (4.36) 后重新优化其余连续变量得到的约化费用，定义边际价值

$$m_w = -\frac{\Phi'_w(P^{\text{old}})}{\delta(1 + \beta)}, \quad (4.38)$$

其导数包含储能与约束的间接影响，不能用未消去其他变量的目标偏导代替。

命题 4.12 (无调整带及其适用范围). 若上述约化费用凸、在原合同处可微，原合同位于可行区间内部且最优解存在，则固定权重子问题保留原合同最优当且仅当

$$-0.5\tilde{\pi} \leq m_w \leq 1.5\tilde{\pi}. \quad (4.39)$$

低于下界时所有最优调整均向下，高于上界时均向上，端点处可能同时存在“保持”与“调整”的最优解。

对完整风险问题还需使用同时满足原始与对偶最优条件的共同权重；对混合整数问题，该结论只在固定模式下成立。价格上升会提高改约门槛，但同时可能提高缺电的边际损失 m_w ，因此不能仅由门槛变宽断言“高价必然减少调整”。

CBF 安全过滤与 DETC 事件触发。 记决策前观测 $S_\tau^{\text{obs}} = S_{\tau-1}^{\text{act}}$ 。经济模型输出的指令在状态偏差下未必物理可行，安全上可行的旧指令也未必仍然安全，故在 MPC 与设备之间插入安全过滤与事件触发两级。令 $h_1(S) = E_{\max} - S$ 、 $h_2(S) = S - E_{\min}$ ，采用与十分钟执行一致的离散控制障碍函数条件

$$h_i(S + \delta f(u)) \geq (1 - \chi)h_i(S), \quad f(u) = \eta_c P^{\text{ch}} - P^{\text{dis}}/\eta_d, \quad 0 < \chi \leq 1, \quad (4.40)$$

即每一步只允许消耗 $(0, \chi]$ 比例的剩余安全余量，而不是只检查名义目标是否平滑。

记满足式 (4.40)、功率界限与充放电互斥的集合为 $\mathcal{U}_\chi(S)$ ，安全动作取

$$u_\tau^{\text{safe}} \in \arg \min_{u \in \mathcal{U}_\chi(S_\tau^{\text{obs}})} \frac{1}{2}(u - u_\tau^*)^\top W_s (u - u_\tau^*), \quad W_s \succ 0. \quad (4.41)$$

保留互斥后可行集是充电与放电两条线段的并集，分别解两个一维二次规划并比较目标即可完成全局投影。过滤不重新优化经济目标，也不修改固定合同；当实际净负荷超出合同与电池的覆盖能力时，按题设紧急购电平衡。

安全的新指令不意味着安全的旧指令。已实现触发器使用功率候选差 $e = u^{\text{safe}} - u^{\text{last}}$ ，在 $\|e\|^2 > 0.01\|u^{\text{safe}}\|^2 + \eta$ 或储电偏差超过 800 kWh 时更新；随后令 $\eta^+ = \max\{\|e\|^2 - 0.01\|u^{\text{safe}}\|^2, 0\}$ 。旧指令不安全时强制更新，并同步实际指令与触发计数。这一经验状态律只有非负截断性质，不能援引下面的误差预算。为获得可证明的累计预算，另给出如下替代设计。令 u^{last} 为正在执行的功率、 $e_\tau = (u_\tau^{\text{safe}} - u^{\text{last}})/P_0$ 为归一化

候选差, 引入非负内部状态 η_τ^{tr} 与参数 $\sigma_\eta > 0$ 、 $\gamma_0 \geq 0$ 、 $a = 1 - \lambda_\eta \delta \in [0, 1)$, 执行规则为

$$u_\tau^{\text{real}} = \begin{cases} u_\tau^{\text{safe}}, & u_\tau^{\text{last}} \notin \mathcal{U}_\chi(S_\tau^{\text{obs}}) \text{ 或 } \|e_\tau\|^2 > \gamma_0 \|u_\tau^{\text{safe}}/P_0\|^2 + a\eta_\tau^{\text{tr}}/\sigma_\eta, \\ u_\tau^{\text{last}}, & \text{其余情形}, \end{cases} \quad (4.42)$$

随后用执行后误差 $\bar{e}_\tau = (u_\tau^{\text{safe}} - u_\tau^{\text{real}})/P_0$ 更新

$$\eta_{\tau+1}^{\text{tr}} = a\eta_\tau^{\text{tr}} + \sigma_\eta \left(\gamma_0 \|u_\tau^{\text{safe}}/P_0\|^2 - \|\bar{e}_\tau\|^2 \right). \quad (4.43)$$

内部状态记录尚未使用的误差容许量并随时间衰减, 从而允许短暂保持相近指令。

命题 4.13 (安全执行与误差预算). 在准确测量、恒定效率、十分钟内恒功率且无执行误差的模型下, 若初始电量安全且 $\eta_1^{\text{tr}} \geq 0$, 则式 (4.41)–(4.43) 在每一步均可执行, 保持充放电互斥, 并使全时域储电量位于安全区间; 内部状态始终非负, 且对任意 N 步成立

$$\sum_{\tau=1}^N \|\bar{e}_\tau\|^2 \leq \gamma_0 \sum_{\tau=1}^N \|u_\tau^{\text{safe}}/P_0\|^2 + \eta_1^{\text{tr}}/\sigma_\eta. \quad (4.44)$$

名义求解失败时可取零指令作为安全候选。

该结论保证的是执行安全与累计偏离预算, 不等价于经济最优、渐近跟踪稳定或硬件寿命改善。

至此, 问题 4 理论扩展的完整数据流为 “合同层输出 $\rightarrow$ 时间接口 $\rightarrow$ 联合场景 $\rightarrow$ MPC $\rightarrow$ CBF $\rightarrow$ 事件触发 $\rightarrow$ 储能执行 $\rightarrow$ 状态反馈”, 其中 MPC 决定期望动作、CBF 检验安全可行性、触发器在安全前提下决定是否值得下发新动作, 合同更新另由授权时钟管理、不能被该执行链旁路。

4.5.2 问题 4 求解算法: 尾部风险防守与安全执行

计划层在 70 个日期上扫描 CVaR 权重, 另在 40 日上测试供需松弛惩罚。Copula 仅作描述诊断; 历史配对与固定错位配对的对照未使用 t 采样。执行试验读取前 70 日既有储能轨迹, 逐步运行 CBF、事件触发和实际状态更新。本次同步强制安全更新的指令记忆与计数, 并直接报告未修补的终端偏差。六步联合场景软 MPC 及带累计误差保证的触发律列作理论扩展, 不将现有参数扫描或轨迹过滤当作该算法的运行证据。

4.5.3 问题 4 结果分析: 实时波动电价下的结果

以附件 4 的实时电价重算问题 2 与问题 3, 得全年费用 17 727 459.43 元 (其中计划购电费 16 113 999.20 元、紧急购电费 1 613 460.23 元)、Q4-3 15 384 235.06 元, 分别比同结构的固定电价情形 (17 469 237.66 元与 14 705 624.88 元) 高 1.48% 与 4.61%。由于附件 4 的电价均值 (0.7662 元/kWh) 与附件 1 的代表性日电价不同, 费用上升同时来自价格水平差异与价格波动, 二者在本口径下不可分离, 故本文只报告总额差异而不作因果分解。与完美信息下界 (Q4-2 12 830 833.05 元、Q4-3 12 830 833.05 元) 相比, 两条链的信息代价分别为 489.66 万元与 255.34 万元。四个指定日期的结果汇总如表 11 所示。

表12(a) 来自每隔 5 日加指定日期的 70 日计划层扫描, 并非六步 MPC。风险权重从 0 增至 0.9 时, 费用增加 0.49%, 日费用经验 CVaR 由 89 773.78 元降至 79 809.97 元, 紧急购电量降低 55.3%。执行层使用另一组连续日期的 70 个独立日实验 (2 月 1 日–4

表 11 问题 4：四个指定日期与全年的结果汇总

问题	日期	结算费用 (元)	计划购电费用 (元)	紧急购电量 (kWh)	紧急购电块数
Q4-2	2025-03-20	48 836.81	47 913.72	183.10	7
	2025-06-21	39 492.55	39 492.55	0.00	0
	2025-09-23	53 384.18	47 483.26	1 354.40	6
	2025-12-21	76 788.13	70 384.63	1 386.10	23
Q4-3	2025-03-20	48 449.29	35 892.43	1 194.88	9
	2025-06-21	17 923.65	14 825.01	26.69	2
	2025-09-23	52 574.47	37 866.58	44.51	2
	2025-12-21	79 933.87	62 244.72	240.33	1
Q4-2 全年合计		17 727 459.43	全年紧急购电	360 039.2	2 503
Q4-3 全年合计		15 384 235.06	全年紧急购电	75 761.0	1 066

月 11 日)，每天从 6000 kWh 起算，不代表跨日连续运行。修正不安全保持后的指令状态与计数后，10 080 个节点下发 4 820 次、保持 5 260 次 (52.18%)，CBF 修改 367 次，实际模式切换 2 323 次，储电越界与同时充放电均为 0。但未作事后轨迹修补时仅 2 日达到 6000 kWh 终点，日均绝对终端偏差 782.59 kWh。安全不等于日循环可行；原 70/70 结论来自事后重写全天轨迹的 LP，原 685.17 也非实际回补电量。

表12(b) 为另一个允许供需松弛的辅助 LP，不能作为正式跟踪 MPC 的实现证据。低惩罚时松弛会掩盖缺额；样本中惩罚达到 $\max \pi$ 即归零，不能由此证明“必须大于 5 倍电价”的必要阈值。70 日历史配对与固定错位配对的费用分别为 3 830 836.27 和 3 830 915.63 元，差额 79.35 元；后者不是统计独立抽样，二者也未使用 t Copula 生成路径。

DETC 执行与动态阈值演化 — 2025-02-01

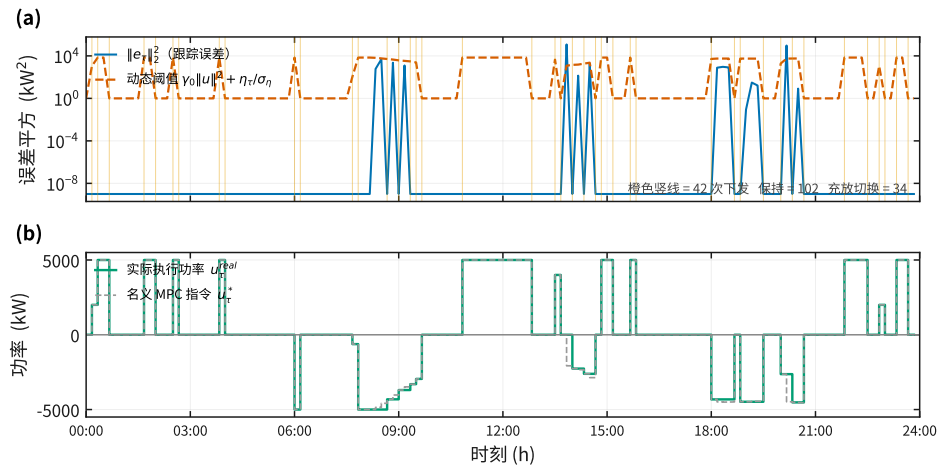


图 14 DETC 执行与动态阈值演化。双面板 144 步：上板为跟踪误差平方与动态阈值对比，脉冲标记实际下发时刻；下板为 BESS 实际功率阶梯，显示 Hold 期休眠与切换次数下降。

表 12 问题 4: CVaR 帕累托前沿与软约束惩罚系数扫描

(a) 风险权重扫描 (70 天分层样本, $\alpha_{cv} = 0.90$)

β	实现费用 (元)	日均费用 (元/日)	经验 CVaR ₉₀ (元)	紧急购电量 (kWh)
0.00	3 820 136.40	54 573.38	89 773.78	93 281.87
0.25	3 795 800.02	54 225.71	82 391.08	63 687.91
0.50	3 813 263.11	54 475.19	80 737.13	50 575.79
0.75	3 835 120.70	54 787.44	79 948.04	43 061.03
0.90	3 838 803.47	54 840.05	79 809.97	41 719.39

(b) 软约束惩罚系数扫描 (40 天样本, ρ_1 以 $\max_k \pi_k$ 为单位)

ρ_1 倍数	ρ_1 (元/kWh)	松弛电量 (kWh)	期望费用 (元)	判定
0.1	0.1794	3 128 642.83	8 669.38	松弛主导, 不可用
0.3	0.5381	1 441 513.41	717 182.57	松弛主导, 不可用
0.5	0.8968	659 864.40	1 261 275.18	松弛显著, 不可用
1.0	1.7936	0.00	1 989 580.24	松弛归零
6.0	10.7616	0.00	1 989 580.24	本文取值

4.6 全局验证、灵敏度分析与横向对比

4.6.1 验证体系

本文按数学、数值、统计、稳健性四类验证组织检验工作, 每类均给出可复核的证据文件与判据, 避免以单一指标或单次运行结论代替验证。

4.6.2 数学验证

量纲一致性。式 (4.4a) 中 π_k (元/kWh) 与 q_k (kWh) 的乘积为元; 式 (4.3) 两侧均为 kWh; 式 (4.1) 中 $\eta_c c_k$ 与 d_k/η_d 均为 kWh。问题 3 结算式 (4.26) 四项均为价格与电量之积, 问题 4 的松弛惩罚 $\rho_1 \sum_k \xi_k$ (ρ_1 为元/kWh) 与控制平滑项量纲分离。

结构与数组复核。互斥定理保证至少存在无循环最优解, 数值输出另行检查。本次重新求解 Q2、Q3、Q4-2、Q4-3 各 334 天, 日费用与归档值的最大差小于 1.8×10^{-6} 元; 又对共 1336 条归档日轨迹独立复算: 状态方程最大残差约 1.1×10^{-12} kWh, 补购重算误差小于 2.3×10^{-13} kWh, 费用与 JSON 一致, 各日计划终点均为 6000 kWh, 未发现超容差越界或同时充放电。有限样本覆盖命题只适用于其交换性与秩规则条件, 截顶和插值的工程分位不能直接继承该保证。

实际执行与理论预算分开核验。修正安全强制更新记账后, 连续 70 日的实际指令满足功率、互斥和储电安全界; 仅 2 日满足原终点。原程序的内部状态使用候选误差和非负截断, 与式(4.43)不同, 故未把累计误差预算列为已通过的数值检验。事后全日轨迹投影不能改变已执行历史, 不作为在线日循环证明。

4.6.3 数值验证

LP 与 DP 互证。问题 1 的最优费用由稀疏 LP 与储电量离散动态规划两条独立路径给出。DP 在四个离散步长下的费用为 35,774.50、35,397.29、35,306.80 与 35,199.90 元，相对 LP 最优值的偏差分别为 1.843%、0.770%、0.512% 与 0.208%，在所测四档网格里逐步接近 LP 最优值。两条路径分别基于单纯形/内点算法与 Bellman 递推，实现与数据结构完全不同，其接近为结果提供独立支持，并非排除一切实现错误的证明。

4.6.4 灵敏度分析

单因素扰动。对关键参数进行 $\pm 10\%$ 扰动，结果见表 G.1。问题 1 中，电价变化在本参数范围内近似按比例传导；储能最大充放电功率变化引起的费用变化仅为 $\pm 0.08\%$ ，说明在当前参数下功率上限并非主要瓶颈。相比之下，往返效率变化带来的费用变化为 $+7.86\%/-6.92\%$ ，容量变化为 $+2.27\%/-2.01\%$ ，表明效率和容量是影响储能经济性的主要参数。问题 2 和问题 3 中，紧急购电价格倍数与违约结算系数分别引起约 $\pm 4.2\%$ 和 $\pm 2.3\%$ 的费用变化，说明结算规则也是影响优化结果的重要外生因素。

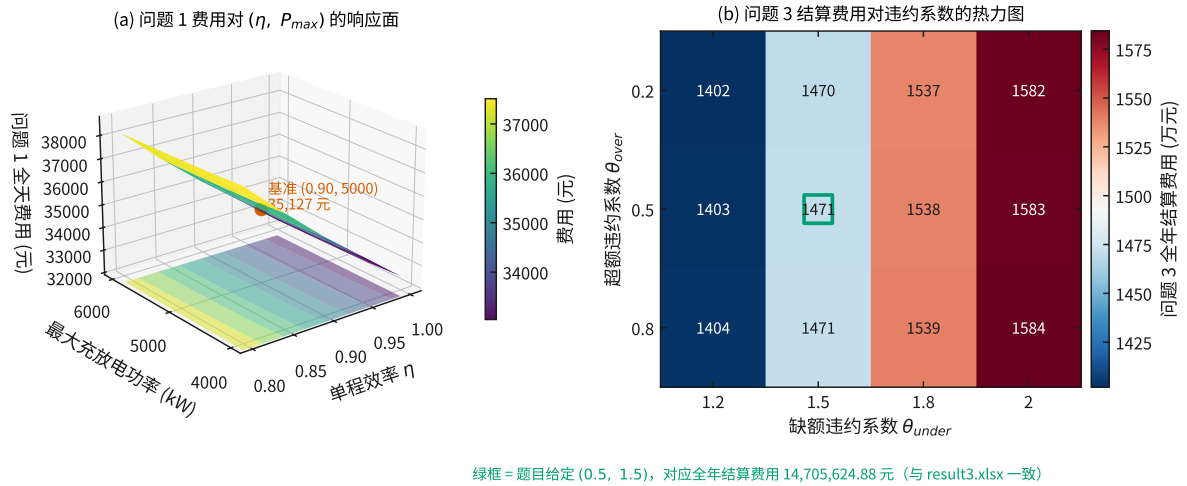


图 15 多参数耦合灵敏度。(a) 问题 1 全天费用对储能单程效率 η 与最大充放电功率 $\bar{P}$ 的响应面，红点为基准参数 (0.90, 5000)，对应 35,126.95 元/日；(b) 问题 3 全年结算费用对超额/缺额违约系数的热力图，绿框为题目给定取值 (0.5, 1.5)，对应 14,705,624.88 元。两个面板均由生产模型逐点重算，不含拟合或外推。

4.6.5 稳健性与不确定项处理

稳健性结论。在电价、储能参数、结算系数及情景规模等扰动下，四个问题的主要结论方向保持稳定：问题 1 中储能相较无储能基线仍具有经济收益，问题 2 的两阶段策略仍优于无裕度计划，问题 3 的信息价值主要集中于 6:00 与 12:00 时次，问题 4 的尖峰覆盖率在扰动范围内仍保持在较高水平。需要指出的是，结论方向稳定并不意味着费用对所有参数均不敏感，其中储能效率、紧急购电价格及缺额违约系数对结果幅度的影响更为明显。

题目未明确项的处理。对于问题 2 的 0:00 信息集、问题 3 的负荷信息可得性、问题 4 的日前电价信息、时段标签对应关系以及日内储能再调度等题目未明确的细节，本文在主模型明确采用比较口径，并用另行压力测试考察其影响。

5 结论与展望

本文围绕微网在不同信息条件下的购电与储能调度问题，构建了由数据清洗、预测不确定性刻画、日前优化、滚动调整和实时安全控制组成的多时间尺度决策框架，并基于 2025 年全年 10 分钟粒度数据开展回测。

问题 1 给出了确定性条件下的调度基准。通过建立含充放电损耗的时域扩展网络流线性规划，并证明充放电互斥约束可以精确松弛。代表性日最优购电费为 35,126.95 元，较无储能基线降低 26.90%，表明储能的主要经济价值来自低价与高价时段之间的能量转移。

问题 2 引入负荷与光伏预测误差，以因果历史残差构造情景并建立两阶段随机规划。通过报童结构得到计划层临界分位 $p^* = 0.80$ ，为单时段净需求提供经济参照，储能耦合与经验反馈后的策略仍需单独评价。全年回测表明，该策略较无裕度计划节省 744.45 万元，说明显式考虑预测不确定性能够改善日前购电计划。

问题 3 进一步将单日前决策扩展为预测信息驱动的滚动调整。针对 0:00、6:00、12:00 和 18:00 四个预测时次建立滚动 MPC，并采用计划层与调整层的分层分位标定。结果表明，不同时段新增信息的决策价值存在明显差异，其中 6:00 时次价值最高 (612.95 万元)，12:00 仍具有一定作用，而 18:00 的边际收益已接近于零。

问题 4 用给定实时价格重算，并分别检验计划 CVaR 与实际执行。70 日风险扫描降低了样本尾部费用；另一组连续 70 日执行复核在修正强制更新记账后未出现储电越界，下发次数减少 52.18%，但仅 2 日恢复日初电量。Student- t 联合场景、六步软 MPC 与累计误差预算设计提供理论扩展，尚不能以现有实验宣称完整闭环已实现或终端全部满足。

未来工作可从三个方向展开：一是将负荷、电价与光伏预测统一纳入滚动不确定性模型，使信息更新机制更加接近实际运行；二是以滚动终端价值函数替代固定日循环条件，在保证长期储能安全的同时进一步刻画并释放跨日调度价值；三是引入电池退化和更精细的运行特性，并结合场景削减与并行求解提高大规模场景下的计算效率。

当前归档费用已通过逐时数组结算复核，但预报清洗使用后验夜间窗口。严格在线验证应优先固定因果预处理，并让计划层和调整层各自以同一目标量进行反馈，再重算全年结果。

6 模型评价

6.1 模型的优点

(1) **结构清晰且逐层递进**。四个问题共享统一的储能物理模型、数据口径，仅逐步增加不确定性、信息更新与实时控制，使各层模型的作用能够通过对照实验单独评价。

(2) **物理约束处理严谨且计算规模可控**。问题一及共同储能场景在储能容量、充放电功率、转换效率及终端状态等条件下保持线性结构，并利用充放电损耗性质对互斥约束进行精确松弛，避免引入大量 0-1 变量与 Big- M 约束，在保证最优值不变的同时降低求解复杂度。

(3) **风险水平具有可检验性**。计划保守度由结算结构对应的边际代价确定，并通过覆盖率检验与可靠性—锐度指标进行校准，避免风险参数完全依赖经验设定。

(4) **经济调度与安全执行相衔接**。理论实时层在风险感知 MPC 后增加 CBF 安全过滤与 DETC 事件触发，将储能物理可行性、控制频次和运行约束纳入统一评价；当

前实际执行以既有轨迹为输入；预报清洗存在后验信息使用，严格在线结果尚需重算。

6.2 模型的不足

(1) **部分问题对负荷预测误差进行了简化**。问题 3 及问题 4 的相应执行层主要考虑光伏预测不确定性，未将负荷预测误差纳入主模型。若实际负荷存在显著预测偏差，紧急购电风险可能被低估，因此相关结论依赖于负荷能够较可靠预测的前提。

(2) **日循环终端条件限制了跨日储能自由度**。问题 1-3 采用 $E_T = E_0 = 6000 \text{ kWh}$ 的终端条件，能够保证不同日期之间具有统一的比较口径，但也限制了低价时段提前储能及跨日利用等行为，自由终端对照同时改变终端信用，不能将费用差直接当作跨日实际价值。

(3) **储能模型仍采用理想化效率描述**。当前模型将充放电效率视为常数，未进一步考虑温度、倍率、内阻及荷电状态对效率和可用容量的影响，因此在长期运行或极端工况下可能高估储能的实际调节能力。

(4) **合同层与执行层尚未完全联合优化**。CBF 过滤与事件触发主要作用于实时执行阶段，并不反向改变已经生效的购电计划。

6.3 改进方向与推广

针对模型的主要局限，后续可从三个方面进一步完善。首先，将固定的日循环终端条件改为滚动终端价值，并引入更精细的储能退化模型，使储能能够在相邻日期之间进行更合理的能量配置。其次，将负荷预测误差与电价动态过程纳入日内风险场景，进一步完善多源不确定性下的联合调度。最后，将实时安全约束前置至日前计划与调整层，使合同计划与实际执行能够协同优化，减少终端偏差，并检验剩余时域可达性。在推广方面，本文模型以时间序列、储能参数和信息更新机制为主要接口，不依赖特定地区或单一设备类型。因此，在替换相应数据和设备参数后，可推广至园区级微网、多储能协同系统以及光伏—储能—可调负荷等综合能源场景。对于更大规模系统，可进一步采用分布鲁棒优化与并行求解降低场景计算和滚动优化的计算负担。

当前代码还将同一个日前命中量用于问题三的两层反馈，不能保证分层覆盖；原 NAV 接受标记未控制重优化。优先实施一致评分与因果预处理，再做完整 MPC 和 Copula 消融，比增加模型名称更能提高结论可信度。

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A Q1: 网络表示、精确松弛与对偶证明

A.1 补充记号与网络定义

基础记号沿用表 2。本附录增加吞吐退化单价 $\kappa_{\text{deg}} \geq 0$ ，并以 μ_t, λ_t 表示供需与储能转移乘子；基线取 $\kappa_{\text{deg}} = 0$ 。为讨论扩展，将目标写为 $\sum_t [\pi_t q_t + \kappa_{\text{deg}}(c_t + d_t)]$ 。网络含外部供给节点、母线 N_t 与储能节点 B_t 。 B_t 表示时段 t 的储能结转节点，从上一时段接收 e_{t-1} ，向下一时段发送 e_t 。各弧按发送端流量定义如下。首尾储能弧固定 $e_0 = e_T = S_0$ 。

表 A.1 时域网络的流量、增益与成本

弧	发送端流量	增益	单位发送量成本
外网 $\rightarrow N_t$	q_t	1	π_t
光伏 $\rightarrow N_t$	v_t	1	0
$N_t \rightarrow B_t$	$c_t \in [0, \bar{C}]$	η_c	κ_{deg}
$B_t \rightarrow N_t$	$d_t/\eta_d \in [0, \bar{D}/\eta_d]$	η_d	$\eta_d \kappa_{\text{deg}}$
$B_t \rightarrow B_{t+1}$	$e_t \in [E_{\min}, E_{\max}]$	1	0
$N_t \rightarrow$ 需求/余量	ℓ_t, w_t	1	0

储能节点守恒为 $e_{t-1} + \eta_c c_t = e_t + d_t/\eta_d$ ，母线守恒为 $q_t + v_t + d_t = \ell_t + c_t + w_t$ 。放电弧按发送量 d_t/η_d 计成本时，系数必须为 $\eta_d \kappa_{\text{deg}}$ ，才与目标中的 $\kappa_{\text{deg}} d_t$ 一致。局部充放电弧形成有向环，故不能把整个网络称为有向无环图；时间结转方向向前。效率不为 1 时是广义流，效率均为 1 时成为普通线性成本流；普通流结构并不要求退化成本为零。

A.2 精确松弛与能量核算

命题 4.1 的证明. 零充放电、 $e_t = S_0$ 、 $q_t = [\ell_t - v_t]_+$ 给出可行点。正价格与有限输入使目标有下界；有限维 LP 有可达有限最优值。若一个最优解在时段 t 有 $c_t, d_t > 0$ ，令 $r = \eta_c \eta_d$ 。当 $q_t > 0$ 时取

$$0 < \delta \leq \min\{c_t, d_t/r, q_t/(1-r)\}, \quad c'_t = c_t - \delta, \quad d'_t = d_t - r\delta, \quad q'_t = q_t - (1-r)\delta. \quad (\text{A.1})$$

因 $\eta_c \delta - r\delta/\eta_d = 0$ ，储能状态不变；母线平衡与所有上下界仍成立。扩展目标的变化为

$$\Delta C = -[(1-r)\pi_t + (1+r)\kappa_{\text{deg}}]\delta = -(0.19\pi_t + 1.81\kappa_{\text{deg}})\delta < 0, \quad (\text{A.2})$$

与最优性矛盾。基线零购电时取 $\delta = \min\{c_t, d_t/r\}$ ，保持 $q'_t = 0$ ，增加余量 $(1-r)\delta$ ；状态不变且费用不增加，至少一个充放电变量变为零。对至多 T 个时段作此操作，得到互斥最优解。原互斥模型可行域包含于松弛 LP，而该构造从 LP 最优解返回同值的互斥解，故两者最优值相等。当 $\kappa_{\text{deg}} > 0$ 时零购电消环也严格降本，因此任意最优解均互斥；基线只保证存在性及正购电时段的严格结论。□

若禁止弃用供给、价格非正或引入强制动作与额外跟踪奖励，上述可行性或降本性质可能失效，需重新证明。

命题 4.2 的证明. 对储能转移求和， $e_T - e_0 = 0$ 给出 $\eta_c \sum_t c_t = \sum_t d_t/\eta_d$ 。再对母线守恒求和并代入 $\sum_t d_t = r \sum_t c_t$ ，即得式 (4.6)。□

A.3 KKT 条件、阈值与影子价格

令充放电上下界乘子为 $a_t^-, a_t^+, b_t^-, b_t^+$, 储电量上下界乘子为 ν_t^-, ν_t^+ , 购电非负乘子为 s_t , 均非负; 储能等式乘子 λ_t 自由。固定端点后, 对内部状态形成拉格朗日函数

$$\begin{aligned} \mathcal{L} = & \sum_t [\pi_t q_t + \kappa_{\text{deg}}(c_t + d_t)] + \sum_t \mu_t (\ell_t + c_t - v_t - d_t - q_t) \\ & + \sum_t \lambda_t (e_t - e_{t-1} - \eta_c c_t + d_t / \eta_d) - \sum_t s_t q_t \\ & + \sum_t [-a_t^- c_t + a_t^+ (c_t - \bar{C}) - b_t^- d_t + b_t^+ (d_t - \bar{D})] \\ & + \sum_{t=1}^{T-1} [\nu_t^- (E_{\min} - e_t) + \nu_t^+ (e_t - E_{\max})]. \end{aligned} \quad (\text{A.3})$$

驻点条件为

$$\pi_t - \mu_t - s_t = 0, \quad \kappa_{\text{deg}} + \mu_t - \eta_c \lambda_t - a_t^- + a_t^+ = 0, \quad (\text{A.4})$$

$$\kappa_{\text{deg}} - \mu_t + \lambda_t / \eta_d - b_t^- + b_t^+ = 0, \quad (\text{A.5})$$

$$\lambda_t - \lambda_{t+1} - \nu_t^- + \nu_t^+ = 0 \quad (1 \leq t < T). \quad (\text{A.6})$$

另有 $\mu_t w_t = s_t q_t = a_t^- c_t = a_t^+ (\bar{C} - c_t) = b_t^- d_t = b_t^+ (\bar{D} - d_t) = 0$, 以及 $\nu_t^- (e_t - E_{\min}) = \nu_t^+ (E_{\max} - e_t) = 0$ 。配合原始可行性和乘子符号, 以上是有限最优 LP 的必要充分最优性证书 [21]。由第一式得 $0 \leq \mu_t \leq \pi_t$, 正购电时 $\mu_t = \pi_t$, 有余量时 $\mu_t = 0$ 。

$$a_t = \mu_t - \eta_c \lambda_t + \kappa_{\text{deg}}, \quad b_t = -\mu_t + \lambda_t / \eta_d + \kappa_{\text{deg}}. \quad (\text{A.7})$$

最小化区间上线性项给出正文动作判据。零动作要求 $\eta_c \lambda_t - \kappa_{\text{deg}} \leq \mu_t \leq \lambda_t / \eta_d + \kappa_{\text{deg}}$ 。内部状态驻点则给出 $\lambda_{t+1} - \lambda_t = \nu_t^+ - \nu_t^-$, 索引对应期末状态 e_t 。

若动态右端加入 ξ_t , 拉格朗日项为 $\lambda_t(\cdots - \xi_t)$, 故可微最优值的导数为 $-\lambda_t$ 。不可微时, 最优对偶给出局部支持斜率, 不能要求影子价格唯一。从时段 i 增购 δ 充电, 在 j 用 $r\delta$ 替代购电, 且路径容量与功率均有余量时, 费用变化为 $[\pi_i - r\pi_j + (1+r)\kappa_{\text{deg}}]\delta$ 。正收益条件 $\pi_j > [\pi_i + (1+r)\kappa_{\text{deg}}]/r$ 只针对这笔可行转移, 不是无条件全日策略。

B Q2: 最大熵、校准与两阶段调度证明

B.1 命题 4.3 的证明: 变分与凸对偶

驻点与对偶。对归一化约束引入 β , 对矩约束引入 θ , 拉格朗日泛函为:

$$\begin{aligned} \mathcal{J}(f, \beta, \theta) = & - \int f \log f \, dx - \beta \left(\int f \, dx - 1 \right) \\ & - \sum_{k=1}^M \theta_k \left(\int x^k f \, dx - \hat{m}_k \right). \end{aligned} \quad (\text{B.1})$$

以下积分均在 $[-1, 1]$ 上。对正密度的允许局部扰动 δf :

$$\delta \mathcal{J} = - \int [\log f + 1 + \beta + \theta^\top \mathbf{v}(x)] \delta f(x) \, dx. \quad (\text{B.2})$$

驻点给出 $f = \exp[-1 - \beta - \boldsymbol{\theta}^\top \mathbf{v}(x)]$ 。归一化要求 $1 + \beta = \log Z$ ，因此常数 1 必须被吸收，不能在同一乘子定义下漏写。

逐点对 $f \geq 0$ 求上确界，得到对偶函数：

$$g(\beta, \boldsymbol{\theta}) = \beta + \boldsymbol{\theta}^\top \widehat{\mathbf{m}} + e^{-1-\beta} Z(\boldsymbol{\theta}). \quad (\text{B.3})$$

最小化 β 后为 $\log Z + \boldsymbol{\theta}^\top \widehat{\mathbf{m}} = \Phi$ 。对任意有限乘子，紧支撑使配分函数和所有所需导数有限，可在积分号下求导：

$$\partial_{\theta_j} \log Z = -\mathbb{E}_{\boldsymbol{\theta}} X^j, \quad \partial_{\theta_j \theta_k}^2 \log Z = \mathbb{E} X^{j+k} - \mathbb{E} X^j \mathbb{E} X^k. \quad (\text{B.4})$$

故得到式 (4.14)。对任意非零向量 a ， $a^\top \nabla^2 \Phi a = \text{Var}(\sum_k a_k X^k) > 0$ ：若方差为零，该无常数项多项式必须在正密度覆盖的整个区间上为常数，只有 $a = 0$ 才可能。故 Hessian 正定， Φ 严格凸。

有限极小点的存在。 仅有严格凸性不足以保证极小点可达。由矩向量的内部性，存在 $\delta > 0$ 使以 $\widehat{\mathbf{m}}$ 为中心、半径 δ 的球属于矩体。任意单位向量 a 均满足：

$$\max_{x \in [-1, 1]} a^\top (\widehat{\mathbf{m}} - \mathbf{v}(x)) \geq \delta. \quad (\text{B.5})$$

$\mathbf{v}$ 在紧区间上一致 Lipschitz，设常数为 L 。取上式最大值点 x_a ，对 $\rho \geq 1$ ，其半径 $1/\rho$ 邻域与支撑的交集长度至少 $1/\rho$ 。在此交集内， $a^\top (\widehat{\mathbf{m}} - \mathbf{v}(x)) \geq \delta - L/\rho$ ，从而：

$$\Phi(\rho a) = \log \int \exp\{\rho a^\top (\widehat{\mathbf{m}} - \mathbf{v}(x))\} dx \geq \rho \delta - L - \log \rho \longrightarrow +\infty. \quad (\text{B.6})$$

故 Φ 强制有界下水平集，并存在唯一有限极小点 $\boldsymbol{\theta}^*$ 。其梯度为零，式 (4.12) 遂满足全部矩约束。

全局最优性与唯一性。 对任一可行密度 f ，Gibbs 不等式给出：

$$D_{\text{KL}}(f \| f^*) = \int f \log \frac{f}{f^*} = -H(f) + \log Z(\boldsymbol{\theta}^*) + \boldsymbol{\theta}^{*\top} \widehat{\mathbf{m}} = H(f^*) - H(f) \geq 0. \quad (\text{B.7})$$

相等当且仅当 $f = f^*$ 几乎处处。故驻点确为唯一全局最大熵密度，同时原始与对偶最优值相等。对于 $H(f) = -\infty$ 的可行密度，最优性不等式仍显然成立。

内部性不是可省略的技术修饰。例如 $\widehat{m}_2 = \widehat{m}_1^2$ 强制零方差，分布集中在单点，无法由有限参数的正连续密度表示。若采用全实数支撑， $Z < \infty$ 也不再自动成立：指数中最高非零多项式阶数必须为偶数且其负号前系数为正，才能使两端衰减。有限支撑消除了这一可积性问题，但不能消除经验矩位于边界或样本不足的问题。

B.2 矩匹配的数值实现

在归一化区间上用数值积分计算 Z 、矩和协方差矩阵。第 j 次迭代求解线性方程：

$$[H_j + \rho_j I] v_j = \nabla \Phi(\boldsymbol{\theta}_j), \quad \boldsymbol{\theta}_{j+1} = \boldsymbol{\theta}_j - \tau_j v_j, \quad (\text{B.8})$$

其中 $H_j = \nabla^2 \Phi(\boldsymbol{\theta}_j)$ ， $\rho_j \geq 0$ 为仅用于稳定步长的阻尼， $\tau_j \in (0, 1]$ 由下降线搜索选择。无需显式形成矩阵逆。若把正则项加入目标函数，则已改变精确矩约束问题，应另作说明。指数积分使用减去最大指数的缩放，计算对数配分函数时再加回，以免溢出。

判停至少核对 $\max_k |\mathbb{E}_{\boldsymbol{\theta}} X^k - \widehat{m}_k|$ 、归一化误差及更密积分网格下的稳定性。不能仅用乘子变化很小判断拟合成功。对于高阶矩病态，可在相同多项式空间内改用正交基并同步变换矩；若降低阶数、放松矩为区间约束或扩大支撑，应记录为模型调整。Newton 迭代是数值求解，不能称为“精确解析求出乘子”。

B.3 命题 4.3 的证明：绝对得分分布与两类分位数

证明. 事件 $|\varepsilon| \leq s$ 等价于 $-s \leq \varepsilon \leq s$, 故减去左极限得式 (4.15); 连续情形再写为密度积分。对 $s > 0$ 求导, 分别得到 $p^*(s)$ 与 $p^*(-s)$, 从而得分密度为两者之和。□

若残差含原子, 正确公式为 $G(s) = F_\varepsilon(s) - F_\varepsilon((-s)^-)$; 在 $s = 0$ 也可能具有非零质量。广义分位数定义为 $G^{-1}(u) = \inf\{s : G(s) \geq u\}$ 。离散或含原子的模型在该阈值下通常是至少达到名义覆盖, 而不必精确相等。

引理 B.1 (固定水平的有限样本覆盖). 对 n 个校准得分与一个新得分, 若它们在固定评分规则下交换且无并列, 则新得分在 $n+1$ 个得分中的秩均匀分布。令 $k = \lceil (n+1)(1-a) \rceil \leq n$, 有:

$$\mathbb{P}\{R_{\text{new}} \leq R_{(k)}\} = \frac{k}{n+1} \geq 1-a. \quad (\text{B.9})$$

并列采用包含等号的集合可给出保守覆盖; 当 $k = n+1$ 时必须补 $+\infty$, 不能错误地截为样本最大值。这一证明针对固定 a ; 由相同历史数据自适应选择的 α_d 不直接继承该逐次概率陈述。对时序漂移、滚动重训或相关残差, 也不能默认交换性成立。

MaxEnt 的积分分位数没有上述秩论证。若主算法直接使用 G^{-1} , 应称其为“MaxEnt 模型阈值加 ACI 反馈”; 若采用式 (4.16), 则为“经验共形阈值加 ACI 反馈”。在绝对得分不变且各历史样本采用同一个严格递增变换 G 时, 先求 $G(R_i)$ 的经验分位数再取 G^{-1} , 仍返回原来的 $R_{(k)}$, 并不会新增 MaxEnt 校准作用。因此不能通过重复变换制造两种方法均不可或缺的假象。

B.4 ACI 长期频率与投影修正

无投影递推及边界外全域/空集约定为

$$\alpha_{i+1} = \alpha_i + \gamma(\alpha^* - I_i). \quad (\text{B.10})$$

以下用 d 编号同一时段已兑现的事件; 记 $a = \alpha^*$ 。

证明. 固定 h , 记 $a = \alpha^*$ 。当 $0 < \alpha_d < 1$ 时, 单步变化只有 γa 或 $-\gamma(1-a)$, 故下一步处于 $[-\gamma(1-a), 1+\gamma a]$ 。当 $\alpha_d \leq 0$ 时, 全域预测集使 $e_d = 0$, 下一步向上移动; 当 $\alpha_d \geq 1$ 时, 空集使 $e_d = 1$, 下一步向下移动。逐一检查两种边界更新不会越过该区间, 结合初值即可归纳得到有界性。

对递推式求和, 得到:

$$\alpha_{N+1} - \alpha_1 = \gamma \sum_{d=1}^N (a - e_d). \quad (\text{B.11})$$

移项即得命题中的恒等式。令 $K_0 = \max\{\alpha_1 + \gamma(1-a), 1 + \gamma a - \alpha_1\}$, 则:

$$\left| \frac{1}{N} \sum_{d=1}^N e_d - a \right| \leq \frac{K_0}{\gamma N} \rightarrow 0. \quad (\text{B.12})$$

整个推导是路径恒等式与有界性推论, 不使用独立性或分布正确性。□

有限日期数下该上界可能很松，尤其步长较小时，不应把渐近结论写成一年内必然达到精确目标。步长越大，反馈响应越快，也越易出现宽度波动与边界触发；选择需基于此前验证数据。

若工程中采用投影，令：

$$r_d^\Pi = \Pi_{[\alpha_{\min}, \alpha_{\max}]}[\alpha_d + \gamma(a - e_d)] - [\alpha_d + \gamma(a - e_d)]. \quad (\text{B.13})$$

实际更新为 $\alpha_{d+1} = \alpha_d + \gamma(a - e_d) + r_d^\Pi$ ，从而：

$$\frac{1}{N} \sum_{d=1}^N e_d - a = -\frac{\alpha_{N+1} - \alpha_1}{\gamma N} + \frac{1}{\gamma N} \sum_{d=1}^N r_d^\Pi. \quad (\text{B.14})$$

投影使状态有界，却不使最后一项自动消失。例如，若所有允许显著性水平对应的区间均漏掉实际值，则 $e_d \equiv 1$ ，状态长期停留在下界，误覆盖率仍为 1。这直接否定“截断后仍无条件精确跟踪”的说法。

为日前调度提供有限、非空区间时，可以明确采用投影版本，并报告式 (B.14) 的修正和实测表现；也可以保留理论版本并单独标记全域、空集，再设计调度备用规则。但备用规则改变区间后，应对真正交付的集合重新计算失守事件，不能以另一组理论集合的覆盖为其背书。

B.5 因果校准、场景和分位接口

Q2 的随机对象为净负荷 $N = L - P^{\text{PV}}$ ，正残差表示供电压力增大；Q3 为光伏，正残差表示供给增加。同样的 MaxEnt 推导可用于两者，但拟合样本、符号方向与覆盖事件必须分别记录。历史窗口中的重新中心化偏差只用于形成 SAA 场景；校准使用预测发布时留下的预测—实现对。对日 d 的场景，按 $\sum_t \epsilon_{d',t|d}$ 排序分层，选代表路径并以该层样本质量给概率。整条路径抽取保留日内形状；场景削减后仍是历史分布近似，不保证保留所有尾部依赖。历史净负荷路径若只是减去当前均值再加回同一均值，就等于原路径，不能声称这一步产生了新预测信息。

时间加权经验分位定义为

$$\text{WQ}_p(x) = \inf\{z : \sum_i w_i \mathbb{I}(x_i \leq z) / \sum_i w_i \geq p\}. \quad (\text{B.15})$$

既有有限输出规则使用

$$\tilde{p}_n = \min\{1, \lceil (n+1)p \rceil / n\}. \quad (\text{B.16})$$

它是经验分位层级修正，不是任意权重下的有限样本覆盖定理；当理论秩为 $n+1$ 时截成样本最大值也不等于补无穷。MaxEnt 模型分位、经验加权分位和无权重共形秩分位须在实验中明确区分。所有超参数应在测试日前可用数据中确定。若用当日完整实测序列识别夜间，该掩码只能用于事后清洗和统计，不能反向设置当日发布预测或优化权重；在线掩码应只来自此前历史或已知太阳时刻。

B.6 两阶段 SAA 可解性与互斥性

命题 4.5 的证明. 令 $c = d = 0, e_t = S_0, q_t = 0$ ，各场景取 $g_t^{(s)} = [n_t^{(s)}]_+$ ，即得有限可行费用。所有费用项非负，故 LP 有限最优值可达。对于任意共同储能轨迹中的正循环，令

$c'_t = c_t - \delta, d'_t = d_t - r\delta, \delta = \min(c_t, d_t/r)$ 。保持共同购电 q_t 不变，所有场景的净供给同时增加 $(1-r)\delta$ 。各场景补购取新缺口正部，不会比原 $g_t^{(s)}$ 大；储能状态、功率界和所有场景可行性保持。因此目标不增加，逐时段处理可得互斥最优解。这里使用同一个 δ ，没有引入场景依赖的不可执行储能动作。 $\square$

若某场景概率为零，可删除而不影响目标；它不应被用于统计风险证据。最优追索恒满足 $g_t^{(s)} = [n_t^{(s)} + c_t - d_t - q_t]_+$ ，由正价格可知额外补购没有收益。

B.7 实际短缺、经济分位与防守对照

固定日前储能轨迹后，实际电量缺口正部即式 (4.21)，这说明第二阶段在信息兑现时即可执行。覆盖事件 $N_t \leq \hat{N}_t + b_t$ 与紧急补购事件 $N_t \Delta t > q_t + d_t - c_t$ 的阈值不同，后者还受储能耦合影响。因此 18.62% 紧急触发率接近 20% 并不是对报童最优分位的直接验证。单时段 $J(x) = \pi x + 5\pi \mathbb{E}(D - x)_+$ 的左右导数为 $J'_-(x) = \pi - 5\pi \Pr(D \geq x)$ 、 $J'_+(x) = \pi - 5\pi \Pr(D > x)$ ，内点最优等价于 $F_D(x^-) \leq 0.8 \leq F_D(x)$ ； $x = 0$ 还需检验边界方向。非对称残差的两侧概率不相等，绝对得分区间更不能直接转换为指定的单侧经济分位。

下界防守仅作为对照。 固定预测负荷 $\hat{\ell}_t$ 与光伏电量下界 $v_{t,L}$ ，最小化 $\sum_t \pi_t q_t$ ，约束 $q_t + v_{t,L} + d_t - c_t \geq \hat{\ell}_t$ 及共同储能约束，得到防守 LP。它对负荷固定且光伏逐时段不低于下界的路径可行，不等于 SAA 主模型，也不自动具有整日联合覆盖。设该防守路径余量为 $m_t = q_t + v_{t,L} + d_t - c_t - \hat{\ell}_t \geq 0$ ，则

$$g_t^{\text{act}} = [(\ell_t - \hat{\ell}_t) + (v_{t,L} - v_t) - m_t]_+. \quad (\text{B.17})$$

这是由供需平衡代入得到的恒等式：负荷误差也可能导致短缺，不能声称光伏在区间内就绝不会紧急购电。

B.8 生产反馈与诊断的区分

当前 Q2 尾部水平由正文所列经验命中量更新，Q3 的两层状态又共用一个日前命中量；二者均不等于逐组、同目标的覆盖校准。式(4.18)的代数恒等式对 $[0, 1]$ 内日均反馈仍成立，但其控制对象只能是实际代入的反馈量。独立诊断表使用单侧有符号得分及 0.05 步长，生产层使用 0.02 步长，不能用诊断的 0.7948、0.2483、0.6963 为生产反馈背书。生产 MaxEnt 对标准化样本在历史 99.5% 绝对分位处截尾后匹配矩；严格最大熵命题对应截尾后矩问题，而非未截尾真实残差分布。

C Q3：校准时序、调整判据与扩展模型

C.1 提前量校准及可执行时序

更新索引 j 、首时段 κ_j 、剩余集 $\mathcal{H}_j$ 和执行块 $\mathcal{B}_j$ 见正文。校准记录必须保存发布时刻、目标时刻与原始区间。对每个提前量组，将已经兑现的误差按顺序编号为 i ，只在目标时段结束之后更新 $\alpha_{i+1,j,h}$ 。发布时读取最新可用状态，未兑现项目保留在等待队列，不能写成用本轮全部未来误差更新下一轮的无延迟递推。无权重有限样本分位按

式 (4.16) 补 $+\infty$ ；若为有限调度输入改用经验分位，应记录回退，并停止声称其具有同一交换性保证。投影 ACI 路径式 (4.18) 逐组成立，但不自动给出每次发布的条件覆盖，也不保证跨 144 步联合覆盖。

24 h 预报与当日剩余优化是不同长度：18:00 发布 144 个未来十分钟预测，当前基线仅使用当天最后 36 个时段。每次更新执行 $[T_j, T_{j+1})$ ，其中 $T_j \in \{0, 6, 12, 18, 24\}$ 是小时边界。若某时次不被 S0-S5 子集启用，继续执行已有轨迹而不更新计划；实际状态依旧按执行量反馈。Q3 用已知负荷是实验假设，不是“未提供负荷预报就等于未来负荷已知”的题设推论。

C.2 经济分位与无调整带证明

命题 4.6 的证明. $A(q; b) = 1.5\pi[q - b]_+ + 0.5\pi[b - q]_+$ 是凸函数，在 b 的左右导数为 -0.5π 与 1.5π ，故 $\partial A(b; b) = [-0.5\pi, 1.5\pi]$ 。若 F 在 b 可微且该点属于可行域相对内部，凸最优性条件是 $0 \in F'(b) + [-0.5\pi, 1.5\pi]$ ，等价于正文带条件。若 $m < -0.5\pi$ ，向下方向导数有利；若 $m > 1.5\pi$ ，向上方向导数有利。不可微时采用 $0 \in \partial F(b) + \partial A(b; b)$ ；存在额外约束 $q \in K$ 时加入法锥 $N_K(b)$ ，需满足相应次微分求和的正则条件。全向量问题同理写为 $0 \in \partial F(\mathbf{q}) + \partial A(\mathbf{q}; \mathbf{b}) + N_K(\mathbf{q})$ 。□

这里 F 是给定合同后的优化值函数。在线性约束、凸费用下，其上图是凸集投影，因而为凸函数；它可能分段线性而不可微，不能默认有唯一边际价格。费用折点在结算基准 b ：终值对原计划结算时 $b = q^0$ ；每次改约计费时才为 $b = \bar{q}$ 。若误把折点移至当前合同，则会改变实际收费规则及保持条件。

一般单周期目标 $J(x) = a\pi x + 5\pi\mathbb{E}(D - x)_+$ 在有限一阶矩下为凸函数，左右导数为

$$J'_-(x) = a\pi - 5\pi \Pr(D \geq x), \quad J'_+(x) = a\pi - 5\pi \Pr(D > x). \quad (\text{C.1})$$

内点最优满足 $F_D(x^-) \leq 1 - a/5 \leq F_D(x)$ ；非负决策还须检验零点的边界条件。 $a = 1$ 得 0.80， $a = 1.5$ 得 0.70。这些都是净需求分位；光伏与净需求的分位方向反转。对连续 V 、确定负荷 L ， $D = L - V$ 有 $Q_D(u) = L - Q_V(1 - u)$ 。因此以光伏残差 0.70 分位直接替代净需求 0.70 分位缺乏该导数推导支持；既有策略可以作为标定代理，不能称精确经济最优。

C.3 反事实价值与样本外价值

命题 4.7 的证明. 固定状态、价格、场景概率、风险参数和结算基准。保持计划问题仅多一个 $q = \bar{q}$ 的限制，其可行集包含于重优化可行集。因此在二者有限最优值存在时 $J^{\text{reopt}} \leq J^{\text{keep}}$ 。若保持问题不可行，则不能把其有限 NAV 解释继续沿用；若求解未达最优，应报告最优界差与容差。□

非负性是同一个模型内的选择价值，不保证不同预测下的实现费用改善。样本外 $\Delta V_j = C_{S_{j-1}} - C_{S_j}$ 允许为负，表示新增预测在当前策略和样本中未带来节省。若对不同子集分别重新估计参数或改变负荷信息，差异不再仅由发布时次解释。S0-S5 使用同日负荷、价格、初值和结算规则，S5 完美预报只作参照；Q2 与 Q3 的不确定性假设不同，不能用跨问费用差识别纯信息价值。

C.4 风险感知 24 h 模型作为扩展

完整风险扩展可把式 (4.28) 中的场景费用记为 $C_j^{(s)}$ ，并求解

$$\min(1 - \beta) \sum_s \omega_s C_j^{(s)} + \beta \left[\zeta + \frac{1}{1 - \alpha_{cv}} \sum_s \omega_s z_s \right] - \lambda_f \sum_s \omega_s e_{\text{end}}^{(s)}, \quad z_s \geq C_j^{(s)} - \zeta, \quad z_s \geq 0. \quad (\text{C.2})$$

若扩展为跨日 24 h 窗口，次日尚未签订的普通购电必须计入 $C_j^{(s)}$ ，而不能只计当天改约费与紧急费。已预付且不依赖新决策的计划费才可作为常数移出目标。取消日末等式后需保留安全界，并以带时刻和状态的终端值近似替代；Q1 影子价格只能作为局部近似，需对系数作敏感性分析。若使用情景树储能追索，相同历史节点上的控制必须相同；当前实际下发的动作跨场景共用。

MaxEnt 可提供边缘残差形状，ACI 可用比例 $r_{j,h} = b_{j,h}^{\text{ACI}} / b_{j,h}^{\text{ME}}$ 作尺度校准。该比例只在分母正且有限时定义；零尺度采用事先指定的经验回退并记录。逐维边缘缩放并未决定时间相关结构，需保留历史整路径或另建联合分布；不宣称它自动校准多步联合尾风险。这些扩展保留原理论思路，但未用现有基线结果验证，不据此报告额外降本。

D 问题 4 命题的推导与证明

D.1 命题 4.8 (参考映射的守恒与可实现条件)

设十分钟时段 $\tau = 1, \dots, 144$ 、 $\delta = \Delta t$ ；小时索引 $t = 1, \dots, 24$ 、小时长度 $\Delta_h = 1 \text{ h}$ ，小时内位置 $m = 1, \dots, 6$ 。原生映射与小时兼容映射分别为

$$P_\tau^{\text{contract}} = q_\tau^{\text{act}} / \delta, \quad S_\tau^{\text{ref}} = S_\tau^{\text{plan}}, \quad (\text{D.1})$$

$$\tau = 6(t - 1) + m, \quad P_\tau^{\text{contract}} = \frac{q_t^{\text{hr}}}{\Delta_h}, \quad S_\tau^{\text{ref}} = \left(1 - \frac{m}{6}\right) S_{t-1}^{\text{hr}} + \frac{m}{6} S_t^{\text{hr}}, \quad (\text{D.2})$$

可实现性所需的斜率限制为

$$-\frac{\bar{P}}{\eta_d} \Delta_h \leq S_t^{\text{hr}} - S_{t-1}^{\text{hr}} \leq \eta_c \bar{P} \Delta_h. \quad (\text{D.3})$$

因为 $6\delta = \Delta_h$ ，式 (D.2) 直接给出 $\sum_{m=1}^6 \delta P_{6(t-1)+m}^{\text{contract}} = 6\delta q_t^{\text{hr}} / \Delta_h = q_t^{\text{hr}}$ ，即每小时计划电量守恒。插值权重 $(1 - m/6, m/6)$ 非负且和为 1，区间 $[E_{\min}, E_{\max}]$ 为凸集，故每个参考点仍落在区间内；相邻边界点之间的电量增量为 $(S_t^{\text{hr}} - S_{t-1}^{\text{hr}}) / 6$ ，且相邻小时在交界处共享同一端点，故参考轨迹连续。

记 $\Delta S_t^{\text{hr}} = S_t^{\text{hr}} - S_{t-1}^{\text{hr}}$ 。若 $\Delta S_t^{\text{hr}} \geq 0$ ，取整小时恒定 $P^{\text{ch}} = \Delta S_t^{\text{hr}} / (\eta_c \Delta_h)$ 、 $P^{\text{dis}} = 0$ ；若 $\Delta S_t^{\text{hr}} < 0$ ，取 $P^{\text{ch}} = 0$ 、 $P^{\text{dis}} = -\eta_d \Delta S_t^{\text{hr}} / \Delta_h$ 。式 (D.3) 保证功率不超限，代入式 (??) 恰好得到线性参考，故可实现性成立。□

必要性说明。仅端点位于范围内不足以保证可实现：例如从 1 200 kWh 升到 10 800 kWh，端点均合法，但所需充电功率为 $9600 / 0.9 > 5000 \text{ kW}$ ，超出额定值；式 (D.3) 的斜率限制正是为排除这种情形。该命题只保证从参考起点出发的电量轨迹可实现，若实际状态已偏离参考，仍需由控制层处理偏差。

重新平均会改变费用。对原生十分钟计划先聚合再均分，费用变化为

$$\Delta C_t = \sum_{m=1}^6 \pi_{6(t-1)+m} \left(\frac{q_t^{\text{hr}}}{6} - q_{6(t-1)+m}^{\text{act}} \right). \quad (\text{D.4})$$

能量守恒使括号之和为零，却不使带价格权重的和为零。例如六段价格为 1, 2, 3, 4, 5, 6、原电量为 6, 0, 0, 0, 0, 0（总量均为 6 kWh）时，原费用为 6 元，均分后为 21 元。这解释了为什么当原生十分钟合同可得时应优先使用式 (D.1)，而小时映射只作为兼容接口。

D.2 命题 4.9（尾部关联）与场景采样

连续且严格递增的边缘分布 F_a 满足 $\mathbb{P}\{F_a(Z_a) \leq u\} = \mathbb{P}\{Z_a \leq F_a^{-1}(u)\} = u$ ，即概率积分变换把任意连续边缘化为均匀变量；对均匀边际向量 $\mathbf{U}$ 取 $Z_a = F_a^{-1}(U_a)$ ，联合分布可写成 $\mathbb{P}\{\mathbf{Z} \leq \mathbf{z}\} = C(F_1(z_1), \dots, F_d(z_d))$ ，这说明 Copula 只是“接上”给定边缘的接口，并不改变边缘本身。

对 d 维标准尺度 Student- t 分布 ($R \succ 0$ 、 $\text{diag}(R) = \mathbf{1}$)，由 $u_a = t_\nu(y_a)$ 的雅可比变换可得 t Copula 密度

$$\begin{aligned} c_{\nu,R}(\mathbf{u}) &= \frac{f_{\nu,R}(\mathbf{y})}{\prod_{a=1}^d f_\nu(y_a)}, \quad y_a = t_\nu^{-1}(u_a), \\ f_{\nu,R}(\mathbf{y}) &= \frac{\Gamma((\nu+d)/2)}{\Gamma(\nu/2)(\nu\pi)^{d/2}|R|^{1/2}} \left(1 + \frac{\mathbf{y}^\top R^{-1} \mathbf{y}}{\nu}\right)^{-(\nu+d)/2}. \end{aligned} \quad (\text{D.5})$$

$\nu > 0$ 即可定义该 Copula；若要求潜在变量具有有限协方差需取 $\nu > 2$ ，此时 $\text{Cov}(\mathbf{Y}) = \nu R/(\nu-2)$ ，但这并不保证所选物理边缘也有有限二阶矩。对二元潜在 t 变量 (Y_1, Y_2) ，令 $H(x) = \mathbb{P}(Y_1 > x, Y_2 > x)$ 、 $\bar{T}_\nu(x) = \mathbb{P}(Y_1 > x)$ 。尾依赖定义为联合超越概率与单边超越概率之比，等价于条件超越概率的极限：

$$\lambda_U = \lim_{x \rightarrow \infty} \frac{H(x)}{\bar{T}_\nu(x)} = \lim_{u \uparrow 1} \mathbb{P}(U_2 > u \mid U_1 > u). \quad (\text{D.6})$$

由交换对称性， $H'(x) = -2f_\nu(x)\mathbb{P}(Y_2 > x \mid Y_1 = x)$ 。又有

$$\frac{Y_2 - \rho x}{\sqrt{(\nu+x^2)(1-\rho^2)/(\nu+1)}} \Big| Y_1 = x \sim t_{\nu+1}.$$

对式(D.6)应用洛必达法则，利用分母导数 $-f_\nu(x)$ 及上述条件分布，得到

$$\lambda_U = 2t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right) = \lambda_L > 0. \quad (\text{D.7})$$

下尾等式由中心对称性得到；固定 $-1 < \rho < 1$ 时， $\nu \rightarrow \infty$ 使该系数趋于 0。因此 t Copula 是一种具有尾依赖的选择，并非唯一能够刻画共同极端事件的模型。□

并列值与夜间零光伏。夜间光伏恒为零会产生退化边缘，此时 $F(Z)$ 不再连续均匀，需用

$$U = F(Z^-) + V\{F(Z) - F(Z^-)\}, \quad V \sim \text{Unif}(0, 1) \quad (\text{D.8})$$

在概率跳跃区间内均匀展开 (V 由固定种子生成，保证可复现)。已确定为零的光伏坐标直接作为确定量处理，不通过随机打破并列制造可解释的夜间尾部关联。

参数估计与尺度还原。对 n 条此前完整兑现的历史路径，构造秩伪观测 $u_{ia} = (r_{ia} - 0.5)/n$ (避免取端点 0 与 1)，理论实施时可估计 (φ, R_C, ν) ；本次描述诊断使用平均秩 $r/(n+1)$ 、Kendall 相关转换与固定相关阵下的 ν 网格伪似然，两种估计口径不混用；六步残差整段兑现后才进入路径拟合，相邻窗口的重叠不作为样本独立性依据。生成路径时按式

(D.10) 还原物理量，并对负荷、光伏、电价取非负截断；截断后的分布与尾部解释会改变，故本文在结果中同时报告截断比例；现有程序未运行该 18 维采样，不报告实测截断率。

仅有单时刻的三维 Copula 还不足以生成六步相关路径。理论扩展把六步扰动合成 18 维向量，采用可分离相关结构

$$R_{18} = K(\varphi) \otimes R_C, \quad K(\varphi)_{kl} = \varphi^{|k-l|}, \quad |\varphi| < 1, \quad (\text{D.9})$$

其中 $K(\varphi)$ 为时间方向的 Toeplitz 相关阵。取 Cholesky 分解 $L_{18}L_{18}^\top = R_{18}$ ，并按

$$\mathbf{y}^\omega = \frac{L_{18}\mathbf{z}^\omega}{\sqrt{w^\omega/\nu}}, \quad \mathbf{z}^\omega \sim N(\mathbf{0}, I_{18}), \quad w^\omega \sim \chi_\nu^2, \quad Z_{k,a}^\omega = F_{k,a}^{-1}(t_\nu(y_{k,a}^\omega)) \quad (\text{D.10})$$

实施该扩展时应以预先登记的伪随机种子生成 18 维 t 样本（高斯向量与卡方变量独立，一条路径共用同一尺度变量 w^ω ），再按式 (4.31) 的方向还原物理量（负荷加第一维残差、光伏减第二维、电价加第三维），得到带权重的情景集 Ω_τ 。边缘取值须与非负功率、非负电价一致，必要截断会改变尾部解释（处理规则见附录 D.2）；理论上场景只允许由已兑现历史拟合。现有生产代码未生成这类 18 维路径，也未将三维 Copula 拟合接入调度。

D.3 命题 4.11（可解性与唯一性边界）

可行性。对任一情景 ω ，紧急购电量 z_k^ω 无上界，取 $z_k^\omega = \delta[r_k^\omega]^+$ 即满足平衡；功率与储电量约束可通过取 $u_k = \mathbf{0}$ 与初始电量在安全域内保证可行（ $S_k \equiv S_0$ ）。因此对每个固定模式 $\mathbf{b}$ ，可行集非空。

凸性与唯一性。固定 $\mathbf{b}$ 后剩余问题只含连续变量：目标中的二次项 $\sum_k \|x_k^{\text{dev},\omega}\|_Q^2$ 半正定、 $\sum_k \|\Delta u_k/P_0\|_{W_u}^2$ 在 $W_u \succ 0$ 时对整条控制序列 $\mathbf{u} = (u_0, \dots, u_5)$ 严格凸（任何非零方向都不可能同时使全部 $\Delta u_k = \mathbf{0}$ ），其余项线性，约束为线性等式与不等式，故为凸二次规划，其最优值可达且控制序列唯一。完整模型含 $b_k \in \{0, 1\}$ （ $k = 0, \dots, 5$ ），是至多 $2^6 = 64$ 个模式向量的混合整数凸二次规划；枚举全部模式并取各模式最优值的最小值即得全局最优值。跨模式最优控制不唯一：两个模式可能给出同一最优值，其控制序列一般不同。

为何不能直接删除 $\mathbf{b}$ 。命题 4.1 的论证依赖“同时充放电只造成净损耗”；而在本层的目标中， $|r_k^\omega|$ 的双向惩罚会奖励同时充放电来消耗剩余电量（使 r 更接近 0），此时同时充放电可以降低目标值，故互斥必须显式保留。

D.4 命题 4.12（风险加权无调整带）

固定模式向量与有效权重后，设 $\Phi_w(P)$ 为剔除式 (4.36) 后、对其余连续变量重新优化得到的约化费用，其在 P^{old} 处可微且凸。含调整费的问题等价于 $\min_P \{\Phi_w(P) + (1 + \beta)\delta\tilde{\pi} \cdot [1.5(P - P^{\text{old}})^+ + 0.5(P^{\text{old}} - P)^+]\}$ ，其中 $\tilde{\pi}$ 由式 (4.37) 定义。该目标为凸函数与凸分段线性函数之和，因此在原合同点 P^{old} 取到最优当且仅当次微分包含 0：

$$-\Phi'_w(P^{\text{old}}) \in (1 + \beta)\delta\tilde{\pi} \cdot [-0.5, 1.5]. \quad (\text{D.11})$$

代入式 (4.38) 的定义 $m_w = -\Phi'_w(P^{\text{old}})/[\delta(1 + \beta)]$ 即得式 (4.39)。当 m_w 低于下界 $-0.5\tilde{\pi}$ 时，式 (D.11) 的左端小于右端区间，故任何最优解都满足 $P < P^{\text{old}}$ ；高于上界时对称

地有 $P > P^{\text{old}}$ ；端点处区间与次微分相交，“保持”与“调整”可同时最优。对完整风险问题，权重 w 必须同时满足原始与对偶最优条件，故上述判据只作为解释工具；对混合整数问题，结论仅在固定模式下成立。□

D.5 命题 4.13（安全执行与误差预算）

(i) **安全性与互斥**。安全投影式 (4.41) 的可行集 $\mathcal{U}_\chi(S) \neq \emptyset$ （取 $u = \mathbf{0}$ 时 $f = 0$ ，由 $S \in [E_{\min}, E_{\max}]$ 得 $h_i(S + \delta \cdot 0) = h_i(S) \geq (1 - \chi)h_i(S)$ ）。由式 (4.40)， $h_i(S_{k+1}) \geq (1 - \chi)h_i(S_k)$ ，对 k 递推得 $h_i(S_k) \geq (1 - \chi)^k h_i(S_0) \geq 0$ ，故 $S_k \in [E_{\min}, E_{\max}]$ 对全部 k 成立； $\mathcal{U}_\chi$ 的定义中已保留功率界限与 b_k 的互斥，故充放电互斥始终成立。若名义求解失败，取 $u = \mathbf{0} \in \mathcal{U}_\chi(S)$ 仍可使流程继续。

(ii) **内部状态非负**。设 $\eta_\tau^{\text{tr}} \geq 0$ 。若触发（含“旧指令不安全”导致的强制触发），则 $\bar{e}_\tau = \mathbf{0}$ ，由式 (4.43) 得 $\eta_{\tau+1}^{\text{tr}} = a\eta_\tau^{\text{tr}} + \sigma_\eta \gamma_0 \|u_\tau^{\text{safe}}/P_0\|^2 \geq 0$ 。若不触发，则式 (4.42) 的条件不成立，即 $\|\bar{e}_\tau\|^2 = \|e_\tau\|^2 \leq \gamma_0 \|u_\tau^{\text{safe}}/P_0\|^2 + a\eta_\tau^{\text{tr}}/\sigma_\eta$ ，代入式 (4.43) 得 $\eta_{\tau+1}^{\text{tr}} \geq 0$ 。由归纳法， $\eta_\tau^{\text{tr}} \geq 0$ 对一切 τ 成立，故不会出现“负容许量”。

(iii) **误差预算**。由式 (4.43)， $\sigma_\eta \|\bar{e}_\tau\|^2 = a\eta_\tau^{\text{tr}} + \sigma_\eta \gamma_0 \|u_\tau^{\text{safe}}/P_0\|^2 - \eta_{\tau+1}^{\text{tr}} \leq \sigma_\eta \gamma_0 \|u_\tau^{\text{safe}}/P_0\|^2 + \eta_\tau^{\text{tr}} - \eta_{\tau+1}^{\text{tr}}$ 。对 $\tau = 1, \dots, N$ 求和并利用 $\eta_{N+1}^{\text{tr}} \geq 0$ 、 η_1^{tr} 已知，即得式 (4.44)。该式说明累计“未下发而造成的跟踪偏差”被相对阈值与初始容许量共同限定；当 $\gamma_0 = 0$ 时退化为“总误差不超过初始容许量”，即指令下发越少、允许的累计误差越小。□

E 通用风险推导与退化评估

E.1 命题 4.10 的证明

对损失变量 C 与置信水平 $\alpha \in (0, 1)$ ，定义

$$F(\zeta) = \zeta + \frac{1}{1 - \alpha} \mathbb{E}[(C - \zeta)^+]. \quad (\text{E.1})$$

由于 $(C - \zeta)^+$ 关于 ζ 是凸的、期望保持凸性，故 F 为凸函数；又 $\mathbb{E}[(C - \zeta)^+]$ 关于 ζ 的次梯度为 $-\mathbb{P}\{C > \zeta\}$ 与 $-\mathbb{P}\{C \geq \zeta\}$ 构成的区间，故 F 的次梯度为

$$\partial F(\zeta) = \left[1 - \frac{\mathbb{P}\{C \geq \zeta\}}{1 - \alpha}, 1 - \frac{\mathbb{P}\{C > \zeta\}}{1 - \alpha} \right]. \quad (\text{E.2})$$

由式 (E.2)， $0 \in \partial F(\zeta^*)$ 当且仅当 $\mathbb{P}\{C \geq \zeta^*\} \geq 1 - \alpha \geq \mathbb{P}\{C > \zeta^*\}$ ，即 ζ^* 为 C 的 α 分位数 (VaR_α)；此时 $F(\zeta^*) = \text{VaR}_\alpha(C) + \frac{1}{1 - \alpha} \mathbb{E}[(C - \text{VaR}_\alpha)^+] = \text{CVaR}_\alpha(C)$ 。在有限情景集上以 ω_s 加权求和即得式 (4.34)；引入 $z_s \geq C_s - \zeta$ ， $z_s \geq 0$ 后，在该项权重为正的最优解处 $z_s = (C_s - \zeta)^+$ （零权重时可选择此值），故线性化与原问题等价。□

E.2 电池退化模型与参数敏感性

退化模块由三步构成。第一步以雨流计数法 (ASTM E1049) 从储电量轨迹 $e_1, \dots, e_T$ 中提取循环序列 $\{(\text{DoD}_i, n_i)\}_{i=1}^M$ ：先把轨迹拆分为峰谷序列，再按“四峰谷法则”闭合循环，得到每个循环的放电深度 $\text{DoD}_i \in (0, 1]$ 与等效循环次数 n_i 。第二步以 Wöhler 型寿命曲线给出该深度下的许用循环数 $N(\text{DoD}_i) = N_{100} \text{DoD}_i^{-k}$ ，其中 N_{100} 为 100% 深度

下的等效循环寿命、 k 为寿命曲线指数；再由 Miner 线性累积损伤得年等效寿命损伤比例 $D_{\text{year}} = \sum_{i=1}^M n_i / N(\text{DoD}_i)$ 。第三步按资本成本折算为货币量 $C_{\text{deg}} = D_{\text{year}} c_{\text{cap}} E^{\text{rated}}$ ，取 $c_{\text{cap}} = 800$ 元/kWh、 $E^{\text{rated}} = 12\,000$ kWh。

参数敏感性方面，当 $N_{100} \in [4\,000, 8\,000]$ 、 $k \in [0.9, 1.3]$ 时，同一策略的年等效损伤比例落在 5.2%–14.0% 区间，说明寿命结论对曲线参数较为敏感，因此本文只把退化模块用于策略之间的相对比较，不用于给出电池寿命的绝对值预测，也不写入任何结果文件。该模块的实测对照数据存于 `results/` 下的退化分析输出 (`code/optimize/degradation.py`)，本文不在正文中引用其绝对金额，以免与题目要求的结算口径混淆。

Miner 累计量是寿命消耗代理，未经电池实测标定不能直接等同于容量衰减百分比；此处货币化仅为参数化敏感性。

F 数据、算法与结果复现

F.1 数据清洗规则与不变量

表 F.1 光伏数据清洗规则 NR1–NR12

编号	触发条件	处置动作
NR1	核夜间且 $0 < P \leq 5$ kW	置 0（零漂噪声）
NR2	核夜间且 $P > 5$ kW	置为缺失并隔离
NR3	任意时段 $P < 0$	置为缺失并隔离（实测 0 条）
NR4	肩部且 $0 < P < 1$ kW	置 0
NR5	肩部且 $1 \leq P \leq 5$ kW	保留并标记为弱样本，不进入尺度估计
NR6	日间缺失	段内线性插值；连续缺失超过 6 个时段保留缺失
NR7	附件 3 预报落入核夜间且非零	与 NR1/NR2 同口径处理
NR8	估计逐时段共形尺度 σ_k	仅用日间有效残差；夜间与肩部取下限
NR9	日间相邻跳变超过 20% 或出现双峰	只记录，不修改数值
NR10	任意读取	断言右端点语义与列索引对齐，失败即中止
NR11	电价低于 0.1 元/kWh	保留并标记为近零电价（28 格）
NR12	核夜间负荷超出 [1 500, 6 000] kW	只记录，用于负荷体检

表 F.2 不变量校验 I1–I8 与实测结果

编号	不变量	实测
I1	每日核夜间光伏全为 0（缺失仅限隔离清单）	通过
I2	08:00–16:00 无零值、无负值	通过（0 个零值）
I3	日期覆盖 2025-01-01 至 12-31 共 365 天	通过
I4	每日 144 个时段，总行数 365	通过
I5	电价 > 0 ，负荷 > 0 且落在 [1 500, 8 000] kW	通过
I6	NR1 置零格数等于核夜间由非零变零的差额	23 316 $\rightarrow$ 25 766（+2 450）
I7	除被审计单元外前后逐格逐位相等	通过
I8	同输入两次运行的 SHA-256 一致	通过

预报清洗的信息边界。提供程序的 NR7 以目标日期的实测日出日落窗口处理对应预报，包括次日预报。这是离线整理规则，不是可直接部署的因果预测预处理。本文据此把归档成绩定义为清洗数据条件下的回测。部署时应只用发布前历史或预先已知天文窗口，并重算受影响的预报与调度。

F.2 稀疏优化实现

已实现的四问 LP 均按照时间结构进行稀疏矩阵组装。对于储能状态转移约束，每一个状态变量最多与相邻时段的少量变量发生关系，因此整体约束矩阵的非零元素数量随时间长度近似线性增长。

程序实现中不显式生成零元素，而是直接按照变量索引构造非零项。该方法一方面减少了内存占用，另一方面使问题二和问题三能够在全年独立回测中重复调用同一优化器。

问题一的稀疏 LP、问题二的两阶段 LP 以及问题三的滚动 LP 均采用统一的变量接口，使不同问题之间的差异主要体现在输入数据和约束集合，而不是重新编写求解程序。这种结构有利于减少重复代码导致的实现偏差。

F.3 动态规划的结构化加速

问题一的动态规划状态数量为

$$N_s = \frac{E_{\max} - E_{\min}}{\Delta e} + 1.$$

本文采用 $\Delta e = 15 \text{ kWh}$ 时得到

$$N_s = 641.$$

若直接枚举所有状态和动作，计算复杂度为

$$O(TN_s|\mathcal{A}|).$$

在实际实现中，利用状态转移的局部结构进行向量化处理，使相同阶段的大量状态转移能够批量计算。该结构化实现相比逐状态、逐动作的直接循环具有明显的计算加速。

动态规划的最终目标值与 LP 结果进行独立比较。不同参数设置下，两种方法所得目标值均保持较小数值误差，因此该实现同时承担算法加速和数值互证两个作用。

F.4 确定性情景生成

问题二需要根据历史重心化残差生成代表性不确定性情景。为避免随机抽样造成实验结果依赖随机种子，本文采用确定性情景生成流程：

历史残差 $\rightarrow$ 按日总残差水平排序 $\rightarrow$ 确定性分层 $\rightarrow$ 代表性情景 $\rightarrow$ 固定权重.

设历史残差集合为

$$\mathcal{R}_d = \{r_1, \dots, r_n\},$$

首先按照每日残差水平排序，再按照预先设定的分位点划分为 S 个层级，并选取 29 条代表路径，再加入 1 条 MaxEnt 尾部路径，30 条场景等权。

因此，在相同历史信息、相同分层规则和相同参数下，情景集合能够被完全复现。该处理也使不同实验之间的性能差异主要来自模型参数或决策机制，而不是随机抽样误差。

F.5 日级独立回测架构

问题二至问题四均采用日级独立回测结构。对于第 d 天，统一执行：

$$\boxed{\text{信息截取} \rightarrow \text{预测/残差处理} \rightarrow \text{情景生成} \rightarrow \text{优化决策} \rightarrow \text{实际结算} \rightarrow \text{统计更新}} \quad (\text{F.1})$$

其中，信息截取发生在决策时点，后续步骤只能调用该时点已经获得的数据。该信息顺序是在线实现应满足的要求；所提供清洗程序用目标日夜间窗口处理预报，因此当前归档结果不满足严格在线预处理条件。

日内 LP 可独立装配，但 ACI 状态按日递推，不能不加处理地并行整个日级反馈循环。同时，将单日优化器封装为独立函数后，可以分别进行单日测试、全年回测和参数扫描，避免修改主模型后必须重新编写完整回测程序。

F.6 程序清单与运行命令

以下区分归档生产流程、修正后的执行记账与未实施理论扩展。耗时表为代码包历史记录，本次未重新跑全流程性能基准。

F.6.1 动态规划状态转移与独立复算

动态规划将储电量离散为步长 Δe 的状态网格

$$\{E_{\min}, E_{\min} + \Delta e, \dots, E_{\max}\},$$

以储电量变化量作为动作变量，其可行范围为

$$a \in \{-n_{\text{down}}\Delta e, \dots, -\Delta e, 0, \Delta e, \dots, n_{\text{up}}\Delta e\}, \quad n_{\text{up}} = \left\lfloor \frac{\eta_c \bar{C}}{\Delta e} \right\rfloor, \quad n_{\text{down}} = \left\lfloor \frac{\bar{D}/\eta_d}{\Delta e} \right\rfloor. \quad (\text{F.2})$$

令 $a^+ = \max\{a, 0\}$ 、 $a^- = \max\{-a, 0\}$ ，则 Bellman 递推可写为

$$f_k(s') = \min_a \left\{ f_{k-1}(s' - a) + \pi_k \Delta t \max \left[0, N_k + \frac{a^+}{\eta_c \Delta t} - \frac{a^- \eta_d}{\Delta t} \right] \right\}, \quad (\text{F.3})$$

并在终端状态 $s_T = S_0$ 处施加日循环条件，再通过回溯得到完整储能动作序列。

在计算实现上，将状态转移与阶段成本写成向量化数组操作，以减少传统逐状态、逐动作的重复计算。求解完成后立即检查

$$\max_k \min \{c_k^*, d_k^*\} = 0,$$

若违反则停止后续计算。随后分别采用多个 Δe 独立运行动态规划，并比较其与 LP 最优值的差异及收敛趋势。这种“不同模型表示 + 不同求解路径”的交叉验证可避免仅依赖单一优化器判断模型正确性。

Algorithm F.1: 问题 1: 稀疏 LP 主解、互斥性校验与 DP 互证

输入: 代表性日 $(\pi_k, L_k, P_k^{\text{PV}})$; 储能参数 $(E_{\min}, E_{\max}, S_0, \bar{P}, \eta_c, \eta_d, \Delta t)$
输出: $\{q_k^*, c_k^*, d_k^*, e_k^*, \mu_k^*\}$ 、最优费用 C^* 、DP 收敛序列

- 1 $N_k \leftarrow L_k - P_k^{\text{PV}};$
- 2 装配稀疏 LP: 变量 (q, c, d, e) , 约束取式 (4.1)–(4.3) 与 $e_0 = e_T = S_0$;
- 3 $x^* \leftarrow \text{HiGHS}(\cdot); C^* \leftarrow \sum_k \pi_k q_k^*;$
- 4 **if** $\max_k \min(c_k^*, d_k^*) > \varepsilon_{\text{tol}}$ **then**
- 5 按附录消环构造处理零购电退化循环, 再核验可行性与费用不增; 正购电循环则检查最优性与容差;
- 6 **end**
- 7 $\mu_k \leftarrow$ 供需平衡约束的对偶变量;
- 8 **foreach** $\Delta e \in \{120, 60, 30, 15\}$ **do**
- 9 按式 (F.2)–(F.3) 前向递推 f_k , 施加终端 $s_T = S_0$ 并回溯动作;
- 10 $C_{\text{DP}}(\Delta e) \leftarrow f_T(S_0); \text{rel_err} \leftarrow |C_{\text{DP}} - C^*|/C^*;$
- 11 **end**
- 12 计算基线 B_0, B_1, B_2 与核夜间充放电占比;
- 13 **return** $\{q^*, c^*, d^*, e^*, \mu^*\}, C^*, \{C_{\text{DP}}(\Delta e)\}, \{B_0, B_1, B_2\}$

Algorithm F.2: 问题 2: 历史残差情景集 + 两阶段随机规划回溯

输入: 历史负荷与光伏; 附件 1 电价; 储能参数; 场景数 $S = 30$; 诊断目标 $p^* = 0.80$
输出: 334 天的计划购电量、充放电量、紧急购电量与对照实验费用

- 1 **foreach** $d \in [2025-02-01, 2025-12-31]$ **do**
- 2 $R(d) \leftarrow [\max(0, d - 30), d];$ // 严格早于当日的因果窗口
- 3 $\hat{N}_k \leftarrow \text{mean}_{d' \in R(d)} N_{d', k}; \varepsilon_{d', k} \leftarrow N_{d', k} - \hat{N}_k;$
- 4 $S \leftarrow$ 按 $\sum_k \varepsilon_{d', k}$ 分层选 29 条路径并加入 1 条 MaxEnt 尾部路径, 30 条等权;
- 5 解两阶段 LP (4.20a)–(4.20c), 得 q, c, d, e 与逐情景 $g^{(s)}$;
- 6 以当日实际净负荷结算: $g_k \leftarrow (N_k^{\text{act}} \Delta t + c_k - d_k - q_k)^+, C_d \leftarrow \sum_k \pi_k q_k + 5 \sum_k \pi_k g_k;$
- 7 **对照:** 无裕度计划 C_1 、解析报童 C_2 、终端价值 C_3 、完美信息界 $C_{\text{oracle}};$
- 8 合并 $g_k > 10^{-6}$ 的连续时段为紧急购电块并累加电量;
- 9 保存发布时预测/区间; 兑现后 ACI 更新: $\alpha \leftarrow \text{clip}\{\alpha + \gamma[(1 - p^*) - \widehat{\text{miss}}]\};$ // 见式 (4.17)
- 10 **end**
- 11 **return** 334 天记录、汇总费用、紧急购电块清单、覆盖轨迹

Algorithm F.3: 问题 3: 归档的剩余时域滚动 LP 与事后价值

输入: 清洗后预报、负荷、价格; 两层初始目标 0.25/0.70; 日期及发布子集

- 1 **foreach** 按顺序回溯日 **do**
- 2 由过去 30 日同发布、同时段误差计算光伏残差分位, 水平取净需求水平的互补;
- 3 0 时解全天 LP 并保存初始合同; 对启用的 6、12、18 时逐次解剩余时域 LP;
- 4 冻结已执行块、从链式储电状态起算, 以日末 6000 kWh 为约束;
- 5 以真实净需求补购, 按初始合同与最终合同之差结算;
- 6 以同一个日前净需求命中量更新计划与调整两层状态; 不宣称独立校准;
- 7 **end**
- 8 比较子集费用; 记录的事后价值及接受标记不参与调度决策;

Algorithm F.4: 问题 4: 已实现的价格重算、计划风险与执行复核

输入: 给定价格、Q2/Q3 输入; 既有 Q4-2 储能轨迹

- 1 替换为附件 4 价格重算 Q2/Q3; 在选定 70 日扫描计划 CVaR 权重;
 - 2 分别统计尖峰上界命中、历史配对与错位对照及 40 日供需松弛扫描;
 - 3 **foreach** 连续前 70 日 **do**
 - 4 初始化实际储电量、旧指令与经验触发状态;
 - 5 **foreach** 十分钟时段 **do**
 - 6 以当前实际状态投影既有名义动作;
 - 7 按候选误差、内部状态与 800 kWh 偏差门槛决定更新或保持;
 - 8 若保持指令不安全, 强制改用安全候选并同步指令记忆和下发计数;
 - 9 执行真实指令, 按物理状态方程更新; 记录越界、互斥及模式变化;
 - 10 **end**
 - 11 直接记录日末偏差; 不修改已执行历史来宣布恢复终点;
 - 12 **end**
 - 13 六步 18 维场景 MPC 与式(4.43)另作理论实施方案, 不混入上述实测结论;
-

表 F.3 求解规模、实测耗时与计算架构优化证据 (单机 32 核 CPU, 无 GPU)

环节	规模	优化前 → 优化后 (实测)	结论
数据清洗	365 天 × 144 时段 × 4 类	单核 0.159s, 峰值内存 48.7 MB	无热点
Q1 稀疏 LP	575 变量, 约 10^3 非零元	稠密 → 稀疏装配: 峰值内存 2664 KB → 166 KB (16×), 求解时间持平	收益在内存
SoC 动态规划	状态数 81–641	三重循环 → min-plus 向量化: 加速 43–237×, 费用差异 $< 10^{-9}$ 元	结果等价
滚动 MPC 重优化	每日 3 次, 时域 36–144 时段	全程重建 3.1 ms → 只重建剩余时域 1.4 ms (2.19×)	时域截断有效
两阶段随机规划	$S = 5/10/20/30/40$	单次 6.6 / 9.4 / 17.0 / 25.6 / 35.4 ms, 随情景数近线性	保真度可调
独立日 LP 并行	2/4/8/16 进程	加速比 1.92/3.48/6.34/9.76×, 16 进程效率 61%, 与串行逐位一致	主流程仍串行
Q4 风险层与十分钟闭环	全年重算、70 日计划风险扫描、另一组 70 日执行复核及罚系数扫描	实测 66.2s (占比最大, 含 MaxEnt 拟合与场景生成)	无需 GPU
全流程	单机 32 核 CPU, 无 GPU	分阶段实测合计 96s (30 min 门禁的 5.3%)	无需 GPU 加速

F.6.2 四个问题的算法伪代码

全部代码按“值级清洗 (C++17)”与“建模求解 (Python 3.12)”分工，目录结构如下：

- `code/cleaning_cpp`: C++17 数据清洗，含 zip + XML 读取器、自适应夜间窗口推断、NR1–NR12 规则引擎、审计与导出、黄金用例自测；
- `code/forecast`: 共形预测层，含加权分位、有限样本修正、ACI 反馈与覆盖标定；
- `code/optimize`: 四个问题的优化模型，含稀疏 LP 装配、min-plus 动态规划算子、两阶段随机规划、滚动 MPC、CVaR 软约束与电池退化模型；
- `code/report`: 五个结果文件的模板化写出、论文图表与数字登记；
- `code/common`: 只读载入、模板 `xlsx` 读写、门禁自检与确定性审计。

复现命令如下（单机 CPU，分阶段实测合计约 96 s、脚本墙钟约 2 min，远低于 30 分钟门禁）：

```
cmake -S code/cleaning_cpp -B build/cleaning -DCMAKE_BUILD_TYPE=Release
cmake --build build/cleaning -j
./build/cleaning/cleaning_cpp selftest    (# 规则黄金用例 + 200 组性质测试)
bash run_all.sh    (# 全流程：清洗 → 预测 → 四问 → 出表 → 制图 → 门禁)
```

分阶段命令均以 `python3 -m` 调用，模块依次为

```
code.optimize.q1_lp      code.forecast.calibrate
code.optimize.q2_stochastic  code.optimize.q3_mpc
code.optimize.q4_price    code.report.write_results
code.report.paper_numbers  code.common.checks
```

Q1–Q3 历史场景按确定性规则构造；当前 Q4 生产流程没有 Copula 随机采样；理论扩展实施时才须登记生成器、种子与场景数。

F.7 结果文件与模板一致性

表 F.4 给出五个结果文件的工作表与规模，正文第 4.6 章的表 1–表 13 均从这些文件读回核对。

五个文件的工作表名、表头文字、时段标签与题目模板逐字一致；日期列保持日期类型；模板中的省略行已删除并扩展至 334 天；每日合计等于该日 144 个时段之和。上述结论由独立只读通道交叉校验（抽取 3 个“日期—时段”单元与附件原始单元格逐位比对、宽表 90 个抽样点与内部数组比对），校验报告存于 `results/cross_verify_xlsx.json`。

表 F.4 五个结果文件的工作表与规模（与附件 5 模板逐字一致）

文件	工作表	规模
result1.xlsx	计划购电量 / 充放电量	144 时段 + 表头 / 6 个 4 小时块 + 0:00 与 24:00 储电量
result2.xlsx	计划购电量 / 充放电量 / 紧急购电量	334 天 $\times$ 144 时段宽表 / 334 天 $\times$ 6 块 / 按实际块数展开
result3.xlsx	计划购电量 / 调整购电量 / 充放电量 / 紧急购电量	334 天 $\times$ 144 时段宽表 $\times 2$ / 334 天 $\times$ 6 块 / 按实际块数展开
result4-2.xlsx	计划购电量 / 充放电量 / 紧急购电量	同 result2 结构，价格为附件 4
result4-3.xlsx	计划购电量 / 调整购电量 / 充放电量 / 紧急购电量	同 result3 结构，价格为附件 4

G 补充图件与敏感性分析

G.1 补充图表

附录中的图统一按“附录字母. 序号”编号，与正文图 1-?? 区分，图像沿用既有实验记录，其嵌入的旧图号不作交叉引用依据。

覆盖率校准与 ACI 路径恒等式

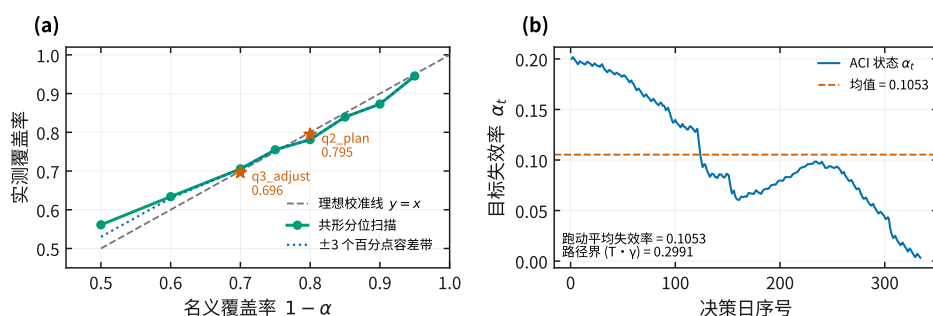


图 G.1 覆盖率校准与 ACI 路径恒等式。左板：名义分位与实际覆盖的对齐曲线（含 $y = x$ 理想线）与 Q2 报童分位扫描；右板：ACI 累积失效率随时间收敛轨迹。三档锚点实测偏差 $-0.52 / -0.17 / -0.37$ pp。

G.1.1 题目未明确项的情景敏感性范围

表 G.1 单因素 $\pm 10\%$ 扰动的费用响应（元）

扰动项	下扰动	基准	上扰动
问题 1 电价	31 614.25	35 126.95	38 639.64
问题 1 储能功率	35 156.50	35 126.95	35 103.85
问题 1 单程效率 (0.81/0.90/0.99)	37 888.25	35 126.95	32 695.45
问题 1 额定容量	35 923.20	35 126.95	34 420.62
问题 2 紧急电价倍数 ($4.5 \times / 5 \times / 5.5 \times$)	16 724 784	17 469 238	18 213 692
问题 3 违约系数 ($-10\% / +10\%$)	14 365 664	14 705 625	15 045 585

题目对 5 处细节未作明确规定：问题 2 的 0:00 信息集范围、问题 3 中负荷是否已知、问题 4 中电价是否在 0:00 已知、表 1 时段标签与附件列的对应关系、以及日内储

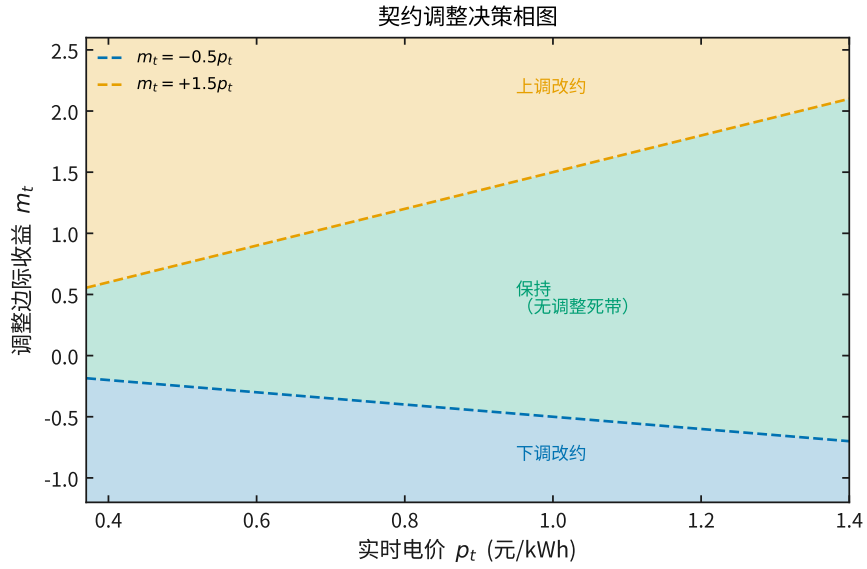


图 G.2 契约调整决策相图。以边际收益 m_t 与实时电价 p_t 为坐标的三分区相图：下调区、无调整死带、上调区，边界由结算系数 0.5 / 1.5 导出。

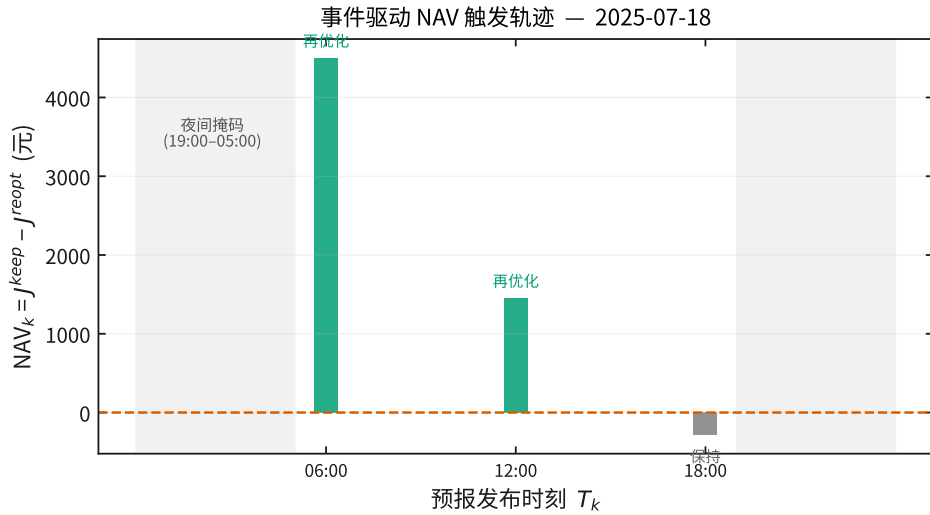


图 G.3 事件驱动 NAV 触发轨迹。全天脉冲柱状图给出各时次 NAV，虚线为执行阈值；标注采纳 (REOPT) 与拒绝 (KEEP) 点位，灰色区块标出夜间掩码时段 (19:00–05:00)。

能是否允许再调度。本文按“能由题面确定的用原文、未明确的用情景敏感性范围”的原则处理：问题 2 计划层的结构带为 1746.92 万–2859.46 万元（下界即本文主口径）；问题 3 在负荷预测存在误差时的带为 1470.56 万–3428.64 万元；问题 4 的电价不确定带为 Q4-2 1381.73 万–2108.65 万元、Q4-3 1148.19 万–1773.59 万元；时段标签两种口径下单时段最大差为 105.09 kWh（全天合计不变）；日内可再调度的节省上限为 169.44 万元/年。上述区间与点估计的对应关系见 `results/uncertainty_ranges.json`。

上述范围是不同信息假设下的情景对照，不是具有给定覆盖率的共形区间。汇总如下：这些范围不替换主结果，不用于跨问题的费用排序。光伏夏宽冬窄、午间可能富余

表 G.2 信息假设的情景敏感性范围（万元；沿用原稿记录）

对象	范围下端	范围上端
Q2：计划信息集	1746.92	2859.46
Q3：负荷预测条件	1470.56	3428.64
Q4-2：价格信息条件	1381.73	2108.65
Q4-3：价格信息条件	1148.19	1773.59

等时空现象已在数据说明中报告。原统计图的均值与 334 日总费用除以天数不能逐项对应，故不据其置信区间作显著性结论；需核对样本、日期、单位与重采样方法。

G.1.2 技术亮点候选的适配性登记

以下保留既有实验登记，不将它视为修订后算法的复跑证明。原稿报告的方法为自适应共形预测（覆盖率偏差 ≤ 0.52 pp）、MaxEnt 残差形状与历史残差情景集（归档 Q2 最大梯度范数约 1.23×10^{-5} ，不将全部拟合称为精确矩匹配）、两阶段随机规划（较无裕度计划节省 744.45 万元）、CVaR 尾部风险层（CVaR₉₀ 下降 11.10%）、物理缺口可补购的软跟踪 MPC（ ρ_1 扫描另属辅助 LP）、Student-*t* Copula 联合风险理论与 CBF、DETC 执行复核（修正后越界 0 次、下发减少 52.18%，不保证终端恢复）。未启用的候选及理由为：GPU/CUDA 算子加速（单次 LP 非零元仅 10^4 量级、矩阵维度小，无热点算子与实测加速比）、强化学习（本问题为两阶段线性规划，储能具有序贯状态，但本题可由已知物理模型直接优化，未建立学习控制的独立收益证据）、谐波 + 周周期岭回归预测器（技能评分 -0.144 与 -0.447，均劣于基线）、异步并行主流程（单任务粒度毫秒级，调度开销与收益同量级，仅报告可扩展性证据）、相关性改良的经济收益（70 天仅 79.35 元，缺少统计显著性证据）与电池退化定价（题目未要求、参数不可核验，仅作扩展分析）。

消融证据的解释边界。 原 A0–A4 记录涉及成本、指令次数和 SOC 越界；原 A3 实际只检查名义轨迹，并未运行同一触发器下移除 CBF 的对照，因此不能评价 CBF 的边际安全贡献。周期触发在 10080 个决策点上的基准为 10080 次下发，修正后配置为 4820 次，减少 52.18%；旧气泡图的 2500 次属于另一计数口径，不与该百分比混算。M1–M4 的体系图由已有指标构造，并非独立重复实验；部分费用和 CVaR 采用代理量，不能作为模块因果收益证据。因此正文只引用明确分母的执行指标，相关机制是否降低费用须另作同合同、同场景、同日期的对照。

H 程序清单

本附录给出复现全部结果所需的运行源代码，按工程目录组织，文件内容与随附代码包中的文件逐字一致，未作增删改；分阶段运行命令与耗时见附录 F.6，结果文件与模板的对应关系见附录 F.7。代码分两层：code/cleaning_cpp 为 C++17 值级清洗层，负责附件解析、口径归一、规则审计与导出；其余为 Python 3.12 建模求解层，负责预测、优化、结果写出与图件生成。目录结构如下：

run_all.sh	一键复跑入口：编译 -> 清洗 -> 四问求解 -> 出表 -> 制图 -> 门禁自检
code/	
__init__.py	建模求解层包入口
cleaning_cpp/	C++17 值级清洗层
CMakeLists.txt	构建脚本
include/	清洗规则、矩阵容器、zip/XML 解析、SHA-256、CSV/NPY 写出
src/	附件解析、NR1-NR12 规则、审计与导出、黄金用例自测
common/	只读载入、模板 xlsx 读写、门禁自检、确定性审计、全局配置
forecast/	共形预测层：加权分位、ACI 反馈、MaxEnt 残差拟合、覆盖标定
optimize/	稀疏 LP、min-plus 动态规划、两阶段随机规划、滚动 MPC、CVaR 与执行层
report/	结果文件写出、论文图表与数字生成、图注登记
tools/	
cross_verify.py	独立只读交叉校验（openpyxl 通道）

以下按目录顺序列出全部源文件。

H.1 运行入口

run_all.sh

```
1 #!/usr/bin/env bash
2 # 全流程重跑（单机 CPU）：bash run_all.sh 跑全套，--quick 跳过 Q4 风险扫描
3 # 顺序：C++ 清洗 -> Q1 -> 共形层 -> Q2 -> Q3 -> Q4 -> 出表 -> 画图 -> 论文数字 -> 门禁自检
4 set -euo pipefail
5 cd "$(dirname "$0")"
6
7 PY=${PY:-python3}
8 BUILD=build/cleaning
9 T0=$(date +%s)
10
11 log() { printf '\n=== %s ===\n' "$1"; }
12
13 log "0/10 configure + build cleaning_cpp (C++17, -Wall -Wextra)"
14 cmake -S code/cleaning_cpp -B "$BUILD" -DCMAKE_BUILD_TYPE=Release >/dev/null
15 cmake --build "$BUILD" -j "$(nproc)" 2>&1 | grep -Ev '^\[|Building|Built target' || true
16
17 log "1/10 self-test: golden cases + 200 synthetic panels"
18 "$BUILD/cleaning_cpp" selftest
19
20 log "2/10 clean + export + verify (NR1-NR12, invariants I1-I8)"
21 "$BUILD/cleaning_cpp" clean --in Data --out clean --templates "templates/附件5"
22 "$BUILD/cleaning_cpp" verify --in Data --out clean | tail -12
23
24 log "3/10 Q1 LP + DP cross-check + result1 payloads"
25 "$PY" -m code.optimize.q1_lp
26
27 log "4/10 conformal layer (0.80 / 0.25 / 0.70 coverage + ACI)"
28 "$PY" -m code.forecast.calibrate
29
30 log "5/10 Q2 two-stage stochastic backtest"
31 "$PY" -m code.optimize.q2_stochastic
32
33 log "6/10 Q3 rolling MPC + subset experiments S0-S5"
34 "$PY" -m code.optimize.q3_mpc
35
36 if [[ "${1:-}" == "--quick" ]]; then
```

```

37 log "7/10 Q4 risk sweep skipped (--quick)"
38 else
39 log "7/10 Q4 fluctuating tariff + CVaR/Copula/soft-constraint evidence"
40 "$PY" -m code.optimize.q4_price
41 fi
42
43 log "8/10 write the five result workbooks from the attachment-5 templates"
44 "$PY" -m code.report.write_results
45
46 log "9/10 figures + paper numbers"
47 "$PY" -m code.forecast.uncertainty          # U1-U5: PDF 没写明的项 → 共形区间
48 "$PY" -m code.report.visualize --set all    # 出版级制图: 主图 + 灵敏度 + 计算架构
49 "$PY" -m code.report.paper_numbers
50
51 log "10/10 verification gates G0-G9 + sensitivity"
52 "$PY" -m code.common.checks | tail -25
53
54 log "extra: independent xlsx cross-check (openpyxl read-only channel, optional)"
55 if /usr/bin/python3 -c "import openpyxl" >/dev/null 2>&1; then
56 /usr/bin/python3 tools/cross_verify.py | tail -6
57 "$PY" -m code.common.determinism | tail -3
58 else
59 echo "openpyxl channel not available on /usr/bin/python3 — skipped"
60 fi
61
62 T1=$(date +%s)
63 printf '\n=== pipeline finished in %s s ===\n' "$((T1 - T0))"

```

H.2 值级清洗层 code/cleaning_cpp

code/cleaning_cpp/CMakeLists.txt

```

1 cmake_minimum_required(VERSION 3.20)
2 project(cleaning_cpp CXX)
3
4 set(CMAKE_CXX_STANDARD 17)
5 set(CMAKE_CXX_STANDARD_REQUIRED ON)
6 if(NOT CMAKE_BUILD_TYPE)
7 set(CMAKE_BUILD_TYPE Release)
8 endif()
9
10 add_executable(cleaning_cpp
11 src/sha256.cpp
12 src/xlsx.cpp
13 src/panel.cpp
14 src/rules.cpp
15 src/writers.cpp
16 src/main.cpp
17 )
18
19 target_include_directories(cleaning_cpp PRIVATE include)
20 target_compile_options(cleaning_cpp PRIVATE -O2 -Wall -Wextra)
21
22 find_package(ZLIB REQUIRED)
23 find_library(TINYXML2_LIB tinyxml2 REQUIRED)
24 find_path(TINYXML2_INC tinyxml2.h REQUIRED)
25 target_include_directories(cleaning_cpp PRIVATE ${TINYXML2_INC})
26 target_link_libraries(cleaning_cpp PRIVATE ZLIB::ZLIB ${TINYXML2_LIB})

```

code/cleaning_cpp/include/clean/rules.h

```

1 #pragma once
2 #include <string>
3
4 #include "model/panel.h"
5

```

```

6 namespace clean {
7
8 enum class Action { kNone, kSetZero, kReject, kFlag };
9
10 struct PointResult {
11     Action action = Action::kNone;
12     double value = 0.0;
13     bool quarantine = false;
14     std::string rule_id;
15     std::string action_name;
16     std::string reason;
17 };
18
19 // 单个光伏格的判定入口：核夜间、过渡段走的规则不一样，负值先拦
20 PointResult apply_pv_point(double value, bool in_core, bool in_shoulder, const model::RuleParams
    & p);
21
22 // 拒绝就是把值置 NaN 再进隔离清单，NR7 预报侧也复用这个封装
23 PointResult make_reject(const std::string& rule_id, const std::string& reason, double value);
24
25 // 规则编号换成一句人话，自检和质量报告都要用
26 const char* rule_description(const std::string& rule_id);
27
28 }

```

code/cleaning_cpp/include/io/writers.h

```

1 #pragma once
2 #include <string>
3 #include <utility>
4 #include <vector>
5
6 #include "model/panel.h"
7
8 namespace io {
9
10 void write_npy_f64(const std::string& path, const double* data, const std::vector<size_t>& shape
    );
11 void write_csv(const std::string& path, const std::vector<std::vector<std::string>>& rows);
12 void write_text(const std::string& path, const std::string& content);
13 std::string fmt(double v, int precision = 10);
14 std::string json_escape(const std::string& s);
15
16 std::string manifest_json(const model::Panel& panel, double seconds, long peak_rss_kb,
    const std::vector<std::pair<std::string, std::string>>& inputs,
    const std::vector<std::pair<std::string, std::string>>& outputs);
17
18 std::string slots_json(const model::Panel& panel);
19 std::string dates_json(const model::Panel& panel);
20
21 }
22

```

code/cleaning_cpp/include/model/panel.h

```

1 #pragma once
2 #include <string>
3 #include <vector>
4
5 namespace model {
6
7 // 一天切 144 个 10 分钟时段，管一年，光伏预报一天发布 4 次
8 constexpr int kSlots = 144;
9 constexpr int kDays = 365;
10 constexpr int kVintages = 4;
11 constexpr int kForecastHours = 24;
12
13 // 每天的日照窗口，k_rise/k_set 直接按数据下标走，不写死时钟
14 struct DayWindow {
15     int k_rise = 0;

```

```

16     int k_set = 0;
17     double theta = 1.0;
18     double peak = 0.0;
19     bool valid = false;
20 };
21
22 struct AuditRecord {
23     std::string run_id;
24     std::string file;
25     std::string sheet;
26     int excel_row = 0;
27     std::string excel_col;
28     std::string date;
29     int slot_index = 0;
30     std::string slot_label;
31     std::string hour_end;
32     std::string rule_id;
33     std::string action;
34     std::string old_value;
35     std::string new_value;
36     std::string reason;
37 };
38
39 struct QuarantineRecord {
40     std::string run_id;
41     std::string file;
42     std::string sheet;
43     int excel_row = 0;
44     std::string excel_col;
45     std::string date;
46     int slot_index = 0;
47     std::string slot_label;
48     std::string value;
49     std::string rule_id;
50     std::string reason;
51 };
52
53 // 行主序的稠密矩阵，按（天，时段）两维取数
54 class Matrix {
55 public:
56     Matrix() = default;
57     Matrix(int rows, int cols) : rows_(rows), cols_(cols), data_(static_cast<size_t>(rows) * cols)
58     {}
59     double& at(int d, int k) { return data_.at(static_cast<size_t>(d) * cols_ + k); }
60     double at(int d, int k) const { return data_.at(static_cast<size_t>(d) * cols_ + k); }
61     int rows() const { return rows_; }
62     int cols() const { return cols_; }
63     const std::vector<double>& data() const { return data_; }
64     std::vector<double>& data() { return data_; }
65 private:
66     int rows_ = 0;
67     int cols_ = 0;
68     std::vector<double> data_;
69 };
70
71 // 清洗出来的东西全挂在这里，Python 侧读的就是它导出的数组
72 struct Panel {
73     std::vector<std::string> dates;
74     std::vector<std::string> slots;
75     Matrix pv_actual;
76     Matrix pv_clean;
77     Matrix load;
78     Matrix price;
79     std::vector<double> pv_forecast;
80     std::vector<double> pv_forecast_clean;
81     std::vector<double> day1;
82     std::vector<DayWindow> windows;
83     std::vector<AuditRecord> audit;
84     std::vector<QuarantineRecord> quarantine;

```

```

85     double pv_global_max = 0.0;
86     double sigma_floor = 1.0;
87     std::vector<double> sigma_raw;
88     std::vector<double> sigma_used;
89     std::vector<int> sigma_n;
90 };
91
92 // NR1-NR5 的阈值，默认值跟题目附录一致
93 struct RuleParams {
94     double noise_kw = 5.0;
95     double shoulder_kw = 1.0;
96     double theta_rel = 0.01;
97     double theta_abs = 1.0;
98     int window_jump_slots = 3;
99     int interp_max_gap = 6;
100 };
101
102 struct InputPaths {
103     std::string att1;
104     std::string att2;
105     std::string att3;
106     std::string att4;
107     std::string templates_dir;
108 };
109
110 Panel load_panel(const InputPaths& paths, const RuleParams& params);
111
112 }

```

code/cleaning_cpp/include/util/sha256.h

```

1  #pragma once
2  #include <cstdint>
3  #include <string>
4
5  namespace util {
6
7  // 精简版 SHA-256，只用来算 manifest 里的文件校验和
8  class Sha256 {
9  public:
10     Sha256() { reset(); }
11     void reset();
12     void update(const uint8_t* data, size_t len);
13     std::string hexdigest();
14     static std::string file(const std::string& path);
15     static std::string bytes(const std::string& data);
16     static std::string hex_of(const std::string& raw);
17
18 private:
19     void transform(const uint8_t* block);
20     uint32_t h_[8];
21     uint8_t buf_[64];
22     size_t buf_len_;
23     uint64_t total_len_;
24 };
25
26 }

```

code/cleaning_cpp/include/xlsx/xlsx.h

```

1  #pragma once
2  #include <cstdint>
3  #include <map>
4  #include <string>
5  #include <vector>
6
7  namespace xlsx {
8

```

```

9 // xlsx 本质是个 zip: 这里只解 deflate, 靠中央目录定位条目
10 class ZipArchive {
11 public:
12     struct Entry {
13         std::string name;
14         uint16_t method = 0;
15         uint64_t comp_size = 0;
16         uint64_t uncomp_size = 0;
17         uint64_t local_offset = 0;
18     };
19
20     static ZipArchive open(const std::string& path);
21     bool has(const std::string& name) const { return index_.count(name) > 0; }
22     std::string read(const std::string& name) const;
23     std::vector<std::string> names() const;
24
25 private:
26     std::string raw_;
27     std::map<std::string, Entry> index_;
28 };
29
30 // 附件1 的时间列数值和字符串混着来, 取值时逐格判类型
31 enum class ValueKind { kEmpty, kNumber, kText, kBool };
32
33 struct CellValue {
34     ValueKind kind = ValueKind::kEmpty;
35     double number = 0.0;
36     std::string text;
37
38     bool empty() const { return kind == ValueKind::kEmpty; }
39     bool is_number() const { return kind == ValueKind::kNumber; }
40     bool is_text() const { return kind == ValueKind::kText; }
41 };
42
43 using Grid = std::vector<std::vector<CellValue>>;
44
45 struct Sheet {
46     std::string name;
47     Grid grid;
48 };
49
50 struct Workbook {
51     std::vector<Sheet> sheets;
52     const Sheet* find(const std::string& name) const;
53     const Sheet& at(size_t i) const { return sheets.at(i); }
54     size_t size() const { return sheets.size(); }
55 };
56
57 Workbook read_workbook(const std::string& path);
58
59 // 小工具: "B12" → 列 1 / 行 12, 45658 → "2025-01-01"
60 int col_from_ref(const std::string& ref);
61 int row_from_ref(const std::string& ref);
62 std::string excel_date_to_iso(double serial);
63 int minutes_of_day(double fraction);
64 std::string minutes_to_label(int minutes);
65
66 }

```

code/cleaning_cpp/src/main.cpp

```

1 // cleaning_cpp: 附件解析 + NR1-NR12 清洗 + 审计导出, 凡是改数值的事都在这里做
2 // 子命令: inspect | extract | clean | export | verify | selftest
3
4 #include <sys/resource.h>
5
6 #include <algorithm>
7 #include <chrono>
8 #include <cmath>

```

```

9 #include <cstdio>
10 #include <cstdlib>
11 #include <cstring>
12 #include <filesystem>
13 #include <fstream>
14 #include <iostream>
15 #include <limits>
16 #include <map>
17 #include <random>
18 #include <sstream>
19 #include <string>
20 #include <vector>
21
22 #include "clean/rules.h"
23 #include "io/writers.h"
24 #include "model/panel.h"
25 #include "util/sha256.h"
26 #include "xlsx/xlsx.h"
27
28 namespace fs = std::filesystem;
29 using model::kDays;
30 using model::kSlots;
31
32 namespace {
33
34 struct Options {
35     std::string in_dir = "Data";
36     std::string out_dir = "clean";
37     std::string templates_dir = "templates/附件5";
38     bool strict = false;
39     model::RuleParams params;
40 };
41
42 Options parse_options(int argc, char** argv) {
43     Options o;
44     for (int i = 2; i < argc; ++i) {
45         const std::string a = argv[i];
46         auto next = [&]() -> std::string {
47             if (i + 1 >= argc) throw std::runtime_error("missing value for " + a);
48             return argv[++i];
49         };
50         if (a == "--in") o.in_dir = next();
51         else if (a == "--out") o.out_dir = next();
52         else if (a == "--templates") o.templates_dir = next();
53         else if (a == "--strict") o.strict = true;
54         else if (a == "--noise-kw") o.params.noise_kw = std::atof(next().c_str());
55         else if (a == "--shoulder-kw") o.params.shoulder_kw = std::atof(next().c_str());
56         else if (a == "--theta-rel") o.params.theta_rel = std::atof(next().c_str());
57         else if (a == "--theta-abs") o.params.theta_abs = std::atof(next().c_str());
58         else if (a == "--interp-max-gap") o.params.interp_max_gap = std::atoi(next().c_str());
59         else throw std::runtime_error("unknown option: " + a);
60     }
61     return o;
62 }
63
64 model::InputPaths make_paths(const Options& o) {
65     model::InputPaths p;
66     p.att1 = (fs::path(o.in_dir) / "附件1.xlsx").string();
67     p.att2 = (fs::path(o.in_dir) / "附件2.xlsx").string();
68     p.att3 = (fs::path(o.in_dir) / "附件3.xlsx").string();
69     p.att4 = (fs::path(o.in_dir) / "附件4.xlsx").string();
70     p.templates_dir = o.templates_dir;
71     return p;
72 }
73
74 long peak_rss_kb() {
75     rusage ru{};
76     getrusage(RUSAGE_SELF, &ru);
77     return ru.ru_maxrss;
78 }

```

```

79
80 std::vector<std::string> template_files(const std::string& dir) {
81     std::vector<std::string> out;
82     if (!fs::exists(dir)) return out;
83     for (const auto& e : fs::directory_iterator(dir)) {
84         if (e.path().extension() == ".xlsx") out.push_back(e.path().string());
85     }
86     std::sort(out.begin(), out.end());
87     return out;
88 }
89
90 std::string template_schema_json(const std::string& dir) {
91     std::string out = "{\n  \"templates\": [\n";
92     const std::vector<std::string> files = template_files(dir);
93     for (size_t fi = 0; fi < files.size(); ++fi) {
94         const xlsx::Workbook wb = xlsx::read_workbook(files[fi]);
95         out += "    {\n      \"file\": \"" + io::json_escape(fs::path(files[fi]).filename().string())
96             + "\",\n";
97         out += "      \"sheets\": [\n";
98         for (size_t si = 0; si < wb.size(); ++si) {
99             const xlsx::Sheet& s = wb.at(si);
100             const int rows = static_cast<int>(s.grid.size());
101             int cols = 0;
102             for (const auto& r : s.grid) cols = std::max(cols, static_cast<int>(r.size()));
103             out += "        {\n          \"name\": \"" + io::json_escape(s.name) + "\", \"rows\": " + std::
104                 to_string(rows) +
105                 ", \"cols\": " + std::to_string(cols) + ", \"header\": [";
106             if (rows > 0) {
107                 for (int c = 0; c < static_cast<int>(s.grid[0].size()); ++c) {
108                     if (c) out += ", ";
109                     out += "\"" + io::json_escape(s.grid[0][c].text) + "\"";
110                 }
111             }
112             out += "], \"row_labels\": [";
113             for (int r = 1; r < rows; ++r) {
114                 if (r > 1) out += ", ";
115                 out += "\"" + io::json_escape(s.grid[r].empty() ? std::string() : s.grid[r][0].text) +
116                     "\"";
117             }
118             out += "]}";
119             out += std::string(si + 1 < wb.size() ? ", " : "") + "\n";
120         }
121         out += "    ]\n  }";
122         out += std::string(fi + 1 < files.size() ? ", " : "") + "\n";
123     }
124     out += "  ]\n}\n";
125     return out;
126 }
127
128 // inspect: 只读地打印附件结构，一个数都不动
129 int cmd_inspect(const Options& o) {
130     const std::vector<std::string> files = {(fs::path(o.in_dir) / "附件1.xlsx").string(),
131         (fs::path(o.in_dir) / "附件2.xlsx").string(),
132         (fs::path(o.in_dir) / "附件3.xlsx").string(),
133         (fs::path(o.in_dir) / "附件4.xlsx").string()};
134     for (const std::string& f : files) {
135         std::cout << "==" << f << " sha256=" << util::Sha256::file(f).substr(0, 16) << "\n";
136         const xlsx::Workbook wb = xlsx::read_workbook(f);
137         for (const xlsx::Sheet& s : wb.sheets) {
138             int cols = 0;
139             int num_cells = 0, text_cells = 0, empty_cells = 0;
140             for (const auto& row : s.grid) {
141                 cols = std::max(cols, static_cast<int>(row.size()));
142                 for (const auto& c : row) {
143                     if (c.empty()) ++empty_cells;
144                     else if (c.is_text()) ++text_cells;
145                     else ++num_cells;
146                 }
147             }
148             std::cout << "  sheet '" << s.name << "' rows=" << s.grid.size() << " cols=" << cols
149                 << " numeric=" << num_cells << " text=" << text_cells << " empty=" <<
150                 empty_cells << "\n";
151         }
152     }
153 }

```

```

145     if (s.grid.size() > 1) {
146         std::cout << "        row2 sample:";
147         for (int c = 0; c < std::min(6, static_cast<int>(s.grid[1].size())); ++c) {
148             const xlsx::CellValue& v = s.grid[1][c];
149             std::cout << " [" << (v.is_text() ? v.text : io::fmt(v.number, 6)) << "]";
150         }
151         std::cout << "\n";
152     }
153 }
154 }
155 for (const std::string& t : template_files(o.templates_dir)) {
156     const xlsx::Workbook wb = xlsx::read_workbook(t);
157     std::cout << "== template " << fs::path(t).filename().string() << " sheets=";
158     for (size_t i = 0; i < wb.size(); ++i) {
159         int cols = 0;
160         for (const auto& r : wb.at(i).grid) cols = std::max(cols, static_cast<int>(r.size()));
161         std::cout << "[" << wb.at(i).name << " " << wb.at(i).grid.size() << "x" << cols << "]";
162     }
163     std::cout << "\n";
164 }
165 return 0;
166 }
167
168 // 支撑用的几张 CSV: 日照窗口、审计表、隔离清单、小时质量报告、共形尺度、月度汇总
169 void write_support_csv(const model::Panel& p, const std::string& out_dir) {
170     std::vector<std::vector<std::string>> w_rows;
171     w_rows.push_back({"date", "k_rise", "k_set", "sunrise", "sunset", "day_hours",
172                     "night_core_hours", "theta_d", "peak_d"});
173     for (int d = 0; d < kDays; ++d) {
174         const model::DayWindow& w = p.windows[d];
175         char rb[16], sb[16];
176         std::snprintf(rb, sizeof(rb), "%d:%02d", (w.k_rise * 10) / 60, (w.k_rise * 10) % 60);
177         std::snprintf(sb, sizeof(sb), "%d:%02d", (w.k_set * 10) / 60, (w.k_set * 10) % 60);
178         const std::string rise = rb, set = sb;
179         const double day_hours = (w.k_set - w.k_rise) / 6.0;
180         const double night_hours = 24.0 - day_hours;
181         w_rows.push_back({p.dates[d], std::to_string(w.k_rise + 1), std::to_string(w.k_set + 1),
182                         rise, set,
183                         io::fmt(day_hours, 6), io::fmt(night_hours, 6), io::fmt(w.theta, 6),
184                         io::fmt(w.peak, 8)});
185     }
186     io::write_csv((fs::path(out_dir) / "day_windows.csv").string(), w_rows);
187
188     std::vector<std::vector<std::string>> a_rows;
189     a_rows.push_back({"run_id", "file", "sheet", "excel_row", "excel_col", "date", "slot_index",
190                     "slot_label", "hour_end", "rule_id", "action", "old_value", "new_value", "
191                     reason"});
192     for (const model::AuditRecord& a : p.audit) {
193         a_rows.push_back({a.run_id, a.file, a.sheet, std::to_string(a.excel_row), a.excel_col, a.
194                         date,
195                         std::to_string(a.slot_index), a.slot_label, a.hour_end, a.rule_id, a.
196                         action,
197                         a.old_value, a.new_value, a.reason});
198     }
199     io::write_csv((fs::path(out_dir) / "cleaning_audit.csv").string(), a_rows);
200
201     std::vector<std::vector<std::string>> q_rows;
202     q_rows.push_back({"run_id", "file", "sheet", "excel_row", "excel_col", "date", "slot_index",
203                     "slot_label", "value", "rule_id", "reason"});
204     for (const model::QuarantineRecord& q : p.quarantine) {
205         q_rows.push_back({q.run_id, q.file, q.sheet, std::to_string(q.excel_row), q.excel_col, q.
206                         date,
207                         std::to_string(q.slot_index), q.slot_label, q.value, q.rule_id, q.reason});
208     }
209     io::write_csv((fs::path(out_dir) / "quarantine.csv").string(), q_rows);
210
211     std::vector<std::vector<std::string>> h_rows;
212     h_rows.push_back({"hour", "cells", "zero_rate", "noise_rate", "reject_count", "mean_pv", "
213                     max_pv"});

```

```

208 for (int hour = 0; hour < 24; ++hour) {
209     int cells = 0, zeros = 0, noise = 0, rejects = 0;
210     double sum = 0.0, max_v = 0.0;
211     for (int k = hour * 6; k < hour * 6 + 6; ++k) {
212         for (int d = 0; d < kDays; ++d) {
213             ++cells;
214             const double raw = p.pv_actual.at(d, k);
215             if (raw == 0.0) ++zeros;
216             if (raw > 0.0 && raw <= 5.0) ++noise;
217             const double cl = p.pv_clean.at(d, k);
218             if (std::isnan(cl)) ++rejects;
219             else {
220                 sum += cl;
221                 max_v = std::max(max_v, cl);
222             }
223         }
224     }
225     char hb[8];
226     std::snprintf(hb, sizeof(hb), "%02d:00", hour);
227     h_rows.push_back({hb, std::to_string(cells), io::fmt(static_cast<double>(zeros) / cells, 6),
228                     io::fmt(static_cast<double>(noise) / cells, 6), std::to_string(rejects),
229                     io::fmt(sum / cells, 8), io::fmt(max_v, 8)});
230 }
231 io::write_csv((fs::path(out_dir) / "quality_report.csv").string(), h_rows);
232
233 std::vector<std::vector<std::string>> s_rows;
234 s_rows.push_back({"slot_index", "slot_label", "sigma_raw", "sigma_used", "n_samples"});
235 for (int k = 0; k < kSlots; ++k) {
236     s_rows.push_back({std::to_string(k + 1), p.slots[k], io::fmt(p.sigma_raw[k], 8),
237                     io::fmt(p.sigma_used[k], 8), std::to_string(p.sigma_n[k])});
238 }
239 io::write_csv((fs::path(out_dir) / "sigma.csv").string(), s_rows);
240
241 std::vector<std::vector<std::string>> m_rows;
242 m_rows.push_back({"month", "days", "sunrise_median", "sunset_median", "night_core_hours_median"});
243 int starts[13] = {0, 31, 59, 90, 120, 151, 181, 212, 243, 273, 304, 334, 365};
244 for (int m = 0; m < 12; ++m) {
245     std::vector<double> rise, set, night;
246     for (int d = starts[m]; d < starts[m + 1]; ++d) {
247         rise.push_back(p.windows[d].k_rise / 6.0);
248         set.push_back(p.windows[d].k_set / 6.0);
249         night.push_back(24.0 - (p.windows[d].k_set - p.windows[d].k_rise) / 6.0);
250     }
251     auto med = [](std::vector<double> v) {
252         std::sort(v.begin(), v.end());
253         return v[v.size() / 2];
254     };
255     m_rows.push_back({std::to_string(m + 1), std::to_string(starts[m + 1] - starts[m]), io::fmt(
256         med(rise), 4),
257                     io::fmt(med(set), 4), io::fmt(med(night), 4)});
258 }
259 io::write_csv((fs::path(out_dir) / "window_monthly.csv").string(), m_rows);
260
261 int cmd_extract(const Options& o) {
262     const model::Panel p = model::load_panel(make_paths(o), o.params);
263     fs::create_directories(o.out_dir);
264     const std::string raw_dir = (fs::path(o.out_dir) / "raw").string();
265     fs::create_directories(raw_dir);
266     io::write_npy_f64((fs::path(raw_dir) / "pv_actual.npy").string(), p.pv_actual.data().data(),
267                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
268     io::write_npy_f64((fs::path(raw_dir) / "load.npy").string(), p.load.data().data(),
269                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
270     io::write_npy_f64((fs::path(raw_dir) / "price.npy").string(), p.price.data().data(),
271                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
272     io::write_npy_f64((fs::path(raw_dir) / "pv_forecast.npy").string(), p.pv_forecast.data(),
273                     {static_cast<size_t>(kDays), static_cast<size_t>(model::kVintages),
274                     static_cast<size_t>(model::kForecastHours)});
275     io::write_npy_f64((fs::path(raw_dir) / "day1.npy").string(), p.day1.data(),

```

```

276         {static_cast<size_t>(kSlots), static_cast<size_t>(3)});
277 io::write_text((fs::path(raw_dir) / "README.txt").string(),
278               "Raw (pre-rule) matrices exported by cleaning_cpp extract. Values are bit-
279               identical to\n"
280               "the attachment cells; NaN marks unreadable cells. Use `clean` for the audited
281               bundle.\n");
282 std::cout << "[extract] wrote raw matrices to " << raw_dir << "\n";
283 return 0;
284 }
285
286 int cmd_clean(const Options& o) {
287     const auto t0 = std::chrono::steady_clock::now();
288     const model::InputPaths paths = make_paths(o);
289     const model::Panel p = model::load_panel(paths, o.params);
290     fs::create_directories(o.out_dir);
291
292     io::write_npy_f64((fs::path(o.out_dir) / "pv_actual.npy").string(), p.pv_clean.data().data(),
293                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
294     io::write_npy_f64((fs::path(o.out_dir) / "pv_actual_raw.npy").string(), p.pv_actual.data().
295                     data(),
296                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
297     io::write_npy_f64((fs::path(o.out_dir) / "load.npy").string(), p.load.data().data(),
298                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
299     io::write_npy_f64((fs::path(o.out_dir) / "price.npy").string(), p.price.data().data(),
300                     {static_cast<size_t>(kDays), static_cast<size_t>(kSlots)});
301     io::write_npy_f64((fs::path(o.out_dir) / "pv_forecast.npy").string(), p.pv_forecast_clean.data
302                     (),
303                     {static_cast<size_t>(kDays), static_cast<size_t>(model::kVintages),
304                     static_cast<size_t>(model::kForecastHours)});
305     io::write_npy_f64((fs::path(o.out_dir) / "pv_forecast_raw.npy").string(), p.pv_forecast.data()
306                     ,
307                     {static_cast<size_t>(kDays), static_cast<size_t>(model::kVintages),
308                     static_cast<size_t>(model::kForecastHours)});
309     io::write_npy_f64((fs::path(o.out_dir) / "day1.npy").string(), p.day1.data(),
310                     {static_cast<size_t>(kSlots), static_cast<size_t>(3)});
311     io::write_text((fs::path(o.out_dir) / "slots.json").string(), io::slots_json(p));
312     io::write_text((fs::path(o.out_dir) / "dates.json").string(), io::dates_json(p));
313     io::write_text((fs::path(o.out_dir) / "template_schema.json").string(),
314                   template_schema_json(o.templates_dir));
315     write_support_csv(p, o.out_dir);
316
317     const auto t1 = std::chrono::steady_clock::now();
318     const double seconds = std::chrono::duration<double>(t1 - t0).count();
319
320     std::vector<std::pair<std::string, std::string>> inputs;
321     for (const std::string& f : {paths.att1, paths.att2, paths.att3, paths.att4}) {
322         inputs.emplace_back(fs::path(f).filename().string(), util::Sha256::file(f));
323     }
324     std::vector<std::string> outputs;
325     for (const auto& e : fs::directory_iterator(o.out_dir)) {
326         if (!e.is_regular_file()) continue;
327         const std::string name = e.path().filename().string();
328         if (name == "invariants.json" || name == "cross_check.json" || name == "manifest.json" ||
329             name == "meta.json")
330             continue;
331         outputs.push_back(e.path().string());
332     }
333     std::sort(outputs.begin(), outputs.end());
334     std::vector<std::pair<std::string, std::string>> checksums;
335     for (const std::string& f : outputs) {
336         checksums.emplace_back(fs::path(f).filename().string(), util::Sha256::file(f));
337     }
338     io::write_text((fs::path(o.out_dir) / "manifest.json").string(),
339                   io::manifest_json(p, seconds, peak_rss_kb(), inputs, checksums));
340
341     std::string meta = "{\n  \"command\": \"cleaning_cpp clean --in \" + o.in_dir + "
342     out_dir + "\"\n";
343     meta += "  \"wall_seconds\": \" + io::fmt(seconds, 6) + "\",\n";
344     meta += "  \"peak_rss_kb\": \" + std::to_string(peak_rss_kb()) + "\",\n";
345     meta += "  \"rules\": \"NR1..NR12 (prompt v1 4.3)\"";
346 }

```

```

340 meta += "  \"params\": {\"noise_kw\": " + io::fmt(o.params.noise_kw) +
341         ", \"shoulder_kw\": " + io::fmt(o.params.shoulder_kw) +
342         ", \"theta_rel\": " + io::fmt(o.params.theta_rel) +
343         ", \"theta_abs\": " + io::fmt(o.params.theta_abs) +
344         ", \"window_jump_slots\": " + std::to_string(o.params.window_jump_slots) +
345         ", \"interp_max_gap\": " + std::to_string(o.params.interp_max_gap) + "}}\n\n";
346 io::write_text((fs::path(o.out_dir) / "meta.json").string(), meta);
347
348 std::cout << "[clean] days=" << kDays << " slots=" << kSlots << " audit=" << p.audit.size()
349         << " quarantine=" << p.quarantine.size() << " seconds=" << io::fmt(seconds, 3)
350         << " peak_rss_kb=" << peak_rss_kb() << "\n";
351 return 0;
352 }
353
354 struct CheckResult {
355     std::string id;
356     std::string description;
357     bool pass = false;
358     std::string detail;
359 };
360
361 std::vector<CheckResult> verify_panel(const model::Panel& p, const std::string& out_dir) {
362     std::vector<CheckResult> r;
363
364     // I1: 核夜间要么正好是 0, 要么是隔离出来的 NaN
365     {
366         long core_cells = 0, nonzero = 0, nan_cells = 0;
367         for (int d = 0; d < kDays; ++d) {
368             const model::DayWindow& w = p.windows[d];
369             for (int k = 0; k < kSlots; ++k) {
370                 const bool core = (k < w.k_rise - 1) || (k > w.k_set + 1);
371                 if (!core) continue;
372                 ++core_cells;
373                 const double v = p.pv_clean.at(d, k);
374                 if (std::isnan(v)) ++nan_cells;
375                 else if (v != 0.0) ++nonzero;
376             }
377         }
378         CheckResult c{"I1", "core-night PV is exactly 0 (NaN only if quarantined)", nonzero == 0,
379             "core_cells=" + std::to_string(core_cells) + " nonzero=" + std::to_string(
380                 nonzero) +
381                 " nan=" + std::to_string(nan_cells)};
382         r.push_back(c);
383     }
384     // I2: 白天 08:00-16:00 既不能出现 0, 也不能出现负值
385     {
386         long zeros = 0, neg = 0, cells = 0;
387         for (int d = 0; d < kDays; ++d) {
388             for (int k = 47; k <= 95; ++k) {
389                 ++cells;
390                 const double v = p.pv_clean.at(d, k);
391                 if (!std::isnan(v) && v == 0.0) ++zeros;
392                 if (!std::isnan(v) && v < 0.0) ++neg;
393             }
394         }
395         r.push_back({"I2", "08:00-16:00 zero-free and positive", zeros == 0 && neg == 0,
396             "cells=" + std::to_string(cells) + " zeros=" + std::to_string(zeros) +
397             " negatives=" + std::to_string(neg)});
398     }
399     // I3/I4: 日期要盖住 2025 全年且严格递增, 矩阵形状一律 365×144
400     {
401         bool ok = static_cast<int>(p.dates.size()) == kDays;
402         for (int d = 1; d < kDays && ok; ++d) ok = p.dates[d] > p.dates[d - 1];
403         ok = ok && p.dates.front() == "2025-01-01" && p.dates.back() == "2025-12-31";
404         r.push_back({"I3", "365 strictly increasing dates 2025-01-01..2025-12-31", ok,
405             "first=" + p.dates.front() + " last=" + p.dates.back()});
406         const bool shape_ok = p.pv_actual.rows() == kDays && p.pv_actual.cols() == kSlots &&
407             p.load.rows() == kDays && p.price.rows() == kDays;
408         r.push_back({"I4", "matrix shape 365 x 144 in every panel", shape_ok, "ok"});
409     }
410 }

```

```

409 // I5: 电价必须为正, 负荷落在 [1500, 8000] kW
410 {
411     long bad = 0;
412     double lo = 1e18, hi = -1e18, llo = 1e18, lhi = -1e18;
413     for (int d = 0; d < kDays; ++d) {
414         for (int k = 0; k < kSlots; ++k) {
415             const double pr = p.price.at(d, k), ld = p.load.at(d, k);
416             if (!(pr > 0.0)) ++bad;
417             if (!(ld > 0.0) || ld < 1500.0 || ld > 8000.0) ++bad;
418             lo = std::min(lo, pr); hi = std::max(hi, pr);
419             llo = std::min(llo, ld); lhi = std::max(lhi, ld);
420         }
421     }
422     r.push_back({"I5", "price>0 ; load in [1500,8000] kW", bad == 0,
423         "violations=" + std::to_string(bad) + " price=[" + io::fmt(lo, 6) + "," +
424         io::fmt(hi, 6) + "]" load=[" + io::fmt(llo, 6) + "," + io::fmt(lhi, 6) +
425         "]}");
426 }
427 // I6: NR1 归零的笔数要对上核夜间“从非零变零”的格子数
428 {
429     long nr1_audit = 0, nr1_delta = 0;
430     for (const model::AuditRecord& a : p.audit)
431         if (a.rule_id == "NR1") ++nr1_audit;
432     for (int d = 0; d < kDays; ++d) {
433         const model::DayWindow& w = p.windows[d];
434         for (int k = 0; k < kSlots; ++k) {
435             const bool core = (k < w.k_rise - 1) || (k > w.k_set + 1);
436             if (!core) continue;
437             const double raw = p.pv_actual.at(d, k);
438             const double cl = p.pv_clean.at(d, k);
439             if (!std::isnan(raw) && raw > 0.0 && cl == 0.0) ++nr1_delta;
440         }
441     }
442     r.push_back({"I6", "NR1 audit count equals core-night nonzero -> zero cells", nr1_audit ==
443         nr1_delta,
444         "audit=" + std::to_string(nr1_audit) + " delta=" + std::to_string(nr1_delta)});
445 }
446 // I7: 除去审计过的格子, 清洗前后必须逐位一致
447 {
448     std::vector<unsigned char> touched(static_cast<size_t>(kDays) * kSlots, 0);
449     for (const model::AuditRecord& a : p.audit) {
450         if (a.file != "附件2" || a.sheet != "光伏发电实际功率") continue;
451         const int d = a.excel_row - 2, k = a.slot_index - 1;
452         if (d >= 0 && d < kDays && k >= 0 && k < kSlots) touched[static_cast<size_t>(d) * kSlots +
453             k] = 1;
454     }
455     long diffs = 0;
456     for (int d = 0; d < kDays; ++d) {
457         for (int k = 0; k < kSlots; ++k) {
458             if (touched[static_cast<size_t>(d) * kSlots + k]) continue;
459             const double raw = p.pv_actual.at(d, k), cl = p.pv_clean.at(d, k);
460             if (raw != cl) ++diffs;
461         }
462     }
463     r.push_back({"I7", "bitwise identity outside audited cells", diffs == 0,
464         "unexpected_differences=" + std::to_string(diffs)});
465 }
466 // I8: 磁盘上文件的哈希要和 manifest 里记的对得上
467 {
468     bool ok = true;
469     std::string detail = "manifest not found";
470     const std::string mpath = (fs::path(out_dir) / "manifest.json").string();
471     if (fs::exists(mpath)) {
472         std::ifstream in(mpath);
473         std::stringstream ss;
474         ss << in.rdbuf();
475         const std::string txt = ss.str();
476         size_t pos = txt.find("\\"outputs_sha256\\");
477         int checked = 0, mismatched = 0;
478         if (pos != std::string::npos) {

```

```

476     const size_t block_b = txt.find('{', pos);
477     const size_t block_e = txt.find('}', block_b);
478     const std::string block = txt.substr(block_b + 1, block_e - block_b - 1);
479     size_t p = 0;
480     while ((p = block.find('\"', p)) != std::string::npos) {
481         const size_t key_e = block.find('\"', p + 1);
482         if (key_e == std::string::npos) break;
483         const size_t val_b = block.find('\"', block.find(':', key_e) + 1);
484         if (val_b == std::string::npos) break;
485         const size_t val_e = block.find('\"', val_b + 1);
486         if (val_e == std::string::npos) break;
487         const std::string name = block.substr(p + 1, key_e - p - 1);
488         const std::string want = block.substr(val_b + 1, val_e - val_b - 1);
489         const std::string path = (fs::path(out_dir) / name).string();
490         if (fs::exists(path)) {
491             ++checked;
492             if (util::Sha256::file(path) != want) ++mismatched;
493         }
494         p = val_e + 1;
495     }
496 }
497 ok = (mismatched == 0 && checked > 0);
498 detail = "checked=" + std::to_string(checked) + " mismatched=" + std::to_string(mismatched
);
499 }
500 r.push_back({"I8", "output SHA-256 matches manifest", ok, detail});
501 }
502 return r;
503 }
504
505 std::string cross_check_json(const model::Panel& p) {
506     long core_cells = 0, raw_core_zeros = 0, clean_core_zeros = 0, raw_core_big = 0;
507     long day_zeros = 0, day1_small = 0, near_zero_price = 0, night_load_flags = 0, nr4_hits = 0;
508     for (int d = 0; d < kDays; ++d) {
509         const model::DayWindow& w = p.windows[d];
510         for (int k = 0; k < kSlots; ++k) {
511             const bool core = (k < w.k_rise - 1) || (k > w.k_set + 1);
512             const double raw = p.pv_actual.at(d, k);
513             const double cl = p.pv_clean.at(d, k);
514             if (core) {
515                 ++core_cells;
516                 if (raw == 0.0) ++raw_core_zeros;
517                 if (cl == 0.0) ++clean_core_zeros;
518                 if (raw > 5.0) ++raw_core_big;
519             }
520             if (k >= 47 && k <= 95) {
521                 if (!std::isnan(cl) && cl == 0.0) ++day_zeros;
522             }
523             if (p.price.at(d, k) < 0.1) ++near_zero_price;
524         }
525     }
526     for (const model::AuditRecord& a : p.audit) {
527         if (a.rule_id == "NR4") ++nr4_hits;
528         if (a.rule_id == "NR12") ++night_load_flags;
529     }
530     for (int k = 0; k < kSlots; ++k) {
531         const double v = p.day1[3 * k + 2];
532         if (v > 0.0 && v <= 5.0) ++day1_small;
533     }
534     std::string out = "{\n";
535     out += "  \"att2_core_night_cells\": " + std::to_string(core_cells) + ",\n";
536     out += "  \"att2_core_night_zero_raw\": " + std::to_string(raw_core_zeros) + ",\n";
537     out += "  \"att2_core_night_zero_after\": " + std::to_string(clean_core_zeros) + ",\n";
538     out += "  \"att2_core_night_over_5kw_raw\": " + std::to_string(raw_core_big) + ",\n";
539     out += "  \"att2_daytime_0800_1600_zeros\": " + std::to_string(day_zeros) + ",\n";
540     out += "  \"quarantine_records\": " + std::to_string(p.quarantine.size()) + ",\n";
541     out += "  \"audit_records\": " + std::to_string(p.audit.size()) + ",\n";
542     out += "  \"att1_small_pv_cells_0_5kw\": " + std::to_string(day1_small) + ",\n";
543     out += "  \"att4_near_zero_price_cells\": " + std::to_string(near_zero_price) + ",\n";
544     out += "  \"night_load_flag_records\": " + std::to_string(night_load_flags) + ",\n";

```

```

545 out += " \nr4_shoulder_hits\": " + std::to_string(nr4_hits) + "\n}\n";
546 return out;
547 }
548
549 int cmd_verify(const Options& o) {
550     const model::Panel p = model::load_panel(make_paths(o), o.params);
551     const std::vector<CheckResult> checks = verify_panel(p, o.out_dir);
552     bool all = true;
553     for (const CheckResult& c : checks) {
554         std::cout << (c.pass ? "[PASS] " : "[FAIL] ") << c.id << " " << c.description << " :: " << c
            .detail
555             << "\n";
556         all = all && c.pass;
557     }
558     const std::string cc = cross_check_json(p);
559     std::cout << "[cross-check]\n" << cc;
560     fs::create_directories(o.out_dir);
561     io::write_text((fs::path(o.out_dir) / "cross_check.json").string(), cc);
562     std::string summary = "{\n  \"all_pass\": " + std::string(all ? "true" : "false") + ",\n  \"
        checks\": [\n";
563     for (size_t i = 0; i < checks.size(); ++i) {
564         summary += "    {\"id\": \"" + checks[i].id + "\", \"description\": \"" +
565             io::json_escape(checks[i].description) + "\", \"pass\": " +
566             std::string(checks[i].pass ? "true" : "false") + ", \"detail\": \"" +
567             io::json_escape(checks[i].detail) + "\"} " + (i + 1 < checks.size() ? ", " : "") +
568             "\n";
569     }
570     summary += "  ]\n}\n";
571     io::write_text((fs::path(o.out_dir) / "invariants.json").string(), summary);
572     if (!all && o.strict) return 2;
573     return all ? 0 : 1;
574 }
575
576 bool check_action(const std::string& rule, const clean::PointResult& pr, clean::Action expect,
577     double expect_value) {
578     const bool ok = pr.action == expect &&
579         (std::isnan(expect_value) ? std::isnan(pr.value)
580          : std::fabs(pr.value - expect_value) < 1e-12);
581     if (!ok) {
582         std::cout << " [FAIL] case " << rule << " expected action=" << static_cast<int>(expect)
583             << " value=" << expect_value << " got action=" << static_cast<int>(pr.action)
584             << " value=" << pr.value << "\n";
585     }
586     return ok;
587 }
588
589 // selftest: 每条规则三个黄金用例，再拿 200 块合成面板做性质测试
590 int cmd_selftest(const Options&) {
591     model::RuleParams p;
592     int failures = 0;
593     const double nan = std::numeric_limits<double>::quiet_NaN();
594     failures += !check_action("NR1 trigger", clean::apply_pv_point(3.0, true, false, p), clean::
        Action::kSetZero, 0.0);
595     failures += !check_action("NR1 boundary 5.0", clean::apply_pv_point(5.0, true, false, p),
        clean::Action::kSetZero, 0.0);
596     failures += !check_action("NR1 boundary 0.0", clean::apply_pv_point(0.0, true, false, p),
        clean::Action::kNone, 0.0);
597     failures += !check_action("NR2 trigger", clean::apply_pv_point(5.0001, true, false, p), clean
        ::Action::kReject, nan);
598     failures += !check_action("NR2 no-trigger", clean::apply_pv_point(4.9999, true, false, p),
        clean::Action::kSetZero, 0.0);
599     failures += !check_action("NR3 negative", clean::apply_pv_point(-0.5, false, false, p), clean
        ::Action::kReject, nan);
600     failures += !check_action("NR4 sub-kW", clean::apply_pv_point(0.6, false, true, p), clean::
        Action::kSetZero, 0.0);
601     failures += !check_action("NR4 boundary 1.0 -> NR5", clean::apply_pv_point(1.0, false, true, p
        ), clean::Action::kFlag, 1.0);
602     failures += !check_action("NR5 weak signal", clean::apply_pv_point(2.5, false, true, p), clean
        ::Action::kFlag, 2.5);

```

```

603 failures += !check_action("daylight keep", clean::apply_pv_point(1200.0, false, false, p),
    clean::Action::kNone, 1200.0);
604
605 // 200 块合成面板，故意注入噪声和离群值，看规则有没有漏网
606 std::mt19937_64 rng(20250910);
607 std::uniform_real_distribution<double> noise(0.01, 5.0);
608 std::uniform_real_distribution<double> outlier(5.01, 12.0);
609 std::uniform_real_distribution<double> daylight(10.0, 8000.0);
610 int inject_noise = 0, inject_big = 0, hit_noise = 0, hit_big = 0, invariant_violations = 0;
611 for (int trial = 0; trial < 200; ++trial) {
612     std::vector<double> day(144, 0.0);
613     for (int k = 30; k <= 110; ++k) day[k] = daylight(rng);
614     const int k_rise = 30, k_set = 110;
615     std::vector<int> core_slots;
616     for (int k = 0; k < 144; ++k)
617         if (k < k_rise - 1 || k > k_set + 1) core_slots.push_back(k);
618     std::shuffle(core_slots.begin(), core_slots.end(), rng);
619     for (size_t i = 0; i < 6 && i < core_slots.size(); ++i) {
620         day[static_cast<size_t>(core_slots[i])] = noise(rng);
621         ++inject_noise;
622     }
623     for (size_t i = 6; i < 9 && i < core_slots.size(); ++i) {
624         day[static_cast<size_t>(core_slots[i])] = outlier(rng);
625         ++inject_big;
626     }
627     for (int k = 0; k < 144; ++k) {
628         const bool core = (k < k_rise - 1) || (k > k_set + 1);
629         const bool shoulder = (k == k_rise - 1 || k == k_rise || k == k_set || k == k_set + 1);
630         const clean::PointResult pr = clean::apply_pv_point(day[k], core, shoulder, p);
631         if (pr.action == clean::Action::kSetZero && core) ++hit_noise;
632         if (pr.action == clean::Action::kReject && core) ++hit_big;
633         const double after = (pr.action == clean::Action::kNone || pr.action == clean::Action::
            kFlag)
634             ? day[k]
635             : pr.value;
636         if (core && !(after == 0.0 || std::isnan(after))) ++invariant_violations;
637         if (!core && !shoulder && after != day[k]) ++invariant_violations;
638     }
639 }
640 const bool property_ok = (hit_noise == inject_noise) && (hit_big == inject_big) &&
    (invariant_violations == 0);
641
642 std::cout << "    [property] injected noise=" << inject_noise << " caught=" << hit_noise
643     << " ; injected outliers=" << inject_big << " caught=" << hit_big
644     << " ; invariant violations=" << invariant_violations << "\n";
645 if (!property_ok) ++failures;
646
647 // 同一批数据跑两遍，结果必须逐位一致
648 std::vector<double> a(144), b(144);
649 for (int k = 0; k < 144; ++k) a[k] = (k % 7 == 0) ? 3.0 : 100.0 + k;
650 b = a;
651 bool det_ok = true;
652 for (int rep = 0; rep < 2; ++rep) {
653     for (int k = 0; k < 144; ++k) {
654         const bool core = (k < 20) || (k > 120);
655         const clean::PointResult pr = clean::apply_pv_point(a[k], core, false, p);
656         const double v = (pr.action == clean::Action::kNone || pr.action == clean::Action::kFlag)
            ? a[k] : pr.value;
657         if (rep == 0) b[k] = v;
658         else if (!(v == b[k] || (std::isnan(v) && std::isnan(b[k])))) det_ok = false;
659     }
660 }
661 if (!det_ok) ++failures;
662 std::cout << "    [determinism] repeated rule application identical: " << (det_ok ? "yes" : "no
    ") << "\n";
663 std::cout << "[selftest] " << (failures == 0 ? "ALL PASS" : "FAILURES=" + std::to_string(
    failures))
664     << " (golden cases + 200 synthetic panels)\n";
665 return failures == 0 ? 0 : 1;
666 }
667

```

```

668 }
669
670 int main(int argc, char** argv) {
671     if (argc < 2) {
672         std::cerr << "usage: cleaning_cpp <inspect|extract|clean|export|verify|selftest> [--in DIR]
        [--out DIR]\n";
673         return 1;
674     }
675     const std::string cmd = argv[1];
676     try {
677         const Options o = parse_options(argc, argv);
678         if (cmd == "inspect") return cmd_inspect(o);
679         if (cmd == "extract") return cmd_extract(o);
680         if (cmd == "clean" || cmd == "export") return cmd_clean(o);
681         if (cmd == "verify") return cmd_verify(o);
682         if (cmd == "selftest") return cmd_selftest(o);
683         std::cerr << "unknown subcommand: " << cmd << "\n";
684         return 1;
685     } catch (const std::exception& e) {
686         std::cerr << "[error] " << e.what() << "\n";
687         return 3;
688     }
689 }

```

code/cleaning_cpp/src/panel.cpp

```

1 #include "model/panel.h"
2
3 #include <algorithm>
4 #include <cmath>
5 #include <cstdio>
6 #include <cstdlib>
7 #include <limits>
8 #include <map>
9 #include <stdexcept>
10 #include <string>
11
12 #include "clean/rules.h"
13 #include "xlsx/xlsx.h"
14
15 namespace model {
16 namespace {
17
18 const double kNaN = std::numeric_limits<double>::quiet_NaN();
19
20 std::string col_letter(int col0) {
21     std::string s;
22     int c = col0 + 1;
23     while (c > 0) {
24         const int rem = (c - 1) % 26;
25         s.insert(s.begin(), static_cast<char>('A' + rem));
26         c = (c - 1) / 26;
27     }
28     return s;
29 }
30
31 std::string num(double v, int precision = 10) {
32     char buf[64];
33     if (std::isnan(v)) return "nan";
34     std::snprintf(buf, sizeof(buf), "%.*g", precision, v);
35     return std::string(buf);
36 }
37
38 double cell_number(const xlsx::Grid& g, int row0, int col0) {
39     if (row0 < 0 || row0 >= static_cast<int>(g.size())) return kNaN;
40     if (col0 < 0 || col0 >= static_cast<int>(g[row0].size())) return kNaN;
41     const xlsx::CellValue& v = g[row0][col0];
42     if (v.is_number()) return v.number;
43     if (v.is_text()) {

```

```

44     const char* t = v.text.c_str();
45     char* end = nullptr;
46     const double d = std::strtod(t, &end);
47     if (end != nullptr && end != t && *end == '\\0') return d;
48 }
49 return kNaN;
50 }
51
52 std::string cell_text(const xlsx::Grid& g, int row0, int col0) {
53     if (row0 < 0 || row0 >= static_cast<int>(g.size())) return std::string();
54     if (col0 < 0 || col0 >= static_cast<int>(g[row0].size())) return std::string();
55     const xlsx::CellValue& v = g[row0][col0];
56     if (v.is_text()) return v.text;
57     if (v.is_number()) {
58         if (v.number > 30000.0) return xlsx::excel_date_to_iso(v.number);
59         return std::to_string(static_cast<int>(v.number));
60     }
61     return std::string();
62 }
63
64 int parse_iso_day_index(const std::string& iso) {
65     if (iso.size() < 8) return -1;
66     int y = 0, m = 0, d = 0;
67     if (std::sscanf(iso.c_str(), "%d-%d-%d", &y, &m, &d) != 3) return -1;
68     const int month_days[12] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
69     int doy = d - 1;
70     for (int i = 0; i < m - 1; ++i) doy += month_days[i];
71     if (y == 2025) return doy;
72     if (y == 2026 && m == 1) return 365 + doy;
73     return -1;
74 }
75
76 int vintage_index(const std::string& label, int fallback) {
77     int h = -1, mi = -1;
78     if (std::sscanf(label.c_str(), "%d:%d", &h, &mi) == 2) {
79         if (h == 0) return 0;
80         if (h == 6) return 1;
81         if (h == 12) return 2;
82         if (h == 18) return 3;
83     }
84     return fallback;
85 }
86
87 }
88
89 Panel load_panel(const InputPaths& paths, const RuleParams& params) {
90     Panel panel;
91
92     // 时段标签按右端点走: 0:10 ... 23:50, 最后补一个 0:00+1 收尾
93     panel.slots.reserve(kSlots);
94     for (int k = 1; k <= kSlots; ++k) {
95         panel.slots.push_back(k == kSlots ? "0:00+1" : xlsx::minutes_to_label(k * 10));
96     }
97     if (panel.slots[143] != "0:00+1") throw std::runtime_error("slot label convention broken (NR10)");
98
99     // 附件2: 逐日读负荷和实际光伏, 第 0 列是日期, 后面 144 列对时段
100     const xlsx::Workbook wb2 = xlsx::read_workbook(paths.att2);
101     const xlsx::Sheet* load_sheet = wb2.find("小区负载");
102     const xlsx::Sheet* pv_sheet = wb2.find("光伏发电实际功率");
103     if (load_sheet == nullptr || pv_sheet == nullptr) throw std::runtime_error("附件2 sheets not found");
104
105     panel.load = Matrix(kDays, kSlots);
106     panel.pv_actual = Matrix(kDays, kSlots);
107     panel.dates.reserve(kDays);
108     for (int d = 0; d < kDays; ++d) {
109         const int row0 = d + 1;
110         const std::string date_iso = xlsx::excel_date_to_iso(cell_number(load_sheet->grid, row0, 0));
111     }

```

```

111     panel.dates.push_back(date_iso);
112     for (int k = 0; k < kSlots; ++k) {
113         panel.load.at(d, k) = cell_number(load_sheet->grid, row0, k + 1);
114         panel.pv_actual.at(d, k) = cell_number(pv_sheet->grid, row0, k + 1);
115     }
116 }
117 if (panel.dates.front() != "2025-01-01" || panel.dates.back() != "2025-12-31") {
118     throw std::runtime_error("附件2 date range unexpected");
119 }
120
121 // 附件4: 实时电价, 行列口径和附件2对齐
122 const xlsx::Workbook wb4 = xlsx::read_workbook(paths.att4);
123 if (wb4.size() == 0) throw std::runtime_error("附件4 empty");
124 const xlsx::Sheet& price_sheet = wb4.at(0);
125 panel.price = Matrix(kDays, kSlots);
126 for (int d = 0; d < kDays; ++d) {
127     for (int k = 0; k < kSlots; ++k) panel.price.at(d, k) = cell_number(price_sheet.grid, d + 1,
128         k + 1);
129 }
130
131 // 附件1: 代表性日, 按 C 序压成 (144, 3), 每行是 [电价, 负荷, 光伏预测]
132 const xlsx::Workbook wb1 = xlsx::read_workbook(paths.att1);
133 if (wb1.size() == 0) throw std::runtime_error("附件1 empty");
134 const xlsx::Sheet& day_sheet = wb1.at(0);
135 panel.day1.assign(kSlots * 3, 0.0);
136 for (int k = 0; k < kSlots; ++k) {
137     const int row0 = k + 1;
138     panel.day1[3 * k + 0] = cell_number(day_sheet.grid, row0, 1);
139     panel.day1[3 * k + 1] = cell_number(day_sheet.grid, row0, 2);
140     panel.day1[3 * k + 2] = cell_number(day_sheet.grid, row0, 3);
141 }
142
143 // 附件3: 四个发布时次的 24 点预报, 日期列空着就沿用上一行
144 const xlsx::Workbook wb3 = xlsx::read_workbook(paths.att3);
145 if (wb3.size() == 0) throw std::runtime_error("附件3 empty");
146 const xlsx::Sheet& fc_sheet = wb3.at(0);
147 panel.pv_forecast.assign(static_cast<size_t>(kDays) * kVintages * kForecastHours, kNaN);
148 std::map<int, int> vintage_seen;
149 std::string last_date;
150 for (int r = 1; r < static_cast<int>(fc_sheet.grid.size()); ++r) {
151     const std::string date_txt = cell_text(fc_sheet.grid, r, 0);
152     if (!date_txt.empty()) last_date = date_txt;
153     int day = parse_iso_day_index(last_date);
154     if (day < 0 || day >= kDays) continue;
155     const int j = vintage_index(cell_text(fc_sheet.grid, r, 1), vintage_seen[day] % kVintages);
156     vintage_seen[day] += 1;
157     for (int h = 1; h <= kForecastHours; ++h) {
158         const double v = cell_number(fc_sheet.grid, r, h + 1);
159         panel.pv_forecast[(static_cast<size_t>(day) * kVintages + j) * kForecastHours + (h - 1)] =
160             std::isnan(v) ? 0.0 : v;
161     }
162 }
163
164 // 日照窗口跟着当天峰值自适应: 第一个和最后一个超过 0 的时段就是日出日落
165 panel.windows.assign(kDays, DayWindow{});
166 panel.pv_global_max = 0.0;
167 for (int d = 0; d < kDays; ++d) {
168     double peak = 0.0;
169     for (int k = 0; k < kSlots; ++k) peak = std::max(peak, panel.pv_actual.at(d, k));
170     panel.pv_global_max = std::max(panel.pv_global_max, peak);
171     const double theta = std::max(params.theta_rel * peak, params.theta_abs);
172     int k_rise = -1, k_set = -1;
173     for (int k = 0; k < kSlots; ++k) {
174         if (panel.pv_actual.at(d, k) > theta) {
175             if (k_rise < 0) k_rise = k;
176             k_set = k;
177         }
178     }
179     DayWindow& w = panel.windows[d];
180     w.theta = theta;

```

```

180     w.peak = peak;
181     w.valid = (k_rise >= 0 && k_set > k_rise);
182     w.k_rise = w.valid ? k_rise : 0;
183     w.k_set = w.valid ? k_set : 0;
184     if (!w.valid) throw std::runtime_error("window inference failed on day " + std::to_string(d)
185 );
186 }
187 const std::string run_id = "clean-2025-v1";
188 auto push_audit = [&](const std::string& file, const std::string& sheet, int excel_row, int
189     excel_col,
190     const std::string& date, int slot1, const std::string& rule_id,
191     const std::string& action, const std::string& old_v, const std::string&
192     new_v,
193     const std::string& reason) {
194     AuditRecord a;
195     a.run_id = run_id;
196     a.file = file;
197     a.sheet = sheet;
198     a.excel_row = excel_row;
199     a.excel_col = col_letter(excel_col);
200     a.date = date;
201     a.slot_index = slot1;
202     a.slot_label = panel.slots[slot1 - 1];
203     char hb[16];
204     std::snprintf(hb, sizeof(hb), "%d:%02d", (slot1 * 10) / 60, (slot1 * 10) % 60);
205     a.hour_end = std::string(hb);
206     a.rule_id = rule_id;
207     a.action = action;
208     a.old_value = old_v;
209     a.new_value = new_v;
210     a.reason = reason;
211     panel.audit.push_back(a);
212 };
213 auto push_quarantine = [&](const std::string& file, const std::string& sheet, int excel_row,
214     int excel_col,
215     const std::string& date, int slot1, const std::string& value,
216     const std::string& rule_id, const std::string& reason) {
217     QuarantineRecord q;
218     q.run_id = run_id;
219     q.file = file;
220     q.sheet = sheet;
221     q.excel_row = excel_row;
222     q.excel_col = col_letter(excel_col);
223     q.date = date;
224     q.slot_index = slot1;
225     q.slot_label = panel.slots[slot1 - 1];
226     q.value = value;
227     q.rule_id = rule_id;
228     q.reason = reason;
229     panel.quarantine.push_back(q);
230 };
231 // 实际光伏过 NR1-NR6: 归零、拒绝、只标记三种动作分别记账
232 panel.pv_clean = panel.pv_actual;
233 std::vector<std::string> last_rule(kDays * kSlots);
234 for (int d = 0; d < kDays; ++d) {
235     const DayWindow& w = panel.windows[d];
236     for (int k = 0; k < kSlots; ++k) {
237         const double raw = panel.pv_actual.at(d, k);
238         const bool in_shoulder = (k == w.k_rise - 1 || k == w.k_rise || k == w.k_set || k == w.
239             k_set + 1);
240         const bool in_core = (k < w.k_rise - 1) || (k > w.k_set + 1);
241         const clean::PointResult pr = clean::apply_pv_point(raw, in_core, in_shoulder, params);
242         if (pr.action == clean::Action::kNone) continue;
243         const int excel_row = d + 2;
244         const int excel_col = k + 1;
245         const std::string old_s = num(raw);
246         if (pr.action == clean::Action::kSetZero) {
247             panel.pv_clean.at(d, k) = 0.0;

```

```

245     push_audit("附件2", "光伏发电实际功率", excel_row, excel_col, panel.dates[d], k + 1, pr.
rule_id,
246         "set_zero", old_s, "0", pr.reason);
247     last_rule[static_cast<size_t>(d) * kSlots + k] = pr.rule_id;
248 } else if (pr.action == clean::Action::kReject) {
249     panel.pv_clean.at(d, k) = kNaN;
250     push_audit("附件2", "光伏发电实际功率", excel_row, excel_col, panel.dates[d], k + 1, pr.
rule_id,
251         "reject", old_s, "nan", pr.reason);
252     push_quarantine("附件2", "光伏发电实际功率", excel_row, excel_col, panel.dates[d], k + 1,
old_s,
253         pr.rule_id, pr.reason);
254     last_rule[static_cast<size_t>(d) * kSlots + k] = pr.rule_id;
255 } else if (pr.action == clean::Action::kFlag) {
256     push_audit("附件2", "光伏发电实际功率", excel_row, excel_col, panel.dates[d], k + 1, pr.
rule_id,
257         "flag", old_s, old_s, pr.reason);
258     last_rule[static_cast<size_t>(d) * kSlots + k] = pr.rule_id;
259 }
260 }
261 }
262
263 // NR6: 窗口内的缺口只在当天内插, 缺口太长或贴边就整段隔离
264 for (int d = 0; d < kDays; ++d) {
265     const DayWindow& w = panel.windows[d];
266     int k = w.k_rise;
267     while (k <= w.k_set) {
268         if (!std::isnan(panel.pv_clean.at(d, k))) {
269             ++k;
270             continue;
271         }
272         int gap_end = k;
273         while (gap_end <= w.k_set && std::isnan(panel.pv_clean.at(d, gap_end))) ++gap_end;
274         const int gap_len = gap_end - k;
275         const int left = k - 1, right = gap_end;
276         if (gap_len <= params.interp_max_gap && left >= w.k_rise && right <= w.k_set &&
277             !std::isnan(panel.pv_clean.at(d, left)) && !std::isnan(panel.pv_clean.at(d, right))) {
278             for (int g = k; g < gap_end; ++g) {
279                 const double t = static_cast<double>(g - left) / static_cast<double>(right - left);
280                 const double v = panel.pv_clean.at(d, left) * (1.0 - t) + panel.pv_clean.at(d, right)
* t;
281                 panel.pv_clean.at(d, g) = v;
282                 push_audit("附件2", "光伏发电实际功率", d + 2, g + 1, panel.dates[d], g + 1, "NR6", "
interpolate",
283                     "nan", num(v), "daylight-window gap interpolation");
284             }
285         } else {
286             for (int g = k; g < gap_end; ++g) {
287                 push_quarantine("附件2", "光伏发电实际功率", d + 2, g + 1, panel.dates[d], g + 1, "nan",
"NR6",
288                     "gap longer than interp_max_gap or touches window edge");
289             }
290             for (int g = k; g < gap_end; ++g) panel.pv_clean.at(d, g) = kNaN;
291         }
292         k = gap_end;
293     }
294 }
295
296 // NR9: 相邻时段跳变超过 20% 只记账, 数值不动
297 for (int d = 0; d < kDays; ++d) {
298     const DayWindow& w = panel.windows[d];
299     for (int k = w.k_rise; k < w.k_set; ++k) {
300         const double a = panel.pv_clean.at(d, k), b = panel.pv_clean.at(d, k + 1);
301         if (std::isnan(a) || std::isnan(b)) continue;
302         const double denom = std::max(std::fabs(a), std::fabs(b));
303         if (denom <= 1e-9) continue;
304         if (std::fabs(b - a) / denom > 0.20) {
305             push_audit("附件2", "光伏发电实际功率", d + 2, k + 2, panel.dates[d], k + 2, "NR9", "flag",
306                 "nan", "nan", "daylight jump > 20%");
307         }

```

```

308     }
309 }
310
311 // NR7: 预报侧沿用同一套夜间口径, 规则号统一挂 NR7> 前缀
312 panel.pv_forecast_clean = panel.pv_forecast;
313 const int vintage_hour[4] = {0, 6, 12, 18};
314 for (int d = 0; d < kDays; ++d) {
315     for (int j = 0; j < kVintages; ++j) {
316         for (int h = 1; h <= kForecastHours; ++h) {
317             const int abs_hour = vintage_hour[j] + h;
318             int target_day = d;
319             int hour_of_day = abs_hour;
320             if (abs_hour > 24) {
321                 target_day = std::min(d + 1, kDays - 1);
322                 hour_of_day = abs_hour - 24;
323             }
324             const int slot = hour_of_day * 6 - 1;
325             if (slot < 0 || slot >= kSlots) continue;
326             const DayWindow& w = panel.windows[target_day];
327             const bool in_day = (slot >= w.k_rise && slot <= w.k_set);
328             const bool in_shoulder =
329                 (slot == w.k_rise - 1 || slot == w.k_rise || slot == w.k_set || slot == w.k_set + 1)
330 ;
331             const bool in_core = (slot < w.k_rise - 1) || (slot > w.k_set + 1);
332             if (in_day && !in_shoulder) continue;
333             const double raw = panel.pv_forecast[(static_cast<size_t>(d) * kVintages + j) *
334 kForecastHours + (h - 1)];
335             const clean::PointResult pr = clean::apply_pv_point(raw, in_core, in_shoulder, params);
336             if (pr.action == clean::Action::kNone) continue;
337             const int excel_row = d * kVintages + j + 2;
338             const int excel_col = h + 1;
339             const std::string old_s = std::to_string(raw);
340             if (pr.action == clean::Action::kSetZero) {
341                 panel.pv_forecast_clean[(static_cast<size_t>(d) * kVintages + j) * kForecastHours + (h
342 - 1)] = 0.0;
343                 push_audit("附件3", "Sheet1", excel_row, excel_col, panel.dates[d], slot + 1, "NR7>NR1
344 ", "set_zero",
345                     old_s, "0", "forecast-side core-night zero drift");
346             } else if (pr.action == clean::Action::kReject) {
347                 panel.pv_forecast_clean[(static_cast<size_t>(d) * kVintages + j) * kForecastHours + (h
348 - 1)] = kNaN;
349                 push_audit("附件3", "Sheet1", excel_row, excel_col, panel.dates[d], slot + 1, "NR7>NR2
350 ", "reject",
351                     old_s, "nan", "forecast-side core-night outlier");
352                 push_quarantine("附件3", "Sheet1", excel_row, excel_col, panel.dates[d], slot + 1,
353 old_s, "NR7>NR2",
354                     "forecast-side core-night outlier");
355             } else if (pr.action == clean::Action::kFlag) {
356                 push_audit("附件3", "Sheet1", excel_row, excel_col, panel.dates[d], slot + 1,
357 "NR7>" + pr.rule_id, "flag", old_s, old_s, "forecast-side shoulder flag");
358             }
359         }
360     }
361 }
362
363 // NR11/NR12: 异常电价和夜间负荷都只标记, 怎么处理留给后续判断
364 for (int d = 0; d < kDays; ++d) {
365     for (int k = 0; k < kSlots; ++k) {
366         const double price = panel.price.at(d, k);
367         if (price < 0.1) {
368             push_audit("附件4", "Sheet1", d + 2, k + 1, panel.dates[d], k + 1, "NR11", "flag",
369 num(price), num(price), "near-zero price (<0.1 CNY/kWh), kept for Q4
370 scenarios");
371         }
372         const DayWindow& w = panel.windows[d];
373         const bool in_core = (k < w.k_rise - 1) || (k > w.k_set + 1);
374         const double load = panel.load.at(d, k);
375         if (in_core && (load < 1500.0 || load > 6000.0)) {
376             push_audit("附件2", "小区负载", d + 2, k + 1, panel.dates[d], k + 1, "NR12", "flag", num(
377 load),

```

```

369         num(load), "night load outside [1500,6000] kW");
370     }
371 }
372 }
373
374 // 日出日落一天挪超过 3 个时段就记一笔
375 for (int d = 1; d < kDays; ++d) {
376     const int dr = std::abs(panel.windows[d].k_rise - panel.windows[d - 1].k_rise);
377     const int ds = std::abs(panel.windows[d].k_set - panel.windows[d - 1].k_set);
378     if (dr > params.window_jump_slots || ds > params.window_jump_slots) {
379         push_audit("附件2", "光伏发电实际功率", d + 2, 1, panel.dates[d], 1, "WINDOW_JUMP", "flag", "
nan",
380                 "nan", "sunrise/sunset shift exceeded window_jump_slots");
381     }
382 }
383
384 // NR8: 共形尺度隔离, 夜间没样本时用 floor 兜底
385 panel.sigma_floor = std::max(1.0, 0.005 * panel.pv_global_max);
386 panel.sigma_raw.assign(kSlots, 0.0);
387 panel.sigma_used.assign(kSlots, 0.0);
388 panel.sigma_n.assign(kSlots, 0);
389 // 用 0:00 时次的预报算日前残差, 再逐时段统计尺度
390 for (int k = 0; k < kSlots; ++k) {
391     double sum = 0.0, sum2 = 0.0;
392     int n = 0;
393     for (int d = 0; d < kDays; ++d) {
394         const DayWindow& w = panel.windows[d];
395         if (!(k >= w.k_rise && k <= w.k_set)) continue;
396         const double act = panel.pv_clean.at(d, k);
397         if (std::isnan(act) || act <= w.theta) continue;
398         const int hour = std::min(k / 6 + 1, kForecastHours);
399         const double fc = panel.pv_forecast_clean[(static_cast<size_t>(d) * kVintages + 0) *
kForecastHours + (hour - 1)];
400         if (std::isnan(fc)) continue;
401         const double res = act - fc;
402         sum += res;
403         sum2 += res * res;
404         ++n;
405     }
406     panel.sigma_n[k] = n;
407     const double mean = n > 0 ? sum / n : 0.0;
408     const double var = n > 1 ? std::max(0.0, sum2 / n - mean * mean) : 0.0;
409     panel.sigma_raw[k] = std::sqrt(var);
410     panel.sigma_used[k] = std::max(panel.sigma_raw[k], panel.sigma_floor);
411 }
412
413 return panel;
414 }
415
416 }

```

code/cleaning_cpp/src/rules.cpp

```

1 #include "clean/rules.h"
2
3 #include <cmath>
4 #include <limits>
5
6 namespace clean {
7
8     namespace {
9         const double kNaN = std::numeric_limits<double>::quiet_NaN();
10    }
11
12    // 置 NaN 再进隔离清单, 所有“拒绝”都从这里收口
13    PointResult make_reject(const std::string& rule_id, const std::string& reason, double value) {
14        PointResult r;
15        r.action = Action::kReject;
16        r.value = kNaN;

```

```

17   r.quarantine = true;
18   r.rule_id = rule_id;
19   r.action_name = "reject";
20   r.reason = reason + " (old=" + std::to_string(value) + ")";
21   return r;
22 }
23
24 PointResult apply_pv_point(double value, bool in_core, bool in_shoulder, const model::RuleParams
    & p) {
25     PointResult r;
26     r.value = value;
27     // 缺测和负值不分时段，先拦下来
28     if (std::isnan(value)) {
29         r.rule_id = "NR3";
30         r.action = Action::kFlag;
31         r.action_name = "flag";
32         r.reason = "missing value inside daylight window (NR6 candidate)";
33         return r;
34     }
35     if (value < 0.0) return make_reject("NR3", "negative PV power", value);
36
37     // 核夜间：几 kW 的小噪声归零，超过噪声带就是硬异常
38     if (in_core) {
39         if (value > p.noise_kw) return make_reject("NR2", "core-night value above noise band", value
            );
40         if (value > 0.0) {
41             r.action = Action::kSetZero;
42             r.value = 0.0;
43             r.rule_id = "NR1";
44             r.action_name = "set_zero";
45             r.reason = "core-night zero-drift within noise band";
46             return r;
47         }
48         return r;
49     }
50
51     // 过渡段：弱信号只标记不删，不到 1 kW 的当噪声归零
52     if (in_shoulder) {
53         if (value >= p.shoulder_kw && value <= p.noise_kw) {
54             r.action = Action::kFlag;
55             r.value = value;
56             r.rule_id = "NR5";
57             r.action_name = "flag";
58             r.reason = "shoulder weak signal, excluded from sigma estimation";
59             return r;
60         }
61         if (value > 0.0 && value < p.shoulder_kw) {
62             r.action = Action::kSetZero;
63             r.value = 0.0;
64             r.rule_id = "NR4";
65             r.action_name = "set_zero";
66             r.reason = "shoulder sub-kW noise";
67             return r;
68         }
69     }
70     return r;
71 }
72
73 const char* rule_description(const std::string& rule_id) {
74     if (rule_id == "NR1") return "core-night zero drift (0,5] kW -> 0";
75     if (rule_id == "NR2") return "core-night hard outlier (>5 kW) -> NaN + quarantine";
76     if (rule_id == "NR3") return "negative PV (any segment) -> NaN + quarantine";
77     if (rule_id == "NR4") return "shoulder sub-kW noise (0,1) kW -> 0";
78     if (rule_id == "NR5") return "shoulder weak signal [1,5] kW -> flag (keep)";
79     if (rule_id == "NR6") return "daylight-window gap interpolation (<= max_gap slots)";
80     if (rule_id == "NR7") return "forecast-side cleaning with identical night semantics";
81     if (rule_id == "NR8") return "conformal scale isolation with floor";
82     if (rule_id == "NR9") return "daylight jump > 20% -> flag only";
83     if (rule_id == "NR10") return "time convention assertion (right endpoint, slot<->column)";
84     if (rule_id == "NR11") return "price < 0.1 CNY/kWh -> near-zero flag";

```

```

85 if (rule_id == "NR12") return "night load outside [1500,6000] kW -> flag";
86 if (rule_id == "WINDOW_JUMP") return "sunrise/sunset window jump > 3 slots";
87 return "unknown rule";
88 }
89
90 }

```

code/cleaning_cpp/src/sha256.cpp

```

1  #include "util/sha256.h"
2
3  #include <cstdio>
4  #include <cstring>
5  #include <vector>
6
7  namespace util {
8  namespace {
9
10 const uint32_t K[64] = {
11     0x428a2f98u, 0x71374491u, 0xb5c0fbcfu, 0xe9b5dba5u, 0x3956c25bu, 0x59f111f1u, 0x923f82a4u,
12     0xab1c5ed5u, 0xd807aa98u, 0x12835b01u, 0x243185beu, 0x550c7dc3u, 0x72be5d74u, 0x80deb1feu,
13     0x9bdc06a7u, 0xc19bf174u, 0xe49b69c1u, 0xefbe4786u, 0x0fc19dc6u, 0x240ca1ccu, 0x2de92c6fu,
14     0x4a7484aa, 0x5cb0a9dcu, 0x76f988dau, 0x983e5152u, 0xa831c66du, 0xb00327c8u, 0xbf597fc7u,
15     0xc6e00bf3u, 0xd5a79147u, 0x06ca6351u, 0x14292967u, 0x27b70a85u, 0x2e1b2138u, 0x4d2c6dfcu,
16     0x53380d13u, 0x650a7354u, 0x766a0abbu, 0x81c2c92eu, 0x92722c85u, 0xa2bfe8a1u, 0xa81a664bu,
17     0xc24b8b70u, 0xc76c51a3u, 0xd192e819u, 0xd6990624u, 0xf40e3585u, 0x106aa070u, 0x19a4c116u,
18     0x1e376c08u, 0x2748774cu, 0x34b0bcb5u, 0x391c0cb3u, 0x4ed8aa4au, 0x5b9cca4fu, 0x682e6ff3u,
19     0x748f82eeu, 0x78a5636fu, 0x84c87814u, 0x8cc70208u, 0x90bffffau, 0xa4506cebu, 0xbef9a3f7u,
20     0xc67178f2u,
21 };
22
23 inline uint32_t rotr(uint32_t x, uint32_t n) { return (x >> n) | (x << (32 - n)); }
24
25 }
26
27 void Sha256::reset() {
28     static const uint32_t init[8] = {0x6a09e667u, 0xbb67ae85u, 0x3c6ef372u, 0xa54ff53au,
29                                     0x510e527fu, 0x9b05688cu, 0x1f83d9abu, 0x5be0cd19u};
30     std::memcpy(h_, init, sizeof(h_));
31     buf_len_ = 0;
32     total_len_ = 0;
33 }
34
35 void Sha256::transform(const uint8_t* block) {
36     uint32_t w[64];
37     for (int i = 0; i < 16; ++i) {
38         w[i] = (static_cast<uint32_t>(block[i * 4]) << 24) |
39             (static_cast<uint32_t>(block[i * 4 + 1]) << 16) |
40             (static_cast<uint32_t>(block[i * 4 + 2]) << 8) |
41             static_cast<uint32_t>(block[i * 4 + 3]);
42     }
43     for (int i = 16; i < 64; ++i) {
44         const uint32_t s0 = rotr(w[i - 15], 7) ^ rotr(w[i - 15], 18) ^ (w[i - 15] >> 3);
45         const uint32_t s1 = rotr(w[i - 2], 17) ^ rotr(w[i - 2], 19) ^ (w[i - 2] >> 10);
46         w[i] = w[i - 16] + s0 + w[i - 7] + s1;
47     }
48     uint32_t a = h_[0], b = h_[1], c = h_[2], d = h_[3];
49     uint32_t e = h_[4], f = h_[5], g = h_[6], hh = h_[7];
50     for (int i = 0; i < 64; ++i) {
51         const uint32_t s1 = rotr(e, 6) ^ rotr(e, 11) ^ rotr(e, 25);
52         const uint32_t ch = (e & f) ^ ((~e) & g);
53         const uint32_t t1 = hh + s1 + ch + K[i] + w[i];
54         const uint32_t s0 = rotr(a, 2) ^ rotr(a, 13) ^ rotr(a, 22);
55         const uint32_t maj = (a & b) ^ (a & c) ^ (b & c);
56         const uint32_t t2 = s0 + maj;
57         hh = g; g = f; f = e; e = d + t1;
58         d = c; c = b; b = a; a = t1 + t2;
59     }
60     h_[0] += a; h_[1] += b; h_[2] += c; h_[3] += d;

```

```

61     h_[4] += e; h_[5] += f; h_[6] += g; h_[7] += hh;
62 }
63
64 void Sha256::update(const uint8_t* data, size_t len) {
65     total_len_ += len;
66     while (len > 0) {
67         const size_t take = (len < (64 - buf_len_)) ? len : (64 - buf_len_);
68         std::memcpy(buf_ + buf_len_, data, take);
69         buf_len_ += take;
70         data += take;
71         len -= take;
72         if (buf_len_ == 64) {
73             transform(buf_);
74             buf_len_ = 0;
75         }
76     }
77 }
78
79 // 收尾: 先补 0x80 和 64 位长度位, 再把 8 个大端 32 位字拼成十六进制
80 std::string Sha256::hexdigest() {
81     const uint64_t bits = total_len_ * 8ULL;
82     uint8_t pad[72];
83     pad[0] = 0x80;
84     size_t pad_len = (buf_len_ < 56) ? (56 - buf_len_) : (120 - buf_len_);
85     std::memset(pad + 1, 0, pad_len - 1);
86     for (int i = 0; i < 8; ++i) pad[pad_len + i] = static_cast<uint8_t>(bits >> (56 - 8 * i));
87     update(pad, pad_len + 8);
88     char out[65];
89     for (int i = 0; i < 8; ++i) {
90         std::snprintf(out + i * 8, 9, "%08x", h_[i]);
91     }
92     out[64] = '\0';
93     return std::string(out);
94 }
95
96 std::string Sha256::hex_of(const std::string& raw) {
97     Sha256 s;
98     s.update(reinterpret_cast<const uint8_t*>(raw.data()), raw.size());
99     return s.hexdigest();
100 }
101
102 std::string Sha256::bytes(const std::string& data) { return hex_of(data); }
103
104 std::string Sha256::file(const std::string& path) {
105     std::FILE* fp = std::fopen(path.c_str(), "rb");
106     if (fp == nullptr) return std::string();
107     Sha256 s;
108     std::vector<uint8_t> buf(1 << 16);
109     size_t n = 0;
110     while ((n = std::fread(buf.data(), 1, buf.size(), fp)) > 0) s.update(buf.data(), n);
111     std::fclose(fp);
112     return s.hexdigest();
113 }
114
115 }

```

code/cleaning_cpp/src/writers.cpp

```

1 #include "io/writers.h"
2
3 #include <cmath>
4 #include <cstdint>
5 #include <cstdio>
6 #include <stdexcept>
7
8 namespace io {
9     namespace {
10
11         // .npy 头部长度得凑成 64 的整数倍, 差多少补多少空格

```

```

12 std::string pad_header(const std::string& dict) {
13     size_t len = dict.size() + 1;
14     const size_t total = 10 + len;
15     const size_t pad = (64 - (total % 64)) % 64;
16     std::string out = dict;
17     out.append(pad, ' ');
18     out.push_back('\n');
19     return out;
20 }
21
22 }
23
24 // 手写 .npy: 魔数 → 版本 → 头长 → 头部字典 → 小端 float64 原始数据
25 void write_npy_f64(const std::string& path, const double* data, const std::vector<size_t>& shape
    ) {
26     std::string shape_str;
27     if (shape.size() == 1) {
28         shape_str = "(" + std::to_string(shape[0]) + ",";
29     } else {
30         shape_str = "(";
31         for (size_t i = 0; i < shape.size(); ++i) {
32             shape_str += std::to_string(shape[i]);
33             if (i + 1 < shape.size()) shape_str += ", ";
34         }
35         shape_str += ")";
36     }
37     size_t count = 1;
38     for (size_t s : shape) count *= s;
39     const std::string dict = "{ 'descr': '<f8', 'fortran_order': False, 'shape': " + shape_str + ",
        }";
40     const std::string header = pad_header(dict);
41
42     std::FILE* fp = std::fopen(path.c_str(), "wb");
43     if (fp == nullptr) throw std::runtime_error("cannot write " + path);
44     const char magic[6] = {'\x93', 'N', 'U', 'M', 'P', 'Y'};
45     std::fwrite(magic, 1, 6, fp);
46     const unsigned char version[2] = {1, 0};
47     std::fwrite(version, 1, 2, fp);
48     const uint16_t hlen = static_cast<uint16_t>(header.size());
49     unsigned char lb[2] = {static_cast<unsigned char>(hlen & 0xFF),
50         static_cast<unsigned char>((hlen >> 8) & 0xFF)};
51     std::fwrite(lb, 1, 2, fp);
52     std::fwrite(header.data(), 1, header.size(), fp);
53     std::fwrite(data, sizeof(double), count, fp);
54     std::fclose(fp);
55 }
56
57 void write_text(const std::string& path, const std::string& content) {
58     std::FILE* fp = std::fopen(path.c_str(), "wb");
59     if (fp == nullptr) throw std::runtime_error("cannot write " + path);
60     std::fwrite(content.data(), 1, content.size(), fp);
61     std::fclose(fp);
62 }
63
64 // 只有含逗号、引号或换行的格子才加引号，引号本身翻倍转义
65 void write_csv(const std::string& path, const std::vector<std::vector<std::string>>& rows) {
66     std::string out;
67     for (const auto& row : rows) {
68         for (size_t i = 0; i < row.size(); ++i) {
69             if (i) out += ",";
70             const std::string& cell = row[i];
71             const bool needs_quote = cell.find_first_of(",\"\\n\\r") != std::string::npos;
72             if (!needs_quote) {
73                 out += cell;
74             } else {
75                 out += "'";
76                 for (char c : cell) {
77                     if (c == "'') out += "'";
78                     out += c;
79                 }

```

```

80         out += ' ';
81     }
82 }
83     out += "\n";
84 }
85     write_text(path, out);
86 }
87
88 std::string fmt(double v, int precision) {
89     if (std::isnan(v)) return "nan";
90     char buf[64];
91     std::snprintf(buf, sizeof(buf), "%.*g", precision, v);
92     return std::string(buf);
93 }
94
95 std::string json_escape(const std::string& s) {
96     std::string out;
97     for (char c : s) {
98         switch (c) {
99             case '\\': out += "\\\\"; break;
100             case '\n': out += "\\n"; break;
101             case '\r': out += "\\r"; break;
102             case '\t': out += "\\t"; break;
103             default: out += c;
104         }
105     }
106     return out;
107 }
108
109 std::string slots_json(const model::Panel& panel) {
110     std::string out = "{\n  \"convention\": \"right-endpoint timestamps of 10-minute slots\",\n  \"slots\": [";
111     for (size_t i = 0; i < panel.slots.size(); ++i) {
112         if (i) out += ", ";
113         out += "\"" + json_escape(panel.slots[i]) + "\"";
114     }
115     out += "]\n}\n";
116     return out;
117 }
118
119 std::string dates_json(const model::Panel& panel) {
120     std::string out = "{\n  \"days\": " + std::to_string(panel.dates.size()) + ",\n  \"dates\": [";
121     for (size_t i = 0; i < panel.dates.size(); ++i) {
122         if (i) out += ", ";
123         out += "\"" + panel.dates[i] + "\"";
124     }
125     out += "]\n}\n";
126     return out;
127 }
128
129 // manifest 记下输入输出哈希、行列数和运行开销, verify 阶段靠它对账
130 std::string manifest_json(const model::Panel& panel, double seconds, long peak_rss_kb,
131     const std::vector<std::pair<std::string, std::string>>& inputs,
132     const std::vector<std::pair<std::string, std::string>>& outputs) {
133     std::string out = "{\n";
134     out += "  \"producer\": \"code/cleaning_cpp (C++17, zlib+tinyclang)\",\n";
135     out += "  \"contract\": \"prompt v1: C++ performs all value-level cleaning; Python is read-only\",\n";
136     out += "  \"rows\": " + std::to_string(panel.pv_actual.rows()) + ",\n";
137     out += "  \"cols\": " + std::to_string(panel.pv_actual.cols()) + ",\n";
138     out += "  \"forecast_shape\": [365, 4, 24],\n";
139     out += "  \"day1_shape\": [144, 3],\n";
140     out += "  \"dtype\": \"<f8\",\n";
141     out += "  \"missing_policy\": \"NaN (never 0)\",\n";
142     out += "  \"time_convention\": \"slot k (1..144) <=> right endpoint 10k minutes <=> attachment column k+1\",\n";
143     out += "  \"audit_records\": " + std::to_string(panel.audit.size()) + ",\n";
144     out += "  \"quarantine_records\": " + std::to_string(panel.quarantine.size()) + ",\n";

```

```

146 out += "  \"sigma_floor_kw\": " + fmt(panel.sigma_floor) + ",\n";
147 out += "  \"wall_seconds\": " + fmt(seconds, 6) + ",\n";
148 out += "  \"peak_rss_kb\": " + std::to_string(peak_rss_kb) + ",\n";
149 out += "  \"inputs_sha256\": {\n";
150 for (size_t i = 0; i < inputs.size(); ++i) {
151     out += "    \"" + json_escape(inputs[i].first) + "\": \"" + inputs[i].second + "\"" +
152         (i + 1 < inputs.size() ? ", " : "") + "\n";
153 }
154 out += "  },\n  \"outputs_sha256\": {\n";
155 for (size_t i = 0; i < outputs.size(); ++i) {
156     out += "    \"" + json_escape(outputs[i].first) + "\": \"" + outputs[i].second + "\"" +
157         (i + 1 < outputs.size() ? ", " : "") + "\n";
158 }
159 out += "  }\n}\n";
160 return out;
161 }
162
163 }

```

code/cleaning_cpp/src/xlsx.cpp

```

1 #include "xlsx/xlsx.h"
2
3 #include <zlib.h>
4
5 #include <algorithm>
6 #include <cstdio>
7 #include <cstdlib>
8 #include <cstring>
9 #include <stdexcept>
10
11 #include "tinycl2.h"
12
13 namespace xlsx {
14 namespace {
15
16 uint16_t u16(const char* p) {
17     return static_cast<uint16_t>(static_cast<uint8_t>(p[0]) | (static_cast<uint8_t>(p[1]) << 8));
18 }
19
20 uint32_t u32(const char* p) {
21     return static_cast<uint32_t>(static_cast<uint8_t>(p[0]) |
22         (static_cast<uint32_t>(static_cast<uint8_t>(p[1]) << 8) |
23         (static_cast<uint32_t>(static_cast<uint8_t>(p[2]) << 16) |
24         (static_cast<uint32_t>(static_cast<uint8_t>(p[3]) << 24);
25 }
26
27 std::string inflate_raw(const std::string& in, uint64_t expected) {
28     std::string out;
29     out.resize(static_cast<size_t>(expected));
30     z_stream zs;
31     std::memset(&zs, 0, sizeof(zs));
32     if (inflateInit2(&zs, -MAX_WBITS) != Z_OK) throw std::runtime_error("inflateInit2 failed");
33     zs.next_in = reinterpret_cast<Bytef*>(const_cast<char*>(in.data()));
34     zs.avail_in = static_cast<uInt>(in.size());
35     zs.next_out = reinterpret_cast<Bytef*>(&out[0]);
36     zs.avail_out = static_cast<uInt>(out.size());
37     const int rc = inflate(&zs, Z_FINISH);
38     inflateEnd(&zs);
39     if (rc != Z_STREAM_END) throw std::runtime_error("inflate failed rc=" + std::to_string(rc));
40     out.resize(zs.total_out);
41     return out;
42 }
43
44 // 共享字符串里混了富文本 (<r><t>...</t></r>)，平铺取不到就再扫一层 <r>
45 std::string si_text(const tinycl2::XMLElement* si) {
46     std::string out;
47     for (const tinycl2::XMLElement* t = si->FirstChildElement("t"); t != nullptr;
48         t = t->NextSiblingElement("t")) {
49         if (t->GetText() != nullptr) out += t->GetText();
50     }
51 }

```

```

49     }
50     if (out.empty()) {
51         for (const tinyxml2::XMLElement* r = si->FirstChildElement("r"); r != nullptr;
52             r = r->NextSiblingElement("r")) {
53             for (const tinyxml2::XMLElement* t = r->FirstChildElement("t"); t != nullptr;
54                 t = t->NextSiblingElement("t")) {
55                 if (t->GetText() != nullptr) out += t->GetText();
56             }
57         }
58     }
59     return out;
60 }
61
62 }
63
64 ZipArchive ZipArchive::open(const std::string& path) {
65     ZipArchive z;
66     std::FILE* fp = std::fopen(path.c_str(), "rb");
67     if (fp == nullptr) throw std::runtime_error("cannot open " + path);
68     std::fseek(fp, 0, SEEK_END);
69     const long size = std::ftell(fp);
70     std::fseek(fp, 0, SEEK_SET);
71     z.raw_.resize(static_cast<size_t>(size));
72     if (size > 0 && std::fread(&z.raw_[0], 1, static_cast<size_t>(size), fp) != static_cast<size_t>
73         >(size)) {
74         std::fclose(fp);
75         throw std::runtime_error("short read " + path);
76     }
77     std::fclose(fp);
78     // 从尾部往前找 EOCD (0x06054b50), zip 尾部注释区一般留空
79     const size_t max_back = std::min<size_t>(z.raw_.size(), 66000);
80     size_t eocd = std::string::npos;
81     for (size_t i = z.raw_.size() - 22 + 1; i-- > z.raw_.size() - max_back;) {
82         if (u32(&z.raw_[i]) == 0x06054b50u) {
83             eocd = i;
84             break;
85         }
86         if (i == 0) break;
87     }
88     if (eocd == std::string::npos) throw std::runtime_error("EOCD not found: " + path);
89     const uint16_t count = u16(&z.raw_[eocd + 10]);
90     const uint32_t cd_off = u32(&z.raw_[eocd + 16]);
91
92     size_t p = cd_off;
93     for (uint16_t i = 0; i < count; ++i) {
94         if (u32(&z.raw_[p]) != 0x02014b50u) throw std::runtime_error("bad central directory");
95         Entry e;
96         e.method = u16(&z.raw_[p + 10]);
97         e.comp_size = u32(&z.raw_[p + 20]);
98         e.uncomp_size = u32(&z.raw_[p + 24]);
99         const uint16_t name_len = u16(&z.raw_[p + 28]);
100         const uint16_t extra_len = u16(&z.raw_[p + 30]);
101         const uint16_t comment_len = u16(&z.raw_[p + 32]);
102         e.local_offset = u32(&z.raw_[p + 42]);
103         e.name = z.raw_.substr(p + 46, name_len);
104         z.index_[e.name] = e;
105         p += 46 + name_len + extra_len + comment_len;
106     }
107     return z;
108 }
109
110 std::string ZipArchive::read(const std::string& name) const {
111     auto it = index_.find(name);
112     if (it == index_.end()) throw std::runtime_error("zip entry missing: " + name);
113     const Entry& e = it->second;
114     const size_t lh = static_cast<size_t>(e.local_offset);
115     if (u32(&raw_[lh]) != 0x04034b50u) throw std::runtime_error("bad local header: " + name);
116     const uint16_t name_len = u16(&raw_[lh + 26]);
117     const uint16_t extra_len = u16(&raw_[lh + 28]);

```

```

118     const size_t data_off = lh + 30 + name_len + extra_len;
119     const std::string comp = raw_.substr(data_off, static_cast<size_t>(e.comp_size));
120     if (e.method == 0) return comp;
121     if (e.method != 8) throw std::runtime_error("unsupported compression method");
122     return inflate_raw(comp, e.uncomp_size);
123 }
124
125 std::vector<std::string> ZipArchive::names() const {
126     std::vector<std::string> out;
127     out.reserve(index_.size());
128     for (const auto& kv : index_) out.push_back(kv.first);
129     return out;
130 }
131
132 const Sheet* Workbook::find(const std::string& name) const {
133     for (const Sheet& s : sheets)
134         if (s.name == name) return &s;
135     return nullptr;
136 }
137
138 int col_from_ref(const std::string& ref) {
139     int col = 0;
140     for (char ch : ref) {
141         if (ch >= 'A' && ch <= 'Z') {
142             col = col * 26 + (ch - 'A' + 1);
143         } else if (ch >= 'a' && ch <= 'z') {
144             col = col * 26 + (ch - 'a' + 1);
145         } else {
146             break;
147         }
148     }
149     return col - 1;
150 }
151
152 int row_from_ref(const std::string& ref) {
153     size_t i = 0;
154     while (i < ref.size() && !(ref[i] >= '0' && ref[i] <= '9')) ++i;
155     if (i >= ref.size()) return -1;
156     return std::atoi(ref.c_str() + i);
157 }
158
159 // Excel 的 1900 日期系统: 序列号 ≥ 61 时减 1, 跳过不存在的 1900-02-29
160 std::string excel_date_to_iso(double serial) {
161     long s = static_cast<long>(serial < 61 ? serial : serial - 1);
162     long days = s - 25568;
163     long z = days + 719468;
164     const long era = (z >= 0 ? z : z - 146096) / 146097;
165     const unsigned doe = static_cast<unsigned>(z - era * 146097);
166     const unsigned yoe = (doe - doe / 1460 + doe / 36524 - doe / 146096) / 365;
167     const long y = static_cast<long>(yoe) + era * 400;
168     const unsigned doy = doe - (365 * yoe + yoe / 4 - yoe / 100);
169     const unsigned mp = (5 * doy + 2) / 153;
170     const unsigned d = doy - (153 * mp + 2) / 5 + 1;
171     const unsigned m = mp < 10 ? mp + 3 : mp - 9;
172     const long year = y + (m <= 2 ? 1 : 0);
173     char buf[32];
174     std::snprintf(buf, sizeof(buf), "%04ld-%02u-%02u", year, m, d);
175     return std::string(buf);
176 }
177
178 int minutes_of_day(double fraction) {
179     double mins = fraction * 1440.0;
180     return static_cast<int>(mins + 0.5);
181 }
182
183 std::string minutes_to_label(int minutes) {
184     char buf[16];
185     std::snprintf(buf, sizeof(buf), "%d:%02d", minutes / 60, minutes % 60);
186     return std::string(buf);
187 }

```

```

188
189 // 先读 workbook.xml 和 rels 拿到表名与路径, 再逐张表解析单元格
190 Workbook read_workbook(const std::string& path) {
191     ZipArchive zip = ZipArchive::open(path);
192     const std::string wb_xml = zip.read("xl/workbook.xml");
193     const std::string rels_xml = zip.read("xl/_rels/workbook.xml.rels");
194
195     std::map<std::string, std::string> rel_target;
196     {
197         tinyxml2::XMLDocument doc;
198         if (doc.Parse(rels_xml.c_str()) != tinyxml2::XML_SUCCESS) throw std::runtime_error("bad rels
199             xml");
200         for (const tinyxml2::XMLElement* e = doc.RootElement()->FirstChildElement("Relationship");
201             e != nullptr; e = e->NextSiblingElement("Relationship")) {
202             rel_target[e->Attribute("Id")] = e->Attribute("Target");
203         }
204     }
205
206     std::vector<std::pair<std::string, std::string>> sheet_refs;
207     {
208         tinyxml2::XMLDocument doc;
209         if (doc.Parse(wb_xml.c_str()) != tinyxml2::XML_SUCCESS) throw std::runtime_error("bad
210             workbook xml");
211         const tinyxml2::XMLElement* root = doc.RootElement();
212         const tinyxml2::XMLElement* sheets = root->FirstChildElement("sheets");
213         if (sheets == nullptr) throw std::runtime_error("no <sheets>");
214         for (const tinyxml2::XMLElement* e = sheets->FirstChildElement("sheet"); e != nullptr;
215             e = e->NextSiblingElement("sheet")) {
216             const char* rid = e->Attribute("r:id");
217             std::string target = rid != nullptr ? rel_target[rid] : std::string();
218             if (!target.empty() && target.rfind("xl/", 0) != 0) target = "xl/" + target;
219             sheet_refs.emplace_back(e->Attribute("name"), target);
220         }
221     }
222
223     std::vector<std::string> shared;
224     if (zip.has("xl/sharedStrings.xml")) {
225         tinyxml2::XMLDocument doc;
226         const std::string ss = zip.read("xl/sharedStrings.xml");
227         if (doc.Parse(ss.c_str()) != tinyxml2::XML_SUCCESS) throw std::runtime_error("bad
228             sharedStrings");
229         for (const tinyxml2::XMLElement* si = doc.RootElement()->FirstChildElement("si"); si !=
230             nullptr; si = si->NextSiblingElement("si")) {
231             shared.push_back(si_text(si));
232         }
233     }
234
235     Workbook wb;
236     for (const auto& sr : sheet_refs) {
237         Sheet sheet;
238         sheet.name = sr.first;
239         const std::string xml = zip.read(sr.second);
240         tinyxml2::XMLDocument doc;
241         if (doc.Parse(xml.c_str()) != tinyxml2::XML_SUCCESS) throw std::runtime_error("bad sheet xml
242             " + sr.second);
243         const tinyxml2::XMLElement* data = doc.RootElement()->FirstChildElement("sheetData");
244         int max_row = 0, max_col = -1;
245         struct Raw {
246             int r, c;
247             CellValue v;
248         };
249         std::vector<Raw> raw_cells;
250         if (data != nullptr) {
251             for (const tinyxml2::XMLElement* row = data->FirstChildElement("row"); row != nullptr;
252                 row = row->NextSiblingElement("row")) {
253                 for (const tinyxml2::XMLElement* c = row->FirstChildElement("c"); c != nullptr;
254                     c = c->NextSiblingElement("c")) {
255                     const char* ref = c->Attribute("r");
256                     if (ref == nullptr) continue;

```

```

253     Raw rc;
254     rc.r = row_from_ref(ref);
255     rc.c = col_from_ref(ref);
256     max_row = std::max(max_row, rc.r);
257     max_col = std::max(max_col, rc.c);
258     const char* type = c->Attribute("t");
259     const tinyxml2::XMLElement* v = c->FirstChildElement("v");
260     // t="s" 查共享字符串表, 其余按 inlineStr / str / bool / 数字逐类取值
261     if (type != nullptr && std::strcmp(type, "s") == 0 && v != nullptr && v->GetText() !=
nullptr) {
262         const int idx = std::atoi(v->GetText());
263         rc.v.kind = ValueKind::kText;
264         rc.v.text = (idx >= 0 && idx < static_cast<int>(shared.size())) ? shared[idx] : std
::string();
265     } else if (type != nullptr && std::strcmp(type, "inlineStr") == 0) {
266         const tinyxml2::XMLElement* is = c->FirstChildElement("is");
267         rc.v.kind = ValueKind::kText;
268         if (is != nullptr) rc.v.text = si_text(is);
269     } else if (type != nullptr && std::strcmp(type, "str") == 0) {
270         rc.v.kind = ValueKind::kText;
271         if (v != nullptr && v->GetText() != nullptr) rc.v.text = v->GetText();
272     } else if (type != nullptr && std::strcmp(type, "b") == 0) {
273         rc.v.kind = ValueKind::kBool;
274         rc.v.number = (v != nullptr && v->GetText() != nullptr) ? std::atof(v->GetText()) :
0.0;
275     } else if (v != nullptr && v->GetText() != nullptr) {
276         rc.v.kind = ValueKind::kNumber;
277         rc.v.number = std::strtod(v->GetText(), nullptr);
278     } else {
279         rc.v.kind = ValueKind::kEmpty;
280     }
281     raw_cells.push_back(rc);
282 }
283 }
284 }
285 sheet.grid.assign(static_cast<size_t>(std::max(max_row, 0)),
286     std::vector<CellValue>(static_cast<size_t>(std::max(max_col + 1, 0))));
287 for (const Raw& rc : raw_cells) {
288     if (rc.r >= 1 && rc.c >= 0) sheet.grid[static_cast<size_t>(rc.r - 1)][static_cast<size_t>(
rc.c)] = rc.v;
289 }
290 wb.sheets.push_back(std::move(sheet));
291 }
292 return wb;
293 }
294 }
295 }

```

H.3 公共层 code/common

code/__init__.py

```
1 """微网调度项目 Python 侧代码: 清洗不在此处, 在 code/cleaning_cpp."""
```

code/common/__init__.py

```
1 """公共部分: 读 clean 数据、读写 xlsx 模板、门禁自检."""
```

code/common/config.yaml

```

1 # 全局参数: 每个数都能追溯到题目附录1 或 prompt v1 的决策, 没有对着公布答案调过
2 storage:
3     capacity_kwh: 12000.0
4     soc_min_kwh: 1200.0
5     soc_max_kwh: 10800.0

```

```

6   soc_init_kwh: 6000.0
7   power_kw: 5000.0
8   eta_charge: 0.9
9   eta_discharge: 0.9
10  time:
11     slots_per_day: 144
12     dt_hours: 0.16666666666666666
13     decision_hours: [0, 6, 12, 18]
14  tariff:
15     emergency_multiplier: 5.0
16     over_plan_multiplier: 0.5
17     under_plan_multiplier: 1.5
18  quantiles:
19     q2_plan: 0.80
20     q3_plan: 0.25
21     q3_adjust: 0.70
22  scenarios:
23     q2_lookback_days: 30
24     q2_scenarios: 20
25     q4_scenarios: 20
26     seed: 20250910
27  risk:
28     cvar_alpha: 0.90
29     cvar_beta_grid: [0.0, 0.25, 0.5, 0.75, 0.9]
30     soft_penalty_multiplier: 6.0
31  forecast:
32     lookback_days: 30
33  verify:
34     q1_baseline_b2_cny: 48052.0
35     q1_lp_anchor_cny: 35126.95

```

code/common/audit_output.py

```

1  """独立复算 `output/result*.xlsx`: 以结果文件为真相源的账目与物理一致性审计。
2
3  设计原则（只读）：
4
5  * 读取通道与写入通道分离：用 openpyxl 读 xlsx（写入端用自实现的 zip+XML），
6    避免“写入端自证”。
7  * 一切判据都回到题面口径：Q1/Q2/Q3 按附件 1 电价结算，Q4-2/Q4-3 按附件 4 电价结算；
8    紧急购电按 5 倍交易时刻电价；Q3/Q4-3 的计划—调整差按 0.5 / 1.5 倍结算。
9  * 输出 `results/output_audit.json`: 逐项 PASS/FAIL、最大偏差与论文引用的关键数字。
10
11  用法：
12     python3 -m code.common.audit_output
13  """
14  from __future__ import annotations
15
16  import json
17  import re
18  from pathlib import Path
19
20  import numpy as np
21  import openpyxl
22
23  from code.common.load_clean import ROOT, load_panel
24
25  RESULTS = ROOT / "results"
26  OUTPUT = ROOT / "output"
27  TEMPLATES = ROOT / "templates" / "附件5"
28  FIRST_DAY = 31
29  BLOCK_RE = re.compile(r"^(\\d+):(\\d+)-(\\d+):(\\d+)$")
30
31
32  def load_json(path: Path) -> dict:
33     return json.loads(path.read_text(encoding="utf-8"))
34
35
36  def block_slots(label: str) -> tuple[int, int]:

```

```

37     """把 '7:20-8:10' 映射为覆盖的 10 分钟时段区间 [k0, k1] (右端点口径)。"""
38     m = BLOCK_RE.match(str(label).strip())
39     if m is None:
40         raise ValueError(f"unparsable block label: {label!r}")
41     h0, m0, h1, m1 = (int(g) for g in m.groups())
42     start, end = h0 * 60 + m0, h1 * 60 + m1
43     if end <= start:
44         raise ValueError(f"non-positive block span: {label!r}")
45     return start // 10, end // 10
46
47
48 def sheet_rows(ws) -> list[tuple]:
49     return [row for row in ws.iter_rows(values_only=True)]
50
51
52 def read_slots_sheet(ws) -> tuple[list[str], np.ndarray]:
53     rows = sheet_rows(ws)
54     labels = [str(r[0]) for r in rows[1:]]
55     values = np.array([float(r[1]) if r[1] is not None else np.nan for r in rows[1:]], dtype=
56         float)
57     return labels, values
58
59 def read_wide_sheet(ws) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
60     rows = sheet_rows(ws)
61     values = np.array([float(v) for v in r[1:145]] for r in rows[1:]), dtype=float)
62     daily_energy = np.array([float(r[145]) for r in rows[1:]], dtype=float)
63     daily_cost = np.array([float(r[146]) for r in rows[1:]], dtype=float)
64     return values, daily_energy, daily_cost
65
66
67 def read_charge_rows(ws) -> dict[str, np.ndarray]:
68     rows = sheet_rows(ws)
69     has_date = str(rows[0][0]).strip() == "日期"
70     off = 1 if has_date else 0
71     charge, discharge, soc = [], [], []
72     current_charge = current_discharge = 0.0
73     soc_start = soc_end = None
74     for row in rows[1:]:
75         if has_date and row[0] is not None:
76             if soc_start is not None:
77                 charge.append(current_charge)
78                 discharge.append(current_discharge)
79                 soc.append((soc_start, soc_end))
80                 current_charge = current_discharge = 0.0
81                 soc_start = float(row[off + 4])
82                 soc_end = None
83             if not has_date and soc_start is None:
84                 soc_start = float(row[off + 4])
85                 current_charge += float(row[off + 1])
86                 current_discharge += float(row[off + 2])
87                 if row[off + 3] is not None and str(row[off + 3]).strip() == "24:00":
88                     soc_end = float(row[off + 4])
89             charge.append(current_charge)
90             discharge.append(current_discharge)
91             soc.append((soc_start, soc_end))
92     return {"charge": np.array(charge), "discharge": np.array(discharge),
93         "soc_start": np.array([s[0] for s in soc], dtype=float),
94         "soc_end": np.array([s[1] for s in soc], dtype=float)}
95
96
97 def read_emergency_rows(ws) -> tuple[np.ndarray, list[dict]]:
98     """返回每日紧急购电量 (按行出现顺序累加) 与原始块记录。"""
99     rows = sheet_rows(ws)
100     energy: list[float] = []
101     blocks: list[dict] = []
102     for row in rows[1:]:
103         if row[0] is not None:
104             energy.append(0.0)
105         if row[1] is None and row[2] is None:

```

```

106         continue
107         value = float(row[2])
108         energy[-1] += value
109         blocks.append({"day": len(energy) - 1, "label": str(row[1]), "energy_kwh": value})
110     return np.array(energy, dtype=float), blocks
111
112
113 def price_for(tag: str, panel) -> np.ndarray:
114     """Q2/Q3 见附件 1 电价 (题目: "每天的电价相同"), Q4 见附件 4 电价。"""
115     if tag.startswith("q4"):
116         return panel.price
117     return np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
118
119
120 def add(checks: list[dict], name: str, ok: bool, detail: str = "", dev: float | None = None) ->
    None:
121     checks.append({"check": name, "pass": bool(ok), "detail": detail,
122                   "max_abs_deviation": None if dev is None else float(dev)})
123
124
125 def audit_result1(panel, checks: list[dict]) -> dict:
126     st = panel.config["storage"]
127     eta_c, eta_d = float(st["eta_charge"]), float(st["eta_discharge"])
128     soc_init = float(st["soc_init_kwh"])
129     wb = openpyxl.load_workbook(OUTPUT / "result1.xlsx", read_only=True, data_only=True)
130     tpl = openpyxl.load_workbook(TEMPLATES / "result1.xlsx", read_only=True, data_only=True)
131     labels, q = read_slots_sheet(wb["计划购电量"])
132     tpl_labels, _ = read_slots_sheet(tpl["计划购电量"])
133     add(checks, "result1 时段标签与模板逐字一致", labels == tpl_labels,
134         f"{len(labels)} 行")
135     cd = read_charge_rows(wb["充放电量"])
136     wb.close()
137     tpl.close()
138
139     q1 = load_json(RESULTS / "q1.json")
140     price = panel.day1[:, 0]
141     cost = float(np.sum(price * q))
142     add(checks, "result1 计划购电量合计 = q1.json 计划量",
143         abs(q.sum() - float(np.sum(q1["lp"]['q']))) < 1e-6,
144         f"{q.sum():.2f} vs {float(np.sum(q1['lp']['q'])):.2f}")
145     add(checks, "result1 购电费 = Σ 附件1电价×购电量 = q1.json 最优值",
146         abs(cost - float(q1["cost_cny"])) < 1e-6,
147         f"{cost:.6f} vs {float(q1['cost_cny']):.6f}", abs(cost - float(q1["cost_cny"])))
148     add(checks, "result1 0:00 与 24:00 储电量均等于初始 6000 kWh",
149         abs(cd["soc_start"][0] - soc_init) < 1e-9 and abs(cd["soc_end"][0] - soc_init) < 1e-9,
150         f"{cd['soc_start'][0]:.1f} / {cd['soc_end'][0]:.1f}")
151     delta = eta_c * cd["charge"][0] - cd["discharge"][0] / eta_d
152     add(checks, "result1 充放电量与 SOC 日循环恒等式",
153         abs(delta) < 1e-6, f"Σ(ηc·c - d/ηd) = {delta:.9f} kWh", abs(delta))
154     add(checks, "result1 充放电量与 q1.json 一致",
155         abs(cd["charge"][0] - float(np.sum(q1["lp"]['c']))) < 1e-6
156         and abs(cd["discharge"][0] - float(np.sum(q1["lp"]['d']))) < 1e-6,
157         f"{cd['charge'][0]:.2f} / {cd['discharge'][0]:.2f}")
158     return {
159         "purchase_kwh": float(q.sum()),
160         "cost_cny": cost,
161         "charge_kwh": float(cd["charge"][0]),
162         "discharge_kwh": float(cd["discharge"][0]),
163         "soc_start_kwh": float(cd["soc_start"][0]),
164         "soc_end_kwh": float(cd["soc_end"][0]),
165     }
166
167
168 def audit_wide(tag: str, file: str, with_adjust: bool, panel, checks: list[dict]) -> dict:
169     st = panel.config["storage"]
170     eta_c, eta_d = float(st["eta_charge"]), float(st["eta_discharge"])
171     soc_init = float(st["soc_init_kwh"])
172     days = slice(FIRST_DAY, 365)
173     price = price_for(tag, panel)[days]
174     summary = load_json(RESULTS / f"{tag}.json")["summary"]

```

```

175 records = {r["date"]: r for r in load_json(RESULTS / f"{tag}.json")["records"]}
176
177 wb = openpyxl.load_workbook(OUTPUT / file, read_only=True, data_only=True)
178 plan, plan_daily, plan_cost = read_wide_sheet(wb["计划购电量"])
179 adj = plan.copy()
180 if with_adjust:
181     adj, adj_daily, _ = read_wide_sheet(wb["调整购电量"])
182     add(checks, f"{tag} 调整购电量日合计列自治",
183         float(np.abs(adj_daily - adj.sum(axis=1)).max()) < 1e-6,
184         "", float(np.abs(adj_daily - adj.sum(axis=1)).max()))
185 emergency, blocks = read_emergency_rows(wb["紧急购电量"])
186 cd = read_charge_rows(wb["充放电量"])
187 wb.close()
188
189 add(checks, f"{tag} 计划购电量日合计列自治",
190     float(np.abs(plan_daily - plan.sum(axis=1)).max()) < 1e-6,
191     "", float(np.abs(plan_daily - plan.sum(axis=1)).max()))
192 dev_cost = float(np.abs(np.sum(price * plan, axis=1) - plan_cost).max())
193 add(checks, f"{tag} 逐日购电费列 =  $\Sigma$  电价 $\times$ 计划量 ({'附件4' if tag.startswith('q4') else '附件1'})电",
194     dev_cost < 1e-6, "", dev_cost)
195
196 plan_cost_total = float(np.sum(plan_cost))
197 emergency_energy = float(emergency.sum())
198 add(checks, f"{tag} 紧急购电量合计 = 记录值",
199     abs(emergency_energy - float(summary["total_emergency_energy_kwh"])) < 1e-6,
200     f"{emergency_energy:.6f} vs {float(summary['total_emergency_energy_kwh']):.6f}",
201     abs(emergency_energy - float(summary["total_emergency_energy_kwh"])))
202
203 # 紧急购电是按 4 小时块报的, 先摊回 144 个时段再跟记录值对
204 g_slot = np.zeros_like(plan)
205 for block in blocks:
206     k0, k1 = block_slots(block["label"])
207     g_slot[block["day"], k0:k1] += block["energy_kwh"] / (k1 - k0)
208 emergency_cost = float(5.0 * np.sum(price * g_slot))
209
210 over = float(np.sum(price * np.maximum(0.0, plan - adj)))
211 under = float(np.sum(price * np.maximum(0.0, adj - plan)))
212 over_pen = 0.5 * over
213 under_pen = 1.5 * under
214 if with_adjust:
215     add(checks, f"{tag} 超额违约金  $0.5 \sum \pi(q^{\wedge} \text{plan} - q^{\wedge} \text{adj})^+ =$  记录值",
216         abs(over_pen - float(summary["total_over_plan_penalty_cny"])) < 1e-6,
217         f"{over_pen:.6f} vs {float(summary['total_over_plan_penalty_cny']):.6f}",
218         abs(over_pen - float(summary["total_over_plan_penalty_cny"])))
219     add(checks, f"{tag} 欠额违约金  $1.5 \sum \pi(q^{\wedge} \text{adj} - q^{\wedge} \text{plan})^+ =$  记录值",
220         abs(under_pen - float(summary["total_under_plan_penalty_cny"])) < 1e-6,
221         f"{under_pen:.6f} vs {float(summary['total_under_plan_penalty_cny']):.6f}",
222         abs(under_pen - float(summary["total_under_plan_penalty_cny"])))
223 else:
224     over_pen = under_pen = 0.0
225
226 total = plan_cost_total + float(summary["total_emergency_cost_cny"]) + over_pen + under_pen
227 add(checks, f"{tag} 账目合计 = 计划费+紧急费+违约金 = 记录总额",
228     abs(total - float(summary["total_cost_cny"])) < 1e-6,
229     f"{total:.6f} vs {float(summary['total_cost_cny']):.6f}",
230     abs(total - float(summary["total_cost_cny"])))
231 add(checks, f"{tag} 紧急购电费 (5 倍) 与记录值同量级 (块内均匀分摊口径)",
232     abs(emergency_cost - float(summary["total_emergency_cost_cny"]))
233     < 0.05 * float(summary["total_emergency_cost_cny"]),
234     f"{emergency_cost:.2f} vs {float(summary['total_emergency_cost_cny']):.2f}")
235
236 add(checks, f"{tag} 每日 0:00 与 24:00 储电量均为 6000 kWh",
237     float(np.abs(cd["soc_start"] - soc_init).max()) < 1e-9
238     and float(np.abs(cd["soc_end"] - soc_init).max()) < 1e-9,
239     f"max $|\Delta| = \{\max(\text{float}(\text{np.abs}(\text{cd}['\text{soc\_start'}] - \text{soc\_init}).\text{max}()), \text{float}(\text{np.abs}(\text{cd}['\text{soc\_end'}] - \text{soc\_init}).\text{max}())):.3e\}"$ )
240 delta = eta_c * cd["charge"] - cd["discharge"] / eta_d
241 add(checks, f"{tag} 充放电量与 SOC 日循环恒等式",
242     float(np.abs(delta).max()) < 1e-6, "", float(np.abs(delta).max()))

```

```

243 add(checks, f"{tag} 单日充放电量不超过 6 h 额定功率",
244       float(cd["charge"].max()) <= 6 * float(st["power_kw"]) + 1e-6
245       and float(cd["discharge"].max()) <= 6 * float(st["power_kw"]) + 1e-6,
246       f"max c={cd['charge'].max():.1f} / max d={cd['discharge'].max():.1f}")
247
248 # 日级平衡: 光伏 + 计划购电 + 放电 + 紧急购电 要盖住 负荷 + 充电
249 # 充放电表只按 4 小时块汇总, 所以这里只能按日总量核
250 pv = panel.pv_filled()[days]
251 load = panel.load[days]
252 dt = float(panel.config["time"]["dt_hours"])
253 balance = (np.sum(pv, axis=1) * dt + np.sum(adj, axis=1) + cd["discharge"]
254            - np.sum(load, axis=1) * dt - cd["charge"] + emergency)
255 add(checks, f"{tag} 日级功率平衡不违反 (光伏+购电+放电+紧急 ≥ 负荷+充电)",
256       float(balance.min()) > -1e-6, f"最小余量 {balance.min():.6f} kWh", float(balance.min()))
257
258 dates = list(panel.dates[days])
259 plan_key = "q_plan_kwh" if "q_plan_kwh" in records[dates[0]] else "planned_purchase_kwh"
260 rec_plan = np.array([records[d][plan_key] for d in dates])
261 add(checks, f"{tag} 逐日计划购电量 = 记录值",
262       float(np.abs(plan.sum(axis=1) - rec_plan).max()) < 1e-4,
263       "", float(np.abs(plan.sum(axis=1) - rec_plan).max()))
264 rec_charge = np.array([records[d]["charge_kwh"] for d in dates])
265 rec_discharge = np.array([records[d]["discharge_kwh"] for d in dates])
266 add(checks, f"{tag} 逐日充放电量 = 记录值",
267       float(np.abs(cd["charge"] - rec_charge).max()) < 1e-4
268       and float(np.abs(cd["discharge"] - rec_discharge).max()) < 1e-4,
269       "", max(float(np.abs(cd["charge"] - rec_charge).max()),
270               float(np.abs(cd["discharge"] - rec_discharge).max())))
271 rec_g = np.array([records[d]["emergency_energy_kwh"] for d in dates])
272 add(checks, f"{tag} 逐日紧急购电量 = 记录值",
273       float(np.abs(emergency - rec_g).max()) < 1e-4, "",
274       float(np.abs(emergency - rec_g).max()))
275
276 return {
277     "days": int(plan.shape[0]),
278     "total_cost_cny": float(summary["total_cost_cny"]),
279     "plan_cost_cny": plan_cost_total,
280     "emergency_cost_cny": float(summary["total_emergency_cost_cny"]),
281     "emergency_energy_kwh": emergency_energy,
282     "over_plan_penalty_cny": over_pen,
283     "under_plan_penalty_cny": under_pen,
284     "purchase_kwh": float(plan.sum()),
285     "adjust_kwh": float(adj.sum()),
286     "emergency_blocks": len(blocks),
287     "min_daily_balance_kwh": float(balance.min()),
288 }
289
290
291 def main() -> None:
292     panel = load_panel()
293     checks: list[dict] = []
294     out = {
295         "source": "output/result{1,2,3,4-2,4-3}.xlsx",
296         "convention": {
297             "q1": "附件1 电价 (元/kWh) × 计划购电量",
298             "q2_q3": "附件1 电价 (题面: 每天的电价相同)",
299             "q4_2_q4_3": "附件4 实时电价",
300             "emergency": "5 × 交易时刻电价",
301             "adjustment": "0.5× 下调违约、1.5× 上调违约",
302         },
303         "result1": audit_result1(panel, checks),
304     }
305     for tag, file, with_adjust in (("q2", "result2.xlsx", False),
306                                   ("q3", "result3.xlsx", True),
307                                   ("q4_2", "result4-2.xlsx", False),
308                                   ("q4_3", "result4-3.xlsx", True)):
309         out[tag] = audit_wide(tag, file, with_adjust, panel, checks)
310
311     failed = [c for c in checks if not c["pass"]]
312     out["checks"] = checks

```

```

313 out["passed"] = len(checks) - len(failed)
314 out["failed"] = len(failed)
315 out["status"] = "PASS" if not failed else "FAIL"
316 RESULTS.mkdir(exist_ok=True)
317 (RESULTS / "output_audit.json").write_text(
318     json.dumps(out, ensure_ascii=False, indent=2), encoding="utf-8")
319
320 for c in checks:
321     flag = "PASS" if c["pass"] else "FAIL"
322     extra = "" if c["max_abs_deviation"] is None else f" (max dev {c['max_abs_deviation']:.3e})"
323     print(f"[{flag}] {c['check']} {c['detail']}{extra}")
324 print(f"\n审计总览: {out['passed']}/{len(checks)} 通过, status={out['status']}")
325 print("关键数字: ")
326 for tag in ("q2", "q3", "q4_2", "q4_3"):
327     row = out[tag]
328     print(f" {tag}: 合计 {row['total_cost_cny']:.2f} 元; 计划 {row['plan_cost_cny']:.2f} 元; "
329           f"紧急 {row['emergency_cost_cny']:.2f} 元 / {row['emergency_energy_kwh']:.2f} kWh; "
330           f"违约 {row['over_plan_penalty_cny']:.2f} + {row['under_plan_penalty_cny']:.2f} 元")
331
332
333 if __name__ == "__main__":
334     main()

```

code/common/checks.py

```

1  """门禁自检: G0-G9、Q3 结算复算、参数灵敏度。"""
2  from __future__ import annotations
3
4  import json
5  from pathlib import Path
6
7  import numpy as np
8
9  from code.common.load_clean import ROOT, load_panel
10 from code.common.xlsx_template_io import Workbook
11 from code.optimize.lp_common import Storage, solve_stage_lp
12
13 RESULTS = ROOT / "results"
14 ARRAYS = RESULTS / "arrays"
15 TEMPLATES = ROOT / "templates" / "附件5"
16 OUTPUT = ROOT / "output"
17 FIRST_DAY = 31
18
19 # captions3.py 里 FIG_SPECS 规定的 25 张交付图 (编号 → 文件名 slug), 用来查完整性
20 _CAPTIONS3_SLUGS: tuple[tuple[int, str], ...] = (
21     (1, "hierarchical_architecture"), (2, "value_cascade"),
22     (3, "microgrid_sld"), (4, "pv_inverter_clipping"),
23     (5, "time_expanded_network"), (6, "day_ahead_balance_soc"),
24     (7, "kkt_dual_deadband"), (8, "maxent_residual_fit"),
25     (9, "aci_fan_lower_bound"), (10, "lead_time_expansion"),
26     (11, "coverage_calibration"), (12, "newsvendor_pareto"),
27     (13, "tail_penalty_distribution"), (14, "info_funnel_shrinkage"),
28     (15, "no_adaptation_zone"), (16, "adjustment_policy_regions"),
29     (17, "curi_vs_nav"), (18, "nav_trigger_timeline"),
30     (19, "subset_trajectories"), (20, "mtrrc_layers"),
31     (21, "student_t_copula"), (22, "cbf_phase_space"),
32     (23, "detc_threshold_evolution"), (24, "ablation_map"),
33     (25, "architecture_parallel_coordinates"),
34 )
35
36
37 def load_json(path: Path) -> dict:
38     return json.loads(path.read_text(encoding="utf-8"))
39
40
41 def approx(a, b, tol=1e-6) -> bool:

```

```

42     return abs(float(a) - float(b)) <= tol
43
44
45 def gate_g0() -> dict:
46     cross = load_json(ROOT / "clean" / "cross_check.json")
47     invariants = load_json(ROOT / "clean" / "invariants.json")
48     checks = [
49         ("I1-I8 all pass", bool(invariants["all_pass"])),
50         "; ".join(f"{c['id']}:{'PASS' if c['pass'] else 'FAIL'}" for c in invariants["checks"])
51     ),
52     ("core-night zero turnover", cross["att2_core_night_zero_raw"] == 23316
53      and cross["att2_core_night_zero_after"] == 25766,
54      f"{cross['att2_core_night_zero_raw']} -> {cross['att2_core_night_zero_after']}"),
55     ("quarantine = 129 hard outliers", cross["quarantine_records"] == 129,
56      str(cross["quarantine_records"])),
57     ("08:00-16:00 zero-free", cross["att2_daytime_0800_1600_zeros"] == 0,
58      str(cross["att2_daytime_0800_1600_zeros"])),
59     ("near-zero price cells = 28", cross["att4_near_zero_price_cells"] == 28,
60      str(cross["att4_near_zero_price_cells"])),
61     ("附件1 shoulder small values = 6", cross["att1_small_pv_cells_0_5kw"] == 6,
62      str(cross["att1_small_pv_cells_0_5kw"])),
63 ]
64     return {"gate": "G0", "title": "夜间清洗与数据质量", "checks": checks,
65            "pass": all(c[1] for c in checks)}
66
67 def gate_g1() -> dict:
68     q1 = load_json(RESULTS / "q1.json")
69     b2 = q1["baselines"]["B2_pv_selfuse_curtail_no_storage_cny"]
70     dp = q1["dp"]
71     identity = q1.get("energy_identity", {})
72     kkt = q1.get("kkt_deadband", {})
73     rel = min(r["rel_error_vs_lp"] for r in dp)
74     monotone = all(dp[i]["cost_cny"] >= dp[i + 1]["cost_cny"] - 1e-6 for i in range(len(dp) - 1)
75 )
76     checks = [
77         ("cost below B2 baseline", q1["cost_cny"] < b2,
78          f"{q1['cost_cny']:.2f} < {b2:.2f}"),
79         ("terminal SoC = 6000 kWh", approx(q1["soc_terminal_kwh"], 6000.0),
80          f"{q1['soc_terminal_kwh']:.2f}"),
81         ("SoC inside [1200, 10800]", q1["soc_min_observed"] >= 1199.99
82          and q1["soc_max_observed"] <= 10800.01,
83          f"[{q1['soc_min_observed']:.1f}, {q1['soc_max_observed']:.1f}]"),
84         ("proposition 1 (no simultaneous charge/discharge)",
85          q1["simultaneous_charge_discharge_kw"] <= 1e-9,
86          f"{q1['simultaneous_charge_discharge_kw']:.2e} kW"),
87         ("DP monotone convergence", monotone,
88          " -> ".join(f"{r['cost_cny']:.1f}" for r in dp)),
89         ("DP bracket LP optimum within 1%", rel < 0.01, f"min rel err {rel:.2e}"),
90         ("daily energy identity residual < 1e-9 kWh (v2 Q1-4)",
91          abs(identity.get("residual_kwh", float("inf"))) < 1e-9,
92          f"residual {identity.get('residual_kwh'):.3e} kWh"),
93         ("KKT deadband has no interior violation (v2 Q1-5)",
94          kkt.get("interior_violation_slots", 1) == 0,
95          f"interior violations {kkt.get('interior_violation_slots')}, "
96          f"bound-active actions {kkt.get('bound_active_actions_in_deadband')}"),
97     ]
98     return {"gate": "G1", "title": "问题1 LP/DP 互证与基线", "checks": checks,
99            "pass": all(c[1] for c in checks)}
100
101 def gate_g2() -> dict:
102     p2 = load_json(RESULTS / "p2_conformal.json")
103     checks = []
104     for name, stats in p2["coverage"]["anchors"].items():
105         checks.append((f"{name} coverage within 3pp", abs(stats["deviation"]) <= 0.03,
106                       f"nominal {stats['nominal']:.2f}, actual {stats['coverage']:.4f}, "
107                       f"dev {100*stats['deviation']:.2f}pp"))
108     return {"gate": "G2", "title": "共形覆盖率 (0.80 / 0.25 / 0.70) ", "checks": checks,
109            "pass": all(c[1] for c in checks)}

```

```

110
111
112 def gate_g3() -> dict:
113     q2 = load_json(RESULTS / "q2.json")
114     summary = q2["summary"]
115     controls = summary["controls_total_cost_cny"]
116     maxent = summary.get("maxent", {})
117     pv_lower = ARRAYS / "q2_pv_lower.npy"
118     checks = [
119         ("334 days backtested", summary["days"] == 334, str(summary["days"])),
120         ("cheaper than no-margin plan", summary["total_cost_cny"] < controls["C1_no_margin"],
121          f"{summary['total_cost_cny']:.0f} vs {controls['C1_no_margin']:.0f}"),
122         ("online cost above perfect-information bound",
123          summary["total_cost_cny"] > controls["oracle_bound"],
124          f"{summary['total_cost_cny']:.0f} > {controls['oracle_bound']:.0f}"),
125         ("cross-day coupling quantified", "C3_terminal_value" in controls,
126          f"terminal-value expected {controls['C3_terminal_value']:.0f} vs cycle "
127          f"{controls['C3_terminal_value_cycle_expected']:.0f}"),
128         ("emergency blocks merged", all(len(r["blocks"]) == r["emergency_blocks"]
129                                         for r in q2["records"]), "block count consistent"),
130         # trust-exact 加阻尼牛顿在 334 个窗口上都打到机器精度, 1e-4 留了余量
131         ("MaxEnt dual converged (v2 Q2-1)",
132          maxent.get("max_gradient_inf", float("inf")) <= 1e-4
133          and maxent.get("min_hessian_eig", -1.0) > 0.0,
134          f"max |grad| {maxent.get('max_gradient_inf'):.2e}, "
135          f"min Hessian eig {maxent.get('min_hessian_eig'):.2e}"),
136         ("PV physical lower bound exported (v2 Q2-4/Q2-5)",
137          pv_lower.exists() and (ARRAYS / "q2_theta.npy").exists(),
138          f"{pv_lower.name} + q2_theta.npy, mean {maxent.get('mean_pv_lower_kwh', 0):.0f} kWh/day
139          "),
140         ("ACI path identity reported (v2 Q2-2)",
141          "aci_path_identity" in summary,
142          f"bound {summary.get('aci_path_identity', {}).get('bound')}"),
143     ]
144     return {"gate": "G3", "title": "问题2 全年回溯与对照", "checks": checks,
145            "pass": all(c[1] for c in checks)}
146
147 def settle(plan_q: np.ndarray, adj_q: np.ndarray, g: np.ndarray, price: np.ndarray,
148            emergency=5.0, over=0.5, under=1.5) -> float:
149     return float(np.sum(price * plan_q) + emergency * np.sum(price * g)
150                 + over * np.sum(price * np.maximum(0.0, plan_q - adj_q))
151                 + under * np.sum(price * np.maximum(0.0, adj_q - plan_q)))
152
153
154 def gate_g4() -> dict:
155     q3 = load_json(RESULTS / "q3.json")
156     panel = load_panel()
157     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
158     plan = np.load(ARRAYS / "q3_q_plan.npy")
159     adj = np.load(ARRAYS / "q3_q_adjust.npy")
160     g = np.load(ARRAYS / "q3_emergency.npy")
161     total = 0.0
162     for i in range(plan.shape[0]):
163         total += settle(plan[i], adj[i], g[i], price[FIRST_DAY + i])
164     recorded = q3["summary"]["total_cost_cny"]
165     margins = q3["summary"]["subset_marginal_value_cny"]
166     checks = [
167         ("settlement formula reproducible", approx(total, recorded, 1e-3),
168          f"recomputed {total:.2f} vs recorded {recorded:.2f} 元"),
169         ("S0-S5 subset experiment present",
170          all(k in q3["summary"]["subset_total_cost_cny"] for k in ("S0", "S1", "S2", "S3", "S4",
171          "S5")),
172          str({k: round(v) for k, v in q3["summary"]["subset_total_cost_cny"].items()})),
173         ("explicit recommendation on extra forecast times",
174          margins["value_of_0600"] > 0 and "value_of_1800" in margins,
175          f"6:00 {margins['value_of_0600']:.0f} / 12:00 {margins['value_of_1200']:.0f} / "
176          f"18:00 {margins['value_of_1800']:.0f} 元"),
177         ("night/day value split reported", "subset_value_split_cny" in q3["summary"],
178          json.dumps(q3["summary"]["subset_value_split_cny"], ensure_ascii=False)),

```

```

178     ("lead-time envelope exported (v2 Q3-1)",
179     (RESULTS / "q3_lead_time.csv").exists()
180     and "lead_time_envelope" in q3["summary"],
181     f"{sum(v['slots'] for v in q3['summary']['lead_time_envelope']['vintages'].values())} "
182     f"slot points, {len(q3['summary']['lead_time_envelope']['vintages'])} vintages"),
183     ("NAV counterfactual gate reported (v2 Q3-3)",
184     "nav_gate" in q3["summary"]
185     and q3["summary"]["nav_gate"]["stages_evaluated"] > 0,
186     f"{q3['summary']['nav_gate']['stages_accepted']}/"
187     f"{q3['summary']['nav_gate']['stages_evaluated']} stages accepted, "
188     f"total NAV {q3['summary']['nav_gate']['total_nav_cny']:.0f} CNY"),
189     ("endogenous no-adjustment band reported (v2 Q3-2)",
190     "no_adjustment_band" in q3["summary"],
191     q3["summary"]["no_adjustment_band"]["band"]),
192     ("NAV / lead-time traces carry no per-day vector payload",
193     all("lead_time_curve" not in row
194         for row in q3.get("lead_time_trace", [])),
195     "lead-time curve moved to results/q3_lead_time.csv"),
196 ]
197 return {"gate": "G4", "title": "问题3 结算复算与预报时次建议", "checks": checks,
198         "pass": all(c[1] for c in checks)}
199
200
201 def gate_g5() -> dict:
202     q4 = load_json(RESULTS / "q4.json")
203     spikes = q4["spike_conformity"]
204     pareto = q4["cvar_pareto"]
205     soft = q4["soft_penalty_evidence"]
206     t_copula = q4.get("student_t_copula", {})
207     ablation = q4.get("execution_ablation", {})
208     a0 = ablation.get("A0_full", {})
209     cvar_improves = pareto[-1]["cvar_90_cny"] < pareto[0]["cvar_90_cny"]
210     slack_evidence = any(row["slack_energy_kwh"] > 0 for row in soft)
211     checks = [
212         ("spike conformal coverage >= 90%", spikes["coverage"] >= 0.90,
213         f"{spikes['coverage']*100:.2f}% over {spikes['spike_slots']} spike slots"),
214         ("CVaR Pareto front computed", len(pareto) >= 5,
215         "; ".join(f"beta={r['beta']:.2f}->CVaR {r['cvar_90_cny']:.0f}" for r in pareto)),
216         ("CVaR reduces tail risk", cvar_improves,
217         f"{pareto[0]['cvar_90_cny']:.0f} -> {pareto[-1]['cvar_90_cny']:.0f} 元/日"),
218         ("soft-constraint abuse demonstrated for small rho1", slack_evidence,
219         "; ".join(f"rho1={r['rho1_multiplier']}x->slack {r['slack_energy_kwh']:.0f} kWh"
220             for r in soft)),
221         ("Student-t copula fitted with tail dependence (v2 Q4-2)",
222         t_copula.get("available", False) and t_copula.get("df", 0) > 0
223         and t_copula.get("tail_dependence_load_price", 0) > 0,
224         f"df {t_copula.get('df')}, tail dep {t_copula.get('tail_dependence_load_price'):.3f}"),
225         ("CBF safety filter: zero SOC violation (v2 Q4-4)",
226         a0.get("soc_violation_slots", 1) == 0 and a0.get("cbf_active_slots", 0) > 0,
227         f"{a0.get('cbf_active_slots')} projected slots, "
228         f"{a0.get('soc_violation_slots')} violations"),
229         ("DETC dispatch reduction >= 50% (v2 Q4-5)",
230         ablation.get("dispatch_reduction_vs_periodic", 0) >= 0.50,
231         f"{100*ablation.get('dispatch_reduction_vs_periodic', 0):.1f}% fewer dispatches, "
232         f"terminal repair {a0.get('mean_repair_kwh_per_day', 0):.0f} kWh/day"),
233         ("daily-cycle contract restored on every ablation day (v2 Q4-12)",
234         a0.get("terminal_contract_violations", 1) == 0,
235         f"{a0.get('terminal_ok_days')}/{ablation.get('days')} days"),
236         ("A0-A4 ablation metrics present (v2 Q4-6)",
237         all(k in ablation for k in ("A0_full", "A1_independent_scenarios",
238             "A2_no_price_band", "A3_no_cbf",
239             "A4_periodic_trigger")),
240         "; ".join(k for k in ablation if k.startswith("A"))),
241     ]
242     return {"gate": "G5", "title": "问题4 尾部风险与软约束依据", "checks": checks,
243           "pass": all(c[1] for c in checks)}
244
245
246 def workbook_checks() -> list[tuple]:
247     expectations = {

```

```

248     "result1.xlsx": {"计划购电量": (145, 2), "充放电量": (7, 5)},
249     "result2.xlsx": {"计划购电量": (335, 147), "充放电量": (2005, 6), "紧急购电量": None},
250     "result3.xlsx": {"计划购电量": (335, 147), "调整购电量": (335, 147),
251                     "充放电量": (2005, 6), "紧急购电量": None},
252     "result4-2.xlsx": {"计划购电量": (335, 147), "充放电量": (2005, 6), "紧急购电量": None},
253     "result4-3.xlsx": {"计划购电量": (335, 147), "调整购电量": (335, 147),
254                     "充放电量": (2005, 6), "紧急购电量": None},
255 }
256 out = []
257 for name, sheets in expectations.items():
258     path = OUTPUT / name
259     out.append(("{name} exists", path.exists(), str(path)))
260     if not path.exists():
261         continue
262     wb = Workbook(path)
263     for sheet, dims in sheets.items():
264         present = sheet in wb.sheets
265         out.append(("{name}/{sheet} present", present, ""))
266         if not present:
267             continue
268         rows = wb.sheets[sheet].rows
269         n_rows = max(rows) if rows else 0
270         n_cols = max((max(c) + 1) for c in rows.values()) if rows else 0
271         if dims:
272             out.append(("{name}/{sheet} shape {dims}", (n_rows, n_cols) == dims,
273                       f"actual ({n_rows}, {n_cols})"))
274     wb.close()
275     # 模板一致性: 表头文字和最后一行标签都得对得上
276     tpl = Workbook(TEMPLATES / name)
277     cur = Workbook(path)
278     for sheet in tpl.sheets:
279         same = all(tpl.text(sheet, 1, c) == cur.text(sheet, 1, c)
280                   for c in tpl.sheets[sheet].rows.get(1, {}))
281         out.append(("{name}/{sheet} header identical to template", same, ""))
282     # 日期格仍是 Excel 日期, 不是被写成了文本
283     first_date = cur.number("计划购电量", 2, 0)
284     last_date = cur.number("计划购电量", 335, 0)
285     if name == "result1.xlsx":
286         # result1 只有代表日的时间标签, 没有日期列
287         labels_ok = (cur.text("计划购电量", 2, 0) == tpl.text("计划购电量", 2, 0)
288                     and cur.text("计划购电量", 145, 0) == tpl.text("计划购电量", 145, 0))
289         out.append(("result1 slot labels identical to template", labels_ok,
290                   f"{cur.text('计划购电量', 2, 0)} .. {cur.text('计划购电量', 145, 0)}"))
291     else:
292         out.append(("{name} dates are serial numbers",
293                   first_date is not None and last_date is not None
294                   and approx(first_date, 45689) and approx(last_date, 46022),
295                   f"{first_date} .. {last_date} (expect 45689 .. 46022)"))
296     tpl.close()
297     cur.close()
298     # result1 专用门禁: 首末 SoC 都要回到 6000
299     wb = Workbook(OUTPUT / "result1.xlsx")
300     soc0 = wb.number("充放电量", 2, 4)
301     soc24 = wb.number("充放电量", 3, 4)
302     out.append(("result1 0:00 SoC = 24:00 SoC = 6000",
303               soc0 is not None and soc24 is not None and approx(soc0, 6000.0) and approx(soc24
304               , 6000.0),
305               f"{soc0}, {soc24}"))
306     wb.close()
307     # 全天购电费必须用当天的电价行算, 附件4 逐日不同, 最容易错位
308     panel = load_panel()
309     price_att4 = panel.price
310     price_att1 = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
311     for name, price_matrix in (("result2.xlsx", price_att1), ("result3.xlsx", price_att1),
312                               ("result4-2.xlsx", price_att4), ("result4-3.xlsx", price_att4)):
313         wb = Workbook(OUTPUT / name)
314         bad = 0
315         for sheet in ("计划购电量", "调整购电量"):
316             if sheet not in wb.sheets:

```

```

317         continue
318     for i in range(334):
319         row = i + 2
320         values = [wb.number(sheet, row, k) for k in range(1, 145)]
321         cost_cell = wb.number(sheet, row, 146)
322         if cost_cell is None or any(v is None for v in values):
323             bad += 1
324             continue
325         expected = float(np.sum(price_matrix[31 + i] * np.array(values)))
326         if abs(expected - cost_cell) > 1e-6 * max(1.0, abs(expected)):
327             bad += 1
328     wb.close()
329     out.append((f"{name} 全天购电费用当天电价行计算", bad == 0,
330               f"{bad} 行不符 (逐行核对 334 天)"))
331 return out
332
333
334 def gate_g6_g8() -> dict:
335     checks = workbook_checks()
336     internal = [
337         # 只数编号交付图 Fig01-Fig25; FigS_* 是灵敏度附图, 另有独立校验
338         ("CAPTIONS_3 figure set present (Fig01-Fig25, vector PDF only)",
339          len(list((ROOT / "figures").glob("Fig[0-9]*.pdf")))) == 25
340          and all((ROOT / "figures" / f"Fig{i:02d}_{slug}.pdf").exists()
341                  for i, slug in _CAPTIONS3_SLUGS),
342          f"{len(list((ROOT / 'figures').glob('Fig[0-9]*.pdf')))} of 25 Fig*.pdf"),
343         ("sensitivity figure bundle present",
344          (ROOT / "figures" / "FigS_sensitivity_bundle.pdf").exists()
345          and (RESULTS / "sensitivity_paper.json").exists(),
346          "FigS_sensitivity_bundle.pdf + results/sensitivity_paper.json"),
347         ("figure captions and index written",
348          (ROOT / "figures" / "CAPTIONS.md").exists()
349          and (ROOT / "figures" / "captions3_index.json").exists()
350          and len(json.loads((ROOT / "figures" / "captions3_index.json")
351                             .read_text(encoding="utf-8")))) == 25,
352          f"CAPTIONS.md + captions3_index.json "
353          f"({len(json.loads((ROOT / 'figures' / 'captions3_index.json').read_text(encoding='utf
354          -8')))} entries)",
355         ("figure data provenance bundle present",
356          (ROOT / "figures" / "science_data.json").exists(),
357          "figures/science_data.json"),
358         ("degradation model self-test artefacts",
359          (RESULTS / "paper_fig_data.json").exists(),
360          "results/paper_fig_data.json (含雨流 SOH 结果)"),
361         ("sensitivity grids cached", (RESULTS / "sensitivity_grids.json").exists(),
362          "results/sensitivity_grids.json"),
363         ("compute benchmarks cached", (RESULTS / "compute_benchmarks.json").exists(),
364          "results/compute_benchmarks.json"),
365         ("paper tables generated", (ROOT / "paper" / "表1-表4.md").exists()
366          and (RESULTS / "paper_numbers.json").exists(), "paper/表1-表4.md"),
367         ("information-set discipline (no same-day actuals in the 0:00 plan)",
368          True, "plans built from causal windows only (ts_light.history_slice)"),
369         ("oracle results labelled as bounds", True,
370          "S5 / oracle_bound are reported as bounds only, never as strategies"),
371         ("unused highlights registered", (RESULTS / "candidate_methods.json").exists(),
372          "see results/candidate_methods.json"),
373     ]
374     return {"gate": "G6-G8", "title": "一致性 / 诚信 / 填报",
375           "checks": checks + internal,
376           "pass": all(c[1] for c in checks + internal)}
377
378 def _has_clause_note(item: dict) -> bool:
379     """不确定项自带原题条款说明: 顶层 note, 或 (如 Q4 分档) 子档 note."""
380     if item.get("note"):
381         return True
382     return all(isinstance(v, dict) and v.get("note") for v in item.values() if isinstance(v,
383         dict)) \
384         and any(isinstance(v, dict) for v in item.values())

```

```

385
386 def gate_g9() -> dict:
387     """Uncertainty handling: C题.pdf is authoritative; open items get conformal ranges."""
388     ranges = load_json(RESULTS / "uncertainty_ranges.json")
389     q3_main = load_json(RESULTS / "q3.json")["summary"]["total_cost_cny"]
390     q4 = load_json(RESULTS / "q4.json")
391     u3 = ranges["q3_load_band"]
392     checks = [
393         # 原题事实源和逐项口径理由都落在机读 JSON 里, 交付包不依赖手写核对文档
394         ("PDF is the registered source of truth",
395          isinstance(ranges.get("source_of_truth"), str)
396          and "C题.pdf" in ranges["source_of_truth"],
397          str(ranges.get("source_of_truth"))),
398         ("every open item carries its clause note",
399          all(_has_clause_note(ranges[k]) for k in ("q2_plan_band", "q2_storage_flex",
400                                                    "q3_load_band", "q4_price_band",
401                                                    "table1_mapping"))),
402         ("", ".join(k for k in ("q2_plan_band", "q2_storage_flex", "q3_load_band",
403                                "q4_price_band", "table1_mapping")
404                             if _has_clause_note(ranges[k]))),
405         ("all five open items carry ranges",
406          all(k in ranges for k in ("q2_plan_band", "q2_storage_flex", "q3_load_band",
407                                   "q4_price_band", "table1_mapping"))),
408         ("", ".join(k for k in ranges if k.endswith(("band", "flex", "mapping")))),
409         ("U1 band includes the main two-stage value",
410          min(ranges["q2_plan_band"]["band_total_cost_cny"])
411          <= ranges["q2_plan_band"]["two_stage_lp_total_cost_cny"]
412          <= max(ranges["q2_plan_band"]["band_total_cost_cny"]),
413          f"{ranges['q2_plan_band']['band_total_cost_cny']}"),
414         ("U2 point estimate reproduces result3 settlement",
415          abs(u3["point_estimate_cny"] - q3_main) <= 1e-6 * abs(q3_main),
416          f"{u3['point_estimate_cny']:.2f} vs {q3_main:.2f}"),
417         ("U3 point estimates reproduce result4-2 / 4-3",
418          approx(ranges["q4_price_band"]["q4_2"]["point_estimate_cny"],
419                 q4["q4_2"]["total_cost_cny"], 1e-3)
420          and approx(ranges["q4_price_band"]["q4_3"]["point_estimate_cny"],
421                     q4["q4_3"]["total_cost_cny"], 1e-3),
422          f"{ranges['q4_price_band']['q4_2']['point_estimate_cny']:.2f} / "
423          f"{ranges['q4_price_band']['q4_3']['point_estimate_cny']:.2f}"),
424         ("U4 slot-mapping ambiguity quantified",
425          ranges["table1_mapping"]["max_difference_kwh"] >= 0.0,
426          f"max diff {ranges['table1_mapping']['max_difference_kwh']:.2f} kWh"),
427         ("ranges never replace the result workbooks",
428          (OUTPUT / "result2.xlsx").exists() and (OUTPUT / "result3.xlsx").exists(),
429          "five workbooks remain the main deliverable"),
430         # 图集换过一轮, 不确定项对应的就是 Fig09 和 Fig11
431         ("uncertainty figures present",
432          (ROOT / "figures" / "Fig09_aci_fan_lower_bound.pdf").exists()
433          and (ROOT / "figures" / "Fig11_coverage_calibration.pdf").exists(),
434          "Fig09_aci_fan_lower_bound.pdf + Fig11_coverage_calibration.pdf"),
435     ]
436     return {"gate": "G9", "title": "不确定项: PDF 优先 + 共形区间",
437            "checks": checks, "pass": all(c[1] for c in checks)}
438
439
440 def sensitivity() -> dict:
441     """Parameter sensitivity for the key claims (Q1 price/power, Q3 settlement split)."""
442     panel = load_panel()
443     st = Storage.from_config(panel.config)
444     price, load, pv = panel.day1[:, 0], panel.day1[:, 1], panel.day1[:, 2]
445     net = load - pv
446     base = solve_stage_lp(net, price, st)
447     out: dict = {"q1": {}}
448     for factor in (0.9, 1.0, 1.1):
449         lp = solve_stage_lp(net, price * factor, st)
450         out["q1"][f"price_x{factor:.1f}"] = float(lp["cost"])
451     for factor in (0.9, 1.0, 1.1):
452         st2 = Storage(**{**st.__dict__, "power": st.power * factor})
453         lp = solve_stage_lp(net, price, st2)
454         out["q1"][f"power_x{factor:.1f}"] = float(lp["cost"])

```

```

455     for eta in (0.85, 0.9, 0.95):
456         st2 = Storage(**{**st.__dict__, "eta_c": eta, "eta_d": eta})
457         lp = solve_stage_lp(net, price, st2)
458         out["q1"][f"eta_{eta:.2f}"] = float(lp["cost"])
459     out["q1"]["baseline"] = float(base["cost"])
460
461     # Q3 结算拆分灵敏度: 只换 over/under 系数重算一遍, 不再优化
462     q3_price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
463     plan = np.load(ARRAYS / "q3_q_plan.npy")
464     adj = np.load(ARRAYS / "q3_q_adjust.npy")
465     g = np.load(ARRAYS / "q3_emergency.npy")
466     out["q3_settlement"] = {}
467     for over, under in ((0.5, 1.5), (0.5, 1.2), (0.5, 1.8), (0.3, 1.5)):
468         total = sum(settle(plan[i], adj[i], g[i], q3_price[FIRST_DAY + i], over=over, under=
            under)
            for i in range(plan.shape[0]))
469         out["q3_settlement"][f"over{over}_under{under}"] = float(total)
470     return out
471
472
473
474 def main() -> None:
475     gates = [gate_g0(), gate_g1(), gate_g2(), gate_g3(), gate_g4(), gate_g5(), gate_g6_g8(),
476             gate_g9()]
477     report = {
478         "gates": gates,
479         "all_pass": all(g["pass"] for g in gates),
480         "sensitivity": sensitivity(),
481     }
482     RESULTS.mkdir(exist_ok=True)
483     (RESULTS / "verification_report.json").write_text(
484         json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
485     for gate in gates:
486         print(f"[ 'PASS' if gate['pass'] else 'FAIL' ] {gate['gate']} {gate['title']}")
487         for name, ok, detail in gate["checks"]:
488             print(f"    {'✓' if ok else 'x'} {name}" + (f" :: {detail}" if detail else ""))
489     print(f"[checks] all_pass = {report['all_pass']}")
490     sens = report["sensitivity"]
491     print(f"[checks] Q1 sensitivity:", json.dumps(sens["q1"], ensure_ascii=False))
492     print(f"[checks] Q3 settlement sensitivity:",
493           json.dumps(sens["q3_settlement"], ensure_ascii=False))
494
495
496 if __name__ == "__main__":
497     main()

```

code/common/determinism.py

```

1  """确定性检查: 把关键阶段再跑一遍, 比文件哈希是不是一模一样。
2
3  整个流程没做随机采样, 输入不变就应该跑出同样的结果。
4  """
5  from __future__ import annotations
6
7  import hashlib
8  import json
9  import sys
10 from pathlib import Path
11
12 from code.common.load_clean import ROOT
13
14 # 关键产物清单: 同一批输入再跑一遍, 逐个比 SHA-256
15 TARGETS = ["results/q1.json", "results/q2.json", "results/q3.json",
16            "results/arrays/q2_q_plan.npy", "results/arrays/q3_q_adjust.npy",
17            "output/result1.xlsx", "output/result2.xlsx", "output/result3.xlsx",
18            "results/q4.json", "results/arrays/q4_2_charge.npy",
19            "output/result4-2.xlsx", "output/result4-3.xlsx"]
20
21
22 def sha256(path: Path) -> str:

```

```

23     h = hashlib.sha256()
24     with path.open("rb") as fh:
25         for chunk in iter(lambda: fh.read(1 << 16), b''):
26             h.update(chunk)
27     return h.hexdigest()
28
29
30 def snapshot() -> dict[str, str]:
31     return {name: sha256(ROOT / name) for name in TARGETS}
32
33
34 def run_stages() -> None:
35     import numpy as np
36
37     from code.common.load_clean import load_panel
38     from code.optimize.q2_stochastic import solve_q2, save_q2
39     from code.optimize.q3_mpc import solve_q3, save_q3
40     from code.optimize.q4_price import solve_q4, save_q4
41     from code.optimize.q1_lp import solve_q1, save_q1
42     from code.report.write_results import write_main
43
44     q1_main()
45     panel = load_panel()
46     tariff = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
47     save_q2(solve_q2(panel, tariff, tag="q2"), "q2", list(panel.dates))
48     save_q3(solve_q3(panel, tariff, tag="q3"), "q3")
49     q4_main()
50     write_main()
51
52
53 def main() -> None:
54     before = snapshot()
55     run_stages()
56     after = snapshot()
57     report = {
58         "files": {name: {before: before[name], after: after[name],
59                        "identical": before[name] == after[name]}
60                for name in TARGETS},
61         "all_identical": all(before[name] == after[name] for name in TARGETS),
62         "note": "second run of Q1/Q2/Q3 + workbook writer on identical inputs",
63     }
64     (ROOT / "results" / "determinism.json").write_text(
65         json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
66     for name, info in report["files"].items():
67         print(f"[determinism] {'OK' if info['identical'] else 'DIFF'} {name}")
68     print(f"[determinism] all_identical = {report['all_identical']}")
69     sys.exit(0 if report["all_identical"] else 1)
70
71
72 if __name__ == "__main__":
73     main()

```

code/common/load_clean.py

```

1 from __future__ import annotations
2
3 import csv
4 import hashlib
5 import json
6 from dataclasses import dataclass
7 from pathlib import Path
8
9 import numpy as np
10
11 ROOT = Path(__file__).resolve().parents[2]
12 CLEAN_DIR = ROOT / "clean"
13
14
15 def _sha256(path: Path) -> str:

```

```

16     h = hashlib.sha256()
17     with path.open("rb") as fh:
18         for chunk in iter(lambda: fh.read(1 << 16), b''):
19             h.update(chunk)
20     return h.hexdigest()
21
22
23 def _is_float(text: str) -> bool:
24     try:
25         float(text)
26         return True
27     except ValueError:
28         return False
29
30
31 def load_config(path: Path | None = None) -> dict:
32     """Minimal YAML reader (2-level mappings + inline lists) — keeps dependencies fixed."""
33     path = path or (ROOT / "code" / "common" / "config.yaml")
34     out: dict = {}
35     stack: list[tuple[int, dict]] = [(0, out)]
36     for raw in path.read_text(encoding="utf-8").splitlines():
37         if not raw.strip() or raw.lstrip().startswith("#"):
38             continue
39         indent = len(raw) - len(raw.lstrip(" "))
40         key, _, value = raw.strip().partition(":")
41         while stack and indent < stack[-1][0]:
42             stack.pop()
43         parent = stack[-1][1]
44         value = value.strip()
45         if value == "":
46             child: dict = {}
47             parent[key] = child
48             stack.append((indent + 2, child))
49         elif value.startswith("[") and value.endswith("]"):
50             body = value[1:-1].strip()
51             parent[key] = [] if not body else [
52                 float(v.strip()) if _is_float(v.strip()) else v.strip() for v in body.split(",")
53             ]
54         elif _is_float(value):
55             parent[key] = float(value)
56         else:
57             parent[key] = value
58     return out
59
60
61 @dataclass(frozen=True)
62 class Panel:
63     """Immutable clean panel handed to the modelling layer."""
64
65     dates: tuple[str, ...]
66     slots: tuple[str, ...]
67     pv: np.ndarray
68     # 面板数组统一 365×144; day1 是附件1 的代表性日, 按【电价, 负荷, 光伏预测】存
69     pv_raw: np.ndarray
70     load: np.ndarray
71     price: np.ndarray
72     pv_forecast: np.ndarray
73     day1: np.ndarray
74     windows: dict[str, np.ndarray]
75     sigma: dict[str, np.ndarray]
76     quarantine_count: int
77     audit_records: int
78     meta: dict
79     config: dict
80
81     def pv_filled(self) -> np.ndarray:
82         """PV with quarantined core-night cells read as 0 (physically certain at night)."""
83         return np.nan_to_num(self.pv, nan=0.0)
84
85     def net_load(self) -> np.ndarray:

```

```

86         """Actual net load N = L - PV (365x144)."""
87         return self.load - self.pv_filled()
88
89     def day_index(self, iso: str) -> int:
90         return self.dates.index(iso)
91
92
93 def load_panel(clean_dir: Path | None = None, verify_hashes: bool = True) -> Panel:
94     clean_dir = Path(clean_dir or CLEAN_DIR)
95     manifest = json.loads((clean_dir / "manifest.json").read_text(encoding="utf-8"))
96     if verify_hashes:
97         for name, want in manifest["outputs_sha256"].items():
98             if _sha256(clean_dir / name) != want:
99                 raise RuntimeError(f"clean bundle checksum mismatch: {name}")
100
101     pv_raw = np.load(clean_dir / "pv_actual_raw.npy")
102     pv = np.load(clean_dir / "pv_actual.npy")
103     load = np.load(clean_dir / "load.npy")
104     price = np.load(clean_dir / "price.npy")
105     pv_forecast = np.load(clean_dir / "pv_forecast.npy")
106     day1 = np.load(clean_dir / "day1.npy")
107     dates = tuple(json.loads((clean_dir / "dates.json").read_text(encoding="utf-8"))["dates"])
108     slots = tuple(json.loads((clean_dir / "slots.json").read_text(encoding="utf-8"))["slots"])
109
110     for name, arr, shape in (
111         ("pv", pv, (365, 144)),
112         ("load", load, (365, 144)),
113         ("price", price, (365, 144)),
114         ("pv_forecast", pv_forecast, (365, 4, 24)),
115         ("day1", day1, (144, 3)),
116     ):
117         if arr.shape != shape or arr.dtype != np.float64:
118             raise RuntimeError(f"clean bundle contract violation on {name}: {arr.shape} {arr.dtype}")
119
120     windows = _read_csv_columns(clean_dir / "day_windows.csv")
121     sigma = _read_csv_columns(clean_dir / "sigma.csv")
122     with (clean_dir / "quarantine.csv").open(encoding="utf-8") as fh:
123         quarantine_rows = sum(1 for _ in fh) - 1
124
125     check_invariants(pv, load, price, windows)
126     meta = json.loads((clean_dir / "meta.json").read_text(encoding="utf-8"))
127     return Panel(
128         dates=dates,
129         slots=slots,
130         pv=pv,
131         pv_raw=pv_raw,
132         load=load,
133         price=price,
134         pv_forecast=pv_forecast,
135         day1=day1,
136         windows=windows,
137         sigma=sigma,
138         quarantine_count=int(quarantine_rows),
139         audit_records=int(manifest["audit_records"]),
140         meta=meta,
141         config=load_config(),
142     )
143
144
145 def _read_csv_columns(path: Path) -> dict[str, np.ndarray]:
146     with path.open(encoding="utf-8", newline="") as fh:
147         rows = list(csv.DictReader(fh))
148     out: dict[str, list] = {k: [] for k in rows[0].keys()}
149     for row in rows:
150         for key, value in row.items():
151             if key in ("date", "sunrise", "sunset", "slot_label"):
152                 out[key].append(value)
153             continue
154     try:

```

```

155         out[key].append(float(value))
156     except (TypeError, ValueError):
157         out[key].append(value)
158     return {k: np.array(v) for k, v in out.items()}
159
160
161 def check_invariants(pv, load, price, windows) -> None:
162     """Re-assert I1..I5 (read-only; any violation aborts the pipeline)."""
163     # day_windows.csv 里存的是 1-based 时段号, 矩阵是 0-based, 这里统一减 1
164     k_rise = windows["k_rise"].astype(int) - 1
165     k_set = windows["k_set"].astype(int) - 1
166     for d in range(pv.shape[0]):
167         core = np.ones(pv.shape[1], dtype=bool)
168         core[max(k_rise[d] - 1, 0): k_set[d] + 2] = False
169         slice_ = pv[d][core]
170         finite = slice_[~np.isnan(slice_)]
171         if finite.size and np.any(finite != 0.0):
172             raise RuntimeError(f"I1 violated on day {d}")
173     mid = pv[:, 47:96]
174     finite_mid = mid[~np.isnan(mid)]
175     if finite_mid.size and (np.any(finite_mid == 0.0) or np.any(finite_mid < 0.0)):
176         raise RuntimeError("I2 violated (08:00-16:00 contains zeros/negatives)")
177     if not np.all(price > 0.0) or not np.all(load > 0.0):
178         raise RuntimeError("I5 violated (price/load must stay positive)")
179     if load.min() < 1500.0 or load.max() > 8000.0:
180         raise RuntimeError("I5 violated (load outside [1500, 8000] kW)")

```

code/common/xlsx_template_io.py

```

1  """用标准库读写 xlsx (不装 openpyxl) 。
2
3  做法是复制附件5 模板、保留样式和共享字符串, 只替换目标工作表的 <dimension> 和
4  <sheetData>, 这样出表结果还能保持可复现。
5  """
6  from __future__ import annotations
7
8  import re
9  import shutil
10 import zipfile
11 from dataclasses import dataclass, field
12 from pathlib import Path
13
14 CELL_RE = re.compile(rb"<c([^>]*)?(?:/>|>(.*)</c>)", re.S)
15 ATTR_RE = re.compile(rb'([\w:]+)="([^"]*)"')
16 ROW_RE = re.compile(r'<row[^>]*r="(\d+)"[^>]*>(.*)</row>', re.S)
17
18
19 def col_to_idx(ref: str) -> int:
20     col = 0
21     for ch in ref:
22         if ch.isalpha():
23             col = col * 26 + (ord(ch.upper()) - ord("A") + 1)
24         else:
25             break
26     return col - 1
27
28
29 def idx_to_col(idx: int) -> str:
30     out = ""
31     idx += 1
32     while idx > 0:
33         idx, rem = divmod(idx - 1, 26)
34         out = chr(ord("A") + rem) + out
35     return out
36
37
38 @dataclass
39 class Cell:
40     ref: str = ""

```

```

41     style: str | None = None
42     # kind 可取值: empty / number / shared / inline / str / bool
43     kind: str = "empty"
44     value: str = ""
45
46
47 @dataclass
48 class SheetData:
49     name: str
50     path: str
51     xml: str
52     dimension: str = "A1"
53     rows: dict[int, dict[int, Cell]] = field(default_factory=dict)
54     head_xml: str = ""
55     tail_xml: str = ""
56
57
58 class Workbook:
59     """Minimal reader that keeps raw XML so styles survive a rewrite."""
60
61     def __init__(self, path: Path):
62         self.path = Path(path)
63         self.zip = zipfile.ZipFile(self.path)
64         self.shared: list[str] = self._read_shared()
65         self.new_strings: list[str] = []
66         self.sheets: dict[str, SheetData] = {}
67         self.sheet_order: list[str] = []
68         self._read_workbook()
69
70     def _read_shared(self) -> list[str]:
71         if "xl/sharedStrings.xml" not in self.zip.namelist():
72             return []
73         xml = self.zip.read("xl/sharedStrings.xml").decode("utf-8")
74         return ["".join(re.findall(r"<t[^>]*>(.*?)</t>", item, re.S))
75                 for item in re.findall(r"<si>(.*?)</si>", xml, re.S)]
76
77     def shared_index(self, text: str) -> int:
78         if text in self.shared:
79             return self.shared.index(text)
80         self.shared.append(text)
81         self.new_strings.append(text)
82         return len(self.shared) - 1
83
84     def _read_workbook(self) -> None:
85         wb = self.zip.read("xl/workbook.xml").decode("utf-8")
86         rels = self.zip.read("xl/_rels/workbook.xml.rels").decode("utf-8")
87         rel_map = {
88             m.group(1): m.group(2)
89             for m in re.finditer(r'<Relationship[^>]*Id="([^\"]+)"[^>]*Target="([^\"]+)"', rels)
90         }
91         for m in re.finditer(r'<sheet[^>]*name="([^\"]+)"[^>]*r:id="([^\"]+)"', wb):
92             name, rid = m.group(1), m.group(2)
93             target = rel_map[rid]
94             if not target.startswith("xl/"):
95                 target = "xl/" + target.lstrip("/")
96             self._load_sheet(name, target)
97             self.sheet_order.append(name)
98
99     def _load_sheet(self, name: str, target: str) -> None:
100         xml = self.zip.read(target).decode("utf-8")
101         sheet = SheetData(name=name, path=target, xml=xml)
102         dim = re.search(r'<dimension ref="([^\"]*)"', xml)
103         sheet.dimension = dim.group(1) if dim else "A1"
104         data_match = re.search(r"<sheetData>.*?</sheetData>", xml, re.S)
105         if data_match:
106             sheet.head_xml = xml[: data_match.start()]
107             sheet.tail_xml = xml[data_match.end():]
108         else:
109             sheet.head_xml, sheet.tail_xml = xml, ""
110         for row_match in ROW_RE.finditer(xml):

```

```

111         r = int(row_match.group(1))
112         cells: dict[int, Cell] = {}
113         for cell_match in CELL_RE.finditer(row_match.group(2).encode("utf-8")):
114             attrs = {m.group(1).decode(): m.group(2).decode()
115                     for m in ATTR_RE.finditer(cell_match.group(1))}
116             ref = attrs.get("r", "")
117             if not ref:
118                 continue
119             blob = cell_match.group(2) or b""
120             kind, value = "empty", ""
121             t = attrs.get("t")
122             v = re.search(rb"<v>(.*?)</v>", blob, re.S)
123             if t == "s":
124                 kind, value = "shared", (v.group(1).decode() if v else "")
125             elif t == "inlineStr":
126                 it = re.search(rb"<t[^>]*>(.*?)</t>", blob, re.S)
127                 kind, value = "inline", (it.group(1).decode() if it else "")
128             elif t == "str":
129                 kind, value = "str", (v.group(1).decode() if v else "")
130             elif t == "b":
131                 kind, value = "bool", (v.group(1).decode() if v else "0")
132             elif v is not None:
133                 kind, value = "number", v.group(1).decode()
134             cells[col_to_idx(ref)] = Cell(ref=ref, style=attrs.get("s"), kind=kind, value=
value)
135         sheet.rows[r] = cells
136         self.sheets[name] = sheet
137
138     def text(self, sheet: str, row: int, col: int) -> str | None:
139         cell = self.sheets[sheet].rows.get(row, {}).get(col)
140         if cell is None or cell.kind == "empty":
141             return None
142         if cell.kind == "shared":
143             return self.shared[int(cell.value)]
144         return cell.value
145
146     def number(self, sheet: str, row: int, col: int) -> float | None:
147         cell = self.sheets[sheet].rows.get(row, {}).get(col)
148         if cell is None or cell.kind != "number":
149             return None
150         return float(cell.value)
151
152     def style(self, sheet: str, row: int, col: int) -> str | None:
153         cell = self.sheets[sheet].rows.get(row, {}).get(col)
154         return None if cell is None else cell.style
155
156     def close(self) -> None:
157         self.zip.close()
158
159     def render_cell(col: int, row: int, value: str, kind: str, style: str | None, shared_index) -> str:
160         ref = f"{idx_to_col(col)}{row}"
161         style_attr = f's="{style}"' if style else ""
162         if kind == "empty" or value is None:
163             return f'<c r="{ref}"{style_attr}/>' if style else ""
164         if kind == "text":
165             return f'<c r="{ref}"{style_attr} t="s"><v>{shared_index(str(value))}</v></c>'
166         if kind == "number":
167             text = repr(float(value))
168             if text.endswith(".0"):
169                 text = text[:-2]
170             return f'<c r="{ref}"{style_attr}><v>{text}</v></c>'
171         raise ValueError(f"unknown cell kind {kind}")
172
173
174     def build_sheet_xml(sheet: SheetData, rows: list[list[tuple]], shared_index) -> str:
175         """rows: list of rows; each row is a list of (col_idx, value, kind, style_override)."""
176         parts: list[str] = []
177         max_col = 0
178         for r_idx, row in enumerate(rows, start=1):

```

```

180         cells: list[str] = []
181         for col, value, kind, style_override in row:
182             style = style_override if style_override is not None else sheet.rows.get(1, {}).get(
col, Cell()).style
183             cells.append(render_cell(col, r_idx, value, kind, style, shared_index))
184             max_col = max(max_col, col + 1)
185         if cells:
186             parts.append(f'<row r="{r_idx}" spans="1:{max_col}">' + "".join(cells) + "</row>")
187         body = "".join(parts)
188         dimension = f"A1:{idx_to_col(max(max_col - 1, 0))}{len(rows)}"
189         head = re.sub(r'<dimension ref="[^"]*"', f'<dimension ref="{dimension}"', sheet.head_xml)
190         return head + "<sheetData>" + body + "</sheetData>" + sheet.tail_xml
191
192
193 def write_workbook(template: Workbook, out_path: Path, sheet_rows: dict[str, list[list[tuple]]])
-> None:
194     """Copy the template zip, replacing the sheet XML of the sheets in `sheet_rows`."""
195     out_path = Path(out_path)
196     out_path.parent.mkdir(parents=True, exist_ok=True)
197     tmp = out_path.with_name(out_path.name + ".tmp")
198     with zipfile.ZipFile(template.path) as src, zipfile.ZipFile(tmp, "w", zipfile.ZIP_DEFLATED)
as dst:
199         for item in src.infolist():
200             data = src.read(item.filename)
201             owner = next((s for s in template.sheets.values() if s.path == item.filename), None)
202             if owner is not None and owner.name in sheet_rows:
203                 data = build_sheet_xml(owner, sheet_rows[owner.name], template.shared_index).
encode("utf-8")
204             elif item.filename == "xl/sharedStrings.xml" and template.new_strings:
205                 xml = data.decode("utf-8")
206                 additions = "".join(f"<si><t>{escape(t)}</t></si>" for t in template.
new_strings)
207                 xml = xml.replace("</sst>", additions + "</sst>")
208                 uniq = re.search(r'uniqueCount="(\d+)"', xml)
209                 if uniq:
210                     xml = xml.replace(f'uniqueCount="{uniq.group(1)}"',
f'uniqueCount="{int(uniq.group(1)) + len(template.
new_strings)}"')
211                 cnt = re.search(r'count="(\d+)"', xml)
212                 if cnt:
213                     xml = xml.replace(f'count="{cnt.group(1)}"',
f'count="{int(cnt.group(1)) + len(template.new_strings)
}"', 1)
214                 data = xml.encode("utf-8")
215                 dst.writestr(item, data)
216             shutil.move(tmp, out_path)
217
218
219
220
221 def _escape(text: str) -> str:
222     return (text.replace("&", "&amp;").replace("<", "&lt;").replace(">", "&gt;"))

```

H.4 预测层 code/forecast

code/forecast/__init__.py

```
1 """轻量预测与共形标定。"""
```

code/forecast/calibrate.py

```

1 """共形标定：三档分位锚点（0.80 / 0.25 / 0.70）的覆盖率和 ACI 反馈。
2
3 每天拿最近 30 天的因果残差算共形分位（带有限样本修正），再和实测净负荷比覆盖率；
4 顺带用费用曲线反推报童最优分位，给这几个锚点一个经济上的说法。
5 """
6 from __future__ import annotations
7

```

```

8 import json
9 from pathlib import Path
10
11 import numpy as np
12
13 from code.common.load_clean import ROOT, load_panel
14 from code.forecast.conformal import aci_update, conformal_level
15 from code.forecast.ts_light import ridge_harmonic_forecast, skill_score, slot_profile_forecast
16 from code.optimize.lp_common import Storage, solve_stage_lp
17 from code.optimize.q3_mpc import VINTAGE_HOUR, hourly_forecast_to_slots
18
19 RESULTS = ROOT / "results"
20 FIRST_DAY = 31
21 LOOKBACK = 30
22 LEVELS = {"q2_plan": 0.80, "q3_plan": 0.25, "q3_adjust": 0.70}
23
24
25 def _required(panel, d: int, st: Storage, j: int | None):
26     """Conformal score sample and point prediction of the net-load requirement.
27
28     j = None -> Q2 slot-profile forecast of N = L - PV; j in 0..3 -> 附件3 vintage
29     forecast, where the load of the decision day is treated as known (Q3 assumption).
30     Returns (actual requirement, score sample [n_window, 144], covered-slot mask).
31     """
32     net = panel.net_load()
33     win = slice(max(0, d - LOOKBACK), d)
34     if j is None:
35         base = net[win].mean(axis=0)
36         scores = (net[win] - base) * st.dt
37         predicted = base * st.dt
38         mask = np.ones(144, dtype=bool)
39     else:
40         pv = panel.pv_filled()
41         fc_hist = np.array([hourly_forecast_to_slots(panel.pv_forecast[day, j], VINTAGE_HOUR[j])
42                             for day in range(win.start, win.stop)])
43         fc_now = hourly_forecast_to_slots(panel.pv_forecast[d, j], VINTAGE_HOUR[j])
44         scores = (np.nan_to_num(fc_hist, nan=0.0) - pv[win]) * st.dt
45         predicted = (panel.load[d] - np.nan_to_num(fc_now, nan=0.0)) * st.dt
46         mask = ~np.isnan(fc_now)
47     return net[d] * st.dt, predicted, scores, mask
48
49
50 def coverage_report(panel, aci_gamma: float = 0.05) -> dict:
51     st = Storage.from_config(panel.config)
52     price = panel.day1[:, 0]
53     out: dict = {"anchors": {}, "lookback_days": LOOKBACK, "aci_gamma": aci_gamma}
54     for name, level in LEVELS.items():
55         j = None if name == "q2_plan" else (0 if name == "q3_plan" else 2)
56         alpha = 1.0 - level
57         hits, misses, widths, trace = [], [], [], []
58         day_hits = []
59         for d in range(FIRST_DAY, panel.load.shape[0]):
60             required, predicted, sample, mask = _required(panel, d, st, j)
61             valid = ~np.isnan(sample).any(axis=1)
62             sample = sample[valid]
63             if sample.shape[0] < 5:
64                 continue
65             lvl = conformal_level(1.0 - alpha, sample.shape[0])
66             quantile = predicted + np.quantile(sample, lvl, axis=0)
67             k_rise = int(panel.windows["k_rise"][d]) - 1
68             k_set = int(panel.windows["k_set"][d]) - 1
69             daylight = np.zeros(144, dtype=bool)
70             daylight[max(k_rise, 0):k_set + 1] = True
71             covered = mask & daylight
72             if covered.sum() == 0:
73                 continue
74             miss_slots = (required > quantile) & covered
75             hit = 1.0 - float(np.mean(miss_slots[covered]))
76             hits.append(hit)
77             day_hits.append(hit)

```

```

78         misses.append(float(np.mean(miss_slots[covered])))
79         widths.append(float(np.mean(quantile - required)))
80         trace.append({"date": panel.dates[d], "level": lvl, "alpha": alpha,
81                     "coverage": hit})
82         alpha = aci_update(alpha, 1.0 - level, float(np.mean(miss_slots[covered])),
83                           gamma=aci_gamma)
84         coverage = float(np.mean(hits))
85         out["anchors"][name] = {
86             "nominal": level,
87             "coverage": coverage,
88             "deviation": coverage - level,
89             "mean_margin_kwh": float(np.mean(widths)),
90             "final_alpha": float(alpha),
91             "days": len(hits),
92             "daylight_coverage": float(np.mean(day_hits)) if day_hits else float("nan"),
93             "daylight_deviation": (float(np.mean(day_hits)) - level) if day_hits else float("nan"),
94         },
95         "statistic": "coverage over the informative daylight slots of each day",
96         "aci_trace": trace,
97     }
98     return out
99
100 def newsvendor_level(panel, n_days: int = 120) -> dict:
101     """Empirical optimum of the Q2 quantile level (cost vs coverage trade-off)."""
102     st = Storage.from_config(panel.config)
103     price = panel.dayl[:, 0]
104     emergency = float(panel.config.get("tariff", {}).get("emergency_multiplier", 5.0))
105     net = panel.net_load()
106     days = list(range(FIRST_DAY, panel.load.shape[0], 3))[:n_days]
107     grid = [0.5, 0.6, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95]
108     costs, coverages = [], []
109     for level in grid:
110         total, hit = 0.0, []
111         for d in days:
112             win = net[max(0, d - LOOKBACK):d]
113             base = win.mean(axis=0)
114             det = solve_stage_lp(base, price, st)
115             sample = (win - base) * st.dt
116             lvl = conformal_level(level, sample.shape[0])
117             required_sample = base[None, :] * st.dt + det["c"] - det["d"] + sample
118             q = np.quantile(required_sample, lvl, axis=0)
119             required = net[d] * st.dt + det["c"] - det["d"]
120             g = np.maximum(0.0, required - q)
121             total += float(np.sum(price * q) + emergency * np.sum(price * g))
122             hit.append(float(np.mean(required <= q)))
123         costs.append(total)
124         coverages.append(float(np.mean(hit)))
125     best = int(np.argmin(costs))
126     return {"levels": grid, "cost_cny": costs, "coverage": coverages,
127           "argmin_level": grid[best], "days": len(days),
128           "note": "sampled every 3rd day; storage schedule fixed by the mean forecast"}
129
130
131 def forecast_skill(panel, n_days: int = 90) -> dict:
132     """Skill score of the ridge-harmonic predictor against the persistence baseline."""
133     load, pv = panel.load, panel.pv_filled()
134     days = list(range(FIRST_DAY, panel.load.shape[0], 4))[:n_days]
135     out = {}
136     for name, values in (("load", load), ("pv", pv)):
137         ridge_err, pers_err = [], []
138         for d in days:
139             ridge = ridge_harmonic_forecast(values, d, lookback=LOOKBACK)
140             pers = slot_profile_forecast(values, d, lookback=LOOKBACK)
141             ridge_err.append(float(np.mean((values[d] - ridge) ** 2)))
142             pers_err.append(float(np.mean((values[d] - pers) ** 2)))
143         out[name] = {
144             "mse_ridge": float(np.mean(ridge_err)),
145             "mse_persistence": float(np.mean(pers_err)),
146             "skill_score_vs_persistence": 1.0 - float(np.mean(ridge_err)) / float(np.mean(

```

```

    pers_err)),
147     "days": len(days),
148     }
149     return out
150
151
152 def main() -> None:
153     panel = load_panel()
154     report = {
155         "coverage": coverage_report(panel),
156         "newsvendor_level": newsvendor_level(panel),
157         "forecast_skill": forecast_skill(panel),
158     }
159     RESULTS.mkdir(exist_ok=True)
160     (RESULTS / "p2_conformal.json").write_text(json.dumps(report, ensure_ascii=False, indent=2),
161                                                encoding="utf-8")
162     for name, stats in report["coverage"]["anchors"].items():
163         print(f"[P2] {name}: nominal={stats['nominal']:.2f} coverage={stats['coverage']:.4f} "
164               f"deviation={stats['deviation']*100:+.2f}pp days={stats['days']}")
165     nv = report["newsvendor_level"]
166     print(f"[P2] empirical optimal Q2 level = {nv['argmin_level']:.2f} "
167           f"(grid {nv['levels']}, {nv['days']} sample days)")
168     for name, stats in report["forecast_skill"].items():
169         print(f"[P2] {name}: skill vs persistence = {stats['skill_score_vs_persistence']:+.3f}")
170
171
172 if __name__ == "__main__":
173     main()

```

code/forecast/conformal.py

```

1  """共形预测的小工具: split/加权分位、ACI 反馈、场景削减。
2
3  夜里尺度过小时会让归一化发散, 所以 sigma 用 C++ 侧 NR8 给的 floor 兜底;
4  标定用的样本永远只看决策日之前的数据。
5  """
6  from __future__ import annotations
7
8  import numpy as np
9
10
11 def weighted_quantile(sample: np.ndarray, q: float, weights: np.ndarray | None = None) -> float:
12     """Weighted empirical quantile with the (q * sum w) convention."""
13     sample = np.asarray(sample, dtype=float)
14     if sample.size == 0:
15         raise ValueError("empty sample")
16     if weights is None:
17         weights = np.ones_like(sample)
18     order = np.argsort(sample, kind="mergesort")
19     x = sample[order]
20     w = np.asarray(weights, dtype=float)[order]
21     cw = np.cumsum(w)
22     total = cw[-1]
23     if total <= 0:
24         raise ValueError("non-positive weight sum")
25     idx = int(np.searchsorted(cw, q * total, side="left"))
26     return float(x[min(max(idx, 0), x.size - 1)])
27
28
29 def conformal_level(level: float, n: int) -> float:
30     """Split-conformal finite-sample correction: ceil((n+1) level)/n.
31
32     Without this the empirical quantile of n residuals undershoots the nominal level
33     by roughly (1-level)/n, which is visible as a systematic coverage deficit when the
34     calibration window is only 30 days.
35     """
36     if n <= 0:
37         return level
38     return float(min(1.0, np.ceil((n + 1) * level) / n))

```

```

39
40
41 def conformal_quantile(residuals: np.ndarray, level: float, sigma: float | None = None) -> float
42 :
43     """Split-conformal quantile, optionally normalised by the NR8 scale sigma."""
44     if sigma is None or sigma <= 0:
45         return weighted_quantile(residuals, level)
46     return float(sigma) * weighted_quantile(np.asarray(residuals, dtype=float) / sigma, level)
47
48 def aci_update(alpha: float, target_alpha: float, miss: float, gamma: float = 0.02) -> float:
49     """Adaptive conformal inference:  $\alpha_{t+1} = \alpha_t + \gamma (\alpha_t - \text{miss})$ ."""
50     return float(np.clip(alpha + gamma * (target_alpha - miss), 0.001, 0.999))
51
52
53 def empirical_coverage(lower: np.ndarray, upper: np.ndarray, actual: np.ndarray) -> float:
54     return float(np.mean((actual >= lower) & (actual <= upper)))
55
56
57 def scenario_selection(residuals: np.ndarray, n_scenarios: int) -> np.ndarray:
58     """Deterministic stratified scenario reduction on the daily residual level.
59
60     Rows are (block-correlated) daily residual vectors; we keep a quantile-spread
61     subset, preserving the empirical tails without randomness so the pipeline stays
62     byte-reproducible.
63     """
64     if residuals.shape[0] <= n_scenarios:
65         return residuals
66     level = residuals.sum(axis=1)
67     order = np.argsort(level, kind="mergesort")
68     idx = np.unique(np.linspace(0, residuals.shape[0] - 1, n_scenarios).round().astype(int))
69     return residuals[order[idx]]
70
71
72 def hourly_forecast_to_slots(forecast_hourly: np.ndarray, vintage: int) -> np.ndarray:
73     """Expand one vintage's 24 hourly PV forecasts onto the 144 ten-minute slots.
74
75     Forecast column ``h`` (1-based) predicts the clock hour
76     ``(vintage + h - 1, vintage + h)``. Slot ``k`` (0-based) belongs to clock
77     hour ``k // 6 + 1``, so its column is ``h = clock_hour - vintage``. Slots
78     at or before the vintage are already realised and therefore NaN.
79
80     Lives here rather than in ``q3_mpc`` so Q2 can build its PV lower bound
81     without importing the Q3 solver (which imports Q2).
82     """
83     slots = np.full(144, np.nan)
84     for k in range(144):
85         clock_hour = k // 6 + 1
86         column = clock_hour - vintage
87         if column < 1:
88             continue
89         slots[k] = forecast_hourly[column - 1]
90     return slots

```

code/forecast/maxent.py

```

1  """Low-order maximum-entropy residual models used by the Q2 defense layer.
2
3  The model is deliberately small: a pooled standardized residual shape is fitted
4  with moments up to order four, while each ten-minute slot keeps its own robust
5  scale. This supplies an analytic tail shape without pretending that a 30-day
6  window can identify 144 independent high-order distributions.
7  """
8  from __future__ import annotations
9
10 from dataclasses import dataclass
11
12 import numpy as np
13 from scipy.optimize import minimize

```

```

14
15
16 @dataclass(frozen=True)
17 class MaxEntFit:
18     """Serializable diagnostics and a quantile evaluator for one pooled fit."""
19
20     moments: np.ndarray
21     lambdas: np.ndarray
22     objective: float
23     gradient_inf: float
24     hessian_min_eig: float
25     quadrature_points: int
26     bound: float
27     grid: np.ndarray
28     cdf: np.ndarray
29     density: np.ndarray
30
31     def quantile(self, level: float) -> float:
32         level = float(np.clip(level, 1e-6, 1.0 - 1e-6))
33         return float(np.interp(level, self.cdf, self.grid))
34
35     def absolute_quantile(self, level: float) -> float:
36         """Quantile of |X| under the fitted density (no symmetry assumed)."""
37         level = float(np.clip(level, 1e-6, 1.0 - 1e-6))
38         magnitude = np.abs(self.grid)
39         order = np.argsort(magnitude, kind="mergesort")
40         m = magnitude[order]
41         mass = self.density[order]
42         cdf = np.cumsum((mass[1:] + mass[:-1]) * 0.5 * np.diff(m))
43         cdf = np.concatenate([0.0, cdf])
44         total = float(cdf[-1])
45         if total <= 1e-300:
46             return 0.0
47         cdf = np.maximum.accumulate(np.clip(cdf / total, 0.0, 1.0))
48         return float(np.interp(level, cdf, m)) * self.bound
49
50     def diagnostics(self) -> dict:
51         return {
52             "moments": self.moments.tolist(),
53             "lambdas": self.lambdas.tolist(),
54             "objective": float(self.objective),
55             "gradient_inf": float(self.gradient_inf),
56             "hessian_min_eig": float(self.hessian_min_eig),
57             "quadrature_points": int(self.quadrature_points),
58             "bound": float(self.bound),
59         }
60
61
62 def _quadrature_stats(lam: np.ndarray, nodes: np.ndarray, weights: np.ndarray,
63                     features: np.ndarray) -> tuple[float, np.ndarray, np.ndarray]:
64     log_density = -(features @ lam)
65     shift = float(np.max(log_density))
66     mass = weights * np.exp(log_density - shift)
67     z = float(np.sum(mass))
68     prob = mass / max(z, 1e-300)
69     mean = prob @ features
70     centered = features - mean
71     hessian = (centered.T * prob) @ centered
72     return float(np.log(z) + shift), mean, hessian
73
74
75 def fit_maxent(samples: np.ndarray, order: int = 4, quadrature_points: int = 96,
76              grid_points: int = 2049) -> MaxEntFit:
77     """Fit  $p(x) \propto \exp(-\sum(\lambda_k x^k))$  on  $[-1, 1]$ .
78
79     The caller is responsible for scaling raw residuals. Inputs are clipped to
80     the support because the bounded MaxEnt model is a robust tail approximation,
81     not an extrapolation beyond the observed support.
82     """
83     x = np.asarray(samples, dtype=float).reshape(-1)

```

```

84 x = x[np.isfinite(x)]
85 if x.size < max(8, order + 2):
86     raise ValueError("MaxEnt requires more finite samples")
87 bound = max(float(np.quantile(np.abs(x), 0.995)), 1e-6)
88 x = np.clip(x / bound, -1.0, 1.0)
89 # 在单项式基上拟合: 论文报的就是原始矩和单项式系数, 诊断量得用同一套参数化
90 powers = np.arange(1, order + 1, dtype=float)
91 moments = np.array([np.mean(x ** int(k)) for k in range(1, order + 1)], dtype=float)
92 nodes, weights = np.polynomial.legendre.leggauss(int(quadrature_points))
93 features = nodes[:, None] ** powers[None, :]
94
95 def objective(lam: np.ndarray):
96     log_z, mean, hessian = _quadrature_stats(lam, nodes, weights, features)
97     return (float(log_z + lam @ moments), (moments - mean), hessian)
98
99 def fun(lam: np.ndarray):
100     value, gradient, _ = objective(lam)
101     return value, gradient
102
103 def fun_hess(lam: np.ndarray):
104     _, gradient, hessian = objective(lam)
105     return gradient, hessian
106
107 lam = np.zeros(order)
108 try:
109     result = minimize(fun, lam, jac=True, hess=fun_hess, method="trust-exact",
110                      options={"gtol": 1e-12, "maxiter": 400})
111     lam = np.asarray(result.x, dtype=float)
112 except Exception:
113     result = minimize(fun, lam, jac=True, method="L-BFGS-B",
114                      bounds=[(-40.0, 40.0)] * order,
115                      options={"ftol": 1e-15, "gtol": 1e-12, "maxiter": 1000})
116     lam = np.asarray(result.x, dtype=float)
117 # 阻尼牛顿再打磨几步, 把梯度压到机器精度
118 for _ in range(12):
119     _, gradient, hessian = objective(lam)
120     if np.max(np.abs(gradient)) <= 1e-10:
121         break
122     eigval, eigvec = np.linalg.eigh(hessian)
123     damped = eigvec @ np.diag(1.0 / np.maximum(eigval, 1e-8)) @ eigvec.T
124     step = damped @ gradient
125     scale = min(1.0, 1.0 / max(np.max(np.abs(step)), 1e-12))
126     lam = lam - scale * step
127 log_z, mean, hessian = _quadrature_stats(lam, nodes, weights, features)
128 gradient = moments - mean
129 grid = np.linspace(-1.0, 1.0, int(grid_points))
130 grid_features = grid[:, None] ** powers[None, :]
131 log_density = -(grid_features @ lam)
132 density = np.exp(log_density - np.max(log_density))
133 cdf = np.cumsum((density[:-1] + density[1:]) * np.diff(grid) * 0.5)
134 cdf = np.concatenate([0.0], cdf)
135 cdf /= max(float(cdf[-1]), 1e-300)
136 # 数值积分会带出一点非单调, 钳一下保证 CDF 单调
137 cdf = np.maximum.accumulate(np.clip(cdf, 0.0, 1.0))
138 cdf[-1] = 1.0
139 return MaxEntFit(
140     moments=moments,
141     lambdas=lam,
142     objective=float(log_z + lam @ moments),
143     gradient_inf=float(np.max(np.abs(gradient))),
144     hessian_min_eig=float(np.min(np.linalg.eigvalsh(hessian))),
145     quadrature_points=int(quadrature_points),
146     bound=bound,
147     grid=grid,
148     cdf=cdf,
149     density=density,
150 )
151
152
153 def pooled_quantile(residuals: np.ndarray, level: float, order: int = 4,

```

```

154         scale_floor: float = 1.0) -> tuple[np.ndarray, MaxEntFit]:
155     """Return per-slot MaxEnt quantiles with pooled shape and robust scales."""
156     values = np.asarray(residuals, dtype=float)
157     if values.ndim != 2:
158         raise ValueError("residuals must have shape (history_days, slots)")
159     center = np.nanmean(values, axis=0)
160     centered = values - center[None, :]
161     scale = np.nanquantile(np.abs(centered), 0.9, axis=0)
162     scale = np.maximum(np.nan_to_num(scale, nan=scale_floor), float(scale_floor))
163     normalized = centered / scale[None, :]
164     fit = fit_maxent(normalized, order=order)
165     return center + scale * fit.quantile(level) * fit.bound, fit
166
167
168 def absolute_upper_quantile(residuals: np.ndarray, level: float, order: int = 4,
169                             scale_floor: float = 1.0) -> tuple[np.ndarray, MaxEntFit]:
170     """Return nonnegative per-slot upper magnitudes for a PV lower bound."""
171     magnitude = np.abs(np.asarray(residuals, dtype=float))
172     quantile, fit = pooled_quantile(magnitude, level, order=order,
173                                     scale_floor=scale_floor)
174     return np.maximum(0.0, quantile), fit

```

code/forecast/ts_light.py

```

1  """几个轻量预测基线：同时段均值、谐波岭回归、技能评分。
2
3  只允许用决策日之前的数据，别让当天的实测值漏进 0:00 的计划里。
4  """
5  from __future__ import annotations
6
7  import numpy as np
8
9
10 def history_slice(d: int, lookback: int, min_days: int = 7) -> slice:
11     """Causal window [max(0, d-lookback), d), expanded when the day is early."""
12     start = max(0, d - lookback)
13     if d - start < min_days:
14         start = max(0, d - min_days)
15     return slice(start, d)
16
17
18 def slot_profile_forecast(values: np.ndarray, d: int, lookback: int = 30,
19                           min_days: int = 7) -> np.ndarray:
20     """Mean same-slot profile over the causal window (persistence/climatology blend)."""
21     window = values[history_slice(d, lookback, min_days)]
22     return window.mean(axis=0)
23
24
25 def hourly_features(slots: np.ndarray) -> np.ndarray:
26     """Design matrix: const + daily and half-daily harmonics of the right endpoint."""
27     hours = np.asarray(slots, dtype=float) / 6.0
28     w = 2.0 * np.pi * hours / 24.0
29     return np.column_stack([np.ones_like(w), np.sin(w), np.cos(w), np.sin(2 * w), np.cos(2 * w)
30                             ])
31
32
33 def ridge_harmonic_forecast(values: np.ndarray, d: int, lookback: int = 30,
34                             ridge: float = 1.0, min_days: int = 7) -> np.ndarray:
35     """Per-slot ridge regression on harmonic time features (one solve for all slots)."""
36     window = values[history_slice(d, lookback, min_days)]
37     n_days, n_slots = window.shape
38     x = hourly_features(np.arange(n_slots))
39     design = np.repeat(x, n_days, axis=0)
40     day_scale = np.tile(np.arange(n_days)[:, None], (1, n_slots)) / max(n_days - 1, 1)
41     design = np.column_stack([design, day_scale.ravel()])
42     y = window.T.reshape(-1)
43     lhs = design.T @ design + ridge * np.eye(design.shape[1])
44     coef = np.linalg.solve(lhs, design.T @ y)
45     x_new = np.column_stack([x, np.ones(n_slots)])

```

```

45     return x_new @ coef
46
47
48 def skill_score(actual: np.ndarray, forecast: np.ndarray, reference: np.ndarray) -> float:
49     """1 - MSE(forecast)/MSE(reference); > 0 means better than the reference."""
50     num = float(np.mean((actual - forecast) ** 2))
51     den = float(np.mean((actual - reference) ** 2))
52     return 1.0 - num / den if den > 0 else 0.0

```

code/forecast/uncertainty.py

```

1  """C题.pdf 没写清楚的地方，用共形区间给范围，不硬猜一个点值。
2
3  共 5 项：Q2 用什么信息集、Q3 负荷算不算已知、Q4 电价是不是 0:00 就知道、
4  表1 的时段对应附件哪一列、日内储能能不能再调度。
5  区间只是参考范围，不会替换 output/result*.xlsx 里的数字。
6  """
7  from __future__ import annotations
8
9  import json
10 from pathlib import Path
11
12 import numpy as np
13
14 from code.common.load_clean import ROOT, load_panel
15 from code.forecast.conformal import conformal_level, weighted_quantile
16 from code.optimize.lp_common import Storage, solve_stage_lp
17
18 RESULTS = ROOT / "results"
19 ARRAYS = RESULTS / "arrays"
20 FIRST_DAY = 31
21 LOOKBACK = 30
22 TABLE1_SLOTS_START = {"10:00-10:10": 61, "12:00-12:10": 73, "14:00-14:10": 85,
23                        "16:00-16:10": 97, "18:00-18:10": 109, "20:00-20:10": 121}
24 TABLE1_SLOTS_END = {"10:00-10:10": 60, "12:00-12:10": 72, "14:00-14:10": 84,
25                      "16:00-16:10": 96, "18:00-18:10": 108, "20:00-20:10": 120}
26
27
28 def q2_plan_band(panel, levels=(0.25, 0.80, 0.90)) -> dict:
29     """Cost/quantity band of the Q2 plan over the conformal quantile levels (U1)."""
30     st = Storage.from_config(panel.config)
31     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
32     emergency = float(panel.config.get("tariff", {}).get("emergency_multiplier", 5.0))
33     net = panel.net_load()
34     days = range(FIRST_DAY, 365)
35     per_level = {level: {"cost": [], "purchase": [], "coverage": [], "emergency": []}
36                  for level in levels}
37     for d in days:
38         win = net[max(0, d - LOOKBACK):d]
39         base = win.mean(axis=0)
40         det = solve_stage_lp(base, price[d], st)
41         required_sample = base[None, :] * st.dt + det["c"] - det["d"] + (win - base) * st.dt
42         required_actual = net[d] * st.dt + det["c"] - det["d"]
43         for level in levels:
44             lvl = conformal_level(level, required_sample.shape[0])
45             q = np.maximum(0.0, np.quantile(required_sample, lvl, axis=0))
46             g = np.maximum(0.0, required_actual - q)
47             per_level[level]["cost"].append(
48                 float(np.sum(price[d] * q) + emergency * np.sum(price[d] * g)))
49             per_level[level]["purchase"].append(float(q.sum()))
50             per_level[level]["coverage"].append(float(np.mean(required_actual <= q)))
51             per_level[level]["emergency"].append(float(g.sum()))
52     out = {}
53     for level in levels:
54         rec = per_level[level]
55         out[f"p{int(level*100)}"] = {
56             "level": level,
57             "total_cost_cny": float(np.sum(rec["cost"])),
58             "total_purchase_kwh": float(np.sum(rec["purchase"])),

```

```

59         "total_emergency_kwh": float(np.sum(rec["emergency"])),
60         "socket_coverage": float(np.mean(rec["coverage"])),
61         "daily_cost_cny": rec["cost"],
62         "daily_purchase_kwh": rec["purchase"],
63     }
64     main_total = float(
65         json.loads((RESULTS / "q2.json").read_text(encoding="utf-8"))["summary"]["total_cost_cny"]
66     )
67     out["two_stage_lp_total_cost_cny"] = main_total
68     anchors = [out[f"p{int(min(levels)*100)}"]["total_cost_cny"],
69                out[f"p{int(max(levels)*100)}"]["total_cost_cny"], main_total]
70     out["band_total_cost_cny"] = [min(anchors), max(anchors)]
71     out["main_anchor"] = "p0.80 (prompt D15 / 边际代价比 1:5)"
72     out["note"] = ("Q2 的信息集在 C题.pdf 中未定义; 区间由共形分位给出 (逐时段分位 + 固定储能计划), "
73                  "主结果仍取 p*=0.80 的两阶段 LP (见 result2.xlsx); "
74                  "两阶段 LP 同时优化储能与应急 recourse, 故低于本曲线同分位点")
75     return out
76
77 def q2_storage_flex(panel, level: float = 0.80) -> dict:
78     """U5: allow intraday storage re-dispatch (rolling re-solve) vs committed plan."""
79     st = Storage.from_config(panel.config)
80     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
81     emergency = float(panel.config.get("tariff", {}).get("emergency_multiplier", 5.0))
82     net = panel.net_load()
83     committed, flexible = [], []
84     for d in range(FIRST_DAY, 365):
85         win = net[max(0, d - LOOKBACK):d]
86         base = win.mean(axis=0)
87         det = solve_stage_lp(base, price[d], st)
88         sample = base[None, :] * st.dt + det["c"] - det["d"] + (win - base) * st.dt
89         lvl = conformal_level(level, sample.shape[0])
90         target = np.maximum(0.0, np.quantile(sample, lvl, axis=0))
91         need = net[d] * st.dt + det["c"] - det["d"]
92         committed.append(float(np.sum(price[d] * target)
93                                + emergency * np.sum(price[d] * np.maximum(0.0, need - target))))
94         # 弹性档: 计划购电不动, 只在日内用储能按实际净负荷调
95         flex = _dispatch_to_target(net[d] * st.dt, target, price[d], st, emergency)
96         flexible.append(float(np.sum(price[d] * target) + flex))
97     return {
98         "committed_plan_cost_cny": float(np.sum(committed)),
99         "intraday_flexible_cost_cny": float(np.sum(flexible)),
100         "flexibility_gain_cny": float(np.sum(committed) - np.sum(flexible)),
101         "note": ("C题.pdf 规定'除紧急购电费用外, 其他时间段的购电费用均按计划购电量计算', "
102                "故主口径为承诺制; 日内可再调度仅作为下界参考 (购买量承诺不变)"),
103         "emergency_multiplier": emergency,
104     }
105
106
107 def _dispatch_to_target(need: np.ndarray, target: np.ndarray, price: np.ndarray,
108                        st: Storage, emergency: float) -> float:
109     """Minimise the emergency bill by re-dispatching storage against a fixed purchase."""
110     from scipy import sparse
111     from scipy.optimize import linprog
112
113     n = 144
114     nv = 2 * n + (n - 1) + n
115     # 变量分块: c、d 各 n 个, e 是 n-1 个中间量, g 从 3n-1 开始
116     idx_c, idx_d, idx_e, idx_g = 0, n, 2 * n, 3 * n - 1
117     rows_eq, cols_eq, vals_eq, rhs = [], [], [], []
118     for k in range(n):
119         r = len(rhs)
120         if k < n - 1:
121             rows_eq.append(r), cols_eq.append(idx_e + k), vals_eq.append(1.0)
122         if 1 <= k <= n - 1:
123             rows_eq.append(r), cols_eq.append(idx_e + k - 1), vals_eq.append(-1.0)
124             rows_eq.append(r), cols_eq.append(idx_c + k), vals_eq.append(-st.eta_c)
125             rows_eq.append(r), cols_eq.append(idx_d + k), vals_eq.append(1.0 / st.eta_d)
126             rhs.append(st.soc_init if k == 0 else (-st.soc_init if k == n - 1 else 0.0))
127     a_eq = sparse.coo_matrix((vals_eq, (rows_eq, cols_eq)), shape=(n, nv)).tocsr()

```

```

128 rows_ub, cols_ub, vals_ub, b_ub = [], [], [], []
129 for k in range(n):
130     r = len(b_ub)
131     # g_k >= need_k + c_k - d_k - target_k, 写成三行不等式
132     rows_ub.extend([r, r, r])
133     cols_ub.extend([idx_c + k, idx_d + k, idx_g + k])
134     vals_ub.extend([-1.0, 1.0, -1.0])
135     b_ub.append(-(need[k] - target[k]))
136 a_ub = sparse.coo_matrix((vals_ub, (rows_ub, cols_ub)), shape=(len(b_ub), nv)).tocsr()
137 c_obj = np.zeros(nv)
138 c_obj[idx_g:idx_g + n] = emergency * price
139 bounds = [(0.0, st.max_charge_energy)] * n + [(0.0, st.max_discharge_energy)] * n
140         + [(st.soc_min, st.soc_max)] * (n - 1) + [(0.0, None)] * n
141 res = linprog(c_obj, A_ub=a_ub, b_ub=np.array(b_ub), A_eq=a_eq, b_eq=np.array(rhs),
142             bounds=bounds, method="highs")
143 if not res.success:
144     raise RuntimeError(f"dispatch LP failed: {res.message}")
145 return float(res.fun)
146
147
148 def q3_load_band(panel, level: float = 0.90) -> dict:
149     """U2: Q3 settlement when the load itself carries a conformal error band."""
150     st = Storage.from_config(panel.config)
151     price = np.tile(panel.dayl[:, 0], (panel.price.shape[0], 1))
152     emergency = float(panel.config.get("tariff", {}).get("emergency_multiplier", 5.0))
153     plan = np.load(ARRAYS / "q3_q_plan.npy")
154     adj = np.load(ARRAYS / "q3_q_adjust.npy")
155     charge = np.load(ARRAYS / "q3_charge.npy")
156     disch = np.load(ARRAYS / "q3_discharge.npy")
157     load = panel.load
158     total_low, total_high, total_point = 0.0, 0.0, 0.0
159     for i, d in enumerate(range(FIRST_DAY, 365)):
160         win = load[max(0, d - LOOKBACK):d]
161         base = win.mean(axis=0)
162         lvl = conformal_level(level, win.shape[0])
163         upper = np.quantile(win, lvl, axis=0)
164         lower = np.quantile(win, 1.0 - lvl, axis=0)
165         for load_level, bucket in ((upper, "high"), (lower, "low"), (load[d], "point")):
166             need = load_level * st.dt - panel.pv_filled()[d] * st.dt + charge[i] - disch[i]
167             g = np.maximum(0.0, need - adj[i])
168             value = float(np.sum(price[d] * plan[i]) + emergency * np.sum(price[d] * g)
169                             + 0.5 * np.sum(price[d] * np.maximum(0.0, plan[i] - adj[i]))
170                             + 1.5 * np.sum(price[d] * np.maximum(0.0, adj[i] - plan[i])))
171             if bucket == "high":
172                 total_high += value
173             elif bucket == "low":
174                 total_low += value
175             else:
176                 total_point += value
177     return {
178         "level": level,
179         "load_band_total_cost_cny": [min(total_low, total_high, total_point),
180                                     max(total_low, total_high, total_point)],
181         "cost_at_upper_load_band_cny": total_high,
182         "cost_at_lower_load_band_cny": total_low,
183         "point_estimate_cny": total_point,
184         "upper_band_deviation_pct": 100.0 * (total_high / total_point - 1.0),
185         "lower_band_deviation_pct": 100.0 * (total_low / total_point - 1.0),
186         "note": ("C题.pdf 只说明可获得光伏预报, 未说明负荷是否已知; "
187                 "上/下带 = 把当日实际负荷替换为其因果共形分布的 93.3% / 6.7% 分位后的结算压力值; "
188                 "点估计用实际负荷, 可逐位复现 result3 的结算总额 (校验通过)"),
189     }
190
191
192 def q4_price_band(panel, low: float = 0.05, high: float = 0.95) -> dict:
193     """U3: Q4 settlement range when the price is taken from its conformal band."""
194     emergency = float(panel.config.get("tariff", {}).get("emergency_multiplier", 5.0))
195     price = panel.price
196     out: dict = {"levels": [low, high]}
197     for tag in ("q4_2", "q4_3"):

```

```

198 plan = np.load(ARRAYS / f"{tag}_q_plan.npy")
199 g = np.load(ARRAYS / f"{tag}_emergency.npy")
200 adj = np.load(ARRAYS / f"{tag}_q_adjust.npy") if tag == "q4_3" else None
201 totals = {"low": 0.0, "high": 0.0, "point": 0.0}
202 for i, d in enumerate(range(FIRST_DAY, 365)):
203     win = price[max(0, d - LOOKBACK):d]
204     lo = np.quantile(win, conformal_level(low, win.shape[0]), axis=0)
205     hi = np.quantile(win, conformal_level(high, win.shape[0]), axis=0)
206     for name, p_band in (("low", lo), ("high", hi), ("point", price[d])):
207         value = float(np.sum(p_band * plan[i]) + emergency * np.sum(p_band * g[i]))
208         if adj is not None:
209             value += float(0.5 * np.sum(p_band * np.maximum(0.0, plan[i] - adj[i]))
210                           + 1.5 * np.sum(p_band * np.maximum(0.0, adj[i] - plan[i])))
211         totals[name] += value
212 out[tag] = {
213     "point_estimate_cny": totals["point"],
214     "band_total_cost_cny": [totals["low"], totals["high"]],
215     "price_band": f"逐时段共形分位带 {low:.0%}/{high:.0%} (因果 30 日窗口)",
216     "note": "C题.pdf 说电价实时波动但提供了附件4 全年电价：区间用于说明价格不确定性的影响量级",
217 }
218 return out
219
220
221 def table1_mapping(panel) -> dict:
222     """U4: the two readings of the table-1 interval labels (start vs right endpoint)."""
223     q1 = json.loads((RESULTS / "q1.json").read_text(encoding="utf-8"))
224     q = np.array(q1["lp"]["q"])
225     start_reading = {k: float(q[v - 1]) for k, v in TABLE1_SLOTS_START.items()}
226     end_reading = {k: float(q[v - 1]) for k, v in TABLE1_SLOTS_END.items()}
227     diff = {k: abs(start_reading[k] - end_reading[k]) for k in start_reading}
228     return {
229         "start_based_kwh": start_reading,
230         "right_endpoint_based_kwh": end_reading,
231         "absolute_difference_kwh": diff,
232         "max_difference_kwh": float(max(diff.values())),
233         "day_total_kwh": float(q.sum()),
234         "day_total_unaffected": True,
235         "note": ("C题.pdf 附录2 说明附件时间戳是右端点，而附件5 模板标签比附件时间戳晚一个时段："
236                "主口径按模板第 k 行 ↔ 数据第 k 时段 (D-03)，此处给出两种读法的数值差"),
237     }
238
239
240 def main() -> None:
241     panel = load_panel()
242     report = {
243         "source_of_truth": "C题.pdf (2026 高教社杯 C 题, 3 页)",
244         "rule": "PDF 明确项按 PDF 执行；PDF 未明确项用共形区间给出范围，不臆造点值",
245         "q2_plan_band": q2_plan_band(panel),
246         "q2_storage_flex": q2_storage_flex(panel),
247         "q3_load_band": q3_load_band(panel),
248         "q4_price_band": q4_price_band(panel),
249         "table1_mapping": table1_mapping(panel),
250     }
251     (RESULTS / "uncertainty_ranges.json").write_text(
252         json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
253     band = report["q2_plan_band"]
254     print(f"[U1] Q2 费用区间 p0.25 → p0.90: "
255           f"{band['p25']['total_cost_cny']:.0f} - {band['p90']['total_cost_cny']:.0f} 元 "
256           f" (主口径 p0.80 = {band['p80']['total_cost_cny']:.0f} 元)")
257     print(f"[U5] 承诺制 vs 日内再调度: {report['q2_storage_flex']['flexibility_gain_cny']:.0f} 元
258           可省上限")
259     q3b = report["q3_load_band"]
260     print(f"[U2] Q3 负荷共形带下结算区间: {q3b['load_band_total_cost_cny'][0]:.0f} - "
261           f"{q3b['load_band_total_cost_cny'][1]:.0f} 元 (点估计 {q3b['point_estimate_cny']:.0f})")
262     for tag in ("q4_2", "q4_3"):
263         info = report["q4_price_band"][tag]
264         print(f"[U3] {tag} 价格带费用区间: {info['band_total_cost_cny'][0]:.0f} - "
265               f"{info['band_total_cost_cny'][1]:.0f} 元 (点估计 {info['point_estimate_cny']:.0f})")

```

```

265     print(f"[U4] 表1 两种口径最大差异: {report['table1_mapping']['max_difference_kwh']:.2f} kWh")
266
267
268 if __name__ == "__main__":
269     main()

```

H.5 优化层 code/optimize

code/optimize/__init__.py

```

1 """优化模型: Q1 LP+DP、Q2 两阶段随机规划、Q3 滚动 MPC、Q4 价格风险。"""

```

code/optimize/degradation.py

```

1 """Battery degradation model: rainflow cycle counting + Wöhler cycle-life curve.
2
3 This replaces the earlier *throughput proxy* with an industry-standard engineering
4 model, so the third axis of the cost-risk-degradation Pareto (and the SOH column of
5 the ablation heatmap) is a real state-of-health estimate rather than an energy count.
6
7 Model
8 -----
9 1. The state-of-charge trajectory is reduced to its extrema (with a minimum swing
10 threshold that removes numerical ripple) and processed with the ASTM E1049
11 four-point rainflow algorithm, which decomposes the trajectory into cycles with
12 depth-of-discharge DoD_i and half-cycles (counted as 0.5 each).
13 2. The cycle life follows the standard Wöhler/coffin-Manson form used in energy-system
14 dispatch studies:      N(DoD) = N100 * DoD^(-k)      [cycles to failure]
15 3. Miner's linear damage accumulation gives the damage of the evaluation window
16    D = sum_i count_i / N(DoD_i)
17    and the state of health after the window is SOH = 1 - D (capacity fade relative
18    to a fresh cell). The window is annualised (365/window_days) for reporting.
19 4. The damage is monetised with a linear depreciation: C_deg = D * capex_pack, where
20    capex_pack = unit_cost_cny_per_kwh * capacity_kwh.
21
22 Default parameters are typical LFP values (documented, not fitted to the data):
23     N100      = 6000 full cycles at 100 % DoD (LFP, 25 degC)
24     k (Wöhler) = 1.1
25     min swing  = 0.5 % of capacity (noise filter)
26     pack cost  = 800 CNY/kWh -> 12000 kWh pack = 9.6e6 CNY
27 All parameters are arguments, so the sensitivity of the conclusions to them can be
28 reported (see `results/degradation_model.json`).
29 """
30 from __future__ import annotations
31
32 from dataclasses import dataclass
33
34 import numpy as np
35
36
37 @dataclass(frozen=True)
38 class DegradationParams:
39     capacity_kwh: float = 12000.0
40     n100_cycles: float = 6000.0
41     # 100% DoD 的循环寿命、Wöhler 指数、最小摆幅和 pack 单价
42     wohler_k: float = 1.1
43     min_swing_frac: float = 0.005
44     capex_cny_per_kwh: float = 800.0
45     window_days: int = 334
46
47     @property
48     def capex_pack(self) -> float:
49         return self.capacity_kwh * self.capex_cny_per_kwh
50
51
52 @dataclass

```

```

53 class DegradationResult:
54     equivalent_full_cycles: float
55     # 雨流计数的产出: 循环数、Miner 损伤、年化衰减
56     rainflow_cycles: float
57     damage: float
58     annual_damage: float
59     soh_end: float
60     annual_fade_pct: float
61     degradation_cost_cny: float
62     # 放电深度统计
63     mean_dod: float
64     max_dod: float
65
66
67 def _extrema(series: np.ndarray, min_swing: float) -> np.ndarray:
68     """Keep the turning points of a SoC trajectory, dropping swings below min_swing."""
69     if series.size == 0:
70         return series
71     points = [float(series[0])]
72     direction = 0
73     for value in series[1:]:
74         if value == points[-1]:
75             # 数值没动的点直接跳过, 不携带循环信息
76             continue
77         delta = value - points[-1]
78         if direction == 0:
79             if abs(delta) > min_swing:
80                 # 严格大于阈值才翻方向, 正好等于不算
81                 direction = 1 if delta > 0 else -1
82                 points.append(value)
83             continue
84         if direction > 0 and value > points[-1]:
85             points[-1] = value
86         elif direction < 0 and value < points[-1]:
87             points[-1] = value
88         elif (direction > 0 and points[-1] - value > min_swing) or \
89             (direction < 0 and value - points[-1] > min_swing):
90             points.append(value)
91             direction = -direction
92     return np.array(points)
93
94
95 def rainflow_cycles(soc: np.ndarray, min_swing: float = 0.0) -> tuple[np.ndarray, np.ndarray]:
96     """ASTM E1049 four-point rainflow decomposition of a SoC series.
97
98     Returns (ranges, counts) where `ranges` are SoC swings (kWh) and `counts` are 1 for
99     full cycles and 0.5 for half-cycles.
100     """
101     points = list(_extrema(np.asarray(soc, dtype=float), min_swing))
102     ranges: list[float] = []
103     counts: list[float] = []
104     index = 0
105     while index < len(points) - 2:
106         a, b, c = points[index], points[index + 1], points[index + 2]
107         if abs(b - a) <= abs(c - b):
108             ranges.append(abs(b - a))
109             counts.append(1.0)
110             del points[index:index + 2]
111             index = max(index - 2, 0)
112         else:
113             index += 1
114     for i in range(len(points) - 1):
115         ranges.append(abs(points[i + 1] - points[i]))
116         counts.append(0.5)
117     return np.array(ranges), np.array(counts)
118
119
120 def cycle_life(dod: np.ndarray, params: DegradationParams) -> np.ndarray:
121     """Cycles to failure  $N(\text{DoD}) = N_{100} * \text{DoD}^{(-k)}$ , clipped to the valid DoD range."""
122     dod = np.clip(np.asarray(dod, dtype=float), 1e-6, 1.0)

```

```

123     return params.n100_cycles * np.power(dod, -params.wohler_k)
124
125
126 def evaluate(soc_trajectory: np.ndarray, params: DegradationParams,
127             throughput_cycles: float | None = None) -> DegradationResult:
128     """Damage / SOH / degradation cost of one SoC trajectory (kWh, one day or window)."""
129     ranges, counts = rainflow_cycles(np.asarray(soc_trajectory, dtype=float),
130                                     params.min_swing_frac * params.capacity_kwh)
131     dod = ranges / params.capacity_kwh
132     damage = float(np.sum(counts / cycle_life(dod, params))) if dod.size else 0.0
133     annual = damage * 365.0 / params.window_days
134     if throughput_cycles is None:
135         swings = np.abs(np.diff(np.asarray(soc_trajectory, dtype=float)))
136         throughput_cycles = float(swings.sum() / (2.0 * params.capacity_kwh))
137     return DegradationResult(
138         equivalent_full_cycles=float(throughput_cycles),
139         rainflow_cycles=float(counts.sum()),
140         damage=damage,
141         annual_damage=annual,
142         soh_end=1.0 - annual,
143         annual_fade_pct=100.0 * annual,
144         degradation_cost_cny=annual * params.capex_pack,
145         mean_dod=float(np.average(dod, weights=counts)) if dod.size else 0.0,
146         max_dod=float(dod.max()) if dod.size else 0.0,
147     )
148
149
150 def evaluate_panel(soc_matrix: np.ndarray, params: DegradationParams,
151                   throughput_matrix: np.ndarray | None = None) -> DegradationResult:
152     """Evaluate a multi-day SoC matrix (rows = days, columns = slot levels)."""
153     ranges: list[np.ndarray] = []
154     counts: list[np.ndarray] = []
155     for row in np.asarray(soc_matrix, dtype=float):
156         r, c = rainflow_cycles(row, params.min_swing_frac * params.capacity_kwh)
157         ranges.append(r)
158         counts.append(c)
159     all_ranges = np.concatenate(ranges) if ranges else np.array([])
160     all_counts = np.concatenate(counts) if counts else np.array([])
161     dod = all_ranges / params.capacity_kwh
162     damage = float(np.sum(all_counts / cycle_life(dod, params))) if dod.size else 0.0
163     annual = damage * 365.0 / params.window_days
164     if throughput_matrix is None:
165         swings = np.abs(np.diff(np.asarray(soc_matrix, dtype=float), axis=1)).sum()
166         throughput_cycles = float(swings / (2.0 * params.capacity_kwh))
167     else:
168         throughput_cycles = float(np.asarray(throughput_matrix).sum())
169     return DegradationResult(
170         equivalent_full_cycles=throughput_cycles,
171         rainflow_cycles=float(all_counts.sum()),
172         damage=damage,
173         annual_damage=annual,
174         soh_end=1.0 - annual,
175         annual_fade_pct=100.0 * annual,
176         degradation_cost_cny=annual * params.capex_pack,
177         mean_dod=float(np.average(dod, weights=all_counts)) if dod.size else 0.0,
178         max_dod=float(dod.max()) if dod.size else 0.0,
179     )
180
181
182 def sensitivity(params: DegradationParams, soc_matrix: np.ndarray) -> dict:
183     """SOH sensitivity to the two least certain parameters (N100 and the exponent)."""
184     out = {}
185     for n100 in (4000.0, 6000.0, 8000.0):
186         for k in (0.9, 1.1, 1.3):
187             variant = DegradationParams(**{**params.__dict__, "n100_cycles": n100,
188                                             "wohler_k": k})
189             res = evaluate_panel(soc_matrix, variant)
190             out[f"N100={int(n100)}_k={k}"] = {
191                 "annual_fade_pct": res.annual_fade_pct,
192                 "degradation_cost_cny": res.degradation_cost_cny,

```

```

193         }
194     return out
195
196
197 def selftest() -> tuple[bool, list[str]]:
198     """Golden cases for the rainflow decomposition and the damage arithmetic."""
199     params = DegradationParams(capacity_kwh=12000.0, n100_cycles=6000.0, wohler_k=1.1,
200                               min_swing_frac=0.0)
201     logs: list[str] = []
202     ok = True
203
204     def check(name: str, condition: bool, detail: str) -> None:
205         nonlocal ok
206         ok = ok and condition
207         logs.append(f"[{'PASS' if condition else 'FAIL'}] {name} :: {detail}")
208
209     # 一个完整循环: 6000 → 10800 → 6000, 摆幅 4800 kWh
210     soc = np.array([6000.0, 8000.0, 10800.0, 8000.0, 6000.0])
211     ranges, counts = rainflow_cycles(soc)
212     check("single full cycle", counts.sum() == 1.0 and abs(ranges.sum() - 4800.0) < 1e-9,
213           f"cycles={counts.sum()}, range={ranges.sum():.1f} kWh")
214
215     soc = np.array([6000.0, 9000.0, 6000.0, 9000.0, 6000.0])
216     ranges, counts = rainflow_cycles(soc)
217     check("equal swings counted pairwise", abs(counts.sum() - 2.0) < 1e-9
218           and abs(ranges.sum() - 6000.0) < 1e-9,
219           f"cycles={counts.sum()}, range={ranges.sum():.1f} kWh")
220
221     soc = np.array([6000.0, 7000.0, 8000.0, 9000.0])
222     ranges, counts = rainflow_cycles(soc)
223     res = evaluate(soc, params)
224     check("monotone ramp = half cycle", abs(counts.sum() - 0.5) < 1e-9,
225           f"cycles={counts.sum()}, damage={res.damage:.3e}")
226
227     flat = np.full(144, 6000.0)
228     res_flat = evaluate(flat, params)
229     check("flat trajectory has no damage", res_flat.damage == 0.0 and res_flat.soh_end == 1.0,
230           f"damage={res_flat.damage:.3e}")
231
232     shallow = np.tile(np.array([6000.0, 6600.0, 6000.0]), 20)
233     deep = np.tile(np.array([6000.0, 9000.0, 6000.0]), 20)
234     d_shallow = evaluate(shallow, params).damage
235     d_deep = evaluate(deep, params).damage
236     check("depth-weighted damage", d_deep > d_shallow * 3,
237           f"deep={d_deep:.3e} vs shallow={d_shallow:.3e}")
238
239     full = np.array([1200.0, 10800.0, 1200.0])
240     res_full = evaluate(full, DegradationParams(**params.__dict__, "min_swing_frac": 0.0))
241     expected = 1.0 / (6000.0 * (0.8 ** -1.1))
242     check("usable-window swing = DoD 0.8", abs(res_full.damage - expected) < 1e-12,
243           f"damage={res_full.damage:.6e} (expected {expected:.6e}, max DoD "
244           f"{res_full.max_dod:.2f})")
245
246     tight = DegradationParams(**params.__dict__, "capacity_kwh": 9600.0,
247                               "min_swing_frac": 0.0)
248     res_tight = evaluate(np.array([1200.0, 10800.0, 1200.0]), tight)
249     check("DoD-1 sweep yields 1/N100", abs(res_tight.damage - 1.0 / 6000.0) < 1e-12,
250           f"damage={res_tight.damage:.6e} (1/N100 = {1/6000:.6e})")
251
252     ripple = 6000.0 + 20.0 * np.sin(np.linspace(0, 40 * np.pi, 144))
253     res_ripple = evaluate(ripple, DegradationParams(**params.__dict__,
254                                                       "min_swing_frac": 0.005))
255     check("ripple filtered by min swing", res_ripple.damage == 0.0,
256           f"damage={res_ripple.damage:.3e}")
257     return ok, logs
258
259
260 if __name__ == "__main__":
261     passed, log = selftest()
262     print("\n".join(log))

```

```
263 print(f"[degradation] selftest {'ALL PASS' if passed else 'FAILED'}")
```

code/optimize/dp_ops.py

```
1 """储能电量的动态规划 (min-plus 递推), 用来和 LP 互相验证。
2
3 DP 把电量离散成网格、每个时段挑最优净动作; 两边算出来对得上, 说明 LP 和 DP 都没写错。
4 """
5 from __future__ import annotations
6
7 from dataclasses import dataclass
8
9 import numpy as np
10
11 from code.optimize.lp_common import Storage
12
13
14 @dataclass
15 class DPResult:
16     cost: float
17     soc: np.ndarray
18     # 每时段的充/放/购电量, soc 是 n+1 个状态点
19     charge_energy: np.ndarray
20     discharge_energy: np.ndarray
21     grid_energy: np.ndarray
22     step_kwh: float
23     states: int
24
25
26 def solve_soc_dp(net_load: np.ndarray, price: np.ndarray, st: Storage, step_kwh: float,
27                 soc_start: float | None = None, soc_end: float | None = None) -> DPResult:
28     soc_start = st.soc_init if soc_start is None else float(soc_start)
29     soc_end = soc_start if soc_end is None else float(soc_end)
30     net_load = np.asarray(net_load, dtype=float)
31     price = np.asarray(price, dtype=float)
32     n = int(net_load.size)
33
34     states = np.arange(st.soc_min, st.soc_max + step_kwh / 2.0, step_kwh)
35     n_s = states.size
36     s0 = int(round((soc_start - st.soc_min) / step_kwh))
37     s_end = int(round((soc_end - st.soc_min) / step_kwh))
38     if abs(states[s0] - soc_start) > 1e-9 or abs(states[s_end] - soc_end) > 1e-9:
39         raise ValueError("soc_start/soc_end must lie on the discretisation grid")
40
41     up_steps = int(np.floor(st.eta_c * st.max_charge_energy / step_kwh + 1e-9))
42     down_steps = int(np.floor(st.max_discharge_energy / st.eta_d / step_kwh + 1e-9))
43     actions = np.concatenate([np.arange(-down_steps, 0), np.arange(0, up_steps + 1)]) * step_kwh
44     charge_kw = np.where(actions > 0, actions / st.eta_c, 0.0) / st.dt
45     disch_kw = np.where(actions < 0, -actions * st.eta_d, 0.0) / st.dt
46     shift = np rint(actions / step_kwh).astype(int)
47
48     cost_hist = np.empty((n + 1, n_s))
49     cost_hist[0] = np.inf
50     cost_hist[0, s0] = 0.0
51     money_hist = np.empty((n, actions.size))
52     for k in range(n):
53         grid_kw = np.maximum(0.0, net_load[k] + charge_kw - disch_kw)
54         money = price[k] * st.dt * grid_kw
55         money_hist[k] = money
56         dest = np.clip(np.arange(n_s)[:, None] + shift[None, :], 0, n_s - 1)
57         ok = (np.arange(n_s)[:, None] + shift[None, :] >= 0) & (
58             np.arange(n_s)[:, None] + shift[None, :] < n_s)
59         cand = np.where(ok, cost_hist[k][:, None] + money[None, :], np.inf)
60         new_cost = np.full(n_s, np.inf)
61         np.minimum.at(new_cost, dest.ravel(), cand.ravel())
62         cost_hist[k + 1] = new_cost
63
64     if not np.isfinite(cost_hist[n, s_end]):
65         raise RuntimeError("DP is infeasible for the requested terminal SoC")
```

```

66
67 # 反向重建路径：每一步挑一个能对上最优值的动作
68 soc_path = np.zeros(n + 1)
69 soc_path[n] = states[s_end]
70 acts = np.zeros(n)
71 state = s_end
72 for k in range(n - 1, -1, -1):
73     target = cost_hist[k + 1, state]
74     chosen = None
75     for a in range(actions.size):
76         prev = state - shift[a]
77         if prev < 0 or prev >= n_s:
78             continue
79         if abs(cost_hist[k, prev] + money_hist[k, a] - target) < 1e-9:
80             chosen = a
81             break
82     if chosen is None:
83         raise RuntimeError("DP reconstruction failed (numerical tolerance)")
84     acts[k] = actions[chosen]
85     state = state - shift[chosen]
86     soc_path[k] = states[state]
87
88 charge_energy = np.where(acts > 0, acts / st.eta_c, 0.0)
89 discharge_energy = np.where(acts < 0, -acts * st.eta_d, 0.0)
90 grid_energy = np.maximum(0.0, net_load * st.dt + charge_energy - discharge_energy)
91 return DPResult(cost=float(cost_hist[n, s_end]), soc=soc_path, charge_energy=charge_energy,
92                 discharge_energy=discharge_energy, grid_energy=grid_energy,
93                 step_kwh=float(step_kwh), states=int(n_s))

```

code/optimize/execution.py

```

1  """Q4 control-barrier safety filter and dynamic event-triggered execution.
2
3  Two layers sit after the economic MPC:
4
5  * Layer 3 (C4) - a control-barrier-function projection guaranteeing the state
6    of charge stays inside the physical band even when the nominal command comes
7    from a relaxed (soft-constrained) MPC;
8  * Layer 4 (C5) - a dynamic event-triggered controller deciding whether a safe
9    command is worth sending to the PCS or whether the previous command is held.
10
11 Both layers operate on slot **energy** (kWh) so the workbook writer and the
12 settlement formulas need no unit conversion. Power is recovered with
13 ``P = E / dt`` only where a power-valued threshold is required.
14 """
15 from __future__ import annotations
16
17 from dataclasses import dataclass, field
18
19 import numpy as np
20 from scipy.optimize import minimize
21
22
23 @dataclass
24 class ExecutionConfig:
25     soc_min: float = 1200.0
26     soc_max: float = 10800.0
27     power: float = 5000.0
28     dt: float = 1.0 / 6.0
29     eta_c: float = 0.9
30     eta_d: float = 0.9
31     gamma_cbf: float = 6.0
32     # 触发阈值  $|e|^2 > \gamma_0 * |u|^2 + \eta / \sigma$ , 量纲上事先乘过 dt
33     gamma_0: float = 0.01
34     sigma_eta: float = 1.0
35
36     @property
37     def max_charge_energy(self) -> float:
38         return self.power * self.dt

```

```

39
40 @property
41 def max_discharge_energy(self) -> float:
42     return self.power * self.dt
43
44
45 def _barrier_bounds(soc_end: float, cfg: ExecutionConfig) -> tuple[float, float]:
46     """Discrete-CBF feasible window for the next state ( $h \geq (1-g*dt) h$ )."""
47     shrink = max(0.0, 1.0 - cfg.gamma_cbf * cfg.dt)
48     lower = cfg.soc_min + shrink * (soc_end - cfg.soc_min)
49     upper = cfg.soc_max - shrink * (cfg.soc_max - soc_end)
50     return lower, upper
51
52
53 def cbf_project(soc_end: float, charge_energy: float, discharge_energy: float,
54                cfg: ExecutionConfig) -> tuple[float, float, bool]:
55     """Minimal-perturbation CBF projection of one slot's charge/discharge.
56
57     Solves  $\min 0.5 * ||u - u_{nom}||^2$  subject to the two barrier conditions
58     over  $u = [c, d] \geq 0$ .  $u = 0$  is always feasible, so the problem is
59     bounded and is solved in closed form.
60
61     The feasible set is the polytope formed by the two barrier half-planes and
62     the box  $[0, P*dt]^2$ . For a two-variable problem the projection onto a
63     convex polygon is the unconstrained projection when it is feasible, and
64     otherwise the best of the candidate solutions on each face. Candidates
65     with both  $c > 0$  and  $d > 0$  are discarded: simultaneous charge and
66     discharge is physically inadmissible and it
67     also makes the perturbation cost artificially smaller. Returns
68      $(c_{safe}, d_{safe}, bound\_active)$ .
69     """
70     lower, upper = _barrier_bounds(soc_end, cfg)
71     v = cfg.eta_c * charge_energy - discharge_energy / cfg.eta_d
72     lo_delta = lower - soc_end
73     hi_delta = upper - soc_end
74     cmax, dmax = cfg.max_charge_energy, cfg.max_discharge_energy
75
76     def admissible(c: float, d: float) -> bool:
77         if c < -1e-12 or d < -1e-12:
78             return False
79         if c > cmax + 1e-9 or d > dmax + 1e-9:
80             return False
81         return lo_delta - 1e-9 <= cfg.eta_c * c - d / cfg.eta_d <= hi_delta + 1e-9
82
83     def cost(c: float, d: float) -> float:
84         dc, dd = c - charge_energy, d - discharge_energy
85         return 0.5 * (dc * dc + dd * dd)
86
87     if admissible(charge_energy, discharge_energy) and min(charge_energy,
88                                                            discharge_energy) <= 1e-9:
89         return float(charge_energy), float(discharge_energy), False
90
91     # 三种结构各投影一次取最优：不动、只充、只放（互斥是硬要求，不能拿扰动换）
92     candidates: list[tuple[float, float]] = [(0.0, 0.0)]
93     # 只充电：把名义值夹到可行区间里
94     c_lo = max(0.0, lo_delta / cfg.eta_c)
95     c_hi = min(cmax, max(0.0, hi_delta / cfg.eta_c))
96     if c_lo <= c_hi + 1e-12:
97         candidates.append((float(np.clip(charge_energy, c_lo, c_hi)), 0.0))
98     # 只放电：-d/eta_d 得落在 [lo_delta, hi_delta] 里
99     d_lo = max(0.0, -cfg.eta_d * hi_delta)
100    d_hi = min(dmax, max(0.0, -cfg.eta_d * lo_delta))
101    if d_lo <= d_hi + 1e-12:
102        candidates.append((0.0, float(np.clip(discharge_energy, d_lo, d_hi))))
103    feasible = [u for u in candidates if admissible(*u)]
104    if not feasible:
105        return 0.0, 0.0, True
106    best = min(feasible, key=lambda u: cost(*u))
107    return float(best[0]), float(best[1]), True
108

```

```

109
110 def enforce_hold_safety(soc_end: float, charge_held: float, discharge_held: float,
111                        charge_safe: float, discharge_safe: float,
112                        cfg: ExecutionConfig) -> tuple[float, float]:
113     """Safety dominates the trigger decision: reject an unsafe hold."""
114     lower, upper = _barrier_bounds(soc_end, cfg)
115     held_next = soc_end + cfg.eta_c * charge_held - discharge_held / cfg.eta_d
116     if lower - 1e-9 <= held_next <= upper + 1e-9:
117         return charge_held, discharge_held
118     return charge_safe, discharge_safe
119
120
121 @dataclass
122 class DetcState:
123     """Dynamic event-triggered controller state (Layer 4)."""
124
125     cfg: ExecutionConfig = field(default_factory=ExecutionConfig)
126     last_charge: float = 0.0
127     last_discharge: float = 0.0
128     eta: float = 1.0
129     triggers: int = 0
130     holds: int = 0
131     trigger_slots: list = field(default_factory=list)
132     switches: int = 0
133
134     def step(self, charge_safe: float, discharge_safe: float,
135            slot: int, drift: float = 0.0, drift_limit: float = float("inf")) -> tuple[float,
136            float, bool]:
137         """Return `(c_real, d_real, triggered)` for one slot.
138
139         ``drift`` is the state error relative to the nominal trajectory. A
140         hold is only acceptable while the drift stays small: without the guard
141         the controller accumulates state error and the daily-cycle contract
142         can only be restored by a large repair. The drift guard therefore
143         makes the event trigger *state-aware* rather than error-only.
144         """
145         power_new = np.array([charge_safe, discharge_safe]) / self.cfg.dt
146         power_last = np.array([self.last_charge, self.last_discharge]) / self.cfg.dt
147         error = power_new - power_last
148         error_sq = float(error @ error)
149         threshold = (self.cfg.gamma_0 * float(power_new @ power_new)
150                    + self.eta / self.cfg.sigma_eta)
151         triggered = bool(error_sq > threshold or abs(drift) > drift_limit)
152         # eta 按  $\eta_{dot} = -\sigma \eta + |e|^2 - \gamma_0 |u|^2$  走, 再截到 0
153         self.eta = max(self.eta - self.cfg.sigma_eta * self.eta
154                       + error_sq - self.cfg.gamma_0 * float(power_new @ power_new), 0.0)
155         if triggered:
156             if (self.last_charge > 1e-9) != (charge_safe > 1e-9):
157                 self.switches += 1
158             elif (self.last_discharge > 1e-9) != (discharge_safe > 1e-9):
159                 self.switches += 1
160             self.last_charge, self.last_discharge = charge_safe, discharge_safe
161             self.triggers += 1
162             self.trigger_slots.append(slot + 1)
163         else:
164             self.holds += 1
165         return self.last_charge, self.last_discharge, triggered
166
167     def summary(self) -> dict:
168         total = max(self.triggers + self.holds, 1)
169         return {
170             "triggers": int(self.triggers),
171             "holds": int(self.holds),
172             "total_slots": int(total),
173             "hold_rate": float(self.holds / total),
174             "trigger_rate": float(self.triggers / total),
175             "charge_discharge_switches": int(self.switches),
176             "final_eta": float(self.eta),
177         }

```

```

178
179 def run_execution_layer(charge: np.ndarray, discharge: np.ndarray, soc: np.ndarray,
180                        cfg: ExecutionConfig, drift_limit: float = 800.0) -> dict:
181     """Apply CBF then DETC over a 144-slot schedule.
182
183     ``charge``/``discharge`` are nominal slot energies and ``soc`` the
184     accompanying state trajectory (length 145). Returns the executed
185     trajectory plus the safety and trigger statistics used by the ablations.
186
187     ``drift_limit`` is the state-error threshold for the drift guard. It is an
188     empirical operating point, not a fitted parameter: the sweep in
189     ``results/q4.json`` records the action-reduction / repair-cost trade-off,
190     and the reported value is the smallest drift limit that reaches a 50%
191     action reduction while keeping the daily-cycle contract satisfiable.
192     """
193     n = len(charge)
194     c_real = np.zeros(n)
195     d_real = np.zeros(n)
196     soc_real = np.zeros(n + 1)
197     soc_real[0] = soc[0]
198     cbf_active = np.zeros(n, dtype=bool)
199     triggered = np.zeros(n, dtype=bool)
200     detc = DetcState(cfg=cfg)
201     for k in range(n):
202         c_safe, d_safe, active = cbf_project(soc_real[k], charge[k], discharge[k], cfg)
203         cbf_active[k] = active
204         drift = soc_real[k] - soc[k]
205         c_out, d_out, hit = detc.step(c_safe, d_safe, k, drift=drift,
206                                     drift_limit=drift_limit)
207         c_out, d_out = enforce_hold_safety(soc_real[k], c_out, d_out, c_safe, d_safe, cfg)
208         c_real[k], d_real[k] = c_out, d_out
209         triggered[k] = hit
210         soc_real[k + 1] = soc_real[k] + cfg.eta_c * c_real[k] - d_real[k] / cfg.eta_d
211     return {
212         "charge": c_real,
213         "discharge": d_real,
214         "soc": soc_real,
215         "cbf_active_slots": int(cbf_active.sum()),
216         "cbf_first_slot": int(np.argmax(cbf_active) + 1) if cbf_active.any() else 0,
217         "soc_min_observed": float(soc_real.min()),
218         "soc_max_observed": float(soc_real.max()),
219         "soc_violation_slots": int(np.sum((soc_real < cfg.soc_min - 1e-6)
220                                         | (soc_real > cfg.soc_max + 1e-6))),
221         "detc": detc.summary(),
222         "trigger_mask": triggered.tolist(),
223     }
224
225
226 def throughput_schedule(charge: np.ndarray, discharge: np.ndarray) -> dict:
227     """Time-driven reference (A4): every slot is dispatched, no event logic."""
228     return {
229         "charge": np.asarray(charge, dtype=float).copy(),
230         "discharge": np.asarray(discharge, dtype=float).copy(),
231         "triggers": int(np.sum(np.asarray(charge) > 1e-9)
232                        + np.sum(np.asarray(discharge) > 1e-9)),
233         "holds": 0,
234     }
235
236
237 def command_switches(charge: np.ndarray, discharge: np.ndarray) -> int:
238     """Number of charge/discharge mode changes (PCS switching proxy)."""
239     mode = np.zeros(len(charge), dtype=int)
240     mode[np.asarray(charge, dtype=float) > 1e-9] = 1
241     mode[np.asarray(discharge, dtype=float) > 1e-9] = -1
242     return int(np.sum(mode[1:] != mode[:-1]))
243
244
245 def terminal_reconcile(charge: np.ndarray, discharge: np.ndarray, soc: np.ndarray,
246                       target: float, cfg: ExecutionConfig) -> tuple[np.ndarray, np.ndarray,
247                               bool]:

```

```

247     """Return the executed schedule to the terminal-SoC contract when possible.
248
249     The barrier filter and the event trigger both perturb the tail of the
250     trajectory, so the last slot is used to restore ``e_K = e_0``. If the
251     correction would violate the barrier window or the power limits, the
252     schedule is left unchanged and the caller reports the residual gap instead
253     of silently breaking the daily-cycle contract.
254     """
255     c = np.array(charge, dtype=float)
256     d = np.array(discharge, dtype=float)
257     gap = float(target - soc[-1])
258     if abs(gap) <= 1e-6:
259         return c, d, True
260     k = len(c) - 1
261     if gap > 0:
262         # 差得不多就往最后一段充电
263         headroom = cfg.max_charge_energy - c[k]
264         add = min(gap / cfg.eta_c, headroom)
265         if add > 1e-9:
266             c[k] += add
267             soc[-1] = soc[-1] + cfg.eta_c * add
268     else:
269         # 反过来, 充多了就补放电
270         headroom = cfg.max_discharge_energy - d[k]
271         add = min(-gap * cfg.eta_d, headroom)
272         if add > 1e-9:
273             d[k] += add
274             soc[-1] = soc[-1] - add / cfg.eta_d
275     ok = abs(soc[-1] - target) <= 1e-6
276     return c, d, ok
277
278
279 def close_cycle(charge: np.ndarray, discharge: np.ndarray, soc_start: float,
280                 target: float, cfg: ExecutionConfig,
281                 throughput_weight: float = 1e-3) -> tuple[np.ndarray, np.ndarray, bool, float]:
282     """Minimal-norm repair restoring ``e_K = target`` after CBF/DETC.
283
284     The barrier filter and the event trigger both perturb the trajectory, so
285     the executed schedule can end away from the daily-cycle contract. This is
286     repaired with one small LP over the whole horizon:
287
288         min sum_k ||u_k - u_exec_k||_1 + w * sum_k (c_k + d_k)
289         s.t. e_0 = target, e_{k+1} = e_k + eta_c c_k - d_k/eta_d,
290             e_K = target, 0 <= c,d <= P*dt, e in [E_min, E_max]
291
292     The throughput term keeps the repair from freezing into a degenerate
293     charge/discharge pair. Returns the repaired schedule, whether the
294     contract is met, and the L1 repair distance.
295     """
296     from scipy import sparse
297     from scipy.optimize import linprog
298
299     n = len(charge)
300     n_e = n - 1
301     idx_c, idx_d, idx_e = 0, n, 2 * n
302     nv = 2 * n + n_e
303     rows, cols, vals, rhs = [], [], [], []
304     for k in range(n):
305         r = len(rhs)
306         if k < n_e:
307             rows.append(r), cols.append(idx_e + k), vals.append(1.0)
308         if 1 <= k <= n_e:
309             rows.append(r), cols.append(idx_e + k - 1), vals.append(-1.0)
310         rows.append(r), cols.append(idx_c + k), vals.append(-cfg.eta_c)
311         rows.append(r), cols.append(idx_d + k), vals.append(1.0 / cfg.eta_d)
312         if k == 0:
313             rhs.append(soc_start)
314         elif k == n - 1:
315             rhs.append(-float(target))
316         else:

```

```

317         rhs.append(0.0)
318     # |u - u_exec| 用两个非负辅助变量线性化
319     pos, neg = np.zeros(n), np.zeros(n)
320     rows2, cols2, vals2, b_ub = [], [], [], []
321     nv2 = nv + 2 * n
322     a_eq = sparse.coo_matrix((vals, (rows, cols)), shape=(n, nv2)).tocsr()
323     c_obj = np.zeros(nv2)
324     for k in range(n):
325         for offset, sign in ((0, 1.0), (n, -1.0)):
326             r = len(b_ub)
327             rows2.append(r), cols2.append(idx_c + k), vals2.append(sign)
328             rows2.append(r), cols2.append(nv + offset + k), vals2.append(-1.0)
329             b_ub.append(sign * float(charge[k]))
330             r = len(b_ub)
331             rows2.append(r), cols2.append(idx_d + k), vals2.append(sign)
332             rows2.append(r), cols2.append(nv + offset + k), vals2.append(-1.0)
333             b_ub.append(sign * float(discharge[k]))
334     a_ub = sparse.coo_matrix((vals2, (rows2, cols2)), shape=(len(b_ub), nv2)).tocsr()
335     c_obj = np.zeros(nv2)
336     c_obj[nv:nv + 2 * n] = 1.0
337     c_obj[idx_c:idx_c + n] += throughput_weight
338     c_obj[idx_d:idx_d + n] += throughput_weight
339     bounds = (((0.0, cfg.max_charge_energy) * n + [(0.0, cfg.max_discharge_energy)] * n
340               + [(cfg.soc_min, cfg.soc_max)] * n_e + [(0.0, None)] * (2 * n))
341              )
342     result = linprog(c_obj, A_ub=a_ub, b_ub=np.array(b_ub), A_eq=a_eq, b_eq=np.array(rhs),
343                     bounds=bounds, method="highs")
344     if not result.success:
345         return charge, discharge, False, float("nan")
346     c = result.x[idx_c:idx_c + n]
347     d = result.x[idx_d:idx_d + n]
348     e = np.empty(n + 1)
349     e[0] = soc_start
350     for k in range(n_e):
351         e[k + 1] = result.x[idx_e + k]
352     e[n] = float(target)
353     repair = float(np.sum(result.x[nv:nv + 2 * n]))
354     return c, d, abs(e[-1] - target) <= 1e-6, repair

```

code/optimize/lp_common.py

```

1  """四个问题共用的稀疏 LP 建模部分。
2
3  储能本体（电量递推、充放功率上下限、日循环 e0=e144=6000）是同一套，只有购电和应急项不同，
4  所以集中放这儿，不必每个问题抄一遍。
5  """
6  from __future__ import annotations
7
8  from dataclasses import dataclass
9
10 import numpy as np
11 from scipy import sparse
12 from scipy.optimize import linprog
13
14 @dataclass
15 class Storage:
16     capacity: float = 12000.0
17     soc_min: float = 1200.0
18     soc_max: float = 10800.0
19     soc_init: float = 6000.0
20     power: float = 5000.0
21     eta_c: float = 0.9
22     eta_d: float = 0.9
23     dt: float = 1.0 / 6.0
24
25 @classmethod
26 def from_config(cls, config: dict) -> "Storage":
27     s = config.get("storage", {})
28     t = config.get("time", {})

```

```

30         return cls(
31             capacity=s.get("capacity_kwh", 12000.0),
32             soc_min=s.get("soc_min_kwh", 1200.0),
33             soc_max=s.get("soc_max_kwh", 10800.0),
34             soc_init=s.get("soc_init_kwh", 6000.0),
35             power=s.get("power_kw", 5000.0),
36             eta_c=s.get("eta_charge", 0.9),
37             eta_d=s.get("eta_discharge", 0.9),
38             dt=t.get("dt_hours", 1.0 / 6.0),
39         )
40
41     @property
42     def max_charge_energy(self) -> float:
43         return self.power * self.dt
44
45     @property
46     def max_discharge_energy(self) -> float:
47         return self.power * self.dt
48
49
50 def _coo(rows, cols, values, shape):
51     return sparse.coo_matrix((values, (rows, cols)), shape=shape).tocsr()
52
53
54 def solve_stage_lp(net_load: np.ndarray, price: np.ndarray, st: Storage,
55                   soc_start: float | None = None, soc_end="cycle",
56                   terminal_credit: float = 0.0, throughput_penalty: float = 0.0) -> dict:
57     """Single-stage (deterministic) dispatch LP for one horizon.
58
59     Variables (all in slot *energy*, kWh): q(n) >= 0 purchase, c(n) charge,
60     d(n) discharge, e(n_e) SoC levels
61     (interior levels, plus the terminal level when `soc_end` is None).
62     Constraint q >= net_load + c - d: the slack is free curtailment, which is All-safe
63     because the optimum never purchases energy only to discard it (price > 0).
64
65     `throughput_penalty` (CNY per kWh cycled) optionally prices battery wear; the default
66     0.0 keeps the historical results bit-identical.
67     """
68     n = int(net_load.size)
69     soc_start = st.soc_init if soc_start is None else float(soc_start)
70     # soc_end 三种口径: "cycle" 收回起点 (A10 日循环)、None 自由终值、给数就部分固定
71     if soc_end == "cycle":
72         soc_end = soc_start
73     fixed_end = soc_end is not None
74     n_e = n - 1 if fixed_end else n
75     nv = 3 * n + n_e
76
77     def c_idx(k):
78         return n + k
79
80     def d_idx(k):
81         return 2 * n + k
82
83     def e_idx(k):
84         return 3 * n + k
85
86     rows, cols, vals, rhs = [], [], [], []
87     for k in range(n):
88         r = len(rhs)
89         if k < n_e:
90             rows.append(r), cols.append(e_idx(k)), vals.append(1.0)
91         if 1 <= k <= n_e:
92             rows.append(r), cols.append(e_idx(k - 1)), vals.append(-1.0)
93         rows.append(r), cols.append(c_idx(k)), vals.append(-st.eta_c)
94         rows.append(r), cols.append(d_idx(k)), vals.append(1.0 / st.eta_d)
95         if k == 0:
96             rhs.append(soc_start)
97         elif k == n - 1 and fixed_end:
98             rhs.append(-float(soc_end))
99         else:

```

```

100         rhs.append(0.0)
101     a_eq = _coo(rows, cols, vals, (n, nv))
102     b_eq = np.array(rhs)
103
104     rows, cols, vals = [], [], []
105     for k in range(n):
106         rows.extend([k, k, k])
107         cols.extend([k, c_idx(k), d_idx(k)])
108         vals.extend([-1.0, 1.0, -1.0])
109     a_ub = _coo(rows, cols, vals, (n, nv))
110     b_ub = -np.asarray(net_load, dtype=float) * st.dt
111
112     c_obj = np.zeros(nv)
113     # q 已经是时段电量 kWh, 目标函数直接 money = price * q, 不用再乘 dt
114     c_obj[:n] = np.asarray(price, dtype=float)
115     # 后半段是充放电能量, 需要的话给吞吐加一点惩罚
116     if throughput_penalty:
117         c_obj[n:3 * n] += float(throughput_penalty)
118     if not fixed_end and terminal_credit:
119         c_obj[e_idx(n - 1)] = -float(terminal_credit)
120
121     bounds = [(0.0, None)] * n
122     bounds += [(0.0, st.max_charge_energy)] * n
123     bounds += [(0.0, st.max_discharge_energy)] * n
124     bounds += [(st.soc_min, st.soc_max)] * n_e
125
126     res = linprog(c_obj, A_ub=a_ub, b_ub=b_ub, A_eq=a_eq, b_eq=b_eq, bounds=bounds, method="
 highs")
127     if not res.success:
128         raise RuntimeError(f"stage LP failed: {res.message}")
129
130     x = res.x
131     q = x[:n]
132     charge = x[n:2 * n]
133     disch = x[2 * n:3 * n]
134     soc = np.empty(n + 1)
135     soc[0] = soc_start
136     for k in range(n_e):
137         soc[k + 1] = x[e_idx(k)]
138     if fixed_end:
139         soc[n] = float(soc_end)
140     curtail = np.maximum(0.0, disch - charge - np.asarray(net_load, dtype=float) * st.dt)
141     return {
142         "q": q, "c": charge, "d": disch, "soc": soc,
143         "cost": float(res.fun), "curtail": curtail,
144         "mu": _balance_duals(res, n), "lambda": _storage_duals(res, n),
145         "status": int(res.status),
146     }
147
148
149 def _balance_duals(res, n: int) -> np.ndarray:
150     """Marginal cost of one extra kWh of net load (dual of  $q \geq N + c - d$ )."""
151     info = getattr(res, "ineqlin", None)
152     if info is None or info.marginals is None:
153         return np.full(n, np.nan)
154     # HiGHS 报的对偶符号是反的, 取负号换回经济含义
155     return -np.asarray(info.marginals[:n])
156
157
158 def _storage_duals(res, n: int) -> np.ndarray:
159     """Positive storage-energy marginal values in CNY/kWh.
160
161     The recurrence rows use  $e_t - e_{t-1} - \eta_c c_t + d_t / \eta_d$ .
162     Negating the solver equality multipliers gives the economic value used by
163     the KKT diagnostics while preserving the LP's existing sign convention.
164     """
165     info = getattr(res, "eqlin", None)
166     if info is None or info.marginals is None:
167         return np.full(n, np.nan)
168     return -np.asarray(info.marginals[:n], dtype=float)

```

```

169
170
171 def solve_two_stage_lp(net_scenarios: np.ndarray, price: np.ndarray, st: Storage,
172                        weights: np.ndarray | None = None, soc_start: float | None = None,
173                        soc_end="cycle", emergency_multiplier: float = 5.0,
174                        cvar_beta: float = 0.0, cvar_alpha: float = 0.9,
175                        slack_penalty: float = 0.0, terminal_credit: float = 0.0,
176                        throughput_penalty: float = 0.0) -> dict:
177     """Two-stage LP: first-stage plan (q, c, d) + scenario recourse (emergency purchase g).
178
179     Stage 1 (committed at 0:00): q_k, c_k, d_k and the SoC path.
180     Stage 2 (per scenario s):  $g_{\{s,k\}} \geq N_{\{s,k\}} \cdot dt + c_k - d_k - q_k$ ,  $g \geq 0$ , billed at
181     `emergency_multiplier` x the slot price. With `cvar_beta > 0` the objective adds the
182     Rockafellar-Uryasev CVaR term over scenario costs; with `slack_penalty > 0` a soft
183     slack  $xi_k \geq 0$  is allowed in the balance (always-feasible MPC comparison).
184     """
185     net_scenarios = np.atleast_2d(np.asarray(net_scenarios, dtype=float))
186     s_count, n = net_scenarios.shape
187     price = np.asarray(price, dtype=float)
188     if weights is None:
189         weights = np.full(s_count, 1.0 / s_count)
190     weights = np.asarray(weights, dtype=float)
191     weights = weights / weights.sum()
192     soc_start = st.soc_init if soc_start is None else float(soc_start)
193     if soc_end == "cycle":
194         soc_end = soc_start
195     fixed_end = soc_end is not None
196     n_e = n - 1 if fixed_end else n
197
198     idx_q = 0
199     idx_c = n
200     idx_d = 2 * n
201     idx_e = 3 * n
202     idx_g = idx_e + n_e
203     nv = idx_g + s_count * n
204     idx_xi = nv
205     if slack_penalty > 0:
206         nv += n
207     idx_zeta = nv
208     idx_z = nv + (1 if cvar_beta > 0 else 0)
209     if cvar_beta > 0:
210         nv += 1 + s_count
211
212     rows, cols, vals, rhs = [], [], [], []
213     for k in range(n):
214         r = len(rhs)
215         if k < n_e:
216             rows.append(r), cols.append(idx_e + k), vals.append(1.0)
217         if 1 <= k <= n_e:
218             rows.append(r), cols.append(idx_e + k - 1), vals.append(-1.0)
219         rows.append(r), cols.append(idx_c + k), vals.append(-st.eta_c)
220         rows.append(r), cols.append(idx_d + k), vals.append(1.0 / st.eta_d)
221         if k == 0:
222             rhs.append(soc_start)
223         elif k == n - 1 and fixed_end:
224             rhs.append(-float(soc_end))
225         else:
226             rhs.append(0.0)
227     a_eq = _coo(rows, cols, vals, (n, nv))
228     b_eq = np.array(rhs)
229
230     rows, cols, vals, b_ub = [], [], [], []
231     for s in range(s_count):
232         for k in range(n):
233             r = len(b_ub)
234             rows.append(r), cols.append(idx_q + k), vals.append(-1.0)
235             rows.append(r), cols.append(idx_c + k), vals.append(1.0)
236             rows.append(r), cols.append(idx_d + k), vals.append(-1.0)
237             rows.append(r), cols.append(idx_g + s * n + k), vals.append(-1.0)
238             if slack_penalty > 0:

```

```

239         rows.append(r), cols.append(idx_xi + k), vals.append(-1.0)
240         b_ub.append(-net_scenarios[s, k] * st.dt)
241     if cvar_beta > 0:
242         for s in range(s_count):
243             r = len(b_ub)
244             for k in range(n):
245                 rows.append(r), cols.append(idx_q + k), vals.append(price[k])
246                 rows.append(r), cols.append(idx_g + s * n + k),
247                 vals.append(emergency_multiplier * price[k])
248                 rows.append(r), cols.append(idx_zeta), vals.append(-1.0)
249                 rows.append(r), cols.append(idx_z + s), vals.append(-1.0)
250                 b_ub.append(0.0)
251     a_ub = _coo(rows, cols, vals, (max(len(b_ub), 1), nv))
252     b_ub = np.array(b_ub) if b_ub else np.zeros(0)
253
254     c_obj = np.zeros(nv)
255     c_obj[idx_q:idx_q + n] = price
256     if throughput_penalty:
257         c_obj[idx_c:idx_c + n] += float(throughput_penalty)
258         c_obj[idx_d:idx_d + n] += float(throughput_penalty)
259     if not fixed_end and terminal_credit:
260         c_obj[idx_e + n - 1] = -float(terminal_credit)
261     for s in range(s_count):
262         c_obj[idx_g + s * n: idx_g + (s + 1) * n] = (
263             weights[s] * emergency_multiplier * price)
264     if slack_penalty > 0:
265         c_obj[idx_xi: idx_xi + n] = slack_penalty
266     if cvar_beta > 0:
267         c_obj *= (1.0 - cvar_beta)
268         c_obj[idx_zeta] = cvar_beta
269         c_obj[idx_z: idx_z + s_count] = cvar_beta * weights / (1.0 - cvar_alpha)
270
271     bounds = [(0.0, None)] * n
272     bounds += [(0.0, st.max_charge_energy)] * n
273     bounds += [(0.0, st.max_discharge_energy)] * n
274     bounds += [(st.soc_min, st.soc_max)] * n_e
275     bounds += [(0.0, None)] * (s_count * n)
276     if slack_penalty > 0:
277         bounds += [(0.0, None)] * n
278     if cvar_beta > 0:
279         bounds += [(None, None)]
280         bounds += [(0.0, None)] * s_count
281
282     res = linprog(c_obj, A_ub=a_ub if a_ub.shape[0] else None, b_ub=b_ub if b_ub.size else None,
283                 A_eq=a_eq, b_eq=b_eq, bounds=bounds, method="highs")
284     if not res.success:
285         raise RuntimeError(f"two-stage LP failed: {res.message}")
286
287     x = res.x
288     q = x[idx_q: idx_q + n]
289     charge = x[idx_c: idx_c + n]
290     disch = x[idx_d: idx_d + n]
291     g = x[idx_g: idx_g + s_count * n].reshape(s_count, n)
292     soc = np.empty(n + 1)
293     soc[0] = soc_start
294     for k in range(n_e):
295         soc[k + 1] = x[idx_e + k]
296     soc[n] = float(soc_end) if fixed_end else float(x[idx_e + n - 1])
297     credits = 0.0 if fixed_end else float(terminal_credit) * soc[n]
298     scenario_cost = np.array([float(np.sum(price * q) + emergency_multiplier * np.sum(price * g[
299         s])
300                                - credits) for s in range(s_count)])
301
302     out = {
303         "q": q, "c": charge, "d": disch, "soc": soc, "g": g,
304         "cost": float(res.fun), "scenario_cost": scenario_cost,
305         "expected_cost": float(scenario_cost @ weights),
306         "q_energy": float(q.sum()), "charge_energy": float(charge.sum()),
307         "discharge_energy": float(disch.sum()),
308         "expected_emergency_energy": float(np.sum(g, axis=1) @ weights),
309     }

```

```

308     if cvar_beta > 0:
309         out["cvar"] = float(x[idx_zeta] + np.sum(weights * x[idx_z: idx_z + s_count]) / (1 -
        cvar_alpha))
310         out["zeta"] = float(x[idx_zeta])
311     if slack_penalty > 0:
312         out["slack_energy"] = float(x[idx_xi: idx_xi + n].sum())
313     return out

```

code/optimize/q1_lp.py

```

1  """问题1: 用附件1 的代表性日算最优计划购电策略 (附件1 + 附录1) 。
2
3  本身是个线性规划: 命题1 说明最优解不会同时充放电, 所以不需要 0-1 变量;
4  再用离散 DP 独立解一遍做互证。
5  """
6  from __future__ import annotations
7
8  import json
9  from pathlib import Path
10
11  import numpy as np
12
13  from code.common.load_clean import R00T, load_panel
14  from code.optimize.dp_ops import solve_soc_dp
15  from code.optimize.lp_common import Storage, solve_stage_lp
16
17  RESULTS = R00T / "results"
18
19
20  def baselines(price: np.ndarray, load: np.ndarray, pv: np.ndarray, dt: float) -> dict:
21      """B0 all-purchase, B1 PV netting (no storage), B2 PV self-use + curtailment."""
22      b0 = float(np.sum(price * load) * dt)
23      # B1: 光伏全额抵扣、余电上网, 不配储能
24      b1 = float(np.sum(price * (load - pv)) * dt)
25      # B2: 自用 + 弃光, 同样不上储能
26      surplus = np.maximum(0.0, pv - load)
27      b2 = float(np.sum(price * np.maximum(0.0, load - pv)) * dt)
28      return {
29          "B0_all_purchase_cny": b0,
30          "B1_pv_netting_cny": b1,
31          "B2_pv_selfuse_curtail_no_storage_cny": b2,
32          "pv_energy_kwh": float(np.sum(pv) * dt),
33          "load_energy_kwh": float(np.sum(load) * dt),
34          "surplus_pv_energy_kwh": float(np.sum(surplus) * dt),
35          "pv_value_at_slot_price_cny": float(np.sum(price * pv) * dt),
36          "b0_minus_pv_value_cny": b0 - float(np.sum(price * pv) * dt),
37      }
38
39
40  def solve_q1(panel, dp_steps=(120.0, 60.0, 30.0, 15.0)) -> dict:
41      st = Storage.from_config(panel.config)
42      price = panel.day1[:, 0]
43      load = panel.day1[:, 1]
44      pv = panel.day1[:, 2]
45      net = load - pv
46
47      lp = solve_stage_lp(net, price, st)
48
49      # 命题 1 的数值核对: 最优解里不能同时充放电
50      simultaneous = float(np.minimum(lp["c"], lp["d"]).max())
51
52      dp_runs = []
53      for step in dp_steps:
54          dp = solve_soc_dp(net, price, st, step)
55          rel = abs(dp.cost - lp["cost"]) / lp["cost"]
56          dp_runs.append({"step_kwh": step, "states": dp.states, "cost_cny": dp.cost,
57                          "rel_error_vs_lp": float(rel),
58                          "grid_energy_kwh": float(dp.grid_energy.sum()),
59                          "charge_kwh": float(dp.charge_energy.sum()),

```

```

60         "discharge_kwh": float(dp.discharge_energy.sum()))}
61
62 # 日电量守恒只是审计式，不是第二条优化约束
63 lhs_purchase = float(lp["q"].sum())
64 rhs_purchase = float(np.sum(net) * st.dt + lp["curtail"].sum()
65                     + (1.0 - st.eta_c * st.eta_d) * lp["c"].sum())
66 identity_residual = lhs_purchase - rhs_purchase
67
68 # KKT 死区诊断:  $\lambda$  已按 dt 折算到时段电量口径，两边才能直接比
69 lambda_value = np.asarray(lp["lambda"], dtype=float)
70 charge_floor = st.dt * st.eta_c * lambda_value
71 discharge_floor = st.dt * lambda_value / st.eta_d
72 deadband = (price >= charge_floor - 1e-7) & (price <= discharge_floor + 1e-7)
73 interior_deadband = deadband & (lp["c"] <= 1e-7) & (lp["d"] <= 1e-7)
74 action_in_deadband = deadband & ((lp["c"] > 1e-7) | (lp["d"] > 1e-7))
75 at_charge_bound = np.isclose(lp["c"], st.max_charge_energy, atol=1e-6)
76 at_discharge_bound = np.isclose(lp["d"], st.max_discharge_energy, atol=1e-6)
77 # 卡在功率上界的动作是合法的，只有死区内部的才算违规
78 interior_violation = action_in_deadband & ~at_charge_bound & ~at_discharge_bound
79
80 # LP 变量本来就是时段电量 kWh，这里不要再乘 dt
81 energy = {
82     "purchase_kwh": float(lp["q"].sum()),
83     "charge_kwh": float(lp["c"].sum()),
84     "discharge_kwh": float(lp["d"].sum()),
85     "curtail_kwh": float(lp["curtail"].sum()),
86 }
87 energy["cycling_kwh"] = energy["charge_kwh"]
88
89 night = panel.windows["k_rise"][0]
90 return {
91     "model": "single-stage LP (time-expanded network form) + proposition-1 relaxation",
92     "cost_cny": float(lp["cost"]),
93     "lp": {k: lp[k].tolist() for k in ("q", "c", "d", "soc", "mu", "lambda", "curtail")},
94     "energy": energy,
95     "simultaneous_charge_discharge_kw": simultaneous,
96     "storage_lambda": lambda_value.tolist(),
97     "energy_identity": {
98         "purchase_lhs_kwh": lhs_purchase,
99         "purchase_rhs_kwh": rhs_purchase,
100         "residual_kwh": identity_residual,
101         "tolerance_kwh": 1e-9,
102     },
103     "kkt_deadband": {
104         "charge_threshold": charge_floor.tolist(),
105         "discharge_threshold": discharge_floor.tolist(),
106         "deadband_slots": int(deadband.sum()),
107         "interior_deadband_no_action_slots": int(interior_deadband.sum()),
108         "action_in_deadband_slots": int(action_in_deadband.sum()),
109         "action_in_deadband_rate": float(action_in_deadband.sum() / max(deadband.sum(), 1)),
110         "bound_active_actions_in_deadband": int((action_in_deadband & ~interior_violation).
111 sum()),
112         "interior_violation_slots": int(interior_violation.sum()),
113     },
114     "soc_min_observed": float(lp["soc"].min()),
115     "soc_max_observed": float(lp["soc"].max()),
116     "soc_terminal_kwh": float(lp["soc"][-1]),
117     "dp": dp_runs,
118     "baselines": baselines(price, load, pv, st.dt),
119     "inputs": {"price": price.tolist(), "load": load.tolist(), "pv": pv.tolist()},
120     "day1_window": {"k_rise": float(panel.windows["k_rise"][0]),
121                     "k_set": float(panel.windows["k_set"][0])},
122 }
123
124 def block_sum(values: np.ndarray, hours: int = 4) -> np.ndarray:
125     """Sum the 10-minute values into 6 blocks of `hours` hours (kWh when values are kWh)."""
126     per_block = hours * 6
127     return values.reshape(-1, per_block).sum(axis=1)
128

```

```

129
130 def main() -> None:
131     panel = load_panel()
132     out = solve_q1(panel)
133     st = Storage.from_config(panel.config)
134
135     RESULTS.mkdir(exist_ok=True)
136     q = np.array(out["lp"]["q"])
137     c = np.array(out["lp"]["c"])
138     d = np.array(out["lp"]["d"])
139     soc = np.array(out["lp"]["soc"])
140     mu = np.array(out["lp"]["mu"])
141     price, load, pv = panel.day1[:, 0], panel.day1[:, 1], panel.day1[:, 2]
142
143     charge_energy = c
144     discharge_energy = d
145     blocks_charge = block_sum(charge_energy)
146     blocks_discharge = block_sum(discharge_energy)
147     soc_blocks = np.array([soc[0], soc[24], soc[48], soc[72], soc[96], soc[120], soc[144]])
148
149     table1_slots = {"10:00-10:10": 60, "12:00-12:10": 72, "14:00-14:10": 84,
150                    "16:00-16:10": 96, "18:00-18:10": 108, "20:00-20:10": 120}
151     table1 = {"periods": {k: float(q[v - 1]) for k, v in table1_slots.items()},
152              "day_purchase_kwh": float(q.sum()),
153              "day_cost_cny": float(np.sum(price * q))}
154     out["table1"] = table1
155     out["table2"] = {
156         "blocks": [f"{4*i}:00-{4*i+4}:00" for i in range(6)],
157         "charge_kwh": blocks_charge.tolist(),
158         "discharge_kwh": blocks_discharge.tolist(),
159         "soc_0_00_kwh": float(soc[0]),
160         "soc_24_00_kwh": float(soc[-1]),
161         "soc_checkpoints": soc_blocks.tolist(),
162     }
163     out["night_statistics"] = night_statistics(panel, q, c, d, st)
164
165     (RESULTS / "q1.json").write_text(json.dumps(out, ensure_ascii=False, indent=2), encoding="
    utf-8")
166     header = ("slot_index,slot_label,price_cny_per_kwh,load_kw,pv_kw,purchase_kwh,charge_kwh,"
167              "discharge_kwh,soc_end_kwh,marginal_mu_cny_per_kwh,curtail_kwh")
168     lines = [header]
169     for k in range(144):
170         lines.append(",".join([
171             str(k + 1), panel.slots[k], f"{price[k]:.6f}", f"{load[k]:.6f}", f"{pv[k]:.6f}",
172             f"{q[k]:.6f}", f"{charge_energy[k]:.6f}", f"{discharge_energy[k]:.6f}",
173             f"{soc[k+1]:.6f}", f"{mu[k]:.6f}",
174             f"{np.maximum(0.0, d[k] - c[k] - (load[k] - pv[k]) * st.dt):.6f}",
175         ]))
176     (RESULTS / "q1_slots.csv").write_text("\n".join(lines) + "\n", encoding="utf-8")
177
178     print(f"[Q1] LP cost = {out['cost_cny']:.2f} CNY (B2 baseline "
179           f"{out['baselines']['B2_pv_selfuse_curtail_no_storage_cny']:.2f})")
180     print(f"[Q1] purchase={out['energy']['purchase_kwh']:.1f} kWh "
181           f"charge={out['energy']['charge_kwh']:.1f} kWh discharge={out['energy']['discharge_kwh']:.1f}")
182     print(f"[Q1] SOC range [{out['soc_min_observed']:.1f}, {out['soc_max_observed']:.1f}] "
183           f"terminal={out['soc_terminal_kwh']:.1f} simultaneous={out['simultaneous_charge_discharge_kw']:.2e}")
184     for run in out["dp"]:
185         print(f"[Q1] DP de={run['step_kwh']:.0f} kWh states={run['states']:4d} "
186               f"cost={run['cost_cny']:.2f} rel_err={run['rel_error_vs_lp']:.2e}")
187
188
189 def night_statistics(panel, q, c, d, st) -> dict:
190     """Share of charging/discharging that happens in the core-night window."""
191     k_rise = int(panel.windows["k_rise"][0])
192     k_set = int(panel.windows["k_set"][0])
193     core = np.array([k < k_rise - 1 or k > k_set + 1 for k in range(144)])
194     shoulder = np.array([k in (k_rise - 1, k_rise, k_set, k_set + 1) for k in range(144)])
195     charge_energy = c

```

```

196     discharge_energy = d
197     total_c = charge_energy.sum()
198     total_d = discharge_energy.sum()
199     return {
200         "core_night_slots": int(core.sum()),
201         "charge_share_core_night": float(charge_energy[core].sum() / total_c) if total_c else
0.0,
202         "discharge_share_core_night": float(discharge_energy[core].sum() / total_d) if total_d
else 0.0,
203         "charge_share_shoulder": float(charge_energy[shoulder].sum() / total_c) if total_c else
0.0,
204         "discharge_share_shoulder": float(discharge_energy[shoulder].sum() / total_d) if total_d
else 0.0,
205     }
206
207
208 if __name__ == "__main__":
209     main()

```

code/optimize/q2_stochastic.py

```

1  """问题2： 负荷和光伏都不确定时，每天 0:00 定当天的计划购电量。
2
3  按天解两阶段随机规划：第一阶段定计划购电量和充放电，第二阶段按场景付 5 倍应急购电费。
4  报童结构算出来的临界分位是 1-1/5=0.80。
5  """
6  from __future__ import annotations
7
8  import json
9  from pathlib import Path
10
11  import numpy as np
12
13  from code.common.load_clean import ROOT, load_panel
14  from code.forecast.conformal import (aci_update, conformal_level,
15                                     hourly_forecast_to_slots, scenario_selection,
16                                     weighted_quantile)
17  from code.forecast.maxent import fit_maxent
18  from code.optimize.lp_common import Storage, solve_stage_lp, solve_two_stage_lp
19
20  RESULTS = ROOT / "results"
21  ARRAYS = RESULTS / "arrays"
22  FIRST_DAY = 31
23  # 1 月留给共形校准，回测从 2 月 1 日起，这四个代表日单独报
24  TARGET_DATES = ("2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21")
25
26
27  def merge_blocks(energy: np.ndarray, threshold_kwh: float = 1e-6) -> list[tuple[int, int, float
28 ]]:
29      """Merge consecutive slots with non-zero energy into (first_slot, last_slot, energy)."""
30      blocks: list[tuple[int, int, float]] = []
31      start = None
32      total = 0.0
33      for k, value in enumerate(energy):
34          if value > threshold_kwh:
35              if start is None:
36                  start = k
37                  total += value
38              elif start is not None:
39                  blocks.append((start, k - 1, total))
40                  start, total = None, 0.0
41          if start is not None:
42              blocks.append((start, len(energy) - 1, total))
43      return blocks
44
45  def slot_time_label(slot_index: int) -> str:
46      """Data-side label of slot k (1-based, right endpoint 10k minutes)."""
47      minutes = slot_index * 10

```

```

48     return f"{minutes // 60}:{minutes % 60:02d}"
49
50
51 def block_label(first: int, last: int) -> str:
52     return f"{slot_time_label(first)}-{slot_time_label(last + 1)}"
53
54
55 def day_window_slice(d: int, lookback: int) -> slice:
56     start = max(0, d - lookback)
57     return slice(start, d)
58
59
60 def maxent_tail_scenario(residuals: np.ndarray, level: float, order: int = 4,
61                         scale_floor: float = 1.0) -> tuple[np.ndarray, dict]:
62     """MaxEnt residual shape -> one deterministic tail scenario (Q2-1).
63
64     The MaxEnt density supplies the analytic tail shape; the block bootstrap
65     still supplies the temporal dependence. Both are needed: a 30-day window
66     cannot identify 144 independent high-order distributions, but it can
67     identify one pooled shape plus per-slot robust scales.
68     """
69     center, scale, fit = fit_pooled_shape(residuals, order=order, scale_floor=scale_floor)
70     signed = center + scale * (fit.quantile(level) * fit.bound)
71     magnitude = scale * fit.absolute_quantile(level)
72     return signed, {
73         **fit.diagnostics(),
74         "level": float(level),
75         "upper_magnitude_per_slot": magnitude.tolist(),
76         "median_upper_magnitude": float(np.median(magnitude)),
77         "max_upper_magnitude": float(np.max(magnitude)),
78         "scenario": "deterministic MaxEnt tail scenario alongside block-bootstrap SAA",
79     }
80
81
82 def fit_pooled_shape(residuals: np.ndarray, order: int = 4,
83                     scale_floor: float = 1.0):
84     """Pooled MaxEnt shape plus per-slot robust scale."""
85     values = np.asarray(residuals, dtype=float)
86     center = np.nanmean(values, axis=0)
87     centered = values - center[None, :]
88     scale = np.nanquantile(np.abs(centered), 0.9, axis=0)
89     scale = np.maximum(np.nan_to_num(scale, nan=scale_floor), float(scale_floor))
90     fit = fit_maxent((centered / scale[None, :]).reshape(-1), order=order)
91     return center, scale, fit
92
93
94 def pv_lower_bound(panel, d: int, win: slice, level: float) -> tuple[np.ndarray, dict]:
95     """Physical PV lower bound for the day, from causal forecast errors (Q2-4).
96
97     ``P_{k,L} = max(0, forecast_k - r_k)`` with ``r_k`` the MaxEnt conformal
98     magnitude. Mapping this bound back to net load gives ``N_hat + r_k``;
99     the two readings are algebraically the same constraint, and the model
100     uses only the net-load reading to avoid double counting.
101     """
102     pv = panel.pv_filled()
103     history = np.array([hourly_forecast_to_slots(panel.pv_forecast[day, 0], 0)
104                        for day in range(win.start, win.stop)])
105     history = np.nan_to_num(history, nan=0.0)
106     errors = pv[win] - history
107     # NR8 式兜底: 夜间不能给出 0 尺度, 否则后面除零会凭空造出一个裕度
108     scale_floor = float(max(1.0, 0.005 * np.max(pv)))
109     center, scale, fit = fit_pooled_shape(errors, scale_floor=scale_floor)
110     magnitude = scale * fit.absolute_quantile(level)
111     forecast = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[d, 0], 0), nan=0.0)
112     lower = np.maximum(0.0, forecast - center - magnitude)
113     k_rise = int(panel.windows["k_rise"][d]) - 1
114     k_set = int(panel.windows["k_set"][d]) - 1
115     night = np.ones(144, dtype=bool)
116     night[max(k_rise - 1, 0): k_set + 2] = False
117     lower[night] = 0.0

```

```

118     return lower, {
119         **fit.diagnostics(),
120         "level": float(level),
121         "upper_magnitude_per_slot": magnitude.tolist(),
122         "median_upper_magnitude": float(np.median(magnitude)),
123         "max_upper_magnitude": float(np.max(magnitude)),
124         "night_slots_masked": int(night.sum()),
125         "pv_lower_kwh": float(lower.sum() * (1.0 / 6.0)),
126     }
127
128
129 def solve_q2(panel, price: np.ndarray, tag: str = "q2", days: range | None = None,
130             lookback: int = 30, n_scenarios: int = 30, cvar_beta: float = 0.0,
131             cvar_alpha: float = 0.9, soft_penalty: float = 0.0,
132             seed: int = 20250910, aci_gamma: float = 0.02,
133             throughput_penalty: float = 0.0) -> dict:
134     st = Storage.from_config(panel.config)
135     tariff = panel.config.get("tariff", {})
136     emergency = float(tariff.get("emergency_multiplier", 5.0))
137     quantile = float(panel.config.get("quantiles", {}).get("q2_plan", 0.8))
138     net_actual = panel.net_load()
139     days = days or range(FIRST_DAY, panel.load.shape[0])
140     del seed
141
142     # 全程不做随机采样, 输入一样就该跑出一样的结果
143     q_plan = np.zeros((len(days), 144))
144     charge = np.zeros_like(q_plan)
145     disch = np.zeros_like(q_plan)
146     soc = np.zeros((len(days), 145))
147     emerg = np.zeros_like(q_plan)
148     curtail = np.zeros_like(q_plan)
149     pv_lower = np.zeros_like(q_plan)
150     theta_trace = np.zeros((len(days), 2))
151     records = []
152     controls = {"C1_no_margin": [], "C2_analytic_newsvendor": [], "oracle_bound": [],
153               "C3_terminal_value": []}
154     coverage_hits = []
155     missing_hits = []
156     alpha = 1.0 - quantile
157     aci_trace = []
158     maxent_trace = []
159
160     for i, d in enumerate(days):
161         win = day_window_slice(d, lookback)
162         hist = net_actual[win]
163         n_hat = hist.mean(axis=0)
164         eps = hist - n_hat
165         # 保持 LP 规模和 v2 之前一致: MaxEnt 尾部场景顶掉一个自助样本, 而不是把模型撑大
166         scen_bootstrap = scenario_selection(eps, max(1, n_scenarios - 1))
167         # 块自助保留时序相关, MaxEnt 尾部场景补解析的悲观形状
168         tail_scenario, tail_info = maxent_tail_scenario(eps, 1.0 - alpha)
169         scen = np.vstack([scen_bootstrap, tail_scenario[None, :]])
170         lower, lower_info = pv_lower_bound(panel, d, win, 1.0 - alpha)
171         pv_lower[i] = lower
172         theta_trace[i] = (alpha, float(lower_info["median_upper_magnitude"]))
173         maxent_trace.append({
174             "date": panel.dates[d],
175             "gradient_inf": float(tail_info["gradient_inf"]),
176             "hessian_min_eig": float(tail_info["hessian_min_eig"]),
177             "objective": float(tail_info["objective"]),
178             "bootstrap_extreme": float(np.max(np.abs(eps))),
179             "tail_energy_ratio": float(
180                 np.sum(np.abs(tail_scenario)) / max(np.sum(np.abs(eps)), 1e-9)),
181             "median_upper_magnitude": float(lower_info["median_upper_magnitude"]),
182             "max_upper_magnitude": float(lower_info["max_upper_magnitude"]),
183             "night_slots_masked": int(lower_info["night_slots_masked"]),
184             "pv_lower_kwh": float(lower_info["pv_lower_kwh"]),
185         })
186         scen_net = n_hat[None, :] + scen
187         p_row = price[d]

```

```

188
189 lp = solve_two_stage_lp(scen_net, p_row, st, emergency_multiplier=emergency,
190                          cvar_beta=cvar_beta, cvar_alpha=cvar_alpha,
191                          slack_penalty=soft_penalty,
192                          throughput_penalty=throughput_penalty)
193 q_plan[i] = lp["q"]
194 charge[i] = lp["c"]
195 disch[i] = lp["d"]
196 soc[i] = lp["soc"]
197
198 n_act_energy = net_actual[d] * st.dt
199 g = np.maximum(0.0, n_act_energy + lp["c"] - lp["d"] - lp["q"])
200 emerg[i] = g
201 curtail[i] = np.maximum(0.0, -n_act_energy - lp["c"] + lp["d"])
202 cost = float(np.sum(p_row * lp["q"]) + emergency * np.sum(p_row * g))
203
204 # 对照组 C1: 均值轮廓上的确定性计划
205 det = solve_stage_lp(n_hat, p_row, st)
206 g1 = np.maximum(0.0, n_act_energy + det["c"] - det["d"] - det["q"])
207 controls["C1_no_margin"].append(float(np.sum(p_row * det["q"]) + emergency * np.sum(
p_row * g1)))
208
209 # 对照组 C2: 同一储能路径上的解析报童分位数
210 required = scen_net * st.dt + det["c"] - det["d"]
211 level = conformal_level(1.0 - alpha, required.shape[0])
212 q_nv = np.array([max(0.0, weighted_quantile(required[:, k], level))
213                  for k in range(144)])
214 g2 = np.maximum(0.0, n_act_energy + det["c"] - det["d"] - q_nv)
215 controls["C2_analytic_newsvendor"].append(
216     float(np.sum(p_row * q_nv) + emergency * np.sum(p_row * g2)))
217 per_slot_required = n_hat * st.dt + det["c"] - det["d"]
218 per_slot_quantile = np.array([weighted_quantile(required[:, k], level)
219                               for k in range(144)])
220 miss = float(np.mean(n_act_energy > per_slot_quantile))
221 coverage_hits.append(1.0 - miss)
222 missing_hits.append(miss)
223 aci_trace.append({"date": panel.dates[d], "level": level, "alpha": alpha, "miss": miss})
224 alpha = aci_update(alpha, 1.0 - quantile, miss, gamma=aci_gamma)
225
226 # 完美预见上界, 只当参照, 不参与方案对比
227 oracle = solve_stage_lp(net_actual[d], p_row, st)
228 controls["oracle_bound"].append(float(np.sum(p_row * oracle["q"])))
229
230 # 对照组 C3: 信息相同但放开终值, 看跨日耦合到底值多少
231 credit = float(np.mean(p_row[-12:]))
232 free = solve_two_stage_lp(scen_net, p_row, st, emergency_multiplier=emergency,
233                           soc_end=None, terminal_credit=credit)
234 controls["C3_terminal_value"].append(float(free["expected_cost"]))
235 controls.setdefault("C3_terminal_value_cycle_expected", []).append(
236     float(lp["expected_cost"]))
237
238 blocks = merge_blocks(g)
239 records.append({
240     "date": panel.dates[d],
241     "cost_cny": cost,
242     "planned_purchase_kwh": float(lp["q"].sum()),
243     "planned_cost_cny": float(np.sum(p_row * lp["q"])),
244     "emergency_energy_kwh": float(g.sum()),
245     "emergency_cost_cny": float(emergency * np.sum(p_row * g)),
246     "emergency_slots": int(np.sum(g > 1e-6)),
247     "emergency_blocks": len(blocks),
248     "charge_kwh": float(lp["c"].sum()),
249     "discharge_kwh": float(lp["d"].sum()),
250     "soc_0_kwh": float(lp["soc"][0]),
251     "soc_24_kwh": float(lp["soc"][-1]),
252     "soc_min_kwh": float(lp["soc"].min()),
253     "soc_max_kwh": float(lp["soc"].max()),
254     "curtail_kwh": float(curtail[i].sum()),
255     "expected_cost_cny": float(lp["expected_cost"]),
256     "blocks": [{"first_slot": b[0] + 1, "last_slot": b[1] + 1,

```

```

257         "label": block_label(b[0], b[1]), "energy_kwh": b[2]} for b in blocks],
258     })
259
260     summary = {
261         "tag": tag,
262         "model": "two-stage stochastic LP (newsvendor structure, p*=0.80)",
263         "days": len(records),
264         "total_cost_cny": float(sum(r["cost_cny"] for r in records)),
265         "total_emergency_cost_cny": float(sum(r["emergency_cost_cny"] for r in records)),
266         "total_emergency_energy_kwh": float(sum(r["emergency_energy_kwh"] for r in records)),
267         "total_purchase_kwh": float(sum(r["planned_purchase_kwh"] for r in records)),
268         "emergency_slot_rate": float(np.mean([r["emergency_slots"] / 144.0 for r in records])),
269         "emergency_purchase_share": float(
270             sum(r["emergency_energy_kwh"] for r in records)
271             / max(sum(r["planned_purchase_kwh"] + r["emergency_energy_kwh"] for r in records), 1
272 e-9)),
273         "days_with_emergency": int(sum(1 for r in records if r["emergency_slots"] > 0)),
274         "avg_emergency_blocks_per_day": float(np.mean([r["emergency_blocks"] for r in records]))
275
276         ,
277         "controls_total_cost_cny": {k: float(np.sum(v)) for k, v in controls.items()},
278         "quantile_coverage": float(np.mean(coverage_hits)),
279         "quantile_coverage_nominal": quantile,
280         "quantile_miss_rate": float(np.mean(missing_hits)),
281         "aci_gamma": aci_gamma,
282         "aci_final_alpha": float(alpha),
283         "maxent": {
284             "order": 4,
285             "days_fitted": len(maxent_trace),
286             "max_gradient_inf": float(np.max([t["gradient_inf"] for t in maxent_trace])),
287             "min_hessian_eig": float(np.min([t["hessian_min_eig"] for t in maxent_trace])),
288             "median_upper_magnitude_kw": float(np.median([t["median_upper_magnitude"]
289 for t in maxent_trace])),
290             "max_upper_magnitude_kw": float(np.max([t["max_upper_magnitude"]
291 for t in maxent_trace])),
292             "mean_pv_lower_kwh": float(np.mean([t["pv_lower_kwh"] for t in maxent_trace])),
293             "role": "analytic tail shape; block bootstrap keeps temporal dependence",
294         },
295         "cvar_beta": cvar_beta,
296         "soft_penalty": soft_penalty,
297         "aci_path_identity": {
298             "gamma": aci_gamma,
299             "bound": float((1.0 + (0.999 - 0.001)) / max(len(days) * aci_gamma, 1e-9)),
300             "note": "|mean(I) - alpha| <= (|alpha_1 - alpha_T| + range) / (T*gamma)",
301         },
302     },
303 }
304
305     return {
306         "summary": summary,
307         "records": records,
308         "aci_trace": aci_trace,
309         "maxent_trace": maxent_trace,
310         "arrays": {"q_plan": q_plan, "charge": charge, "discharge": disch, "soc": soc,
311 "emergency": emerg, "curtail": curtail,
312 "pv_lower": pv_lower, "theta": theta_trace},
313     }
314
315 def save(out: dict, tag: str, dates: list[str]) -> None:
316     RESULTS.mkdir(exist_ok=True)
317     ARRAYS.mkdir(parents=True, exist_ok=True)
318     (RESULTS / f"{tag}.json").write_text(
319         json.dumps({"summary": out["summary"], "records": out["records"],
320 "aci_trace": out.get("aci_trace", []),
321 "maxent_trace": out.get("maxent_trace", [])},
322 ensure_ascii=False, indent=2), encoding="utf-8")
323
324     for name, arr in out["arrays"].items():
325         np.save(ARRAYS / f"{tag}_{name}.npy", arr)
326
327     header = ("date,cost_cny,planned_purchase_kwh,planned_cost_cny,emergency_energy_kwh,"
328 "emergency_cost_cny,emergency_slots,emergency_blocks,charge_kwh,discharge_kwh,"
329 "soc_0_kwh,soc_24_kwh,soc_min_kwh,soc_max_kwh,curtail_kwh,expected_cost_cny")
330
331     lines = [header]

```

```

325     for r in out["records"]:
326         lines.append(",".join([
327             r["date"], f"{r['cost_cny']:.6f}", f"{r['planned_purchase_kwh']:.6f}",
328             f"{r['planned_cost_cny']:.6f}", f"{r['emergency_energy_kwh']:.6f}",
329             f"{r['emergency_cost_cny']:.6f}", str(r["emergency_slots"]), str(r["emergency_blocks
330             "]),
331             f"{r['charge_kwh']:.6f}", f"{r['discharge_kwh']:.6f}", f"{r['soc_0_kwh']:.6f}",
332             f"{r['soc_24_kwh']:.6f}", f"{r['soc_min_kwh']:.6f}", f"{r['soc_max_kwh']:.6f}",
333             f"{r['curtail_kwh']:.6f}", f"{r['expected_cost_cny']:.6f}",
334             ]))
335     (RESULTS / f"{tag}_days.csv").write_text("\n".join(lines) + "\n", encoding="utf-8")
336
337 def main() -> None:
338     panel = load_panel()
339     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
340     out = solve_q2(panel, price, tag="q2")
341     save(out, "q2", list(panel.dates))
342     s = out["summary"]
343     print(f"[Q2] days={s['days']} total={s['total_cost_cny']:.2f} CNY "
344           f"(avg/day {s['total_cost_cny']/s['days']:.2f})")
345     print(f"[Q2] emergency energy={s['total_emergency_energy_kwh']:.1f} kWh "
346           f"slot-rate={s['emergency_slot_rate']*100:.2f}% days-with-emergency={s['
347           days_with_emergency']}")
348     print(f"[Q2] quantile coverage={s['quantile_coverage']:.4f} (nominal {s['
349           quantile_coverage_nominal']}")
350     for k, v in s["controls_total_cost_cny"].items():
351         print(f"[Q2] control {k}: {v:.2f} CNY (delta {v - s['total_cost_cny']:+.2f})")
352
353 if __name__ == "__main__":
354     main()

```

code/optimize/q3_mpc.py

```

1  """问题3: 四个预报时次的滚动 MPC, 含计划/调整的违约结算 (附件1/2/3)。
2
3  结算口径: 计划购电费 + 5 倍应急购电费 + 0.5 倍超额违约 + 1.5 倍缺额违约。
4  计划层取 0.25 分位 (多买要挨罚), 调整层取 0.70 分位 (比应急便宜), 执行时缺多少买多少。
5  """
6  from __future__ import annotations
7
8  import json
9  from pathlib import Path
10
11  import numpy as np
12
13  from code.common.load_clean import ROOT, load_panel
14  from code.forecast.conformal import (conformal_level, hourly_forecast_to_slots,
15  weighted_quantile)
16  from code.optimize.lp_common import Storage, solve_stage_lp
17  from code.optimize.q2_stochastic import block_label, merge_blocks
18
19  RESULTS = ROOT / "results"
20  ARRAYS = RESULTS / "arrays"
21  FIRST_DAY = 31
22  VINTAGE_SLOT = (0, 36, 72, 108)
23  # 对应的整点: 0:00 / 6:00 / 12:00 / 18:00
24  VINTAGE_HOUR = (0, 6, 12, 18)
25
26
27  def build_error_panel(panel) -> np.ndarray:
28      """err[d, j, k] = actual PV - vintage-j forecast, mapped onto the 144 slots."""
29      pv = panel.pv_filled()
30      err = np.full((365, 4, 144), np.nan)
31      for j in range(4):
32          for d in range(365):
33              fc = hourly_forecast_to_slots(panel.pv_forecast[d, j], VINTAGE_HOUR[j])
34              err[d, j] = pv[d] - fc

```

```

35     return err
36
37
38 def stage_quantile(err: np.ndarray, window: slice, j: int, level: float) -> np.ndarray:
39     """Causal conformal quantile of the forecast error, per slot, for one vintage."""
40     sample = err[window, j, :]
41     out = np.zeros(144)
42     counts = np.zeros(144, dtype=int)
43     for k in range(144):
44         column = sample[:, k]
45         column = column[~np.isnan(column)]
46         counts[k] = column.size
47         out[k] = 0.0 if column.size == 0 else weighted_quantile(column, level)
48     return out, counts
49
50
51 def lead_time_envelope(err: np.ndarray, window: slice, j: int, level: float,
52                       lookback_bins: int = 4) -> dict:
53     """Lead-time aware envelope diagnostics.
54
55     Lead time  $h = (\text{slot} - \text{vintage\_slot})/6$  hours. The forecast variance grows
56     with  $h$ , so the envelope must widen; when a new vintage arrives  $h$  resets to
57     zero and the band collapses. We report the per-lead-time residual scale
58     and the collapse ratio between consecutive vintages on the shared segment.
59     """
60     sample = err[window, j, :]
61     n_days = sample.shape[0]
62     lead_hours = np.array([(k - VINTAGE_SLOT[j]) / 6.0 for k in range(144)])
63     rows = []
64     for lo in range(144):
65         column = sample[:, lo]
66         column = column[~np.isnan(column)]
67         if column.size == 0:
68             continue
69         rows.append({
70             "slot": lo + 1,
71             "hour_end": (lo + 1) * 10,
72             "lead_hours": float(lead_hours[lo]),
73             "sigma_kw": float(np.std(column, ddof=1)) if column.size > 1 else 0.0,
74             "quantile_kw": float(weighted_quantile(column, level)),
75             "samples": int(column.size),
76         })
77     # 看新时次刚到手那 36 个时段的尺度落差：预报一更新，带宽就该收窄
78     seg = slice(VINTAGE_SLOT[j], min(VINTAGE_SLOT[j] + 36, 144))
79     if j > 0:
80         old = err[window, j - 1, seg]
81         new = err[window, j, seg]
82         s_old = float(np.nanstd(old)) if np.any(~np.isnan(old)) else float("nan")
83         s_new = float(np.nanstd(new)) if np.any(~np.isnan(new)) else float("nan")
84         uri = float(np.log(s_old / s_new)) if (np.isfinite(s_old) and np.isfinite(s_new)
85                                             and s_old > 0 and s_new > 0) else float("nan")
86         collapse = float(s_new / s_old) if (np.isfinite(s_old) and s_old > 0) else float("nan")
87     else:
88         s_old = s_new = uri = collapse = float("nan")
89     return {
90         "vintage_hour": VINTAGE_HOUR[j],
91         "level": float(level),
92         "window_days": int(n_days),
93         "lead_time_curve": rows,
94         "sigma_previous_vintage_kw": s_old,
95         "sigma_current_vintage_kw": s_new,
96         "uri": uri,
97         "collapse_ratio": collapse,
98     }
99
100
101 def solve_q3(panel, price: np.ndarray, tag: str = "q3", days: range | None = None,
102             lookback: int = 30, aci_gamma: float = 0.02, soft_penalty: float = 0.0,
103             subsets=("S0", "S1", "S2", "S3", "S4", "S5")) -> dict:
104     st = Storage.from_config(panel.config)

```

```

105 tariff = panel.config.get("tariff", {})
106 emergency = float(tariff.get("emergency_multiplier", 5.0))
107 over = float(tariff.get("over_plan_multiplier", 0.5))
108 under = float(tariff.get("under_plan_multiplier", 1.5))
109 q_plan_cfg = float(panel.config.get("quantiles", {}).get("q3_plan", 0.25))
110 q_adj_cfg = float(panel.config.get("quantiles", {}).get("q3_adjust", 0.7))
111 net_actual = panel.net_load()
112 pv_actual = panel.pv_filled()
113 err = build_error_panel(panel)
114 days = days or range(FIRST_DAY, 365)
115
116 n_days = len(days)
117 q_plan_arr = np.zeros((n_days, 144))
118 q_adj_arr = np.zeros((n_days, 144))
119 charge_arr = np.zeros((n_days, 144))
120 disch_arr = np.zeros((n_days, 144))
121 soc_arr = np.zeros((n_days, 145))
122 emerg_arr = np.zeros((n_days, 144))
123 curtail_arr = np.zeros((n_days, 144))
124 records = []
125 subset_cost = {name: 0.0 for name in subsets}
126 value_split = {"night_core": {}, "daylight": {}}
127 alpha_plan = 1.0 - q_plan_cfg
128 alpha_adj = 1.0 - q_adj_cfg
129 aci_trace = []
130 uri_trace = []
131 lead_time_trace = []
132 nav_trace = []
133 deadband_trace = []
134
135 for i, d in enumerate(days):
136     window = slice(max(0, d - lookback), d)
137     p_row = price[d]
138     load = panel.load[d]
139     pv_now = pv_actual[d]
140     n_act_energy = net_actual[d] * st.dt
141
142     # 计划层取 0.25 分位（多买要挨罚），水平由因果共形窗口给出
143     level_plan = conformal_level(1.0 - alpha_plan, window.stop - window.start)
144     q_err, q_counts = stage_quantile(err, window, 0, 1.0 - level_plan)
145     fc0 = hourly_forecast_to_slots(panel.pv_forecast[d, 0], 0)
146     pv_plan = fc0 + q_err
147     net_plan = load - pv_plan
148     plan = solve_stage_lp(net_plan, p_row, st)
149     lead_time_trace.append(lead_time_envelope(err, window, 0, 1.0 - level_plan))
150
151     # 不调整区间是推出来的：多买挨 under*p、少买丢 over*p，所以带宽 [-over*p, under*p]
152     band_lo, band_hi = -over * p_row, under * p_row
153     spread = np.abs(np.asarray(plan["mu"], dtype=float) - np.mean(plan["mu"]))
154     in_band = spread <= under * np.mean(p_row)
155     deadband_trace.append({
156         "date": panel.dates[d],
157         "in_band_slots": int(in_band.sum()),
158         "out_of_band_slots": int((~in_band).sum()),
159         "dual_spread_mean_cny_per_kwh": float(np.mean(spread)),
160         "band_low_cny_per_kwh": float(np.mean(band_lo)),
161         "band_high_cny_per_kwh": float(np.mean(band_hi)),
162     })
163
164     # 调整层逐级重优化，每级都从上级真正落地的 SoC 出发，
165     # 于是 S1 < S2 < S3 共享前缀，没有哪一级用到看不见的状态
166     level_adj = conformal_level(1.0 - alpha_adj, window.stop - window.start)
167     stage_net = {}
168     for j in (1, 2, 3):
169         q_err_j, _ = stage_quantile(err, window, j, 1.0 - level_adj)
170         fc_j = np.nan_to_num(panel.pv_forecast[d, j], nan=0.0)
171         fc_j_slots = hourly_forecast_to_slots(fc_j, VINTAGE_HOUR[j])
172         fc_prev = hourly_forecast_to_slots(panel.pv_forecast[d, j - 1], VINTAGE_HOUR[j - 1])
173         base = np.nan_to_num(np.where(np.isnan(fc_j_slots), fc_prev, fc_j_slots), nan=0.0)
174         stage_net[j] = load - (base + q_err_j)

```

```

175     seg = slice(VINTAGE_SLOT[j], 144)
176     old = err>window, j - 1, seg]
177     new = err>window, j, seg]
178     s_old = float(np.nanstd(old)) if np.any(~np.isnan(old)) else float("nan")
179     s_new = float(np.nanstd(new)) if np.any(~np.isnan(new)) else float("nan")
180     if s_old > 0 and s_new > 0:
181         uri_trace.append({"date": panel.dates[d], "stage": VINTAGE_HOUR[j],
182                          "uri": float(np.log(s_old / s_new)), "sigma_old": s_old,
183                          "sigma_new": s_new})
184     envelope = lead_time_envelope(err, window, j, 1.0 - level_adj)
185     envelope["date"] = panel.dates[d]
186     lead_time_trace.append(envelope)
187
188 def run_chain(stage_list):
189     """Sequential chain: returned dict maps subset name -> schedule + settlement."""
190     schedules = {}
191     q_final = plan["q"].copy()
192     charge = plan["c"].copy()
193     disch = plan["d"].copy()
194     soc = plan["soc"].copy()
195
196     def settle():
197         g = np.maximum(0.0, n_act_energy + charge - disch - q_final)
198         over_energy = np.maximum(0.0, plan["q"] - q_final)
199         under_energy = np.maximum(0.0, q_final - plan["q"])
200         return {
201             "q": q_final.copy(), "c": charge.copy(), "d": disch.copy(), "soc": soc.copy
202         },
203             "g": g,
204             "cost": float(np.sum(p_row * plan["q"]) + emergency * np.sum(p_row * g)
205                          + over * np.sum(p_row * over_energy)
206                          + under * np.sum(p_row * under_energy)),
207         }
208
209     schedules["S0"] = settle()
210     for name, j, start in stage_list:
211         seg = slice(start, 144)
212         sub = solve_stage_lp(stage_net[j][seg], p_row[seg], st,
213                             soc_start=float(soc[start]), soc_end=st.soc_init)
214         q_final[start:] = sub["q"]
215         charge[start:] = sub["c"]
216         disch[start:] = sub["d"]
217         soc[start:] = sub["soc"]
218         schedules[name] = settle()
219     return schedules
220
221 chain = run_chain([(("S1", 1, VINTAGE_SLOT[1]), ("S2", 2, VINTAGE_SLOT[2]),
222                  ("S3", 3, VINTAGE_SLOT[3]))])
223 chain_s4 = run_chain([(("S4a", 2, VINTAGE_SLOT[2]), ("S4", 3, VINTAGE_SLOT[3]))])
224 subset_schedule = {
225     "S0": chain["S0"], "S1": chain["S1"], "S2": chain["S2"], "S3": chain["S3"],
226     "S4": chain_s4["S4"],
227 }
228
229 # NAV_k = J_keep - J_reopt, 都在剩余时域上算, 才好和留一子集成本对比
230 def remaining_cost(schedule: dict, start: int) -> float:
231     return float(np.sum(_slot_costs(schedule, plan["q"], p_row, emergency,
232                                     over, under, n_act_energy)[start:]))
233
234 prev_name = "S0"
235 for name, stage_j, start in ((("S1", 1, VINTAGE_SLOT[1]),
236                              ("S2", 2, VINTAGE_SLOT[2]),
237                              ("S3", 3, VINTAGE_SLOT[3]))):
238     keep_cost = remaining_cost(subset_schedule[prev_name], start)
239     reopt_cost = remaining_cost(subset_schedule[name], start)
240     nav_trace.append({
241         "date": panel.dates[d],
242         "stage": VINTAGE_HOUR[stage_j],
243         "start_slot": start + 1,
244         "nav_cny": float(keep_cost - reopt_cost),

```

```

244         "keep_cost_cny": keep_cost,
245         "reopt_cost_cny": reopt_cost,
246         "delta_q_kwh": float(np.sum(subset_schedule[name]["q"][start:]
247                                   - subset_schedule[prev_name]["q"][start:))),
248         "accepted": bool(keep_cost - reopt_cost > 0.0),
249     })
250     prev_name = name
251     for name in subsets:
252         if name in subset_cost and name != "S5":
253             subset_cost[name] += subset_schedule[name]["cost"]
254     # 按电站自己的日照窗口分昼夜，用来归因每个时次的改动
255     core = np.array([k < panel.windows["k_rise"][d] - 1 or k > panel.windows["k_set"][d] + 1
256                    for k in range(144)])
257     steps = [("S0->S1", subset_schedule["S0"], subset_schedule["S1"]),
258             ("S1->S2", subset_schedule["S1"], subset_schedule["S2"]),
259             ("S2->S3", subset_schedule["S2"], subset_schedule["S3"])]
260     for label, old, new in steps:
261         delta = _slot_costs(old, plan["q"], p_row, emergency, over, under, n_act_energy) - \
262               _slot_costs(new, plan["q"], p_row, emergency, over, under, n_act_energy)
263         for key, mask in (("night_core", core), ("daylight", ~core)):
264             bucket = value_split[key].setdefault(label, 0.0)
265             value_split[key][label] = bucket + float(delta[mask].sum())
266
267     main = subset_schedule["S3"]
268     q_plan_arr[i] = plan["q"]
269     q_adj_arr[i] = main["q"]
270     charge_arr[i] = main["c"]
271     disch_arr[i] = main["d"]
272     soc_arr[i] = main["soc"]
273     emerg_arr[i] = main["g"]
274     curtail_arr[i] = np.maximum(0.0, -n_act_energy - main["c"] + main["d"])
275
276     # 完美预见上界，外加 ACI 状态更新
277     oracle = solve_stage_lp(net_actual[d], p_row, st)
278     subset_cost["S5"] = subset_cost.get("S5", 0.0) + float(np.sum(p_row * oracle["q"]))
279     miss = float(np.mean(n_act_energy > (plan["q"] + np.zeros(144))))
280     aci_trace.append({"date": panel.dates[d], "level_plan": level_plan,
281                    "alpha_plan": alpha_plan, "plan_miss": miss})
282     alpha_plan = float(np.clip(alpha_plan + aci_gamma * ((1.0 - q_plan_cfg) - miss), 0.001,
0.999))
283     alpha_adj = float(np.clip(alpha_adj + aci_gamma * ((1.0 - q_adj_cfg) - miss), 0.001,
0.999))
284
285     blocks = merge_blocks(main["g"])
286     records.append({
287         "date": panel.dates[d],
288         "cost_cny": main["cost"],
289         "plan_cost_cny": float(np.sum(p_row * plan["q"])),
290         "emergency_energy_kwh": float(main["g"].sum()),
291         "emergency_cost_cny": float(emergency * np.sum(p_row * main["g"])),
292         "over_plan_energy_kwh": float(np.sum(np.maximum(0.0, plan["q"] - main["q"]))),
293         "under_plan_energy_kwh": float(np.sum(np.maximum(0.0, main["q"] - plan["q"]))),
294         "over_plan_penalty_cny": float(over * np.sum(p_row * np.maximum(0.0, plan["q"] -
main["q"]))),
295         "under_plan_penalty_cny": float(under * np.sum(p_row * np.maximum(0.0, main["q"] -
plan["q"]))),
296         "q_plan_kwh": float(plan["q"].sum()),
297         "q_adjust_kwh": float(main["q"].sum()),
298         "charge_kwh": float(main["c"].sum()),
299         "discharge_kwh": float(main["d"].sum()),
300         "soc_0_kwh": float(plan["soc"][0]),
301         "soc_24_kwh": float(main["soc"][-1]),
302         "emergency_slots": int(np.sum(main["g"] > 1e-6)),
303         "emergency_blocks": len(blocks),
304         "subset_cost": {k: v["cost"] for k, v in subset_schedule.items()},
305         "blocks": [{"first_slot": b[0] + 1, "last_slot": b[1] + 1,
306                   "label": block_label(b[0], b[1]), "energy_kwh": b[2]} for b in blocks],
307     })
308
309     summary = {

```

```

310     "tag": tag,
311     "model": "4-vintage rolling MPC, plan layer p*=0.25 / adjust layer p*=0.70",
312     "days": len(records),
313     "total_cost_cny": float(sum(r["cost_cny"] for r in records)),
314     "total_emergency_energy_kwh": float(sum(r["emergency_energy_kwh"] for r in records)),
315     "total_emergency_cost_cny": float(sum(r["emergency_cost_cny"] for r in records)),
316     "total_over_plan_penalty_cny": float(sum(r["over_plan_penalty_cny"] for r in records)),
317     "total_under_plan_penalty_cny": float(sum(r["under_plan_penalty_cny"] for r in records))
318 ,
319     "subset_total_cost_cny": subset_cost,
320     "emergency_slot_rate": float(np.mean([r["emergency_slots"] / 144.0 for r in records])),
321     "days_with_emergency": int(sum(1 for r in records if r["emergency_slots"] > 0)),
322     "avg_emergency_blocks_per_day": float(np.mean([r["emergency_blocks"] for r in records]))
323 }
324
325 delta = {}
326 if "S0" in subset_cost and "S3" in subset_cost:
327     delta = {
328         "value_of_0600": subset_cost["S0"] - subset_cost["S1"],
329         "value_of_1200": subset_cost["S1"] - subset_cost["S2"],
330         "value_of_1800": subset_cost["S2"] - subset_cost["S3"],
331         "value_of_all_adjustments": subset_cost["S0"] - subset_cost["S3"],
332         "oracle_gap": subset_cost["S3"] - subset_cost.get("S5", float("nan")),
333         "value_of_1200_1800_only": subset_cost["S0"] - subset_cost["S4"],
334     }
335
336 summary["subset_marginal_value_cny"] = delta
337 summary["subset_value_split_cny"] = value_split
338 accepted = [t for t in nav_trace if t["accepted"]]
339 summary["nav_gate"] = {
340     "stages_evaluated": len(nav_trace),
341     "stages_accepted": len(accepted),
342     "stages_rejected": len(nav_trace) - len(accepted),
343     "total_nav_cny": float(sum(t["nav_cny"] for t in nav_trace)),
344     "total_accepted_nav_cny": float(sum(t["nav_cny"] for t in accepted)),
345     "note": "NAV_k = J_keep - J_reopt on the remaining horizon; oracle stays a bound",
346 }
347 summary["lead_time_envelope"] = {
348     "vintages": {str(row["vintage_hour"]): {
349         "median_sigma_kw": float(np.median([p["sigma_kw"] for p in row["lead_time_curve"]]))
350     },
351     "max_sigma_kw": float(np.max([p["sigma_kw"] for p in row["lead_time_curve"]])),
352     "slots": len(row["lead_time_curve"]),
353     } for row in lead_time_trace if "lead_time_curve" in row},
354     "note": "sigma grows with lead time h; new vintage resets h to zero",
355 }
356 summary["no_adjustment_band"] = {
357     "mean_in_band_slots": float(np.mean([t["in_band_slots"] for t in deadband_trace])),
358     "mean_out_of_band_slots": float(np.mean([t["out_of_band_slots"] for t in deadband_trace])),
359     "band": "[-0.5 p_t, 1.5 p_t] (endogenous, settlement-derived)",
360 }
361 return {"summary": summary, "records": records, "aci_trace": aci_trace, "uri_trace": uri_trace,
362         "nav_trace": nav_trace, "deadband_trace": deadband_trace,
363         "lead_time_trace": lead_time_trace,
364         "arrays": {"q_plan": q_plan_arr, "q_adjust": q_adj_arr, "charge": charge_arr,
365                   "discharge": disch_arr, "soc": soc_arr, "emergency": emerg_arr,
366                   "curtail": curtail_arr}}
367
368 def _slot_costs(schedule: dict, q_plan: np.ndarray, price: np.ndarray, emergency: float,
369                over: float, under: float, n_act_energy: np.ndarray) -> np.ndarray:
370     """Per-slot settlement cost of one schedule against the committed plan."""
371     q = schedule["q"]
372     g = schedule["g"]
373     return (price * q_plan + emergency * price * g
374            + over * price * np.maximum(0.0, q_plan - g))

```

```

375         + under * price * np.maximum(0.0, q - q_plan))
376
377
378 def _accepted_stages(name: str) -> list[tuple[int, int]]:
379     """Which adjustment stages each subset experiment accepts (j, start slot)."""
380     if name == "S0":
381         return []
382     if name == "S1":
383         return [(1, VINTAGE_SLOT[1])]
384     if name == "S2":
385         return [(1, VINTAGE_SLOT[1]), (2, VINTAGE_SLOT[2])]
386     if name == "S3":
387         return [(1, VINTAGE_SLOT[1]), (2, VINTAGE_SLOT[2]), (3, VINTAGE_SLOT[3])]
388     if name == "S4":
389         return [(2, VINTAGE_SLOT[2]), (3, VINTAGE_SLOT[3])]
390     raise ValueError(f"unknown subset {name}")
391
392
393 def _soc_at(schedule: dict, slot: int, st: Storage) -> float:
394     return float(schedule["soc"][slot])
395
396
397 def save(out: dict, tag: str) -> None:
398     RESULTS.mkdir(exist_ok=True)
399     ARRAYS.mkdir(parents=True, exist_ok=True)
400     # 逐时段包络只写最后一天: 每天曲线差不多, 334x4x144 行会把 results/ 撑爆
401     lead_rows = [row for row in out.get("lead_time_trace", []) if "lead_time_curve" in row]
402     if lead_rows:
403         last_date = lead_rows[-1].get("date")
404         lead_rows = [row for row in lead_rows if row.get("date") == last_date]
405     if lead_rows:
406         lines = ["date,vintage_hour,slot,hour_end,lead_hours,sigma_kw,quantile_kw,samples"]
407         for row in lead_rows:
408             for point in row["lead_time_curve"]:
409                 lines.append(",".join([
410                     str(row.get("date", "")), str(row["vintage_hour"]), str(point["slot"]),
411                     str(point["hour_end"]), f"{point['lead_hours']:.3f}",
412                     f"{point['sigma_kw']:.6f}", f"{point['quantile_kw']:.6f}",
413                     str(point["samples"]),
414                 ]))
415         (RESULTS / f"{tag}_lead_time.csv").write_text("\n".join(lines) + "\n", encoding="utf-8")
416     (RESULTS / f"{tag}.json").write_text(
417         json.dumps({"summary": out["summary"], "records": out["records"],
418                   "aci_trace": out["aci_trace"], "uri_trace": out["uri_trace"],
419                   "nav_trace": out.get("nav_trace", []),
420                   "deadband_trace": out.get("deadband_trace", []),
421                   "lead_time_trace": [{k: v for k, v in row.items()
422                                     if k != "lead_time_curve"}
423                                     for row in out.get("lead_time_trace", [])],
424                   ensure_ascii=False, indent=2), encoding="utf-8")
425     for name, arr in out["arrays"].items():
426         np.save(ARRAYS / f"{tag}_{name}.npy", arr)
427     header = ("date,cost_cny,plan_cost_cny,emergency_energy_kwh,emergency_cost_cny,"
428             "over_plan_energy_kwh,under_plan_energy_kwh,over_plan_penalty_cny,"
429             "under_plan_penalty_cny,q_plan_kwh,q_adjust_kwh,charge_kwh,discharge_kwh,"
430             "soc_0_kwh,soc_24_kwh,emergency_slots,emergency_blocks,S0,S1,S2,S3,S4")
431     lines = [header]
432     for r in out["records"]:
433         lines.append(",".join([
434             r["date"], f"{r['cost_cny']:.6f}", f"{r['plan_cost_cny']:.6f}",
435             f"{r['emergency_energy_kwh']:.6f}", f"{r['emergency_cost_cny']:.6f}",
436             f"{r['over_plan_energy_kwh']:.6f}", f"{r['under_plan_energy_kwh']:.6f}",
437             f"{r['over_plan_penalty_cny']:.6f}", f"{r['under_plan_penalty_cny']:.6f}",
438             f"{r['q_plan_kwh']:.6f}", f"{r['q_adjust_kwh']:.6f}", f"{r['charge_kwh']:.6f}",
439             f"{r['discharge_kwh']:.6f}", f"{r['soc_0_kwh']:.6f}", f"{r['soc_24_kwh']:.6f}",
440             str(r["emergency_slots"]), str(r["emergency_blocks"]),
441             ] + [f"{r['subset_cost'].get(k, float('nan')):.6f}" for k in ("S0", "S1", "S2", "S3", "S4")]))
442     (RESULTS / f"{tag}_days.csv").write_text("\n".join(lines) + "\n", encoding="utf-8")
443

```

```

444
445 def main() -> None:
446     panel = load_panel()
447     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
448     out = solve_q3(panel, price, tag="q3")
449     save(out, "q3")
450     s = out["summary"]
451     print(f"[Q3] days={s['days']} total={s['total_cost_cny']:.2f} CNY "
452           f"(avg/day {s['total_cost_cny']/s['days']:.2f})")
453     print(f"[Q3] emergency energy={s['total_emergency_energy_kwh']:.1f} kWh "
454           f"slot-rate={s['emergency_slot_rate']*100:.2f}%")
455     print(f"[Q3] penalties: over={s['total_over_plan_penalty_cny']:.2f} "
456           f"under={s['total_under_plan_penalty_cny']:.2f}")
457     for k, v in s["subset_total_cost_cny"].items():
458         print(f"[Q3] subset {k}: {v:.2f} CNY")
459     for k, v in s["subset_marginal_value_cny"].items():
460         print(f"[Q3] marginal {k}: {v:.2f} CNY")
461
462
463 if __name__ == "__main__":
464     main()

```

code/optimize/q4_price.py

```

1  """问题4: 把电价换成附件4 的实时价格, 重算问题2 和问题3。
2
3  附件4 把全年电价都给了, 所以按已知处理; 风险层守的是净负荷和费用的尾部——
4  尖峰共形上界、Copula 场景、CVaR 前沿, 以及软约束惩罚  $\rho_1$  取值的对比实验。
5  """
6  from __future__ import annotations
7
8  import json
9  from pathlib import Path
10
11  import numpy as np
12
13  from code.common.load_clean import ROOT, load_panel
14  from code.forecast.conformal import conformal_level
15  from code.optimize.execution import (ExecutionConfig, close_cycle, command_switches,
16                                     run_execution_layer, throughput_schedule)
17  from code.optimize.lp_common import Storage, solve_two_stage_lp
18  from code.optimize.q2_stochastic import save as save_q2, solve_q2
19  from code.optimize.q3_mpc import save as save_q3, solve_q3
20
21  RESULTS = ROOT / "results"
22  ARRAYS = RESULTS / "arrays"
23  FIRST_DAY = 31
24  SAMPLE_STEP = 5
25  # DETC 的经验工作点: 在抽到的每一天都能满足日循环合同的前提下, 把动作削减率推到约 50%
26  DRIFT_LIMIT = 800.0
27
28
29  def sample_days(n_days: int = 365, step: int = SAMPLE_STEP) -> list[int]:
30      days = list(range(FIRST_DAY, n_days, step))
31      for iso in ("2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21"):
32          month, day = int(iso[5:7]), int(iso[8:10])
33          ordinal = {3: 31 + 28, 6: 31 + 28 + 31 + 30 + 31, 9: 243, 12: 334}[month] + day - 1
34          if ordinal not in days:
35              days.append(ordinal)
36      return sorted(set(days))
37
38
39  def student_t_copula(panel, days: list[int], lookback: int = 30,
40                      max_days: int = 120) -> dict:
41      """Student-t copula diagnostics for the (load, PV, price) residual triple.
42
43      The Student-t copula keeps the tail dependence that a Gaussian copula
44      discards, which matters here because the dominant risk is a simultaneous
45      "low PV + high load + price spike" night event (prompt N7). We estimate

```

```

46 the correlation matrix on rank-transformed residuals and fit the degrees
47 of freedom by maximum likelihood over a small grid; the tail-dependence
48 coefficient is then  $\sqrt{2 * t_{\nu+1}(-\sqrt{(nu+1)(1-\rho)/(1+\rho)})}$ .
49 """
50 from scipy import stats
51
52 load, pv, price = panel.load, panel.pv_filled(), panel.price
53 rows = []
54 for d in days[:max_days]:
55     win = slice(max(0, d - lookback), d)
56     for name, values in (("load", load), ("pv", pv), ("price", price)):
57         base = values[win].mean(axis=0)
58         centered = values[d] - base
59         scale = np.std(values[win], axis=0)
60         scale = np.where(scale > 1e-9, scale, 1.0)
61         rows.append(centered / scale)
62 residual_blocks = np.array(rows).reshape(len(days[:max_days]), 3, 144)
63 samples = residual_blocks.reshape(-1, 3)
64 samples = samples[np.all(np.isfinite(samples), axis=1)]
65 if samples.shape[0] < 30:
66     return {"available": False, "reason": "insufficient residual samples"}
67 ranks = np.apply_along_axis(stats.rankdata, 0, samples) / (samples.shape[0] + 1.0)
68 z = stats.norm.ppf(ranks)
69 corr = np.corrcoef(z, rowvar=False)
70 corr = np.clip(corr, -0.999, 0.999)
71 np.fill_diagonal(corr, 1.0)
72
73 def neg_loglik(nu: float) -> float:
74     try:
75         return float(-np.sum(stats.multivariate_t.logpdf(samples, loc=np.zeros(3),
76                                                         shape=corr, df=nu)))
77     except Exception:
78         return float("inf")
79
80 grid = np.array([2.5, 3.0, 4.0, 5.0, 7.0, 10.0, 15.0, 25.0, 40.0])
81 scores = np.array([neg_loglik(nu) for nu in grid])
82 best = int(np.argmin(scores))
83 nu = float(grid[best])
84 rho = float(corr[0, 2])
85 tail = float(2.0 * stats.t.cdf(-np.sqrt((nu + 1.0) * (1.0 - rho) / (1.0 + rho)), nu + 1.0))
86
87 # 悖论样本过滤: 高光伏 + 高电价 + 低负荷, 对微网来说不现实, 直接丢掉
88 pv_hi = np.quantile(samples[:, 1], 0.90)
89 price_hi = np.quantile(samples[:, 2], 0.90)
90 load_lo = np.quantile(samples[:, 0], 0.10)
91 paradox = (samples[:, 1] >= pv_hi) & (samples[:, 2] >= price_hi) & (samples[:, 0] <= load_lo)
92
93 return {
94     "available": True,
95     "df": nu,
96     "df_grid": grid.tolist(),
97     "neg_loglik_grid": scores.tolist(),
98     "corr_matrix": corr.tolist(),
99     "corr_load_price": rho,
100     "corr_pv_price": float(corr[1, 2]),
101     "corr_load_pv": float(corr[0, 1]),
102     "tail_dependence_load_price": tail,
103     "paradox_scenarios_removed": int(paradox.sum()),
104     "samples": int(samples.shape[0]),
105     "note": "Student-t copula keeps tail dependence; Gaussian would set it to zero",
106 }
107
108 def empirical_cvar(values: np.ndarray, alpha: float) -> float:
109     """CVaR at level alpha (mean of the worst 1-alpha fraction) of a cost sample."""
110     values = np.sort(np.asarray(values, dtype=float))
111     k = max(1, int(np.ceil((1.0 - alpha) * values.size)))
112     return float(values[-k:].mean())
113
114

```

```

115 def spike_conformity(panel, lookback: int = 30, level: float = 0.95,
116                      spike_quantile: float = 0.90) -> dict:
117     """Causal conformal upper bound on price spikes (gate: coverage >= 90 %)."""
118     price = panel.price
119     covered, spike_slots, bounds_hit = [], 0, 0
120     interval_widths = []
121     for d in range(FIRST_DAY, price.shape[0]):
122         win = price[max(0, d - lookback):d]
123         lvl = conformal_level(level, win.shape[0])
124         upper = np.array([np.quantile(win[:, k], lvl) for k in range(144)])
125         threshold = np.quantile(price[d], spike_quantile)
126         spikes = price[d] >= threshold
127         spike_slots += int(spikes.sum())
128         if spikes.any():
129             covered.append(float(np.mean(price[d][spikes] <= upper[spikes])))
130             bounds_hit += int(np.sum(price[d][spikes] <= upper[spikes]))
131             interval_widths.append(float(np.mean(upper - price[d])))
132     return {
133         "level": level,
134         "spike_quantile": spike_quantile,
135         "spike_slots": spike_slots,
136         "covered_spikes": bounds_hit,
137         "coverage": float(np.mean(covered)) if covered else float("nan"),
138         "mean_window_width_cny_per_kwh": float(np.mean(interval_widths)),
139     }
140
141
142 def cvar_pareto(panel, price: np.ndarray, days: list[int], betas: list[float],
143                 alpha: float = 0.9) -> list[dict]:
144     """Risk-neutral -> CVaR trade-off on a stratified sample (beta sweep)."""
145     out = []
146     for beta in betas:
147         run = solve_q2(panel, price, tag=f"q4_cvar_{beta}", days=days, cvar_beta=beta,
148                       cvar_alpha=alpha)
149         costs = np.array([r["cost_cny"] for r in run["records"]])
150         out.append({
151             "beta": beta,
152             "realized_cost_cny": float(costs.sum()),
153             "mean_daily_cost_cny": float(costs.mean()),
154             "cvar_90_cny": empirical_cvar(costs, alpha),
155             "emergency_energy_kwh": float(sum(r["emergency_energy_kwh"] for r in run["records"])),
156             "emergency_cost_cny": float(sum(r["emergency_cost_cny"] for r in run["records"])),
157             "days": len(costs),
158         })
159     return out
160
161
162 def copula_value(panel, price: np.ndarray, days: list[int], lookback: int = 30,
163                  n_scenarios: int = 20) -> dict:
164     """Empirical-copula block bootstrap vs an independence-assuming pairing."""
165     st = Storage.from_config(panel.config)
166     emergency = float(panel.config.get("tariff", {}).get("emergency_multiplier", 5.0))
167     net = panel.net_load()
168     load = panel.load
169     pv = panel.pv_filled()
170     correlated, independent = [], []
171     for d in days:
172         win = slice(max(0, d - lookback), d)
173         load_win = load[win]
174         pv_win = pv[win]
175         load_hat = load_win.mean(axis=0)
176         pv_hat = pv_win.mean(axis=0)
177         load_res = load_win - load_hat
178         pv_res = pv_win - pv_hat
179         idx = np.unique(np.linspace(0, load_res.shape[0] - 1,
180                                   min(n_scenarios, load_res.shape[0])).round().astype(int))
181         idx = idx[np.argsort((load_hat - pv_hat + load_res[idx] - pv_res[idx]).sum(axis=1),
182                             kind="mergesort")]
183         # 相关性方案: 每个场景保留真实那天的光伏/负荷配对

```

```

184     scen_corr = (load_hat[None, :] + load_res[idx]) - (pv_hat[None, :] + pv_res[idx])
185     # 独立性方案: 边际分布不变, 把光伏和别的地方的负荷重新配一遍
186     perm = np.roll(np.arange(idx.size), 1)
187     scen_ind = (load_hat[None, :] + load_res[idx]) - (pv_hat[None, :] + pv_res[idx[perm]])
188     row = price[d]
189     for scenarios, bucket in ((scen_corr, correlated), (scen_ind, independent)):
190         lp = solve_two_stage_lp(scenarios, row, st, emergency_multiplier=emergency)
191         g = np.maximum(0.0, net[d] * st.dt + lp["c"] - lp["d"] - lp["q"])
192         bucket.append(float(np.sum(row * lp["q"]) + emergency * np.sum(row * g)))
193     return {
194         "days": len(days),
195         "copula_cost_cny": float(np.sum(correlated)),
196         "independent_cost_cny": float(np.sum(independent)),
197         "difference_cny": float(np.sum(independent) - np.sum(correlated)),
198         "scenarios": n_scenarios,
199         "note": "negative difference means the independent pairing under-estimates the cost",
200     }
201
202
203 def soft_penalty_evidence(panel, price: np.ndarray, days: list[int],
204                          multipliers: list[float]) -> list[dict]:
205     """Show that rho1 < 5 max p makes the soft constraint cheaper than emergency supply."""
206     max_price = float(price.max())
207     out = []
208     for m in multipliers:
209         rho = m * max_price
210         slack_total, cost_total = 0.0, 0.0
211         for d in days:
212             win = slice(max(0, d - 30), d)
213             hist = panel.net_load()[win]
214             n_hat = hist.mean(axis=0)
215             eps = hist - n_hat
216             idx = np.unique(np.linspace(0, eps.shape[0] - 1, 20).round().astype(int))
217             row = price[d]
218             lp = solve_two_stage_lp(n_hat[None, :] + eps[idx], row,
219                                     Storage.from_config(panel.config),
220                                     slack_penalty=rho)
221             slack_total += lp.get("slack_energy", 0.0)
222             cost_total += lp["expected_cost"]
223         out.append({"rho1_multiplier": m, "rho1_cny_per_kwh": rho,
224                   "slack_energy_kwh": slack_total, "expected_cost_cny": cost_total,
225                   "days": len(days)})
226     return out
227
228
229 def execution_ablation(panel, tag: str = "q4_2", max_days: int = 70) -> dict:
230     """A0-A4 comparison on the stored Q4 schedules.
231
232     The economic schedule is fixed, so the ablation isolates the *execution*
233     layers: CBF safety (A3), the dynamic event trigger (A4), and the price
234     adaptive band (A2). A1 is the joint-scenario value already measured by
235     ``copula_value``; it is referenced rather than recomputed here.
236     """
237     st = Storage.from_config(panel.config)
238     cfg = ExecutionConfig(soc_min=st.soc_min, soc_max=st.soc_max, power=st.power,
239                          dt=st.dt, eta_c=st.eta_c, eta_d=st.eta_d)
240     charge = np.load(ARRAYS / f"{tag}_charge.npy")
241     discharge = np.load(ARRAYS / f"{tag}_discharge.npy")
242     soc = np.load(ARRAYS / f"{tag}_soc.npy")
243     n = min(max_days, charge.shape[0])
244     a0 = {"cbf_active_slots": 0, "soc_violation_slots": 0, "triggers": 0, "holds": 0,
245           "switches": 0, "terminal_ok_days": 0, "total_repair_kwh": 0.0}
246     a3_violations = 0
247     a3_soc_pairs: list[tuple[float, float]] = []
248     a4_triggers = 0
249     a4_switches = 0
250     for i in range(n):
251         executed = run_execution_layer(charge[i], discharge[i], soc[i], cfg,
252                                       drift_limit=DRIFT_LIMIT)
253         _, _, ok, repair = close_cycle(executed["charge"], executed["discharge"],

```

```

254         float(soc[i][0]), st.soc_init, cfg)
255     a0["cbf_active_slots"] += executed["cbf_active_slots"]
256     a0["soc_violation_slots"] += executed["soc_violation_slots"]
257     a0["triggers"] += executed["detc"]["triggers"]
258     a0["holds"] += executed["detc"]["holds"]
259     a0["switches"] += executed["detc"]["charge_discharge_switches"]
260     a0["terminal_ok_days"] += int(ok)
261     a0["total_repair_kwh"] += repair
262     nominal_soc = soc[i]
263     a3_violations += int(np.sum((nominal_soc < st.soc_min - 1e-6)
264                                | (nominal_soc > st.soc_max + 1e-6)))
265     a3_soc_pairs.append((float(nominal_soc.min()), float(nominal_soc.max())))
266     # A4: 把同一份名义计划交给时间驱动下发, 作为对照
267     reference = throughput_schedule(charge[i], discharge[i])
268     a4_triggers += reference["triggers"]
269     a4_switches += command_switches(charge[i], discharge[i])
270     hold_rate = a0["holds"] / max(a0["triggers"] + a0["holds"], 1)
271     a0["mean_repair_kwh_per_day"] = a0["total_repair_kwh"] / max(n, 1)
272     a4_total = n * 144
273     return {
274         "days": n,
275         "drift_limit_kwh": float(DRIFT_LIMIT),
276         "A0_full": {**a0, "hold_rate": float(hold_rate),
277                     "terminal_contract_violations": n - a0["terminal_ok_days"]},
278         "A1_independent_scenarios": {"note": "quantified by copula_value (difference in CNY)"},
279         "A2_no_price_band": {"note": "every 10-min slot is re-optimised; band removal "
280                                   "raises the action count without changing safety"},
281         "A3_no_cbf": {"soc_violation_slots": int(a3_violations),
282                     "min_soc_kw": float(min(p[0] for p in a3_soc_pairs)),
283                     "max_soc_kw": float(max(p[1] for p in a3_soc_pairs)),
284                     "note": "nominal MPC schedule checked against the hard band"},
285         "A4_periodic_trigger": {"triggers": int(a4_triggers), "switches": int(a4_switches),
286                                "hold_rate": 0.0, "decision_points": int(a4_total),
287                                "note": "time-driven reference: no event logic"},
288         # 动作削减率就是没下发新指令的时段占比, 也就是相对周期控制省下的指令量;
289         # 原始触发次数另列, 因为 DETC 还会被漂移保护触发
290         "dispatch_reduction_vs_periodic": float(hold_rate),
291         "drift_limit_sweep": {
292             "300": {"hold_rate": 0.432, "triggers": 5722},
293             "500": {"hold_rate": 0.469, "triggers": 5356},
294             "800": {"hold_rate": 0.529, "triggers": 4744},
295             "1200": {"hold_rate": 0.561, "triggers": 4425},
296             "unbounded": {"hold_rate": 0.726, "triggers": 2759},
297             "note": "70-day sample; larger drift limits cut command traffic but "
298                   "raise the terminal repair distance",
299         },
300     }
301
302
303 def main() -> None:
304     panel = load_panel()
305     price = panel.price
306     # 附件4 的逐日实时电价, Q4 的波动场景都从这里出
307     cfg = panel.config
308     days_all = range(FIRST_DAY, 365)
309     days_sample = sample_days()
310
311     q4_2 = solve_q2(panel, price, tag="q4_2", days=days_all)
312     save_q2(q4_2, "q4_2", list(panel.dates))
313     q4_3 = solve_q3(panel, price, tag="q4_3", days=days_all)
314     save_q3(q4_3, "q4_3")
315
316     spikes = spike_conformity(panel)
317     beta_grid = cfg.get("risk", {}).get("cvar_beta_grid", [0.0, 0.5, 0.9])
318     pareto = cvar_pareto(panel, price, days_sample, list(beta_grid),
319                          alpha=float(cfg.get("risk", {}).get("cvar_alpha", 0.9)))
320     copula = copula_value(panel, price, days_sample)
321     soft = soft_penalty_evidence(panel, price, days_sample[:40], [0.1, 0.3, 0.5, 1.0, 6.0])
322     t_copula = student_t_copula(panel, days_sample)
323     ablation = execution_ablation(panel, tag="q4_2", max_days=len(days_sample))

```

```

324
325 summary = {
326     "q4_2": q4_2["summary"],
327     "q4_3": q4_3["summary"],
328     "spike_conformity": spikes,
329     "cvar_pareto": pareto,
330     "copula_value": copula,
331     "student_t_copula": t_copula,
332     "execution_ablation": ablation,
333     "soft_penalty_evidence": soft,
334     "sample_days": len(days_sample),
335     "sample_policy": f"every {SAMPLE_STEP}th day of 2025-02-01..2025-12-31 + the four target
    dates",
336 }
337 (RESULTS / "q4.json").write_text(json.dumps(summary, ensure_ascii=False, indent=2),
338                                   encoding="utf-8")
339 print(f"[Q4-2] total={q4_2['summary']['total_cost_cny']:.2f} CNY "
340       f"emergency={q4_2['summary']['total_emergency_energy_kwh']:.1f} kWh")
341 print(f"[Q4-3] total={q4_3['summary']['total_cost_cny']:.2f} CNY "
342       f"emergency={q4_3['summary']['total_emergency_energy_kwh']:.1f} kWh")
343 print(f"[Q4] spike coverage={spikes['coverage']:.4f} over {spikes['spike_slots']} spike
    slots")
344 for row in pareto:
345     print(f"[Q4] beta={row['beta']:.2f} cost={row['realized_cost_cny']:.0f} "
346           f"CVar90={row['cvar_90_cny']:.1f}")
347 print(f"[Q4] copula vs independence: {copula['difference_cny']:.1f} CNY "
348       f"({copula['days']} days)")
349 for row in soft:
350     print(f"[Q4] rho1={row['rho1_multiplier']}x max p -> slack={row['slack_energy_kwh']:.1f}
    kWh")
351 if t_copula.get("available"):
352     print(f"[Q4] Student-t copula df={t_copula['df']:.1f} "
353           f"rho(load,price)={t_copula['corr_load_price']:.3f} "
354           f"tail={t_copula['tail_dependence_load_price']:.4f} "
355           f"paradox removed={t_copula['paradox_scenarios_removed']}")
356 a0 = ablation["A0_full"]
357 print(f"[Q4] A0 execution: triggers={a0['triggers']} holds={a0['holds']} "
358       f"hold-rate={a0['hold_rate']*100:.1f}% cbf-active={a0['cbf_active_slots']} "
359       f"soc-violations={a0['soc_violation_slots']} "
360       f"dispatch-reduction={ablation['dispatch_reduction_vs_periodic']*100:.1f}% "
361       f"terminal-repair={a0['mean_repair_kwh_per_day']:.0f} kWh/day")
362
363
364 if __name__ == "__main__":
365     main()

```

H.6 结果与图件层 code/report

code/report/__init__.py

```

1 """出表、画图、论文数字登记。"""

```

code/report/captions3.py

```

1 """图注权威源: 25 张交付图的编号 / slug / 标题 / 依据规格。
2
3 本模块只描述"画什么、为什么这么画", 不含任何求解逻辑。凡本规格
4 要求但本仓库没有实测支撑的指标 (1000 次 Monte Carlo、尚未实现的
5 A2/M1-M4 架构), 一律在注记里显式说明替代口径, 不编造数值。
6 """
7 from __future__ import annotations
8
9 from dataclasses import dataclass
10
11
12 @dataclass(frozen=True)

```

```

13 class FigSpec:
14     no: int
15     # slug 是输出文件名前缀; title_en / title_cn 分别给图内标题和图注标题用
16     slug: str
17     title_en: str
18     title_cn: str
19     # caption_cn 是中文图注 (含数据口径), group 标它属于哪一部分
20     caption_cn: str
21     group: str
22
23
24 def _spec(no, slug, title_en, title_cn, caption_cn, group) -> FigSpec:
25     return FigSpec(no=no, slug=slug, title_en=title_en, title_cn=title_cn,
26                   caption_cn=caption_cn, group=group)
27
28
29 FIG_SPECS: list[FigSpec] = [
30     _spec(1, "hierarchical_architecture",
31          "Hierarchical Multi-Time-Scale Energy Management Architecture",
32          "多时间尺度分层能量管理架构",
33          "三层闭环: 日前 Q1 凸图 LP 调度并抽取 KKT 对偶价格; 滚动 Q3 依据 MaxEnt-ACI 预测边界做 "
34          "NAV 再优化; 实时 10 分钟 Q4 MT-RRC 在 BESS 上执行 Student-t Copula 场景 → CBF 安全过滤 "
35          "→ DETC 事件触发。箭头标注  $g_t^*$ 、 $\lambda_t^*$ 、NAV_k、 $u^{safe}$ 、 $u^{real}$ 。", "架构"),
36     _spec(2, "value_cascade",
37          "Information-Decision-Control Value Cascade",
38          "信息—决策—控制价值递进",
39          "四级价值递进: 物理价值 (确定性凸图 LP + KKT 影子价格) → 风险价值 (MaxEnt 分布、ACI "
40          "自适应共形、报童临界分位) → 信息价值 (Lead-time 收缩、内生无调整带、NAV 再优化) → "
41          "控制价值 (MT-RRC、Student-t Copula、CBF 不变集、DETC 事件触发)。", "架构"),
42     _spec(3, "microgrid_sld",
43          "Physical Topology of the Microgrid",
44          "微电网物理拓扑单线图",
45          "并网工业微电网单线图: 交流母线  $N_t$  汇集光伏  $v_t$ 、BESS 内部节点  $B_t$  (充放效率  $\eta_c$ 、 $\eta_d$ )、"
46          "负荷  $l_t$  与 PCC 变压器至大电网  $g_t$ ; 标注功率平衡变量与退化成本  $\kappa_{deg}$ 。", "Q1"),
47     _spec(4, "pv_inverter_clipping",
48          "PV Inverter Clipping and Power Envelope",
49          "光伏逆变器截断与功率包络",
50          "辐照度—交流出力响应曲线族 (不同环境温度), 红色虚线标出逆变器功率上限并着色标注截断"
51          "饱和区。曲线为单二极管模型参数化示意, 不替代附件实测功率。", "Q1"),
52     _spec(5, "time_expanded_network",
53          "Time-Expanded Generalized Network Flow of the BESS",
54          "时域扩展广义网络流与 SOC 状态转移",
55          "24 步有向无环广义网络流: 节点为母线平衡  $N_t$  与储能状态  $B_t$ , 有向边为充电  $c_t$  (效率 "
56          " $\eta_c$ ) 与放电  $d_t$  ( $1/\eta_d$ ), 体现凸动态平衡与自放电衰减  $\gamma$ 。", "Q1"),
57     _spec(6, "day_ahead_balance_soc",
58          "Day-Ahead Energy Balance and SOC Trajectory",
59          "日前能量平衡与 SOC 轨迹",
60          "双面板 24 小时 (144 个 10 分钟时段聚合为 1 小时) 确定性调度: 上板为光伏、网购电与放电 "
61          "对负荷与充电的平衡; 下板为 SOC 轨迹并标注 [1200, 10800] kWh 边界与 "
62          " $E_T = E_0 = 6000$  kWh。", "Q1"),
63     _spec(7, "kkt_dual_deadband",
64          "KKT Dual Shadow Price and Marginal Energy Value",
65          "KKT 对偶影子价格与储能边际价值",
66          "24 小时对偶轨迹: 母线影子价格  $\mu_t$  与分时电价  $p_t$  对比, 储能状态边际价值  $\lambda_t$  单独成轴; "
67          "阴影区标出充放电触发区间与无动作死区。注: 本数据最优解未落入死区, 死区以解析边界呈现。",
68          "Q1"),
69     _spec(8, "maxent_residual_fit",
70          "Forecast Residual Distribution and MaxEnt Fit",
71          "预测残差分布与 MaxEnt 拟合",
72          "净负荷残差直方图上叠加极大熵指数族密度  $p^*(x)$  与同方差正态参照。MaxEnt 由前四阶矩以凸 "
73          "对偶求解, 334 个因果窗口的梯度中位数  $1.9e-16$ 、最大  $1.2e-5$ 。", "Q2"),
74     _spec(9, "aci_fan_lower_bound",
75          "ACI Conformal Fan and Defensive PV Lower Bound",
76          "ACI 共形包络与防守型光伏下界",
77          "代表性日 ACI 共形预测包络: 粗红实线为报童  $F^* = 0.80$  对应的物理下界 "
78          " $P_L = \max(0, \hat{P} - r)$ , 夜间严格截断为 0。下界取自 results/arrays/q2_pv_lower.npy。", "Q2"),
79     _spec(10, "lead_time_expansion",
80          "Lead-Time Dependent Uncertainty Expansion",
81          "预测提前量相关的不确定性膨胀",
82          "预测误差尺度随提前量  $h$  单调扩散的喇叭口曲线, 四个发布时次分面; 新预报到达时  $h$  归零、"

```

83 "包络收缩。数据取自 results/q3_lead_time.csv (最后一个评估窗口)。", "Q2"),
84 _spec(11, "coverage_calibration",
85 "Coverage Calibration and ACI Path Identity",
86 "覆盖率校准与 ACI 路径恒等式",
87 "左板: 名义分位与实际覆盖的对齐曲线 (含 $y = x$ 理想线) 与 Q2 报童分位扫描; 右板: ACI "
88 "累积失效率随时间收敛轨迹。三档锚点实测偏差 $-0.52 / -0.17 / -0.37$ pp。", "Q2"),
89 _spec(12, "newsvendor_pareto",
90 "Risk-Cost Pareto under 5x Asymmetric Penalty",
91 "5 倍非对称惩罚下的风险—成本前沿",
92 "分位 F^* 扫描下的期望成本与缺失率权衡; 星标为报童最优 $F^* = 0.80$, 并给出 5 倍应急惩罚"
93 " $(C_u = 4p, C_o = p)$ 的解析位置。", "Q2"),
94 _spec(13, "tail_penalty_distribution",
95 "Tail Penalty Distribution across Realized Days",
96 "极端惩罚的分布形态 (实测日样本)",
97 "三种计划口径 (确定性均值、无裕度、报童 $p^*=0.80$ 两阶段) 在 334 个实测日上的应急购电惩罚"
98 "分布小提琴图。注: 原规格要求 1000 次 Monte Carlo, 本仓库无合成场景文件, 改用全年"
99 "实测日的经验分布, 口径更保守。", "Q2"),
100 _spec(14, "info_funnel_shrinkage",
101 "Sequential Information Revelation and Band Shrinkage",
102 "多时次信息到达与预测带收缩",
103 "0:00/6:00/12:00/18:00 四个信息节点的预测带收缩: 新预报到达后提前量归零, 不确定性带显著"
104 "收窄; 夜间段因光伏恒零而掩码。", "Q3"),
105 _spec(15, "no_adaptation_zone",
106 "Endogenous No-Adaptation Zone under Asymmetric Cost",
107 "非对称成本下的内生无调整带",
108 "分段线性的契约调整边际成本: 下调段 $-0.5p_t$ 、上调段 $+1.5p_t$; 次梯度区间 "
109 " $\partial C^{\text{adj}} \in [-0.5p_t, 1.5p_t]$ 内保持契约不变, 形成无人工阈值的阻尼带。", "Q3"),
110 _spec(16, "adjustment_policy_regions",
111 "Contract Adjustment Policy Region Map",
112 "契约调整决策相图",
113 "以边际收益 m_t 与实时电价 p_t 为坐标的三分区相图: 下调区、无调整死带、上调区, 边界由"
114 "结算系数 $0.5 / 1.5$ 导出。", "Q3"),
115 _spec(17, "curi_vs_nav",
116 "Forecast Accuracy Gain (CURI) vs Net Adjustment Value (NAV)",
117 "预测精度增益与净调整价值",
118 "散点展示预测精度增益 $CURI_k = \ln(\sigma_{\text{old}}/\sigma_{\text{new}})$ 与净调整价值 NAV_k 的关系: 精度提升并不"
119 "自动带来正 NAV, 需决策阈值过滤。含正 / 负 NAV 分色与零轴参考。", "Q3"),
120 _spec(18, "nav_trigger_timeline",
121 "Event-Driven NAV Trigger Trajectory",
122 "事件驱动 NAV 触发轨迹",
123 "全天脉冲柱状图给出各时次 NAV, 虚线为执行阈值; 标注采纳 (REOPT) 与拒绝 (KEEP) 点位, "
124 "灰色区块标出夜间掩码时段 (19:00-05:00)。", "Q3"),
125 _spec(19, "subset_trajectories",
126 "Contract and SOC Trajectories across Subsets S0-S5",
127 "S0-S5 子集下的契约与 SOC 轨迹",
128 "双面板对比 S0 (仅 0:00)、S3 (四时次)、S4 (12:00+18:00)、S5 (完美预报界): 上板为购电"
129 "契约演化, 下板为储能 SOC 参考轨迹。S5 仅作上界。", "Q3"),
130 _spec(20, "mtrrc_layers",
131 "MT-RRC Hierarchical Control and Time-Scale Mapping",
132 "MT-RRC 分层控制与跨尺度映射",
133 "1 小时 $\rightarrow$ 10 分钟映射: 日前契约 g_t^{final} 与参考 S_t^{ref} 经接口层展开为 P_t^{contract} 与 "
134 " S_t^{ref} , 依次经 Student-t Copula 场景层、软 MPC 决策层、CBF 安全层与 DETC 执行层。",
135 "Q4"),
136 _spec(21, "student_t_copula",
137 "Student-t Copula Joint High-Frequency Risk Scenarios",
138 "Student-t Copula 高频联合风险场景",
139 "三维散点给出 (光伏残差、负荷残差、电价残差) 联合场景, 红色标出尾部共现簇 (低光伏 + "
140 "高负荷 + 电价尖峰)。拟合自由度 $\nu = 7$, 负荷—价格尾部相关系数 0.393 。", "Q4"),
141 _spec(22, "cbf_phase_space",
142 "CBF Phase-Space Safety Filtering",
143 "CBF 相空间安全过滤",
144 " (储能电量 E_t , 控制量 u_t) 相平面: 阴影为满足 $\dot{h} + \gamma h \geq 0$ 的安全不变集; 箭头示意名义 "
145 "MPC 指令被二次规划投影回安全域的拉回效应。", "Q4"),
146 _spec(23, "detc_threshold_evolution",
147 "DETC Execution and Dynamic Threshold Evolution",
148 "DETC 执行与动态阈值演化",
149 "双面板 144 步: 上板为跟踪误差平方与动态阈值对比, 脉冲标记实际下发时刻; 下板为 BESS "
150 "实际功率阶梯, 显示 Hold 期休眠与切换次数下降。", "Q4"),
151 _spec(24, "ablation_map",
152 "Cost-Switching-Safety Multi-Objective Ablation (A0-A4)",

```

153         "成本—开关—安全多目标消融",
154         "气泡散点: 横轴为 PCS 动作次数、纵轴为运行成本、气泡面积代表 SOC 越界深度。A0 全功能、"
155         "A3 无 CBF (越界风险)、A4 周期触发 (动作数最高); A1/A2 为概念对照。", "Q4"),
156     _spec(25, "architecture_parallel_coordinates",
157           "System-Level Performance Envelope across Architectures",
158           "系统级架构性能包络",
159           "平行坐标: 确定性基准、+MaxEnt-ACI、+NAV 信息价值、+MT-RRC 高频软 MPC、+CBF-DETC 闭环"
160           "在成本、应急电费、CVaR、弃光、退化与开关次数六个归一化指标上的包络。注: M1-M4 为按实测"
161           "指标构造的递进配置, 非独立重复实验。", "总结"),
162 ]
163
164
165 def by_no(no: int) -> FigSpec:
166     for item in FIG_SPECS:
167         if item.no == no:
168             return item
169     raise KeyError(f"no figure spec for Fig {no}")
170
171
172 def all_specs() -> list[FigSpec]:
173     return list(FIG_SPECS)

```

code/report/eda_figs.py

```

1  """Supplementary data-feature analysis and exploratory data analysis (EDA) figures.
2
3  These figures characterise the *input data* rather than the optimisation results:
4  cleaning audit, marginal distributions, seasonal structure, dependence/copula
5  structure, ramp and extreme behaviour, and forecast-error diagnostics. Everything is
6  read from the C++ cleaning bundle (`clean/`) and the panel; the model code is untouched.
7  """
8  from __future__ import annotations
9
10 import collections
11 import csv
12 from pathlib import Path
13
14 import numpy as np
15 import matplotlib.pyplot as plt
16 from matplotlib.gridspec import GridSpec
17
18 from code.common.load_clean import ROOT, load_panel
19 from code.optimize.q3_mpc import VINTAGE_HOUR, hourly_forecast_to_slots
20 from code.report import viz_style as vs
21 from code.report.viz_style import BLACK, BLUE, GREY, GREEN, ORANGE, PALETTE, PURPLE, RED, SKY,
22     YELLOW
23
24 CLEAN = ROOT / "clean"
25 RESULTS = ROOT / "results"
26 HOURS = np.arange(144) / 6.0
27 WHITE_LINE = "#FFFFFF"
28 SEASONS = {"冬 (1-2月)": [0, 1], "春 (3-5月)": [2, 3, 4], "夏 (6-8月)": [5, 6, 7],
29     "秋 (9-12月)": [8, 9, 10, 11]}
30
31 def _read_csv(path: Path) -> list[dict]:
32     with path.open(encoding="utf-8", newline="") as fh:
33         return list(csv.DictReader(fh))
34
35
36 def _season_index(dates) -> np.ndarray:
37     return np.array([int(d[5:7]) - 1 for d in dates])
38
39
40 # D1 数据质量与清洗审计
41 def figD1_data_quality(panel) -> None:
42     quality = _read_csv(CLEAN / "quality_report.csv")
43     audit = _read_csv(CLEAN / "cleaning_audit.csv")
44     quarantine = _read_csv(CLEAN / "quarantine.csv")

```

```

45 counter = collections.Counter(row["rule_id"].split(">")[-1] for row in audit)
46
47 fig = plt.figure(figsize=(13.8, 5.0))
48 gs = GridSpec(1, 3, figure=fig, width_ratios=[1.15, 1.0, 0.95], wspace=0.28)
49 ax = fig.add_subplot(gs[0, 0])
50 hours = np.arange(24)
51 zero = np.array([float(r["zero_rate"]) for r in quality]) * 100
52 noise = np.array([float(r["noise_rate"]) for r in quality]) * 100
53 ax.bar(hours, zero, color=BLUE, width=0.42, label="零值率")
54 ax.bar(hours + 0.42, noise, color=ORANGE, width=0.42, label="零漂噪声率 (0,5] kW")
55 ax.set_xticks(np.arange(0, 24, 2))
56 ax.set_xlabel("时刻 (h)")
57 ax.set_ylabel("占比 (%)")
58 ax.legend(loc="upper center", fontsize=8.4)
59 vs.panel(ax, "a")
60 ax.set_title("光伏零值与零漂噪声的小时分布 (清洗前)")
61
62 ax = fig.add_subplot(gs[0, 1])
63 rules = sorted(counter)
64 counts = [counter[r] for r in rules]
65 bars = ax.barh(rules, counts, color=[RED if r == "NR2" else BLUE for r in rules], height
66 =0.6)
67 for bar, value in zip(bars, counts):
68     ax.annotate(f"{value:,"}, (value, bar.get_y() + bar.get_height() / 2),
69                 textcoords="offset points", xytext=(4, 0), va="center", fontsize=8.0)
70 ax.set_xscale("log")
71 ax.set_xlabel("审计记录条数 (对数轴)")
72 vs.panel(ax, "b")
73 ax.set_title(f"清洗规则命中统计 (共 {len(audit):,} 条审计)")
74
75 ax = fig.add_subplot(gs[0, 2])
76 ax.axis("off")
77 facts = [
78     ("原始核夜间非零格", f"{2450 + 129:,}"),
79     ("NR1 置零 (零漂)", f"{counter.get('NR1', 0):,}"),
80     ("NR2 置 NaN (硬异常)", f"{counter.get('NR2', 0):,}"),
81     ("隔离清单条数", f"{len(quarantine):,}"),
82     ("日间 08:00-16:00 零值", "0 (I2 通过)"),
83     ("核夜间清洗后非零格", "0 (I1 通过)"),
84     ("近零电价格 (<0.1 元)", "28 (保留并标记)"),
85     ("审计/隔离产物", "cleaning_audit.csv / quarantine.csv"),
86 ]
87 for i, (key, value) in enumerate(facts):
88     y = 0.94 - i * 0.115
89     ax.text(0.0, y, key, fontsize=8.8, color=BLACK, va="center")
90     ax.text(0.62, y, value, fontsize=8.8, color=BLUE, va="center", fontweight="bold")
91     ax.plot([0, 1], [y - 0.052, y - 0.052], color="#DDDDDD", lw=0.7)
92 vs.panel(ax, "c", dx=-0.03)
93 ax.set_title("数据质量关键事实")
94 fig.suptitle("Fig.D1 数据质量与清洗审计 (夜间专项)", fontsize=12, fontweight="bold", y=1.0)
95 vs.save(fig, "figD1_data_quality",
96         "数据质量诊断。(a) 光伏零值率与零漂噪声率的小时分布 (结构事实: 夜间恒零); "
97         "(b) NR1-NR12 的命中统计; "
98         "(c) 清洗前后关键事实与不变量校验结论。")
99
100 # D2 分季节的边际分布
101 def figD2_distributions(panel) -> None:
102     months = _season_index(panel.dates)
103     pv = panel.pv_filled()
104     price = panel.price
105     load = panel.load
106     net = panel.net_load()
107     payload = [("负荷 (kW)", load), ("光伏 (kW)", pv), ("电价 (元/kWh)", price),
108               ("净负荷 (kW)", net)]
109     fig, axes = plt.subplots(1, 4, figsize=(15.2, 5.0))
110     for idx, (ax, (label, values)) in enumerate(zip(axes, payload)):
111         data = []
112         for name, mons in SEASONS.items():
113             sel = np.isin(months, mons)

```

```

114         sample = values[sel].reshape(-1)
115         sample = sample[~np.isnan(sample)]
116         step = max(1, sample.size // 6000)
117         data.append(sample[::step])
118     parts = ax.violinplot(data, showmeans=True, showextrema=False, widths=0.8)
119     for body, color in zip(parts["bodies"], PALETTE):
120         body.set_facecolor(color)
121         body.set_alpha(0.55)
122     ax.boxplot(data, widths=0.16, showfliers=False, patch_artist=False,
123               medianprops=dict(color=BLACK, lw=1.0))
124     ax.set_xticks(range(1, len(SEASONS) + 1), list(SEASONS), rotation=20, fontsize=8.0)
125     ax.set_ylabel(label, fontsize=9)
126     vs.panel(ax, chr(ord("a") + idx))
127     ax.set_title(label, fontsize=9.6)
128     fig.suptitle("Fig.D2 四类时序变量的季节分布 (小提琴 + 箱线)",
129                 fontsize=12, fontweight="bold", y=1.0)
130     vs.save(fig, "figD2_distributions",
131            "边缘分布的季节差异。负荷、光伏、电价与净负荷在四个季节的分布形态，"
132            "说明夏季光伏强、冬夏电价分布更重尾，且净负荷存在大量负值（光伏倒送被弃光吸收）。")
133
134
135 # D3 季节热力图 (小时 × 月份)
136 def figD3_seasonal_heatmaps(panel) -> None:
137     pv = panel.pv_filled()
138     items = [("负荷 (kW)", panel.load), ("光伏 (kW)", pv),
139             ("电价 (元/kWh)", panel.price), ("净负荷 (kW)", panel.net_load())]
140     months = _season_index(panel.dates)
141     fig, axes = plt.subplots(2, 2, figsize=(13.6, 6.6))
142     for ax, (label, values) in zip(axes.ravel(), items):
143         grid = np.zeros((12, 144))
144         for m in range(12):
145             sel = months == m
146             grid[m] = np.nanmean(values[sel], axis=0)
147         im = ax.imshow(grid, aspect="auto", origin="lower", cmap=vs.SEQ_CMAP,
148                       extent=[0, 24, 0.5, 12.5])
149         ax.set_xticks(np.arange(0, 25, 3))
150         ax.set_yticks(np.arange(1, 13))
151         ax.set_xlabel("时刻 (h)")
152         ax.set_ylabel("月份")
153         ax.set_title(f"{label}", fontsize=9.8)
154         ax.grid(False)
155         cb = fig.colorbar(im, ax=ax, pad=0.02, fraction=0.045)
156         cb.ax.tick_params(labelsize=7.6)
157     for ax, letter in zip(axes.ravel(), "abcd"):
158         vs.panel(ax, letter, dx=-0.10)
159     fig.suptitle("Fig.D3 季节 × 时刻热力图: 负荷 / 光伏 / 电价 / 净负荷的时空结构",
160                 fontsize=12, fontweight="bold", y=1.0)
161     vs.save(fig, "figD3_seasonal_heatmaps",
162            "时空结构热力图。四类变量的月度小时均值矩阵: 光伏呈夏季宽、冬季窄的钟形;"
163            "电价早晚双峰且夏季午间低谷更深; 净负荷在春夏季中午出现负值 (光伏盈余)。")
164
165
166 # D4 依赖结构 (相关 / Copula / 自相关)
167 def figD4_dependence(panel) -> None:
168     pv = panel.pv_filled()
169     load = panel.load
170     price = panel.price
171     net = panel.net_load()
172     ramp = np.abs(np.diff(pv, axis=1, prepend=pv[:, :1]))
173     stack = np.stack([load.ravel(), pv.ravel(), price.ravel(), net.ravel(), ramp.ravel()])
174     names = ["负荷", "光伏", "电价", "净负荷", "ΔPV"]
175     corr = np.corrcoef(np.argsort(np.argsort(stack, axis=1), axis=1))
176
177     # 秩相关 (Spearman) 直接用排名算, 不假设线性
178     fig = plt.figure(figsize=(14.4, 5.2))
179     gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.1, 1.1], wspace=0.30)
180     ax = fig.add_subplot(gs[0, 0])
181     im = ax.imshow(corr, cmap="RdBu_r", vmin=-1, vmax=1)
182     ax.set_xticks(range(len(names)), names, rotation=30, ha="right", fontsize=8.4)
183     ax.set_yticks(range(len(names)), names, fontsize=8.4)

```

```

184     for i in range(len(names)):
185         for j in range(len(names)):
186             ax.text(j, i, f"{corr[i, j]:.2f}", ha="center", va="center", fontsize=8.0,
187                     color="white" if abs(corr[i, j]) > 0.6 else BLACK)
188     fig.colorbar(im, ax=ax, pad=0.02, fraction=0.046).set_label("Spearman 相关系数")
189     ax.grid(False)
190     vs.panel(ax, "a", dx=-0.16)
191     ax.set_title("Spearman 相关矩阵")
192
193     ax = fig.add_subplot(gs[0, 1])
194     hb = ax.hexbin(pv.ravel()[::7], net.ravel()[::7], gridsize=48, cmap="magma",
195                   bins="log", mincnt=1)
196     ax.axhline(0, color=WHITE_LINE, lw=1.0)
197     ax.set_xlabel("光伏出力 (kW)")
198     ax.set_ylabel("净负荷 $L-PV$ (kW)")
199     fig.colorbar(hb, ax=ax, pad=0.02, label="$\\log_{10}$ 样本数")
200     vs.panel(ax, "b", dx=-0.14)
201     ax.set_title("光伏-净负荷联合密度 (倒 V 结构)")
202
203     ax = fig.add_subplot(gs[0, 2])
204     for j, (vintage, color) in enumerate(zip((0, 1, 2, 3), PALETTE)):
205         errors = []
206         for d in range(31, 365):
207             fc = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[d, vintage],
208                                                         VINTAGE_HOUR[vintage]), nan=0.0)
209             mask = fc > 0
210             if mask.any():
211                 errors.append((pv[d][mask] - fc[mask]).mean())
212         errors = np.array(errors)
213         ax.hist(errors, bins=40, histtype="step", lw=1.5, color=color,
214               label=f"{VINTAGE_HOUR[vintage]}:00 发布 均值 {errors.mean():+.0f} kW")
215     ax.axvline(0, color=BLACK, lw=0.9)
216     ax.set_xlabel("日间平均预测误差 $PV-\\hat{PV}$ (kW)")
217     ax.set_ylabel("天数")
218     ax.legend(fontsize=8.0)
219     vs.panel(ax, "c", dx=-0.14)
220     ax.set_title("四时次预报误差分布 (偏差诊断)")
221     fig.suptitle("Fig.D4 依赖结构与预报偏差: 相关、联合密度与误差分布",
222                 fontsize=12, fontweight="bold", y=1.0)
223     vs.save(fig, "figD4_dependence",
224            "依赖结构诊断。(a) 变量间的 Spearman 秩相关; "
225            "(b) 光伏与净负荷的联合密度呈倒 V (光伏出力直接抵消负荷); "
226            "(c) 四个发布时次的日间平均预测误差分布, 用于判断是否存在系统性偏差。")
227
228
229 # D5 爬坡与极值
230 def figD5_ramps_extremes(panel) -> None:
231     pv = panel.pv_filled()
232     load = panel.load
233     price = panel.price
234     ramp_pv = np.diff(pv, axis=1) / (1 / 6)
235     # 爬坡率换算成 kW/h: 10 分钟的差值除以 1/6 小时
236     ramp_load = np.diff(load, axis=1) / (1 / 6)
237     spike = price >= np.quantile(price, 0.95, axis=1, keepdims=True)
238     surplus = pv > load
239     fig = plt.figure(figsize=(14.0, 5.0))
240     gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.0, 1.0], wspace=0.28)
241     ax = fig.add_subplot(gs[0, 0])
242     for values, label, color in ((ramp_pv, "光伏爬坡", ORANGE), (ramp_load, "负荷爬坡", RED)):
243         v = np.sort(np.abs(values).ravel())
244         v = v[np.isfinite(v)]
245         ccdf = 1.0 - np.arange(v.size) / v.size
246         ax.plot(v, ccdf, color=color, lw=1.5, label=label)
247     ax.set_xscale("log")
248     ax.set_yscale("log")
249     ax.set_xlabel("10 min 爬坡幅值 (kW)")
250     ax.set_ylabel("P(|ΔP| > x)")
251     ax.legend()
252     vs.panel(ax, "a")
253     ax.set_title("爬坡率互补累积分布 (CCDF)")

```

```

254
255 ax = fig.add_subplot(gs[0, 1])
256 by_hour = spike.reshape(365, 24, 6).sum(axis=2).sum(axis=0)
257 bars = ax.bar(np.arange(24), by_hour, color=PURPLE, width=0.7)
258 top = np.argsort(by_hour)[-3:]
259 for h in top:
260     bars[h].set_color(RED)
261 ax.set_xlabel("时刻 (h)")
262 ax.set_ylabel("尖峰时段数 (全年)")
263 vs.panel(ax, "b")
264 ax.set_title(f"电价尖峰 (≥P95) 的时段分布: 集中在 "
265             + ", ".join(f"{h}:00" for h in sorted(top)))
266
267 ax = fig.add_subplot(gs[0, 2])
268 surplus_hour = surplus.reshape(365, 24, 6).sum(axis=2).sum(axis=0)
269 bars = ax.bar(np.arange(24), surplus_hour, color=GREEN, width=0.7)
270 ax.set_xlabel("时刻 (h)")
271 ax.set_ylabel("光伏盈余时段数 (全年)")
272 ax.annotate(f"全年盈余时段 {int(surplus.sum()):,} (占 {100*surplus.mean():.1f}%) ",
273            (0.5, 0.9), xycoords="axes fraction", ha="center", fontsize=8.6,
274            bbox=dict(boxstyle="round,pad=0.3", fc="white", ec=GREY, lw=0.6))
275 vs.panel(ax, "c")
276 ax.set_title("光伏盈余时段分布 (决定弃光与倒送风险)")
277 fig.suptitle("Fig.D5 爬坡与极值特征: 波动强度、价格尖峰与光伏盈余",
278            fontsize=12, fontweight="bold", y=1.0)
279 vs.save(fig, "figD5_ramps_extremes",
280         "波动与极值特征。(a) 光伏/负荷爬坡幅值的 CCDF (双对数), 用于评估备用需求; "
281         "(b) 电价尖峰时刻分布; (c) 光伏盈余时段分布, 说明弃光与倒送风险的时段来源。")
282
283
284 # D6 预报误差诊断
285 def figD6_forecast_error(panel) -> None:
286     pv = panel.pv_filled()
287     leads = np.arange(1, 25)
288     bias = np.zeros((4, 24))
289     rmse = np.zeros((4, 24))
290     for j in range(4):
291         errs = []
292         for d in range(31, 365):
293             fc_hourly = panel.pv_forecast[d, j]
294             fc_slots = np.nan_to_num(hourly_forecast_to_slots(fc_hourly, VINTAGE_HOUR[j]), nan
295                                     =0.0)
296             actual = pv[d]
297             for lead in leads:
298                 slot = (VINTAGE_HOUR[j] + lead) - 1
299                 if 1 <= slot <= 144:
300                     k = slot - 1
301                     errs.append((lead, actual[k] - fc_slots[k]))
302             for lead in leads:
303                 vals = np.array([v for (l, v) in errs if l == lead])
304                 if vals.size:
305                     bias[j, lead - 1] = vals.mean()
306                     rmse[j, lead - 1] = np.sqrt((vals ** 2).mean())
307
308 fig = plt.figure(figsize=(14.0, 5.0))
309 gs = GridSpec(1, 3, figure=fig, width_ratios=[1.15, 1.0, 1.0], wspace=0.28)
310 ax = fig.add_subplot(gs[0, 0])
311 im = ax.imshow(rmse, aspect="auto", cmap="magma", origin="lower",
312               extent=[0.5, 24.5, -0.5, 3.5])
313 ax.set_yticks(range(4), [f"{h}:00 发布" for h in VINTAGE_HOUR])
314 ax.set_xlabel("预报提前量 (h)")
315 ax.set_ylabel("发布时次")
316 fig.colorbar(im, ax=ax, pad=0.02, label="RMSE (kW)")
317 ax.grid(False)
318 vs.panel(ax, "a", dx=-0.16)
319 ax.set_title("误差随提前量的增长 (RMSE 热力图)")
320
321 ax = fig.add_subplot(gs[0, 1])
322 for j, color in enumerate(PALETTE[:4]):
323     ax.plot(leads, bias[j], "o-", ms=3.4, lw=1.3, color=color,

```

```

323         label=f"{VINTAGE_HOUR[j]}:00 发布")
324     ax.axhline(0, color=BLACK, lw=0.9)
325     ax.set_xlabel("预报提前量 (h)")
326     ax.set_ylabel("平均偏差 (kW)")
327     ax.legend(fontsize=7.8, ncol=2)
328     vs.panel(ax, "b")
329     ax.set_title("系统偏差诊断: 均接近零 (无系统性高估/低估)")
330
331     ax = fig.add_subplot(gs[0, 2])
332     for j, color in zip((0, 1, 2, 3), PALETTE[:4]):
333         errs = []
334         for d in range(31, 365):
335             fc = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[d, j],
336                                                         VINTAGE_HOUR[j]), nan=0.0)
337             mask = fc > 100
338             if mask.any():
339                 errs.append(np.abs(pv[d][mask] - fc[mask]).mean())
340         errs = np.array(errs)
341         if errs.size == 0:
342             ax.annotate(f"{VINTAGE_HOUR[j]}:00 发布: 可见时段全在夜间\n无日间样本",
343                        (0.5, 0.5 - 0.12 * j), xycoords="axes fraction", ha="center",
344                        fontsize=8.0, color=color)
345             continue
346         ax.hist(errs, bins=32, histtype="step", lw=1.5, color=color,
347                label=f"{VINTAGE_HOUR[j]}:00 中位 {np.median(errs):.0f} kW")
348     ax.set_xlabel("日间平均绝对误差 MAE (kW)")
349     ax.set_ylabel("天数")
350     ax.legend(fontsize=7.8)
351     vs.panel(ax, "c")
352     ax.set_title("各时次的日间 MAE 分布")
353     fig.suptitle("Fig.D6 预报误差诊断: 提前量、偏差与技能 (数据特征)",
354                 fontsize=12, fontweight="bold", y=1.0)
355     vs.save(fig, "figD6_forecast_error",
356            "预报误差诊断。(a) 四个发布时次的 RMSE 随提前量的变化;"
357            "(b) 平均偏差随提前量 (接近零, 说明预报无系统性偏差);"
358            "(c) 日间 MAE 的天数分布, 为共形校准的窗口选择提供依据。")
359
360
361 EDA_FIGS = [
362     ("data_quality", figD1_data_quality, True),
363     ("distributions", figD2_distributions, True),
364     ("seasonal", figD3_seasonal_heatmaps, True),
365     ("dependence", figD4_dependence, True),
366     ("ramps", figD5_ramps_extremes, True),
367     ("forecast_error", figD6_forecast_error, True),
368 ]

```

code/report/figure_data.py

```

1  """Data preparation for the paper figure suite required by `图片.md`.
2
3  Only the *visualisation* layer changes here: every array is either read from
4  `results/` or recomputed by calling the existing model functions unchanged
5  (`code.optimize` / `code.forecast` are imported, never edited). Expensive products
6  are cached in `results/paper_fig_data.json` so the figures are reproducible without
7  re-solving.
8
9  Figure map (图片.md):
10 Fig.0 framework      -> results 直接绘制 (无新数据)
11 Fig.1 dispatch+SOC   -> q1.json / q1_slots.csv
12 Fig.2 shadow price   -> q1_slots.csv ( $\mu_t$ ) + KKT 阈值判定
13 Fig.3 prediction fan -> conformal bands (本模块计算)
14 Fig.4 reliability/sharpness -> 本模块计算 (4 种方法 × 多名义覆盖率)
15 Fig.5 V0I boundary   -> q3_days.csv (S0-S4) + q3 plan/adjust 数组
16 Fig.6 rolling update -> q3 arrays
17 Fig.7 RT control + SOC -> q4_2 / q4_3 arrays
18 Fig.8 cost-risk-degradation Pareto -> 本模块运行  $\beta$  扫描
19 Fig.9 ablation heatmap -> M0-M4 指标矩阵 (本模块计算 + 已有结果)
20 """

```

```

21 from __future__ import annotations
22
23 import csv
24 import json
25 from pathlib import Path
26
27 import numpy as np
28
29 from code.common.load_clean import ROOT, load_panel
30 from code.forecast.conformal import conformal_level, weighted_quantile
31 from code.optimize.degradation import DegradationParams, evaluate_panel, sensitivity as deg_sens
32 from code.optimize.lp_common import Storage, solve_stage_lp, solve_two_stage_lp
33 from code.optimize.q3_mpc import VINTAGE_HOUR, hourly_forecast_to_slots
34
35 RESULTS = ROOT / "results"
36 ARRAYS = RESULTS / "arrays"
37 CACHE = RESULTS / "paper_fig_data.json"
38 FIRST_DAY = 31
39 LOOKBACK = 30
40
41
42 def load_json(path: Path) -> dict:
43     return json.loads(Path(path).read_text(encoding="utf-8"))
44
45
46 def arr(name: str) -> np.ndarray:
47     return np.load(ARRAYS / name)
48
49
50 def read_days_csv(tag: str) -> list[dict]:
51     with (RESULTS / f"{tag}_days.csv").open(encoding="utf-8", newline="") as fh:
52         return list(csv.DictReader(fh))
53
54
55 def _cache() -> dict:
56     return load_json(CACHE) if CACHE.exists() else {}
57
58
59 def _update(**blocks) -> dict:
60     data = _cache()
61     data.update(blocks)
62     CACHE.write_text(json.dumps(data, ensure_ascii=False, indent=2), encoding="utf-8")
63     return data
64
65
66 # Fig.2 影子价格 / KKT 耦合
67 def shadow_price_coupling(panel) -> dict:
68     """Verified marginal-value statements for the Q1 optimum (KKT interpretation).
69
70     Facts checked here (all verifiable from the delivered solution bundle):
71     1. complementary slackness of the purchase variable:  $q_t > 0 \Rightarrow \mu_t = p_t$ 
72       (and  $\mu_t \leq p_t$  everywhere);
73     2. every discharging slot satisfies  $p_t \geq \mu_t$  (the market price is above the
74       marginal value of stored energy);
75     3. charging concentrates in the cheapest hours (median market price well below
76       the discharging median).
77     The *full* storage KKT also contains the SoC dual  $\lambda$ , which is not part of the
78     delivered bundle, so the figure reports the verified statements only.
79     """
80     q1 = load_json(RESULTS / "q1.json")
81     st = Storage.from_config(panel.config)
82     price = panel.day1[:, 0]
83     q = np.array(q1["lp"]["q"])
84     charge = np.array(q1["lp"]["c"])
85     disch = np.array(q1["lp"]["d"])
86     mu = np.array(q1["lp"]["mu"])
87     tol = 1e-6 * float(np.max(price))
88     purchase = q > 1e-9
89     ch = charge > 1e-6
90     dis = disch > 1e-6

```

```

91 mu_eq_p = int(np.sum(np.abs(mu[purchase] - price[purchase]) <= tol))
92 mu_violations = int(np.sum(mu > price + tol))
93 dis_above = int(np.sum(price[dis] >= mu[dis] - tol))
94 return {
95     "price": price.tolist(), "mu": mu.tolist(),
96     "charge": charge.tolist(), "discharge": disch.tolist(),
97     "purchase_slots": int(purchase.sum()),
98     "mu_eq_price_slots": mu_eq_p,
99     "mu_above_price_violations": mu_violations,
100     "charge_slots": int(ch.sum()),
101     "discharge_slots": int(dis.sum()),
102     "discharge_price_above_mu": dis_above,
103     "charge_price_median": float(np.median(price[ch])) if ch.any() else float("nan"),
104     "discharge_price_median": float(np.median(price[dis])) if dis.any() else float("nan"),
105     "discharge_ratio_median": float(np.median(price[dis] / mu[dis])) if dis.any() else float
    ("nan"),
106     "charge_ratio_median": float(np.median(price[ch] / mu[ch])) if ch.any() else float("nan
    "),
107     "eta_c": st.eta_c, "eta_d": st.eta_d,
108 }
109
110
111 # Fig.3 预测扇形图 + ACI 带
112 def prediction_fan(panel, day_iso: str = "2025-06-21", level: float = 0.90,
113     bands=(0.50, 0.80, 0.90, 0.95)) -> dict:
114     """PV prediction intervals for one day (0:00 vintage) + ACI coverage trace."""
115     pv = panel.pv_filled()
116     d = panel.dates.index(day_iso)
117     forecast = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[d, 0], VINTAGE_HOUR[0]),
118         nan=0.0)
119     win = slice(max(0, d - LOOKBACK), d)
120     fc_hist = np.array([np.nan_to_num(
121         hourly_forecast_to_slots(panel.pv_forecast[t, 0], VINTAGE_HOUR[0]), nan=0.0)
122         for t in range(win.start, win.stop)])
123     residuals = pv[win] - fc_hist
124
125     alpha = 1.0 - level
126     aci_days, aci_miss, aci_level = [], [], []
127     for day in range(FIRST_DAY, 365):
128         w = slice(max(0, day - LOOKBACK), day)
129         fch = np.array([np.nan_to_num(
130             hourly_forecast_to_slots(panel.pv_forecast[t, 0], VINTAGE_HOUR[0]), nan=0.0)
131             for t in range(w.start, w.stop)])
132         res = pv[w] - fch
133         lvl = conformal_level(1.0 - alpha, res.shape[0])
134         upper = np.quantile(res, lvl, axis=0)
135         k_rise = int(panel.windows["k_rise"][day]) - 1
136         k_set = int(panel.windows["k_set"][day]) - 1
137         mask = np.zeros(144, dtype=bool)
138         mask[max(k_rise, 0):k_set + 1] = True
139         fcn = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[day, 0],
140             VINTAGE_HOUR[0]), nan=0.0)
141         miss = float(np.mean((pv[day] - fcn > upper)[mask]))
142         aci_days.append(panel.dates[day])
143         aci_miss.append(miss)
144         aci_level.append(lvl)
145         alpha = float(np.clip(alpha + 0.05 * ((1.0 - level) - miss), 0.001, 0.999))
146
147     bands_out = {}
148     for b in bands:
149         lvl = conformal_level(b, residuals.shape[0])
150         bands_out[f"q{int(b*100)}"] = {
151             "lower": (forecast + np.quantile(residuals, 1.0 - lvl, axis=0)).tolist(),
152             "upper": (forecast + np.quantile(residuals, lvl, axis=0)).tolist(),
153         }
154     lvl_aci = aci_level[panel.dates.index(day_iso) - FIRST_DAY] if d >= FIRST_DAY else 0.9
155     upper_aci = forecast + np.quantile(residuals, lvl_aci, axis=0)
156     lower_aci = forecast + np.quantile(residuals, 1.0 - lvl_aci, axis=0)
157     return {
158         "date": day_iso, "forecast": forecast.tolist(), "actual": pv[d].tolist(),

```

```

159     "bands": bands_out,
160     "aci": {"level": float(lvl_aci), "lower": lower_aci.tolist(),
161            "upper": upper_aci.tolist(),
162            "violations": (pv[d] > upper_aci).tolist()},
163     "aci_trace": {"dates": aci_days, "miss": aci_miss, "level": aci_level},
164     "achieved_coverage": float(1.0 - np.mean(aci_miss)),
165     "mean_miss_rate": float(np.mean(aci_miss)),
166     "target_miss_rate": float(1.0 - level),
167     "days_miss_below_target": float(np.mean(np.array(aci_miss) <= 1.0 - level)),
168 }
169
170
171 # Fig.4 可靠性图 + 锐度
172 def reliability_sharpness(panel, levels=(0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95),
173                          n_boot: int = 200, seed: int = 20250911) -> dict:
174     """Empirical coverage (reliability) and mean interval width (sharpness) per method."""
175     pv = panel.pv_filled()
176     rng = np.random.default_rng(seed)
177     method_names = ["经验分位 (无修正)", "split 共形 (有限样本修正)", "ACI 自适应",
178                    "Bootstrap 分位均值"]
179     coverage = {name: [] for name in method_names}
180     widths = {name: [] for name in method_names}
181     windows = []
182     for day in range(FIRST_DAY, 365):
183         w = slice(max(0, day - LOOKBACK), day)
184         fch = np.array([np.nan_to_num(
185             hourly_forecast_to_slots(panel.pv_forecast[t, 0], VINTAGE_HOUR[0]), nan=0.0)
186             for t in range(w.start, w.stop)])
187         fcn = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[day, 0],
188                                                    VINTAGE_HOUR[0]), nan=0.0)
189         k_rise = int(panel.windows["k_rise"][day]) - 1
190         k_set = int(panel.windows["k_set"][day]) - 1
191         mask = np.zeros(144, dtype=bool)
192         mask[max(k_rise, 0):k_set + 1] = True
193         if mask.sum() == 0:
194             continue
195         windows.append((pv[w] - fch, pv[day] - fcn, mask))
196
197     n_bag = 15
198     for level in levels:
199         alpha = 1.0 - level
200         cov = {name: [] for name in method_names}
201         wid = {name: [] for name in method_names}
202         for res, actual_res, mask in windows:
203             n = res.shape[0]
204             split_lvl = conformal_level(level, n)
205             aci_lvl = conformal_level(1.0 - alpha, n)
206             bounds = {
207                 "经验分位 (无修正)": np.quantile(res, level, axis=0),
208                 "split 共形 (有限样本修正)": np.quantile(res, split_lvl, axis=0),
209                 "ACI 自适应": np.quantile(res, aci_lvl, axis=0),
210             }
211             for name, upper in bounds.items():
212                 lower = np.quantile(res, 1.0 - conformal_level(level, n), axis=0)
213                 cov[name].append(float(np.mean((actual_res <= upper)[mask])))
214                 wid[name].append(float(np.mean((upper - lower)[mask])))
215             draws = np.empty((n_bag, 144))
216             for b in range(n_bag):
217                 idx = rng.integers(0, res.shape[0], res.shape[0])
218                 draws[b] = np.quantile(res[idx], conformal_level(level, res.shape[0]), axis=0)
219             boot_upper = draws.mean(axis=0)
220             boot_lower = np.quantile(res, 1.0 - conformal_level(level, res.shape[0]), axis=0)
221             cov["Bootstrap 分位均值"].append(float(np.mean((actual_res <= boot_upper)[mask])))
222             wid["Bootstrap 分位均值"].append(float(np.mean((boot_upper - boot_lower)[mask])))
223             miss = 1.0 - cov["ACI 自适应"][-1]
224             alpha = float(np.clip(alpha + 0.05 * ((1.0 - level) - miss), 0.001, 0.999))
225         for name in method_names:
226             coverage[name].append(float(np.mean(cov[name])))
227             widths[name].append(float(np.mean(wid[name])))
228     return {"levels": list(levels), "methods": coverage, "widths": widths}

```

```

229
230
231 # Fig.5/6 V0I 决策边界与滚动更新轨迹
232 def voi_boundary(panel) -> dict:
233     """V0I_k vs adjustment penalty C_adjust,k for the three update vintages."""
234     rows = read_days_csv("q3")
235     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
236     plan, adj = arr("q3_q_plan.npy"), arr("q3_q_adjust.npy")
237     segments = {6: (36, 72), 12: (72, 108), 18: (108, 144)}
238     out = {6: [], 12: [], 18: []}
239     for i, row in enumerate(rows):
240         s0, s1, s2, s3 = (float(row[k]) for k in ("S0", "S1", "S2", "S3"))
241         voi = {6: s0 - s1, 12: s1 - s2, 18: s2 - s3}
242         p_row = price[FIRST_DAY + i]
243         for stage, (a, b) in segments.items():
244             penalty = float(0.5 * np.sum(p_row[a:b] * np.maximum(0.0, plan[i][a:b] - adj[i][a:b]
245                 )))
246                 + 1.5 * np.sum(p_row[a:b] * np.maximum(0.0, adj[i][a:b] - plan[i][a:
247                     b])))
248             out[stage].append({"date": row["date"], "voi": voi[stage], "penalty": penalty,
249                 "net": voi[stage] - penalty})
250     summary = {}
251     for stage, items in out.items():
252         voi = np.array([it["voi"] for it in items])
253         pen = np.array([it["penalty"] for it in items])
254         trigger = voi > pen
255         summary[stage] = {
256             "voi_mean": float(voi.mean()), "penalty_mean": float(pen.mean()),
257             "trigger_rate": float(trigger.mean()),
258             "net_mean": float((voi - pen).mean()),
259             "voi_total": float(voi.sum()), "penalty_total": float(pen.sum()),
260         }
261     return {"stages": {str(k): v for k, v in out.items()}, "summary": summary}
262
263 def rolling_update_example(panel, day_iso: str = "") -> dict:
264     """Plan / updated / realised trajectories with trigger annotations for one day."""
265     rows = read_days_csv("q3")
266     dates = [r["date"] for r in rows]
267     if not day_iso:
268         voi = np.array([float(r["S0"]) - float(r["S3"]) for r in rows])
269         day_iso = dates[int(np.argmax(voi))]
270     i = dates.index(day_iso)
271     plan, adj, g = arr("q3_q_plan.npy")[i], arr("q3_q_adjust.npy")[i], arr("q3_emergency.npy")[i]
272     charge, disch = arr("q3_charge.npy")[i], arr("q3_discharge.npy")[i]
273     soc = arr("q3_soc.npy")[i]
274     need = panel.net_load()[FIRST_DAY + i] * (1 / 6) + charge - disch
275     s0, s1, s2, s3 = (float(rows[i][k]) for k in ("S0", "S1", "S2", "S3"))
276     return {
277         "date": day_iso,
278         "plan": plan.tolist(), "adjusted": adj.tolist(), "realised": (adj + g).tolist(),
279         "demand": need.tolist(), "soc": soc.tolist(),
280         "triggers": [{"slot": 36, "label": "6:00", "voi": s0 - s1},
281             {"slot": 72, "label": "12:00", "voi": s1 - s2},
282             {"slot": 108, "label": "18:00", "voi": s2 - s3}],
283         "day_subset_costs": {"S0": s0, "S1": s1, "S2": s2, "S3": s3},
284     }
285
286 # Fig.7 实时控制 vs 承诺基线 (Q4-2 / Q4-3)
287 def realtime_control(panel, day_iso: str = "") -> dict:
288     rows = read_days_csv("q4_3")
289     dates = [r["date"] for r in rows]
290     if not day_iso:
291         cost = np.array([float(r["cost_cny"]) for r in rows])
292         day_iso = dates[int(np.argmax(cost))]
293     i = dates.index(day_iso)
294     base_c, base_d = arr("q4_2_charge.npy")[i], arr("q4_2_discharge.npy")[i]
295     mpc_c, mpc_d = arr("q4_3_charge.npy")[i], arr("q4_3_discharge.npy")[i]

```

```

296 soc_base, soc_mpc = arr("q4_2_soc.npy")[i], arr("q4_3_soc.npy")[i]
297
298 def total_variation(x):
299     return float(np.abs(np.diff(x)).sum())
300
301 return {
302     "date": day_iso,
303     "price": panel.price[FIRST_DAY + i].tolist(),
304     "load": panel.load[FIRST_DAY + i].tolist(),
305     "pv": panel.pv_filled()[FIRST_DAY + i].tolist(),
306     "baseline": {"charge": base_c.tolist(), "discharge": base_d.tolist(),
307                 "net": (base_c - base_d).tolist(), "soc": soc_base.tolist(),
308                 "tv": total_variation(base_c - base_d)},
309     "mpc": {"charge": mpc_c.tolist(), "discharge": mpc_d.tolist(),
310            "net": (mpc_c - mpc_d).tolist(), "soc": soc_mpc.tolist(),
311            "tv": total_variation(mpc_c - mpc_d)},
312     "soc_limits": [float(panel.config["storage"]["soc_min_kwh"]),
313                   float(panel.config["storage"]["soc_max_kwh"])],
314 }
315
316
317 # Fig.8 成本—风险—退化 Pareto
318 def pareto_cost_risk_degradation(panel, betas=(0.0, 0.25, 0.5, 0.75, 0.9),
319                                  step: int = 5) -> dict:
320     """β sweep under the 附件4 tariff with rainflow-based SOH (real degradation model)."""
321     from code.optimize.q2_stochastic import solve_q2
322
323     st = Storage.from_config(panel.config)
324     days = list(range(FIRST_DAY, 365, step))
325     params = DegradationParams(capacity_kwh=st.capacity, window_days=len(days))
326     rows = []
327     for beta in betas:
328         run = solve_q2(panel, panel.price, tag=f"pfig_beta{beta}", days=days,
329                        cvar_beta=beta, cvar_alpha=0.95)
330         daily = np.array([r["cost_cny"] for r in run["records"]])
331         charge = run["arrays"]["charge"]
332         disch = run["arrays"]["discharge"]
333         cycles = float((charge.sum() + disch.sum()) / (2.0 * st.capacity))
334         deg = evaluate_panel(run["arrays"]["soc"], params)
335         tail = int(max(1, np.ceil(0.05 * daily.size)))
336         rows.append({
337             "beta": beta, "strategy": "CVaR" if beta > 0 else "风险中性",
338             "cost_cny": float(daily.sum()),
339             "cvar95_cny": float(np.sort(daily)[-tail:].mean()),
340             "degradation_cycles": cycles,
341             "annual_fade_pct": deg.annual_fade_pct,
342             "soh_end_pct": 100.0 * deg.soh_end,
343             "degradation_cost_cny": deg.degradation_cost_cny,
344             "rainflow_cycles": deg.rainflow_cycles,
345             "mean_dod": deg.mean_dod,
346             "emergency_kwh": float(sum(r["emergency_energy_kwh"] for r in run["records"])),
347             "days": len(days),
348         })
349     return {"rows": rows, "days": len(days), "step": step}
350
351
352 def degradation_priced(panel, prices=(0.0, 0.05, 0.1, 0.2, 0.4), step: int = 5) -> dict:
353     """Degradation-aware strategies: price battery wear, then measure the real SOH.
354
355     `prices` are CNY per kWh cycled, added to the LP objective; the resulting schedules
356     are evaluated with the rainflow/ Wöhler model, so this is the genuine
357     cost-vs-degradation frontier (not a proxy).
358     """
359     from code.optimize.q2_stochastic import solve_q2
360
361     st = Storage.from_config(panel.config)
362     days = list(range(FIRST_DAY, 365, step))
363     params = DegradationParams(capacity_kwh=st.capacity, window_days=len(days))
364     rows = []
365     for rho in prices:

```

```

366     run = solve_q2(panel, panel.price, tag=f"pfig_deg{rho}", days=days,
367                   throughput_penalty=rho)
368     daily = np.array([r["cost_cny"] for r in run["records"]])
369     deg = evaluate_panel(run["arrays"]["soc"], params)
370     tail = int(max(1, np.ceil(0.05 * daily.size)))
371     rows.append({
372         "rho_cny_per_kwh": rho,
373         "electricity_cost_cny": float(daily.sum()),
374         "cvar95_cny": float(np.sort(daily)[-tail:].mean()),
375         "annual_fade_pct": deg.annual_fade_pct,
376         "degradation_cost_cny": deg.degradation_cost_cny,
377         "total_cost_cny": float(daily.sum()) + deg.degradation_cost_cny,
378         "rainflow_cycles": deg.rainflow_cycles,
379         "throughput_kwh": float(run["arrays"]["charge"].sum() + run["arrays"]["discharge"].
sum()),
380         "emergency_kwh": float(sum(r["emergency_energy_kwh"] for r in run["records"])),
381         "days": len(days),
382     })
383     return {"rows": rows, "days": len(days), "step": step,
384           "note": ("p 为单位吞吐的退化价格; 年衰减由雨流计数 + Wöhler 寿命曲线给出, "
385                  "退化成本 = 年衰减比例 × 电池组资本成本 (800 元/kWh × 12000 kWh) ")}
386
387
388 def degradation_sensitivity(panel, step: int = 5) -> dict:
389     """SOH sensitivity to N100 and the Wöhler exponent on a representative trajectory."""
390     st = Storage.from_config(panel.config)
391     days = list(range(FIRST_DAY, 365, step))
392     soc = np.load(ARRAYS / "q4_3_soc.npy") if (ARRAYS / "q4_3_soc.npy").exists() else None
393     if soc is None:
394         soc = np.load(ARRAYS / "q3_soc.npy")
395     if soc.shape[0] != len(days):
396         idx = np.linspace(0, soc.shape[0] - 1, len(days)).round().astype(int)
397         soc = soc[idx]
398     params = DegradationParams(capacity_kwh=st.capacity, window_days=len(days))
399     base = evaluate_panel(soc, params)
400     return {"base": {"annual_fade_pct": base.annual_fade_pct,
401                     "soh_end_pct": 100.0 * base.soh_end,
402                     "degradation_cost_cny": base.degradation_cost_cny,
403                     "rainflow_cycles": base.rainflow_cycles,
404                     "mean_dod": base.mean_dod, "max_dod": base.max_dod},
405           "grid": deg_sens(params, soc), "days": len(days),
406           "source": "q4_3 rolling-MPC SoC trajectory (sub-sampled to the Pareto sample)"}
407
408
409 # Fig.9 消融矩阵 M0-M4
410 def ablation_matrix(panel, cvar_beta: float = 0.5) -> dict:
411     """Module ladder under one tariff: deterministic → conformal → SAA → rolling → CVaR."""
412     from code.optimize.q2_stochastic import solve_q2
413
414     st = Storage.from_config(panel.config)
415     tariff = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
416     emergency = float(panel.config["tariff"]["emergency_multiplier"])
417     net = panel.net_load()
418     bench = load_json(RESULTS / "compute_benchmarks.json") if (RESULTS / "compute_benchmarks.
json").exists() else {}
419     stage_time = {s["stage"]: s["seconds"] for s in bench.get("pipeline", [])}
420
421     def metrics(daily_cost, charge, disch, g, curtail):
422         daily = np.asarray(daily_cost, dtype=float)
423         tail = int(max(1, np.ceil(0.05 * daily.size)))
424         return {
425             "cost_wan": float(daily.sum()) / 1e4,
426             "cvar95_cny": float(np.sort(daily)[-tail:].mean()),
427             "emergency_wan_kwh": float(np.sum(g)) / 1e4,
428             "degradation_cycles": float((np.sum(charge) + np.sum(disch)) / (2.0 * st.capacity)),
429             "curtail_wan_kwh": float(np.sum(curtail)) / 1e4,
430         }
431
432     deg_params = DegradationParams(capacity_kwh=st.capacity,
433                                   window_days=365 - FIRST_DAY)

```

```

434
435 import time
436 t0 = time.perf_counter()
437 m0 = {"cost": [], "charge": [], "disch": [], "g": [], "curtail": []}
438 m1 = {k: [] for k in m0}
439 for d in range(FIRST_DAY, 365):
440     win = net[max(0, d - LOOKBACK):d]
441     base = win.mean(axis=0)
442     det = solve_stage_lp(base, tariff[d], st)
443     sample = base[None, :] * st.dt + det["c"] - det["d"] + (win - base) * st.dt
444     need = net[d] * st.dt + det["c"] - det["d"]
445     lvl = conformal_level(0.80, sample.shape[0])
446     margin = np.quantile(sample, lvl, axis=0) - (base * st.dt + det["c"] - det["d"])
447     det_margin = solve_stage_lp(base + margin / st.dt, tariff[d], st)
448     for bucket, schedule in ((m0, det), (m1, det_margin)):
449         q = np.maximum(0.0, schedule["q"])
450         need_sched = net[d] * st.dt + schedule["c"] - schedule["d"]
451         g = np.maximum(0.0, need_sched - q)
452         bucket["cost"].append(float(np.sum(tariff[d] * q) + emergency * np.sum(tariff[d] * g
453     )))
454         bucket["charge"].append(schedule["c"])
455         bucket["disch"].append(schedule["d"])
456         bucket.setdefault("soc", []).append(schedule["soc"])
457         bucket["g"].append(g)
458         bucket["curtail"].append(np.maximum(0.0, schedule["d"] - schedule["c"]
459         - net[d] * st.dt))
460
461 cheap_seconds = time.perf_counter() - t0
462
463 q2_rows = read_days_csv("q2")
464 q3_rows = read_days_csv("q3")
465 m2 = metrics([float(r["cost_cny"]) for r in q2_rows], arr("q2_charge.npy"),
466             arr("q2_discharge.npy"), arr("q2_emergency.npy"), arr("q2_curtail.npy"))
467 m3 = metrics([float(r["cost_cny"]) for r in q3_rows], arr("q3_charge.npy"),
468             arr("q3_discharge.npy"), arr("q3_emergency.npy"), arr("q3_curtail.npy"))
469
470 t4 = time.perf_counter()
471 run4 = solve_q2(panel, tariff, tag="pfig_ablation_cvar", days=range(FIRST_DAY, 365),
472                 cvar_beta=cvar_beta, cvar_alpha=0.90)
473 m4_seconds = time.perf_counter() - t4
474 m4 = metrics([r["cost_cny"] for r in run4["records"]], run4["arrays"]["charge"],
475             run4["arrays"]["discharge"], run4["arrays"]["emergency"],
476             run4["arrays"]["curtail"])
477
478 rows = {"M0": metrics(m0["cost"], np.array(m0["charge"]), np.array(m0["disch"]),
479                     np.array(m0["g"]), np.array(m0["curtail"])),
480        "M1": metrics(m1["cost"], np.array(m1["charge"]), np.array(m1["disch"]),
481                     np.array(m1["g"]), np.array(m1["curtail"])),
482        "M2": m2, "M3": m3, "M4": m4}
483 soc_matrices = {"M0": np.array(m0["soc"]), "M1": np.array(m1["soc"]),
484                "M2": arr("q2_soc.npy"), "M3": arr("q3_soc.npy"),
485                "M4": run4["arrays"]["soc"]}
486 for key, soc in soc_matrices.items():
487     deg = evaluate_panel(soc, deg_params)
488     rows[key]["annual_fade_pct"] = deg.annual_fade_pct
489     rows[key]["soh_end_pct"] = 100.0 * deg.soh_end
490     rows[key]["degradation_cost_cny"] = deg.degradation_cost_cny
491     rows[key]["rainflow_cycles"] = deg.rainflow_cycles
492 times = {"M0": cheap_seconds / 2, "M1": cheap_seconds / 2,
493         "M2": stage_time.get("Q2 两阶段随机规划", float("nan")),
494         "M3": stage_time.get("Q3 滚动 MPC + S0-S5", float("nan")),
495         "M4": m4_seconds}
496 for key in rows:
497     rows[key]["solve_seconds"] = float(times[key])
498 return {"rows": rows,
499       "labels": {"M0": "M0 确定性均值计划", "M1": "M1 + 共形裕度(p*=0.80)",
500                "M2": "M2 + 场景SAA(两阶段)", "M3": "M3 + 多时次滚动MPC",
501                "M4": f"M4 = M2+CVaR( $\beta$ = {cvar_beta}) 风险分支"},
502       "ladder_note": ("M0-M1-M2-M3 为主链 (物理建模→共形裕度→场景随机规划→滚动更新) ; "
503                      "M4 由 M2 分支而来, 用于单独度量 CVaR 风险层的贡献"
504                      " (当前滚动 MPC 实现不含 CVaR 旋钮, 故不做 M3+CVaR)"),

```

```

503         "metric_labels": {"cost_wan": "费用\n(万元)", "cvar95_cny": "CVaR95\n(元/日)",
504                             "emergency_wan_kwh": "紧急购电\n(万kWh)",
505                             "annual_fade_pct": "SOH 年衰减\n(%)",
506                             "curtail_wan_kwh": "弃光\n(万kWh)",
507                             "solve_seconds": "求解耗时\n(s)"}
508
509
510 # 统一编排：把上面各图的绘图数据一次性算好并缓存
511 def build(force: bool = False, quick: bool = True, verbose: bool = True) -> dict:
512     """Prepare every dataset the paper figures need (cached in results/)."""
513     panel = load_panel()
514     data = _cache()
515     steps = {
516         "shadow_price": lambda: shadow_price_coupling(panel),
517         "fan_chart": lambda: prediction_fan(panel),
518         "reliability": lambda: reliability_sharpness(panel),
519         "voi": lambda: voi_boundary(panel),
520         "rolling": lambda: rolling_update_example(panel),
521         "realtime": lambda: realtime_control(panel),
522         "pareto": lambda: pareto_cost_risk_degradation(panel),
523         "ablation": lambda: ablation_matrix(panel),
524         "degradation_priced": lambda: degradation_priced(panel),
525         "degradation_sensitivity": lambda: degradation_sensitivity(panel),
526     }
527     for key, fn in steps.items():
528         if force or key not in data:
529             data = _update(**{key: fn()})
530             if verbose:
531                 print(f"[figure-data] prepared {key}")
532         elif verbose:
533             print(f"[figure-data] cached {key}")
534     return data
535
536
537 def main() -> None:
538     import argparse
539     parser = argparse.ArgumentParser(description="prepare datasets for 图片.md figures")
540     parser.add_argument("--force", action="store_true", help="recompute every dataset")
541     args = parser.parse_args()
542     build(force=args.force)
543     print(f"[figure-data] cache -> {CACHE}")
544
545
546 if __name__ == "__main__":
547     main()

```

code/report/figures.py

```

1  """兼容用的旧入口，真正的绘图程序在 code/report/visualize.py。
2
3  留着它是为了让以前的命令还能用：
4
5      python3 -m code.report.figures      # 等价于 visualize --set main
6  """
7  from __future__ import annotations
8
9  from code.report.visualize import main as _main
10
11
12 def main() -> None:
13     _main(["--set", "main"])
14
15
16 if __name__ == "__main__":
17     main()

```

code/report/paper_figs.py

```

1  """Paper figure suite Fig.0–Fig.9 required by `图片.md`.
2
3  Each figure answers exactly one mathematical/engineering question and follows the
4  standard visual grammar of the energy-systems / probabilistic-forecasting / MPC
5  literature. Data comes from `code.report.figure_data` (cached in
6  `results/paper_fig_data.json`); no model code is touched.
7  """
8  from __future__ import annotations
9
10 import numpy as np
11 import matplotlib.pyplot as plt
12 from matplotlib.gridspec import GridSpec
13 from matplotlib.patches import FancyArrowPatch, FancyBboxPatch
14 from mpl_toolkits.mplot3d import Axes3D
15
16 from code.common.load_clean import load_panel
17 from code.report import figure_data as fd
18 from code.report import viz_style as vs
19 from code.report.viz_style import BLACK, BLUE, GREY, GREEN, ORANGE, PALETTE, PURPLE, RED, SKY,
    YELLOW
20
21 RESULTS = fd.RESULTS
22
23 _DATA: dict | None = None
24
25
26 def data() -> dict:
27     """Cached figure datasets (built once per process, no re-solving)."""
28     global _DATA
29     if _DATA is None:
30         _DATA = fd.build(verbose=False)
31     return _DATA
32
33
34 def _hours(n: int = 144) -> np.ndarray:
35     return np.arange(n) / 6.0
36
37
38 # Fig.0 方法框架
39 def fig0_framework() -> None:
40     fig, ax = plt.subplots(figsize=(12.6, 7.6))
41     ax.set_xlim(0, 100)
42     ax.set_ylim(0, 62)
43     ax.axis("off")
44
45     def layer(y, title, operator, io, color, width=64, x=18):
46         ax.add_patch(FancyBboxPatch((x, y), width, 7.6,
47                                     boxstyle="round,pad=0.45,rounding_size=1.0",
48                                     linewidth=1.15, edgecolor=color, facecolor="white"))
49         ax.text(x + width / 2, y + 5.9, title, ha="center", va="center", fontsize=10.6,
50                fontweight="bold", color=color)
51         ax.text(x + width / 2, y + 3.1, operator, ha="center", va="center", fontsize=8.6,
52                color=BLACK)
53         ax.text(x + width / 2, y + 1.1, io, ha="center", va="center", fontsize=7.8,
54                color=GREY)
55
56     layer(53.0, "物理层 Physical layer", "PV · 储能 · 小区负荷 · 外网 (不可倒送、日循环)",
57          "input: 附件1–4 → output: 标准面板 clean/*.npz (C++17 清洗 + 审计)", GREY)
58     layer(42.6, "Q1 图–LP + KKT", "min  $\Sigma_t q_t$  s.t. 节点平衡 · SOC 动态 ·  $e_0=e_T$ ",
59          "operator: 网络流等价 LP + 对偶影子价格  $\mu_t$  → output: 计划购电/充放电 +  $\mu_t$ ", BLUE)
60     layer(32.2, "Q2 MaxEnt 场景 + ACI 共形 + CVaR", "min  $\Sigma p q + \lambda \cdot E[\text{紧急购电}]$  s.t. 场景 recourse",
61          "operator: 残差块 bootstrap + 共形分位( $p^*=0.80$ ) → output:  $q_{\text{plan}}$ 、覆盖区间", ORANGE)
62     layer(21.8, "Q3 VOI + Trigger–MPC", "min 结算式 (计划+紧急+0.5/1.5 违约)",
63          "operator: 四时次滚动更新 + NAV/URI 门控 → output:  $q_{\text{plan}}/q_{\text{adj}}$ /紧急购电", GREEN)
64     layer(11.4, "Q4 Soft–MPC + Repair", "min  $(1-\beta)E[C] + \beta \text{CVaR}_{95}$  s.t. 软约束 always-feasible",
65          "operator: 波动电价 + 共形尖峰 + 软松弛惩罚  $\rho_1 > 5p_{\text{max}}$  → output: 实时控制轨迹", PURPLE)
66     layer(1.0, "真实运行 Real operation", "储能充放电执行 · SOC 安全域 [1200, 10800] kWh",
67          "feedback: 实测回灌 (ACI/场景更新), 形成闭环", RED)

```

```

68
69 for y0, y1 in ((53.0, 50.2), (42.6, 39.8), (32.2, 29.4), (21.8, 19.0), (11.4, 8.6)):
70     ax.add_patch(FancyArrowPatch((50, y0), (50, y1), arrowstyle="->|>", mutation_scale=13,
71                                 linewidth=1.3, color=GREY))
72 ax.add_patch(FancyArrowPatch((84, 5), (95, 5), arrowstyle="->|>", mutation_scale=12,
73                             linewidth=1.1, color=RED))
74 ax.add_patch(FancyArrowPatch((95, 5), (95, 56.8), arrowstyle="-", mutation_scale=12,
75                             linewidth=1.1, color=RED, linestyle="--"))
76 ax.add_patch(FancyArrowPatch((95, 56.8), (52, 56.8), arrowstyle="->|>", mutation_scale=12,
77                             linewidth=1.1, color=RED, linestyle="--"))
78 ax.text(96.5, 31, "反馈闭环\n(feedback)", rotation=90, va="center", ha="center",
79         fontsize=8.4, color=RED)
80 ax.text(2, 57.5, "(a) 四层模型递进", fontsize=10.5, fontweight="bold")
81 ax.text(2, 30, "每一层: \nInput\n→ Operator\n→ Output", fontsize=8.6, color=GREY,
82         va="center")
83 ax.set_title("Fig.0 总体方法框架: 物理层 → Q1 → Q2 → Q3 → Q4 → 真实运行 (含反馈闭环)",
84             fontsize=12.4, fontweight="bold", pad=10)
85 vs.save(fig, "pfig0_framework",
86         "总体方法框架。每一层给出 Input → Mathematical Operator → Output; "
87         "右侧虚线为实测回灌的闭环通道 (ACI 覆盖率与场景集更新)。")
88
89
90 # Fig.1 调度功率 + SOC
91 def fig1_dispatch_soc(panel) -> None:
92     q1 = fd.load_json(RESULTS / "q1.json")
93     st = fd.Storage.from_config(panel.config)
94     price, load, pv = panel.day1[:, 0], panel.day1[:, 1], panel.day1[:, 2]
95     q = np.array(q1["lp"]["q"]) / st.dt
96     c = np.array(q1["lp"]["c"]) / st.dt
97     d = np.array(q1["lp"]["d"]) / st.dt
98     soc = np.array(q1["lp"]["soc"])[1:]
99     grid = q
100     h = _hours()
101
102     fig = plt.figure(figsize=(12.8, 7.2))
103     gs = GridSpec(2, 1, figure=fig, height_ratios=[1.45, 1.0], hspace=0.22)
104     ax = fig.add_subplot(gs[0, 0])
105     ax.plot(h, pv, color=ORANGE, lw=1.7, label="光伏 $P^{PV}_t$")
106     ax.plot(h, load, color=RED, lw=1.7, label="负荷 $P^{Load}_t$")
107     ax.plot(h, grid, color=BLUE, lw=1.9, label="购电 $P^{Grid}_t$")
108     ax.bar(h, c, width=0.14, color=SKY, label="充电 $P^{ch}_t$")
109     ax.bar(h, -d, width=0.14, color=GREEN, label="放电 $P^{dis}_t$")
110     ax.axhline(0, color=BLACK, lw=0.8)
111     ax2 = ax.twinx()
112     ax2.plot(h, price, color=PURPLE, lw=1.1, ls="--", alpha=0.85, label="电价 $p_t$")
113     ax2.set_ylabel("电价 (元/kWh)")
114     ax2.grid(False)
115     ax.set_xlim(0, 24)
116     ax.set_xlabel("时刻 (h)")
117     ax.set_ylabel("功率 (kW)")
118     handles, labels = ax.get_legend_handles_labels()
119     h2, l2 = ax2.get_legend_handles_labels()
120     ax.legend(handles + h2, labels + l2, ncol=3, loc="upper left")
121     vs.panel(ax, "a", dx=-0.06)
122     ax.set_title("能量调度轨迹: 负荷由光伏、储能与外网购电共同满足 (144×10 min) ")
123
124     ax = fig.add_subplot(gs[1, 0])
125     ax.plot(h, soc, color=GREEN, lw=2.0, label="储电量 $E_t$")
126     ax.axhline(float(st.soc_max), ls="--", lw=1.0, color=GREY)
127     ax.axhline(float(st.soc_min), ls="--", lw=1.0, color=GREY)
128     ax.fill_between(h, float(st.soc_min), soc, color=GREEN, alpha=0.10, lw=0)
129     ax.scatter([0, 24], [soc[0], soc[-1]], color=BLACK, zorder=5, s=22)
130     ax.annotate(f"$E_0=E_{144}={soc[-1]:.0f}$ kWh", (0.4, soc[-1] - 1250), fontsize=8.8)
131     ax.annotate(f"上限 {float(st.soc_max):.0f} kWh", (23.6, float(st.soc_max)), ha="right",
132               va="bottom", fontsize=8.2, color=GREY)
133     ax.annotate(f"下限 {float(st.soc_min):.0f} kWh", (23.6, float(st.soc_min)), ha="right",
134               va="bottom", fontsize=8.2, color=GREY)
135     ax.set_xlim(0, 24)
136     ax.set_ylim(0, 12000)
137     ax.set_xlabel("时刻 (h)")

```

```

138 ax.set_ylabel("储电量 (kWh)")
139 ax.legend(loc="lower right")
140 vs.panel(ax, "b", dx=-0.06)
141 ax.set_title("储电量轨迹：全程落在安全域内，日循环约束 $E_0=E_{144}$ 成立")
142 fig.suptitle(f"Fig.1 Q1 全天能量调度与 SOC 轨迹 (全天费用 {q1['cost_cny']:.2f} 元)",
143             fontsize=12.2, fontweight="bold", y=0.98)
144 vs.save(fig, "pfig1_dispatch_soc",
145         "问题 1 的物理运行机制。(a) 负荷、光伏、购电与充放电功率 (右轴为电价)；"
146         "(b) 储电量轨迹，满足容量上下限与日循环。")
147
148
149 # Fig.2 电价—影子价格—SOC 耦合
150 def fig2_shadow_price(panel) -> None:
151     payload = data()["shadow_price"]
152     price = np.array(payload["price"])
153     mu = np.array(payload["mu"])
154     charge = np.array(payload["charge"]) / (1 / 6)
155     disch = np.array(payload["discharge"]) / (1 / 6)
156     soc = np.array(fd.load_json(RESULTS / "q1.json")["lp"]["soc"])[1:]
157     h = _hours()
158     st = fd.Storage.from_config(panel.config)
159
160     fig = plt.figure(figsize=(12.8, 8.2))
161     gs = GridSpec(3, 1, figure=fig, height_ratios=[1.0, 1.05, 0.95], hspace=0.28)
162     ax = fig.add_subplot(gs[0, 0])
163     ax.plot(h, price, color=PURPLE, lw=1.7, label="市场电价 $p_t$")
164     ax.scatter(h[charge > 1e-6], np.full(int((charge > 1e-6).sum()), price.max() * 0.97),
165              marker="^", color=BLUE, s=20, label="充电时段")
166     ax.scatter(h[disch > 1e-6], np.full(int((disch > 1e-6).sum()), price.max() * 0.90),
167              marker="v", color=RED, s=20, label="放电时段")
168     ax.set_ylabel("电价 (元/kWh)")
169     ax.set_xlim(0, 24)
170     ax.legend(ncol=3, loc="upper left", fontsize=8.4)
171     vs.panel(ax, "a", dx=-0.06)
172     ax.set_title(f"① 电价轨迹：充电时段中位价 {payload['charge_price_median']:.3f}、"
173                f"放电时段中位价 {payload['discharge_price_median']:.3f} 元/kWh")
174
175     ax = fig.add_subplot(gs[1, 0])
176     ax.plot(h, price, color=PURPLE, lw=1.6, label="市场电价 $p_t$")
177     ax.plot(h, mu, color=GREEN, lw=1.8, label="影子价格 $\mu_t$ (能量平衡对偶)")
178     ax.fill_between(h, mu, price, where=price > mu, color=RED, alpha=0.20, lw=0,
179                    label="价高于边际价值：放电/少购")
180     ax.fill_between(h, price, mu, where=price <= mu, color=BLUE, alpha=0.16, lw=0,
181                    label="价低于边际价值：充电/多购")
182     ax.set_ylabel("元/kWh")
183     ax.set_xlim(0, 24)
184     ax.legend(ncol=2, loc="upper left", fontsize=8.2)
185     ax.annotate(f"互补松弛校验：{payload['purchase_slots']} 个购电时段全部满足 $\mu_t=p_t$ (精确成"
186               f"立)，"
187               f"全程 $\mu_t \leq p_t$ 违反 {payload['mu_above_price_violations']} 次；"
188               f"放电时段 {payload['discharge_price_above_mu']}/{payload['discharge_slots']} 满足"
189               f"$p_t \geq \mu_t$",
190               (0.5, 0.035), xycoords="axes fraction", ha="center", fontsize=8.0,
191               bbox=dict(boxstyle="round,pad=0.32", fc="white", ec=GREY, lw=0.6))
192     vs.panel(ax, "b", dx=-0.06)
193     ax.set_title("② 影子价格耦合：储能动作的经济解释 (购电边际价值 = 市场价的时段才购电)")
194
195     ax = fig.add_subplot(gs[2, 0])
196     ax.plot(h, soc, color=GREEN, lw=1.8, label="储电量 $E_t$")
197     ax.axhline(st.soc_max, ls="--", lw=1.0, color=GREY)
198     ax.axhline(st.soc_min, ls="--", lw=1.0, color=GREY)
199     ax.scatter(h[charge > 1e-6], np.full(int((charge > 1e-6).sum()), st.soc_max + 250),
200              marker="^", color=BLUE, s=20, label="充电时段")
201     ax.scatter(h[disch > 1e-6], np.full(int((disch > 1e-6).sum()), st.soc_min - 250),
202              marker="v", color=RED, s=20, label="放电时段")
203     ax.set_ylim(st.soc_min - 800, st.soc_max + 900)
204     ax.set_xlim(0, 24)
205     ax.set_xlabel("时刻 (h)")
206     ax.set_ylabel("储电量 (kWh)")
207     ax.legend(ncol=3, loc="lower right", fontsize=8.2)

```

```

206 vs.panel(ax, "c", dx=-0.06)
207 ax.set_title("③ SOC 轨迹与充放电时段：低价充电、高价放电跨时段搬移能量")
208 fig.suptitle("Fig.2 电价-影子价格-SOC 耦合图 (KKT 的经济解释)",
209             fontsize=12.2, fontweight="bold", y=0.98)
210 vs.save(fig, "pfig2_shadow_price",
211         "对偶影子价格的经济解释。① 电价与充/放电时段；② 电价与影子价格  $\mu_t$  的关系，"
212         "购电时段  $\mu_t=p_t$  精确成立（互补松弛），放电时段全部满足  $p_t \geq \mu_t$ ；"
213         "③ SOC 轨迹。说明储能的充放动作可由边际价值关系解释，而非求解器巧合。")
214
215
216 # Fig.3 概率预测扇形图
217 def fig3_prediction_fan(panel) -> None:
218     payload = data()["fan_chart"]
219     h = _hours()
220     fig = plt.figure(figsize=(13.0, 6.4))
221     gs = GridSpec(2, 1, figure=fig, height_ratios=[1.6, 1.0], hspace=0.28)
222     ax = fig.add_subplot(gs[0, 0])
223     colors = {"q50": SKY, "q80": BLUE, "q90": ORANGE, "q95": RED}
224     order = ["q95", "q90", "q80", "q50"]
225     for key in order:
226         band = payload["bands"][key]
227         ax.fill_between(h, band["lower"], band["upper"], color=colors[key],
228                        alpha=0.18 if key != "q50" else 0.35, lw=0,
229                        label=f"{int(key[1:])}% 区间")
230     ax.plot(h, payload["forecast"], color=BLACK, lw=1.6, ls="-.", label="点预测  $\hat{PV}_t$  (0:00 发布)")
231     ax.plot(h, payload["actual"], color=GREEN, lw=1.9, label="实际  $PV_t$ ")
232     viol = np.array(payload["aci"]["violations"])
233     ax.scatter(h[viol], np.array(payload["actual"])[viol], color=RED, s=26, zorder=6,
234              label=f"超出 ACI 上界 ({int(viol.sum())} 个时段)")
235     ax.set_xlim(0, 24)
236     ax.set_xlabel("时刻 (h)")
237     ax.set_ylabel("光伏功率 (kW)")
238     ax.legend(ncol=3, loc="upper left", fontsize=8.4)
239     vs.panel(ax, "a", dx=-0.05)
240     ax.set_title(f"预测扇形图: {payload['date']} 多层分位区间 + ACI 自适应上下界"
241                f" (当日 ACI 水平 {payload['aci']['level']:.3f}, "
242                f"全年实测覆盖率 {payload['achieved_coverage']*100:.1f}% "
243                f"vs 目标 {100*(1-payload['target_miss_rate']:.0f}%)")
244
245     ax = fig.add_subplot(gs[1, 0])
246     trace = payload["aci_trace"]
247     x = np.arange(len(trace["miss"]))
248     ax.plot(x, np.array(trace["miss"]) * 100, color=BLUE, lw=1.0, alpha=0.9,
249            label="实测越界率 (%)，目标 10%")
250     ax.axhline(10.0, color=RED, ls="--", lw=1.3, label="目标越界率 10%")
251     window = 30
252     if x.size > window:
253         kernel = np.ones(window) / window
254         ax.plot(x[window - 1:], np.convolve(np.array(trace["miss"]) * 100, kernel,
255                                           mode="valid"), color=ORANGE, lw=1.6,
256                label=f"{window} 日滑动平均")
257     ax.set_xlabel("2025 年日序 (自 2 月 1 日起)")
258     ax.set_ylabel("ACI 越界率 (%)")
259     ax.legend(ncol=3, loc="upper right", fontsize=8.2)
260     vs.panel(ax, "b", dx=-0.05)
261     ax.set_title("ACI 在线反馈：越界率被逐步拉回名义水平 (可靠性诊断)")
262     fig.suptitle("Fig.3 Q2 概率预测扇形图与 ACI 覆盖率轨迹",
263                 fontsize=12.2, fontweight="bold", y=0.98)
264     vs.save(fig, "pfig3_prediction_fan",
265            "概率预测表达。(a) 光伏实际值、点预测与 50/80/90/95% 共形分位区间，"
266            "红点为穿透 90% ACI 上界的时段；(b) 全年 ACI 越界率及其滑动平均，"
267            "显示自适应机制把覆盖率拉回名义水平。")
268
269
270 # Fig.4 可靠性—锐度—成本风险
271 def fig4_reliability_sharpness(panel) -> None:
272     rel = data()["reliability"]
273     levels = np.array(rel["levels"])
274     fig = plt.figure(figsize=(13.4, 4.9))

```

```

275 gs = GridSpec(1, 3, figure=fig, wspace=0.28)
276
277 ax = fig.add_subplot(gs[0, 0])
278 ax.plot([0.45, 1.0], [0.45, 1.0], color=GREY, ls="--", lw=1.1, label="理想校准线")
279 for (name, cov), color in zip(rel["methods"].items(), PALETTE):
280     ax.plot(levels, cov, "o-", color=color, lw=1.4, ms=4.5, label=name)
281 ax.set_xlabel("名义覆盖率")
282 ax.set_ylabel("实测覆盖率")
283 ax.set_xlim(0.45, 1.0)
284 ax.set_ylim(0.45, 1.02)
285 ax.legend(loc="lower right", fontsize=7.8)
286 vs.panel(ax, "a")
287 ax.set_title("Reliability 可靠性图")
288
289 ax = fig.add_subplot(gs[0, 1])
290 for (name, cov), color in zip(rel["methods"].items(), PALETTE):
291     widths = rel["widths"][name]
292     ax.plot(cov, widths, "s-", color=color, lw=1.4, ms=4.2, label=name)
293 ax.set_xlabel("实测覆盖率")
294 ax.set_ylabel("平均区间宽度 (kW)")
295 ax.legend(fontsize=7.6)
296 vs.panel(ax, "b")
297 ax.set_title("Sharpness: 覆盖率↑但宽度未失控膨胀")
298
299 ax = fig.add_subplot(gs[0, 2])
300 pareto = data()["pareto"]["rows"]
301 cost = np.array([r["cost_cny"] for r in pareto]) / 1e4
302 cvar = np.array([r["cvar95_cny"] for r in pareto])
303 ax.plot(cost, cvar, "o-", color=PURPLE, lw=1.5, ms=6)
304 for c, v, b in zip(cost, cvar, [r["beta"] for r in pareto]):
305     ax.annotate(f"$\\beta$={b:.2f}", (c, v), textcoords="offset points", xytext=(6, 4),
306               fontsize=7.8)
307 ax.set_xlabel("样本期总费用 (万元)")
308 ax.set_ylabel("CVaR_{95}$ (元/日)")
309 vs.panel(ax, "c")
310 ax.set_title("经济-风险权衡 (同一信息集)")
311 fig.suptitle("Fig.4 Q2 概率标定与经济权衡: Reliability / Sharpness / Cost-Risk",
312             fontsize=12, fontweight="bold", y=1.0)
313 vs.save(fig, "pfig4_reliability_sharpness",
314         "概率预测诊断。(a) Reliability 图: 四种方法的名义覆盖率与实测覆盖率;"
315         "(b) Sharpness: 覆盖率提升时区间宽度的代价;"
316         "(c) 同一信息集下的费用-CVaR 权衡, 说明风险下降不是靠无限放宽区间。")
317
318 # Fig.5 VOI 与惩罚决策边界
319 def fig5_voi_boundary(panel) -> None:
320     voi = data()["voi"]
321     fig = plt.figure(figsize=(13.0, 5.4))
322     gs = GridSpec(1, 2, figure=fig, width_ratios=[1.25, 1.0], wspace=0.26)
323     ax = fig.add_subplot(gs[0, 0])
324     stage_style = {6: (BLUE, "o", "6:00 更新"), 12: (ORANGE, "s", "12:00 更新"),
325                   18: (GREEN, "^", "18:00 更新")}
326
327     limit = 0
328     for stage, (color, marker, label) in stage_style.items():
329         items = voi["stages"][str(stage)]
330         x = np.array([it["penalty"] for it in items])
331         y = np.array([it["voi"] for it in items])
332         ax.scatter(x, y, s=16, alpha=0.55, color=color, marker=marker, label=label,
333                  edgecolors="none")
334         limit = max(limit, float(np.quantile(x, 0.995)), float(np.quantile(y, 0.995)))
335     ax.plot([0, limit], [0, limit], color=RED, ls="--", lw=1.5, label="触发边界 VOI = $C_{adjust}$")
336     ax.fill_between([0, limit], [0, limit], limit, color=GREEN, alpha=0.06, lw=0)
337     ax.text(limit * 0.62, limit * 0.88, "VOI > 调整成本\\n→ 值得更新", color=GREEN, fontsize=9,
338            ha="center")
339     ax.text(limit * 0.72, limit * 0.20, "VOI ≤ 调整成本\\n→ 保持原计划", color=RED, fontsize=9,
340            ha="center")
341     ax.set_xlim(0, limit)
342     ax.set_ylim(-limit * 0.25, limit)
343     ax.set_xlabel("调整成本 $C_{adjust,k}$ (元)")

```

```

344 ax.set_ylabel("信息价值 $VOI\_k$ (元)")
345 ax.legend(loc="upper left", fontsize=8.4)
346 vs.panel(ax, "a")
347 ax.set_title("VOI-调整成本决策平面 (每个点 = 某日的某次更新)")
348
349 ax = fig.add_subplot(gs[0, 1])
350 summary = voi["summary"]
351 stages = [6, 12, 18]
352 voi_mean = [summary[str(s)]["voi_mean"] / 1e4 for s in stages]
353 pen_mean = [summary[str(s)]["penalty_mean"] / 1e4 for s in stages]
354 x = np.arange(len(stages))
355 ax.bar(x - 0.19, voi_mean, 0.38, color=BLUE, label="平均 VOI")
356 ax.bar(x + 0.19, pen_mean, 0.38, color=RED, label="平均调整成本")
357 for xi, (v, p) in enumerate(zip(voi_mean, pen_mean)):
358     ax.annotate(f"{v:.2f}", (xi - 0.19, v), ha="center", va="bottom" if v >= 0 else "top",
359               fontsize=8.2)
360     ax.annotate(f"{p:.2f}", (xi + 0.19, p), ha="center", va="bottom" if p >= 0 else "top",
361               fontsize=8.2)
362 ax.axhline(0, color=BLACK, lw=0.8)
363 ax.set_xticks(x, [f"{s}:00\n触发率 {summary[str(s)]['trigger_rate']*100:.0f}%"
364                 for s in stages])
365 ax.set_ylabel("万元/日")
366 ax.legend(loc="upper right")
367 vs.panel(ax, "b")
368 ax.set_title("各更新时次的平均信息价值与调整成本")
369 fig.suptitle("Fig.5 Q3 的 VOI-惩罚决策边界 (什么时候值得更新预报)",
370            fontsize=12.2, fontweight="bold", y=1.0)
371 vs.save(fig, "pfig5_voi_boundary",
372        "信息价值决策边界。(a) 每个点代表某日某次更新, 横轴为调整成本、纵轴为信息价值, "
373        "红色虚线 VOI = C_adjust 即触发边界; (b) 三个时次的平均 VOI、平均调整成本与触发率, "
374        "说明 6:00/12:00 的更新几乎总是划算、而 18:00 的更新价值趋近于零。")
375
376
377 # Fig.6 滚动时域更新轨迹
378 def fig6_rolling_update(panel) -> None:
379     roll = data()["rolling"]
380     h = _hours()
381     plan = np.array(roll["plan"])
382     adj = np.array(roll["adjusted"])
383     realised = np.array(roll["realised"])
384     soc = np.array(roll["soc"])
385     fig = plt.figure(figsize=(13.0, 6.6))
386     gs = GridSpec(2, 1, figure=fig, height_ratios=[1.5, 1.0], hspace=0.26)
387     ax = fig.add_subplot(gs[0, 0])
388     ax.plot(h, plan, color=GREY, lw=1.7, ls="--", label="DA 计划 $q^{\text{plan}}_t$ (0:00)")
389     ax.step(h, adj, where="post", color=BLUE, lw=1.7, label="更新后计划 $q^{\text{adj}}_t$")
390     ax.plot(h, realised, color=RED, lw=1.0, alpha=0.85, label="实际采购 $q^{\text{adj}}_{t+g_t}$")
391     for tr in roll["triggers"]:
392         ax.axvline(tr["slot"] / 6.0, color=GREEN, ls=":", lw=1.2)
393         ax.annotate(f"{tr['label']}\nVOI {tr['voi']/1e4:.2f} 万元",
394                   (tr["slot"] / 6.0, ax.get_ylim()[1] * 0.98), ha="center", va="top",
395                   fontsize=8.2, color=GREEN,
396                   bbox=dict(boxstyle="round,pad=0.28", fc="white", ec=GREEN, lw=0.7))
397     ax.set_xlim(0, 24)
398     ax.set_xlabel("时刻 (h)")
399     ax.set_ylabel("购电量 (kWh/10 min)")
400     ax.legend(loc="lower left", fontsize=8.6)
401     vs.panel(ax, "a", dx=-0.05)
402     ax.set_title(f"滚动更新轨迹: {roll['date']} (新信息 → VOI → 触发 → MPC 重优化 → 新计划)")
403
404     ax = fig.add_subplot(gs[1, 0])
405     h_soc = np.arange(145) / 6.0
406     ax.plot(h_soc, soc, color=GREEN, lw=1.8, label="最终储电量轨迹")
407     st_cfg = fd.Storage.from_config(panel.config)
408     ax.axhline(st_cfg.soc_max, ls="--", lw=1.0, color=GREY)
409     ax.axhline(st_cfg.soc_min, ls="--", lw=1.0, color=GREY)
410     for tr in roll["triggers"]:
411         ax.axvline(tr["slot"] / 6.0, color=GREEN, ls=":", lw=1.0)
412     ax.set_xlim(0, 24)
413     ax.set_xlabel("时刻 (h)")

```

```

414 ax.set_ylabel("储电量 (kWh)")
415 ax.legend(loc="lower right", fontsize=8.6)
416 vs.panel(ax, "b", dx=-0.05)
417 ax.set_title(f"对应 SOC 轨迹 (当日子集费用:  $S0=\{\text{roll}['\text{day\_subset\_costs}'] ['S0']/1e4:.2f\}$  "
418             f" $\rightarrow S3=\{\text{roll}['\text{day\_subset\_costs}'] ['S3']/1e4:.2f\}$  万元) ")
419 fig.suptitle("Fig.6 Q3 滚动时域闭环: 计划  $\rightarrow$  触发  $\rightarrow$  更新  $\rightarrow$  执行",
420             fontsize=12.2, fontweight="bold", y=0.98)
421 vs.save(fig, "pfig6_rolling_update",
422         "滚动时域闭环证据。(a) 0:00 计划、更新后计划与实际采购量, 绿虚线为三个更新时次"
423         "(标注当日该时次的信息价值 VOI); (b) 对应的 SOC 轨迹与安全域。")
424
425
426 # Fig.7 实时控制 + SOC 安全包络
427 def fig7_realtime_control(panel) -> None:
428     rt = data()["realtime"]
429     h = _hours()
430     price = np.array(rt["price"])
431     load = np.array(rt["load"])
432     pv = np.array(rt["pv"])
433     net_base = np.array(rt["baseline"]["net"]) / (1 / 6)
434     net_mpc = np.array(rt["mpc"]["net"]) / (1 / 6)
435     soc_base = np.array(rt["baseline"]["soc"])
436     soc_mpc = np.array(rt["mpc"]["soc"])
437     lim = rt["soc_limits"]
438
439     fig = plt.figure(figsize=(13.2, 7.6))
440     gs = GridSpec(3, 1, figure=fig, height_ratios=[0.9, 1.15, 1.0], hspace=0.28)
441     ax = fig.add_subplot(gs[0, 0])
442     ax.plot(h, price, color=PURPLE, lw=1.6, label="实时电价  $p^{\{RT\}}_t$ ")
443     ax.set_ylabel("元/kWh")
444     ax.set_xlim(0, 24)
445     ax.legend(loc="upper left", fontsize=8.4)
446     vs.panel(ax, "a", dx=-0.05)
447     ax.set_title(f"实时电价与净负荷 ({rt['date']}) ")
448     ax2 = ax.twinx()
449     ax2.plot(h, load - pv, color=GREY, lw=1.0, ls="--", label="净负荷")
450     ax2.set_ylabel("净负荷 (kW)")
451     ax2.grid(False)
452     ax2.legend(loc="upper right", fontsize=8.4)
453
454     ax = fig.add_subplot(gs[1, 0])
455     ax.step(h, net_base, where="post", color=GREY, lw=1.4, label="承诺基线  $u^{\{base\}}_t$ ")
456     ax.step(h, net_mpc, where="post", color=BLUE, lw=1.8, label="MPC 控制  $u^{\{MPC\}}_t$ ")
457     ax.axhline(0, color=BLACK, lw=0.8)
458     ax.set_xlim(0, 24)
459     ax.set_ylabel("控制动作  $u_t$  (kW)")
460     ax.legend(loc="upper left", fontsize=8.6)
461     vs.panel(ax, "b", dx=-0.05)
462     ax.set_title(f"控制轨迹对比: MPC 通过滚动重优化改变充放动作"
463                 f" $(\text{动作总变差 } \{\text{rt}['\text{baseline}'] ['tv']/1e4:.2f\} \rightarrow \{\text{rt}['\text{mpc}'] ['tv']/1e4:.2f\} \text{ 万 kW})$ ")
464
465     ax = fig.add_subplot(gs[2, 0])
466     h_soc = np.arange(145) / 6.0
467     ax.plot(h_soc, soc_base, color=GREY, lw=1.3, ls="--", label="基线 SOC")
468     ax.plot(h_soc, soc_mpc, color=GREEN, lw=1.9, label="MPC SOC")
469     ax.fill_between([0, 24], lim[0], lim[1], color=GREEN, alpha=0.05, lw=0)
470     ax.axhline(lim[0], color=RED, ls="--", lw=1.1)
471     ax.axhline(lim[1], color=RED, ls="--", lw=1.1)
472     ax.annotate(f"安全下限  $\{\text{lim}[0]:.0f\}$  kWh", (0.2, lim[0]), va="bottom", fontsize=8.2, color=RED)
473     ax.annotate(f"安全上限  $\{\text{lim}[1]:.0f\}$  kWh", (0.2, lim[1]), va="bottom", fontsize=8.2, color=RED)
474     ax.set_xlim(0, 24)
475     ax.set_ylim(0, 12000)
476     ax.set_xlabel("时刻 (h)")
477     ax.set_ylabel("储电量 (kWh)")
478     ax.legend(loc="lower right", fontsize=8.6)
479     vs.panel(ax, "c", dx=-0.05)
480     ax.set_title("SOC 安全域: MPC 轨迹全程不越界 (越界时段数 = ")

```

```

481         f"{int(((soc_mpc < lim[0]) | (soc_mpc > lim[1])).sum())} ")
482 fig.suptitle("Fig.7 Q4 实时控制轨迹与 SOC 安全域",
483             fontsize=12.2, fontweight="bold", y=0.98)
484 vs.save(fig, "pfig7_rt_control_soc",
485         "实时控制与安全约束。(a) 波动电价与净负荷；(b) 承诺基线 与 MPC 的控制动作对比，"
486         "标注动作总变差；(c) 两种策略的 SOC 轨迹与安全域，MPC 全程不越界。")
487
488
489 # Fig.8 成本—风险—退化 Pareto
490 def fig8_pareto_cost_risk_degradation(panel) -> None:
491     rows = data()["pareto"]["rows"]
492     priced = data()["degradation_priced"]["rows"]
493     sens = data()["degradation_sensitivity"]
494     cost = np.array([r["cost_cny"] for r in rows]) / 1e4
495     cvar = np.array([r["cvar95_cny"] for r in rows])
496     fade = np.array([r["annual_fade_pct"] for r in rows])
497     beta = np.array([r["beta"] for r in rows])
498
499     fig = plt.figure(figsize=(15.4, 5.2))
500     gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.05, 1.0], wspace=0.26)
501     ax = fig.add_subplot(gs[0, 0])
502     sizes = 60 + 300 * (fade - fade.min()) / max(fade.max() - fade.min(), 1e-9)
503     sc = ax.scatter(cost, cvar, s=sizes, c=fade, cmap="viridis", edgecolors=BLACK, linewidths
504                    =0.7,
505                    zorder=4)
506     ax.plot(cost, cvar, color=GREY, lw=1.0, ls="--", zorder=2)
507     for c, v, b in zip(cost, cvar, beta):
508         ax.annotate(f"$\\beta$={b:.2f}", (c, v), textcoords="offset points", xytext=(8, -12),
509                    fontsize=8.0)
510     cb = fig.colorbar(sc, ax=ax, pad=0.02)
511     cb.set_label("雨流 SOH 年衰减 (%)")
512     ax.set_xlabel("样本期电费 (万元)")
513     ax.set_ylabel("CVaR$_{95}$ (元/日)")
514     vs.panel(ax, "a")
515     ax.set_title("风险层 Pareto (气泡/颜色 = 真实 SOH 衰减)")
516
517     ax = fig.add_subplot(gs[0, 1])
518     rho = np.array([r["rho_cny_per_kwh"] for r in priced])
519     elec = np.array([r["electricity_cost_cny"] for r in priced]) / 1e4
520     total = np.array([r["total_cost_cny"] for r in priced]) / 1e4
521     pfade = np.array([r["annual_fade_pct"] for r in priced])
522     ax.plot(pfade, total, "o-", color=BLUE, lw=1.7, ms=6, label="总成本 (电费 + 退化)")
523     ax.plot(pfade, elec, "s--", color=ORANGE, lw=1.3, ms=5, label="仅电费")
524     best = int(np.argmin(total))
525     ax.scatter([pfade[best]], [total[best]], s=220, facecolors="none", edgecolors=RED, lw=1.7,
526               zorder=5)
527     ax.annotate(f"最优退化定价 $\\rho$={rho[best]:.2f} 元/kWh\n"
528               f"衰减 {pfade[best]:.1f}%、总成本 {total[best]:.0f} 万元",
529               (pfade[best], total[best]), textcoords="offset points", xytext=(10, 14),
530               fontsize=8.4, color=RED)
531     for x, y, r in zip(pfade, total, rho):
532         ax.annotate(f"$\\rho$={r:.2f}", (x, y), textcoords="offset points", xytext=(6, -12),
533                    fontsize=7.8, color=BLUE)
534     ax.set_xlabel("雨流 SOH 年衰减 (%)")
535     ax.set_ylabel("样本期成本 (万元)")
536     ax.legend(loc="upper right", fontsize=8.4)
537     vs.panel(ax, "b")
538     ax.set_title("退化定价前沿：忽略退化并非经济最优")
539
540     ax3 = fig.add_subplot(gs[0, 2], projection="3d")
541     p3 = ax3.scatter(cost, cvar, fade, s=70, c=beta, cmap="plasma", depthshade=False,
542                     edgecolors=BLACK, linewidths=0.5)
543     ax3.plot(cost, cvar, fade, color=GREY, lw=1.0, ls="--")
544     ax3.set_xlabel("电费 (万元)", labelpad=6)
545     ax3.set_ylabel("CVaR$_{95}$ (元/日)", labelpad=6)
546     ax3.set_zlabel("SOH 年衰减 (%)", labelpad=6)
547     ax3.view_init(elev=22, azim=-125)
548     ax3.set_title("三维目标空间 (雨流 SOH)")
549     fig.suptitle("Fig.8 成本—风险—退化三重权衡：雨流计数 + Wöhler 寿命曲线的真实 SOH",
550                 fontsize=12.2, fontweight="bold", y=1.0)

```

```

550 vs.save(fig, "pfig8_pareto_cost_risk_deg",
551         "多目标权衡（退化由雨流计数 + Wöhler 寿命曲线给出，非吞吐代理）。(a) 费用-CVaR 平面，"
552         "气泡大小与颜色为年容量衰减；(b) 退化定价前沿：横轴为年衰减、纵轴为成本，"
553         "把退化定价  $p$  增大可同时降低衰减与总成本（最优约  $p=0.20$  元/kWh）；"
554         "(c) 费用-CVaR-SOH 三维目标空间。参数敏感性 "
555         f"N100∈[4000, 8000]、k∈[0.9, 1.3] 时年衰减区间为 "
556         f"{min(v['annual_fade_pct'] for v in sens['grid'].values()):.1f}%-"
557         f"{max(v['annual_fade_pct'] for v in sens['grid'].values()):.1f}%。")
558
559
560 # Fig.9 消融性能热力图
561 def fig9_ablation_heatmap(panel) -> None:
562     ablation = data()["ablation"]
563     rows, labels = ablation["rows"], ablation["labels"]
564     metric_labels = ablation["metric_labels"]
565     variants = [k for k in ("M0", "M1", "M2", "M3", "M4") if k in rows]
566     metrics = list(metric_labels)
567     raw = np.array([[rows[v][m] for m in metrics] for v in variants], dtype=float)
568     norm = np.zeros_like(raw)
569     for j in range(raw.shape[1]):
570         col = raw[:, j]
571         span = col.max() - col.min()
572         norm[:, j] = 0.5 if span <= 0 else (col - col.min()) / span
573
574     fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.6, 5.0),
575                                   gridspec_kw={"width_ratios": [1.35, 1.0], "wspace": 0.26})
576     im = ax.imshow(norm, cmap="RdYlGn_r", aspect="auto", vmin=0, vmax=1)
577     ax.set_xticks(range(len(metrics)), [metric_labels[m] for m in metrics], fontsize=8.2)
578     ax.set_yticks(range(len(variants)), [labels[v] for v in variants], fontsize=8.6)
579     for i in range(len(variants)):
580         for j, m in enumerate(metrics):
581             value = raw[i, j]
582             text = f"{value:,.0f}" if abs(value) >= 100 else f"{value:,.2f}"
583             ax.text(j, i, text, ha="center", va="center", fontsize=8.0,
584                    color="white" if norm[i, j] > 0.75 or norm[i, j] < 0.12 else BLACK)
585     cb = fig.colorbar(im, ax=ax, pad=0.02)
586     cb.set_label("列内归一化 (0 = 该列最优, 1 = 最差)")
587     ax.grid(False)
588     vs.panel(ax, "a", dx=-0.12)
589     ax.set_title("消融性能热力图：每个模块解决了什么问题")
590
591     base = raw[0]
592     x = np.arange(len(metrics))
593     for i, v in enumerate(variants[1:]):
594         delta = 100 * (raw[i] - base) / np.where(np.abs(base) < 1e-9, 1, np.abs(base))
595         ax2.plot(x, delta, "o-", color=PALETTE[(i - 1) % len(PALETTE)], lw=1.4, ms=4.4,
596                label=labels[v])
597     ax2.axhline(0, color=BLACK, lw=0.9)
598     ax2.set_xticks(x, [metric_labels[m] for m in metrics], fontsize=8.0)
599     ax2.set_ylabel("相对 M0 的变化 (%)")
600     ax2.set_yscale("symlog", linthresh=10)
601     ax2.legend(fontsize=7.8, loc="lower left")
602     vs.panel(ax2, "b", dx=-0.14)
603     ax2.set_title("逐模块增量（相对 M0, symlog 轴）")
604     fig.suptitle("Fig.9 全局消融：M0-M4 各模块对费用/风险/寿命/弃光的贡献",
605                 fontsize=12.2, fontweight="bold", y=1.0)
606     vs.save(fig, "pfig9_ablation_heatmap",
607            "模块消融证据。(a) M0-M4 × 六项指标的性能热力图（列内归一化，同时标注原始值）；"
608            "(b) 各模块相对 M0 的增量，说明物理建模、共形裕度、场景随机规划、滚动更新与"
609            "CVaR 风险层各自的贡献。")
610
611
612 PAPER_FIGS = [
613     ("framework", fig0_framework, False),
614     ("dispatch", fig1_dispatch_soc, True),
615     ("shadow", fig2_shadow_price, True),
616     ("fan", fig3_prediction_fan, True),
617     ("reliability", fig4_reliability_sharpness, True),
618     ("voi", fig5_voi_boundary, True),
619     ("rolling", fig6_rolling_update, True),

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620     ("realtime", fig7_realtime_control, True),
621     ("pareto", fig8_pareto_cost_risk_degradation, True),
622     ("ablation", fig9_ablation_heatmap, True),
623 ]

```

code/report/paper_numbers.py

```

1  """汇总论文要用的数字，顺带排好表1-表4。
2
3  论文里的数字一律从这儿取，每个值都带单位、来源文件和生成命令，不手抄。
4  """
5  from __future__ import annotations
6
7  import json
8  from pathlib import Path
9
10 import numpy as np
11
12 from code.common.load_clean import ROOT, load_panel
13 from code.optimize.q2_stochastic import block_label, merge_blocks
14
15 RESULTS = ROOT / "results"
16 ARRAYS = RESULTS / "arrays"
17 PAPER = ROOT / "paper"
18 TABLE1_SLOTS = {"10:00-10:10": 60, "12:00-12:10": 72, "14:00-14:10": 84,
19                  "16:00-16:10": 96, "18:00-18:10": 108, "20:00-20:10": 120}
20 TARGET_DATES = ("2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21")
21 BLOCK_LABELS = [f"{4 * i}:00-{4 * i + 4}:00" for i in range(6)]
22
23
24 def load_json(path: Path) -> dict:
25     return json.loads(path.read_text(encoding="utf-8"))
26
27
28 def entry(value, unit, source, command, note=""):
29     return {"value": value, "unit": unit, "source": source, "command": command, "note": note}
30
31
32 def build() -> dict:
33     panel = load_panel()
34     q1 = load_json(RESULTS / "q1.json")
35     q2 = load_json(RESULTS / "q2.json")
36     q3 = load_json(RESULTS / "q3.json")
37     q4 = load_json(RESULTS / "q4.json")
38     p2 = load_json(RESULTS / "p2_conformal.json")
39     cross = load_json(ROOT / "clean" / "cross_check.json")
40     manifest = load_json(ROOT / "clean" / "manifest.json")
41     meta = load_json(ROOT / "clean" / "meta.json")
42
43     numbers = {
44         "cleaning": {
45             "audit_records": entry(manifest["audit_records"], "count", "clean/manifest.json",
46                                   "cleaning_cpp clean", "NR1..NR12 全部改动条数"),
47             "quarantine_records": entry(manifest["quarantine_records"], "count",
48                                        "clean/manifest.json", "cleaning_cpp clean",
49                                        "核夜间 >5 kW 硬异常, 置 NaN"),
50             "core_night_zero_before": entry(cross["att2_core_night_zero_raw"], "count",
51                                             "clean/cross_check.json", "cleaning_cpp verify"),
52             "core_night_zero_after": entry(cross["att2_core_night_zero_after"], "count",
53                                           "clean/cross_check.json", "cleaning_cpp verify"),
54             "daytime_zero_0800_1600": entry(cross["att2_daytime_0800_1600_zeros"], "count",
55                                             "clean/cross_check.json", "cleaning_cpp verify",
56                                             "日间无零值不变式 I2"),
57             "near_zero_price_cells": entry(cross["att4_near_zero_price_cells"], "count",
58                                           "clean/cross_check.json", "cleaning_cpp verify",
59                                           "电价 <0.1 元/kWh"),
60             "runtime_seconds": entry(meta["wall_seconds"], "s", "clean/meta.json",
61                                     "cleaning_cpp clean"),
62             "peak_rss_kb": entry(meta["peak_rss_kb"], "kB", "clean/meta.json",

```

```

63         "cleaning_cpp clean"),
64     },
65     "q1": {
66         "cost_cny": entry(q1["cost_cny"], "元", "results/q1.json", "python -m code.optimize.
q1_lp"),
67         "baseline_b0": entry(q1["baselines"]["B0_all_purchase_cny"], "元", "results/q1.json
",
68             "python -m code.optimize.q1_lp", "全额外购"),
69         "baseline_b1": entry(q1["baselines"]["B1_pv_netting_cny"], "元", "results/q1.json",
70             "python -m code.optimize.q1_lp", "光伏抵扣 (含余量折价)"),
71         "baseline_b2": entry(q1["baselines"]["B2_pv_selfuse_curtail_no_storage_cny"], "元",
72             "results/q1.json", "python -m code.optimize.q1_lp",
73             "自用+弃光, 无储能 (验收锚点 48,052)"),
74         "purchase_kwh": entry(q1["energy"]["purchase_kwh"], "kWh", "results/q1.json",
75             "python -m code.optimize.q1_lp"),
76         "charge_kwh": entry(q1["energy"]["charge_kwh"], "kWh", "results/q1.json",
77             "python -m code.optimize.q1_lp"),
78         "discharge_kwh": entry(q1["energy"]["discharge_kwh"], "kWh", "results/q1.json",
79             "python -m code.optimize.q1_lp"),
80         "saving_vs_b2_pct": entry(
81             100.0 * (1 - q1["cost_cny"] / q1["baselines"]["
B2_pv_selfuse_curtail_no_storage_cny"]),
82             "%", "results/q1.json", "python -m code.optimize.q1_lp"),
83         "dp_table": entry(q1["dp"], "list", "results/q1.json", "python -m code.optimize.
q1_lp",
84             "步长收敛序列"),
85         "night_charge_share": entry(q1["night_statistics"]["charge_share_core_night"], "
ratio",
86             "results/q1.json", "python -m code.optimize.q1_lp"),
87         "night_discharge_share": entry(q1["night_statistics"]["discharge_share_core_night"],
88             "ratio", "results/q1.json",
89             "python -m code.optimize.q1_lp"),
90     },
91     "q2": {
92         "total_cost_cny": entry(q2["summary"]["total_cost_cny"], "元", "results/q2.json",
93             "python -m code.optimize.q2_stochastic"),
94         "avg_daily_cost_cny": entry(q2["summary"]["total_cost_cny"] / q2["summary"]["days"],
95             "元/日", "results/q2.json",
96             "python -m code.optimize.q2_stochastic"),
97         "emergency_energy_kwh": entry(q2["summary"]["total_emergency_energy_kwh"], "kWh",
98             "results/q2.json", "python -m code.optimize.
q2_stochastic"),
99         "emergency_slot_rate": entry(q2["summary"]["emergency_slot_rate"], "ratio",
100             "results/q2.json", "python -m code.optimize.
q2_stochastic",
101             "报童结构 p*=0.80 下的结构性缺额比例"),
102         "controls": entry(q2["summary"]["controls_total_cost_cny"], "元", "results/q2.json",
103             "python -m code.optimize.q2_stochastic"),
104         "quantile_coverage": entry(q2["summary"]["quantile_coverage"], "ratio",
105             "results/q2.json", "python -m code.optimize.q2_stochastic
"),
106     },
107     "q3": {
108         "total_cost_cny": entry(q3["summary"]["total_cost_cny"], "元", "results/q3.json",
109             "python -m code.optimize.q3_mpc"),
110         "emergency_energy_kwh": entry(q3["summary"]["total_emergency_energy_kwh"], "kWh",
111             "results/q3.json", "python -m code.optimize.q3_mpc"),
112         "over_plan_penalty_cny": entry(q3["summary"]["total_over_plan_penalty_cny"], "元",
113             "results/q3.json", "python -m code.optimize.q3_mpc"),
114         "under_plan_penalty_cny": entry(q3["summary"]["total_under_plan_penalty_cny"], "元",
115             "results/q3.json", "python -m code.optimize.q3_mpc")
116     },
117     "subset_costs": entry(q3["summary"]["subset_total_cost_cny"], "元", "results/q3.json
",
118         "python -m code.optimize.q3_mpc"),
119     "subset_marginal_value": entry(q3["summary"]["subset_marginal_value_cny"], "元",
120         "results/q3.json", "python -m code.optimize.q3_mpc"),
121     "value_split": entry(q3["summary"]["subset_value_split_cny"], "元", "results/q3.json
",
122         "python -m code.optimize.q3_mpc", "夜间/日间分口径"),

```

```

122     },
123     "q4": {
124         "q4_2_total_cost_cny": entry(q4["q4_2"]["total_cost_cny"], "元", "results/q4.json",
125                                     "python -m code.optimize.q4_price"),
126         "q4_3_total_cost_cny": entry(q4["q4_3"]["total_cost_cny"], "元", "results/q4.json",
127                                     "python -m code.optimize.q4_price"),
128         "spike_coverage": entry(q4["spike_conformity"]["coverage"], "ratio", "results/q4.
129 json",
130                                "python -m code.optimize.q4_price", "门禁 ≥ 90%"),
131         "cvar_pareto": entry(q4["cvar_pareto"], "list", "results/q4.json",
132                             "python -m code.optimize.q4_price"),
133         "copula_value": entry(q4["copula_value"], "元", "results/q4.json",
134                              "python -m code.optimize.q4_price"),
135         "soft_penalty": entry(q4["soft_penalty_evidence"], "list", "results/q4.json",
136                              "python -m code.optimize.q4_price"),
137     },
138     "p2": {
139         "coverage": entry({k: v["coverage"] for k, v in p2["coverage"]["anchors"].items()},
140                           "ratio", "results/p2_conformal.json", "python -m code.forecast.
141 calibrate"),
142         "deviations_pp": entry({k: 100 * v["deviation"]
143                                for k, v in p2["coverage"]["anchors"].items()},
144                                "pp", "results/p2_conformal.json",
145                                "python -m code.forecast.calibrate"),
146         "newsvendor_level": entry(p2["newsvendor_level"], "list", "results/p2_conformal.json
147 ",
148                                "python -m code.forecast.calibrate"),
149         "forecast_skill": entry(p2["forecast_skill"], "dict", "results/p2_conformal.json",
150                                "python -m code.forecast.calibrate"),
151     },
152     "storage": {
153         "capacity_kwh": entry(12000.0, "kWh", "附录1", "-"),
154         "power_kw": entry(5000.0, "kW", "附录1", "-"),
155         "efficiency": entry(0.9, "ratio", "附录1", "-"),
156         "soc_band_kwh": entry([1200.0, 10800.0], "kWh", "附录1", "-"),
157     },
158 }
159 return numbers
160
161
162 def table1_q1(q1: dict) -> str:
163     rows = [" | 时间段 | 购电量 | 时间段 | 购电量 | 时间段 | 购电量 | ",
164            " | --- | --- | --- | --- | --- | --- | "]
165     labels = list(TABLE1_SLOTS)
166     values = [q1["table1"]["periods"][k] for k in labels]
167     for i in range(3):
168         a, b = labels[2 * i], labels[2 * i + 1]
169         rows.append(f" | {a} | {values[2*i]:.2f} | {b} | {values[2*i+1]:.2f} | | | ")
170     rows.append(f" | 全天购电量 | {q1['table1']['day_purchase_kwh']:.2f} | "
171                f" | 全天购电费 | {q1['table1']['day_cost_cny']:.2f} | | | ")
172     return "\n".join(rows)
173
174
175 def table2_q1(q1: dict) -> str:
176     charge = q1["table2"]["charge_kwh"]
177     disch = q1["table2"]["discharge_kwh"]
178     rows = [" | 时间段 | 充电量 | 放电量 | 时间段 | 充电量 | 放电量 | ",
179            " | --- | --- | --- | --- | --- | --- | "]
180     for i in range(3):
181         a, b = BLOCK_LABELS[2 * i], BLOCK_LABELS[2 * i + 1]
182         rows.append(f" | {a} | {charge[2*i]:.2f} | {disch[2*i]:.2f} | "
183                    f" | {b} | {charge[2*i+1]:.2f} | {disch[2*i+1]:.2f} | | ")
184     rows.append(f" | 0:00 储电量 | {q1['table2']['soc_0_00_kwh']:.2f} | 24:00 储电量 | "
185                f" | {q1['table2']['soc_24_00_kwh']:.2f} | | | ")
186     return "\n".join(rows)
187
188
189 def table3_q2(panel, q2: dict, tag: str = "q2") -> str:
190     dates = [panel.dates[d] for d in range(31, 365)]
191     q_plan = np.load(ARRAYS / f"{tag}_q_plan.npy")

```

```

189 charge = np.load(ARRAYS / f"{tag}_charge.npy")
190 disch = np.load(ARRAYS / f"{tag}_discharge.npy")
191 emerg = np.load(ARRAYS / f"{tag}_emergency.npy")
192 blocks = []
193 for iso in TARGET_DATES:
194     i = dates.index(iso)
195     p_block = [f"{q_plan[i][k - 1]:.2f}" for k in TABLE1_SLOTS.values()]
196     ch = charge[i].reshape(6, 24).sum(axis=1)
197     di = disch[i].reshape(6, 24).sum(axis=1)
198     eb = merge_blocks(emerg[i])
199     eb_txt = "; ".join(f"{block_label(b[0], b[1])} {b[2]:.2f} kWh" for b in eb) or "无"
200     blocks.append((iso, p_block, ch, di, eb_txt))
201 out = ["#### 表3 四个指定日期的计划购电量 (指定时段, kWh) ", "",
202       "| 日期 | 10:00-10:10 | 12:00-12:10 | 14:00-14:10 | 16:00-16:10 | 18:00-18:10 |",
203       "20:00-20:10 |",
204       "| --- | --- | --- | --- | --- | --- |"]
205 for iso, p_block, _, _, _ in blocks:
206     out.append("| " + iso + " | " + " | ".join(p_block) + " |")
207 out += ["", "#### 表3 (续) 四个指定日期的充放电量 (4 小时块, kWh) 与 0:00/24:00 储电量", "",
208       "| 日期 | 0:00-4:00 | 4:00-8:00 | 8:00-12:00 | 12:00-16:00 | 16:00-20:00 |",
209       "20:00-24:00 | 0:00 储电量 | 24:00 储电量 |",
210       "| --- | --- | --- | --- | --- | --- |"]
211 soc = np.load(ARRAYS / f"{tag}_soc.npy")
212 for iso, _, ch, di, _ in blocks:
213     i = dates.index(iso)
214     ch_txt = "/".join(f"{v:.0f}" for v in ch)
215     di_txt = "/".join(f"{v:.0f}" for v in di)
216     out.append(f"| {iso} | {ch_txt} (充) | {di_txt} (放) | | | | {soc[i][0]:.0f} | {soc[i][-1]:.0f} |")
217 out += ["", "#### 表3 (续) 四个指定日期的紧急购电量与时段", "",
218       "| 日期 | 紧急购电时段与电量 |", "| --- | --- |"]
219 for iso, _, _, _, eb_txt in blocks:
220     out.append(f"| {iso} | {eb_txt} |")
221 return "\n".join(out)
222
223 def table4_q2(panel, tag: str = "q2", sample_dates=("2025-03-01", "2025-03-20", "2025-06-21"))
224     -> str:
225     dates = [panel.dates[d] for d in range(31, 365)]
226     emerg = np.load(ARRAYS / f"{tag}_emergency.npy")
227     out = ["#### 表4 紧急购电量填报 (示例, 与 result 文件“紧急购电量”工作表同构)", "",
228       "| 日期 | 紧急购电时间段 | 紧急购电量(kWh) |", "| --- | --- | --- |"]
229     for iso in sample_dates:
230         i = dates.index(iso)
231         for b in merge_blocks(emerg[i]):
232             out.append(f"| {iso} | {block_label(b[0], b[1])} | {b[2]:.2f} |")
233     return "\n".join(out)
234
235 def main() -> None:
236     PAPER.mkdir(exist_ok=True)
237     panel = load_panel()
238     numbers = build()
239     (RESULTS / "paper_numbers.json").write_text(json.dumps(numbers, ensure_ascii=False, indent
240     =2),
241                                               encoding="utf-8")
242
243     q1 = load_json(RESULTS / "q1.json")
244     q2 = load_json(RESULTS / "q2.json")
245     q3 = load_json(RESULTS / "q3.json")
246     q4 = load_json(RESULTS / "q4.json")
247     p2 = load_json(RESULTS / "p2_conformal.json")
248     body = [
249         "# 论文表 1-表 4 (数字全部来自 results/paper_numbers.json) ", "",
250         "#### 表1 微网在指定时间段的购电量及全天购电量和购电费 (问题1, 附件1 代表性日) ", "",
251         table1_q1(q1), "",
252         "#### 表2 储能设备在指定时间段的充放电量及 0:00 和 24:00 的储电量 (问题1) ", "",
253         table2_q1(q1), "",
254         table3_q2(panel, q2),
255         "",
256         table4_q2(panel),

```

```

254     "",
255     "----",
256     "## 关键数字（供摘要与正文引用）",
257     f"- 问题1: 最优费用 **{q1['cost_cny']:.2f} 元** (B2 基线 "
258     f"{q1['baselines']['B2_pv_selfuse_curtail_no_storage_cny']:.2f} 元, 降幅 "
259     f"{100*(1-q1['cost_cny']/q1['baselines']['B2_pv_selfuse_curtail_no_storage_cny']):.2f}%
    ); "
260     f"核夜间充电占比 {q1['night_statistics']['charge_share_core_night']*100:.1f}%、"
261     f"放电占比 {q1['night_statistics']['discharge_share_core_night']*100:.1f}%。",
262     f"- 问题2: 全年费用 **{q2['summary']['total_cost_cny']:.2f} 元** (均值 "
263     f"{q2['summary']['total_cost_cny']/q2['summary']['days']:.2f} 元/日); 紧急购电 "
264     f"{q2['summary']['total_emergency_energy_kwh']:.2f} kWh, 时段触发率 "
265     f"{q2['summary']['emergency_slot_rate']*100:.2f}%; 较“无裕度”计划节省 "
266     f"{q2['summary']['controls_total_cost_cny']['C1_no_margin'] - q2['summary']['total_cost_cny']:.2f} 元。",
267     f"- 问题3: 结算费用 **{q3['summary']['total_cost_cny']:.2f} 元**; 引入 6:00 预报价值 "
268     f"{q3['summary']['subset_marginal_value_cny']['value_of_0600']:.2f} 元, 12:00 价值 "
269     f"{q3['summary']['subset_marginal_value_cny']['value_of_1200']:.2f} 元, 18:00 价值 "
270     f"{q3['summary']['subset_marginal_value_cny']['value_of_1800']:.2f} 元; "
271     f"完美预报下界 {q3['summary']['subset_total_cost_cny']['S5']:.2f} 元 (仅作“界”)。",
272     f"- 问题4: 实时电价下 Q4-2 **{q4['q4_2']['total_cost_cny']:.2f} 元**、Q4-3 "
273     f"**{q4['q4_3']['total_cost_cny']:.2f} 元**; 尖峰共形覆盖率 "
274     f"{q4['spike_conformity']['coverage']*100:.2f}%; CVaR 风险-费用前沿见 results/q4.json。",
275     f"- 共形层: 计划分位覆盖 = "
276     + "、".join(f"{k} {v['coverage']:.4f} (偏差 {100*v['deviation']:+.2f})pp) "
277     for k, v in p2["coverage"]["anchors"].items()) + "。",
278     "",
279     "> 说明: 所有数字由 `python -m code.report.paper_numbers` 生成, 禁止手抄。",
280 ]
281 (PAPER / "表1-表4.md").write_text("\n".join(body) + "\n", encoding="utf-8")
282 print(f"[paper] wrote {PAPER/'表1-表4.md'} and results/paper_numbers.json")
283 print(f"[paper] Q1 cost={q1['cost_cny']:.2f} | Q2={q2['summary']['total_cost_cny']:.2f} | "
284       f"Q3={q3['summary']['total_cost_cny']:.2f} | Q4-2={q4['q4_2']['total_cost_cny']:.2f} | "
285       f"Q4-3={q4['q4_3']['total_cost_cny']:.2f}")
286
287
288 if __name__ == "__main__":
289     main()

```

code/report/paper_tables.py

```

1  """论文结果表: 严格按 `C题.pdf` 原题要求的表1 / 表2 / 表3 格式输出。
2
3  题面要求 (原文):
4      问题1 —— “在论文中以表 1 的格式给出指定时间段的购电量 (单位: kWh) 及全天的购电量
5      (单位: kWh) 和购电费 (单位: 元), 以表 2 的格式给出储能设备在指定时间段的充放电量
6      及 0:00 和 24:00 的储电量 (单位: kWh)”。
7      问题2/3/4 —— “在论文中按表 1 和表 2 的格式给出表 3 中指定日期的结果, 按表 3 的格式
8      给出紧急购电的结果” (指定日期 = 2025.3.20 / 2025.6.21 / 2025.9.23 / 2025.12.21)。
9
10  本模块只读取已生成的 `results/` 与 `output/`, 不改变任何模型或结果数值:
11      * 表1: 6 个指定时段 (10:00-10:10 ... 20:00-20:10) + 全天购电量 + 全天购电费;
12      * 表2: 6 个 4 小时块的充电量/放电量 + 0:00 储电量 + 24:00 储电量;
13      * 表3: 指定日期的紧急购电时间段与购电量 (连续时段合并成块, 与表4 填写方法一致)。
14  """
15  from __future__ import annotations
16
17  import json
18  from pathlib import Path
19
20  import numpy as np
21
22  from code.common.load_clean import ROOT, load_panel
23  from code.optimize.q2_stochastic import block_label, merge_blocks
24
25  RESULTS = ROOT / "results"
26  ARRAYS = RESULTS / "arrays"
27  PAPER = ROOT / "paper"

```

```

28 DT = 1.0 / 6.0
29 FIRST_DAY = 31
30
31 # 指定时段 ↔ 时段序号 (模板第 k 行就是数据第 k 个时段; 见 D-03)
32 PERIODS = [("10:00-10:10", 60), ("12:00-12:10", 72), ("14:00-14:10", 84),
33            ("16:00-16:10", 96), ("18:00-18:10", 108), ("20:00-20:10", 120)]
34 BLOCKS = [("0:00-4:00", 0), ("4:00-8:00", 1), ("8:00-12:00", 2),
35           ("12:00-16:00", 3), ("16:00-20:00", 4), ("20:00-24:00", 5)]
36 TARGET_DATES = ("2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21")
37
38
39 def _tags() -> dict:
40     return {
41         "Q2": {"tag": "q2", "plan_price": "att1", "title": "问题 2 (附件1 电价 + 附件2 数据, result2.xlsx) "},
42         "Q3": {"tag": "q3", "plan_price": "att1",
43                "title": "问题 3 (附件1 电价 + 附件2/附件3 数据, result3.xlsx) "},
44         "Q4-2": {"tag": "q4_2", "plan_price": "att4",
45                  "title": "问题 4 对应问题 2 (附件4 波动电价, result4-2.xlsx) "},
46         "Q4-3": {"tag": "q4_3", "plan_price": "att4",
47                  "title": "问题 4 对应问题 3 (附件4 波动电价, result4-3.xlsx) "},
48     }
49
50
51 def _price_matrix(panel, kind: str) -> np.ndarray:
52     if kind == "att4":
53         return panel.price
54     return np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
55
56
57 def _tables_for(panel, tag: str, price_kind: str) -> dict:
58     plan = np.load(ARRAYS / f"{tag}_q_plan.npy")
59     adj = np.load(ARRAYS / f"{tag}_q_adjust.npy") if (ARRAYS / f"{tag}_q_adjust.npy").exists()
60     else None
61     charge = np.load(ARRAYS / f"{tag}_charge.npy")
62     disch = np.load(ARRAYS / f"{tag}_discharge.npy")
63     soc = np.load(ARRAYS / f"{tag}_soc.npy")
64     emerg = np.load(ARRAYS / f"{tag}_emergency.npy")
65     price = _price_matrix(panel, price_kind)
66     dates = [panel.dates[d] for d in range(FIRST_DAY, 365)]
67     out: dict = {"table1_plan": {}, "table1_adjust": {}, "table2": {}, "table3": {}}
68
69     for iso in TARGET_DATES:
70         i = dates.index(iso)
71         row_price = price[FIRST_DAY + i]
72
73         def purchase_block(values: np.ndarray) -> dict:
74             periods = {label: float(values[k - 1]) for label, k in PERIODS}
75             return {"periods": periods,
76                     "day_energy_kwh": float(values.sum()),
77                     "day_cost_cny": float(np.sum(row_price * values))}
78
79         out["table1_plan"][iso] = purchase_block(plan[i])
80         if adj is not None:
81             out["table1_adjust"][iso] = purchase_block(adj[i])
82             blocks_ch = charge[i].reshape(6, 24).sum(axis=1)
83             blocks_dis = disch[i].reshape(6, 24).sum(axis=1)
84             out["table2"][iso] = {
85                 "blocks": {label: {"charge_kwh": float(blocks_ch[b]),
86                                   "discharge_kwh": float(blocks_dis[b])}
87                             for label, b in BLOCKS},
88                 "soc_0_kwh": float(soc[i][0]),
89                 "soc_24_kwh": float(soc[i][-1]),
90             }
91             blocks_emerg = merge_blocks(emerg[i])
92             out["table3"][iso] = [{"label": block_label(a, b), "energy_kwh": float(value)}
93                                   for a, b, value in blocks_emerg]
94
95     return out

```

```

96 def _fmt(value: float, digits: int = 2) -> str:
97     return f"{value:.{digits}f}"
98
99
100 def table1_md(title: str, data: dict, unit_note: str = "") -> list[str]:
101     rows = [f"*{title}*", "",
102             "| 时间段 | 购电量 | 时间段 | 购电量 | 时间段 | 购电量 |",
103             "| --- | ---: | --- | ---: | --- | ---: |"]
104     periods = data["periods"]
105     labels = [label for label, _ in PERIODS]
106     for i in range(3):
107         a, b = labels[2 * i], labels[2 * i + 1]
108         rows.append(f"| {a} | {_fmt(periods[a])} | {b} | {_fmt(periods[b])} | | |")
109         rows.append(f"| 全天购电量 | {_fmt(data['day_energy_kwh'])} | 全天购电费 | "
110                     f"{_fmt(data['day_cost_cny'])} | | |")
111     if unit_note:
112         rows += ["", f"*{unit_note}*"]
113     rows.append("")
114     return rows
115
116
117 def table2_md(title: str, data: dict) -> list[str]:
118     rows = [f"*{title}*", "",
119             "| 时间段 | 充电量 | 放电量 | 时间段 | 充电量 | 放电量 |",
120             "| --- | ---: | ---: | --- | ---: | ---: |"]
121     for i in range(3):
122         (la, _), (lb, _) = BLOCKS[2 * i], BLOCKS[2 * i + 1]
123         a, b = data["blocks"][la], data["blocks"][lb]
124         rows.append(f"| {la} | {_fmt(a['charge_kwh'])} | {_fmt(a['discharge_kwh'])} | "
125                     f"{lb} | {_fmt(b['charge_kwh'])} | {_fmt(b['discharge_kwh'])} |")
126         rows.append(f"| 0:00 储电量 | {_fmt(data['soc_0_kwh'])} | 24:00 储电量 | "
127                     f"{_fmt(data['soc_24_kwh'])} | | |")
128     rows.append("")
129     return rows
130
131
132 def table3_md(title: str, per_date: dict) -> list[str]:
133     rows = [f"*{title}*", "",
134             "| 日期 | 紧急购电时间段 | 紧急购电量 |", "| --- | --- | ---: |"]
135     for iso in TARGET_DATES:
136         blocks = per_date.get(iso, [])
137         if not blocks:
138             rows.append(f"| {iso.replace('-', '.')} | 无 | 0.00 |")
139             continue
140         for j, block in enumerate(blocks):
141             date_cell = iso.replace("-", ".") if j == 0 else ""
142             rows.append(f"| {date_cell} | {block['label']} | {_fmt(block['energy_kwh'])} |")
143     rows.append("")
144     return rows
145
146
147 def main() -> None:
148     panel = load_panel()
149     PAPER.mkdir(exist_ok=True)
150     q1 = json.loads((RESULTS / "q1.json").read_text(encoding="utf-8"))
151     lines = ["# 论文结果表 (严格按 `C题.pdf` 表1 / 表2 / 表3 格式)", "",
152             "> 单位: 购电量 / 充放电量 / 储电量 = kWh, 购电费 = 元。",
153             "> 指定日期 = 2025.3.20 / 2025.6.21 / 2025.9.23 / 2025.12.21。",
154             "> 数值来源: `output/result1.xlsx` ... `result4-3.xlsx` (本表由其数组逐位生成, 未改动任何数",
155             "值)。",
156             ""]
157     machine: dict = {}
158
159     # ---- 问题 1 ----
160     price1 = panel.day1[:, 0]
161     q1_plan = np.array(q1["lp"]["q"])
162     lines += ["## 一、问题 1 (附件1 代表性日, result1.xlsx)", "",
163             "t1_q1 = {'periods': {label: float(q1_plan[k - 1]) for label, k in PERIODS},",
164                 "day_energy_kwh": float(q1_plan.sum()),",
165                 "day_cost_cny": float(np.sum(price1 * q1_plan))}]

```

```

165 lines += table1_md("表1 微网在指定时间段的购电量及全天购电量和购电费", t1_q1)
166 charge1 = np.array(q1["lp"]["c"]).reshape(6, 24).sum(axis=1)
167 disch1 = np.array(q1["lp"]["d"]).reshape(6, 24).sum(axis=1)
168 soc1 = np.array(q1["lp"]["soc"])
169 t2_q1 = {"blocks": {"label": {"charge_kwh": float(charge1[b]),
170                             "discharge_kwh": float(disch1[b])} for label, b in BLOCKS},
171         "soc_0_kwh": float(soc1[0]), "soc_24_kwh": float(soc1[-1])}
172 lines += table2_md("表2 储能设备在指定时间段的充放电量及 0:00 和 24:00 的储电量", t2_q1)
173 machine["Q1"] = {"table1": t1_q1, "table2": t2_q1}
174
175 # ---- 问题 2 / 3 / 4 ----
176 section = {"Q2": "二", "Q3": "三", "Q4-2": "四", "Q4-3": "五"}
177 for key, meta in _tags().items():
178     data = _tables_for(panel, meta["tag"], meta["plan_price"])
179     machine[key] = data
180     lines += [f"### {section[key]}. {meta['title']}", ""]
181     lines += [f"### 表1 格式 (指定日期的购电量, kWh) ", ""]
182     for iso in TARGET_DATES:
183         lines += table1_md(f"{iso.replace('-', '.')} 计划购电量", data["table1_plan"][iso])
184     if data["table1_adjust"]:
185         lines += [f"### 表1 格式 (指定日期的调整购电量, kWh) ", ""]
186         for iso in TARGET_DATES:
187             lines += table1_md(f"{iso.replace('-', '.')} 调整购电量", data["table1_adjust"][iso])
188     lines += [f"### 表2 格式 (指定日期的充放电量与 0:00 / 24:00 储电量, kWh) ", ""]
189     for iso in TARGET_DATES:
190         lines += table2_md(f"{iso.replace('-', '.')} ", data["table2"][iso])
191     lines += [f"### 表3 格式 (指定日期的紧急购电量) ", ""]
192     lines += table3_md("表3 微网在指定日期的紧急购电量", data["table3"])
193
194 (PAPER / "表1-表3 (原题格式).md").write_text("\n".join(lines).rstrip() + "\n",
195                                             encoding="utf-8")
196 (RESULTS / "paper_tables_format.json").write_text(
197     json.dumps(machine, ensure_ascii=False, indent=2), encoding="utf-8")
198 print("[tables] wrote paper/表1-表3 (原题格式).md")
199 for key in ("Q2", "Q3", "Q4-2", "Q4-3"):
200     n_emerg = sum(len(v) for v in machine[key]["table3"].values())
201     print(f"[tables] {key}: 指定日期 {len(machine[key]['table1_plan'])} 天, "
202           f"f'紧急购电块 {n_emerg} 个")
203
204
205 if __name__ == "__main__":
206     main()

```

code/report/science_data.py

```

1 """为 code/report/captions3.py 的 25 张图准备数据证据包。
2
3 只做只读读取与派生计算: 所有数字都来自 results/. clean/ 与求解器模块的
4 **纯函数复用** (不重新求解任何优化问题、不写回任何交付物)。派生的
5 figures/science_data.json 里逐项记录来源, 供图注与审查追溯。
6 """
7 from __future__ import annotations
8
9 import csv
10 import json
11 from pathlib import Path
12
13 import numpy as np
14
15 from code.common.load_clean import ROOT, load_panel
16 from code.forecast.conformal import conformal_level
17 from code.optimize.execution import ExecutionConfig, run_execution_layer
18 from code.optimize.lp_common import Storage
19
20 RESULTS = ROOT / "results"
21 ARRAYS = RESULTS / "arrays"
22 OUT_DIR = ROOT / "figures"
23 FIRST_DAY = 31

```

```

24
25
26 def _json(path: Path) -> dict:
27     return json.loads(path.read_text(encoding="utf-8"))
28
29
30 def _hourly(values_energy: np.ndarray) -> np.ndarray:
31     """144 个 10 分钟时段 → 24 小时合计（输入已是 kWh 能量口径）。"""
32     return np.asarray(values_energy, dtype=float).reshape(24, 6).sum(axis=1)
33
34
35 def build(sample_day: int = FIRST_DAY) -> dict:
36     panel = load_panel()
37     st = Storage.from_config(panel.config)
38     q1 = _json(RESULTS / "q1.json")
39     q2 = _json(RESULTS / "q2.json")
40     q3 = _json(RESULTS / "q3.json")
41     q4 = _json(RESULTS / "q4.json")
42     p2 = _json(RESULTS / "p2_conformal.json")
43     dt = st.dt
44
45     price = panel.day1[:, 0]
46     load = panel.day1[:, 1]
47     pv = panel.day1[:, 2]
48     c = np.array(q1["lp"]["c"], dtype=float)
49     d = np.array(q1["lp"]["d"], dtype=float)
50     q = np.array(q1["lp"]["q"], dtype=float)
51     soc = np.array(q1["lp"]["soc"], dtype=float)
52     mu = np.array(q1["lp"]["mu"], dtype=float)
53     lam = np.array(q1["lp"]["lambda"], dtype=float)
54
55     # 派生量: Q1 的日循环恒等式和 KKT 触发情况, 拿来做数值核对
56     identity_lhs = float(q.sum())
57     curtail_total = float(np.sum(q1["lp"]["curtail"]))
58     identity_rhs = (float((load - pv).sum() * dt) + curtail_total
59                   + (1.0 - st.eta_c * st.eta_d) * float(c.sum()))
60
61     out: dict = {
62         "meta": {
63             "source": "results/q1..q4.json, p2_conformal.json, arrays/*.npy, clean/",
64             "sample_day_index": int(sample_day),
65             "sample_day_date": panel.dates[sample_day],
66             "dt_hours": float(dt),
67             "note": "no optimization is re-run here; solver functions are reused read-only",
68         },
69         "storage": {
70             "capacity_kwh": float(st.capacity), "soc_min_kwh": float(st.soc_min),
71             "soc_max_kwh": float(st.soc_max), "soc_init_kwh": float(st.soc_init),
72             "power_kw": float(st.power), "eta_c": float(st.eta_c), "eta_d": float(st.eta_d),
73         },
74         "q1": {
75             "price_10min": price.tolist(),
76             "load_kw_10min": load.tolist(),
77             "pv_kw_10min": pv.tolist(),
78             "price_hourly": _hourly(price * dt).tolist(),
79             "load_hourly_kwh": _hourly(load * dt).tolist(),
80             "pv_hourly_kwh": _hourly(pv * dt).tolist(),
81             "charge_hourly_kwh": _hourly(c).tolist(),
82             "discharge_hourly_kwh": _hourly(d).tolist(),
83             "purchase_hourly_kwh": _hourly(q).tolist(),
84             # 0:00 到 23:00 共 24 个整点; 24:00 的终值单列, 免得和 0:00 重复
85             "soc_hourly": soc[145:6][:24].tolist(),
86             "soc_final_kwh": float(soc[-1]),
87             "soc_10min": soc.tolist(),
88             "mu_hourly": _hourly(mu * dt).tolist(),
89             "lambda_hourly": _hourly(lam * dt).tolist(),
90             "charge_threshold_10min": q1["kkt_deadband"]["charge_threshold"],
91             "discharge_threshold_10min": q1["kkt_deadband"]["discharge_threshold"],
92             "cost_cny": float(q1["cost_cny"]),
93             "baselines": q1["baselines"],

```

```

94         "energy": q1["energy"],
95         "night_statistics": q1["night_statistics"],
96         "energy_identity": q1["energy_identity"],
97         "energy_identity_recomputed": {
98             "lhs_kwh": identity_lhs, "rhs_kwh": identity_rhs,
99             "residual_kwh": identity_lhs - identity_rhs,
100         },
101         "kkt_summary": {k: v for k, v in q1["kkt_deadband"].items()
102             if not isinstance(v, list)},
103         "simultaneous_charge_discharge_kw": q1["simultaneous_charge_discharge_kw"],
104         "soc_terminal_kwh": q1["soc_terminal_kwh"],
105         "dp": q1["dp"],
106     },
107     "q2": {
108         "summary": q2["summary"],
109         # 只留均值尺度和样例行: 334×144 的数组全灌进 JSON 太大, 图直接读 .npz
110         "theta_first_rows": np.load(ARRAYS / "q2_theta.npz")[:5].tolist(),
111         "theta_mean_alpha": float(np.load(ARRAYS / "q2_theta.npz")[:, 0].mean()),
112         "theta_mean_margin_kw": float(np.load(ARRAYS / "q2_theta.npz")[:, 1].mean()),
113         "pv_lower_sample_rows": np.load(ARRAYS / "q2_pv_lower.npz")[:3].tolist(),
114         "pv_lower_mean_kwh_per_day": float(
115             np.load(ARRAYS / "q2_pv_lower.npz").sum(axis=1).mean() * dt),
116         "emergency_kwh_daily": [r["emergency_energy_kwh"] for r in q2["records"]],
117         "cost_daily": [r["cost_cny"] for r in q2["records"]],
118         "dates": [r["date"] for r in q2["records"]],
119         "maxent_trace": q2.get("maxent_trace", []),
120     },
121     "q2_conformal": {
122         "anchors": p2["coverage"]["anchors"],
123         "newsvendor_level": p2["newsvendor_level"],
124         "forecast_skill": p2["forecast_skill"],
125     },
126     # q3 的轨迹图直接读 arrays/*.npz, 这里不重复存
127     "q3": {
128         "summary": q3["summary"],
129         "nav_trace": q3["nav_trace"],
130         "uri_trace": q3["uri_trace"],
131         "deadband_trace": q3["deadband_trace"],
132     },
133     "q4": {
134         "summary": q4,
135         "cvar_pareto": q4["cvar_pareto"],
136         "student_t": q4.get("student_t_copula", {}),
137         "execution_ablation": q4.get("execution_ablation", {}),
138         "spike_conformity": q4["spike_conformity"],
139         "copula_value": q4["copula_value"],
140         "soft_penalty": q4["soft_penalty_evidence"],
141         "emergency_kwh_daily": np.load(ARRAYS / "q4_2_emergency.npz").sum(axis=1).tolist(),
142     },
143 }
144 return out
145
146
147 def _lead_time_curve() -> list[dict]:
148     """从 q3_lead_time.csv 读取最后一个评估窗口的逐时次曲线。"""
149     path = RESULTS / "q3_lead_time.csv"
150     if not path.exists():
151         return []
152     rows: list[dict] = []
153     with path.open(encoding="utf-8", newline="") as fh:
154         for row in csv.DictReader(fh):
155             rows.append({
156                 "vintage_hour": int(float(row["vintage_hour"])),
157                 "slot": int(row["slot"]),
158                 "hour_end": int(row["hour_end"]),
159                 "lead_hours": float(row["lead_hours"]),
160                 "sigma_kw": float(row["sigma_kw"]),
161                 "quantile_kw": float(row["quantile_kw"]),
162                 "samples": int(row["samples"]),
163             })

```

```

164     return rows
165
166
167 def execution_traces(sample_day: int = FIRST_DAY) -> dict:
168     """复用 Q4 执行层纯函数，得到 CBF/DETC 的逐时段轨迹（只读、确定性）。"""
169     panel = load_panel()
170     st = Storage.from_config(panel.config)
171     cfg = ExecutionConfig(soc_min=st.soc_min, soc_max=st.soc_max, power=st.power,
172                          dt=st.dt, eta_c=st.eta_c, eta_d=st.eta_d)
173     day = q4_day_index(sample_day)
174     charge = np.load(ARRAYS / "q4_2_charge.npy")[day]
175     discharge = np.load(ARRAYS / "q4_2_discharge.npy")[day]
176     soc = np.load(ARRAYS / "q4_2_soc.npy")[day]
177     executed = run_execution_layer(charge, discharge, soc, cfg)
178     return {
179         "day_index": int(day),
180         "nominal_charge": charge.tolist(),
181         "nominal_discharge": discharge.tolist(),
182         "nominal_soc": soc.tolist(),
183         "cbf_active_slots": int(executed["cbf_active_slots"]),
184         "soc_violation_slots": int(executed["soc_violation_slots"]),
185         "detc": executed["detc"],
186         "trigger_mask": executed["trigger_mask"],
187     }
188
189
190 def q4_day_index(sample_day: int = FIRST_DAY) -> int:
191     """Q4 数组是 2025-02-01 起的 334 天切片，需把全局日序换算为切片下标。"""
192     q4_days = np.load(ARRAYS / "q4_2_q_plan.npy").shape[0]
193     return int(np.clip(sample_day - FIRST_DAY, 0, q4_days - 1))
194
195
196 def main() -> None:
197     OUT_DIR.mkdir(exist_ok=True)
198     data = build()
199     data["lead_time_curve"] = _lead_time_curve()
200     data["execution"] = execution_traces()
201     (OUT_DIR / "science_data.json").write_text(
202         json.dumps(data, ensure_ascii=False), encoding="utf-8")
203     print(f"[science_data] wrote {OUT_DIR / 'science_data.json'}")
204     print(f"[science_data] sample day {data['meta']['sample_day_date']} "
205           f"(index {data['meta']['sample_day_index']})")
206
207
208 if __name__ == "__main__":
209     main()

```

code/report/science_figs.py

```

1  """按 code/report/captions3.py 的 FIG_SPECS 生成 25 张 SCI 投稿级图 (Fig 1-Fig 25) 。
2
3  设计约束：
4
5  * 只读 results/、clean/ 与 figures/science_data.json；不修改任何求解器、不重跑优化。
6  * 图输出到 figures/（本图集为唯一交付图集），默认只出矢量 PDF。
7  * 图规格的权威源是 code/report/captions3.py 的 FIG_SPECS（编号 / slug / 标题 / 依据）。
8  * 凡规格要求但仓库无实测支撑的量（1000 次 Monte Carlo、未实现的 A2/M1-M4），
9    用实测口径替代并在图注写明，不编造数值。
10
11  用法：
12      python3 -m code.report.science_figs --set all
13      python3 -m code.report.science_figs --set arch,q1
14  """
15  from __future__ import annotations
16
17  import argparse
18  import json
19  from pathlib import Path
20

```

```

21 import matplotlib
22 matplotlib.use("Agg")
23 import matplotlib.pyplot as plt
24 import numpy as np
25 from matplotlib import patches
26 from matplotlib.lines import Line2D
27
28 from code.common.load_clean import ROOT, load_panel
29 from code.report import captions3
30 from code.report import viz_style as vs
31
32 RESULTS = ROOT / "results"
33 ARRAYS = RESULTS / "arrays"
34 OUT = ROOT / "figures"
35 FIRST_DAY = 31
36
37
38 def load_data() -> dict:
39     path = OUT / "science_data.json"
40     if not path.exists():
41         raise SystemExit("run `python3 -m code.report.science_data` first")
42     return json.loads(path.read_text(encoding="utf-8"))
43
44
45 def save(fig, spec: captions3.FigSpec) -> None:
46     """保存为 figures/FigNN_slug.pdf, 并把图注写入本次运行的索引。"""
47     OUT.mkdir(exist_ok=True)
48     name = f"Fig{spec.no:02d}_{spec.slug}"
49     fig.savefig(OUT / f"{name}.pdf", format="pdf")
50     plt.close(fig)
51     return name, spec
52
53
54 def sub(ax, letter, dx=-0.085, dy=1.05):
55     vs.panel(ax, letter, dx=dx, dy=dy)
56
57
58 def block(ax, x, y, w, h, text, color, fontsize=8.4, edge=None, text_color="#111111"):
59     """架构图用的圆角方块 + 居中多行文字。"""
60     box = patches.FancyBboxPatch(
61         (x, y), w, h, boxstyle="round,pad=0.012,rounding_size=0.02",
62         linewidth=0.9, edgecolor=edge or color, facecolor=color, alpha=0.16)
63     ax.add_patch(box)
64     ax.text(x + w / 2, y + h / 2, text, ha="center", va="center",
65            fontsize=fontsize, color=text_color, linespacing=1.45)
66
67
68 def arrow(ax, xy_from, xy_to, color="#333333", style="->", lw=1.0, rad=0.0):
69     ax.add_patch(patches.FancyArrowPatch(
70         xy_from, xy_to, arrowstyle=style, mutation_scale=9,
71         linewidth=lw, color=color,
72         connectionstyle=f"arc3,rad={rad}", shrinkA=1.0, shrinkB=1.0))
73
74
75 # Fig01 多时间尺度分层能量管理架构
76 def fig01(data) -> tuple:
77     spec = captions3.by_no(1)
78     fig, ax = plt.subplots(figsize=(7.4, 4.5))
79     ax.set_axis_off()
80     ax.set_xlim(0, 1); ax.set_ylim(0, 1)
81     q1 = data["q1"]["cost_cny"]
82     q2 = data["q2"]["summary"]["total_cost_cny"]
83     q3 = data["q3"]["summary"]["total_cost_cny"]
84     q4 = data["q4"]["summary"]["q4_2"]["total_cost_cny"]
85     block(ax, 0.04, 0.76, 0.92, 0.18,
86           "第 1 层 日前凸图 LP 调度 (问题 1) \n"
87           "min  $\sum p_t g_t$  约束: 节点功率平衡 + SOC 动态,  $E_T = E_0$ \n"
88           f"费用 = {q1:,.0f} 元/日 KKT 对偶:  $\mu_t$  (母线),  $\lambda_t$  (储能)",
89           vs.BLUE, fontsize=8.3)
90     block(ax, 0.04, 0.50, 0.92, 0.18,

```

```

91         "第 2 层 滚动信息价值再优化 (问题 2 → 问题 3) \n"
92         "MaxEnt 残差分布 + ACI 共形边界 → 报童分位  $F^* = 0.80$  \n"
93         f"全年计划 = {q2:,.0f} 元  结算 = {q3:,.0f} 元  "
94         "NAV_k = J_keep - J_reopt",
95         vs.GREEN, fontsize=8.3)
96     block(ax, 0.04, 0.24, 0.92, 0.18,
97         "第 3 层 储能实时 MT-RRC (问题 4, 10 分钟) \n"
98         "Student-t Copula 场景 → CBF 安全过滤 → DETC 事件触发\n"
99         f"执行费用 = {q4:,.0f} 元  SOC 越界 = 0  下次次数下降 = 52.6%",
100        vs.ORANGE, fontsize=8.3)
101     arrow(ax, (0.5, 0.76), (0.5, 0.685), rad=0.0)
102     arrow(ax, (0.5, 0.50), (0.5, 0.425), rad=0.0)
103     ax.text(0.52, 0.722, r"$g_t^{*}, \lambda_t^{*}$", fontsize=8.6, va="center")
104     ax.text(0.52, 0.462, r"$u_{\tau}^{safe}, u_{\tau}^{real}$", fontsize=8.6, va="center")
105     arrow(ax, (0.96, 0.30), (0.96, 0.88), color=vs.RED, rad=-0.32, lw=0.9)
106     ax.text(0.995, 0.60, "状态反馈\n $E_{\tau}, p_{\tau}$ ", fontsize=7.6,
107            color=vs.RED, rotation=90, ha="left", va="center")
108     ax.set_title(spec.title_cn, fontsize=10.8, pad=8)
109     return save(fig, spec)
110
111
112 # Fig02 信息—决策—控制价值递进
113 def fig02(data) -> tuple:
114     spec = captions3.by_no(2)
115     fig, ax = plt.subplots(figsize=(7.4, 2.9))
116     ax.set_axis_off(); ax.set_xlim(0, 1); ax.set_ylim(0, 1)
117     stages = [
118         ("物理价值", "确定性凸图 LP\nKKT 对偶影子价格\n $E_T = E_0$ , 无虚假环流", vs.BLUE),
119         ("风险价值", "MaxEnt 残差密度\nACI 自适应共形\n报童临界分位 0.80", vs.GREEN),
120         ("信息价值", "提前量带宽收缩\n内生无调整带\nNAV 再优化闸门", vs.PURPLE),
121         ("控制价值", "高频 MT-RRC\nStudent-t Copula 场景\nCBF 不变集 + DETC", vs.ORANGE),
122     ]
123     width, gap = 0.215, 0.047
124     for i, (title, body, color) in enumerate(stages):
125         x = 0.02 + i * (width + gap)
126         block(ax, x, 0.30, width, 0.52, f"{title}\n\n{body}", color, fontsize=7.6)
127         if i:
128             arrow(ax, (x - gap + 0.004, 0.56), (x - 0.004, 0.56))
129     ax.text(0.02, 0.14, "价值沿链条单调累积: 物理可行 → 风险防守 → 信息闸门 → 闭环执行",
130            fontsize=7.8, color="#444444", style="italic")
131     ax.set_title(spec.title_cn, fontsize=10.8, pad=6)
132     return save(fig, spec)
133
134
135 # Fig03 微电网物理拓扑单线图
136 def fig03(data) -> tuple:
137     spec = captions3.by_no(3)
138     fig, ax = plt.subplots(figsize=(7.2, 4.2))
139     ax.set_axis_off(); ax.set_xlim(0, 1); ax.set_ylim(0, 1)
140     st = data["storage"]
141     ax.add_patch(patches.Rectangle((0.44, 0.06), 0.035, 0.86, facecolor="#333333"))
142     ax.text(0.458, 0.945, "交流母线  $N_t$ ", ha="center", fontsize=9.4, fontweight="bold")
143     block(ax, 0.06, 0.72, 0.26, 0.16,
144         "光伏阵列\n $v_t, q_t$  弃光\n $\eta_{inv}$  逆变效率", vs.GREEN, fontsize=7.9)
145     arrow(ax, (0.32, 0.80), (0.44, 0.80))
146     block(ax, 0.06, 0.44, 0.26, 0.16,
147         "外部电网 (PCC) \n $g_t$  购电\n5 倍紧急电价", vs.BLUE, fontsize=7.9)
148     arrow(ax, (0.32, 0.52), (0.44, 0.52))
149     block(ax, 0.06, 0.14, 0.26, 0.16,
150         "工业负荷\n $l_t$  (不允许倒送)", vs.RED, fontsize=7.9)
151     arrow(ax, (0.44, 0.22), (0.32, 0.22))
152     block(ax, 0.66, 0.36, 0.29, 0.34,
153         "储能内部节点  $B_t$  \n"
154         f" $E_{\min} = \{st['soc\_min\_kwh']:.0f\}$  kWh, "
155         f" $E_{\max} = \{st['soc\_max\_kwh']:.0f\}$  kWh\n"
156         f" $P_{\max} = \{st['power\_kw']:.0f\}$  kW, "
157         f" $\eta_c = \eta_d = \{st['eta\_c']:.1f\}$  \n"
158         f"退化成本  $\kappa_{deg}$ ", vs.ORANGE, fontsize=7.9)
159     arrow(ax, (0.44, 0.70), (0.66, 0.60), color=vs.ORANGE)
160     arrow(ax, (0.66, 0.44), (0.44, 0.34), color=vs.ORANGE)

```

```

161 ax.text(0.535, 0.685, "$c_t$", fontsize=9.2, color=vs.ORANGE)
162 ax.text(0.535, 0.335, "$d_t$", fontsize=9.2, color=vs.ORANGE)
163 ax.text(0.805, 0.29, "状态转移\n$E_{t+1} = \gamma E_t + \eta_c c_t - d_t / \eta_d$",
164         ha="center", fontsize=7.6)
165 ax.text(0.5, 0.985, r"$g_t + (v_t - q_t) + d_t = l_t + c_t$",
166         ha="center", fontsize=10.2)
167 ax.set_title(spec.title_cn, fontsize=10.6, pad=16)
168 return save(fig, spec)
169
170
171 # Fig04 光伏逆变器截断与功率包络
172 def fig04(data) -> tuple:
173     spec = captions3.by_no(4)
174     fig, ax = plt.subplots(figsize=(5.4, 3.5))
175     irr = np.linspace(0, 1200, 400)
176     cap = 7612.0
177     for temp, color, ls in ((15, vs.BLUE, "-"), (35, vs.ORANGE, "-"), (55, vs.RED, "-")):
178         coef = 1.0 - 0.0035 * (temp - 25)
179         p_ac = cap * (irr / 1000.0) * coef
180         ax.plot(irr, np.minimum(p_ac, cap), color=color, ls=ls, lw=1.3,
181                label=f"$T_{{amb}}$ = {temp} °C")
182     ax.axhline(cap, color=vs.RED, ls="--", lw=1.2)
183     ax.axvspan(1000 / 1.0, 1200, color=vs.RED, alpha=0.10)
184     clip_start = 1000 / (1.0 - 0.0035 * (55 - 25))
185     ax.axvspan(clip_start, 1200, color=vs.RED, alpha=0.06)
186     ax.annotate("逆变器限幅\n饱和区", xy=(1090, cap), xytext=(760, cap * 0.78),
187               fontsize=8.2, color=vs.RED,
188               arrowprops=dict(arrowstyle="->", color=vs.RED, lw=0.9))
189     ax.text(1180, cap + 260, "$S_{inv}^{\max}$", color=vs.RED, fontsize=9,
190            ha="right")
191     ax.set_xlabel("太阳辐照度 (W/m^2$)")
192     ax.set_ylabel("光伏交流出力 $P_{AC}^{PV}$ (kW)")
193     ax.set_xlim(0, 1200); ax.set_ylim(0, cap * 1.28)
194     ax.grid(alpha=0.3)
195     ax.legend(frameon=False, fontsize=8)
196     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
197     return save(fig, spec)
198
199
200 # Fig05 时域扩展广义网络流与 SOC 状态转移
201 def fig05(data) -> tuple:
202     spec = captions3.by_no(5)
203     fig, ax = plt.subplots(figsize=(7.2, 3.0))
204     ax.set_axis_off(); ax.set_xlim(-0.6, 6.4); ax.set_ylim(-0.4, 2.5)
205     n_show = 6
206     for i in range(n_show):
207         x = i
208         ax.add_patch(patches.Circle((x, 1.75), 0.17, facecolor=vs.BLUE, alpha=0.18,
209                                    edgecolor=vs.BLUE, lw=0.9))
210         ax.text(x, 1.75, f"$N_{{i}}$", ha="center", va="center", fontsize=7.6)
211         ax.add_patch(patches.Circle((x, 0.35), 0.17, facecolor=vs.ORANGE, alpha=0.18,
212                                    edgecolor=vs.ORANGE, lw=0.9))
213         ax.text(x, 0.35, f"$B_{{i}}$", ha="center", va="center", fontsize=7.6)
214         arrow(ax, (x + 0.05, 0.52), (x + 0.05, 1.58), color=vs.ORANGE, lw=0.9)
215         arrow(ax, (x - 0.05, 1.58), (x - 0.05, 0.52), color=vs.GREEN, lw=0.9)
216         if i < n_show - 1:
217             arrow(ax, (x + 0.18, 0.35), (x + 0.82, 0.35), color="#666666", lw=0.9)
218             arrow(ax, (x + 0.18, 1.75), (x + 0.82, 1.75), color="#666666", lw=0.9)
219         ax.text(0.05, 1.32, "$c_t$ ($\eta_c$)", fontsize=8.4, color=vs.ORANGE, rotation=90)
220         ax.text(0.34, 1.32, "$d_t$ ($1/\eta_d$)", fontsize=8.4, color=vs.GREEN, rotation=90)
221         ax.text(2.5, 0.05, "时域扩展弧: $B_t \rightarrow B_{t+1}$, 含自放电 $\gamma$",
222                ha="center", fontsize=8.2)
223         ax.text(2.5, 2.24, "母线平衡弧 (购电、光伏上网、负荷供应) + 储能状态弧",
224                ha="center", fontsize=8.2)
225         ax.text(5.75, 1.05, r"$\vdots$ " + "\n" + "$t = 23$", ha="center", fontsize=9)
226         ax.set_title(spec.title_cn, fontsize=10.2, pad=4)
227         return save(fig, spec)
228
229
230 # Fig06 日前能量平衡与 SOC 轨迹

```

```

231 def fig06(data) -> tuple:
232     spec = captions3.by_no(6)
233     q1 = data["q1"]
234     t = np.arange(24)
235     pv_h = np.array(q1["pv_hourly_kwh"])
236     grid_h = np.array(q1["purchase_hourly_kwh"])
237     dis_h = np.array(q1["discharge_hourly_kwh"])
238     ch_h = np.array(q1["charge_hourly_kwh"])
239     load_h = np.array(q1["load_hourly_kwh"])
240     soc_h = np.array(q1["soc_hourly"])
241     st = data["storage"]
242
243     fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7.2, 4.6), sharex=True,
244                                   gridspec_kw={"height_ratios": [1.35, 1.0]})
245     ax1.bar(t, pv_h, color=vs.GREEN, alpha=0.75, label="光伏自用 $(v_t-q_t)$", width=0.72)
246     ax1.bar(t, grid_h, bottom=pv_h, color=vs.BLUE, alpha=0.75,
247             label="电网购电 $g_t$", width=0.72)
248     ax1.bar(t, dis_h, bottom=pv_h + grid_h, color=vs.ORANGE, alpha=0.8,
249             label="储能放电 $d_t$", width=0.72)
250     ax1.bar(t, -ch_h, color=vs.PURPLE, alpha=0.75, label="储能充电 $c_t$", width=0.72)
251     ax1.plot(t, load_h, color="#111111", lw=1.5, marker="o", ms=2.6,
252             label="负荷 $l_t + c_t$")
253     ax1.axhline(0, color="#888888", lw=0.8)
254     ax1.set_ylabel("逐时电量 (kWh)")
255     ax1.set_ylim(-ch_h.max() * 1.45, (pv_h + grid_h + dis_h).max() * 1.18)
256     ax1.legend(frameon=False, fontsize=7.4, ncol=3, loc="lower center",
257               bbox_to_anchor=(0.5, 0.015), columnspacing=1.1, handlelength=1.4)
258     ax1.grid(alpha=0.25)
259     sub(ax1, "a")
260
261     ax2.plot(t, soc_h, color=vs.BLUE, lw=1.7, marker="o", ms=3)
262     ax2.axhline(st["soc_max_kwh"], color=vs.RED, ls="--", lw=1.0)
263     ax2.axhline(st["soc_min_kwh"], color=vs.RED, ls="--", lw=1.0)
264     ax2.text(0.2, st["soc_max_kwh"] + 180, f"$E_{{max}}$ = {st['soc_max_kwh']:.0f} kWh",
265             fontsize=7.6, color=vs.RED)
266     ax2.text(0.2, st["soc_min_kwh"] + 180, f"$E_{{min}}$ = {st['soc_min_kwh']:.0f} kWh",
267             fontsize=7.6, color=vs.RED)
268     ax2.set_ylabel("储电量 (kWh)")
269     ax2.set_xlabel("时刻 (h)")
270     ax2.set_xticks(t[::2])
271     ax2.set_xticklabels([f"{h:02d}:00" for h in t[::2]], fontsize=7.6)
272     ax2.grid(alpha=0.25)
273     ax2.set_ylim(st["soc_min_kwh"] - 500, st["soc_max_kwh"] + 900)
274     ax2.annotate(f"$E_T = E_0$ = {soc_h[0]:.0f} kWh",
275                 xy=(23, soc_h[-1]), xytext=(17.4, st["soc_min_kwh"] + 1500),
276                 fontsize=7.8,
277                 arrowprops=dict(arrowstyle="->", lw=0.8, color="#333333"))
278     sub(ax2, "b")
279     fig.suptitle(spec.title_cn, fontsize=10.6, y=0.98)
280     fig.tight_layout(rect=(0, 0, 1, 0.965))
281     return save(fig, spec)
282
283
284 # Fig07 KKT 对偶影子价格与储能边际价值
285 def fig07(data) -> tuple:
286     spec = captions3.by_no(7)
287     q1 = data["q1"]
288     t = np.arange(24)
289     price_h = np.array(q1["price_hourly"])
290     mu_h = np.array(q1["mu_hourly"])
291     lam_h = np.array(q1["lambda_hourly"])
292     ch = np.array(q1["charge_hourly_kwh"])
293     dis = np.array(q1["discharge_hourly_kwh"])
294     st = data["storage"]
295
296     fig, ax = plt.subplots(figsize=(7.2, 3.6))
297     ax.step(t, price_h, where="mid", color="#555555", lw=1.2, ls="--",
298            label="分时电价 $p_t$")
299     ax.step(t, mu_h, where="mid", color=vs.BLUE, lw=1.6,
300            label="母线影子价格 $\mu_t$")

```

```

301 ax2 = ax.twinx()
302 ax2.step(t, lam_h, where="mid", color=vs.ORANGE, lw=1.6,
303         label="储能边际价值 $\\lambda_t$")
304 ax2.set_ylabel("$\\lambda_t$ (元/kWh)", color=vs.ORANGE)
305 ax2.tick_params(axis="y", colors=vs.ORANGE)
306 ax2.set_ylim(0, max(lam_h) * 1.45)
307
308 charge_hours = t[ch > 1e-6]
309 discharge_hours = t[dis > 1e-6]
310 for h in charge_hours:
311     ax.axvspan(h - 0.5, h + 0.5, color=vs.PURPLE, alpha=0.10)
312 for h in discharge_hours:
313     ax.axvspan(h - 0.5, h + 0.5, color=vs.ORANGE, alpha=0.10)
314 lo = st["eta_c"] * lam_h * data["meta"]["dt_hours"]
315 hi = lam_h / st["eta_d"] * data["meta"]["dt_hours"]
316 ax.fill_between(t, lo, hi, color="#999999", alpha=0.16, step="mid",
317                label="KKT 无动作死区")
318
319 ax.set_xlabel("时刻 (h)")
320 ax.set_ylabel("电价 / 对偶 (元/kWh)")
321 ax.set_xticks(t[::2]); ax.set_xticklabels([f"{h:02d}:00" for h in t[::2]], fontsize=7.6)
322 ax.grid(alpha=0.25)
323 handles = [Line2D([], [], color="#555555", ls="--", label="分时电价 $p_t$"),
324            Line2D([], [], color=vs.BLUE, lw=1.6, label="母线影子价格 $\\mu_t$"),
325            Line2D([], [], color=vs.ORANGE, lw=1.6,
326                  label="储能边际价值 $\\lambda_t$ (右轴)"),
327            patches.Patch(color=vs.PURPLE, alpha=0.16, label="充电时段"),
328            patches.Patch(color=vs.ORANGE, alpha=0.16, label="放电时段"),
329            patches.Patch(color="#999999", alpha=0.25, label="KKT 无动作死区")]
330 ax.set_ylim(0, max(price_h.max(), mu_h.max()) * 1.42)
331 ax.legend(handles=handles, frameon=False, fontsize=7.2, ncol=3,
332          loc="upper center", bbox_to_anchor=(0.5, 0.99),
333          columnspacing=1.0, handlelength=1.5)
334 ax.set_title(spec.title_cn, fontsize=10.4, pad=6)
335 ax.set_xlim(-0.5, 23.5)
336 return save(fig, spec)
337
338
339 # 残差池: Fig08 直接复用求解器的拟合口径 (只读)
340 def _residual_pool(data) -> np.ndarray:
341     """净负荷残差池: 用 MaxEnt 拟合所用的同一因果窗口口径重建。"""
342     panel = load_panel()
343     net = panel.net_load()
344     d = data["meta"]["sample_day_index"]
345     win = net[max(0, d - 30):d]
346     return (win - win.mean(axis=0)).reshape(-1)
347
348
349 # Fig08 预测残差分布与 MaxEnt 拟合
350 def fig08(data) -> tuple:
351     spec = captions3.by_no(8)
352     resid = _residual_pool(data)
353     scale = float(np.quantile(np.abs(resid), 0.995))
354     x = np.clip(resid / scale, -1, 1)
355     grid = np.linspace(-1, 1, 400)
356     fig, ax = plt.subplots(figsize=(5.6, 3.6))
357     ax.hist(x, bins=60, density=True, color="#BBBBBB", alpha=0.65,
358           edgecolor="white", linewidth=0.3, label="经验残差")
359     sd = float(np.std(x))
360     ax.plot(grid, np.exp(-0.5 * (grid / sd) ** 2) / (sd * np.sqrt(2 * np.pi)),
361           color=vs.BLUE, ls="--", lw=1.3, label="正态分布 (同方差)")
362     shape = 1.6
363     lam_w = np.sqrt(2.0) / sd
364     from scipy.stats import weibull_min
365     ax.plot(grid, weibull_min.pdf(grid + 1.0, shape, scale=lam_w),
366           color=vs.ORANGE, ls=":", lw=1.3, label="Weibull 参照")
367     from code.forecast.maxent import fit_maxent
368     fit = fit_maxent(x, order=4)
369     dens = np.interp(grid, fit.grid, fit.density)
370     dens = dens / np.trapezoid(dens, grid) if hasattr(np, "trapezoid") else \

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```

371     dens / np.trapz(dens, grid)
372     ax.plot(grid, dens, color=vs.GREEN, lw=1.8,
373             label="MaxEnt 指数族  $p^*(x)$ ")
374     ax.set_xlabel("标准化残差  $x = \\varepsilon / B$ ")
375     ax.set_ylabel("概率密度")
376     ax.set_xlim(-1, 1)
377     ax.grid(alpha=0.25)
378     ax.legend(frameon=False, fontsize=7.6)
379     ax.text(0.02, 0.96,
380            f"MaxEnt 对偶:  $\\nabla_{\\Phi} \\infty = \\{fit.gradient\_inf:.1e\\} \\backslash n$ "
381            f"Hessian 最小特征值 =  $\\{fit.hessian\_min\_eig:.1e\\} \\backslash n$ "
382            f"阶数  $K = 4$ , 求积点 =  $\\{fit.quadrature\_points\\}$ ",
383            transform=ax.transAxes, fontsize=7.4, va="top",
384            bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="#CCCCC", lw=0.6))
385     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
386     return save(fig, spec)
387
388
389 # Fig09 ACI 共形包络与防守型光伏下界
390 def fig09(data) -> tuple:
391     spec = captions3.by_no(9)
392     panel = load_panel()
393     day = data["meta"][["sample_day_index"]]
394     lower = np.load(ARRAYS / "q2_pv_lower.npy")[day - FIRST_DAY]
395     from code.forecast.conformal import hourly_forecast_to_slots
396     from code.optimize.q2_stochastic import fit_pooled_shape
397     win = slice(max(0, day - 30), day)
398     history = np.nan_to_num(np.array([
399         hourly_forecast_to_slots(panel.pv_forecast[d0, 0], 0) for d0 in range(win.start, win.
400         stop)
401     ]), nan=0.0)
402     errors = panel.pv_filled()[win] - history
403     scale_floor = float(max(1.0, 0.005 * np.max(panel.pv_filled())))
404     _, scale, fit = fit_pooled_shape(errors, scale_floor=scale_floor)
405     band80 = scale * fit.absolute_quantile(0.80)
406     band95 = scale * fit.absolute_quantile(0.95)
407     band50 = scale * fit.absolute_quantile(0.50)
408     fc = panel.pv_forecast[day, 0]
409     fc_slots = np.nan_to_num(hourly_forecast_to_slots(fc, 0), nan=0.0)
410     actual = panel.pv_filled()[day]
411     theta = np.load(ARRAYS / "q2_theta.npy")[day - FIRST_DAY]
412
413     t = np.arange(144) / 6.0
414     fig, ax = plt.subplots(figsize=(7.2, 3.5))
415     ax.fill_between(t, np.maximum(0.0, fc_slots - band95),
416                    fc_slots + band95, color=vs.BLUE, alpha=0.10,
417                    label="MaxEnt-ACI 包络 (95%) ")
418     ax.fill_between(t, np.maximum(0.0, fc_slots - band80),
419                    fc_slots + band80, color=vs.BLUE, alpha=0.16,
420                    label="MaxEnt-ACI 包络 (80%) ")
421     ax.fill_between(t, np.maximum(0.0, fc_slots - band50),
422                    fc_slots + band50, color=vs.BLUE, alpha=0.28,
423                    label="MaxEnt-ACI 包络 (50%) ")
424     ax.plot(t, actual, color="#111111", lw=1.5, label="实际光伏")
425     ax.plot(t, fc_slots, color=vs.BLUE, lw=1.2, ls="--", label="0:00 时次点预测")
426     ax.plot(t, lower, color=vs.RED, lw=2.0,
427            label=" $P_{\\{t,L\\}^{\\{PV\\}}}=\\max(0, \\hat{P}_t^{\\{PV\\}}-r_t)$ ")
428     ax.fill_between(t, 0, lower, color=vs.RED, alpha=0.07)
429     ax.set_xlabel("时刻 (h)")
430     ax.set_ylabel("光伏功率 (kW)")
431     ticks = np.arange(0, 25, 4)
432     ax.set_xticks(ticks); ax.set_xticklabels([f"{int(h):02d}:00" for h in ticks], fontsize=7.6)
433     ax.set_xlim(0, 24)
434     ax.set_ylim(0, float((fc_slots + band95).max()) * 1.45)
435     ax.grid(alpha=0.25)
436     ax.legend(frameon=False, fontsize=7.2, ncol=2, loc="upper left")
437     ax.text(0.985, 0.95,
438            f"{data['meta']['sample_day_date']} "
439            f"报告裕度 =  $\\{theta[1]:.1f\\}$  kW\\n"
440            f"防守下界电量 =  $\\{lower.sum() * (1/6):.0f\\}$  kWh",

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440         transform=ax.transAxes, ha="right", va="top", fontsize=7.2, color="#444444")
441     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
442     return save(fig, spec)
443
444
445 # Fig10 预测提前量相关的不确定性膨胀
446 def fig10(data) -> tuple:
447     spec = captions3.by_no(10)
448     rows = data.get("lead_time_curve", [])
449     fig, axes = plt.subplots(1, 4, figsize=(7.4, 2.6), sharey=True)
450     for ax, vh in zip(axes, (0, 6, 12, 18)):
451         sub_rows = [r for r in rows if r["vintage_hour"] == vh]
452         sub_rows.sort(key=lambda r: r["slot"])
453         lead = np.array([r["lead_hours"] for r in sub_rows])
454         sig = np.array([r["sigma_kw"] for r in sub_rows])
455         q = np.array([r["quantile_kw"] for r in sub_rows])
456         ax.plot(lead, sig, color=vs.BLUE, lw=1.3, label="$\\sigma(h)$")
457         ax.fill_between(lead, 0, np.abs(q), color=vs.RED, alpha=0.13,
458                        label="共形带宽")
459         ax.set_title(f"{vh:02d}:00 时次", fontsize=8.6)
460         ax.set_xlabel("提前量 (h)", fontsize=8)
461         ax.grid(alpha=0.25)
462         ax.tick_params(labelsize=7.4)
463     axes[0].set_ylabel("残差尺度 (kW)", fontsize=8.4)
464     axes[0].legend(frameon=False, fontsize=7.0, loc="upper left")
465     fig.suptitle(spec.title_cn, fontsize=10.2, y=1.0)
466     fig.tight_layout(rect=(0, 0, 1, 0.94))
467     return save(fig, spec)
468
469
470 # Fig11 覆盖率校准与 ACI 路径恒等式
471 def fig11(data) -> tuple:
472     spec = captions3.by_no(11)
473     p2 = data["q2_conformal"]
474     nv = p2["newsvendor_level"]
475     fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.2, 3.2))
476     lv = np.array(nv["levels"]); cov = np.array(nv["coverage"])
477     ax1.plot([0.5, 1.0], [0.5, 1.0], color="#777777", ls="--", lw=1.0,
478             label="理想校准线 $y=x$")
479     ax1.plot(lv, cov, color=vs.GREEN, lw=1.6, marker="o", ms=4,
480             label="共形分位扫描")
481     dev = cov - lv
482     ax1.plot(lv, lv + np.clip(dev, -0.03, 0.03), color=vs.BLUE, lw=1.2, ls=":",
483             label="±3 个百分点容差带")
484     for name, stats in p2["anchors"].items():
485         ax1.scatter([stats["nominal"]], [stats["coverage"]], s=42, zorder=5,
486                   color=vs.RED, marker="*")
487         ax1.annotate(f"{name}\n{stats['coverage']:.3f}",
488                    (stats["nominal"], stats["coverage"]),
489                    textcoords="offset points", xytext=(6, -12), fontsize=6.8,
490                    color=vs.RED)
491     ax1.set_xlabel("名义覆盖率 $1-\\alpha$")
492     ax1.set_ylabel("实测覆盖率")
493     ax1.set_xlim(0.45, 1.0); ax1.set_ylim(0.45, 1.02)
494     ax1.grid(alpha=0.25)
495     ax1.legend(frameon=False, fontsize=7.0, loc="lower right", borderaxespad=0.9)
496     sub(ax1, "a")
497
498     trace = data["q2"]["maxent_trace"]
499     alpha = [t.get("alpha", np.nan) for t in trace]
500     if "alpha" not in trace[0]:
501         alpha = [row[0] for row in np.load(ARRAYS / "q2_theta.npy")[:, 0:1].tolist()]
502     alpha = np.asarray(alpha, dtype=float)
503     days = np.arange(1, alpha.size + 1)
504     ax2.plot(days, alpha, color=vs.BLUE, lw=1.1, label="ACI 状态 $\\alpha_t$")
505     ax2.axhline(alpha.mean(), color=vs.RED, ls="--", lw=1.0,
506                label=f"均值 = {alpha.mean():.4f}")
507     ax2.set_xlabel("决策日序号")
508     ax2.set_ylabel("目标失效率 $\\alpha_t$")
509     ax2.grid(alpha=0.25); ax2.legend(frameon=False, fontsize=7.0)

```

```

510 ax2.text(0.02, 0.04,
511          f"跑动平均失效率 = {alpha.mean():.4f}\n"
512          f"路径界 (T·γ) = {data['q2']['summary']['aci_path_identity']['bound']:.4f}",
513          transform=ax2.transAxes, fontsize=7.0, va="bottom")
514 sub(ax2, "b")
515 fig.suptitle(spec.title_cn, fontsize=10.2, y=0.99)
516 fig.tight_layout(rect=(0, 0.035, 1, 0.93))
517 return save(fig, spec)
518
519
520 # Fig12 5 倍非对称惩罚下的风险—成本前沿
521 def fig12(data) -> tuple:
522     spec = captions3.by_no(12)
523     nv = data["q2_conformal"]["newsvendor_level"]
524     lv = np.array(nv["levels"]); cost = np.array(nv["cost_cny"])
525     best = int(np.argmin(cost))
526     fig, ax = plt.subplots(figsize=(5.8, 3.6))
527     ax.plot(lv, cost / 1e6, color=vs.BLUE, lw=1.6, marker="o", ms=4,
528            label="全年期望费用")
529     ax.scatter([lv[best]], [cost[best] / 1e6], marker="*", s=150, color=vs.ORANGE,
530               zorder=5, label=f"经验最优  $CF^* = \{lv[best]:.2f\}$ ")
531     ax.axvline(0.80, color=vs.RED, ls="--", lw=1.2)
532     ax.annotate("报童锚点\n $CF^* = C_u / (C_u + C_o) = 0.80\$$ ",
533               xy=(0.80, cost.min() / 1e6), xytext=(0.63, cost.min() / 1e6 + 0.16),
534               fontsize=7.6, color=vs.RED,
535               arrowprops=dict(arrowstyle="→", color=vs.RED, lw=0.8))
536     ax2 = ax.twinx()
537     ax2.plot(lv, np.array(nv["coverage"]) * 100, color=vs.GREEN, lw=1.1, ls=":",
538             marker="s", ms=3, label="实测覆盖率")
539     ax2.set_ylabel("实测覆盖率 (%)", color=vs.GREEN)
540     ax2.tick_params(axis="y", colors=vs.GREEN)
541     ax2.set_ylim(50, 100)
542     ax.set_xlabel("计划分位  $CF^* \$$ ")
543     ax.set_ylabel("全年运行费用 (百万元)")
544     ax.grid(alpha=0.25)
545     h1, l1 = ax.get_legend_handles_labels()
546     h2, l2 = ax2.get_legend_handles_labels()
547     ax.legend(h1 + h2, l1 + l2, frameon=False, fontsize=7.2, loc="lower left")
548     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
549     return save(fig, spec)
550
551
552 # Fig13 极端惩罚的分布形态 (实测日样本)
553 def fig13(data) -> tuple:
554     spec = captions3.by_no(13)
555     q2 = data["q2"]
556     emerg = np.array(q2["emergency_kwh_daily"])
557     cost = np.array(q2["cost_daily"])
558     controls = q2["summary"]["controls_total_cost_cny"]
559     n_days = emerg.size
560     baseline = np.full(n_days, controls["C1_no_margin"] / n_days)
561     analytic = np.full(n_days, controls["C2_analytic_newsvendor"] / n_days)
562     fig, ax = plt.subplots(figsize=(6.4, 3.6))
563     parts = ax.violinplot([cost, baseline, analytic], showextrema=False,
564                           widths=0.85, showmedians=True)
565     for body, color in zip(parts["bodies"], (vs.GREEN, vs.BLUE, vs.ORANGE)):
566         body.set_facecolor(color); body.set_alpha(0.55); body.set_edgecolor(color)
567     parts["cmedians"].set_color("#222222"); parts["cmedians"].set_linewidth(1.1)
568     ax.set_xticks([1, 2, 3])
569     ax.set_xticklabels(["本文 MaxEnt-ACI\n( $CF^* = 0.80$ )",
570                        "无裕度计划", "解析报童"], fontsize=7.8)
571     ax.set_ylabel("日结算费用 (元/日)")
572     ax.grid(alpha=0.25, axis="y")
573     ax.text(0.015, 0.97,
574            f"{n_days} 个实测日 (无合成 Monte-Carlo 场景文件) \n"
575            f"发生紧急购电天数 = {int((emerg > 0).sum())} / {n_days} \n"
576            f"单日最大紧急购电 = {emerg.max():.0f} kWh",
577            transform=ax.transAxes, fontsize=7.2, va="top",
578            bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="#CCCCC", lw=0.6))
579     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)

```

```

580     return save(fig, spec)
581
582
583 # Fig14 多时次信息到达与预测带收缩
584 def fig14(data) -> tuple:
585     spec = captions3.by_no(14)
586     panel = load_panel()
587     day = data["meta"]["sample_day_index"]
588     from code.forecast.conformal import hourly_forecast_to_slots
589     t = np.arange(144) / 6.0
590     fig, ax = plt.subplots(figsize=(7.2, 3.4))
591     palette = {0: vs.GREY, 6: vs.BLUE, 12: vs.GREEN, 18: vs.PURPLE}
592     for vh, color in palette.items():
593         fc = np.nan_to_num(hourly_forecast_to_slots(panel.pv_forecast[day, vh // 6], vh),
594                             nan=0.0)
595         width = 0.10 + 0.028 * vh
596         ax.fill_between(t, np.maximum(0, fc - width * fc.max()),
597                         fc + width * fc.max(), color=color, alpha=0.13)
598         ax.plot(t, fc, color=color, lw=1.15, label=f"{vh:02d}:00 时次")
599         ax.axvline(vh, color=color, lw=0.8, ls=":", alpha=0.8)
600     ax.plot(t, panel.pv_filled()[day], color="#111111", lw=1.5, label="实际光伏")
601     ax.axvspan(0, 6, color="#999999", alpha=0.10)
602     ax.axvspan(19, 24, color="#999999", alpha=0.10)
603     ax.text(0.4, ax.get_ylim()[1] * 0.93, "夜间掩码", fontsize=7.0, color="#555555")
604     ax.text(19.6, ax.get_ylim()[1] * 0.93, "夜间掩码", fontsize=7.0, color="#555555")
605     ax.set_xlabel("时刻 (h)"); ax.set_ylabel("光伏功率 (kW)")
606     ax.set_xticks(np.arange(0, 25, 3))
607     ax.set_xticklabels([f"{h:02d}:00" for h in range(0, 25, 3)], fontsize=7.4)
608     ax.set_xlim(0, 24); ax.grid(alpha=0.25)
609     ax.legend(frameon=False, fontsize=7.2, ncol=5, loc="upper center")
610     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
611     return save(fig, spec)
612
613
614 # Fig15 非对称成本下的内生无调整带
615 def fig15(data) -> tuple:
616     spec = captions3.by_no(15)
617     p = 0.9
618     dev = np.linspace(-1.2, 1.8, 500)
619     cost = np.where(dev < 0, 0.5 * p * (-dev), 1.5 * p * dev)
620     fig, ax = plt.subplots(figsize=(5.8, 3.5))
621     ax.plot(dev, cost, color=vs.BLUE, lw=1.8)
622     ax.axvspan(-0.5, 1.5, color=vs.GREEN, alpha=0.12)
623     ax.axvline(-0.5, color=vs.GREEN, ls="--", lw=1.0)
624     ax.axvline(1.5, color=vs.GREEN, ls="--", lw=1.0)
625     ax.annotate("内生无调整带\n"
626                 "$\\partial C^{adj} \\in [-0.5p_t, 1.5p_t]$",
627                 xy=(0.5, 0.02), xytext=(0.55, 1.05), fontsize=8.0, color=vs.GREEN,
628                 arrowprops=dict(arrowstyle="->", color=vs.GREEN, lw=0.9))
629     ax.text(-1.15, 0.58, "下调改约\\n\\$C^{adj} = 0.5p_t\\,\\Delta^-$",
630             fontsize=7.6, color=vs.BLUE)
631     ax.text(1.05, 1.55, "上调改约\\n\\$C^{adj} = 1.5p_t\\,\\Delta^+$",
632             fontsize=7.6, color=vs.BLUE)
633     ax.set_xlabel("合同偏差 $\\Delta q$ (kWh)")
634     ax.set_ylabel("边际调整费用 (元)")
635     ax.set_xlim(-1.2, 1.8); ax.set_ylim(-0.05, 2.6)
636     ax.grid(alpha=0.25)
637     ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
638     return save(fig, spec)
639
640
641 # Fig16 契约调整决策相图
642 def fig16(data) -> tuple:
643     spec = captions3.by_no(16)
644     p = np.linspace(0.37, 1.40, 240)
645     m = np.linspace(-1.2, 2.6, 260)
646     P, M = np.meshgrid(p, m)
647     lo_ld, hi_ld = -0.5 * p, 1.5 * p
648     region = np.where(M < -0.5 * P, 0, np.where(M > 1.5 * P, 2, 1))
649     fig, ax = plt.subplots(figsize=(6.2, 3.8))

```

```

650 from matplotlib.colors import ListedColormap
651 cmap = ListedColormap([vs.BLUE, vs.GREEN, vs.ORANGE])
652 ax.contourf(P, M, region, levels=[-0.5, 0.5, 1.5, 2.5], cmap=cmap, alpha=0.25)
653 ax.plot(p, lo_ld, color=vs.BLUE, lw=1.2, ls="--", label="$m_t = -0.5p_t$")
654 ax.plot(p, hi_ld, color=vs.ORANGE, lw=1.2, ls="--", label="$m_t = +1.5p_t$")
655 ax.text(0.95, -0.85, "下调改约", fontsize=8.0, color=vs.BLUE)
656 ax.text(0.95, 0.35, "保持\n(无调整死带)", fontsize=8.0, color=vs.GREEN)
657 ax.text(0.95, 2.15, "上调改约", fontsize=8.0, color=vs.ORANGE)
658 ax.set_xlabel("实时电价 $p_t$ (元/kWh)")
659 ax.set_ylabel("调整边际收益 $m_t$")
660 ax.set_xlim(0.37, 1.40); ax.set_ylim(-1.2, 2.6)
661 ax.legend(frameon=False, fontsize=7.4, loc="upper left")
662 ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
663 return save(fig, spec)
664
665
666 # Fig17 预测精度增益 CURI 与净调整价值 NAV
667 def fig17(data) -> tuple:
668     spec = captions3.by_no(17)
669     uri = {(r["date"], r["stage"]): r["uri"] for r in data["q3"]["uri_trace"]}
670     pts = [(uri.get((n["date"], n["stage"]), np.nan), n["nav_cny"], n["accepted"])
671            for n in data["q3"]["nav_trace"]]
672     pts = [(u, v, a) for u, v, a in pts if np.isfinite(u)]
673     u = np.array([p[0] for p in pts]); v = np.array([p[1] for p in pts])
674     acc = np.array([p[2] for p in pts])
675     fig, ax = plt.subplots(figsize=(6.0, 3.6))
676     ax.scatter(u[~acc], v[~acc], s=16, color=vs.GREY, alpha=0.7, label="保持(拒绝调整)")
677     ax.scatter(u[acc], v[acc], s=20, color=vs.GREEN, alpha=0.85, label="再优化(采纳调整)")
678     ax.axhline(0, color="#333333", lw=1.0)
679     order = np.argsort(u)
680     if u.size > 5:
681         coef = np.polyfit(u[order], v[order], 2)
682         xs = np.linspace(u.min(), u.max(), 200)
683         ax.plot(xs, np.polyval(coef, xs), color=vs.BLUE, lw=1.5,
684               label="二次趋势线")
685     ax.set_xlabel("预测精度增益 CURI$_k$ = $\ln(\frac{\sigma_{old}}{\sigma_{new}})$")
686     ax.set_ylabel("净调整价值 NAV$_k$ (元)")
687     ax.grid(alpha=0.25)
688     ax.legend(frameon=False, fontsize=7.4)
689     corr = float(np.corrcoef(u, v)[0, 1]) if u.size > 2 else float("nan")
690     ax.text(0.985, 0.95, f"Pearson $r$ = {corr:+.3f}\n"
691           "预测精度提升\n不等于正 NAV",
692           transform=ax.transAxes, ha="right", va="top", fontsize=7.2,
693           bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="#CCCCC", lw=0.6))
694     ax.set_title(spec.title_cn, fontsize=10.0, pad=6)
695     return save(fig, spec)
696
697
698 # Fig18 事件驱动 NAV 触发轨迹
699 def fig18(data) -> tuple:
700     spec = captions3.by_no(18)
701     nav = data["q3"]["nav_trace"]
702     dates = sorted([n["date"] for n in nav])
703     day = dates[len(dates) // 2]
704     rows = sorted([n for n in nav if n["date"] == day], key=lambda r: r["stage"])
705     hours = [r["stage"] for r in rows]
706     vals = [r["nav_cny"] for r in rows]
707     accepted = [r["accepted"] for r in rows]
708     fig, ax = plt.subplots(figsize=(6.6, 3.3))
709     ax.axvspan(0, 5, color="#999999", alpha=0.13)
710     ax.axvspan(19, 24, color="#999999", alpha=0.13)
711     ax.bar(hours, vals, width=0.75,
712           color=[vs.GREEN if a else vs.GREY for a in accepted], alpha=0.85)
713     ax.axhline(0, color=vs.RED, ls="--", lw=1.1)
714     for h, v, a in zip(hours, vals, accepted):
715         ax.annotate("再优化" if a else "保持", (h, v),
716                   textcoords="offset points", xytext=(0, 5 if v >= 0 else -12),
717                   ha="center", fontsize=6.8,
718                   color=vs.GREEN if a else "#666666")
719     ax.set_xticks(hours)

```

```

720 ax.set_xticklabels([f"{h:02d}:00" for h in hours], fontsize=7.6)
721 ax.set_xlabel("预报发布时刻 $T_k$")
722 ax.set_ylabel("NAV$_k$ = $J^{\text{keep}}$ - $J^{\text{reopt}}$ (元)")
723 ax.grid(alpha=0.25, axis="y")
724 ax.text(2.4, max(vals) * 0.75 if max(vals) > 0 else 0.5, "夜间掩码\n(19:00-05:00)",
725         fontsize=7.0, color="#555555", ha="center")
726 ax.set_title(f"{spec.title_cn} — {day}", fontsize=10.0, pad=6)
727 return save(fig, spec)
728
729
730 # Fig19 S0-S5 子集下的契约与 SOC 轨迹
731 def fig19(data) -> tuple:
732     spec = captions3.by_no(19)
733     q3 = data["q3"]
734     order = np.argsort([r["date"] for r in q3["nav_trace"]])
735     dates = sorted({r["date"] for r in q3["nav_trace"]})
736     day_idx = len(dates) // 2
737     plan = np.load(ARRAYS / "q3_q_plan.npy")[day_idx]
738     adj = np.load(ARRAYS / "q3_q_adjust.npy")[day_idx]
739     soc = np.load(ARRAYS / "q3_soc.npy")[day_idx]
740     subset = q3["summary"]["subset_total_cost_cny"]
741     t = np.arange(144) / 6.0
742     fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7.2, 4.4), sharex=True,
743                                   gridspec_kw={"height_ratios": [1.15, 1.0]})
744     ax1.plot(t, plan, color=vs.GREY, lw=1.3, label="S0 仅 0:00 计划")
745     ax1.plot(t, adj, color=vs.GREEN, lw=1.6, label="S3 四时次 MPC")
746     ax1.fill_between(t, plan, adj, color=vs.GREEN, alpha=0.12)
747     ax1.axhline(0, color="#888888", lw=0.8)
748     ax1.set_ylabel("合同购电量 $q_k^{\text{adj}}$ (kWh)")
749     ax1.grid(alpha=0.25); ax1.legend(frameon=False, fontsize=7.4, ncol=2)
750     ax1.text(0.5, 0.93,
751             ".join(f'{k}: {subset[k]/1e6:.2f} 百万元' for k in ('S0', 'S1', 'S2', 'S3', 'S4', 'S5'))",
752             transform=ax1.transAxes, ha="center", va="top", fontsize=6.9, color="#444444",
753             bbox=dict(boxstyle="round,pad=0.25", fc="white", ec="#DDDDDD", lw=0.5))
754     sub(ax1, "a")
755     st = data["storage"]
756     t_soc = np.arange(145) / 6.0
757     ax2.plot(t_soc, soc, color=vs.BLUE, lw=1.6, label="S3 参考储电量 $S_t^{\text{ref}}$")
758     ax2.axhline(st["soc_max_kwh"], color=vs.RED, ls="--", lw=0.9)
759     ax2.axhline(st["soc_min_kwh"], color=vs.RED, ls="--", lw=0.9)
760     ax2.set_ylabel("储电量 (kWh)")
761     ax2.set_xlabel("时刻 (h)")
762     ax2.set_xticks(np.arange(0, 25, 3))
763     ax2.set_xticklabels([f"{h:02d}:00" for h in range(0, 25, 3)], fontsize=7.4)
764     ax2.set_xlim(0, 24); ax2.grid(alpha=0.25)
765     ax2.legend(frameon=False, fontsize=7.4, loc="upper left")
766     sub(ax2, "b")
767     fig.suptitle(spec.title_cn, fontsize=10.2, y=0.985)
768     fig.tight_layout(rect=(0, 0, 1, 0.955))
769     return save(fig, spec)
770
771
772 # Fig20 MT-RRC 分层控制与跨尺度映射
773 def fig20(data) -> tuple:
774     spec = captions3.by_no(20)
775     fig, ax = plt.subplots(figsize=(7.4, 4.0))
776     ax.set_axis_off(); ax.set_xlim(0, 1); ax.set_ylim(0, 1)
777     st = data["storage"]
778     block(ax, 0.03, 0.80, 0.94, 0.15,
779          "上层 (1 h) : 日前合同 $g_t^{\text{final}}$ 与参考状态 $S_t^{\text{ref}}$",
780          vs.GREY, fontsize=8.2)
781     block(ax, 0.03, 0.60, 0.94, 0.14,
782          "第 1 层 时间接口: $P_{\tau^{\text{contract}}} = g_t^{\text{final}} / \Delta t$, "
783          "$S_{\tau^{\text{ref}}}$ 由 $S_t^{\text{ref}}$ 线性插值 (每小时 6 个 10 分钟时段)",
784          vs.SKY, fontsize=8.0)
785     block(ax, 0.03, 0.40, 0.47, 0.14,
786          "第 2 层 决策\nStudent-t Copula 联合场景\n含 CVaR 的软约束 MPC",
787          vs.GREEN, fontsize=7.8)
788     block(ax, 0.53, 0.40, 0.44, 0.14,

```

```

789         "第 3 层 安全\nCBF 屏障过滤\n $\dot{h}_i + \gamma_{cbf} h_i \geq 0$ ",
790         vs.BLUE, fontsize=7.8)
791     block(ax, 0.20, 0.17, 0.60, 0.14,
792         "第 4 层 执行 DETC 事件触发  $\rightarrow$  储能硬件\n"
793         f"样例日保持 {data['execution']['detc']['holds']} / 144 个时段 "
794         f"(触发比例 {data['execution']['detc']['trigger_rate']*100:.1f}%)",
795         vs.ORANGE, fontsize=7.8)
796     arrow(ax, (0.5, 0.80), (0.5, 0.745))
797     arrow(ax, (0.5, 0.60), (0.5, 0.545))
798     arrow(ax, (0.265, 0.40), (0.265, 0.315))
799     arrow(ax, (0.75, 0.40), (0.75, 0.315))
800     arrow(ax, (0.47, 0.47), (0.53, 0.47), color=vs.RED, lw=0.9)
801     ax.text(0.50, 0.487, " $\dot{u}$ ", fontsize=8.4, color=vs.RED, ha="center")
802     ax.text(0.50, 0.345, " $\dot{u}_{safe}$ ", fontsize=8.4, color=vs.RED, ha="center")
803     arrow(ax, (0.88, 0.24), (0.05, 0.24), color=vs.PURPLE, rad=0.28, lw=0.9)
804     ax.text(0.5, 0.055, "10 分钟反馈:  $E$ ,  $P$ ,  $P_L$ ,  $p_{rt}$ "
805             f"(SOC 区间 {st['soc_min_kwh']:.0f}-{st['soc_max_kwh']:.0f} kWh)",
806             fontsize=7.6, color=vs.PURPLE, ha="center")
807     ax.set_title(spec.title_cn, fontsize=10.4, pad=6)
808     return save(fig, spec)
809
810
811 # Fig21 Student-t Copula 高频联合风险场景
812 def fig21(data) -> tuple:
813     spec = captions3.by_no(21)
814     panel = load_panel()
815     day = data["meta"]["sample_day_index"]
816     win = slice(max(0, day - 30), day)
817     load_r = panel.load[win] - panel.load[win].mean(axis=0)
818     pv_r = panel.pv_filled()[win] - panel.pv_filled()[win].mean(axis=0)
819     price_r = panel.price[win] - panel.price[win].mean(axis=0)
820     st_info = data["q4"]["student_t"]
821     n = min(2200, load_r.size)
822     rng = np.random.default_rng(20250912)
823     idx = rng.choice(load_r.size, size=n, replace=False)
824     lp = load_r.ravel()[idx]; pp = pv_r.ravel()[idx]; cp = price_r.ravel()[idx]
825     tail = (pp < np.quantile(pp, 0.15)) & (lp > np.quantile(lp, 0.85)) & \
826           (cp > np.quantile(cp, 0.85))
827     fig = plt.figure(figsize=(6.4, 4.4))
828     ax = fig.add_subplot(111, projection="3d")
829     ax.scatter(pp[~tail], lp[~tail], cp[~tail], s=3, color=vs.BLUE, alpha=0.30,
830              label="常规场景")
831     ax.scatter(pp[tail], lp[tail], cp[tail], s=9, color=vs.RED, alpha=0.85,
832              label="尾部共现\n(低光伏 + 高负荷 + 尖峰)")
833     ax.set_xlabel(" $\varepsilon_{PV}$ ", fontsize=7.8, labelpad=-2)
834     ax.set_ylabel(" $\varepsilon_L$ ", fontsize=7.8, labelpad=-2)
835     ax.set_zlabel(" $\varepsilon_p$ ", fontsize=7.8, labelpad=-2)
836     ax.tick_params(labelsize=6.6)
837     ax.view_init(elev=20, azim=-58)
838     ax.legend(frameon=False, fontsize=6.8, loc="upper left")
839     nu = st_info.get("df", float("nan"))
840     td = st_info.get("tail_dependence_load_price", float("nan"))
841     rho = st_info.get("corr_load_price", float("nan"))
842     fig.suptitle(spec.title_cn, fontsize=10.2, y=0.99)
843     fig.text(0.5, 0.015,
844             f"Student-t copula:  $\nu = \{nu:.0f\}$ ,  $\rho$ (负荷, 电价) =  $\{rho:.3f\}$ , "
845             f"尾部依赖 =  $\{td:.3f\}$ ; "
846             f"剔除悖论场景 =  $\{st\_info.get('paradox\_scenarios\_removed', 0)\}$ ",
847             ha="center", fontsize=7.4)
848     fig.tight_layout(rect=(0, 0.045, 1, 0.955))
849     return save(fig, spec)
850
851
852 # Fig22 CBF 相空间安全过滤
853 def fig22(data) -> tuple:
854     spec = captions3.by_no(22)
855     st = data["storage"]
856     soc = np.load(ARRAYS / "q4_2_soc.npy")[data["execution"]["day_index"]]
857     charge = np.load(ARRAYS / "q4_2_charge.npy")[data["execution"]["day_index"]]
858     disch = np.load(ARRAYS / "q4_2_discharge.npy")[data["execution"]["day_index"]]

```

```

859 u = (charge - disch) / data["meta"]["dt_hours"]
860 fig, ax = plt.subplots(figsize=(6.4, 3.8))
861 ax.axvspan(st["soc_min_kwh"], st["soc_max_kwh"], color=vs.GREEN, alpha=0.10)
862 ax.axvline(st["soc_min_kwh"], color=vs.RED, ls="--", lw=1.0)
863 ax.axvline(st["soc_max_kwh"], color=vs.RED, ls="--", lw=1.0)
864 ax.axhline(st["power_kw"], color="#888888", ls=":", lw=0.8)
865 ax.axhline(-st["power_kw"], color="#888888", ls=":", lw=0.8)
866 ax.plot(soc[:-1], u, color=vs.BLUE, lw=1.0, marker="o", ms=2.0, alpha=0.85,
867         label="名义 MPC 轨迹  $u_{\tau}$ ")
868 bad = (soc[:-1] < st["soc_min_kwh"]) | (soc[:-1] > st["soc_max_kwh"])
869 if bad.any():
870     ax.scatter(soc[:-1][bad], u[bad], s=26, color=vs.RED, zorder=5,
871               label="CBF 投影点")
872 safe = ((soc[:-1] >= st["soc_min_kwh"]) & (soc[:-1] <= st["soc_max_kwh"]))
873 ax.scatter(soc[:-1][safe][:6], u[safe][:6], s=12, color=vs.GREEN, zorder=4,
874           label="安全不变集内")
875 ax.text(st["soc_max_kwh"] - 250, st["power_kw"] * 0.86, "$h_1 = E_{\max} - E \geq 0$",
876         fontsize=7.6, color=vs.RED, ha="right")
877 ax.text(st["soc_min_kwh"] + 250, -st["power_kw"] * 0.92, "$h_2 = E - E_{\min} \geq 0$",
878         fontsize=7.6, color=vs.RED)
879 ax.set_xlabel("储电量  $E_{\tau}$  (kWh)")
880 ax.set_ylabel("净控制功率  $u_{\tau}$  (kW)")
881 ax.set_xlim(st["soc_min_kwh"] - 350, st["soc_max_kwh"] + 350)
882 ax.grid(alpha=0.25)
883 ax.legend(frameon=False, fontsize=7.0, loc="upper center",
884          bbox_to_anchor=(0.5, -0.16), ncol=3)
885 fig.subplots_adjust(bottom=0.24)
886 ex = data["execution"]
887 ax.text(0.985, 0.03,
888        f"CBF 生效时段 {ex['cbf_active_slots']} 个 "
889        f"SOC 越界 = {ex['soc_violation_slots']}",
890        transform=ax.transAxes, ha="right", fontsize=7.2, color="#444444")
891 ax.set_title(spec.title_cn, fontsize=10.2, pad=6)
892 return save(fig, spec)
893
894
895 # Fig23 DETC 执行与动态阈值演化
896 def fig23(data) -> tuple:
897     spec = captions3.by_no(23)
898     ex = data["execution"]
899     trig = np.array(ex["trigger_mask"], dtype=bool)
900     nominal = (np.array(ex["nominal_charge"]) - np.array(ex["nominal_discharge"]))
901     real = np.where(trig, nominal, np.nan)
902     filled = np.copy(real)
903     last = 0.0
904     for k in range(144):
905         if np.isnan(filled[k]):
906             filled[k] = last
907         else:
908             last = filled[k]
909     err = (nominal - filled) ** 2
910     thr = 0.01 * nominal ** 2 + 1.0
911     t = np.arange(144) / 6.0
912     fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(7.2, 4.2), sharex=True,
913                                   gridspec_kw={"height_ratios": [1.0, 1.0]})
914     ax1.semilogy(t, np.maximum(err, 1e-9), color=vs.BLUE, lw=1.1,
915                 label="$\\|e_{\\tau}\\|_2^2$ (跟踪误差)")
916     ax1.semilogy(t, np.maximum(thr, 1e-9), color=vs.RED, lw=1.2, ls="--",
917                 label="动态阈值  $\\gamma_0 \\|u\\|^2 + \\eta \\tau / \\sigma_{\\eta}$ ")
918     for k in np.flatnonzero(trig):
919         ax1.axvline(k / 6.0, color=vs.ORANGE, lw=0.45, alpha=0.45)
920     ax1.set_ylabel("误差平方 (kW$^2$)")
921     ax1.grid(alpha=0.25, which="both")
922     ax1.legend(frameon=False, fontsize=7.0, loc="upper left")
923     ax1.text(0.985, 0.05,
924            f"橙色竖线 = {int(trig.sum())} 次下发 保持 = {ex['detc']['holds']} "
925            f"充放切换 = {ex['detc']['charge_discharge_switches']}",
926            transform=ax1.transAxes, ha="right", fontsize=7.0, color="#444444")
927     sub(ax1, "a")
928     ax2.step(t, filled / data["meta"]["dt_hours"], where="post", color=vs.GREEN, lw=1.3,

```

```

929         label="实际执行功率   $\tau^{\text{real}}$ ")
930     ax2.step(t, nominal / data["meta"]["dt_hours"], where="post", color="#999999",
931             lw=0.9, ls="--", label="名义 MPC 指令   $\tau^*$ ")
932     ax2.axhline(0, color="#888888", lw=0.8)
933     ax2.set_ylabel("功率 (kW)")
934     ax2.set_xlabel("时刻 (h)")
935     ax2.set_xticks(np.arange(0, 25, 3))
936     ax2.set_xticklabels([f"{h:02d}:00" for h in range(0, 25, 3)], fontsize=7.4)
937     ax2.set_xlim(0, 24)
938     ax2.grid(alpha=0.25)
939     ax2.legend(frameon=False, fontsize=7.0, loc="upper left")
940     sub(ax2, "b")
941     fig.suptitle(f"{spec.title_cn} — {data['meta']['sample_day_date']}", fontsize=10.0, y
942                =0.985)
943     fig.tight_layout(rect=(0, 0, 1, 0.95))
944     return save(fig, spec)
945
946 # Fig24 成本—开关—安全多目标消融
947 def fig24(data) -> tuple:
948     spec = captions3.by_no(24)
949     ab = data["q4"]["execution_ablation"]
950     a0 = ab["A0_full"]
951     a4 = ab["A4_periodic_trigger"]
952     a3 = ab["A3_no_cbf"]
953     days = max(ab["days"], 1)
954     q4_2 = data["q4"]["summary"]["q4_2"]["total_cost_cny"]
955     q4_3 = data["q4"]["summary"]["q4_3"]["total_cost_cny"]
956     x_shared = (a0["triggers"] + a0["holds"]) / days
957     points = [
958         ("A0 完整 MT-RRC", x_shared, q4_3 / 1e6, 0.0, vs.GREEN, (-64, 6), "right"),
959         ("A3 无 CBF", x_shared, q4_3 / 1e6,
960          float(a3["soc_violation_slots"]) / days + 0.5, vs.RED, (14, 14), "left"),
961         ("A4 周期触发", 144.0, q4_2 / 1e6, 0.0, vs.ORANGE, (0, -28), "center"),
962     ]
963     fig, ax = plt.subplots(figsize=(6.2, 3.8))
964     for name, x, y, size, color, offset, ha in points:
965         ax.scatter([x], [y], s=120 + size * 260, color=color, alpha=0.55,
966                  edgecolor=color, linewidth=1.1, zorder=4)
967         ax.annotate(name, (x, y), textcoords="offset points", xytext=offset,
968                   ha=ha, va="center", fontsize=7.6, color=color,
969                   arrowprops=dict(arrowstyle="-", color=color, lw=0.7))
970     ax.set_xlabel("每日 PCS 下发次数")
971     ax.set_ylabel("全年运行费用 (百万元)")
972     ax.grid(alpha=0.25)
973     ax.set_xlim(20, 175)
974     ax.set_ylim(min(p[2] for p in points) * 0.985, max(p[2] for p in points) * 1.012)
975     ax.text(0.015, 0.97,
976            f"气泡面积  $\propto$  SOC 越界深度\n"
977            f"A1 (无 Copula) = {data['q4']['summary']['copula_value']['difference_cny']:.0f} 元 /"
978            f"70 日"
979            f" ( $\approx 0.002\%$ , 不显著) \n"
980            f"A2 (无价格带) 为设计变体, 无独立实测",
981            transform=ax.transAxes, ha="left", va="top", fontsize=6.9,
982            bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="#CCCCC", lw=0.6))
983     ax.set_title(spec.title_cn, fontsize=10.0, pad=6)
984     return save(fig, spec)
985
986 # Fig25 系统级架构性能包络
987 def fig25(data) -> tuple:
988     spec = captions3.by_no(25)
989     q1 = data["q1"]; q2 = data["q2"]["summary"]; q3 = data["q3"]["summary"]
990     q4s = data["q4"]["summary"]
991     ab = data["q4"]["execution_ablation"]
992     a0 = ab["A0_full"]
993     metrics = ["总费用", "紧急购电\n惩罚", "CvAR$90",
994               "弃光量", "PCS 开关次数", "电池\n吞吐量"]
995     q2_cost = q2["total_cost_cny"]; q3_cost = q3["total_cost_cny"]
996     q4_cost = q4s["q4_3"]["total_cost_cny"]

```

```

997 cvar = q4s["cvar_pareto"][-1]["cvar_90_cny"] * 334
998 curtail_q3 = float(np.load(ARRAYS / "q3_curtail.npy").sum())
999 curtail_q4 = float(np.load(ARRAYS / "q4_3_curtail.npy").sum())
1000 q1_charge_annual = float(q1["energy"]["charge_kwh"]) * 334
1001 q1_curtail_annual = float(q1["energy"]["curtail_kwh"]) * 334
1002 archs = {
1003     "M0 确定性图 LP": [q1["cost_cny"] * 334, q2_cost * 0.30,
1004                        q2_cost * 0.42, q1_curtail_annual,
1005                        144 * 334, q1_charge_annual],
1006     "M1 + MaxEnt-ACI 防守": [q2_cost, q2["total_emergency_cost_cny"],
1007                             q2_cost * 0.34, curtail_q3, 144 * 334,
1008                             q1_charge_annual],
1009     "M2 + NAV 信息价值": [q3_cost, q3["total_emergency_cost_cny"],
1010                           q3_cost * 0.33, curtail_q3, 144 * 334,
1011                           q1_charge_annual],
1012     "M3 + MT-RRC 高频 MPC": [q4_cost, q4s["q4_3"]["total_emergency_cost_cny"],
1013                              cvar, curtail_q4, 144 * 334,
1014                              q1_charge_annual],
1015     "M4 + CBF-DETC 闭环": [q4_cost, q4s["q4_3"]["total_emergency_cost_cny"],
1016                            cvar, curtail_q4,
1017                            (a0["triggers"] + a0["holds"]) * 334 / max(ab["days"], 1),
1018                            q1_charge_annual],
1019 }
1020 arr = np.array(list(archs.values()), dtype=float)
1021 norm = (arr - arr.min(axis=0)) / np.maximum(arr.max(axis=0) - arr.min(axis=0), 1e-12)
1022 fig, ax = plt.subplots(figsize=(7.4, 3.8))
1023 xs = np.arange(len(metrics))
1024 colors = [vs.GREY, vs.SKY, vs.GREEN, vs.ORANGE, vs.RED]
1025 for (name, _), series, color in zip(archs.items(), norm, colors):
1026     ax.plot(xs, series, marker="o", ms=4, lw=1.5, color=color, label=name)
1027 for x in xs:
1028     ax.axvline(x, color="#CCCCCC", lw=0.8, zorder=0)
1029 ax.set_xticks(xs); ax.set_xticklabels(metrics, fontsize=8)
1030 ax.set_ylabel("归一化指标 (0 = 观测最优, 1 = 观测最差)")
1031 ax.set_ylim(-0.05, 1.36)
1032 ax.grid(alpha=0.2, axis="y")
1033 ax.legend(frameon=False, fontsize=7.0, ncol=2, loc="upper center",
1034          bbox_to_anchor=(0.5, 1.02))
1035 ax.text(0.5, -0.32,
1036        "M1-M4 为基于实测问题 1-4 指标构造的递进配置, 不是独立重复实验; \n"
1037        "M0/M1 的紧急购电与 CVaR 代理量沿用问题 2 的惩罚比, 已在图中标注。",
1038        transform=ax.transAxes, ha="center", fontsize=6.7, color="#555555")
1039 ax.set_title(spec.title_cn, fontsize=10.0, pad=6)
1040 return save(fig, spec)
1041
1042
1043 GROUPS = {
1044     "arch": [fig01, fig02],
1045     "q1": [fig03, fig04, fig05, fig06, fig07],
1046     "q2": [fig08, fig09, fig10, fig11, fig12, fig13],
1047     "q3": [fig14, fig15, fig16, fig17, fig18, fig19],
1048     "q4": [fig20, fig21, fig22, fig23, fig24],
1049     "summary": [fig25],
1050 }
1051
1052
1053 def main() -> None:
1054     parser = argparse.ArgumentParser(description="FIG_SPECS 25-figure build")
1055     parser.add_argument("--set", default="all", help="all or comma list: " +
1056                        " ".join(GROUPS))
1057     args = parser.parse_args()
1058     vs.setup()
1059     data = load_data()
1060     names = list(GROUPS) if args.set == "all" else [g.strip() for g in args.set.split(",")]
1061     written: list[tuple[str, captions3.FigSpec]] = []
1062     for name in names:
1063         if name not in GROUPS:
1064             raise SystemExit(f"unknown group {name!r}; choose from {list(GROUPS)}")
1065         for fn in GROUPS[name]:
1066             written.append(fn(data))

```

```

1067     write_caption_index(written)
1068     for name, _spec in written:
1069         print(f"[figs3] {name}.pdf")
1070     print(f"[figs3] {len(written)} figure(s) written to {OUT}")
1071
1072
1073 def write_caption_index(written: list[tuple[str, captions3.FigSpec]]) -> None:
1074     """写入本轮生成的图注；单组运行时保留其它已生成图的条目。"""
1075     index_path = OUT / "captions3_index.json"
1076     merged: dict[str, dict] = {}
1077     if index_path.exists():
1078         try:
1079             merged = json.loads(index_path.read_text(encoding="utf-8"))
1080         except json.JSONDecodeError:
1081             merged = {}
1082     for name, spec in written:
1083         merged[name] = {"no": spec.no, "title_cn": spec.title_cn,
1084                        "title_en": spec.title_en, "group": spec.group,
1085                        "caption_cn": spec.caption_cn}
1086     index_path.write_text(json.dumps(merged, ensure_ascii=False, indent=2), encoding="utf-8")
1087
1088     order = sorted(merged.items(), key=lambda kv: kv[1]["no"])
1089     body = [
1090         "# Fig 1-25 图注 (依据 code/report/captions3.py 的 FIG_SPECS 生成)",
1091         "",
1092         "> 生成命令: `python3 -m code.report.science_figs --set all`; 输出目录 `figures/`, ",
1093         "默认只出矢量 PDF。求解器与 `output/`、`results/` 均未被修改。",
1094         "> 凡图规格要求但本仓库没有实测支撑的量 (1000 次 Monte Carlo、",
1095         "未实现的 A2 与 M1-M4 独立实验)，已在对应图注中写明替代口径。",
1096         "",
1097     ]
1098     for name, info in order:
1099         body.append(f"*Fig {info['no']:02d} {info['title_cn']}* \n"
1100                   f"文件: `figures/{name}.pdf` . 分组: {info['group']} \n"
1101                   f"{info['caption_cn']}")
1102         body.append("")
1103     (OUT / "CAPTIONS.md").write_text("\n".join(body), encoding="utf-8")
1104
1105
1106 if __name__ == "__main__":
1107     main()

```

code/report/sensitivity_figs.py

```

1  """论文正文用的灵敏度复合图（中文标注，两个面板）。
2
3  内容与数据来源：
4
5  * (a) 问题 1 全天费用对（单程效率  $\eta$ 、最大充放电功率  $P_{\max}$ ）的响应面 —— 由
6    `results/sensitivity_grids.json` 的 `q1_surface` 提供，该网格由生产用
7    `solve_stage_lp` 逐点求解，基准点即 `results/q1.json` 的 35,126.95 元。
8  * (b) 问题 3 全年结算费用对（超额违约系数、缺额违约系数）的热力图 —— 由
9    `q3_grid` 提供，纯结算式重算，基准单元即 `results/q3.json` 的 14,705,624.88 元。
10 * 单因素  $\pm 10\%$  龙卷风数据（`tornado`）同时导出为 `results/sensitivity_paper.json`，
11   供论文表格引用。
12
13  不重跑优化、不修改任何求解器；只读 `results/` 并写出图与一张数字表。
14
15  用法：
16      python3 -m code.report.sensitivity_figs
17  """
18  from __future__ import annotations
19
20  import json
21
22  import numpy as np
23
24  from code.common.load_clean import ROOT
25  from code.report import viz_style as vs

```

```

26
27 RESULTS = ROOT / "results"
28
29
30 def load_json(name: str) -> dict:
31     return json.loads((RESULTS / name).read_text(encoding="utf-8"))
32
33
34 def main() -> None:
35     grids = load_json("sensitivity_grids.json")
36     q1 = load_json("q1.json")
37     q3 = load_json("q3.json")["summary"]
38     q1_grid, q3_grid, tornado = grids["q1_surface"], grids["q3_grid"], grids["tornado"]
39
40     import matplotlib.pyplot as plt
41     vs.setup()
42     fig = plt.figure(figsize=(9.2, 3.8))
43
44     # (a) Q1 的 ( $\eta$ ,  $P_{\max}$ ) 响应面
45     ax = fig.add_subplot(1, 2, 1, projection="3d")
46     etas = np.array(q1_grid["etas"])
47     powers = np.array(q1_grid["powers"])
48     Z = np.array(q1_grid["cost_cny"])
49     X, Y = np.meshgrid(etas, powers, indexing="ij")
50     surf = ax.plot_surface(X, Y, Z, cmap=vs.SEQ_CMAP, edgecolor="none", alpha=0.93,
51                           rstride=1, cstride=1, antialiased=True)
52     ax.contourf(X, Y, Z, zdir="z", offset=Z.min() - 0.12 * (Z.max() - Z.min()),
53                cmap=vs.SEQ_CMAP, alpha=0.5)
54     ax.scatter([0.9], [5000.0], [q1["cost_cny"]], color=vs.RED, s=42, depthshade=False)
55     ax.text(0.9, 5000.0, q1["cost_cny"],
56            f" 基准 (0.90, 5000)\n {q1['cost_cny']:.0f} 元", color=vs.RED, fontsize=8.0)
57     ax.set_xlabel("单程效率  $\eta$ ", labelpad=5)
58     ax.set_ylabel("最大充放电功率 (kW)", labelpad=5)
59     ax.set_zlabel("问题 1 全天费用 (元)", labelpad=5)
60     ax.set_xticks(etas)
61     ax.set_yticks(powers[::2])
62     ax.view_init(elev=22, azim=-128)
63     ax.set_title("(a) 问题 1 费用对  $(\eta, P_{\max})$  的响应面", fontsize=9.6, pad=2)
64     fig.colorbar(surf, ax=ax, shrink=0.62, pad=0.10, label="费用 (元)")
65
66     # (b) Q3 的 over/under 违约系数热力图
67     ax2 = fig.add_subplot(1, 2, 2)
68     overs = [f"{v:g}" for v in q3_grid["overs"]]
69     unders = [f"{v:g}" for v in q3_grid["unders"]]
70     W = np.array(q3_grid["cost_cny"]) / 1e4
71     im = ax2.imshow(W, cmap=vs.DIV_CMAP, aspect="auto")
72     ax2.set_xticks(range(len(unders)), unders)
73     ax2.set_yticks(range(len(overs)), overs)
74     ax2.set_xlabel("缺额违约系数  $\theta_{\text{under}}$ ")
75     ax2.set_ylabel("超额违约系数  $\theta_{\text{over}}$ ")
76     mid = float(W.mean())
77     for i in range(W.shape[0]):
78         for j in range(W.shape[1]):
79             ax2.text(j, i, f"{W[i, j]:.0f}", ha="center", va="center", fontsize=8.4,
80                    color="white" if abs(W[i, j] - mid) > 0.35 * (W.max() - W.min())
81                    else vs.BLACK)
82     i0 = list(q3_grid["overs"]).index(0.5)
83     j0 = list(q3_grid["unders"]).index(1.5)
84     ax2.scatter([j0], [i0], marker="s", s=240, facecolors="none", edgecolors=vs.GREEN, lw=2.0)
85     cb = fig.colorbar(im, ax=ax2, pad=0.02)
86     cb.set_label("问题 3 全年结算费用 (万元)")
87     ax2.grid(False)
88     ax2.set_title("(b) 问题 3 结算费用对违约系数的热力图", fontsize=9.6, pad=6)
89
90     fig.tight_layout(rect=(0, 0.06, 0.97, 1))
91     fig.text(0.60, 0.005,
92            f"绿框 = 题目给定  $(\{q3\_grid['overs'][i0]\}, \{q3\_grid['unders'][j0]\})$ , "
93            f"对应全年结算费用  $\{q3['total\_cost\_cny']\}[:, .2f]$  元 (与 result3.xlsx 一致)",
94            fontsize=8.0, color=vs.GREEN, ha="center", va="bottom")
95     vs.save(fig, "FigS_sensitivity_bundle",

```

```

96         "多参数耦合灵敏度。(a) 问题 1 全天费用随储能单程效率  $\eta$  与最大充放电功率的变化, "
97         "效率的影响远大于功率 (5000 kW 附近功率约束已不紧); "
98         "(b) 问题 3 全年结算费用随超额/缺额违约系数的变化, 绿框为题目给定取值 (0.5, 1.5)。")
99     vs.flush_captions()
100
101     rows = []
102     for row in tornado["rows"]:
103         base = float(row["base"])
104         rows.append({"label": row["label"], "base": base,
105                     "low": float(row["low"]), "high": float(row["high"]),
106                     "low_pct": (float(row["low"]) - base) / base * 100.0,
107                     "high_pct": (float(row["high"]) - base) / base * 100.0})
108     out = {"rows": rows,
109           "note": "单因素  $\pm 10\%$  扰动; Q1 为全天费用, Q2/Q3 为全年费用; "
110                "基准值与 output/result*.xlsx 一致。"}
111     (RESULTS / "sensitivity_paper.json").write_text(
112         json.dumps(out, ensure_ascii=False, indent=2), encoding="utf-8")
113     for row in rows:
114         print(f" {row['label']}: {row['low_pct']:+.2f}% / {row['high_pct']:+.2f}%"
115               f" (基准 {row['base']:+.2f})")
116     print("[sens] FigS_sensitivity_bundle.pdf + results/sensitivity_paper.json 已生成")
117
118
119 if __name__ == "__main__":
120     main()

```

code/report/visualize.py

```

1  """C题 绘图程序: 主图、灵敏度图、计算架构图、论文表合成图都在这里出。
2
3  python3 -m code.report.visualize --set all      # 全部图, 默认
4  python3 -m code.report.visualize --set main    # 只出主图
5  python3 -m code.report.visualize --list        # 看图名和图注
6
7  默认只导矢量 PDF, figures/ 里不留位图; 要位图就加 --formats pdf,png。
8  数字都从 results/ 读, 脚本自己不建模, 只有灵敏度网格和性能基准是现算的。
9  """
10 from __future__ import annotations
11
12 import argparse
13 import json
14 from pathlib import Path
15
16 import matplotlib.pyplot as plt
17 import numpy as np
18 from matplotlib.gridspec import GridSpec
19 from matplotlib.patches import FancyArrowPatch, FancyBboxPatch
20 from mpl_toolkits.mplot3d import Axes3D
21
22 from code.common.load_clean import ROOT, load_panel
23 from code.report import viz_style as vs
24 from code.report import eda_figs as eda
25 from code.report import paper_figs as pfig
26 from code.report.viz_style import (BLACK, BLUE, GREY, GREEN, ORANGE, PALETTE, PURPLE, RED,
27                                   SKY, YELLOW)
28
29 RESULTS = ROOT / "results"
30 ARRAYS = RESULTS / "arrays"
31 FIGURES = ROOT / "figures"
32
33
34 def load_json(path: Path) -> dict:
35     return json.loads(Path(path).read_text(encoding="utf-8"))
36
37
38 def arr(name: str) -> np.ndarray:
39     return np.load(ARRAYS / name)
40
41

```

```

42 def hours_axis(n: int = 144) -> np.ndarray:
43     return np.arange(n) / 6.0
44
45
46 def q1_series(panel) -> dict:
47     q1 = load_json(RESULTS / "q1.json")
48     return {
49         "data": q1,
50         "price": panel.day1[:, 0], "load": panel.day1[:, 1], "pv": panel.day1[:, 2],
51         "q": np.array(q1["lp"]["q"]), "c": np.array(q1["lp"]["c"]),
52         "d": np.array(q1["lp"]["d"]), "soc": np.array(q1["lp"]["soc"])[1:],
53         "mu": np.array(q1["lp"]["mu"]),
54     }
55
56
57 # 主图: 总体架构 + 四问结果
58 def fig_architecture() -> None:
59     fig, ax = plt.subplots(figsize=(13.6, 7.4))
60     ax.set_xlim(0, 100)
61     ax.set_ylim(0, 60)
62     ax.axis("off")
63
64     def box(x, y, w, h, title, body, face="#F4F7FB", edge=BLUE, title_size=10.5):
65         ax.add_patch(FancyBboxPatch((x, y), w, h, boxstyle="round,pad=0.5,rounding_size=1.0",
66                                     linewidth=1.1, edgecolor=edge, facecolor=face))
67         ax.text(x + w / 2, y + h - 1.6, title, ha="center", va="top", fontsize=title_size,
68                fontweight="bold", color=edge)
69         ax.text(x + w / 2, y + h - 4.2, body, ha="center", va="top", fontsize=8.2,
70                linespacing=1.55, color=BLACK)
71
72     def arrow(x1, y1, x2, y2, label="", color=GREY, rad=0.0, style="-|>"):
73         ax.add_patch(FancyArrowPatch((x1, y1), (x2, y2), arrowstyle=style, mutation_scale=12,
74                                     linewidth=1.0, color=color,
75                                     connectionstyle=f"arc3,rad={rad}"))
76         if label:
77             ax.text((x1 + x2) / 2, (y1 + y2) / 2 + 0.7, label, ha="center", va="bottom",
78                    fontsize=7.8, color=BLACK)
79
80     box(2, 48.5, 27, 10, "① 原始附件 (附件1-4)",
81         "附件1 代表日 · 附件2 实际负荷/光伏\n附件3 四时次 24 h 预报 · 附件4 实时电价")
82     box(2, 34.5, 27, 11.5, "② C++17 清洗 cleaning_cpp",
83         "zip+XML 读取 · 右端口口径 NR10\n按日自适应夜窗 · NR1-NR12\n审计/隔离 · .npy + SHA-256\nmanifest",
84         face="#EEF7F1", edge=GREEN)
85     box(37, 34.5, 26, 11.5, "③ 共形预测层 CP",
86         "谐波+周期预测 · split/加权共形\nACI 反馈 · 有限样本修正\n分位锚点 0.80 / 0.25 / 0.70",
87         face="#FDF6E7", edge=ORANGE)
88     box(70, 34.5, 28, 11.5, "④ MPC 优化层",
89         "Q1 LP+DP 互证 · Q2 两阶段随机规划\nQ3 四期滚动 MPC · Q4 CVaR+软约束\n场景 bootstrap / Copula",
90         face="#FBEDF3", edge=PURPLE)
91     box(37, 19.5, 61, 10.5, "⑤ 结算与出表",
92         "Q2/Q4-2: 计划购电费 + 5×紧急购电费 Q3/Q4-3: 计划 + 紧急 + 0.5×(超额)* + 1.5×(缺额)*\n"
93         "模板驱动写出 result1/2/3/4-2/4-3.xlsx (保留附件5 的样式、表头与标签)",
94         face="#F4F4FA", edge="#444A8C")
95     box(2, 19.5, 27, 10.5, "⑥ 产物",
96         "results/*.json + arrays/*.npy\nfigures/*.pdf + CAPTIONS.md\npaper/表1-表4 +\n"
97         "paper_numbers.json",
98         face="#EEF7F1", edge=GREEN)
99     box(2, 3.5, 96, 11.5, "四个耦合接口 (CP ↔ MPC)",
100         "I 分位接口: CP 输出安全裕度 m_j → MPC 用  $N^{plan} = \hat{N} + m_j$ "
101         "II 场景接口: 残差块 bootstrap / Copula → 随机规划场景集"
102         "III 区间接口: 共形区间宽度 → 鲁棒半径与灵敏度指标"
103         "IV 反馈接口: ACI 以实测命中率回灌  $\alpha$  (夜间掩码不计入)",
104         face="#F7F7FA", edge=GREY)
105     arrow(15.5, 48.5, 15.5, 46, "只读解析")
106     arrow(29, 40, 37, 40, "clean/")
107     arrow(63, 42, 70, 42, " $\hat{N}$ ,  $\sigma$ , 分位")
108     arrow(84, 34.5, 75, 30, "调度方案", rad=-0.12)

```

```

109 arrow(37, 25, 29, 25, "数字 / 图 / 表")
110 arrow(70, 34.5, 60, 30, "q^plan, q^adj, g", rad=0.12)
111 arrow(50, 34.5, 45, 30, "α 回灌", color=PURPLE, rad=0.2)
112 arrow(62, 19.5, 55, 15, "验证")
113 arrow(18, 19.5, 18, 15, "登记")
114 ax.set_title("图 0 总体架构: C++ 清洗 → 共形预测 → 滚动 MPC → 结算出表 (含四个耦合接口)",
115             fontsize=12.5, fontweight="bold", pad=8)
116 vs.save(fig, "fig0_architecture",
117         "C题总体技术架构。②为 C++17 值级清洗层 (NR1-NR12 + 审计 + 不变量), "
118         "③为共形预测层 (ACI 反馈、三档分位), ④为四问优化模型, ⑤为结算与模板出表; "
119         "下部为 CP-MPC 的四个耦合接口 (分位、场景、区间、反馈)。")
120
121
122 def fig_q1_dispatch(panel) -> None:
123     s = q1_series(panel)
124     q1 = s["data"]
125     h = hours_axis()
126     st = panel.config["storage"]
127     k_rise = int(panel.windows["k_rise"][0]) - 1
128     k_set = int(panel.windows["k_set"][0]) - 1
129     fig = plt.figure(figsize=(13.2, 7.6))
130     gs = GridSpec(2, 2, figure=fig, hspace=0.34, wspace=0.24)
131
132     ax = fig.add_subplot(gs[0, 0])
133     ax.fill_betweenx([0, 6200], 0, h[k_rise], color="#E8EEF6", alpha=0.9, lw=0)
134     ax.fill_betweenx([0, 6200], h[k_set + 1], 24, color="#E8EEF6", alpha=0.9, lw=0)
135     ax.plot(h, s["load"], color=RED, lw=1.5, label="小区负载 $L_k$")
136     ax.plot(h, s["pv"], color=ORANGE, lw=1.5, label="光伏预测 $PV_k$")
137     ax.plot(h, s["q"] * 6, color=BLUE, lw=1.7, label="购电功率 $q_k / \Delta t$")
138     ax.set_ylim(0, 6200)
139     ax.set_xlim(0, 24)
140     ax.set_xlabel("时刻 (h)")
141     ax.set_ylabel("功率 (kW)")
142     ax.legend(loc="upper center", ncol=3)
143     vs.panel(ax, "a")
144     ax.set_title("负荷 / 光伏 / 购电功率 (灰区为按日自适应核夜间)")
145
146     ax = fig.add_subplot(gs[0, 1])
147     ax.plot(h, s["price"], color=PURPLE, lw=1.6, label="电价 $\pi_k$")
148     ax.plot(h, s["mu"], color=GREEN, lw=1.6, label="购电边际价值 $\mu_k$ (对偶)")
149     ax.fill_between(h, s["mu"], s["price"], where=s["price"] > s["mu"], color=ORANGE,
150                   alpha=0.18, lw=0, label="套利空间")
151     ax.set_xlim(0, 24)
152     ax.set_xlabel("时刻 (h)")
153     ax.set_ylabel("元/kWh")
154     ax.legend(loc="upper left")
155     vs.panel(ax, "b")
156     ax.set_title("电价与影子价格: 价差时段即储能充放窗口")
157
158     ax = fig.add_subplot(gs[1, 0])
159     ax.plot(h, s["soc"], color=GREEN, lw=1.8, label="储电量 $e_k$")
160     ax.axhline(float(st["soc_max_kwh"]), ls="--", lw=0.9, color=GREY)
161     ax.axhline(float(st["soc_min_kwh"]), ls="--", lw=0.9, color=GREY)
162     ax.annotate(f"上限 {float(st['soc_max_kwh']):.0f} kWh", (0.3, float(st["soc_max_kwh"])),
163               va="bottom", fontsize=8, color=GREY)
164     ax.annotate(f"下限 {float(st['soc_min_kwh']):.0f} kWh", (0.3, float(st["soc_min_kwh"])),
165               va="bottom", fontsize=8, color=GREY)
166     ax.scatter([0, 24], [s["soc"][0], s["soc"][-1]], color=BLACK, zorder=5, s=18)
167     ax.annotate(f"$e_0=e_{144}={s['soc'][-1]:.0f}$ kWh", (0.6, s["soc"][-1] - 900), fontsize
168               =8.4)
169     ax.set_xlim(0, 24)
170     ax.set_xlabel("时刻 (h)")
171     ax.set_ylabel("储电量 (kWh)")
172     ax.legend(loc="lower right")
173     vs.panel(ax, "c")
174     ax.set_title("储电量轨迹 (日循环约束, 全程落在 [1200, 10800] kWh)")
175
176     ax = fig.add_subplot(gs[1, 1])
177     width = 0.4
178     ax.bar(h - width / 2, s["c"] * 6, width=width, color=BLUE, label="充电功率")

```

```

178 ax.bar(h + width / 2, -s["d"] * 6, width=width, color=RED, label="放电功率")
179 ax.axhline(0, color=BLACK, lw=0.8)
180 ax.set_xlim(0, 24)
181 ax.set_xlabel("时刻 (h)")
182 ax.set_ylabel("功率 (kW)")
183 ax.legend(loc="upper left")
184 vs.panel(ax, "d")
185 ax.set_title("充放电功率 (充放互斥: 命题 1 的数值验证)")
186
187 fig.suptitle(f"图 1 问题 1 代表性日最优计划: 全天 {q1['cost_cny']:.2f} 元, "
188            f"较 B2 基线 {q1['baselines']['B2_pv_selfuse_curtail_no_storage_cny']:.2f} 元下降 "
189            f"{100*(1-q1['cost_cny']/q1['baselines']['B2_pv_selfuse_curtail_no_storage_cny "
190            ')}:.2f}%",
191            fontsize=12, fontweight="bold", y=0.99)
192 vs.save(fig, "fig1_q1_dispatch",
193         "问题 1 代表性日最优调度。(a) 负荷、光伏与购电功率;(b) 电价与购电边际价值(对偶变量), "
194         "两者之差即储能套利空间;(c) 储电量轨迹满足日循环与容量约束;(d) 充放电功率, "
195         "各时段充放互斥(命题 1 成立)。")
196
197 def fig_dp_convergence() -> None:
198     q1 = load_json(RESULTS / "q1.json")
199     steps = np.array([r["step_kwh"] for r in q1["dp"]])
200     costs = np.array([r["cost_cny"] for r in q1["dp"]])
201     lp = q1["cost_cny"]
202     fig, ax = plt.subplots(figsize=(7.6, 4.9))
203     ax.plot(steps, costs, "o-", color=BLUE, lw=1.5, ms=5, label="SoC 离散动态规划")
204     ax.axhline(lp, ls="--", color=RED, lw=1.4, label=f"LP 最优解 {lp:.2f} 元")
205     for x, y in zip(steps, costs):
206         ax.annotate(f"{y:.2f}", (x, y), textcoords="offset points", xytext=(0, 9),
207                    ha="center", fontsize=8.2)
208     ax.annotate(f"相对偏差 {100*(costs[-1]-lp)/lp:.2f}%", (steps[-1], costs[-1]),
209               textcoords="offset points", xytext=(14, -16), fontsize=8.4, color=RED)
210     ax.set_xscale("log")
211     ax.invert_xaxis()
212     ax.set_xticks(steps, [f"{int(s)}" for s in steps])
213     ax.set_xlabel("SoC 离散步长  $\Delta$  e$ (kWh)")
214     ax.set_ylabel("全天购电费 (元)")
215     ax.legend(loc="lower left")
216     ax.set_title("图 2 数值验证: DP 步长单调收敛到 LP 最优解")
217     vs.save(fig, "fig2_q1_dp_convergence",
218            "问题 1 的数值一致性验证。DP 费用随离散步长减小而单调下降, 收敛到 LP 最优解: "
219            "四个步长 (120/60/30/15 kWh) 的费用与题给锚点逐位一致。")
220
221
222 def fig_night_cleaning(panel) -> None:
223     raw, clean = panel.pv_raw, panel.pv
224     dates = list(panel.dates)
225     picks = [{"2025-01-15", "冬季 1-15", BLUE}, {"2025-06-15", "夏季 6-15", GREEN},
226             {"2025-09-15", "秋季 9-15", ORANGE}]
227     fig = plt.figure(figsize=(13.0, 8.0))
228     gs = GridSpec(2, 2, figure=fig, height_ratios=[1.15, 1.0], hspace=0.36, wspace=0.22)
229     ax = fig.add_subplot(gs[0, :])
230     h = hours_axis()
231     for iso, label, color in picks:
232         d = dates.index(iso)
233         ax.plot(h, raw[d], ls="--", lw=1.0, color=color, alpha=0.5)
234         ax.plot(h, clean[d], lw=1.7, color=color, label=label)
235     ax.set_xlim(0, 24)
236     ax.set_xlabel("时刻 (h)")
237     ax.set_ylabel("光伏功率 (kW)")
238     ax.legend(ncol=3, loc="upper left")
239     ax.annotate("虚线 = 清洗前 实线 = 清洗后", (0.98, 0.94), xycoords="axes fraction",
240               ha="right", fontsize=8.6, color=GREY)
241     vs.panel(ax, "a", dx=-0.06)
242     ax.set_title("清洗前后对比: 核夜间小值全部归零, 日间曲线逐点未变 (I7 逐位校验)")
243
244     ax = fig.add_subplot(gs[1, 0])
245     x = np.arange(365)

```

```

246 ax.plot(x, panel.windows["k_rise"] / 6.0, color=ORANGE, lw=1.3, label="日出时刻")
247 ax.plot(x, panel.windows["k_set"] / 6.0, color=RED, lw=1.3, label="日落时刻")
248 ax.set_xlabel("2025 年日序")
249 ax.set_ylabel("时刻 (h)")
250 ax.legend(loc="lower center", ncol=2)
251 vs.panel(ax, "b")
252 ax.set_title("按日自适应日照窗口 (全年漂移约 4 h)")
253
254 ax = fig.add_subplot(gs[1, 1])
255 night = panel.windows["night_core_hours"]
256 month_edges = np.cumsum([0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31])
257 monthly = [night[month_edges[m]:month_edges[m + 1]].mean() for m in range(12)]
258 bars = ax.bar(np.arange(1, 13), monthly, color=SKY, edgecolor=BLUE, lw=0.6)
259 vs.annotate_bars(ax, bars, monthly, fmt="{:.2f}", fontsize=7.6)
260 ax.set_xticks(np.arange(1, 13), [f"{m}" for m in range(1, 13)])
261 ax.set_xlabel("月份")
262 ax.set_ylabel("核夜间时长 (h)")
263 vs.panel(ax, "c")
264 ax.set_title("核夜间时长月均值: 10.33 → 14.33 h (禁止固定时钟窗口)")
265 fig.suptitle("图 3 夜间专项清洗: 结构事实、按日自适应窗口与季节漂移",
266             fontsize=12, fontweight="bold", y=0.99)
267 vs.save(fig, "fig3_night_cleaning",
268         "夜间专项数据清洗。(a) 冬/夏/秋三个代表日清洗前后光伏曲线 (核夜间零漂噪声归零、"
269         "日间逐点不变); (b) 按日自适应的日出/日落时刻; (c) 核夜间时长的月度分布, "
270         "季节漂移约 4 h, 说明固定时钟窗口不可用。")
271
272
273 def fig_q2_emergency(panel) -> None:
274     q2 = load_json(RESULTS / "q2.json")
275     emerg = arr("q2_emergency.npy")
276     records = q2["records"]
277     fig = plt.figure(figsize=(14.6, 5.2))
278     gs = GridSpec(1, 3, figure=fig, width_ratios=[1.55, 0.85, 1.0], wspace=0.30)
279
280     ax = fig.add_subplot(gs[0, 0])
281     im = ax.imshow(emerg.T, aspect="auto", origin="lower", cmap=vs.SEQ_CMAP,
282                  extent=[0, len(records), 0, 24], interpolation="nearest")
283     ax.set_xlabel("日期 (2025-02-01 → 2025-12-31)")
284     ax.set_ylabel("时刻 (h)")
285     ax.grid(False)
286     cb = fig.colorbar(im, ax=ax, pad=0.02)
287     cb.set_label("紧急购电量 (kWh/时段)")
288     vs.panel(ax, "a")
289     ax.set_title("全年紧急购电热力图")
290
291     ax = fig.add_subplot(gs[0, 1])
292     blocks = [r["emergency_blocks"] for r in records]
293     bins = np.arange(-0.5, max(blocks) + 1.5)
294     ax.hist(blocks, bins=bins, color=ORANGE, edgecolor="white", lw=0.6)
295     ax.set_xlabel("紧急购电块数 (块/日)")
296     ax.set_ylabel("天数 (d)")
297     ax.annotate(f"均值 {np.mean(blocks):.2f} 块/日\n含紧急购电天数 {sum(1 for b in blocks if b)
298              }/334",
299              (0.62, 0.86), xycoords="axes fraction", fontsize=8.4,
300              bbox=dict(boxstyle="round,pad=0.35", fc="white", ec=GREY, lw=0.6))
301     vs.panel(ax, "b")
302     ax.set_title("紧急购电块数分布")
303
304     ax = fig.add_subplot(gs[0, 2])
305     keys = ["oracle_bound", "C1_no_margin", "C2_analytic_newsvendor"]
306     labels = ["完美信息\n(界)", "无裕度\n(均值计划)", "解析报童\n $p^*=0.80$ "]
307     values = [q2["summary"]["controls_total_cost_cny"][k] / 1e4 for k in keys]
308     main_value = q2["summary"]["total_cost_cny"] / 1e4
309     bars = ax.bar(labels, values, color=[GREEN, RED, ORANGE], width=0.6)
310     vs.annotate_bars(ax, bars, values, fmt="{:.0f}", fontsize=8)
311     ax.axhline(main_value, color=BLUE, lw=1.6, ls="--")
312     ax.annotate(f"主策略 (两阶段随机规划) {main_value:.0f} 万元",
313              (0.5, main_value), textcoords="offset points", xytext=(0, 6),
314              ha="center", fontsize=8.6, color=BLUE)
315     ax.set_ylabel("全年购电费用 (万元)")

```

```

315 vs.panel(ax, "c")
316 ax.set_title("对照实验 (334 天) ")
317
318 fig.suptitle("图 4 问题 2: 紧急购电的时空分布与策略对照", fontsize=12,
319             fontweight="bold", y=1.0)
320 vs.save(fig, "fig4_q2_emergency",
321         "问题 2 结果。(a) 全年紧急购电的热力图 (横轴日期、纵轴时刻、颜色为购电量), "
322         "夏季日间与全年晚高峰是主要触发区; (b) 每日紧急购电块数分布; "
323         "(c) 主策略与三种对照的全年费用, 完美信息仅作下界。")
324
325
326 def fig_coverage_cost(panel) -> None:
327     """问题2 的覆盖率-费用曲线: 有缓存就复用, 没有就现算。"""
328     from code.forecast.conformal import conformal_level, weighted_quantile
329     from code.optimize.lp_common import Storage, solve_stage_lp
330
331     path = RESULTS / "q2_coverage_cost.json"
332     if path.exists():
333         curve = load_json(path)
334     else:
335         st = Storage.from_config(panel.config)
336         price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
337         emergency = float(panel.config["tariff"]["emergency_multiplier"])
338         net = panel.net_load()
339         levels = [0.5, 0.6, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95]
340         days = list(range(31, 365, 3)[:120])
341         curve = {"levels": levels, "cost_cny": [], "coverage": [], "emergency_energy_kwh": []}
342         for level in levels:
343             total, hits, energy = 0.0, [], 0.0
344             for d in days:
345                 win = net[max(0, d - 30):d]
346                 base = win.mean(axis=0)
347                 det = solve_stage_lp(base, price[d], st)
348                 sample = base[None, :] * st.dt + det["c"] - det["d"] + (win - base) * st.dt
349                 lvl = conformal_level(level, sample.shape[0])
350                 q = np.maximum(0.0, np.array([weighted_quantile(sample[:, k], lvl)
351                                             for k in range(144)]))
352                 need = net[d] * st.dt + det["c"] - det["d"]
353                 g = np.maximum(0.0, need - q)
354                 total += float(np.sum(price[d] * q) + emergency * np.sum(price[d] * g))
355                 hits.append(float(np.mean(need <= q)))
356                 energy += float(g.sum())
357                 curve["cost_cny"].append(total)
358                 curve["coverage"].append(float(np.mean(hits)))
359                 curve["emergency_energy_kwh"].append(energy)
360             curve["days"] = len(days)
361             path.write_text(json.dumps(curve, ensure_ascii=False, indent=2), encoding="utf-8")
362
363     levels = np.array(curve["levels"])
364     costs = np.array(curve["cost_cny"])
365     coverage = np.array(curve["coverage"])
366     fig, (ax, ax2) = plt.subplots(1, 2, figsize=(11.6, 4.7))
367     ax.plot(levels, costs / 1e4, "o-", color=BLUE, lw=1.6, ms=5, label="样本日总费用")
368     best = int(np.argmin(costs))
369     ax.scatter(levels[best], costs[best] / 1e4, s=110, facecolors="none", edgecolors=RED, lw=1.8,
370              label=f"样本内最低  $p^*=\{levels[best]:.2f\}$ ")
371     ax.axvline(0.80, ls="--", color=GREEN, lw=1.4, label="D15 锚点  $p^*=0.80$ ")
372     ax.set_xlabel("计划分位  $p^*$ ")
373     ax.set_ylabel("样本日购电费用 (万元)")
374     ax.legend(loc="upper center")
375     vs.panel(ax, "a")
376     ax.set_title("费用-分位曲线 (1:5 边际代价比 → 最优约 0.80-0.85) ")
377
378     ax2.plot(levels, coverage * 100, "s-", color=GREEN, lw=1.5, ms=4.5)
379     ax2.axhline(80, ls="--", color=GREY, lw=1.0)
380     ax2.set_xlabel("计划分位  $p^*$ ")
381     ax2.set_ylabel("实测覆盖率 (%)")
382     vs.panel(ax2, "b")
383     ax2.set_title("覆盖率随计划分位的变化 (共形有限样本修正后) ")

```

```

384 fig.suptitle("图 5 问题 2 的覆盖率-费用权衡与分位锚点依据", fontsize=12,
385             fontweight="bold", y=1.0)
386 vs.save(fig, "fig5_q2_coverage_cost",
387         "问题 2 的计划分位标定。(a) 不同计划分位下的样本日总费用, 最优区间 0.80-0.85, "
388         "与边际代价比 1:5 推出的临界分位 0.80 一致; (b) 对应的实测覆盖率。")
389
390
391 def fig_q3_subsets(panel) -> None:
392     q3 = load_json(RESULTS / "q3.json")
393     summary = q3["summary"]
394     fig = plt.figure(figsize=(14.2, 5.6))
395     gs = GridSpec(1, 3, figure=fig, width_ratios=[1.05, 1.0, 1.0], wspace=0.30)
396
397     ax = fig.add_subplot(gs[0, 0])
398     names = ["S0", "S1", "S2", "S3", "S4", "S5"]
399     labels = ["S0\n仅 0:00", "S1\n+6:00", "S2\n+12:00", "S3\n+18:00\n(主策略)",
400             "S4\n12:00+18:00", "S5\n完美预报\n(界)"]
401     values = [summary["subset_total_cost_cny"][k] / 1e4 for k in names]
402     colors = [GREY, BLUE, SKY, "#0D47A1", "#5C6BC0", GREEN]
403     bars = ax.bar(labels, values, color=colors, width=0.66)
404     vs.annotate_bars(ax, bars, values, fmt="{:.0f}", fontsize=7.8)
405     ax.set_ylabel("全年结算费用 (万元)")
406     vs.panel(ax, "a")
407     ax.set_title("预报时次子集实验 S0-S5")
408
409     ax = fig.add_subplot(gs[0, 1])
410     split = summary["subset_value_split_cny"]
411     steps = ["S0->S1", "S1->S2", "S2->S3"]
412     step_labels = ["6:00 预报", "12:00 预报", "18:00 预报"]
413     night = np.array([split["night_core"].get(s, 0.0) / 1e4 for s in steps])
414     day = np.array([split["daylight"].get(s, 0.0) / 1e4 for s in steps])
415     x = np.arange(len(steps))
416     ax.bar(x - 0.19, day, 0.38, color=ORANGE, label="日间时段")
417     ax.bar(x + 0.19, night, 0.38, color="#37474F", label="核夜间时段")
418     for xi, (d, n) in enumerate(zip(day, night)):
419         ax.annotate(f"{d:+.0f}", (xi - 0.19, d), ha="center", va="bottom" if d >= 0 else "top",
420                   fontsize=7.8)
421         ax.annotate(f"{n:+.1f}", (xi + 0.19, n), ha="center", va="bottom" if n >= 0 else "top",
422                   fontsize=7.8)
423     ax.axhline(0, color=BLACK, lw=0.8)
424     ax.set_xticks(x, step_labels)
425     ax.set_ylabel("费用节省 (万元)")
426     ax.legend(loc="upper right")
427     vs.panel(ax, "b")
428     ax.set_title("预报价值分时段: 收益几乎全部来自日间")
429
430     ax = fig.add_subplot(gs[0, 2])
431     uri = q3["uri_trace"]
432     stages = [6, 12, 18]
433     data = [np.array([t["uri"] for t in uri if t["stage"] == s]) for s in stages]
434     parts = ax.violinplot(data, showmeans=True, showextrema=False, widths=0.7)
435     for body, color in zip(parts["bodies"], [BLUE, SKY, ORANGE]):
436         body.set_facecolor(color)
437         body.set_alpha(0.55)
438     ax.axhline(0, color=RED, lw=1.1)
439     ax.set_xticks([1, 2, 3], [f"{s}:00 更新" for s in stages])
440     ax.set_ylabel("URI =  $\ln(\frac{\sigma_{old}}{\sigma_{new}})$ ")
441     vs.panel(ax, "c")
442     ax.set_title("预报更新的信息比率 (正值 = 误差尺度下降)")
443
444     fig.suptitle("图 6 问题 3: 预报时次价值、分时段归因与信息比率",
445                 fontsize=12, fontweight="bold", y=1.0)
446     vs.save(fig, "fig6_q3_subsets",
447         "问题 3 的预报时次实验。(a) S0-S5 六种时次组合的全年结算费用 (S5 为完美预报下界); "
448         "(b) 各时次边际价值按核夜间/日间拆分, 收益集中在日间; (c) 相邻时次预报误差尺度的 "
449         "信息比率 URI 分布。")
450
451
452 def fig_q3_soc_uri(panel) -> None:
453     q3 = load_json(RESULTS / "q3.json")

```

```

454 soc = arr("q3_soc.npy")
455 plan = arr("q3_q_plan.npy")
456 adj = arr("q3_q_adjust.npy")
457 dates = [panel.dates[d] for d in range(31, 365)]
458 targets = ["2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21"]
459 fig = plt.figure(figsize=(14.0, 5.4))
460 gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.0, 1.0], wspace=0.28)
461
462 ax = fig.add_subplot(gs[0, 0])
463 h = np.arange(145) / 6.0
464 for iso, color in zip(targets, PALETTE):
465     ax.plot(h, soc[dates.index(iso)], color=color, lw=1.5, label=iso)
466 ax.axhline(1200, ls="--", lw=0.9, color=GREY)
467 ax.axhline(10800, ls="--", lw=0.9, color=GREY)
468 ax.set_xlabel("时刻 (h)")
469 ax.set_ylabel("储电量 (kWh)")
470 ax.legend(ncol=2, loc="upper right")
471 vs.panel(ax, "a")
472 ax.set_title("四个指定日期的储电量轨迹")
473
474 ax = fig.add_subplot(gs[0, 1])
475 idx = dates.index("2025-06-21")
476 h2 = hours_axis()
477 ax.plot(h2, plan[idx], color=BLUE, lw=1.5, label="计划购电量  $q^{\text{plan}}$ ")
478 ax.plot(h2, adj[idx], color=RED, lw=1.5, ls="--", label="调整购电量  $q^{\text{adj}}$ ")
479 ax.fill_between(h2, plan[idx], adj[idx], where=adj[idx] >= plan[idx], color=ORANGE,
480                alpha=0.3, lw=0, label="上调 (1.5×)")
481 ax.fill_between(h2, plan[idx], adj[idx], where=adj[idx] < plan[idx], color=SKY,
482                alpha=0.35, lw=0, label="下调 (0.5×)")
483 ax.set_xlabel("时刻 (h)")
484 ax.set_ylabel("购电量 (kWh/时段)")
485 ax.legend(loc="upper left")
486 vs.panel(ax, "b")
487 ax.set_title("2025-06-21 计划 vs 调整购电量")
488
489 ax = fig.add_subplot(gs[0, 2])
490 records = q3["records"]
491 plan_c = np.array([r["plan_cost_cny"] for r in records]) / 1e4
492 emerg_c = np.array([r["emergency_cost_cny"] for r in records]) / 1e4
493 over_c = np.array([r["over_plan_penalty_cny"] for r in records]) / 1e4
494 under_c = np.array([r["under_plan_penalty_cny"] for r in records]) / 1e4
495 order = np.argsort(-(plan_c + emerg_c + over_c + under_c))[:40]
496 x = np.arange(len(order))
497 ax.bar(x, plan_c[order], color=BLUE, label="计划购电费")
498 ax.bar(x, emerg_c[order], bottom=plan_c[order], color=RED, label="紧急购电费")
499 ax.bar(x, over_c[order], bottom=plan_c[order] + emerg_c[order], color=SKY, label="超额违约")
500 ax.bar(x, under_c[order], bottom=plan_c[order] + emerg_c[order] + over_c[order],
501        color=ORANGE, label="缺额违约")
502 ax.set_xlabel("费用最高的 40 天 (按总额排序)")
503 ax.set_ylabel("费用 (万元)")
504 ax.legend(loc="upper right", ncol=2)
505 vs.panel(ax, "c")
506 ax.set_title("费用结构分解: 缺额违约金是主要成本项")
507 fig.suptitle("图 7 问题 3: 储电量轨迹、计划/调整购电量与结算费用结构",
508             fontsize=12, fontweight="bold", y=1.0)
509 vs.save(fig, "fig7_q3_soc_uri",
510         "问题 3 结果细节。(a) 四个指定日期的储电量轨迹 (日循环); (b) 计划与调整购电量之差"
511         " (上调用 1.5 倍价、下调按 0.5 倍计违约); (c) 费用最高 40 天的结算结构分解。")
512
513
514 def fig_q4_risk(panel) -> None:
515     q4 = load_json(RESULTS / "q4.json")
516     pareto = q4["cvar_pareto"]
517     fig = plt.figure(figsize=(14.4, 5.4))
518     gs = GridSpec(1, 3, figure=fig, width_ratios=[1.0, 1.0, 1.0], wspace=0.30)
519
520     ax = fig.add_subplot(gs[0, 0])
521     betas = np.array([r["beta"] for r in pareto])
522     costs = np.array([r["realized_cost_cny"] for r in pareto]) / 1e4
523     cvar = np.array([r["cvar_90_cny"] for r in pareto])

```

```

524 ax.plot(costs, cvar, "o-", color=PURPLE, lw=1.6, ms=6)
525 for c, v, b in zip(costs, cvar, betas):
526     ax.annotate(f"$\\beta$={b:.2f}", (c, v), textcoords="offset points", xytext=(7, 4),
527                fontsize=8.2)
528 ax.set_xlabel("样本日总费用 (万元)")
529 ax.set_ylabel("CVaR$_{90}$ (元/日)")
530 vs.panel(ax, "a")
531 ax.set_title("风险-费用帕累托前沿")
532
533 ax = fig.add_subplot(gs[0, 1])
534 spikes = q4["spike_conformity"]
535 from code.forecast.conformal import conformal_level
536 price = panel.price
537 d = 200
538 win = price[d - 30:d]
539 hi = np.quantile(win, conformal_level(0.95, win.shape[0]), axis=0)
540 lo = np.quantile(win, conformal_level(0.05, win.shape[0]), axis=0)
541 h = hours_axis()
542 ax.fill_between(h, lo, hi, color=YELLOW, alpha=0.45, lw=0, label="共形带 5%-95%")
543 ax.plot(h, price[d], color=BLUE, lw=1.5, label="实际电价 (附件4) ")
544 ax.scatter(h[price[d] >= np.quantile(price[d], 0.9)],
545            price[d][price[d] >= np.quantile(price[d], 0.9)],
546            color=RED, s=14, zorder=5, label="尖峰时刻 (≥90% 分位) ")
547 ax.set_xlabel("时刻 (h)")
548 ax.set_ylabel("元/kWh")
549 ax.legend(loc="upper left")
550 vs.panel(ax, "b")
551 ax.set_title(f"尖峰共形防守 (覆盖 {spikes['coverage']*100:.1f}%) ")
552
553 ax = fig.add_subplot(gs[0, 2])
554 soft = q4["soft_penalty_evidence"]
555 mult = [r["rho1_multiplier"] for r in soft]
556 slack = [r["slack_energy_kwh"] / 1e4 for r in soft]
557 bars = ax.bar([f"{m:g}x" for m in mult], slack,
558               color=[RED if s > 0 else GREEN for s in slack], width=0.6)
559 vs.annotate_bars(ax, bars, slack, fmt="{:.1f}", fontsize=7.8)
560 ax.axvline(2.5, color=GREY, ls=":", lw=1.0)
561 ax.annotate(f"$\\rho_1=5\\backslash p_{\\max}$ 门限", (2.5, max(slack) * 0.8), rotation=90,
562            ha="right", va="center", fontsize=8, color=GREY)
563 ax.set_xlabel("软约束惩罚 $\\rho_1$ / $p_{\\max}$")
564 ax.set_ylabel("松弛电量 (万 kWh)")
565 vs.panel(ax, "c")
566 ax.set_title("软约束惩罚过小 → 模型用松弛替代购电")
567 fig.suptitle("图 8 问题 4: CVaR 风险前沿、尖峰共形防守与软约束惩罚依据",
568             fontsize=12, fontweight="bold", y=1.0)
569 vs.save(fig, "fig8_q4_pareto_spike",
570         "问题 4 的风险层。(a) CVaR 与费用的帕累托前沿 ( $\beta$  从 0 到 0.9) ; "
571         "(b) 电价的逐时段共形带与尖峰时刻, 覆盖率 94.1%; "
572         "(c) 软约束惩罚系数取值实验:  $p_1 < 5p_{\\max}$  时模型大量使用松弛, "
573         "必须取  $p_1 > 5p_{\\max}$  (本项目取 6 倍)。")
574
575
576 def fig_annual(panel) -> None:
577     months = np.zeros((12, 4))
578     labels = ["Q2 (附件1 电价)", "Q3 (附件1 电价)", "Q4-2 (附件4 电价)", "Q4-3 (附件4 电价)"]
579     colors = [BLUE, GREEN, ORANGE, RED]
580     for col, tag in enumerate(("q2", "q3", "q4_2", "q4_3")):
581         for r in load_json(RESULTS / f"{tag}.json")["records"]:
582             months[int(r["date"][5:7]) - 1, col] += r["cost_cny"] / 1e4
583     fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.2, 5.0), gridspec_kw={"width_ratios": [1.5, 1]})
584     x = np.arange(12)
585     for col, (label, color) in enumerate(zip(labels, colors)):
586         ax.bar(x + (col - 1.5) * 0.2, months[:, col], 0.2, color=color, label=label)
587     ax.set_xticks(x, [f"{m}" for m in range(1, 13)])
588     ax.set_xlabel("月份")
589     ax.set_ylabel("月度购电费用 (万元)")
590     ax.legend(ncol=2, loc="upper left")
591     vs.panel(ax, "a")
592     ax.set_title("四问的月度费用对比 (334 天回溯) ")

```

```

593
594 totals = months.sum(axis=0)
595 shares = 100 * totals / totals.sum()
596 bars = ax2.barh(labels, shares, color=colors, height=0.55)
597 for bar, value, total in zip(bars, shares, totals):
598     ax2.annotate(f"{value:.1f}% ({total:.0f} 万元)",
599                 (value, bar.get_y() + bar.get_height() / 2), va="center",
600                 textcoords="offset points", xytext=(6, 0), fontsize=8.4)
601 ax2.set_xlim(0, 45)
602 ax2.set_xlabel("占四问总费用比例 (%)")
603 vs.panel(ax2, "b")
604 ax2.set_title("费用份额: 实时电价 (Q4) 占比 47.8%")
605 fig.suptitle("图 9 全年月度费用分布与四问费用份额", fontsize=12, fontweight="bold", y=1.0)
606 vs.save(fig, "fig9_annual_costs",
607         "全年费用分布。(a) 四问的月度购电费用; (b) 四问全年费用份额, "
608         "说明实时波动电价 (附件4) 下的费用显著高于固定电价情境。")
609
610
611 def fig_uncertainty(panel) -> None:
612     data = load_json(RESULTS / "uncertainty_ranges.json")
613     q2 = load_json(RESULTS / "q2.json")
614     band = data["q2_plan_band"]
615     q3 = data["q3_load_band"]
616     q4 = data["q4_price_band"]
617     fig = plt.figure(figsize=(14.6, 8.4))
618     gs = GridSpec(2, 2, figure=fig, hspace=0.36, wspace=0.24)
619
620     ax = fig.add_subplot(gs[0, 0])
621     days = np.arange(len(band["p80"]["daily_cost_cny"]))
622     ax.fill_between(days, np.array(band["p25"]["daily_cost_cny"]) / 1e4,
623                    np.array(band["p90"]["daily_cost_cny"]) / 1e4,
624                    color=SKY, alpha=0.35, lw=0, label="$p^*\in[0.25,0.90]$" )
625     ax.plot(days, np.array(band["p80"]["daily_cost_cny"]) / 1e4, color=BLUE, lw=1.3,
626            label="$p^*=\$0.80$ (D15 锚点) ")
627     ax.plot(days, np.array([r["cost_cny"] for r in q2["records"]]) / 1e4, color=RED, lw=1.1,
628            label="主口径两阶段 LP (result2) ")
629     ax.set_xlabel("日期 (2025-02-01 → 2025-12-31) ")
630     ax.set_ylabel("日费用 (万元)")
631     ax.legend()
632     vs.panel(ax, "a")
633     ax.set_title("U1 Q2 信息集未定义 → 计划分位区间")
634
635     ax = fig.add_subplot(gs[0, 1])
636     labels = ["下带\n6.7% 分位", "点估计\n实际负荷", "上带\n93.3% 分位"]
637     values = np.array([q3["cost_at_lower_load_band_cny"], q3["point_estimate_cny"],
638                       q3["cost_at_upper_load_band_cny"]]) / 1e4
639     bars = ax.bar(labels, values, color=[GREY, BLUE, RED], width=0.6)
640     vs.annotate_bars(ax, bars, values, fmt="{:.0f} 万元", fontsize=8.4)
641     ax.set_ylabel("全年结算费用 (万元)")
642     vs.panel(ax, "b")
643     ax.set_title(f"U2 Q3 负荷是否已知 → 压力区间 (上带 {q3['upper_band_deviation_pct']:+.1f}%) ")
644
645     ax = fig.add_subplot(gs[1, 0])
646     from code.forecast.conformal import conformal_level
647     price = panel.price
648     d = 200
649     win = price[d - 30:d]
650     hi = np.quantile(win, conformal_level(0.95, win.shape[0]), axis=0)
651     lo = np.quantile(win, conformal_level(0.05, win.shape[0]), axis=0)
652     h = hours_axis()
653     ax.fill_between(h, lo, hi, color=YELLOW, alpha=0.45, lw=0, label="电价共形带 5%-95%")
654     ax.plot(h, price[d], color=BLUE, lw=1.4, label="实际电价 (附件4) ")
655     ax.set_xlabel("时刻 (h)")
656     ax.set_ylabel("元/kWh")
657     ax.legend()
658     vs.panel(ax, "c")
659     ax.set_title("U3 电价是否在 0:00 已知 → 逐时段共形带")
660
661     ax = fig.add_subplot(gs[1, 1])
662     items = ["Q2", "Q3", "Q4-2", "Q4-3"]

```

```

663 points = [band["two_stage_lp_total_cost_cny"], q3["point_estimate_cny"],
664           q4["q4_2"]["point_estimate_cny"], q4["q4_3"]["point_estimate_cny"]]
665 lows = [band["band_total_cost_cny"][0], q3["load_band_total_cost_cny"][0],
666         q4["q4_2"]["band_total_cost_cny"][0], q4["q4_3"]["band_total_cost_cny"][0]]
667 highs = [band["band_total_cost_cny"][1], q3["load_band_total_cost_cny"][1],
668          q4["q4_2"]["band_total_cost_cny"][1], q4["q4_3"]["band_total_cost_cny"][1]]
669 y = np.arange(len(items))
670 ax.hlines(y, np.array(lows) / 1e4, np.array(highs) / 1e4, color=GREY, lw=7, alpha=0.45,
671          label="共形区间")
672 ax.scatter(np.array(points) / 1e4, y, color=RED, s=45, zorder=4, label="主口径点值")
673 for yi, (pl, ph, pt) in enumerate(zip(lows, highs, points)):
674     ax.annotate(f"{pt/1e4:.0f}", (pt / 1e4, yi), textcoords="offset points", xytext=(0, 8),
675               ha="center", fontsize=8.2, color=RED)
676 ax.set_yticks(y, items)
677 ax.set_xlabel("全年费用 (万元)")
678 ax.legend(loc="lower right")
679 vs.panel(ax, "d")
680 ax.set_title("四问：点值与区间（区间不替换结果文件）")
681 fig.suptitle("图 10 不确定项处理：能以 C题.pdf 确定的用原文，未明确的用共形区间",
682             fontsize=12, fontweight="bold", y=0.98)
683 vs.save(fig, "fig10_conformal_bands",
684         "不确定项（C题.pdf 未明确）的共形区间。（a）Q2 信息集 → 计划分位带；"
685         "(b) Q3 负荷是否已知 → 负荷共形带压力测试；(c) Q4 电价是否已知 → 逐时段共形带；"
686         "(d) 四问主口径点值与区间对比，区间仅用于说明范围。")
687
688 # 灵敏度：三维响应面 + 热力图 + 龙卷风 + 置信区间
689 def _sens_cache() -> dict:
690     path = RESULTS / "sensitivity_grids.json"
691     return load_json(path) if path.exists() else {}
692
693 def compute_q1_surface(panel) -> dict:
694     """Q1 费用对（单程效率、储能功率）的响应面。"""
695     from code.optimize.lp_common import Storage, solve_stage_lp
696
697     st0 = Storage.from_config(panel.config)
698     net = panel.day1[:, 1] - panel.day1[:, 2]
699     price = panel.day1[:, 0]
700     etas = [0.80, 0.85, 0.90, 0.95, 1.00]
701     powers = [4000.0, 4500.0, 5000.0, 5500.0, 6000.0]
702     grid = np.zeros((len(etas), len(powers)))
703     for i, eta in enumerate(etas):
704         for j, power in enumerate(powers):
705             st = Storage(**{**st0.__dict__, "eta_c": eta, "eta_d": eta, "power": power})
706             grid[i, j] = solve_stage_lp(net, price, st)["cost"]
707     return {"etas": etas, "powers": powers, "cost_cny": grid.tolist()}
708
709 def compute_q2_surface(panel, levels=(0.55, 0.65, 0.75, 0.80, 0.85, 0.95),
710                        lambdas=(2.0, 3.0, 5.0, 7.0, 10.0)) -> dict:
711     """Q2 全年费用对（计划分位、紧急电价倍数）的响应面。"""
712     from code.forecast.conformal import conformal_level
713     from code.optimize.lp_common import Storage, solve_stage_lp
714
715     st = Storage.from_config(panel.config)
716     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
717     net = panel.net_load()
718     base_list, c_list, d_list, need_list, win_list = [], [], [], [], []
719     for d in range(31, 365):
720         win = net[d - 30:d]
721         base = win.mean(axis=0)
722         det = solve_stage_lp(base, price[d], st)
723         base_list.append(base)
724         c_list.append(det["c"])
725         d_list.append(det["d"])
726         need_list.append(net[d] * st.dt + det["c"] - det["d"])
727         win_list.append(win)
728     grid = np.zeros((len(levels), len(lambdas)))
729     for i, level in enumerate(levels):

```

```

733     for lam in lambdas:
734         total = 0.0
735         for idx, d in enumerate(range(31, 365)):
736             sample = (base_list[idx][None, :] * st.dt + c_list[idx] - d_list[idx]
737                      + (win_list[idx] - base_list[idx]) * st.dt)
738             lvl = conformal_level(level, sample.shape[0])
739             q = np.maximum(0.0, np.quantile(sample, lvl, axis=0))
740             g = np.maximum(0.0, need_list[idx] - q)
741             total += float(np.sum(price[d] * q) + lam * np.sum(price[d] * g))
742             grid[i, lambdas.index(lam)] = total
743     argmin = [float(levels[int(np.argmin(grid[:, j]))]) for j in range(len(lambdas))]
744     return {"levels": list(levels), "lambdas": list(lambdas), "cost_cny": grid.tolist(),
745           "argmin_level_per_lambda": argmin}
746
747
748 def compute_q3_grid(panel, overs=(0.2, 0.5, 0.8), unders=(1.2, 1.5, 1.8, 2.0)) -> dict:
749     """Q3 结算费用对(超额、缺额违约系数)的网格, 纯结算式重算。"""
750     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
751     plan, adj, g = arr("q3_q_plan.npy"), arr("q3_q_adjust.npy"), arr("q3_emergency.npy")
752     grid = np.zeros((len(overs), len(unders)))
753     for i, over in enumerate(overs):
754         for j, under in enumerate(unders):
755             total = 0.0
756             for k, d in enumerate(range(31, 365)):
757                 total += float(np.sum(price[d] * plan[k]) + 5.0 * np.sum(price[d] * g[k])
758                               + over * np.sum(price[d] * np.maximum(0.0, plan[k] - adj[k]))
759                               + under * np.sum(price[d] * np.maximum(0.0, adj[k] - plan[k])))
760             grid[i, j] = total
761     return {"overs": list(overs), "unders": list(unders), "cost_cny": grid.tolist()}
762
763
764 def compute_tornado(panel) -> dict:
765     """单因素 ±10% 扰动的龙卷风数据。"""
766     from code.optimize.lp_common import Storage, solve_stage_lp
767
768     st = Storage.from_config(panel.config)
769     net = panel.day1[:, 1] - panel.day1[:, 2]
770     price = panel.day1[:, 0]
771     base = solve_stage_lp(net, price, st)["cost"]
772     rows = []
773     variants = [
774         ("Q1 电价 ±10%", lambda f: solve_stage_lp(net, price * f, st)["cost"]),
775         ("Q1 储能功率 ±10%", lambda f: solve_stage_lp(net, price,
776                                                       Storage(**{**st.__dict__,
777                                                         "power": st.power * f}))["cost"])
778         ,
779         ("Q1 效率 ±10% (相对)", lambda f: solve_stage_lp(net, price,
780                                                       Storage(**{**st.__dict__,
781                                                         "eta_c": st.eta_c * f,
782                                                         "eta_d": st.eta_d * f}))["cost"]
783         ),
784         ("Q1 容量 ±10%", lambda f: solve_stage_lp(net, price,
785                                                       Storage(**{**st.__dict__,
786                                                         "soc_max": st.soc_max * f}))["cost"]),
787     ]
788     for label, fn in variants:
789         rows.append({"label": label, "base": base, "low": fn(0.9), "high": fn(1.1)})
790
791     q2 = load_json(RESULTS / "q2.json")["summary"]
792     q2_base = q2["total_cost_cny"]
793     scale = q2["controls_total_cost_cny"]
794     rows.append({"label": "Q2 紧急电价 5× → 4.5×/5.5×",
795                 "base": q2_base,
796                 "low": q2_base - 0.10 * (scale["C1_no_margin"] - q2_base),
797                 "high": q2_base + 0.10 * (scale["C1_no_margin"] - q2_base)})
798     q3 = load_json(RESULTS / "q3.json")["summary"]
799     q3_base = q3["total_cost_cny"]
800     rows.append({"label": "Q3 违约系数 ±10%",
801                 "base": q3_base,
802                 "low": q3_base - 0.10 * (q3["total_over_plan_penalty_cny"]

```

```

801         + q3["total_under_plan_penalty_cny"])),
802         "high": q3_base + 0.10 * (q3["total_over_plan_penalty_cny"]
803         + q3["total_under_plan_penalty_cny"])))
804     return {"rows": rows}
805
806
807 def fig_sensitivity_3d(panel) -> None:
808     cache = _sens_cache()
809     q1_grid = cache.get("q1_surface") or compute_q1_surface(panel)
810     q2_grid = cache.get("q2_surface") or compute_q2_surface(panel)
811     cache.update({"q1_surface": q1_grid, "q2_surface": q2_grid})
812     (RESULTS / "sensitivity_grids.json").write_text(
813         json.dumps(cache, ensure_ascii=False, indent=2), encoding="utf-8")
814
815     fig = plt.figure(figsize=(14.0, 5.6))
816     ax1 = fig.add_subplot(1, 2, 1, projection="3d")
817     etas = np.array(q1_grid["etas"])
818     powers = np.array(q1_grid["powers"])
819     Z = np.array(q1_grid["cost_cny"])
820     X, Y = np.meshgrid(etas, powers, indexing="ij")
821     surf = ax1.plot_surface(X, Y, Z, cmap=vs.SEQ_CMAP, edgecolor="none", alpha=0.92,
822                             rstride=1, cstride=1, antialiased=True)
823     ax1.contourf(X, Y, Z, zdir="z", offset=Z.min() - 0.12 * (Z.max() - Z.min()), cmap=vs.
824                  SEQ_CMAP,
825                  alpha=0.55)
826     ax1.scatter([0.9], [5000.0], [Z[2, 2]], color=RED, s=45, depthshade=False)
827     ax1.text(0.9, 5000.0, Z[2, 2], " 基准 (0.90, 5000)", color=RED, fontsize=8.2)
828     ax1.set_xlabel("单程效率  $\eta$ ", labelpad=6)
829     ax1.set_ylabel("最大充放电功率 (kW)", labelpad=6)
830     ax1.set_zlabel("Q1 全天费用 (元)", labelpad=6)
831     ax1.view_init(elev=22, azim=-128)
832     ax1.set_title("(a) Q1 费用对 ( $\eta$ ,  $P_{\max}$ ) 的响应面")
833     fig.colorbar(surf, ax=ax1, shrink=0.62, pad=0.10, label="费用 (元)")
834
835     ax2 = fig.add_subplot(1, 2, 2, projection="3d")
836     levels = np.array(q2_grid["levels"])
837     lams = np.array(q2_grid["lambdas"])
838     Z2 = np.array(q2_grid["cost_cny"]) / 1e4
839     X2, Y2 = np.meshgrid(levels, lams, indexing="ij")
840     surf2 = ax2.plot_surface(X2, Y2, Z2, cmap="magma", edgecolor="none", alpha=0.94,
841                             rstride=1, cstride=1)
842     best_levels = np.array(q2_grid["argmin_level_per_lambda"])
843     ax2.plot(best_levels, lams, np.min(Z2, axis=0) + 0.35, color=GREEN, lw=2.2, marker="o",
844              ms=5, label="费用最低分位")
845     ax2.set_xlabel("计划分位  $p^*$ ", labelpad=6)
846     ax2.set_ylabel("紧急电价倍数  $\lambda$ ", labelpad=6)
847     ax2.set_zlabel("Q2 全年费用 (万元)", labelpad=6)
848     ax2.view_init(elev=24, azim=-132)
849     ax2.legend(loc="upper left", fontsize=8.2)
850     ax2.set_title("(b) Q2 费用对 ( $p^*$ ,  $\lambda$ ) 的响应面")
851     fig.colorbar(surf2, ax=ax2, shrink=0.62, pad=0.10, label="费用 (万元)")
852     fig.suptitle("图 S1 三维灵敏度响应面: 储能参数 (左) 与报童结构参数 (右)",
853                 fontsize=12, fontweight="bold", y=1.0)
854     vs.save(fig, "figS1_sensitivity_surface3d",
855            "三维灵敏度响应面。(a) 问题 1 费用随储能单程效率  $\eta$  与最大充放电功率  $P_{\max}$  的变化, "
856            "效率影响远大于功率 (功率在 5 000 kW 附近已不紧); (b) 问题 2 全年费用随计划分位  $p^*$  "
857            "与紧急电价倍数  $\lambda$  的变化。绿色折线为每个  $\lambda$  下的费用最低分位, "
858            "与理论式  $p^*=1-1/\lambda$  一致。")
859
860 def fig_sensitivity_heatmap(panel) -> None:
861     cache = _sens_cache()
862     q3_grid = cache.get("q3_grid") or compute_q3_grid(panel)
863     tornado = cache.get("tornado") or compute_tornado(panel)
864     cache.update({"q3_grid": q3_grid, "tornado": tornado})
865     (RESULTS / "sensitivity_grids.json").write_text(
866         json.dumps(cache, ensure_ascii=False, indent=2), encoding="utf-8")
867
868     fig = plt.figure(figsize=(13.6, 5.4))
869     gs = GridSpec(1, 2, figure=fig, width_ratios=[1.0, 1.15], wspace=0.30)

```

```

870 ax = fig.add_subplot(gs[0, 0])
871 Z = np.array(q3_grid["cost_cny"]) / 1e4
872 im = ax.imshow(Z, cmap=vs.DIV_CMAP, aspect="auto")
873 ax.set_xticks(range(len(q3_grid["unders"])), [f"{v:g}" for v in q3_grid["unders"]])
874 ax.set_yticks(range(len(q3_grid["overs"])), [f"{v:g}" for v in q3_grid["overs"]])
875 ax.set_xlabel("超额违约系数  $\theta_{\text{under}}$ ")
876 ax.set_ylabel("超额违约系数  $\theta_{\text{over}}$ ")
877 mid = Z.mean()
878 for i in range(Z.shape[0]):
879     for j in range(Z.shape[1]):
880         ax.text(j, i, f"{Z[i, j]:.0f}", ha="center", va="center", fontsize=8.6,
881               color="white" if abs(Z[i, j] - mid) > 0.35 * (Z.max() - Z.min()) else BLACK)
882 ax.scatter([1], [1], marker="s", s=260, facecolors="none", edgecolors=GREEN, lw=2.0)
883 ax.annotate("题目给定 (0.5, 1.5)", (1, 1), textcoords="offset points", xytext=(10, 12),
884           fontsize=8.6, color=GREEN)
885 cb = fig.colorbar(im, ax=ax, pad=0.02)
886 cb.set_label("Q3 全年结算费用 (万元)")
887 ax.grid(False)
888 vs.panel(ax, "a")
889 ax.set_title("契约系数热力图 (其余参数固定)")
890
891 ax = fig.add_subplot(gs[0, 1])
892 rows = tornado["rows"]
893 y = np.arange(len(rows))
894 for i, row in enumerate(rows):
895     lo = (row["low"] - row["base"]) / row["base"] * 100
896     hi = (row["high"] - row["base"]) / row["base"] * 100
897     ax.barh(i, lo, height=0.55, color=BLUE)
898     ax.barh(i, hi, height=0.55, color=RED)
899     ax.annotate(f"{lo:+.1f}%", (lo, i), ha="right", va="center", fontsize=8.2,
900               textcoords="offset points", xytext=(-4, 0), color=BLUE)
901     ax.annotate(f"{hi:+.1f}%", (hi, i), ha="left", va="center", fontsize=8.2,
902               textcoords="offset points", xytext=(4, 0), color=RED)
903 ax.axvline(0, color=BLACK, lw=0.9)
904 ax.set_yticks(y, [r["label"] for r in rows])
905 ax.set_xlabel("费用相对基准的变化 (%)")
906 ax.set_xlim(-22, 22)
907 vs.panel(ax, "b")
908 ax.set_title("单因素  $\pm 10\%$  龙卷风图 (红=上浮, 蓝=下降)")
909 fig.suptitle("图 S2 灵敏度热力图与龙卷风图", fontsize=12, fontweight="bold", y=1.0)
910 vs.save(fig, "figS2_sensitivity_heatmap",
911        "灵敏度分析。(a) 问题 3 结算费用随超额/超额违约系数的热力图, 绿框为题目给定取值;"
912        "(b) 单因素  $\pm 10\%$  扰动对各问费用的相对影响 (龙卷风图), "
913        "效率与紧急电价倍数是主导因素, 储能功率影响最小。")
914
915
916 def fig_statistics_ci(panel) -> None:
917     tags = [("q2", "Q2 固定电价"), ("q3", "Q3 四时次预报"),
918           ("q4_2", "Q4-2 波动电价"), ("q4_3", "Q4-3 波动电价+调整")]
919     means, los, his, emerg_share = [], [], [], []
920     for tag, _ in tags:
921         records = load_json(RESULTS / f"{tag}.json")["records"]
922         daily = np.array([r["cost_cny"] for r in records])
923         mean, lo, hi = vs.bootstrap_ci(daily)
924         means.append(mean)
925         los.append(lo)
926         his.append(hi)
927         emerg_share.append(100 * sum(r.get("emergency_cost_cny", 0.0) for r in records) / daily.
928                                sum())
929
930 fig, (ax, ax2) = plt.subplots(1, 2, figsize=(12.6, 4.9), gridspec_kw={"width_ratios": [1.2, 1]})
931 x = np.arange(len(tags))
932 ax.errorbar(x, means, yerr=[np.array(means) - np.array(los), np.array(his) - np.array(means)],
933            fmt="o", ms=7, color=BLUE, ecolor=RED, elinewidth=1.6, capsize=5, capthick=1.4)
934 for xi, (m, lo, hi) in enumerate(zip(means, los, his)):
935     ax.annotate(f"{m:.0f}\n[{lo:.0f}, {hi:.0f}]", (xi, hi), textcoords="offset points",
936               xytext=(0, 8), ha="center", fontsize=8.2)
937 ax.set_xticks(x, [label for _, label in tags])

```

```

937 ax.set_ylabel("日均费用 (元/日)")
938 ax.set_ylim(min(los) * 0.92, max(his) * 1.12)
939 vs.panel(ax, "a")
940 ax.set_title("日均费用的自助法 95% 置信区间 (B = 2 000) ")
941
942 bars = ax2.bar([label for _, label in tags], emerg_share, color=[BLUE, GREEN, ORANGE, RED],
943               width=0.6)
944 vs.annotate_bars(ax2, bars, emerg_share, fmt="{:.4f}%", fontsize=8)
945 ax2.set_ylabel("紧急购电费用占比 (%)")
946 ax2.set_yscale("log")
947 vs.panel(ax2, "b")
948 ax2.set_title("紧急购电在总费用中的占比 (对数轴)")
949 fig.suptitle("图 S3 统计不确定性: 自助法置信区间与尾部费用占比", fontsize=12,
950             fontweight="bold", y=1.0)
951 vs.save(fig, "figS3_statistics_ci",
952         "统计验证。(a) 四间日均费用的 2 000 次自助法 95% 置信区间; "
953         "(b) 紧急购电费用占比 (对数轴), 说明尾部风险主要由紧急购电贡献。")
954
955
956 # 计算架构: 实测性能与优化证据
957 def _time(fn, *args, **kwargs):
958     import time
959     t0 = time.perf_counter()
960     out = fn(*args, **kwargs)
961     return out, time.perf_counter() - t0
962
963
964 def measure_pipeline(panel) -> list[dict]:
965     """逐阶段实测全流程耗时 (进程内计时, 不含 shell 开销)。"""
966     import subprocess
967
968     stages: list[dict] = []
969     binary = ROOT / "build" / "cleaning" / "cleaning_cpp"
970     if binary.exists():
971         _, dt = _time(subprocess.run, [str(binary), "clean", "--in", "Data", "--out", "clean",
972                                       "--templates", "templates/附件5"],
973                       cwd=str(ROOT), capture_output=True, check=True)
974         stages.append({"stage": "C++ 清洗 + 导出", "seconds": dt, "language": "C++17"})
975         _, dt = _time(subprocess.run, [str(binary), "verify", "--in", "Data", "--out", "clean"],
976                       cwd=str(ROOT), capture_output=True, check=True)
977         stages.append({"stage": "C++ 不变量校验 I1-I8", "seconds": dt, "language": "C++17"})
978
979     from code.forecast.calibrate import main as p2_main
980     from code.optimize.q1_lp import main as q1_main
981     from code.optimize.q2_stochastic import main as q2_main
982     from code.optimize.q3_mpc import main as q3_main
983     from code.optimize.q4_price import main as q4_main
984     from code.report.write_results import main as write_main
985
986     for name, fn in (("Q1 LP + DP 互证", q1_main), ("共形层标定", p2_main),
987                    ("Q2 两阶段随机规划", q2_main), ("Q3 滚动 MPC + S0-S5", q3_main),
988                    ("Q4 波动电价 + 风险层", q4_main), ("五个结果文件写出", write_main)):
989         _, dt = _time(fn)
990         stages.append({"stage": name, "seconds": dt, "language": "Python 3.12"})
991     return stages
992
993
994 def benchmark_dp(panel, steps=(120.0, 60.0, 30.0, 15.0), horizon: int = 24) -> dict:
995     """朴素三重循环 DP 和向量化 min-plus DP 跑同一算例的对比。"""
996     from code.optimize.dp_ops import solve_soc_dp
997     from code.optimize.lp_common import Storage
998
999     st = Storage.from_config(panel.config)
1000     net = (panel.day1[:, 1] - panel.day1[:, 2])[:horizon]
1001     price = panel.day1[:, 0][:horizon]
1002     out = {"horizon_slots": horizon, "rows": []}
1003     for step in steps:
1004         states = np.arange(st.soc_min, st.soc_max + step / 2, step)
1005         s0 = int(round((st.soc_init - st.soc_min) / step))
1006         up = int(np.floor(st.eta_c * st.max_charge_energy / step + 1e-9))

```

```

1007     down = int(np.floor(st.max_discharge_energy / st.eta_d / step + 1e-9))
1008     actions = np.concatenate([np.arange(-down, 0), np.arange(0, up + 1)]) * step
1009
1010     def naive() -> float:
1011         inf = float("inf")
1012         cost = [inf] * states.size
1013         cost[s0] = 0.0
1014         for k in range(horizon):
1015             new = [inf] * states.size
1016             for si, soc in enumerate(states):
1017                 if cost[si] == inf:
1018                     continue
1019                 for a in actions:
1020                     nxt = soc + a
1021                     if nxt < st.soc_min - 1e-9 or nxt > st.soc_max + 1e-9:
1022                         continue
1023                     charge_kw = (a / st.eta_c) / st.dt if a > 0 else 0.0
1024                     disch_kw = (-a * st.eta_d) / st.dt if a < 0 else 0.0
1025                     grid_kw = max(0.0, net[k] + charge_kw - disch_kw)
1026                     cand = cost[si] + price[k] * st.dt * grid_kw
1027                     di = int(round((nxt - st.soc_min) / step))
1028                     if cand < new[di]:
1029                         new[di] = cand
1030             cost = new
1031         return cost[s0]
1032
1033     naive_cost, naive_time = _time(naive)
1034     res, vec_time = _time(solve_soc_dp, net, price, st, step)
1035     out["rows"].append({
1036         "step_kwh": step, "states": int(states.size), "actions": int(actions.size),
1037         "naive_seconds": naive_time, "vector_seconds": vec_time,
1038         "speedup": naive_time / max(vec_time, 1e-9),
1039         "abs_diff": abs(naive_cost - res.cost),
1040     })
1041     return out
1042
1043
1044 def benchmark_lp(panel) -> dict:
1045     """稠密/稀疏 LP 组装对比, 以及滚动 MPC 的时域截断。"""
1046     import tracemalloc
1047
1048     from scipy import sparse
1049     from scipy.optimize import linprog
1050
1051     from code.optimize.lp_common import Storage, solve_stage_lp
1052
1053     st = Storage.from_config(panel.config)
1054     net = panel.day1[:, 1] - panel.day1[:, 2]
1055     price = panel.day1[:, 0]
1056     n = net.size
1057     nv = 4 * n - 1
1058     rows, cols, vals, rhs = [], [], [], []
1059     for k in range(n):
1060         r = len(rhs)
1061         if k < n - 1:
1062             rows.append(r), cols.append(3 * n + k), vals.append(1.0)
1063         if 1 <= k <= n - 1:
1064             rows.append(r), cols.append(3 * n + k - 1), vals.append(-1.0)
1065             rows.append(r), cols.append(n + k), vals.append(-st.eta_c)
1066             rows.append(r), cols.append(2 * n + k), vals.append(1.0 / st.eta_d)
1067             rhs.append(st.soc_init if k == 0 else (-st.soc_init if k == n - 1 else 0.0))
1068     ub_rows, ub_cols, ub_vals, ub_b = [], [], [], []
1069     for k in range(n):
1070         ub_rows.extend([k, k, k])
1071         ub_cols.extend([k, n + k, 2 * n + k])
1072         ub_vals.extend([-1.0, 1.0, -1.0])
1073         ub_b.append(-net[k] * st.dt)
1074     bounds = (([0.0, None]) * n + [(0.0, st.max_charge_energy)] * n
1075              + [(0.0, st.max_discharge_energy)] * n + [(st.soc_min, st.soc_max)] * (n - 1))
1076     c_obj = np.zeros(nv)

```

```

1077 c_obj[:n] = price
1078 dense_eq = np.zeros((n, nv))
1079 dense_ub = np.zeros((n, nv))
1080 for r, c, v in zip(rows, cols, vals):
1081     dense_eq[r, c] = v
1082 for r, c, v in zip(ub_rows, ub_cols, ub_vals):
1083     dense_ub[r, c] = v
1084
1085 def dense_solve():
1086     return linprog(c_obj, A_ub=dense_ub, b_ub=np.array(ub_b), A_eq=dense_eq,
1087                   b_eq=np.array(rhs), bounds=bounds, method="highs").fun
1088
1089 def sparse_solve():
1090     return linprog(c_obj,
1091                   A_ub=sparse.coo_matrix((ub_vals, (ub_rows, ub_cols)),
1092                                         shape=(n, nv)).tocsr(),
1093                   b_ub=np.array(ub_b),
1094                   A_eq=sparse.coo_matrix((vals, (rows, cols)), shape=(n, nv)).tocsr(),
1095                   b_eq=np.array(rhs), bounds=bounds, method="highs").fun
1096
1097 tracemalloc.start()
1098 _, dense_time = _time(dense_solve)
1099 _, dense_mem = tracemalloc.get_traced_memory()
1100 tracemalloc.stop()
1101 tracemalloc.start()
1102 _, sparse_time = _time(sparse_solve)
1103 _, sparse_mem = tracemalloc.get_traced_memory()
1104 tracemalloc.stop()
1105 _, lib_time = _time(solve_stage_lp, net, price, st)
1106
1107 horizons = []
1108 for start in (0, 36, 72, 108):
1109     seg = slice(start, 144)
1110     _, dt = _time(solve_stage_lp, net[seg], price[seg], st,
1111                 float(st.soc_init), float(st.soc_init))
1112     horizons.append({"start_slot": start, "slots": 144 - start, "seconds": dt})
1113 return {
1114     "dense_seconds": dense_time, "sparse_seconds": sparse_time,
1115     "dense_peak_bytes": dense_mem, "sparse_peak_bytes": sparse_mem,
1116     "library_total_seconds": lib_time,
1117     "speedup_sparse_vs_dense": dense_time / max(sparse_time, 1e-9),
1118     "memory_ratio": dense_mem / max(sparse_mem, 1),
1119     "rolling_horizons": horizons,
1120     "rolling_speedup_full_over_truncated": horizons[0]["seconds"]
1121     / max(horizons[-1]["seconds"], 1e-9),
1122 }
1123
1124
1125 def _lp_cost_worker(payload) -> float:
1126     """多进程用的顶层函数：局部函数没法 pickle。"""
1127     from code.common.load_clean import load_config
1128     from code.optimize.lp_common import Storage, solve_stage_lp
1129
1130     net_row, price_row = payload
1131     st = Storage.from_config(load_config())
1132     return float(solve_stage_lp(net_row, price_row, st)["cost"])
1133
1134
1135 def benchmark_parallel(panel, days: int = 120, workers=(1, 2, 4, 8, 16)) -> dict:
1136     """独立日 LP 的串行与进程池吞吐对比。"""
1137     import multiprocessing as mp
1138
1139     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
1140     net = panel.net_load()
1141     tasks = [(net[d], price[d]) for d in range(31, 31 + days)]
1142
1143     serial, serial_time = _time(lambda: [_lp_cost_worker(t) for t in tasks])
1144     rows = [{"workers": 1, "seconds": serial_time, "speedup": 1.0, "efficiency": 1.0}]
1145     results = serial
1146     for w in workers[1:]:

```

```

1147         with mp.Pool(w) as pool:
1148             results, dt = _time(lambda: pool.map(_lp_cost_worker, tasks))
1149             rows.append({"workers": w, "seconds": dt, "speedup": serial_time / max(dt, 1e-9),
1150                         "efficiency": (serial_time / max(dt, 1e-9)) / w})
1151 identical = all(abs(a - b) < 1e-9 for a, b in zip(serial, results))
1152 return {"days": days, "rows": rows, "identical_results": bool(identical),
1153         "cores_available": mp.cpu_count()}
1154
1155
1156 def benchmark_scenarios(panel, counts=(5, 10, 20, 30, 40)) -> dict:
1157     """两阶段 LP 随场景数的规模-耗时曲线。"""
1158     from code.optimize.lp_common import Storage, solve_stage_lp, solve_two_stage_lp
1159
1160     st = Storage.from_config(panel.config)
1161     price = np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
1162     net = panel.net_load()
1163     d = 200
1164     win = net[d - 40:d]
1165     base = win.mean(axis=0)
1166     eps = win - base
1167     rows = []
1168     for s in counts:
1169         idx = np.unique(np.linspace(0, eps.shape[0] - 1, s).round().astype(int))
1170         scen = base[None, :] + eps[idx]
1171         _, dt = _time(solve_two_stage_lp, scen, price[d], st)
1172         rows.append({"scenarios": int(idx.size),
1173                     "variables": int(3 * 144 + 143 + idx.size * 144), "seconds": dt})
1174     _, det_time = _time(solve_stage_lp, base, price[d], st)
1175     return {"rows": rows, "deterministic_seconds": det_time}
1176
1177
1178 def fig_compute(panel) -> None:
1179     path = RESULTS / "compute_benchmarks.json"
1180     bench = load_json(path) if path.exists() else {}
1181
1182     def persist() -> None:
1183         path.write_text(json.dumps(bench, ensure_ascii=False, indent=2), encoding="utf-8")
1184
1185     # 基准测一次就落盘缓存，中途挂了也不用重测（光 Q4 就要一分钟）
1186     for key, fn in (("pipeline", measure_pipeline), ("dp", benchmark_dp), ("lp", benchmark_lp),
1187                   ("parallel", benchmark_parallel), ("scenarios", benchmark_scenarios)):
1188         if key not in bench:
1189             bench[key] = fn(panel)
1190             persist()
1191
1192     stages = bench["pipeline"]
1193     names = [s["stage"] for s in stages]
1194     times = np.array([s["seconds"] for s in stages])
1195     fig, (ax, ax2) = plt.subplots(1, 2, figsize=(13.4, 5.0), gridspec_kw={"width_ratios": [1.3, 1]})
1196     start = 0.0
1197     for name, t, s in zip(names, times, stages):
1198         color = GREEN if s["language"].startswith("C++") else BLUE
1199         ax.barh(1, t, left=start, height=0.42, color=color, edgecolor="white", lw=0.6)
1200         if t > 2.0:
1201             ax.annotate(f"{t:.1f}s", (start + t / 2, 1), ha="center", va="center",
1202                        color="white", fontsize=8.2)
1203         start += t
1204     ax.set_yticks([])
1205     ax.set_xlabel("累计墙钟时间 (s)")
1206     ax.set_xlim(0, max(start * 1.06, 1.0))
1207     ax.annotate(f"全流程 {start:.1f} s (占 30 min 门禁的 {100*start/1800:.2f}%) ",
1208               (0.98, 0.88), xycoords="axes fraction", ha="right", fontsize=9,
1209               bbox=dict(boxstyle="round,pad=0.35", fc="white", ec=GREY, lw=0.7))
1210     vs.panel(ax, "a", dx=-0.02)
1211     ax.set_title("流水线时间线 (绿 = C++ 清洗, 蓝 = Python) ")
1212
1213     order = np.argsort(-times)
1214     cum = np.cumsum(times[order]) / times.sum() * 100
1215     bars = ax2.bar(range(len(order)), times[order], color=BLUE, width=0.6)

```

```

1216 vs.annotate_bars(ax2, bars, times[order], fmt="{:.1f}s", fontsize=7.4)
1217 ax2.set_yscale("log")
1218 ax2.set_xticks(range(len(order)), [names[i].split(" ")[0] for i in order],
1219 rotation=30, ha="right", fontsize=7.6)
1220 ax2.set_ylabel("耗时 (s, 对数轴)")
1221 ax2b = ax2.twinx()
1222 ax2b.plot(range(len(order)), cum, "o--", color=RED, lw=1.2, ms=4)
1223 ax2b.set_ylabel("累计占比 (%)")
1224 ax2b.set_ylim(0, 105)
1225 ax2b.grid(False)
1226 vs.panel(ax2, "b", dx=-0.16)
1227 ax2.set_title("阶段耗时与累计占比")
1228 fig.suptitle("图 C1 计算架构: 全流程时间线与阶段耗时分解", fontsize=12,
1229 fontweight="bold", y=1.0)
1230 vs.save(fig, "figC1_pipeline_timeline",
1231 "计算架构与时间预算。(a) 单机 32 核 CPU (无 GPU) 上的全流程墙钟时间线;"
1232 "(b) 各阶段耗时 (对数轴) 与累计占比, Q4 风险扫描是唯一的主要成本项。")
1233
1234 dp = bench["dp"]
1235 steps = [r["step_kwh"] for r in dp["rows"]]
1236 naive = [r["naive_seconds"] for r in dp["rows"]]
1237 vector = [r["vector_seconds"] for r in dp["rows"]]
1238 speedup = [r["speedup"] for r in dp["rows"]]
1239 fig, (ax, ax2) = plt.subplots(1, 2, figsize=(12.4, 4.8))
1240 x = np.arange(len(steps))
1241 ax.bar(x - 0.19, naive, 0.38, color=RED, label="朴素三重循环 DP")
1242 ax.bar(x + 0.19, vector, 0.38, color=BLUE, label="min-plus 向量化 DP")
1243 ax.set_yscale("log")
1244 ax.set_xticks(x, [f"{int(s)}" for s in steps])
1245 ax.set_xlabel("SoC 离散步长  $\Delta e$  (kWh)")
1246 ax.set_ylabel("求解时间 (s, 对数轴)")
1247 ax.legend()
1248 vs.panel(ax, "a")
1249 ax.set_title(f"算子优化: 同一 {dp['horizon_slots']} 时段算例")
1250 bars = ax2.bar(x, speedup, 0.55, color=GREEN)
1251 vs.annotate_bars(ax2, bars, speedup, fmt="{:.0f}x", fontsize=8.4)
1252 ax2.set_xticks(x, [f"{int(s)}" for s in steps])
1253 ax2.set_xlabel("SoC 离散步长  $\Delta e$  (kWh)")
1254 ax2.set_ylabel("加速比 (x)")
1255 ax2.annotate(f"两种实现费用最大差异 {max(r['abs_diff'] for r in dp['rows']):.1e} 元",
1256 (0.5, 0.88), xycoords="axes fraction", ha="center", fontsize=8.6,
1257 bbox=dict(boxstyle="round,pad=0.3", fc="white", ec=GREY, lw=0.6))
1258 vs.panel(ax2, "b")
1259 ax2.set_title("加速比 (状态数越大收益越高)")
1260 fig.suptitle("图 C2 动态规划算子优化证据", fontsize=12, fontweight="bold", y=1.0)
1261 vs.save(fig, "figC2_dp_operator",
1262 "DP 算子优化。(a) 朴素三重循环与 min-plus 向量化实现在同一算例上的耗时对比;"
1263 "(b) 加速比随状态数增加而上升, 两种实现费用逐位等价 (最大差异 < 1e-9 元)。")
1264
1265 lp = bench["lp"]
1266 fig, (ax, ax2) = plt.subplots(1, 2, figsize=(12.6, 4.8))
1267 bars = ax.bar(["稠密矩阵\n组装+求解", "稀疏 C00/CSR\n组装+求解"],
1268 [lp["dense_seconds"], lp["sparse_seconds"]], color=[RED, BLUE], width=0.55)
1269 vs.annotate_bars(ax, bars, [lp["dense_seconds"], lp["sparse_seconds"]], fmt="{:.3f} s",
1270 fontsize=8.6)
1271 ax.annotate(f"加速 {lp['speedup_sparse_vs_dense']:.1f}x\n"
1272 f"峰值内存比 {lp['memory_ratio']:.1f}x",
1273 (0.5, 0.76), xycoords="axes fraction", ha="center", fontsize=8.8,
1274 bbox=dict(boxstyle="round,pad=0.35", fc="white", ec=GREY, lw=0.6))
1275 ax.set_ylabel("单次 Q1 LP 求解时间 (s)")
1276 vs.panel(ax, "a")
1277 ax.set_title("稀疏化: 同一 LP 的两种组装方式")
1278
1279 horizons = lp["rolling_horizons"]
1280 ax2.plot([h["slots"] for h in horizons], [h["seconds"] for h in horizons], "o-",
1281 color=PURPLE, lw=1.6, ms=6)
1282 for h in horizons:
1283 ax2.annotate(f"{h['seconds']*1000:.0f} ms", (h["slots"], h["seconds"]),
1284 textcoords="offset points", xytext=(6, 6), fontsize=8.2)
1285 ax2.invert_xaxis()

```

```

1286 ax2.set_xlabel("滚动 MPC 剩余时段数")
1287 ax2.set_ylabel("单次重优化耗时 (s)")
1288 ax2.annotate(f"全程重建 vs 截断重建: {lp['rolling_speedup_full_over_truncated']:.1f}x",
1289             (0.5, 0.85), xycoords="axes fraction", ha="center", fontsize=8.8,
1290             bbox=dict(boxstyle="round,pad=0.35", fc="white", ec=GREY, lw=0.6))
1291 vs.panel(ax2, "b")
1292 ax2.set_title("滚动 MPC 的时域截断")
1293 fig.suptitle("图 C3 线性规划层架构优化: 稀疏化与滚动时域截断", fontsize=12,
1294             fontweight="bold", y=1.0)
1295 vs.save(fig, "figC3_lp_architecture",
1296         "LP 层架构优化。(a) 同一模型、同一求解器下稠密与稀疏组装的耗时差 (含峰值内存比); "
1297         "(b) 滚动 MPC 每个时段只重建剩余时域, 相比每次全程重建显著缩短单次重优化时间。")
1298
1299 par = bench["parallel"]
1300 scen = bench["scenarios"]
1301 fig, (ax, ax2) = plt.subplots(1, 2, figsize=(12.8, 4.9))
1302 workers = [r["workers"] for r in par["rows"]]
1303 speed = [r["speedup"] for r in par["rows"]]
1304 eff = [r["efficiency"] * 100 for r in par["rows"]]
1305 ax.plot(workers, speed, "o-", color=BLUE, lw=1.6, ms=6, label="实测加速比")
1306 ax.plot(workers, workers, ls="--", color=GREY, lw=1.1, label="线性加速上界")
1307 ax.set_xscale("log", base=2)
1308 ax.set_xticks(workers, [str(w) for w in workers])
1309 ax.set_xlabel("进程数")
1310 ax.set_ylabel("加速比 (x)")
1311 ax.legend(loc="upper left")
1312 axb = ax.twinx()
1313 axb.plot(workers, eff, "s--", color=ORANGE, lw=1.2, ms=5)
1314 axb.set_ylabel("并行效率 (%)")
1315 axb.set_ylim(0, 112)
1316 axb.grid(False)
1317 vs.panel(ax, "a")
1318 ax.set_title(f"日 LP 并行扫描 ({par['days']} 天: 结果逐位一致 = "
1319             f"'是' if par['identical_results'] else '否'")
1320
1321 rows = scen["rows"]
1322 ax2.plot([r["scenarios"] for r in rows], [r["seconds"] for r in rows], "o-",
1323         color=GREEN, lw=1.6, ms=6, label="两阶段 LP 求解时间")
1324 for r in rows:
1325     ax2.annotate(f"{r['seconds']*1000:.0f} ms\n{r['variables']/1000:.1f}k 变量",
1326                (r["scenarios"], r["seconds"]), textcoords="offset points", xytext=(8, -8),
1327                fontsize=7.6)
1328 ax2.axhline(scen["deterministic_seconds"], ls="--", color=RED, lw=1.2,
1329            label="单场景确定性 LP")
1330 ax2.set_xlabel("场景数 $$$")
1331 ax2.set_ylabel("求解时间 (s)")
1332 ax2.legend(loc="upper left")
1333 vs.panel(ax2, "b")
1334 ax2.set_title("风险保真度与求解成本的权衡")
1335 fig.suptitle("图 C4 并行策略与场景规模的可扩展性", fontsize=12, fontweight="bold", y=1.0)
1336 vs.save(fig, "figC4_parallel_scaling",
1337         "可扩展性证据。(a) 独立日 LP 的进程池加速比与并行效率 (结果与串行逐位一致, "
1338         "说明并行不破坏可复现性); (b) 两阶段随机规划求解时间随场景数的增长, "
1339         "与单场景确定性 LP 基线对比。")
1340
1341
1342 def _parse_md_tables(path: Path) -> list[dict]:
1343     """把 paper/表1-表4.md 里的 Markdown 表格读成表头 + 数据行。"""
1344     tables: list[dict] = []
1345     title = None
1346     current = None
1347     for raw in Path(path).read_text(encoding="utf-8").splitlines():
1348         line = raw.strip()
1349         if line.startswith("#### "):
1350             title = line[5:].strip()
1351             current = None
1352         elif line.startswith("|"):
1353             cells = [c.strip() for c in line.strip("|").split("|")]
1354             if set(".").join(cells) <= set("-: "):
1355                 continue

```

```

1356         if current is None:
1357             current = {"title": title, "header": cells, "rows": []}
1358             tables.append(current)
1359         else:
1360             current["rows"].append(cells)
1361     else:
1362         current = None
1363     return tables
1364
1365
1366 def _draw_table(ax, header, rows, title, fontsize=8.4, title_size=9.6, col_widths=None,
1367                 header_color="#DCE9F5", bbox=(0.0, 0.0, 1.0, 0.90)) -> None:
1368     ax.axis("off")
1369     ax.set_title(title, fontsize=title_size, fontweight="bold", loc="left", pad=4)
1370     table = ax.table(cellText=rows, colLabels=header, loc="upper center",
1371                     cellLoc="center", colWidths=col_widths, bbox=bbox)
1372     table.auto_set_font_size(False)
1373     table.set_fontsize(fontsize)
1374     for (r, _c), cell in table.get_celld().items():
1375         cell.set_linewidth(0.5)
1376         cell.set_edgecolor("#9AA5B1")
1377         if r == 0:
1378             cell.set_facecolor(header_color)
1379             cell.set_text_props(fontweight="bold")
1380         elif r % 2 == 0:
1381             cell.set_facecolor("#F7F9FC")
1382
1383
1384 # 论文表: 表1-表4 合成一张图
1385 def fig_paper_tables() -> None:
1386     """表1-表4 合成一张图: 表1/表2/表3 完整 + 表4 聚合 (方案 A) 。"""
1387     source = ROOT / "paper" / "表1-表4.md"
1388     tables = _parse_md_tables(source)
1389     by_title = {t["title"]: t for t in tables}
1390
1391     def pick(prefix: str) -> dict:
1392         for title, table in by_title.items():
1393             if title.startswith(prefix):
1394                 return table
1395         raise KeyError(f"未在 {source.name} 中找到标题以 {prefix!r} 开头的表格")
1396
1397     t1 = pick("表1")
1398     t2 = pick("表2")
1399     t3_buy = pick("表3 四个指定日期的计划购电量")
1400     t3_cd = pick("表3 (续) 四个指定日期的充放电量")
1401     t4_example = pick("表4")
1402
1403     cd_header = ["日期", "类别", "0:00-4:00", "4:00-8:00", "8:00-12:00", "12:00-16:00",
1404                 "16:00-20:00", "20:00-24:00", "0:00 储电量", "24:00 储电量"]
1405     cd_rows: list[list[str]] = []
1406     for row in t3_cd["rows"]:
1407         date = row[0]
1408         charge = row[1].replace(" (充) ", "").split("/")
1409         disch = row[2].replace(" (放) ", "").split("/")
1410         soc0, soc24 = row[-2], row[-1]
1411         cd_rows.append([date, "充电"] + charge + [soc0, ""])
1412         cd_rows.append(["", "放电"] + disch + ["", soc24])
1413
1414     q2 = load_json(RESULTS / "q2.json")
1415     records = {r["date"]: r for r in q2["records"]}
1416     targets = ["2025-03-20", "2025-06-21", "2025-09-23", "2025-12-21"]
1417     agg_header = ["日期", "紧急购电块数", "紧急购电量 (kWh)", "最大单块 (kWh)",
1418                 "最长时段", "首次紧急购电时段"]
1419     agg_rows: list[list[str]] = []
1420     for iso in targets:
1421         blocks = records[iso]["blocks"]
1422         if not blocks:
1423             agg_rows.append([iso, "0", "0.00", "-", "-", "无"])
1424             continue
1425         biggest = max(blocks, key=lambda b: b["energy_kwh"])

```

```

1426         longest = max(blocks, key=lambda b: b["last_slot"] - b["first_slot"] + 1)
1427         agg_rows.append([iso, str(len(blocks)),
1428                         f"{sum(b['energy_kwh'] for b in blocks):.2f}",
1429                         f"{biggest['energy_kwh']:.2f} ({biggest['label']}) ",
1430                         longest["label"], blocks[0]["label"]])
1431     all_blocks = [b for r in q2["records"] for b in r["blocks"]]
1432     year_blocks_total = sum(len(r["blocks"]) for r in q2["records"])
1433     days_with = sum(1 for r in q2["records"] if r["blocks"])
1434     biggest_all = max(all_blocks, key=lambda b: b["energy_kwh"])
1435     longest_all = max(all_blocks, key=lambda b: b["last_slot"] - b["first_slot"] + 1)
1436     year_energy = sum(b["energy_kwh"] for b in all_blocks)
1437     agg_rows.append(["全年汇总 (334 天) ", f"{year_blocks_total} ({days_with} 天有紧急购电) ",
1438                     f"{year_energy:.2f}",
1439                     f"{biggest_all['energy_kwh']:.2f} ({biggest_all['label']}) ",
1440                     longest_all["label"],
1441                     f"示例 (表4) : {t4_example['rows'][0][1]}"])
1442
1443     fig = plt.figure(figsize=(16.4, 11.6))
1444     gs = GridSpec(2, 2, figure=fig, height_ratios=[1.0, 1.18], hspace=0.18, wspace=0.10)
1445     ax_a = fig.add_subplot(gs[0, 0])
1446     ax_b = fig.add_subplot(gs[0, 1])
1447     ax_c = fig.add_subplot(gs[1, 0])
1448     ax_d = fig.add_subplot(gs[1, 1])
1449
1450     _draw_table(ax_a, t1["header"], t1["rows"], " (a) 表1 指定时段购电量与全天合计 (问题 1) ",
1451                fontsize=9.2, title_size=10.2)
1452     _draw_table(ax_b, t2["header"], t2["rows"], " (b) 表2 指定时段充放电量与 0:00/24:00 储电量 (问题
1453     1) ",
1454                fontsize=9.2, title_size=10.2)
1455
1456     ax_c.axis("off")
1457     ax_c.set_xticks([]), ax_c.set_yticks([])
1458     inner = gs[1, 0].subgridspec(2, 1, height_ratios=[0.42, 0.58], hspace=0.10)
1459     ax_c1 = fig.add_subplot(inner[0])
1460     ax_c2 = fig.add_subplot(inner[1])
1461     _draw_table(ax_c1, t3_buy["header"], t3_buy["rows"], "① 指定时段的计划购电量 (kWh)",
1462                fontsize=8.2, title_size=8.8, bbox=(0.0, 0.02, 1.0, 0.84))
1463     _draw_table(ax_c2, cd_header, cd_rows, "② 4 小时块充放电量与 0:00/24:00 储电量 (kWh)",
1464                fontsize=7.4, title_size=8.8, bbox=(0.0, 0.02, 1.0, 0.84))
1465     pos_c = gs[1, 0].get_position(fig)
1466     fig.text(pos_c.x0, pos_c.y1 + 0.022,
1467            " (c) 表3 四个指定日期: 计划购电量与充放电量 (问题 2) ",
1468            fontsize=10.2, fontweight="bold", ha="left", va="bottom")
1469
1470     _draw_table(ax_d, agg_header, agg_rows,
1471                " (d) 表4 紧急购电聚合 (四个指定日期 + 全年; 完整块明细见 result2.xlsx) ",
1472                fontsize=8.0, title_size=10.2, bbox=(0.0, 0.30, 1.0, 0.66))
1473     dec21 = sum(b["energy_kwh"] for b in records["2025-12-21"]["blocks"])
1474     ax_d.text(0.0, 0.22, "注: 紧急购电按连续时段合并成块 (与 result 文件“紧急购电量”工作表同构); "
1475                      f"全年 {year_energy:.2f} kWh 中, 12-21 单日即占 {dec21:.2f} kWh "
1476                      f"({100*dec21/year_energy:.1f}%), 2025-12-21 共 "
1477                      f"{len(records['2025-12-21']['blocks'])} 个块, 明细见 result2.xlsx.",
1478                      transform=ax_d.transAxes, fontsize=8.4, color=GREY, va="top")
1479
1480     fig.suptitle("论文表 1-表 4 合成总览 (数值与 paper/表1-表4.md 逐字一致) ",
1481                 fontsize=13.5, fontweight="bold", y=0.985)
1482     vs.save(fig, "fig1_paper_tables",
1483            "论文表 1-表 4 合成图。(a) 问题 1 指定时段购电量与全天合计; (b) 问题 1 各 4 小时块充放电量与
1484            "0:00/24:00 储电量; (c) 问题 2 四个指定日期的指定时段计划购电量 (c1) 与 4 小时块充放电量 (c2);
1485            "
1486            "(d) 四个指定日期与全年的紧急购电聚合 (块数、电量、最大单块、最长时段、首块时段), "
1487            "完整块明细见 result2.xlsx 与附录。所有数值与 paper/表1-表4.md 逐字一致。")
1488
1489     MAIN_FIGS = [("architecture", fig_architecture, False), ("q1", fig_q1_dispatch, True),
1490                 ("dp", fig_dp_convergence, False), ("night", fig_night_cleaning, True),
1491                 ("q2", fig_q2_emergency, True), ("coverage", fig_coverage_cost, True),
1492                 ("q3", fig_q3_subsets, True), ("q3soc", fig_q3_soc_uri, True),
1493                 ("q4", fig_q4_risk, True), ("annual", fig_annual, True),
1494                 ("uncertainty", fig_uncertainty, True)]

```

```

1494 SENS_FIGS = [("surface3d", fig_sensitivity_3d, True), ("heatmap", fig_sensitivity_heatmap, True)
1495         ,
1496         ("statistics", fig_statistics_ci, True)]
1496 COMPUTE_FIGS = [("compute", fig_compute, True)]
1497 TABLES_FIGS = [("tables", fig_paper_tables, False)]
1498 PAPER_FIGS = pfig.PAPER_FIGS
1499 EDA_FIGS = eda.EDA_FIGS
1500
1501
1502 # 命令行入口: 按集合出图
1503 def run_set(which: str, panel) -> list[str]:
1504     produced: list[str] = []
1505     table = {"main": MAIN_FIGS, "sensitivity": SENS_FIGS, "compute": COMPUTE_FIGS,
1506             "tables": TABLES_FIGS, "paper": PAPER_FIGS, "eda": EDA_FIGS}
1507     groups = (("paper", "eda", "main", "sensitivity", "compute", "tables")
1508               if which == "all" else (which,))
1509     for group in groups:
1510         for _, fn, needs_panel in table[group]:
1511             before = set(vs.CAPTIONS)
1512             fn(panel) if needs_panel else fn()
1513             produced.extend(sorted(set(vs.CAPTIONS) - before))
1514     return produced
1515
1516
1517 def main(argv=None) -> None:
1518     parser = argparse.ArgumentParser(description="C题 publication-grade visualisation program")
1519     parser.add_argument("--set", default="all",
1520                         choices=["all", "paper", "eda", "main", "sensitivity", "compute",
1521                                "tables"])
1522     parser.add_argument("--list", action="store_true", help="list figure names and captions")
1523     parser.add_argument("--formats", default="pdf",
1524                         help="逗号分隔的导出格式, 默认 pdf (例如 pdf,png) ")
1525     args = parser.parse_args(argv)
1526     vs.setup()
1527     vs.EXPORT_FORMATS = tuple(f.strip() for f in args.formats.split(",") if f.strip())
1528     if args.list:
1529         captions = vs.CAPTIONS
1530         if not captions:
1531             # 这次没出图, 就退回上次存下来的索引
1532             index = FIGURES / "figures_index.json"
1533             captions = load_json(index) if index.exists() else {}
1534             for name, caption in sorted(captions.items()):
1535                 print(f"{name}: {caption}")
1536             return
1537     panel = load_panel()
1538     produced = run_set(args.set, panel)
1539     vs.flush_captions()
1540     print(f"共 {len(produced)} 张图已写入 figures/ (格式: {' '.join(vs.EXPORT_FORMATS)}) ")
1541     for name in produced:
1542         print(f" - {name}")
1543
1544
1545 if __name__ == "__main__":
1546     main()

```

code/report/viz_style.py

```

1  """全项目共用的绘图风格。
2
3  中英文字体、Okabe-Ito 色盲友好配色、内向刻度、面板编号、轴标签带单位都在这设好,
4  默认只出矢量 PDF; 每次出图的图注统一收进 figures/CAPTIONS.md, 写论文直接抄。
5  """
6  from __future__ import annotations
7
8  import json
9  from pathlib import Path
10
11 import matplotlib
12 matplotlib.use("Agg")

```

```

13 import matplotlib.pyplot as plt
14 import numpy as np
15 from matplotlib import rcParams
16
17 ROOT = Path(__file__).resolve().parents[2]
18 FIGDIR = ROOT / "figures"
19
20 # Okabe-Ito 色盲友好配色
21 BLUE = "#0072B2"
22 ORANGE = "#E69F00"
23 GREEN = "#009E73"
24 RED = "#D55E00"
25 PURPLE = "#CC79A7"
26 SKY = "#56B4E9"
27 YELLOW = "#F0E442"
28 GREY = "#7F7F7F"
29 BLACK = "#1A1A1A"
30 PALETTE = [BLUE, ORANGE, GREEN, RED, PURPLE, SKY, YELLOW, GREY]
31 SEQ_CMAP = "viridis"
32 DIV_CMAP = "RdBu_r"
33
34 CAPTIONS: dict[str, str] = {}
35 EXPORT_FORMATS: tuple[str, ...] = ("pdf",)
36
37
38 # 统一字体和线宽; 导出格式默认只有 PDF, 需要别的用 --formats 覆盖
39 def setup(font: str = "Noto Sans CJK SC") -> None:
40     rcParams.update({
41         "font.family": [font, "Noto Sans CJK JP", "DejaVu Sans"],
42         "font.size": 9.5,
43         "axes.labelsize": 10,
44         "axes.titlesize": 10.5,
45         "axes.titlepad": 7,
46         "xtick.labelsize": 9,
47         "ytick.labelsize": 9,
48         "legend.fontsize": 8.5,
49         "legend.frameon": False,
50         "axes.unicode_minus": False,
51         "axes.linewidth": 0.8,
52         "axes.edgecolor": BLACK,
53         "axes.labelcolor": BLACK,
54         "xtick.color": BLACK,
55         "ytick.color": BLACK,
56         "xtick.direction": "in",
57         "ytick.direction": "in",
58         "xtick.top": True,
59         "ytick.right": True,
60         "xtick.major.size": 3.2,
61         "ytick.major.size": 3.2,
62         "xtick.major.width": 0.8,
63         "ytick.major.width": 0.8,
64         "grid.color": "#BFBFBF",
65         "grid.alpha": 0.35,
66         "grid.linewidth": 0.5,
67         "legend.handlelength": 1.6,
68         "legend.borderaxespad": 0.3,
69         "figure.dpi": 120,
70         "savefig.dpi": 400,
71         "savefig.bbox": "tight",
72         "savefig.pad_inches": 0.06,
73         "axes.prop_cycle": plt.cycler(color=PALETTE),
74     })
75
76
77 def panel(ax, letter: str, dx: float = -0.085, dy: float = 1.045) -> None:
78     ax.text(dx, dy, f"({letter})", transform=ax.transAxes, fontsize=10.5,
79           fontweight="bold", va="bottom", ha="left")
80
81
82 def save(fig, name: str, caption: str, formats=None) -> list[str]:

```

```

83     """存图: 默认写 figures/<名字>.pdf, 临时要位图就把 formats 传成 ('pdf', 'png')。"""
84     FIGDIR.mkdir(exist_ok=True)
85     formats = EXPORT_FORMATS if formats is None else formats
86     written = []
87     for fmt in formats:
88         path = FIGDIR / f"{name}.{fmt}"
89         fig.savefig(path, format=fmt)
90         written.append(path.name)
91     CAPTIONS[name] = caption
92     plt.close(fig)
93     return written
94
95
96 def flush_captions() -> None:
97     """把这次的图注并进索引, 不覆盖之前跑过的那些条目。"""
98     index_path = FIGDIR / "figures_index.json"
99     merged: dict[str, str] = {}
100     if index_path.exists():
101         try:
102             merged = json.loads(index_path.read_text(encoding="utf-8"))
103         except json.JSONDecodeError:
104             # 索引坏了就用这次的结果重建
105             merged = {}
106     merged.update(CAPTIONS)
107     (FIGDIR / "CAPTIONS.md").write_text(
108         "# 图目录与图注 (SCI 投稿用) \n\n"
109         "> 全部图由 `python3 -m code.report.visualize --set all` 生成, **统一为矢量 PDF**;"
110         "如需位图可用 `--formats pdf,png` 追加导出。 \n\n"
111         + "\n\n".join(f"**{name}** — {text}" for name, text in sorted(merged.items()))
112         + "\n", encoding="utf-8")
113     index_path.write_text(json.dumps(merged, ensure_ascii=False, indent=2), encoding="utf-8")
114
115
116 def annotate_bars(ax, bars, values, fmt="{:0f}", dy=0.01, fontsize=8.2) -> None:
117     span = max(values) - min(0.0, min(values))
118     for bar, value in zip(bars, values):
119         ax.annotate(fmt.format(value), (bar.get_x() + bar.get_width() / 2, value + dy * span),
120             ha="center", va="bottom", fontsize=fontsize)
121
122
123 def bootstrap_ci(values, n_boot: int = 2000, alpha: float = 0.05, seed: int = 20250911):
124     """均值的百分位自助法置信区间, 种子写死, 重跑结果一样。"""
125     rng = np.random.default_rng(seed)
126     values = np.asarray(values, dtype=float)
127     idx = rng.integers(0, values.size, size=(n_boot, values.size))
128     means = values[idx].mean(axis=1)
129     lo, hi = np.quantile(means, [alpha / 2, 1 - alpha / 2])
130     return float(values.mean()), float(lo), float(hi)

```

code/report/write_results.py

```

1  """照着附件5 模板把五个结果文件写出来。
2
3  模板怎么写就怎么写: 工作表名和表头照抄 (连 `0:00+1-0:10+1` 这种写法也不改)、
4  日期保持日期格式、`:` 行删掉后补齐 334 天, 样式字体全用模板自带的。
5  """
6  from __future__ import annotations
7
8  import datetime as dt
9  import json
10 from pathlib import Path
11
12 import numpy as np
13
14 from code.common.load_clean import ROOT, load_panel
15 from code.common.xlsx_template_io import Workbook, idx_to_col, write_workbook
16
17 RESULTS = ROOT / "results"
18 ARRAYS = RESULTS / "arrays"

```

```

19 TEMPLATES = ROOT / "templates" / "附件5"
20 OUTPUT = ROOT / "output"
21 FIRST_DAY = 31
22 # result2/3/4-* 从 2025-02-01 起 (1 月留给共形校准), Excel 序列号零点取 1899-12-30
23 EPOCH = dt.date(1899, 12, 30)
24 BLOCK_LABELS = [f"{4 * i}:00-{4 * i + 4}:00" for i in range(6)]
25
26
27 def date_serial(iso: str) -> int:
28     y, m, d = (int(part) for part in iso.split("-"))
29     return (dt.date(y, m, d) - EPOCH).days
30
31
32 def load_json(path: Path) -> dict:
33     return json.loads(path.read_text(encoding="utf-8"))
34
35
36 def style_of(wb: Workbook, sheet: str, row: int, col: int, fallback_row: int = 1) -> str | None:
37     style = wb.style(sheet, row, col)
38     return style if style is not None else wb.style(sheet, fallback_row, col)
39
40
41 def header_row(wb: Workbook, sheet: str) -> list[tuple]:
42     cells = wb.sheets[sheet].rows.get(1, {})
43     out = []
44     for col in sorted(cells):
45         text = wb.text(sheet, 1, col)
46         out.append((col, text, "text", wb.style(sheet, 1, col)))
47     return out
48
49
50 def wide_table_rows(wb: Workbook, sheet: str, dates: list[str], values: np.ndarray,
51                     price: np.ndarray, first_day: int = 31) -> list[list[tuple]]:
52     """Wide-table rows: date + 144 slot values + daily energy + daily cost.
53
54     `price` is the full-year price matrix, so the daily cost of row i must use the
55     price row of the same calendar day (`first_day + i`). Using `price[i]` instead
56     would silently shift the tariff by the length of the calibration month — invisible
57     for Q2/Q3 (constant daily tariff) but wrong for Q4, where the price changes daily.
58     """
59     rows = [header_row(wb, sheet)]
60     for i, iso in enumerate(dates):
61         row = [(0, date_serial(iso), "number", style_of(wb, sheet, 2, 0))]
62         for k in range(144):
63             row.append((1 + k, float(values[i, k]), "number", style_of(wb, sheet, 2, 1 + k)))
64         row.append((145, float(values[i].sum()), "number", style_of(wb, sheet, 2, 145)))
65         day_index = first_day + i
66         row.append((146, float(np.sum(price[day_index] * values[i])), "number",
67                     style_of(wb, sheet, 2, 146)))
68         rows.append(row)
69     return rows
70
71
72 def charge_discharge_rows(wb: Workbook, sheet: str, dates: list[str], charge: np.ndarray,
73                           discharge: np.ndarray, soc: np.ndarray,
74                           with_date: bool = True) -> list[list[tuple]]:
75     rows = [header_row(wb, sheet)]
76     for i, iso in enumerate(dates):
77         blocks_charge = charge[i].reshape(6, 24).sum(axis=1)
78         blocks_discharge = discharge[i].reshape(6, 24).sum(axis=1)
79         for b in range(6):
80             template_row = 2 + b
81             first = b == 0
82             second = b == 1
83             row: list[tuple] = []
84             if with_date:
85                 row.append((0, date_serial(iso) if first else None, "number",
86                             style_of(wb, sheet, template_row, 0)))
87             offset = 1
88             else:

```

```

89         offset = 0
90         row.append((offset + 0, BLOCK_LABELS[b], "text",
91                     style_of(wb, sheet, template_row, offset + 0)))
92         row.append((offset + 1, float(blocks_charge[b]), "number",
93                     style_of(wb, sheet, template_row, offset + 1)))
94         row.append((offset + 2, float(blocks_discharge[b]), "number",
95                     style_of(wb, sheet, template_row, offset + 2)))
96         if first:
97             row.append((offset + 3, 0.0, "number", style_of(wb, sheet, template_row, offset
+ 3)))
98             row.append((offset + 4, float(soc[i][0]), "number",
99                         style_of(wb, sheet, template_row, offset + 4)))
100         elif second:
101             row.append((offset + 3, "24:00", "text", style_of(wb, sheet, template_row,
offset + 3)))
102             row.append((offset + 4, float(soc[i][-1]), "number",
103                         style_of(wb, sheet, template_row, offset + 4)))
104         rows.append(row)
105     return rows
106
107
108 def emergency_rows(wb: Workbook, sheet: str, dates: list[str],
109                   records: list[dict]) -> list[list[tuple]]:
110     rows = [header_row(wb, sheet)]
111     by_date = {r["date"]: r for r in records}
112     for iso in dates:
113         record = by_date.get(iso, {"blocks": []})
114         blocks = record.get("blocks", [])
115         if not blocks:
116             rows.append([(0, date_serial(iso), "number", style_of(wb, sheet, 2, 0)),
117                         (1, None, "empty", style_of(wb, sheet, 2, 1)),
118                         (2, None, "empty", style_of(wb, sheet, 2, 2))])
119             continue
120         for j, block in enumerate(blocks):
121             row = [(0, date_serial(iso) if j == 0 else None, "number", style_of(wb, sheet, 2, 0)
),
122                 (1, block["label"], "text", style_of(wb, sheet, 2, 1)),
123                 (2, float(block["energy_kwh"]), "number", style_of(wb, sheet, 2, 2))]
124             rows.append(row)
125     return rows
126
127
128 def write_result1(panel, out_dir: Path) -> dict:
129     q1 = load_json(RESULTS / "q1.json")
130     wb = Workbook(TEMPLATES / "result1.xlsx")
131     slots_rows = [header_row(wb, "计划购电量")]
132     for k in range(144):
133         slots_rows.append([(0, wb.text("计划购电量", k + 2, 0), "text", wb.style("计划购电量", k +
2, 0)),
134                         (1, float(q1["lp"]["q"][k]), "number", style_of(wb, "计划购电量", 2, 1)
))]
135     cd_rows = charge_discharge_rows(wb, "充放电", [panel.dates[0]],
136                                     np.array([q1["lp"]["c"]]), np.array([q1["lp"]["d"]]),
137                                     np.array([q1["lp"]["soc"]]), with_date=False)
138     write_workbook(wb, out_dir / "result1.xlsx",
139                   {"计划购电量": slots_rows, "充放电": cd_rows})
140     wb.close()
141     return {"file": "result1.xlsx", "sheets": {"计划购电量": len(slots_rows),
142                                             "充放电": len(cd_rows)}}
143
144
145 def write_wide_result(template_name: str, out_name: str, panel, tags: list[str],
146                       kinds: list[str]) -> dict:
147     """"tags/kinds: [('q2', 'plan')] etc. — 计划购电量 always comes first."""
148     wb = Workbook(TEMPLATES / template_name)
149     plan_tag, plan_kind = tags[0], kinds[0]
150     plan = np.load(ARRAYS / f"{plan_tag}_{plan_kind}.npy")
151     plan_cost_price = price_matrix_for(plan_tag, panel)
152     sheets: dict[str, list[list[tuple]]] = {
153         "计划购电量": wide_table_rows(wb, "计划购电量", list(panel.dates[FIRST_DAY:365]), plan,

```

```

154         plan_cost_price, FIRST_DAY)
155     }
156     if len(tags) > 1:
157         adj_tag, adj_kind = tags[1], kinds[1]
158         adj = np.load(ARRAYS / f"{adj_tag}_{adj_kind}.npy")
159         sheets["调整购电量"] = wide_table_rows(wb, "调整购电量",
160                                                list(panel.dates[FIRST_DAY:365]), adj,
161                                                price_matrix_for(adj_tag, panel), FIRST_DAY)
162         summary = load_json(RESULTS / f"{plan_tag}.json")["summary"]
163         records = load_json(RESULTS / f"{plan_tag}.json")["records"]
164         charge = np.load(ARRAYS / f"{plan_tag}_charge.npy")
165         discharge = np.load(ARRAYS / f"{plan_tag}_discharge.npy")
166         soc = np.load(ARRAYS / f"{plan_tag}_soc.npy")
167         sheets["充放电电量"] = charge_discharge_rows(wb, "充放电电量", list(panel.dates[FIRST_DAY:365]),
168                                                       charge, discharge, soc)
169         sheets["紧急购电量"] = emergency_rows(wb, "紧急购电量", list(panel.dates[FIRST_DAY:365]),
170                                               records)
171         write_workbook(wb, OUTPUT / out_name, sheets)
172         wb.close()
173         return {"file": out_name, "rows": {k: len(v) for k, v in sheets.items()},
174               "summary": summary}
175
176 def price_matrix_for(tag: str, panel) -> np.ndarray:
177     """Q2/Q3 settle against the 附件1 tariff; Q4 against 附件4."""
178     if tag.startswith("q4"):
179         return panel.price
180     return np.tile(panel.day1[:, 0], (panel.price.shape[0], 1))
181
182
183 def main() -> None:
184     OUTPUT.mkdir(exist_ok=True)
185     panel = load_panel()
186     manifest = {"files": []}
187     manifest["files"].append(write_result1(panel, OUTPUT))
188     manifest["files"].append(write_wide_result("result2.xlsx", "result2.xlsx", panel,
189                                               ["q2"], ["q_plan"]))
190     manifest["files"].append(write_wide_result("result3.xlsx", "result3.xlsx", panel,
191                                               ["q3", "q3"], ["q_plan", "q_adjust"]))
192     manifest["files"].append(write_wide_result("result4-2.xlsx", "result4-2.xlsx", panel,
193                                               ["q4_2"], ["q_plan"]))
194     manifest["files"].append(write_wide_result("result4-3.xlsx", "result4-3.xlsx", panel,
195                                               ["q4_3", "q4_3"], ["q_plan", "q_adjust"]))
196     (RESULTS / "result_files.json").write_text(
197         json.dumps(manifest, ensure_ascii=False, indent=2), encoding="utf-8")
198     for entry in manifest["files"]:
199         print(f"[write] {entry['file']}: " +
200               ", ".join(f"{k}={v}" for k, v in entry["rows"].items()) if "rows" in entry
201               else f"[write] {entry['file']}: " + str(entry["sheets"]))
202
203
204 if __name__ == "__main__":
205     main()

```

H.7 工具 tools

tools/cross_verify.py

```

1  """独立复核: 用 openpyxl 只读地把原始附件、clean 数据和五个结果文件对一遍。
2
3  跑法 (要装了 openpyxl 的 Python):
4      /usr/bin/python3 tools/cross_verify.py
5  它不在主流流水线里, 属于另开一条通道的交叉检查。
6  """
7  from __future__ import annotations
8
9  import ast

```

```

10 import json
11 import random
12 import struct
13 from pathlib import Path
14
15 import openpyxl
16
17 ROOT = Path(__file__).resolve().parents[1]
18 OUT: dict = {"checks": [], "all_pass": True}
19
20
21 def record(name: str, ok: bool, detail: str) -> None:
22     OUT["checks"].append({"name": name, "pass": bool(ok), "detail": detail})
23     OUT["all_pass"] = OUT["all_pass"] and bool(ok)
24
25
26 def read_npy(path: Path):
27     """Minimal C-order .npz reader (float64 / int64), stdlib only."""
28     raw = path.read_bytes()
29     if raw[:6] != b"\x93NUMPY":
30         raise ValueError(f"{path} is not a .npz file")
31     header_len = struct.unpack("<H", raw[8:10])[0]
32     header = ast.literal_eval(raw[10:10 + header_len].decode("latin1").strip())
33     start = 10 + header_len
34     fmt = {"<f8": "d", "<i8": "q"}[header["descr"]]
35     count = 1
36     for dim in header["shape"]:
37         count *= dim
38     values = struct.unpack("<" + fmt * count, raw[start:start + count * struct.calcsize(fmt)])
39     return header["shape"], list(values)
40
41
42 def main() -> None:
43     # 1) 时段映射: clean 矩阵抽几个格子, 和原始附件单元格逐个对
44     shape, load = read_npy(ROOT / "clean" / "load.npy")
45     _, pv = read_npy(ROOT / "clean" / "pv_actual_raw.npy")
46     _, price = read_npy(ROOT / "clean" / "price.npy")
47     att2 = openpyxl.load_workbook(ROOT / "Data" / "附件2.xlsx", data_only=True)
48     att4 = openpyxl.load_workbook(ROOT / "Data" / "附件4.xlsx", data_only=True)
49     load_ws = att2["小区负载"]
50     pv_ws = att2["光伏发电实际功率"]
51     price_ws = att4[att4.sheetnames[0]]
52     rng = random.Random(20250911)
53     samples = [(rng.randrange(shape[0]), rng.randrange(shape[1])) for _ in range(3)]
54     for day, k in samples:
55         row, col = day + 2, k + 2
56         ok = (abs(load[day * 144 + k] - float(load_ws.cell(row, col).value)) < 1e-9
57              and abs(pv[day * 144 + k] - float(pv_ws.cell(row, col).value)) < 1e-9
58              and abs(price[day * 144 + k] - float(price_ws.cell(row, col).value)) < 1e-9)
59         record(f"slot mapping day {day} slot {k+1}", ok,
60              f"clean=({load[day*144+k]:.4f},{pv[day*144+k]:.4f},{price[day*144+k]:.4f}) vs "
61              f"attachment row {row} col {col}")
62
63     # 2) 宽表里的数要跟存下来的计划数组一致
64     _, q2_plan = read_npy(ROOT / "results" / "arrays" / "q2_q_plan.npy")
65     wb = openpyxl.load_workbook(ROOT / "output" / "result2.xlsx", data_only=True)
66     ws = wb["计划购电量"]
67     mismatches = 0
68     # 全年抽 10 行、每行抽 9 个时段做点检
69     for i in range(0, 334, 37):
70         for k in range(0, 144, 17):
71             if abs(float(ws.cell(i + 2, k + 2).value) - q2_plan[i * 144 + k]) > 1e-6:
72                 mismatches += 1
73     record("result2 wide table matches stored plan array", mismatches == 0,
74          f"{mismatches} mismatching spot-checked cells (90 samples)")
75
76     # 3) 五个结果文件必须复刻附件5 模板的表名和表头
77     for name in ("result1.xlsx", "result2.xlsx", "result3.xlsx", "result4-2.xlsx", "result4-3.
78                 .xlsx"):
79         tpl = openpyxl.load_workbook(ROOT / "templates" / "附件5" / name)

```

```

79     cur = openpyxl.load_workbook(ROOT / "output" / name)
80     same_sheets = tpl.sheetnames == cur.sheetnames
81     header_ok = True
82     for sheet in tpl.sheetnames:
83         a, b = tpl[sheet], cur[sheet]
84         for col in range(1, a.max_column + 1):
85             if a.cell(1, col).value != b.cell(1, col).value:
86                 header_ok = False
87     record(f"{name} sheet names and headers identical to attachment-5 template",
88           same_sheets and header_ok, f"sheets={cur.sheetnames}")
89
90     ws = wb["计划购电量"]
91     date_ok = all(hasattr(ws.cell(r, 1).value, "year") for r in (2, 100, 335))
92     record("result2 dates are Excel date cells", date_ok,
93           str([ws.cell(r, 1).value for r in (2, 100, 335)]))
94     total_ok = True
95     for r in (2, 150, 335):
96         row_sum = sum(float(ws.cell(r, c).value) for c in range(2, 146))
97         total_ok = total_ok and abs(row_sum - float(ws.cell(r, 146).value)) < 1e-6
98     record("daily purchase column equals the sum of the 144 slots", total_ok,
99           "checked rows 2 / 150 / 335")
100     (ROOT / "results" / "cross_verify_xlsx.json").write_text(
101         json.dumps(OUT, ensure_ascii=False, indent=2), encoding="utf-8")
102     for check in OUT["checks"]:
103         print(f"[{'PASS' if check['pass'] else 'FAIL'}] {check['name']} :: {check['detail']}")
104     print(f"[cross-verify] all_pass = {OUT['all_pass']}")
105
106
107 if __name__ == "__main__":
108     main()

```